FRM PART II • 2026 • EXAM 25 NOVEMBER 2026

Market Risk

Notes + Question Workbook

P1 • P2 • P3, in order of importance

72 learning objectives from 18 readings, banded by the Master LO Map v3. P4 is excluded. Built from Vardeez's notes, GARP 2026 Market Risk, SchweserNotes 2026 Book 1, the Tier A and B–D Recall Workbooks and the MR Question Bank.

How to use this book

- The book runs **P1, then P2, then P3**. Inside each part, LOs follow syllabus order. The syllabus index at the front lists all 72 LOs in GARP order with their band and page.
- Each LO is one block: **focus box** (why it sits in its band, Vardeez's pointer and the reported exam shapes), **notes** limited to that focus, a **formula card** or comparison table, **exam traps**, then **6 to 10 practice questions**. **Solutions** sit at the end of each block, so attempt first.
- LO wording is verbatim from the official GARP 2026 learning objectives, as listed in the Master LO Map.
- Type tags: CALC numerical questions expected; CONCEPT statement questions (only the reported workbook calculations appear); BOTH a mix set by the reported shapes and Vardeez's focus.
- Question sources, in priority order: Tier workbook (August 2026 recall and mapped shapes), GARP-tagged bank questions (practice exams, GARP book, mocks), other bank questions, and *New* questions written in GARP style from the readings where an LO was short. Every number has been recomputed.
- Green-edged boxes mark corrections to the source notes or to a bank key. Figures are drawn for black-and-white printing.

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MR-6 • Validating Bank Holding Companies' Value-at-Risk Models for Market Risk

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MR-10 • Financial Correlation Modeling: Bottom-Up Approaches

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MR-11 • Regression Hedging and Principal Component Analysis

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MR-18 • Fundamental Review of the Trading Book

MR-18 a P1 CONCEPT 276
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MARKET RISK • PART

P1

Most important

Tier A recall topics and the concepts Vardeez flags in Very Important chapters. 30 LOs from 9 readings.

MR-1 • Estimating Market Risk Measures: An Introduction and Overview • Vardeez: VERY IMPORTANT

MR-1 b

P1 CALC

Estimate VaR using a parametric approach for both normal and lognormal asset price distributions.

Workbook: A13 Lognormal VaR • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: lognormal VaR came up in the August 2026 sitting ("straightforward formula, 2Q").
Vardeez: calculation of normal and lognormal VaR.

Reported exam shapes

- **S1** Choose between two volatilities (16% rising to 18% six weeks ago): which regime governs the VaR horizon.
- **S2** Direct lognormal VaR from the formula, the plain form seen in August.
- **S3** Keep the lognormal form $1-e^{\mu-z\sigma}$ distinct from the normal form $-\mu+z\sigma$. GARP places the normal answer right next to the lognormal one.

What the questions look like: almost all numerical. You are given μ, σ, a horizon and a confidence level, sometimes with extra inputs (beta, share count, portfolio value when a percentage is asked) that you do not need.

1. What "parametric" means here

A parametric approach *assumes a distribution* for the data, estimates its parameters (mean and standard deviation), and reads VaR off the fitted distribution's quantile. Historical simulation (MR-1 a) assumes nothing; the parametric route assumes everything. The single decision that drives the formula is **which variable is assumed normal**:

Assumed normal	Implies	Formula
P/L (in currency)	P/L normal	Normal VaR, currency
Arithmetic returns r_t	P_t normal	Normal VaR $\times P_{t-1}$
Geometric returns R_t	$\ln P_t$ normal, P_t **lognormal**	Lognormal VaR

Arithmetic versus geometric returns

$$r_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}} \qquad R_t = \ln\left(\frac{P_t + D_t}{P_{t-1}}\right) = \ln(1 + r_t)$$

- **Arithmetic:** interim payments are *not* reinvested. Not appropriate for long horizons.
- **Geometric:** interim payments are *continuously reinvested*; since $P_t = P_{t-1}e^{R_t}$, the price **can never be negative**. Geometric returns add over time.
- $R_t \le r_t$ always, since $\ln(1+r) \le r$, with equality only at $r = 0$. Example: $r = 5\% \Rightarrow R = 4.88\%$; $R = 5\% \Rightarrow r = 5.13\%$. The two are close for short periods and diverge over long ones.

2. Normal VaR

VaR is the loss at the cut-off in the left tail. The quantile itself is negative (a loss), but VaR is *reported as a positive number*.

$$\boxed{\text{VaR}_\alpha = -\mu_{P/L} + \sigma_{P/L}\, z_\alpha} \qquad \boxed{\text{VaR}_\alpha = (-\mu_r + \sigma_r\, z_\alpha)\, P_{t-1}}$$

Use the first when the data are in currency (P/L), the second when they are percentage returns. With the same underlying assumptions, P/L, L/P and arithmetic-return VaR are the same thing in different units.

- **P/L.** $\mu = \$15\text{m}$, $\sigma = \$10\text{m}$: $\text{VaR}_{95} = -15 + 10(1.645) = \$1.45\text{m}$; $\text{VaR}_{99} = -15 + 10(2.326) = \$8.26\text{m}$.
- **Returns.** $P_{t-1} = \$100$, $\mu_r = 10\%$, $\sigma_r = 20\%$: $\text{VaR}_{95} = (-0.10 + 0.20 \times 1.645) \times 100 = \$22.90$; $\text{VaR}_{99} = \$36.52$.

Drawback. A normal distribution for P_t gives a positive probability to the asset price becoming negative. This is the reason for the lognormal approach.

3. Lognormal VaR

If geometric returns are $N(\mu_R, \sigma_R)$, then $\ln P_t$ is normal and P_t is lognormal: **right-skewed, bounded below by zero**. Work in three moves:

$$\begin{aligned} R^* &= \mu_R - \sigma_R\, z_\alpha && \text{critical log return} \\ P^* &= P_{t-1}\, \mathrm{e}^{R^*} && \text{critical portfolio value} \\ \text{VaR}_\alpha &= P_{t-1} - P^* = \boxed{P_{t-1}\left(1 - \mathrm{e}^{\mu_R - \sigma_R z_\alpha}\right)} \end{aligned}$$

Set $P_{t-1} = 1$ when VaR is asked as a percentage of portfolio value.

Example. $\mu_R = 10\%$, $\sigma_R = 20\%$, $P_{t-1} = \$100$:

$$\text{VaR}_{95} = 100\left(1 - \mathrm{e}^{-0.229}\right) = \$20.47 \qquad \text{VaR}_{99} = 100\left(1 - \mathrm{e}^{-0.3652}\right) = \$30.59$$

Properties to remember

- Lognormal VaR **can never exceed the portfolio value**: you cannot lose more than 100%.
- With the *same* μ and σ, lognormal VaR is **smaller** than normal VaR, because $1 - \mathrm{e}^{-x} < x$ for $x > 0$.
- Standardised case ($\mu = 0,\ \sigma = 1$): 95% lognormal VaR $= 1 - \mathrm{e}^{-1.645} = 0.807$, against the normal 1.645.

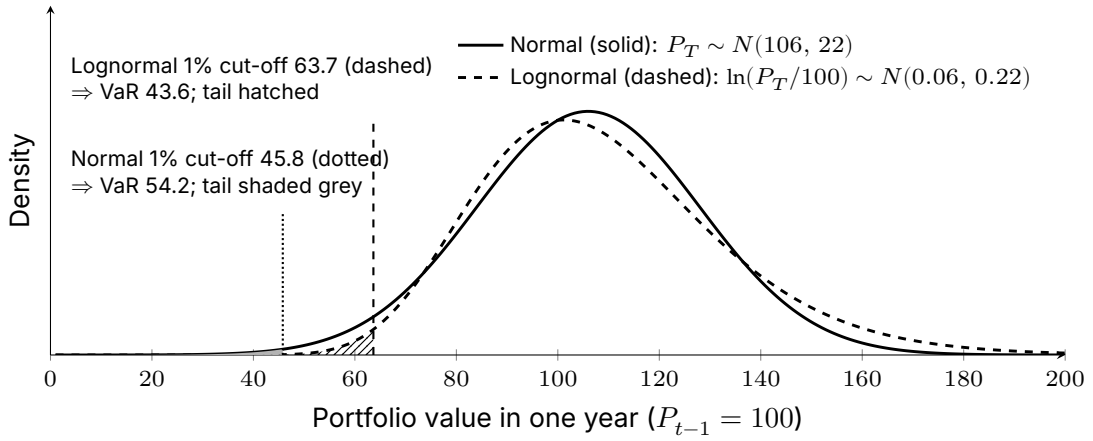


Figure 1. The same inputs ($\mu = 6\%$, $\sigma = 22\%$, one year, 99%) read through the two assumptions (these are the numbers of Question MR-1 b.6). The lognormal curve is skewed right and stops at zero, so its 1% cut-off sits higher and the VaR is smaller.

4. Normal versus lognormal: when the gap matters

- **Short horizons, realistic parameters: "much the same".** Dowd's Example 1.5: annual $\mu = 10\%$, $\sigma = 40\%$, 1-day 95%: normal 4.12%, lognormal 4.04%.
- **Long horizons, high volatility: material.** The gap grows with the horizon and with the confidence level (Figure 2). Dowd: use geometric returns for longer periods.
- **GARP's twist.** When the stem gives *separate* parameter sets for arithmetic and geometric returns, each VaR uses its own set. Lognormal VaR can then come out *higher*, simply because the geometric volatility given is higher (Question MR-1 b.4).

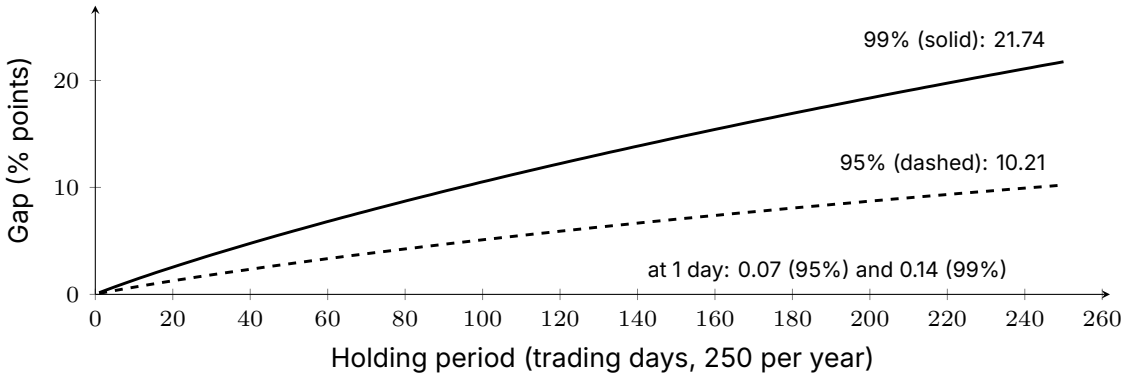


Figure 2. Gap between normal and lognormal VaR for $\mu = 12\%$, $\sigma = 37\%$ (the inputs of Question MR-1 b.5). Negligible at one day, about 22 percentage points at one year and 99%.

5. Inputs that change the answer

Horizon scaling

Given annual parameters and h trading days (use the day count in the stem, 250 or 252):

$$\mu_h = \mu \cdot \frac{h}{250} \qquad \sigma_h = \sigma\sqrt{\frac{h}{250}}$$

The mean scales with time; the volatility scales with the square root of time.

Which volatility? (shape S1)

VaR is a *forecast* for the coming holding period, so σ must describe the regime that governs that period. If volatility rose from 16% to 18% six weeks ago and has stayed there, the 18% regime is current and has lasted longer than a 10-day horizon: use 18%. This is the volatility-weighting logic of MR-2 c: older, calmer data "understate the changes we can expect to see tomorrow".

Reverse-solving for the volatility

From $V = 1 - \mathrm{e}^{\mu_R - \sigma_R z}$: $\quad \ln(1-V) = \mu_R - \sigma_R z \quad \Rightarrow \quad \sigma_R = \dfrac{\mu_R - \ln(1-V)}{z_\alpha}.$

6. Reading a VaR number

- **Positive VaR (a loss).** 99% VaR of \$5,000: "99% confident the loss will not exceed \$5,000", or equivalently "a 1% chance the loss will be \$5,000 or more".
- **Negative VaR (a profit).** Both statements reverse. 99% VaR of $-\$5{,}000$: "99% confident the profit will be at least \$5,000", or "a 1% chance the profit will be less than \$5,000".

Correction to the source notes. Vardeez's Example 4 ($\mu = \$20$m, $\sigma = \$10$m, 5% VaR $= -\$3.50$m) marks the answer as B, "5% probability of earnings of at least \$3.50m". That is the wrong tail: only 5% of outcomes fall *below* $+\$3.50$m. The correct choice is D, "95% probability of earnings of at least \$3.50m", which is the key in Schweser's Module Quiz 1.1. It is worked as Question MR-1 b.2.

Formula card: MR-1 b

Normal VaR, P/L	$\mathrm{VaR} = -\mu_{P/L} + \sigma_{P/L}\, z_\alpha$
Normal VaR, returns	$\mathrm{VaR} = (-\mu_r + \sigma_r z_\alpha)\, P_{t-1}$
Lognormal VaR	$R^* = \mu_R - \sigma_R z_\alpha, \quad \mathrm{VaR} = P_{t-1}\,(1 - \mathrm{e}^{R^*})$
Horizon scaling	$\mu_h = \mu\, h/250, \quad \sigma_h = \sigma\sqrt{h/250}$
Sum of independent P/Ls	$\mu_P = \sum \mu_i, \quad \sigma_P^2 = \sum \sigma_i^2$ (variances add, not standard deviations)
Reverse-solve	$\sigma_R = (\mu_R - \ln(1-V))\,/z_\alpha$
One-tailed z	90%: 1.282 95%: 1.645 99%: 2.326 (two-tailed traps: 1.96, 2.576)

	Normal VaR	Lognormal VaR
Assumes normal	P/L or arithmetic returns	Geometric (log) returns
Price distribution	Normal: can go negative	Lognormal: bounded at zero, right-skewed
Formula	$-\mu + \sigma z$	$1 - \mathrm{e}^{\mu - \sigma z}$
Maximum VaR	Unbounded	Below P_{t-1} (never more than 100%)
Same μ, σ	Larger	Smaller; gap grows with horizon and confidence
Best for	Short horizons	Long horizons (Dowd)

Exam traps

- **Wrong formula twin.** Plugging the inputs into $-\mu + \sigma z$ when lognormal is asked (or the reverse). The result is slightly larger, and GARP puts it next to the correct option.
- **Stale volatility.** Using the old regime, or an average across both, when the stem says volatility changed recently. This understates VaR.
- **$\sqrt{t}$ errors.** Scaling σ by h/T instead of $\sqrt{h/T}$, or μ by $\sqrt{h/T}$ instead of h/T. Also stopping at the 1-day figure when 10-day is asked.
- **Dropped mean.** Using $1 - \mathrm{e}^{-\sigma z}$ or σz alone. This overstates VaR when $\mu > 0$.
- **Wrong critical value.** 1.645 for a 99% question, 2.326 for 95%, or a two-tailed 1.96 or 2.576.
- **Sign inside the exponent.** The critical return is $\mu_R - \sigma_R z$, a left-tail value. Writing $\ln(1+V)$ in a reverse-solve flips the tail.
- **Adding standard deviations.** For independent positions the variances add; summing the σ's is the perfect-correlation case.
- **Negative VaR read as a loss.** A negative VaR is a profit, so both the confidence statement and the tail statement reverse.

Practice questions: MR-1 b

Question MR-1 b.1

Bank Q37 • AnalystPrep Q.3036 (rebuilt)

A portfolio $P = A + B$ consists of two independent, normally distributed positions with $A \sim N(0.5,\ 1)$ and $B \sim N(3,\ 15)$, where the parameters are the mean and the variance of profit/loss. What is the 95% parametric (normal) VaR of the portfolio?

A. 6.58

B. 4.87

C. 3.08

D. 1.54

Question MR-1 b.2

Bank Q34 • Schweser Module Quiz 1.1 Q2

Assume the profit/loss distribution for XYZ is normally distributed with an annual mean of \$20 million and a standard deviation of \$10 million. The 5% VaR is calculated and interpreted as which of the following statements?

A. 5% probability of losses of at least \$3.50 million.

B. 5% probability of earnings of at least \$3.50 million.

C. 95% probability of losses of at least \$3.50 million.

D. 95% probability of earnings of at least \$3.50 million.

Question MR-1 b.3

Workbook A13.2 • S2, S3 • Aug-2026 • Map #14

A risk consultant at an Indian mutual fund is reporting VaR as a percentage of portfolio value. The fund's geometric returns are normally distributed with an annualised mean of 12% and an annualised volatility of 30%. The fund is worth INR 8,000 million and its benchmark beta is 1.05. Assuming 250 trading days per year, which of the following is closest to the 10-day 95% lognormal VaR, expressed as a percentage of portfolio value?

A. 3.03%

B. 8.96%

C. 9.39%

D. 12.61%

Question MR-1 b.4

Bank Q31 • GARP PE1 Q2 • Mock 3 Q78

A risk manager is estimating the market risk of a portfolio under two assumptions: normally distributed arithmetic returns (normal VaR) and normally distributed geometric returns (log-normal VaR). The manager gathers the following data on the portfolio's daily returns:

Annualized average of arithmetic returns	16%
Annualized standard deviation of arithmetic returns	27%
Annualized average of geometric returns	13%
Annualized standard deviation of geometric returns	29%
Current portfolio value	EUR 5,200,000
Trading days in a year	252

Assuming both daily arithmetic returns and daily geometric returns are serially independent, which of the following statements is correct?

A. The 1-day 95% normal VaR is 2.12% and the 1-day 95% lognormal VaR is 2.26%.

B. The 1-day 95% normal VaR is 2.80% and the 1-day 95% lognormal VaR is 2.96%.

C. The 1-day 95% normal VaR is 2.73% and the 1-day 95% lognormal VaR is 2.91%.

D. The 1-day 95% normal VaR is 3.27% and the 1-day 95% lognormal VaR is 3.47%.

Question MR-1 b.5

Bank Q30 • BT P2.T5.707.3

A risk manager is estimating the market risk of a portfolio using both the normal distribution and the lognormal distribution assumptions. The manager gathers the following data on the portfolio:

- Annual mean: 12.0%
- Annual volatility: 37.0%
- Current portfolio value: USD 1.0 million, due to 12,500 shares at a current price of USD 80.00 per share
- Trading days in a year: 250

Which of the following statements is correct?

A. Lognormal 95% VaR is greater than normal 95% VaR at the one-day holding period by about 2.43%.

B. Lognormal 95% VaR is less than normal 95% VaR at the one-year (250 days) holding period by about 3.90%.

C. Lognormal 99% VaR is greater than normal 99% VaR at the one-day holding period by about 4.04%.

D. Lognormal 99% VaR is less than normal 99% VaR at the one-year (250 days) holding period by about 21.75%.

Question MR-1 b.6

Workbook A13.4 • S2, S3 • Aug-2026

The CRO of an Australian pension fund asks a risk analyst to compare normal and lognormal VaR for a long-horizon equity mandate worth AUD 120 million. Annual returns have a mean of 6% and a volatility of 22%, and dividends are reinvested. The CRO wants the figure under the assumption that geometric returns are normal. Which of the following is closest to the 1-year 99% VaR?

A. AUD 31.3 million

B. AUD 43.6 million

C. AUD 48.1 million

D. AUD 54.2 million

Question MR-1 b.7 Workbook A13.3 • S1, S3 • Map #14

A market risk analyst at a US asset manager is computing the 10-day 99% VaR of a USD 50 million equity portfolio for the weekly risk report. Geometric returns are assumed normally distributed, dividends are reinvested and the desk uses 250 trading days per year. The desk's volatility model reports:

- Annualised mean of geometric returns: 8%
- Annualised volatility of geometric returns: 16% for most of the past year, increasing to 18% six weeks ago and unchanged since

The analyst must choose between the two volatilities. Which of the following is closest to the 10-day 99% lognormal VaR?

A. USD 3.44 million

B. USD 3.87 million

C. USD 4.03 million

D. USD 4.28 million

Question MR-1 b.8 Workbook A13.1 • S3 • self-generated

A model validator at a Singapore insurer is reviewing an equity sleeve whose reported 1-year 99% lognormal VaR is 35% of portfolio value. The sleeve's geometric returns have an annual mean of 7%. Using $z_{0.99} = 2.326$, which of the following is closest to the annual volatility of geometric returns implied by the reported VaR?

A. 15.9%

B. 18.1%

C. 18.5%

D. 21.5%

Solutions • MR-1 b

Solution MR-1 b.1 — Answer: C

Bank Q37 • AnalystPrep Q.3036

Step 1 – Distribution of the portfolio P/L

A sum of independent normals is normal: $\mu_P = 0.5 + 3 = 3.5$. Independence means zero covariance, so the *variances* add: $\sigma_P^2 = 1 + 15 = 16$, $\sigma_P = 4$.
$\Rightarrow$ The portfolio P/L is $N(3.5,\ 4)$.

Step 2 – Normal VaR

$$\text{VaR}_{95} = -\mu_P + z_{0.95}\,\sigma_P = -3.5 + 1.645 \times 4 = -3.5 + 6.58 = 3.08$$

$\Rightarrow$ 5% of the time the portfolio loses 3.08 or more.

Interpretation

The expected profit of 3.5 offsets part of the 6.58 tail term. The question tests two things at once: that variances (not standard deviations) add for independent positions, and that the mean enters VaR.

Why the other options are wrong

A is incorrect. Dropped the mean: $1.645 \times 4 = 6.58$.
B is incorrect. Added the standard deviations, $\sqrt{1} + \sqrt{15} = 1 + 3.873 = 4.87$ (the perfect-correlation case), and reported that as the VaR with no z or mean.
D is incorrect. Treated P as the average $(A + B)/2$: mean 1.75, variance 4, $\sigma = 2$, so $-1.75 + 1.645 \times 2 = 1.54$. The portfolio is the sum, not the average.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. •
Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.2 Estimating Normal VaR

Solution MR-1 b.2 — Answer: D

Bank Q34 • Schweser Module Quiz 1.1 Q2

Step 1 – Normal VaR from P/L

$$\text{VaR}_{5\%} = -\mu_{P/L} + z\,\sigma_{P/L} = -20 + 1.65 \times 10 = -\$3.50 \text{ million}$$

$\Rightarrow$ The VaR is negative, so the 5th percentile of P/L is a *profit*: $20 - 16.5 = +\$3.50$ million.

Step 2 – Apply the alternate (profit) interpretation

$\Pr(P/L < +3.50) = 5\%$, so $\Pr(P/L \geq +3.50) = 95\%$.
$\Rightarrow$ With 95% probability, earnings are at least \$3.50 million.

Interpretation

A negative VaR means even the bad tail is profitable. Both the confidence statement and the tail statement flip from losses to profits. (Vardeez's notes key this example as B; see the correction box in Section 6.)

Why the other options are wrong

A is incorrect. Ignored the sign and read $-\$3.50$m as a \$3.50m loss.
B is incorrect. Wrong tail: only 5% of outcomes fall *below* \$3.50m; earnings of at least \$3.50m happen 95% of the time.
C is incorrect. Wrong sign and wrong tail: at the 95% level there are no losses here at all.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. •

Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.2; SchweserNotes 2026 Book 1, Reading 1, LO 1.b

Solution MR-1 b.3

Answer: B

Workbook A13.2 • S2, S3 • Aug-2026

Step 1 – Scale the mean to 10 days

$$\mu_{10} = 0.12 \times \frac{10}{250} = 0.0048$$

$\Rightarrow$ Expected 10-day log return of 0.48%: the mean scales linearly with time.

Step 2 – Scale the volatility to 10 days

$$\sigma_{10} = 0.30 \times \sqrt{\frac{10}{250}} = 0.30 \times 0.2 = 0.06$$

$\Rightarrow$ 10-day volatility of 6%: volatility scales with $\sqrt{t}$.

Step 3 – Critical geometric return

$$R^* = 0.0048 - 0.06 \times 1.645 = -0.0939$$

Step 4 – Lognormal VaR as a fraction ($P_{t-1} = 1$)

$$\text{VaR} = 1 - \text{e}^{-0.0939} = 0.0896 = 8.96\%$$

$\Rightarrow$ Over 10 days there is a 5% chance the fund loses more than 8.96% of its value (about INR 717 million).

Interpretation

Reporting as a percentage makes the number comparable across funds of different size. The portfolio value and beta are not needed for a percentage VaR. Scaling volatility linearly (1.48%) or scaling the mean by $\sqrt{10/250}$ (7.20%) are the other classic slips on this template.

Why the other options are wrong

A is incorrect. Computed the 1-day VaR and stopped: $1 - \text{e}^{0.12/250 - (0.30/\sqrt{250}) \times 1.645} = 3.03\%$.
C is incorrect. Applied the normal (arithmetic) formula to the same inputs: $-0.0048 + 0.06 \times 1.645 = 9.39\%$. Close, but it is the normal VaR, not the lognormal VaR.
D is incorrect. Used the 99% critical value 2.326 instead of 1.645: $1 - \text{e}^{0.0048 - 0.06 \times 2.326} = 12.61\%$.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. •
Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.3, eqs. 1.14 and 1.16, Example 1.5

Solution MR-1 b.4

Answer: C

Bank Q31 • GARP PE1 Q2

Step 1 – Scale the arithmetic inputs to one day

$$\mu_a = \frac{0.16}{252} = 0.000635, \qquad \sigma_a = \frac{0.27}{\sqrt{252}} = 0.017008$$

Step 2 – Scale the geometric inputs to one day

$$\mu_g = \frac{0.13}{252} = 0.000516, \qquad \sigma_g = \frac{0.29}{\sqrt{252}} = 0.018268$$

$\Rightarrow$ Each VaR uses its own return series: arithmetic for normal VaR, geometric for lognormal VaR.

Step 3 – Normal VaR

$$\text{VaR}_N = -0.000635 + 1.645 \times 0.017008 = 0.027344 = 2.73\%$$

Step 4 – Lognormal VaR

$$\text{VaR}_{LN} = 1 - \text{e}^{0.000516 - 1.645 \times 0.018268} = 1 - \text{e}^{-0.029534} = 2.91\%$$

Interpretation

Lognormal VaR comes out *higher* here only because the geometric volatility given (29%) is higher than the arithmetic one (27%). With identical inputs it would be lower. The portfolio value is not needed for percentage VaR (in currency: about EUR 142,190 and EUR 151,340).

Why the other options are wrong

A is incorrect. Used the 90% quantile 1.282 instead of 1.645: 2.12% and 2.26%.
B is incorrect. Ignored the daily mean: $1.645\,\sigma_a = 2.80\%$ and $1 - \text{e}^{-1.645\,\sigma_g} = 2.96\%$.
D is incorrect. Used the two-tailed 1.96 instead of the one-tailed 1.645: 3.27% and 3.47%.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. •
Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.2 and 1.3, eqs. 1.14 and 1.16

Solution MR-1 b.5 **Answer: D**

Bank Q30 • BT P2.T5.707.3

Step 1 – Set up both formulas for horizon T (in years)

$$\text{VaR}_N = -0.12\,T + z\,(0.37\sqrt{T}), \qquad \text{VaR}_{LN} = 1 - \text{e}^{0.12\,T - z\,(0.37\sqrt{T})}$$

Step 2 – One day ($T = 1/250$)

95%: normal 3.80%, lognormal 3.73% (gap 0.07%). 99%: normal 5.40%, lognormal 5.25% (gap 0.14%).
$\Rightarrow$ At one day lognormal VaR is *less* than normal VaR, and only slightly. A and C are wrong on direction.

Step 3 – One year ($T = 1$)

95%: normal $-0.12 + 1.645 \times 0.37 = 48.87\%$; lognormal $1 - \text{e}^{0.12 - 0.6087} = 38.65\%$; gap 10.21%.
99%: normal $-0.12 + 2.326 \times 0.37 = 74.06\%$; lognormal $1 - \text{e}^{0.12 - 0.8606} = 52.32\%$; gap 21.74%.
$\Rightarrow$ The gap grows with both the horizon and the confidence level.

Interpretation

With identical μ and σ, lognormal VaR is always below normal VaR because $1 - \text{e}^{-x} < x$. At one year and 99% the difference is about 21.75 percentage points (Figure 2). The share count and price are a red herring: they only restate the USD 1 million value.

Why the other options are wrong

A is incorrect. Wrong direction and size: at one day, 95%, lognormal is *below* normal, by about 0.07%.
B is incorrect. Right direction, wrong size: the one-year 95% gap is 10.21%, not 3.90%.
C is incorrect. Wrong direction: at one day, 99%, lognormal (5.25%) is below normal (5.40%), a gap of about 0.14%.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.3, Example 1.5

Solution MR-1 b.6 — Answer: B

Workbook A13.4 • S2, S3 • Aug-2026

Step 1 – Critical geometric return (1-year horizon, no scaling needed)

$$R^* = \mu_R - \sigma_R z_{0.99} = 0.06 - 0.22 \times 2.326 = -0.45172$$

$\Rightarrow$ The 1st percentile log return over one year is -45.17%.

Step 2 – Critical portfolio value ratio

$$\mathrm{e}^{R^*} = \mathrm{e}^{-0.45172} = 0.63653$$

$\Rightarrow$ In the 1% worst case the portfolio keeps about 63.7% of its value.

Step 3 – Lognormal VaR

$$\text{VaR} = 120 \times (1 - 0.63653) = \text{AUD } 43.6 \text{ million}$$

Step 4 – Contrast with the normal VaR

$$\text{Normal VaR} = 120 \times (-0.06 + 0.22 \times 2.326) = \text{AUD } 54.2 \text{ million}$$

$\Rightarrow$ Over one year with 22% volatility the gap is about AUD 10.6 million: the lognormal caps losses because the price cannot fall below zero (Figure 1).

Interpretation

For long horizons the choice of distribution matters materially; the "much the same" result of Dowd's Example 1.5 is a short-horizon result. Dowd's advice is to use geometric returns when dealing with longer periods.

Why the other options are wrong

A is incorrect. Used $z_{0.95} = 1.645$: $120 \times (1 - \mathrm{e}^{0.06 - 0.22 \times 1.645}) = 31.3$.
C is incorrect. Dropped the mean: $120 \times (1 - \mathrm{e}^{-0.22 \times 2.326}) = 48.1$.
D is incorrect. Used the normal formula, which is what the CRO did not ask for: $120 \times (0.22 \times 2.326 - 0.06) = 54.2$.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.1 Geometric Return Data; 1.3, eqs. 1.14 and 1.16, Examples 1.4 and 1.5

Solution MR-1 b.7 — Answer: B

Workbook A13.3 • S1, S3

Step 1 – Choose the volatility

The VaR is a forecast for the next 10 trading days. The 18% regime has held for about 30 trading days, longer than the horizon, and is the current state of the market, so it governs the forecast. Dowd's volatility-weighting argument makes the same point: when volatility has risen, older data "understate the changes we can expect to see tomorrow".

Step 2 – Scale to 10 days

$$\mu_{10} = 0.08 \times \frac{10}{250} = 0.0032, \qquad \sigma_{10} = 0.18 \times \sqrt{\frac{10}{250}} = 0.036$$

Step 3 – Critical geometric return

$$R^* = \mu_{10} - \sigma_{10} z_{0.99} = 0.0032 - 0.036 \times 2.326 = -0.080536$$

$\Rightarrow$ On the worst 1% of 10-day periods, the log return is at or below -8.05%.

Step 4 – Lognormal VaR

$$\text{VaR} = 50 \times (1 - \text{e}^{-0.080536}) = 50 \times 0.077378 = \text{USD } 3.87 \text{ million}$$

$\Rightarrow$ A 1% chance of losing more than about USD 3.87 million (7.74% of the portfolio) over the next 10 days.

Interpretation

The volatility choice moves VaR by about USD 0.43 million, far more than the choice between the normal and lognormal formulas at this horizon (USD 0.16 million). Reporting the stale-regime figure would understate the risk the desk is running now.

Why the other options are wrong

A is incorrect. Used the stale 16% volatility (scope/condition error): $\sigma_{10} = 0.032$, $R^* = 0.0032 - 0.032 \times 2.326 = -0.071232$, VaR $= 50(1 - \text{e}^{-0.071232}) = 3.44$.

C is incorrect. Correct volatility, but applied the normal (arithmetic) formula: $50 \times (-0.0032 + 0.036 \times 2.326) = 4.03$.

D is incorrect. Used the two-tailed 99% critical value 2.576 for a one-tailed VaR: $50 \times (1 - \text{e}^{0.0032 - 0.036 \times 2.576}) = 4.28$.

LO: MR-1 b; MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.3, eq. 1.16 and Example 1.5; Ch.4, 2.4 Volatility-weighted Historical Simulation

Solution MR-1 b.8 **Answer: D**

Workbook A13.1 • S3

Step 1 – Write the lognormal VaR as a fraction

$$V = 1 - \text{e}^{\mu_R - \sigma_R z_\alpha} \quad \Rightarrow \quad \ln(1 - V) = \mu_R - \sigma_R z_\alpha$$

Step 2 – Take the log of the surviving value

$$\ln(1 - 0.35) = \ln(0.65) = -0.43078$$

$\Rightarrow$ The critical geometric return R^* is -43.08%: a 35% price fall corresponds to a log return of -43%.

Step 3 – Solve for σ_R

$$\sigma_R = \frac{\mu_R - \ln(1 - V)}{z_{0.99}} = \frac{0.07 + 0.43078}{2.326} = 0.2153$$

$\Rightarrow$ The reported VaR is consistent with roughly 21.5% annual volatility of log returns.

Interpretation

If the validator's own volatility estimate differs materially from 21.5%, the reported VaR is inconsistent with the model's stated inputs and should be challenged.

Why the other options are wrong

A is incorrect. Wrong tail: treated the 35% move as a gain and used $\ln(1+V) = \ln 1.35 = 0.30010$ as the size of the critical log return instead of $-\ln(1-V) = 0.43078$: $(0.07 + 0.30010)/2.326 = 15.9\%$.

B is incorrect. Reverse-solved the normal formula $V = -\mu + \sigma z$: $\sigma = (0.35 + 0.07)/2.326 = 18.1\%$.

C is incorrect. Dropped the mean: $\sigma = 0.43078/2.326 = 18.5\%$.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.3 Estimating Lognormal VaR, eq. 1.16

MR-2 • Non-Parametric Approaches • Vardeez: VERY IMPORTANT

MR-2 a

P1 BOTH

Apply the bootstrap historical simulation approach to estimate coherent risk measures.

Workbook: A19 Non-Parametric VaR • Tier A • B2 Expected Shortfall and Coherence • Tier B

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A19 (non-parametric VaR) was in the August 2026 sitting ("age-weighted and volatility-weighted HS"), and the reported "bootstrap HS property question" sits in this LO.
Vardeez: know the bootstrapping HS method.

Reported exam shapes

- **A19·S3** Bootstrap HS property question: with replacement, mean of the resample estimates, cannot exceed the largest historical loss, assumes independence.
- **B2·S1** ES from a sample and its bootstrap resamples: exactly which losses enter the tail average, 4, 5 or 6.
- **B2·S4** Coherence: ES is subadditive, VaR is not (worked as Question MR-2 c.8).

What the questions look like: mostly "which statement is correct" about the procedure, plus one numerical where you build the bootstrap VaR or ES from a few resamples. The trap in the numerical is tail membership: how many losses go into the ES average.

1. Starting point: basic historical simulation

Non-parametric approaches estimate risk measures without assuming a distribution for P/L: they "let the P/L data speak for themselves". The core assumption is that **the near future will be sufficiently like the recent past**, so the historical distribution is a fair forecast of tomorrow's.

With n P/L observations and confidence level α:

- The tail holds $(1-\alpha)\,n$ losses.
- **VaR** is the $[(1-\alpha)n+1]$th highest loss: the highest loss *outside* the tail. (1,000 observations at 95%: 50 in the tail, VaR is the 51st highest loss.)
- **ES** is the average of the $(1-\alpha)n$ losses *in* the tail, which are the losses beyond the VaR.

Dowd's illustration (1,000 standard normal observations): basic HS gives 95% VaR $= 1.704$ and ES $= 2.196$, against the true values 1.645 and 2.063.

2. The bootstrap procedure

The bootstrap is a simple refinement of basic HS: **resample from the existing data set with replacement**.

1. Draw a resample of the same size as the original data, at random and with replacement (the same day can be drawn twice, some days not at all).
2. Compute the risk measure on that resample in the usual way: the resample VaR, and the resample ES as "the average of the losses in each resample exceeding the resample VaR".
3. Return the data to the pool and repeat many times (say $B = 1{,}000$ resamples).

4. The **best estimate is the mean of the resample estimates**:

$$\widehat{\mathrm{VaR}}^{B} = \frac{1}{B}\sum_{b=1}^{B}\mathrm{VaR}^{(b)} \qquad \widehat{\mathrm{ES}}^{B} = \frac{1}{B}\sum_{b=1}^{B}\mathrm{ES}^{(b)}$$

The same loop works for *any* coherent risk measure: compute it on each resample and average. That is what the LO means by "estimate coherent risk measures".

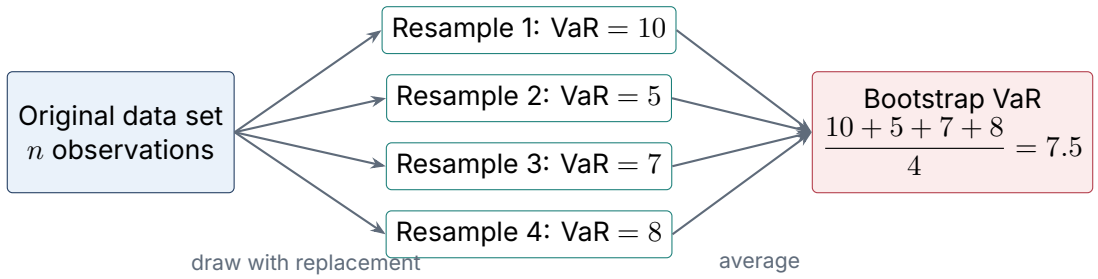


Figure 1. Vardeez's example (USD million). One data set is resampled four times with replacement; each resample gives its own VaR and the bootstrap estimate is their simple average. In practice B is in the hundreds or thousands.

3. Why bother: precision and confidence intervals

- **More accurate.** A bootstrapped estimate will often be more accurate than the raw sample estimate. Dowd's 1,000 resamples give 95% VaR ≈ 1.669 and ES ≈ 2.114, much closer to the true 1.645 and 2.063 than basic HS (1.704 and 2.196).
- **Measures its own precision.** The spread of the resample estimates gauges the standard error. The *percentile interval* reads the bounds off the quantiles of the bootstrapped estimates (Dowd: 90% interval for VaR $[1.554, 1.797]$, for ES $[1.986, 2.271]$). The bias-corrected and accelerated (BCa) method refines the end-points. Bootstrap estimates have little bias but often substantial variance.
- **ES behaves better than VaR.** The bootstrapped ES histogram is smoother than the VaR histogram, because each ES is itself an average of tail losses.
- **Intuitive and easy to apply**, on a spreadsheet if necessary.

4. What the bootstrap cannot do

- **It stays inside the data.** Every resample is built from observed losses, so no bootstrapped VaR or ES can exceed the *largest historical loss*. (Volatility-weighted HS and filtered HS can; see MR-2 c.)
- **It assumes independence over time.** "Perhaps the main limitation of the bootstrap is that standard bootstrap procedures presuppose that observations are independent over time." Resampling single days destroys volatility clustering. Remedies: bootstrap GARCH residuals (this is filtered HS, at the cost of model risk), use a block bootstrap, or make selection probabilities depend on recently drawn observations.
- It inherits every weakness of the historical sample (quiet sample $\Rightarrow$ VaR too low; see MR-2 d).

Worked example: bootstrapped ES, counting the tail

Resamples of 100 losses each, 95% ES. Tail size $(1 - 0.95) \times 100 = 5$: the ES of each resample averages the **5** largest losses, and the 95% VaR is the 6th largest. If a resample's ten largest losses are $9.8, 8.1, 7.4, 6.9, 6.9, 6.2, 5.6, \ldots$, then VaR $= 6.2$ and ES $= (9.8 + 8.1 + 7.4 + 6.9 + 6.9)/5 = 7.82$. The repeated 6.9 is legitimate under resampling with replacement. Averaging 4 losses gives the 96% ES; averaging 6 drags the VaR observation into the tail. The bootstrap answer is then the mean of the resample ESs (Question MR-2 a.4).

Formula card: MR-2 a

Tail size	$(1-\alpha)\,n$ losses
HS VaR	the $[(1-\alpha)n+1]$th highest loss (highest loss outside the tail)
HS ES	average of the $(1-\alpha)n$ largest losses
Bootstrap estimate	$\hat{\theta}^B = \frac{1}{B}\sum_b \theta^{(b)}$ for θ = VaR, ES or any coherent measure
Percentile interval	$\left[\hat{\theta}^B_{(q)},\ \hat{\theta}^B_{(1-q)}\right]$ from the quantiles of the resample estimates
Book check	basic HS 1.704 / 2.196; bootstrap 1.669 / 2.114; true 1.645 / 2.063 (95% VaR / ES)

Exam traps

- **"Without replacement".** Always *with* replacement; that is what lets a resample differ from the original.
- **Wrong aggregator.** The best estimate is the *mean* of the resample estimates, not the sum, the maximum, the median, or the single original-sample figure.
- **Intermediate result.** One resample's ES is not the bootstrap ES.
- **Off-by-one on tail membership.** At 95% with 100 losses, the ES averages 5 losses; 4 or 6 is wrong.
- **Sibling swap on the maximum loss.** Basic HS and the bootstrap are capped by the largest historical loss; HW and FHS are not.
- **"Resampling fixes clustering".** False: the standard bootstrap assumes independence over time.

Practice questions: MR-2 a

Question MR-2 a.1

Bank Q84 • BT P2.T5.22.3.1

Which of the following is an essential difference between basic historical simulation and bootstrap historical simulation; i.e., one of them involves this feature or element, but the other does not?

A. Random number generator

B. Non-parametric methodology

C. Parametric distributional assumption

D. Actual observed (i.e., historical) loss data

Question MR-2 a.2

Bank Q85 • AP Mock 3 Q26

A risk manager wishes to use the bootstrap historical simulation approach to estimate VaR. What is the main reason for using historical returns to forecast VaR in this approach?

A. The distribution of returns is expected to change over time.

B. The distribution of returns will remain the same in the past and in the future.

C. The historical returns are always normally distributed.

D. The historical returns have no impact on future returns.

Question MR-2 a.3

Bank Q92 • BT P2.T5.710.1 (key corrected)

Betty has 1,000 daily P&L observations and applies the bootstrap historical simulation approach to estimate the 95% VaR and expected shortfall (ES). Which of the following statements about the procedure is TRUE?

A. Because resampling adds randomness, the bootstrap can produce a VaR or ES estimate larger than the worst loss in the original dataset.

B. The best VaR estimate is the VaR computed once from the original dataset; the resamples are used only to assess its accuracy.

C. Each resample is drawn without replacement so that no observation appears twice within a resample.

D. The best estimates of VaR and ES are the averages of the VaRs and ES values computed from the many resampled datasets.

Question MR-2 a.4

Workbook B2.1 • S1 • Map #12

A market risk analyst at a Canadian asset manager estimates the 1-day 95% expected shortfall of an equity portfolio using bootstrapped historical simulation. Each resample contains 100 daily P/L observations drawn with replacement from the historical sample. For illustration, only three resamples are used:

- ES computed from resample 1: CAD 7.10 million; ES computed from resample 2: CAD 7.32 million
- Ten largest losses in resample 3, in CAD million, ranked: 9.8, 8.1, 7.4, 6.9, 6.9, 6.2, 5.6, 5.3, 5.0, 4.8

Which of the following is closest to the bootstrapped estimate of the 1-day 95% ES?

A. CAD 7.32 million

B. CAD 7.41 million

C. CAD 7.49 million

D. CAD 7.82 million

Question MR-2 a.5

New • built from GARP Ch.4 (Dowd 2.2) • S1

A risk analyst at a Hong Kong bank estimates the 1-day 97.5% VaR and expected shortfall of a HKD 2 billion trading book using bootstrapped historical simulation. The historical window holds 500 daily P/L observations, and each resample contains 200 observations drawn with replacement. For illustration, three resamples are used. The seven largest losses in each resample (HKD million) are:

Rank	1	2	3	4	5	6	7
Resample 1	12.4	10.9	9.8	9.1	8.7	8.2	7.9
Resample 2	13.1	11.0	9.5	9.5	8.4	8.0	7.6
Resample 3	11.8	10.2	9.9	8.9	8.6	7.7	7.5

Using the convention that VaR is the highest loss outside the tail, which of the following is closest to the bootstrapped 97.5% VaR and ES?

A. VaR HKD 7.97 million; ES HKD 9.76 million

B. VaR HKD 7.97 million; ES HKD 10.12 million

C. VaR HKD 8.20 million; ES HKD 10.30 million

D. VaR HKD 8.57 million; ES HKD 10.51 million

Question MR-2 a.6

Workbook A19.5 • S2, S3, S4, S5 • Map #28

A model validator at a Canadian bank is reviewing a proposal to replace the bank's basic HS engine (500 daily observations) with a refined non-parametric method. The current EWMA volatility forecast is 1.6% a day. One observation in the window, 20 days old, is a return of -3.0% recorded on a day whose volatility forecast was 1.2%. Which of the following statements should the validator include in the report?

A. Under volatility-weighted HS, the -3.0% return enters the adjusted series as -2.25%, because historical returns are scaled down when the current volatility forecast is above the forecast for the day they were observed.

B. Under bootstrapped HS, each resample is drawn from the 500 observations with replacement, the best VaR estimate is the mean of the resample VaRs, and each resample ES is the average of the losses beyond that resample's VaR.

C. Under filtered HS, the standardised return of -2.5 is bootstrapped and rescaled by the 1.6% forecast to -4.0%, but the resulting VaR and ES still cannot exceed the largest loss in the historical sample.

D. Under bootstrapped HS, the standard resampling procedure remains reliable when returns are dependent over time, because each observation is drawn at random from the full window of 500 observations.

Solutions • MR-2 a

Solution MR-2 a.1

Answer: A

Bank Q84 • BT P2.T5.22.3.1

Step 1 – What basic HS does

Basic HS orders the single historical sample and reads off the quantile. Nothing is random: run it twice on the same data and you get the same VaR.

Step 2 – What the bootstrap adds

The bootstrap draws many resamples at random, with replacement, from that sample. Random draws need a random number generator.

$\Rightarrow$ Random resampling is the one element present in the bootstrap and absent from basic HS.

Interpretation

Both methods are non-parametric and both use only actual historical data; the bootstrap simply re-uses that data many times in random combinations and averages the results.

Why the other options are wrong

B is incorrect. Both approaches are non-parametric, so this does not distinguish them.

C is incorrect. Neither approach makes a parametric distributional assumption.

D is incorrect. Both approaches rely entirely on actual historical loss data.

LO: MR-2 a: Apply the bootstrap historical simulation approach to estimate coherent risk measures. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.2 Bootstrapped Historical Simulation

Solution MR-2 a.2

Answer: B

Bank Q85 • AP Mock 3 Q26

Step 1 – What the bootstrap treats as the forecast distribution

Bootstrap HS resamples observed returns and uses their empirical distribution as tomorrow's distribution.

Step 2 – The assumption that makes this valid

That only works if the return distribution is the same in the future as in the sample period, i.e. the near future is sufficiently like the recent past.

$\Rightarrow$ Using past returns as the forecast presupposes an unchanged distribution.

Interpretation

This is the core assumption of every non-parametric method, and also its main weakness: if the distribution shifts, the estimate lags (MR-2 d).

Why the other options are wrong

A is incorrect. A changing distribution is the main criticism of historical methods, not their rationale.

C is incorrect. Bootstrap HS is non-parametric; it assumes no distribution, normal or otherwise.

D is incorrect. If past returns had no bearing on future returns, historical data would be useless for forecasting.

LO: MR-2 a: Apply the bootstrap historical simulation approach to estimate coherent risk measures. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.1 and 2.2

Solution MR-2 a.3 — Answer: D

Bank Q92 • BT P2.T5.710.1

Step 1 – The resampling rule

Each resample is drawn from the 1,000 observations *with* replacement and the data are returned to the pool after each draw, so observations can repeat.

Step 2 – The estimator

A VaR and an ES are computed on each resample; the best estimates for the original data set are the averages of those resample figures.

$\Rightarrow$ D describes the procedure correctly.

Interpretation

Because every resample contains only observed losses, the bootstrap is still capped by the worst historical loss. Only methods that rescale returns (volatility-weighted HS, filtered HS) can break that cap.

Why the other options are wrong

A is incorrect. False. Randomness only recombines observed values; no resample can contain a loss worse than the historical worst. This confuses the bootstrap with filtered HS.

B is incorrect. False. Using the single original-sample VaR is plain HS; in the bootstrap the mean of the resample VaRs is the best estimate.

C is incorrect. False. Sampling is with replacement, so the same observation can appear twice.

LO: MR-2 a: Apply the bootstrap historical simulation approach to estimate coherent risk measures. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.2 Bootstrapped Historical Simulation

Solution MR-2 a.4 — Answer: B

Workbook B2.1 • S1

Step 1 – Size of the tail

$$(1-\alpha)\,n = (1-0.95)\times 100 = 5 \text{ losses}$$

$\Rightarrow$ Five of the 100 resampled losses lie in the 5% tail. The 95% VaR is the 6th largest loss, CAD 6.2 million, the highest loss outside the tail.

Step 2 – ES of resample 3

$$\text{ES}^{(3)} = \frac{9.8+8.1+7.4+6.9+6.9}{5} = \frac{39.1}{5} = 7.82$$

$\Rightarrow$ Average of the losses exceeding the resample VaR. The repeated 6.9 is legitimate: resampling with replacement can draw the same day twice.

Step 3 – Bootstrap point estimate

$$\widehat{\text{ES}}^{B} = \frac{7.10+7.32+7.82}{3} = \frac{22.24}{3} = 7.41$$

Interpretation

The bootstrapped 95% ES is about CAD 7.41 million: on the worst 5% of days the portfolio is expected to lose that amount on average. Averaging across resamples is what makes the bootstrap estimate more accurate than a single raw sample estimate (Dowd: 2.114 against 2.196 for basic HS).

Why the other options are wrong

A is incorrect. Averaged six losses, pulling the VaR observation into the tail: $(39.1 + 6.2)/6 = 7.55$, then $(7.10 + 7.32 + 7.55)/3 = 7.32$.
C is incorrect. Averaged only four losses, treating the 5th largest loss as the VaR (that is the 96% VaR): $32.2/4 = 8.05$, then $(7.10 + 7.32 + 8.05)/3 = 7.49$. This is the reported "4 or 5" trap.
D is incorrect. Stopped at the intermediate result: the ES of resample 3 alone, 7.82, instead of the mean over all resamples.

LO: MR-1 c; MR-2 a: Apply the bootstrap historical simulation approach to estimate coherent risk measures. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.2 (footnote 2) and 1.4; Ch.4, 2.2 Basic and Bootstrapped Historical Simulation

Solution MR-2 a.5

Answer: B

New • built from GARP Ch.4 (Dowd 2.2)

Step 1 – Size of the tail in each resample

$$(1-\alpha)\,n = (1-0.975) \times 200 = 5 \text{ losses}$$

$\Rightarrow$ The ES averages the 5 largest losses of each resample; the VaR is the 6th largest, the highest loss outside the tail.

Step 2 – Resample VaRs and their mean

$$\text{VaR}^{(1)} = 8.2, \quad \text{VaR}^{(2)} = 8.0, \quad \text{VaR}^{(3)} = 7.7 \qquad \Rightarrow \qquad \widehat{\text{VaR}}^{B} = \frac{8.2 + 8.0 + 7.7}{3} = 7.97$$

Step 3 – Resample ESs and their mean

$$\text{ES}^{(1)} = \frac{50.9}{5} = 10.18, \qquad \text{ES}^{(2)} = \frac{51.5}{5} = 10.30, \qquad \text{ES}^{(3)} = \frac{49.4}{5} = 9.88$$

$$\widehat{\text{ES}}^{B} = \frac{10.18 + 10.30 + 9.88}{3} = \frac{30.36}{3} = 10.12$$

$\Rightarrow$ On the worst 2.5% of days the book is expected to lose about HKD 10.1 million; the VaR threshold is about HKD 8.0 million.

Interpretation

The ES is above the VaR, as it must be, because it averages only losses beyond the VaR. The 500-observation window and the HKD 2 billion book size are not needed: what matters is the resample size (200) and the confidence level. The repeated 9.5 in resample 2 is a normal feature of drawing with replacement.

Why the other options are wrong

A is incorrect. Included the VaR observation in the ES (6 losses): ES $= (59.1/6 + 59.5/6 + 57.1/6)/3 = 9.76$.
C is incorrect. Took the most conservative resample instead of averaging: the largest VaR (8.2, resample 1) and the largest ES (10.30, resample 2).
D is incorrect. Off by one the other way: VaR as the 5th largest loss, $(8.7 + 8.4 + 8.6)/3 = 8.57$, and ES over only 4 losses, $(10.55 + 10.775 + 10.20)/3 = 10.51$.

LO: MR-2 a: Apply the bootstrap historical simulation approach to estimate coherent risk measures. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.2 Bootstrapped Historical Simulation; Ch.3, 1.2 and 1.4

Solution MR-2 a.6 **Answer: B**

Workbook A19.5 • S2, S3, S4, S5

Step 1 – The correct statement

Dowd: a bootstrap "involves resampling from our existing data set with replacement"; each resample gives a VaR, "our 'best' estimate [is] the mean of these resample-based estimates", and each resample ES is "the average of the losses in each resample exceeding the resample VaR". B states all three properties.

Step 2 – The numbers behind A and C

$$\text{HW: } r^* = \frac{\sigma_T}{\sigma_t} r = \frac{1.6}{1.2} \times (-3.0\%) = -4.0\%; \quad \text{FHS: } \frac{-3.0\%}{1.2\%} = -2.5, \quad -2.5 \times 1.6\% = -4.0\%$$

$\Rightarrow$ Both semi-parametric methods inflate the old return to -4.0% because volatility has risen since it was observed; that rescaling is exactly what lets them produce losses beyond the historical maximum.

Interpretation

The bootstrap improves the precision of the estimate and gives confidence intervals, but it stays inside the historical data: it cannot exceed the largest historical loss and, in standard form, it assumes independence. HW and FHS are the refinements that respond to the change in volatility; FHS also extends "easily" to ES, and using ES is Dowd's partial response to the criticism that FHS VaR pays too little attention to extremes.

Why the other options are wrong

A is incorrect. Polarity flip (the documented GARP trap). Returns are increased when the current forecast is above the forecast for day t: the return enters as -4.0%, not -2.25% (which is the inverted ratio 1.2/1.6).

C is incorrect. Sibling swap. "Cannot exceed the largest historical loss" is the constraint of basic HS (and of the plain bootstrap). FHS, like HW, "allows us to get VaR and ES estimates that can exceed the maximum historical loss".

D is incorrect. Scope/condition error. "Perhaps the main limitation of the bootstrap is that standard bootstrap procedures presuppose that observations are independent over time"; remedies are GARCH residuals, blocks, or time-dependent selection probabilities.

LO: MR-2 a; MR-2 c; MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.2 Bootstrapped Historical Simulation; 2.4 Volatility-weighted and Filtered Historical Simulation; Appendix 2, Time Dependency and the Bootstrap

MR-2 • Non-Parametric Approaches • Vardeez: VERY IMPORTANT

MR-2 c

P1 CONCEPT

Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches.

Workbook: A19 Non-Parametric VaR • Tier A • August 2026 recall • B2 Tier B

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: age-weighted and volatility-weighted HS came up in the August 2026 sitting.
Vardeez: volatility-weighted and filtered HS methods.

Reported exam shapes

- **A19·S1** Age-weighted HS (BRW): single weights, cumulative weights and the weighted VaR read-off.
- **A19·S2** Volatility-weighted HS (HW), with the polarity trap: returns are scaled *up* when current volatility is above the historical level.
- **A19·S4** Differences among the non-parametric approaches.
- **A19·S5** Expected shortfall in filtered historical simulation. **B2·S3** ES within FHS.

What the questions look like: the LO verb is "compare and contrast", so most questions are "which statement is correct" across the four approaches. The August paper also had the workbook's age-weight and HW calculations, which are included here as they were reported.

1. Why plain HS needs fixing

Basic HS gives every observation in the window the same weight $1/n$ and every older observation zero. Dowd's problems with this:

- The $1/n$ weight is hard to justify: why should a 249-day-old return count as much as yesterday's?
- Risk estimates are slow to respond to major events, and the i.i.d. assumption ignores volatility clustering.
- **Ghost effects.** A big loss keeps VaR high for exactly n days, then VaR falls the day it drops out of the window. That fall is an artefact of the window, not a change in risk.

Each refinement below keeps the historical data but *reweights or rescales* it. They are **semi-parametric**: non-parametric HS plus a parametric add-on (a decay rule, a GARCH forecast, a correlation matrix).

2. Age-weighted HS (Boudoukh, Richardson and Whitelaw, BRW)

Weight observations by how old they are. Yesterday's observation gets $w(1)$, and each further day back multiplies the weight by λ:

$$\boxed{w(i) = \frac{\lambda^{i-1}(1-\lambda)}{1-\lambda^n}} \qquad w(1) = \frac{1-\lambda}{1-\lambda^n}, \qquad \sum_{i=1}^{k} w(i) = \frac{1-\lambda^k}{1-\lambda^n}$$

i is the age in days ($i = 1$ is yesterday), $0 < \lambda < 1$ is the decay factor, n the window. The denominator $1-\lambda^n$ makes the n weights sum to 1.

- λ **close to 1** means **slow decay**; λ far from 1 means fast decay (only very recent data matter); $\lambda \to 1$ gives back equal-weight HS.
- **VaR read-off.** Rank P/L from the worst loss, attach each loss's age weight, accumulate, and stop where the cumulative weight first exceeds the tail probability $1 - \alpha$ (interpolate if allowed).

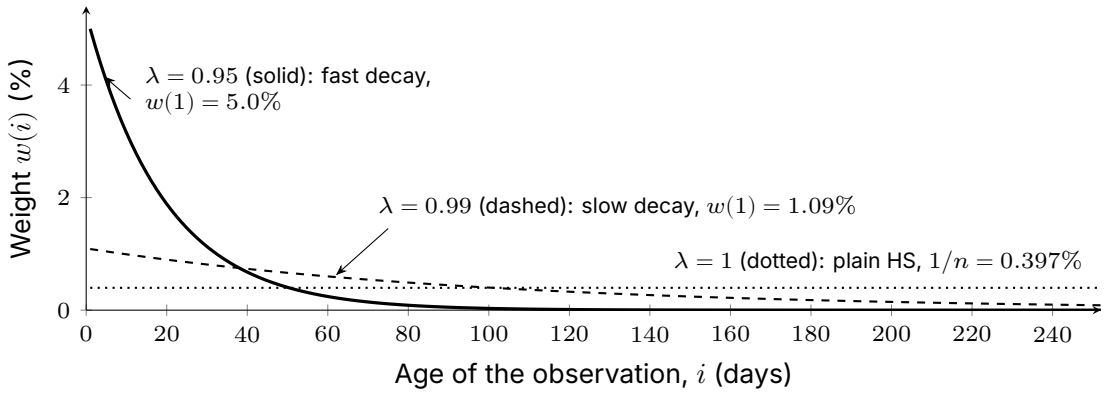


Figure 2. Age weights on a 252-day window. At $\lambda = 0.95$ yesterday carries 5% and the weight has effectively vanished by day 60, so VaR is driven by about two months of data. At $\lambda = 0.99$ yesterday carries 1.09% and the oldest observation still has 8% of yesterday's weight. At $\lambda = 1$ the line is flat: ordinary HS.

Attractions of age-weighting	Drawbacks
Generalises HS (equal weights are the case $\lambda \to 1$)	Reduces the effective sample size
More responsive to large recent losses; better with loss clusters	A run of big profits or losses can distort the risk profile
Reduces ghost effects: an observation leaves with weight $\lambda^n w(1) < 1/n$	VaR can still be insufficiently responsive (Pritsker)
Lets the window grow over time, improving efficiency	Ad hoc: only $\lambda = 1$ is consistent with a return process

3. Volatility-weighted HS (Hull and White, HW)

Weight observations by *volatility*, not by age. Replace each historical return by a volatility-adjusted return:

$$\boxed{r^*_{t,i} = \frac{\sigma_{T,i}}{\sigma_{t,i}}\, r_{t,i}} \qquad \sigma_{T,i} = \text{current (most recent) forecast}, \quad \sigma_{t,i} = \text{forecast that applied on day } t$$

then estimate VaR or ES from the adjusted series in the ordinary equal-weight way.

- **Polarity.** Returns are **increased when current volatility is above** the volatility on the day they were observed, and decreased when it is below. Dowd's intuition: if volatility is 1.5% today and was 1% a month ago, month-old data "understate the changes we can expect to see tomorrow". (The same logic picks the current volatility regime in Question MR-1 b.7.)
- **Advantages.** Takes account of volatility changes in a natural, direct way (age-weighting does it "in a rather arbitrary and restrictive way"); sensitive to current GARCH or EWMA information; **VaR and ES can exceed the largest historical loss**; HW's evidence shows better VaR than BRW; can be combined with age-weighting and with bootstrap or order-statistics confidence intervals.

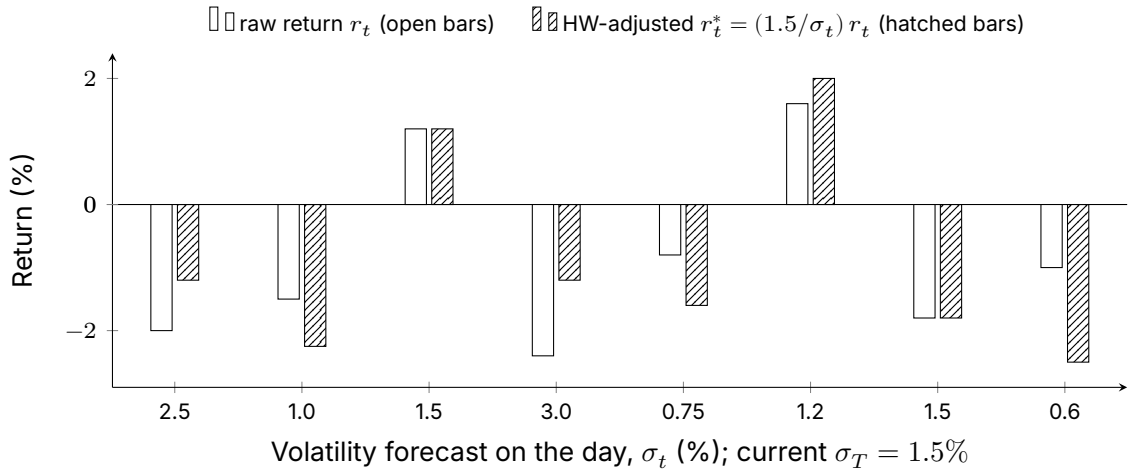


Figure 3. Eight illustrative days. Returns from volatile days ($\sigma_t = 2.5\%, 3.0\%$) shrink; returns from calm days ($\sigma_t = 1.0\%, 0.75\%, 0.6\%$) grow, and the worst adjusted loss (-2.5%) comes from a day whose raw loss was only -1.0%. Days with $\sigma_t = \sigma_T$ are unchanged.

Correction to the source notes. Vardeez lists "higher VaR estimates: the volatility-adjusted returns enable VaR estimates that are higher than those obtained from using the historical data set alone" as an advantage of HW. That is too strong. HW VaR is higher only when current volatility is above the volatility of the period that drives the tail; when volatility has fallen, HW VaR is *lower* than basic HS (Question MR-2 c.4: 0.51 against 0.65). Dowd's actual point is that HW *can* produce VaR and ES above the maximum historical loss, which basic HS cannot.

4. Correlation-weighted HS

A generalisation of HW: adjust historical returns to reflect *current correlations* as well as current volatilities (Duffie and Pan's suggestion). With Choleski matrices A (historical correlation matrix) and $\tilde{A}$ (current one):

$$R = A\varepsilon, \qquad \tilde{R} = \tilde{A}\varepsilon \qquad \Rightarrow \qquad \tilde{R} = \tilde{A}A^{-1}R$$

The historical returns are "multiplied" by a matrix built from the revised correlation matrix, so the adjusted returns carry the current correlation structure. Dowd's Example 2.1: moving from a historical correlation of 0.3 to a current one of 0.9 turns the second row of the adjustment matrix into $[0.7629,\ 0.4569]$. It is more involved than volatility-weighting, and it reintroduces a correlation matrix, so it does not escape the dimensionality problems of parametric methods.

5. Filtered historical simulation (FHS)

A **semi-parametric bootstrap** (Barone-Adesi et al.): combine HS with a conditional volatility model.

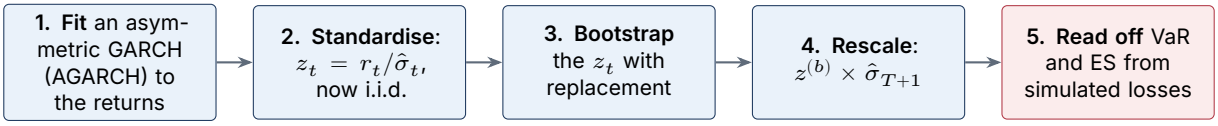


Figure 4. FHS in five steps. Standardising removes the volatility clustering, which is what makes the bootstrap step valid; rescaling by tomorrow's forecast puts today's volatility back in.

- **Multi-asset portfolios:** bootstrap whole vectors of standardised returns (all assets on the same day), which keeps the cross-sectional correlation structure *without* a variance-covariance matrix.

- **Attractions:** combines non-parametric HS with a sophisticated volatility model; fast, even for large portfolios; VaR and ES **can exceed the maximum historical loss**; can be combined with confidence-interval methods; can simulate volatility paths over multi-day horizons.
- **ES in FHS (shape S5):** FHS "can also be extended easily to allow for the estimation of ES as well as VaR". Pritsker's criticism is that FHS VaR may not pay enough attention to extreme observations or time-varying correlations; "a partial response to the first point would be to use ES instead of VaR".

6. The four approaches side by side

	What it reweights or rescales, and how	Key point	Beyond max historical loss?
Age-weighted (BRW)	Probability weights by age: $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$	λ near 1 decays slowly; $\lambda = 1$ is plain HS; reduces ghost effects	No
Volatility-weighted (HW)	Each return scaled by σ_T/σ_t, then equal weights	Up when vol has risen, down when it has fallen	Yes
Correlation-weighted	Returns multiplied by $\tilde{A}A^{-1}$ to impose current correlations	Generalises HW to correlations; more involved	Yes
Filtered (FHS)	GARCH-standardised returns bootstrapped and rescaled by tomorrow's forecast	Semi-parametric bootstrap; extends easily to ES	Yes

Formula card: MR-2 c (for the workbook calculations)

Age weight	$w(i) = \lambda^{i-1}w(1), \quad w(1) = (1-\lambda)/(1-\lambda^n)$
Cumulative age weight	$\sum_{i=1}^{k} w(i) = (1-\lambda^k)/(1-\lambda^n)$
Age-weighted VaR	accumulate weights down the loss ranking; VaR where the total first exceeds $1-\alpha$
HW adjusted return	$r_t^* = (\sigma_T/\sigma_t)\, r_t$, then equal-weight HS on r^*
Equal-weight HS VaR	the $[(1-\alpha)n+1]$th largest loss
Correlation weighting	$\tilde{R} = \tilde{A}A^{-1}R$ (Choleski factors)
FHS	$z_t = r_t/\hat{\sigma}_t$; simulated return $= z^{(b)}\hat{\sigma}_{T+1}$

Exam traps

- **Volatility polarity (the documented GARP trap).** HW scales returns *up* when current volatility is above the volatility on the day observed. The reversed statement and the inverted ratio σ_t/σ_T are the distractors.
- **Dropped normaliser.** Using $\lambda^{i-1}(1-\lambda)$ without dividing by $1-\lambda^n$, or raw λ^{i-1} as probabilities.
- **Exponent off by one.** The i-day-old observation uses λ^{i-1}; yesterday carries the full $w(1)$. And the normaliser uses n, not i.

- **Stopping early.** In age-weighted HS keep accumulating until the cumulative weight first *exceeds* $1 - \alpha$.
- **Sibling swap on the maximum loss.** Basic HS and the bootstrap are capped by the largest loss; HW, correlation-weighted and FHS are not.
- **Mislabelled methods.** A hard cutoff with equal weights is plain HS, not age-weighting. FHS "filters" through a volatility model; it does not trim data by size or weekday.
- **"No dimensionality problem"** is true of plain HS but not of correlation-weighting, which needs full correlation matrices.

Practice questions: MR-2 c

Question MR-2 c.1

Bank Q118 • GARP PE2 Q15 • Mock 4 Q16

An analyst at a commercial bank is evaluating how the bank applies historical simulation (HS) to estimate VaR and ES. The analyst focuses on the approaches used for weighting past return observations, including the age-weighted, volatility-weighted, correlation-weighted, and filtered HS approaches. Which of the following statements is correct regarding the given weighting approach?

A. The age-weighted approach typically specifies that observations that occurred after the cutoff date are given an equal weight while observations that occurred before the cutoff date are given a weight of zero.

B. The volatility-weighted approach adjusts historical returns in the sample by increasing them if the historical volatility forecast was higher than the current volatility forecast, and decreasing them if the historical volatility forecast was lower than the current volatility forecast.

C. The correlation-weighted approach uses matrix multiplication to adjust historical portfolio returns so that these returns reflect current volatilities and current correlations.

D. The filtered HS approach allows the historical returns data to be trimmed or customized based on day of the week, magnitude of return, or any other characteristic of the data.

Question MR-2 c.2

Workbook A19.1 • S1 • Aug-2026

A junior risk analyst at a Singapore asset manager is implementing the Boudoukh, Richardson and Whitelaw age-weighted historical simulation approach on a window of the most recent 100 daily returns of an SGD 300 million fixed income portfolio. The decay parameter is $\lambda = 0.98$. Which of the following is closest to the probability weight assigned to the return observed 12 days ago?

A. 0.0160

B. 0.0185

C. 0.0231

D. 0.2482

Question MR-2 c.3

Workbook A19.2 • S1 • Aug-2026

A market risk manager at a UK bank estimates 1-day 95% VaR for a GBP trading book using age-weighted historical simulation with $n = 100$ daily observations and $\lambda = 0.96$. The nine largest historical losses, ranked from worst, and their ages in days are:

Rank	1	2	3	4	5	6	7	8	9
Loss (GBP million)	7.4	6.9	6.1	5.6	5.2	4.8	4.5	4.3	4.0
Age (days)	85	3	58	10	47	72	6	30	21

The bank takes VaR to be the first ranked loss at which the cumulative probability weight exceeds 5% (no interpolation). Which of the following is closest to the age-weighted 95% VaR?

A. GBP 4.8 million

B. GBP 5.6 million

C. GBP 6.1 million

D. GBP 6.9 million

Question MR-2 c.4

Workbook A19.3 • S2 • Aug-2026 • Map #28

A risk analyst at a US hedge fund applies the Hull and White volatility-weighted historical simulation approach to a USD 25 million single-asset position. The asset's beta to the S&P 500 is 1.3. The most recent EWMA forecast of daily volatility is 1.2%. The ten daily returns in the window and the EWMA volatility forecast that applied on each day are:

Day	1	2	3	4	5	6	7	8	9	10
Return (%)	−3.1	1.2	−2.6	−0.9	1.8	−2.2	0.7	−1.7	0.4	−1.2
Vol forecast (%)	2.4	1.5	2.0	1.1	1.6	1.3	1.0	0.9	1.2	1.4

Using the convention that VaR is the highest loss outside the tail, which of the following is closest to the 1-day 90% volatility-weighted HS VaR?

A. USD 0.51 million

B. USD 0.57 million

C. USD 0.65 million

D. USD 1.08 million

Question MR-2 c.5

Bank Q112 • BT L2.T5.79.2

In regard to Dowd's correlation-weighted historical simulation, EACH of the following is TRUE EXCEPT:

A. The covariance/correlation matrix "add-on" to the non-parametric historical simulation makes this a semi-parametric approach.

B. It can be combined with volatility-weighted historical simulation.

C. The weighting takes into account information about both volatilities and correlations.

D. Because actual historical returns are used, the "curse of dimensionality" created by the need for many pairwise correlations is avoided.

Question MR-2 c.6 Bank Q113 • BT L2.T5.80.1

Filtered historical simulation includes EACH of the following steps EXCEPT:

A. Specify a conditional volatility model, e.g., GARCH(1,1).

B. Convert actual historical returns into volatility-standardized returns.

C. Bootstrap (simulate) from the series of standardized returns.

D. Fit a parametric VaR model to the standardized returns.

Question MR-2 c.7 Bank Q117 • BT P2.T5.22.4.2

Peter manages a large portfolio and wants to generate a one-day value at risk (VaR), but he has the following preferences with respect to the approach:

- He would like the approach to incorporate the cross-sectional correlation structure on a given day.
- He would like to be able to incorporate autocorrelations in a long-term forecast that will extend over at least 20 trading days.
- If the current regime suggests, he would like to be able to produce VaR estimates that exceed the greatest loss in the historical sample.

Given these preferences, which of the following approaches is BEST?

A. Basic HS

B. Bootstrap HS

C. Filtered HS

D. Volatility-weighted HS

Question MR-2 c.8 Workbook B2.3 • S3, S4 • Map #12

The CRO of a European bank is merging a credit trading desk, whose losses are heavily skewed, with a rates desk. A junior risk analyst is preparing a report recommending that the bank replace its basic historical simulation VaR with an expected shortfall estimated by filtered historical simulation (FHS). Which of the following statements is correct for the analyst to include in the report?

A. Because VaR is subadditive for skewed loss distributions, the combined 99% VaR of the two desks cannot exceed the sum of their stand-alone VaRs.

B. Because FHS bootstraps from the historical data set, its VaR and ES estimates, like those of basic HS, cannot exceed the largest historical loss.

C. Because ES averages all tail losses, its estimators have smaller standard errors than VaR estimators, and the advantage grows as tails become heavier.

D. Because ES is subadditive, the combined ES of the two desks cannot exceed the sum of their stand-alone ESs, and FHS can be extended to estimate ES.

Solutions • MR-2 c

Solution MR-2 c.1 — Answer: C

Bank Q118 • GARP PE2 Q15

Step 1 – Test each approach against its definition

Age-weighting gives geometrically declining weights $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$; there is no hard cutoff with equal weights. Volatility-weighting scales each return by σ_T/σ_t. Correlation-weighting multiplies historical returns by a matrix built from the revised variance-covariance matrix. FHS standardises returns with a GARCH model and bootstraps them.

Step 2 – Pick the statement that matches

Correlation-weighting updates the diagonal (variances) and the off-diagonal (covariances and correlations), so the adjusted returns reflect current volatilities *and* current correlations.
$\Rightarrow$ C is correct: correlation-weighting is the matrix generalisation of HW.

Interpretation

GARP builds each wrong option from one of the other three approaches with a single element changed: a description of plain HS labelled as age-weighting, a polarity flip on HW, and a mislabelled FHS.

Why the other options are wrong

A is incorrect. Describes the traditional equal-weight window with a cutoff, i.e. plain HS. Age-weighting uses exponentially declining weights.
B is incorrect. Polarity flip. Returns are increased when the *current* forecast is above the historical one, not when the historical forecast was higher.
D is incorrect. Confuses FHS with data screening. FHS filters returns through a volatility model; it does not trim observations by weekday or size.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Weighted Historical Simulation

Solution MR-2 c.2 — Answer: B

Workbook A19.1 • S1

Step 1 – Weight on the most recent observation, $w(1)$

$$w(1) = \frac{1-\lambda}{1-\lambda^n} = \frac{1-0.98}{1-0.98^{100}} = \frac{0.02}{1-0.13262} = \frac{0.02}{0.86738} = 0.023058$$

$\Rightarrow$ Yesterday's return carries a probability weight of about 2.31%, versus 1% under equal weighting.

Step 2 – Decay to age 12

$$\lambda^{i-1} = 0.98^{11} = 0.80073$$

$\Rightarrow$ An observation 12 days old keeps about 80% of the weight of yesterday's observation.

Step 3 – Age weight, eq. 2.2

$$w(12) = \lambda^{11}\, w(1) = 0.80073 \times 0.023058 = 0.018463$$

$\Rightarrow$ The 12-day-old return counts as a 1.85% probability mass, almost twice its equal-weight value of 1%.

Interpretation

With $\lambda = 0.98$ recent observations dominate the tail estimate, so a large recent loss raises VaR more than under basic HS. The portfolio value is irrelevant: weights depend only on λ, n and age.

Why the other options are wrong

A is incorrect. Dropped the normaliser $1 - \lambda^n$: $0.98^{11} \times 0.02 = 0.0160$. These weights would not sum to 1 over a finite window.

C is incorrect. Reported $w(1) = 0.0231$, the weight on yesterday's observation (an intermediate result), rather than decaying it to age 12.

D is incorrect. Reported the cumulative weight on the 12 most recent observations, $(1 - 0.98^{12})/(1 - 0.98^{100}) = 0.2482$, instead of the weight on the single 12-day-old observation.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Age-weighted Historical Simulation, eq. 2.2

Solution MR-2 c.3

Answer: B

Workbook A19.2 • S1

Step 1 – Weight on the most recent observation

$$w(1) = \frac{1 - 0.96}{1 - 0.96^{100}} = \frac{0.04}{1 - 0.016870} = \frac{0.04}{0.98313} = 0.040686$$

Step 2 – Age weights of the largest losses, $w(i) = 0.96^{i-1} \times 0.040686$

$$w(85) = 0.96^{84} \times 0.040686 = 0.00132 \qquad w(3) = 0.96^{2} \times 0.040686 = 0.03750$$
$$w(58) = 0.96^{57} \times 0.040686 = 0.00397 \qquad w(10) = 0.96^{9} \times 0.040686 = 0.02818$$

$\Rightarrow$ The worst loss is 85 days old and counts for only 0.13%; the 3-day-old loss counts for 3.75%.

Step 3 – Cumulate down the ranking

After rank 1: 0.00132; after rank 2: 0.03882; after rank 3: 0.04279; after rank 4: $0.04279 + 0.02818 = 0.07096 > 0.05$.

$\Rightarrow$ The 5% tail is exhausted at rank 4: the first three losses sit inside the tail and the 10-day-old loss of GBP 5.6 million is where the cumulative weight first exceeds 5%.

Step 4 – Age-weighted VaR

$$\text{VaR}_{95\%} = \text{GBP } 5.6 \text{ million}$$

Interpretation

Two recent large losses (ages 3 and 10) carry heavy weights, so the age-weighted VaR (5.6) is above the equal-weight HS VaR (4.8). This is the BRW attraction: VaR is more responsive to recent large losses and clusters. Reversing the age ordering (heaviest weights on the oldest data) would give 4.3.

Why the other options are wrong

A is incorrect. Used equal-weight HS: each loss has weight 1/100, the tail holds 5 losses, and VaR is the 6th largest loss, 4.8.

C is incorrect. Stopped one rank early, at the last loss inside the tail: after rank 3 the cumulative weight is only 4.28%, so GBP 6.1 million has not yet reached the 5% tail probability.

D is incorrect. Used the raw decay factors λ^{i-1} as probabilities, dropping the $(1-\lambda)/(1-\lambda^n)$ scaling: $0.96^{84} = 0.0323$, then $+0.96^2 = 0.9216$ exceeds 5% at rank 2, giving 6.9.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Age-weighted Historical Simulation, eq. 2.2

Solution MR-2 c.4

Answer: A

Workbook A19.3 • S2

Step 1 – Volatility-adjusted returns, eq. 2.3

$$r_t^* = \frac{\sigma_T}{\sigma_t}\, r_t = \frac{1.2\%}{\sigma_t}\, r_t$$

Day 1: $\frac{1.2}{2.4}(-3.1\%) = -1.550\%$ Day 3: $\frac{1.2}{2.0}(-2.6\%) = -1.560\%$

Day 4: $\frac{1.2}{1.1}(-0.9\%) = -0.982\%$ Day 6: $\frac{1.2}{1.3}(-2.2\%) = -2.031\%$

Day 8: $\frac{1.2}{0.9}(-1.7\%) = -2.267\%$ Day 10: $\frac{1.2}{1.4}(-1.2\%) = -1.029\%$

(The four positive returns stay positive after scaling and do not enter the left tail.)
$\Rightarrow$ Day 1's -3.1% happened when volatility was 2.4%, twice today's level, so it is halved; Day 8's -1.7% happened in a calm 0.9% regime and is scaled up to -2.27%.

Step 2 – Locate the VaR observation

With 10 observations and a 10% tail, one observation sits in the tail, so VaR is the 2nd largest adjusted loss. Ranked adjusted losses: 2.267%, 2.031%, 1.560%, 1.550%, ...
$\Rightarrow$ The 90% VaR return is -2.031% (Day 6).

Step 3 – Convert to currency

$$\text{VaR}_{90\%} = 0.02031 \times 25 = \text{USD } 0.508 \text{ million}$$

Interpretation

Because current volatility (1.2%) is below the levels prevailing on the worst raw days, volatility-weighting lowers VaR relative to basic HS (USD 0.65 million). The HW approach makes the estimate appropriately sensitive to current volatility. The beta is irrelevant.

Why the other options are wrong

B is incorrect. Took the largest adjusted loss (inside the tail) instead of the highest loss outside it: $0.02267 \times 25 = 0.57$.
C is incorrect. Ignored the volatility adjustment and used basic HS: 2nd largest raw loss $2.6\% \times 25 = 0.65$.
D is incorrect. Inverted the ratio, scaling by σ_t/σ_T (the polarity trap): Day 3 becomes $-2.6\% \times 2.0/1.2 = -4.333\%$ (2nd largest after Day 1's -6.2%), giving $0.04333 \times 25 = 1.08$.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Volatility-weighted Historical Simulation, eq. 2.3

Solution MR-2 c.5 Answer: D

Bank Q112 • BT L2.T5.79.2

Step 1 – Check the true statements

The correlation-matrix add-on to non-parametric HS makes it semi-parametric (A). It builds on volatility-weighting and can be combined with it (B). The revised matrix carries both variances on the diagonal and correlations off the diagonal, so it uses both (C).

Step 2 – The false statement

To impose current correlations, the method needs the historical and current correlation matrices for every pair of assets, and their Choleski factors.

$\Rightarrow$ The curse of dimensionality is *not* avoided; D is the exception.

Interpretation

"No covariance matrices, no curse of dimensionality" is an advantage of *plain* non-parametric HS (MR-2 d). Correlation-weighting gives part of that advantage up in exchange for current correlations. FHS keeps it, because it bootstraps whole standardised return vectors instead of using a matrix.

Why the other options are wrong

A is incorrect. True: the parametric correlation adjustment on top of HS makes the approach semi-parametric.

B is incorrect. True: it is a generalisation of volatility-weighted HS and can be used with it.

C is incorrect. True: the updated matrix holds both variances and correlations, which makes it richer than volatility-weighting alone.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Correlation-weighted Historical Simulation, eqs. 2.4 and 2.5

Solution MR-2 c.6 Answer: D

Bank Q113 • BT L2.T5.80.1

Step 1 – The FHS steps

(1) Fit a conditional volatility model such as an (asymmetric) GARCH. (2) Divide each realised return by its volatility forecast to get standardised, i.i.d. returns. (3) Bootstrap the standardised returns and rescale them by tomorrow's volatility forecast. (4) Read VaR and ES off the simulated losses.

Step 2 – Find the step that is not there

VaR is read from the bootstrapped empirical distribution. No parametric distribution is fitted to the standardised returns.

$\Rightarrow$ D is the exception.

Interpretation

FHS is semi-parametric because the *volatility* is modelled parametrically, while the *shape* of the return distribution stays empirical. Fitting a parametric VaR model would throw away the non-parametric half.

Why the other options are wrong

A is incorrect. This is step 1 of FHS.

B is incorrect. Standardising returns by their conditional volatility is step 2.

C is incorrect. Bootstrapping the standardised returns is the simulation step.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Filtered Historical

Simulation, eq. 2.6

Solution MR-2 c.7 Answer: C

Bank Q117 • BT P2.T5.22.4.2

Step 1 – Requirement 3: VaR above the worst historical loss

This needs returns rescaled to current volatility. Basic HS and bootstrap HS only reuse observed losses, so they fail.

Step 2 – Requirement 1: same-day cross-sectional correlation

Kept by any method that uses all assets' returns from the same date together: basic, bootstrap, volatility-weighted and filtered HS all pass.

Step 3 – Requirement 2: multi-day forecast with autocorrelation

Projecting volatility dynamics over 20 or more days needs a time-series volatility model. Only FHS combines a GARCH model with bootstrapped standardised returns, so it can simulate a volatility path forward.

$\Rightarrow$ Only filtered HS meets all three requirements.

Interpretation

Volatility-weighted HS is the near-miss: it can exceed the historical maximum and keeps same-day correlation, but it only rescales by today's forecast and has no model for how volatility evolves over the next 20 days.

Why the other options are wrong

A is incorrect. Basic HS cannot produce a VaR beyond the worst observed loss and has no multi-day dynamics; it meets only the correlation requirement.

B is incorrect. Bootstrap HS is capped by the worst historical loss and treats draws as independent, ignoring autocorrelation and volatility clustering.

D is incorrect. Volatility-weighted HS meets requirements 1 and 3 but has no volatility model to project dynamics over a 20-day horizon.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Volatility-weighted and Filtered Historical Simulation

Solution MR-2 c.8 Answer: D

Workbook B2.3 • S3, S4

Step 1 – Coherence

A coherent risk measure satisfies monotonicity, positive homogeneity, translation invariance and subadditivity, $\rho(X + Y) \leq \rho(X) + \rho(Y)$. ES is coherent; VaR "may not always satisfy the subadditivity condition" (ORR-19), and "may not behave in a sub-additive fashion" (Gregory, CR-14).

Step 2 – FHS and ES

FHS bootstraps GARCH-standardised returns and rescales them by tomorrow's volatility forecast; Dowd notes that it "can also be extended easily to allow for the estimation of ES as well as VaR". Using ES is also the suggested response to Pritsker's finding that FHS VaR may not pay enough attention to extreme observations.

Interpretation

D is correct. For a merger involving a skewed credit book, ES is the measure that guarantees the combined number cannot exceed the sum of the parts, so desk limits and capital allocation stay

consistent with the firm-wide figure.

Why the other options are wrong

A is incorrect. Scope/condition error. VaR is subadditive for elliptical distributions such as the normal; non-subadditivity "can occur when assets in portfolios have very skewed loss distributions", exactly the credit desk here.

B is incorrect. Sibling swap. Being capped by the largest historical loss is a property of basic HS; FHS, like the Hull and White approach, scales returns by current volatility and "allows us to get VaR and ES estimates that can exceed the maximum historical loss".

C is incorrect. Polarity flip. Yamai and Yoshiba found ES estimators had much bigger standard errors than VaR for particularly heavy-tailed distributions, because ES is more prone to large but infrequent losses.

LO: MR-2 c; ORR-19 a; CR-14 j: Describe the approaches used and the challenges faced in modeling derivatives risk.

• **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 and footnotes 13 and 15; Ch.3, 1.5; BCBS (2009), Range of Practices, risk measures section; Gregory, The xVA Challenge 4e, Ch.2

MR-2 • Non-Parametric Approaches • Vardeez: VERY IMPORTANT

MR-2 d

P1 CONCEPT

Identify advantages and disadvantages of non-parametric estimation methods.

Workbook: A19 Non-Parametric VaR • Tier A

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A19 covers this LO, and the reported "difference in non-parametric approaches" question lives here and in MR-2 c.
Vardeez: no separate pointer; his advantages and disadvantages table is reproduced below.

Reported exam shapes

- **A19·S4** Differences among the non-parametric approaches, including what each one fixes and what it cannot do.

What the questions look like: "which is an advantage", "each of the following EXCEPT", or a short scenario (new market, short calm sample, very high confidence level) asking which concern applies or which method fits. Learn the two lists as pairs, because GARP flips one into the other.

1. The two lists

Advantages	Disadvantages
Intuitive and conceptually simple	Results are very (often completely) dependent on the historical data set
No parametric assumptions: **fat tails, skewness** and other non-normal features are accommodated automatically	A **quiet** sample gives VaR and ES that are **too low**; a volatile sample gives estimates that are too high
Can accommodate any type of position, including derivatives	Slow to reflect **shifts** (e.g. a permanent change in exchange rate risk) and sudden turbulence
Easy to implement, even on a spreadsheet	Extreme losses that are unlikely to recur can dominate the estimate
No covariance matrices, no curse of dimensionality	**Ghost (shadow) effects**
Data are (often) readily available (Bloomberg, in-house marks)	No allowance for **plausible events that did not occur** in the sample
Results are easy to report and communicate to senior managers, supervisors and rating agencies	Constrained by the **largest loss** in the data; poorly suited to extremes with small or medium samples
Confidence intervals are easy to produce	Need enough history: hard for new markets or instruments
Can be refined with semi-parametric add-ons (BRW, HW, FHS, correlation-weighting)	

2. Remedies and the window length

- **Most weaknesses have a refinement.** Problems with volatility, turbulence and correlations are eased by semi-parametric adjustments such as HW, FHS and correlation weighting. Ghost effects: age-weighting and letting the sample grow over time (MR-2 c).
- **The maximum-loss constraint never fully goes away.** Sophisticated versions (HW, FHS) relax it, but non-parametric estimates remain constrained by the largest loss in a way that parametric estimates are not.

- **Window length.** Rule of thumb: at least 500 daily observations (two years). A longer window worsens aged-data problems, lengthens the distortion from unlikely-to-recur events and ghost effects, and drowns out current news, so estimates respond less to current conditions.
- **Never rely on them alone.** Complement non-parametric estimates with stress testing, which covers the plausible events the sample does not contain.

Exam traps

- **Direction of the sample bias.** Quiet sample $\Rightarrow$ too *low*; volatile sample $\Rightarrow$ too *high*. GARP flips this.
- **Real data $\neq$ accurate.** Using actual events does not make the estimate more accurate than a parametric one; accuracy depends on the sample.
- **Longer window is not a free lunch.** More precision, less responsiveness.
- **Advantage dressed as disadvantage.** "Hard to communicate" and "needs covariance matrices" are false; "depends on historical data" is the central weakness, not a strength.
- **Resampling or reweighting does not create new tail events.** The bootstrap and age-weighting only reuse the observed losses.

Practice questions: MR-2 d

Question MR-2 d.1

Bank Q128 • GARP PE1 Q53

A risk analyst at an investment bank is evaluating the historical simulation approach to estimating ES. Which of the following statements accurately describes an advantage of the historical simulation approach to estimating ES?

A. Since the data reflect actual historical events, an ES estimate derived from this data is more accurate than an estimate derived from using parametric methods.

B. The precision of ES estimates and their responsiveness to new market observations can typically be enhanced by increasing the length of the sample period.

C. Major shifts in markets, their structure, or the factors that influence them can be ignored since any effects of these shifts will be included in historical datasets.

D. Features such as fat tails and skewness are easily accommodated without making assumptions about the distribution.

Question MR-2 d.2

Bank Q129 • GARP Mock 2 Q69

A hedge fund manager has asked her risk analyst to incorporate nonparametric methods into their statistical modeling program. The analyst, used to parametric methods, is concerned about using nonparametric methods. Which of the following statements would accurately reflect the analyst's concerns?

A. Historical data is a relatively small input to the calculations.

B. Model outputs tend to exhibit positive skewness and fat tails.

C. These methods may be less reliable for newer markets or instruments.

D. Volatile data periods lead to value at risk (VaR) and expected shortfall (ES) estimates that are too low.

Question MR-2 d.3 Bank Q124 • BT L2.T5.81.1 (rebuilt)

In regard to non-parametric estimation methods, EACH of the following is an ADVANTAGE EXCEPT:

A. Intuitive and easy to implement on a spreadsheet.

B. Avoids the dimension problems and complex variance-covariance matrices of parametric methods.

C. Good at allowing for plausible events that did not actually occur in the sample period.

D. Not hindered by skewness or fat tails in the data.

Question MR-2 d.4 Bank Q125 • BT L2.T5.81.2 (rebuilt)

In regard to non-parametric estimation methods, EACH of the following is a DISADVANTAGE EXCEPT:

A. Difficult to communicate to senior management.

B. Cannot easily be applied to new instruments or markets that lack sufficient history.

C. More constrained than parametric methods by the largest loss in the historical dataset.

D. Difficulty detecting abrupt shifts (e.g., a permanent change in exchange rate risk) during the sample period.

Question MR-2 d.5 Bank Q122 • AnalystPrep Q.6465 (rebuilt)

A risk manager uses bootstrap historical simulation on a small loss dataset in which the worst observed daily loss is 4.8%. Which of the following is the key limitation of the approach in this setting?

A. Resamples cannot contain a loss larger than 4.8%, so the method cannot reflect plausible losses worse than any observed.

B. Resamples require the returns to be normally distributed.

C. The bootstrap estimate is less precise than the raw historical simulation estimate.

D. Each resample is drawn without replacement, which shrinks the sample size.

Question MR-2 d.6 Bank Q126 • BT P2.T5.22.4.3 (rebuilt)

A risk manager has only 63 daily P/L observations for a new portfolio, all from a low-volatility regime. The board wants VaR at 99.5%, 99.9% and 99.99% confidence. Other portfolios are reported with non-parametric methods because they are intuitive and simple. Which approach is most appropriate here?

A. Basic historical simulation, since it is intuitive and consistent with the other portfolios.

B. Bootstrap historical simulation with many resamples to generate additional tail observations.

C. Age-weighted historical simulation to emphasize the most recent observations.

D. A parametric (distributional) approach, because the sample is too short and too calm to support non-parametric estimates at such confidence levels.

Question MR-2 d.7 Workbook A19.4 • S1 • Aug-2026

A model validator at an Indian insurer is reviewing the age-weighted historical simulation engine used for an INR 12,000 million equity portfolio. The engine uses $n = 250$ daily observations and $\lambda = 0.99$. To judge how concentrated the model is on recent data, the validator wants the total probability weight carried by the 50 most recent observations. Which of the following is closest?

A. 0.200

B. 0.395

C. 0.430

D. 0.544

Solutions • MR-2 d

Solution MR-2 d.1

Answer: D

Bank Q128 • GARP PE1 Q53

Step 1 – What HS assumes

Historical simulation uses the empirical P/L distribution and imposes no distributional form.

Step 2 – Consequence for ES

Any fat tails or skewness present in the data pass straight into the ES estimate.
$\Rightarrow$ D is a genuine advantage.

Interpretation

Each wrong option overstates a property of the historical data: accuracy, the effect of a longer window, and the treatment of structural shifts. These are all listed by Dowd as weaknesses or trade-offs, not advantages.

Why the other options are wrong

A is incorrect. Real data do not guarantee accuracy; the estimate depends on whether the window was calm or turbulent, and on its length.
B is incorrect. A longer window raises precision but *lowers* responsiveness to new observations; the two trade off.
C is incorrect. Structural shifts cannot be ignored: HS reflects them only slowly and has difficulty detecting them.

LO: MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.5 Advantages and Disadvantages of Non-parametric Methods

Solution MR-2 d.2

Answer: C

Bank Q129 • GARP Mock 2 Q69

Step 1 – What non-parametric methods need

They rely critically on a sufficient historical sample of the positions' risk factors.

Step 2 – Where that fails

New markets and new instruments have little or no history, so the estimates are unreliable.
$\Rightarrow$ C is a legitimate concern.

Interpretation

The other options each reverse a fact: historical data is the *main* input, fat tails in the output come from the data rather than being a defect, and the volatile-period bias goes the other way.

Why the other options are wrong

A is incorrect. History is the main input to non-parametric methods, not a small one.
B is incorrect. The methods impose no shape; skewness and fat tails in the output simply reflect the data and are an advantage.
D is incorrect. Polarity flip: volatile periods make VaR and ES too *high*; quiet periods make them too low.

LO: MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.5 Advantages and Disadvantages of Non-parametric Methods

Solution MR-2 d.3 — Answer: C

Bank Q124 • BT L2.T5.81.1

Step 1 – Check the advantages

Intuitive and spreadsheet-friendly (A), no covariance matrices or dimension problems (B), not hindered by skewness or fat tails (D): all are on Dowd's list of advantages.

Step 2 – The odd one out

Non-parametric estimates come only from the sample, so events that did not occur in the window have no influence.

$\Rightarrow$ "Allowing for plausible events that did not occur" is a listed *disadvantage*; C is the exception.

Interpretation

This is why non-parametric methods are always paired with stress testing: the stress scenarios supply the plausible events the sample does not contain.

Why the other options are wrong

A is incorrect. A true advantage: simple to build and run on a spreadsheet.

B is incorrect. A true advantage: no variance-covariance matrix is needed, so there is no curse of dimensionality.

D is incorrect. A true advantage: the empirical distribution carries skewness and fat tails automatically.

LO: MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.5

Solution MR-2 d.4 — Answer: A

Bank Q125 • BT L2.T5.81.2

Step 1 – Check the disadvantages

Need for sufficient history (B), the maximum-loss constraint, which binds more tightly than for parametric methods (C), and difficulty handling shifts during the sample period (D) are all on Dowd's list.

Step 2 – The odd one out

Dowd lists "results that are easy to report and communicate to senior managers" as an *advantage*.

$\Rightarrow$ A is the exception.

Interpretation

GARP-style EXCEPT questions often take one item from the opposite list and phrase it negatively. Learning the advantages and disadvantages as matched pairs makes the odd one out easy to spot.

Why the other options are wrong

B is incorrect. A genuine disadvantage: non-parametric methods need enough historical data.

C is incorrect. A genuine disadvantage: estimates are constrained by the largest historical loss in a way parametric estimates are not.

D is incorrect. A genuine disadvantage: structural shifts are hard to detect and slow to show up.

LO: MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.5

Solution MR-2 d.5 — Answer: A

Bank Q122 • AnalystPrep Q.6465

Step 1 – What a resample can contain

Bootstrapping draws only from the observations in the data set, with replacement.

Step 2 – Consequence

No resample can contain a loss beyond the largest observed one (4.8%), and events absent from the sample cannot appear.

$\Rightarrow$ The bootstrap cannot reflect plausible losses worse than any observed: A.

Interpretation

This is the maximum-loss constraint of non-parametric methods. The bootstrap improves precision and gives confidence intervals, but it cannot extrapolate; HW and FHS relax the constraint by rescaling returns.

Why the other options are wrong

B is incorrect. Bootstrap HS is non-parametric and makes no normality assumption.

C is incorrect. Polarity flip: the bootstrap estimate is usually *more* precise than the raw HS estimate.

D is incorrect. Resampling is with replacement, and each resample has the same size as the original.

LO: MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.2 and 2.5

Solution MR-2 d.6 — Answer: D

Bank Q126 • BT P2.T5.22.4.3

Step 1 – How much probability one observation carries

$$\frac{1}{63} = 1.59\%$$

Step 2 – Compare with the required tail probabilities

The tails asked for are 0.5%, 0.1% and 0.01%. Expected observations beyond them: $63 \times 0.005 = 0.32$, $63 \times 0.001 = 0.06$, $63 \times 0.0001 = 0.006$.

$\Rightarrow$ All three VaRs lie beyond the worst observed loss, which no non-parametric method can reach, and the calm regime would understate risk anyway.

Interpretation

A distributional assumption (possibly with EVT tail modelling) is needed to extrapolate. This question combines three disadvantages at once: too little data, a quiet sample, and the maximum-loss constraint.

Why the other options are wrong

A is incorrect. Plain HS cannot resolve probabilities below $1/63 = 1.59\%$ and reflects only the quiet regime; consistency with other portfolios does not fix that.

B is incorrect. Bootstrapping resamples existing observations and never creates losses beyond the sample maximum.

C is incorrect. Age-weighting only reweights the same 63 observations; it does not extend the tail or fix the short, calm sample.

LO: MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.5

Solution MR-2 d.7 **Answer: C**

Workbook A19.4 • S1

Step 1 – Normalising constant

$$1 - \lambda^n = 1 - 0.99^{250} = 1 - 0.081059 = 0.918941$$

$\Rightarrow$ An infinite EWMA would leave about 8.1% of weight beyond day 250; the BRW normaliser rescales so the 250 in-sample weights sum to 1.

Step 2 – Un-normalised weight on the 50 most recent observations

$$\sum_{i=1}^{50} \lambda^{i-1}(1-\lambda) = 1 - \lambda^{50} = 1 - 0.99^{50} = 1 - 0.605006 = 0.394994$$

Step 3 – Normalise

$$\sum_{i=1}^{50} w(i) = \frac{1-\lambda^{50}}{1-\lambda^{250}} = \frac{0.394994}{0.918941} = 0.4298$$

$\Rightarrow$ The most recent 20% of the window carries about 43% of the probability mass.

Interpretation

Age-weighting concentrates information in recent data, which raises responsiveness but reduces the effective sample size (a BRW drawback). It also explains the reduced ghost effect: the oldest observation leaves with weight $\lambda^{249}w(1) = 0.00089$, far below $1/250 = 0.004$.

Why the other options are wrong

A is incorrect. Used equal weights: $50/250 = 0.200$.

B is incorrect. Dropped the normaliser and reported the intermediate result $1 - \lambda^{50} = 0.395$ (Step 2), so the 250 weights would sum to only 0.919.

D is incorrect. Multiplied $w(1)$ by 50, ignoring decay: $50 \times 0.0108821 = 0.544$.

LO: MR-2 c; MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Age-weighted Historical Simulation, eq. 2.2

MR-3 • Parametric Approaches (II): Extreme Value • Vardeez: IMPORTANT

MR-3 b

P1 CONCEPT

Describe extreme value theory (EVT) and its use in risk management.

Workbook: A17 Extreme Value Theory: GEV vs POT • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: a GEV theory question came up in the August 2026 sitting (Map #25, 5/16).
Vardeez: EVT and the 3 general cases of the GEV distribution. Do not go too deep into calculations here.

Reported exam shapes

- **S1** GEV theory: Fisher-Tippett, three parameters, the Frechet, Gumbel and Weibull cases.
- **S5** Shape parameter $\xi > 0$ means fat tails.
- **S6** Operational-risk framing of EVT and block maxima.

What the questions look like: "which of the following is correct" statements. The distractors flip the polarity of ξ (attaching "lighter tails" or "Weibull" to a positive tail index), swap the GEV and POT attributes, or make absolute claims (EVT needs a known parent, EVT is free of judgement).

1. Why extremes need their own theory

Extreme events (market crashes, failures of major institutions, financial crises, natural catastrophes) are low-probability, high-impact events. Estimating their risk is hard for one reason: **they are rare**, so there are few extreme observations, and estimates are very uncertain, especially when we need risks *beyond* the range of the historical data (an unprecedented fall).

- A distribution fitted to the whole data set accommodates the many central observations, not the sparse extremes: good for the centre, poor for the tails.
- Extremes are governed by **extreme-value theorems**, not by central limit theorems. Distributions justified by central limit theory (above all the normal) are exactly what EVT says not to use for extremes.
- The standard analogy is hydrology: how high to build dikes and sea walls, with even less data and quantiles well outside the sample.

EVT is the branch of applied statistics tailor-made for this problem. Unlike "central tendency" statistics, it focuses on the extremes themselves and uses extreme-value theorems to choose the distribution and estimate its parameters.

2. Fisher-Tippett and the GEV distribution

Take iid losses X from an *unknown* parent distribution $F(x)$. Draw a sample of size n and let M_n be its maximum. If n is large, M_n is an extreme value. The **Fisher-Tippett theorem** says that as n gets large, the distribution of M_n converges to the **generalised extreme value (GEV)** distribution:

$$H_{\xi,\mu,\sigma}(x) = \begin{cases} \exp\left[-\left(1+\xi\,\dfrac{x-\mu}{\sigma}\right)^{-1/\xi}\right] & \xi \neq 0 \\ \exp\left[-\exp\left(-\dfrac{x-\mu}{\sigma}\right)\right] & \xi = 0 \end{cases} \qquad \text{with } 1+\xi\,\frac{x-\mu}{\sigma} > 0.$$

Three parameters

- μ, the **location** parameter: the central tendency of M_n.
- σ, the **scale** parameter: the dispersion of M_n.
- ξ, the **tail index**: the shape, or heaviness, of the tail. It decides which case of the GEV applies.

μ and σ are *related to, but not the same as*, the mean and standard deviation. The same theory works for minima: multiply the data by -1.

> **Correction to the source notes.** Vardeez's notes print the $\xi = 0$ (Gumbel) case as $\exp\left[-\exp\left(\frac{x-\mu}{\sigma}\right)\right]$. The inner exponent needs a minus sign: $\exp\left[-\exp\left(-\frac{x-\mu}{\sigma}\right)\right]$, as in GARP eq. 3.1 and the Tier A workbook. Without it the function falls as x rises and is not a distribution function.

3. The three cases of the GEV (shapes S1 and S5)

Tail index	GEV becomes	Tails	Parent distributions (domain of attraction)
$\xi > 0$	Frechet	**Heavy**: power-law decay	Levy, Student t, Pareto. The relevant case for financial returns; estimates are typically positive but below 0.35.
$\xi = 0$	Gumbel	**Light**: exponential decay	Normal, lognormal.
$\xi < 0$	Weibull	**Lighter than normal**; finite upper bound	Rare in finance: few return series are this light-tailed.

The parent's **domain of attraction** decides the case: extremes drawn from a t or Levy parent tend to the Frechet, extremes from a normal or lognormal parent tend to the Gumbel. Financial risk management therefore works with the first two cases.

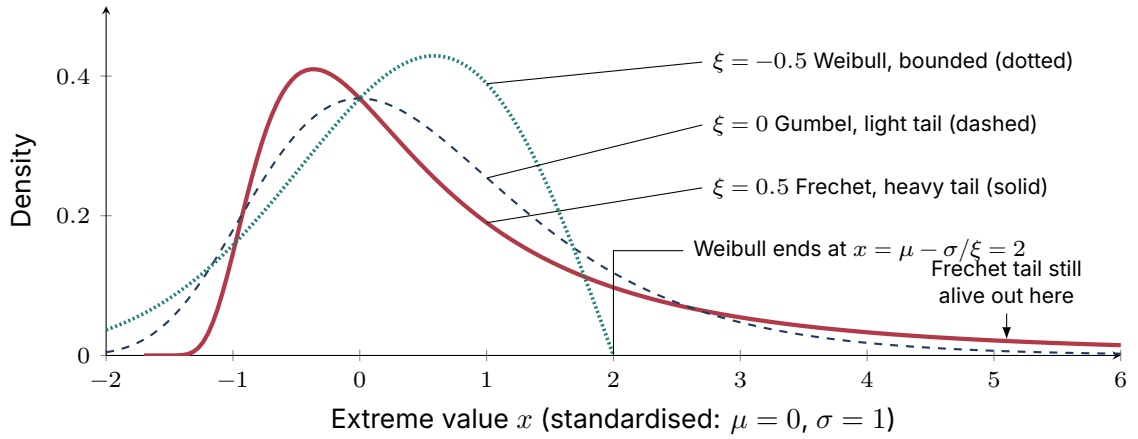


Figure 1. The tail index is the tail. At $\xi < 0$ the distribution has a finite upper bound and simply stops, which is why it is no use for financial losses. At $\xi = 0$ the tail decays exponentially. At $\xi > 0$ it decays as a power, so very large values remain possible. Both the Frechet and the Gumbel are skewed to the right; the Frechet more so.

Frechet versus Gumbel. Frechet quantiles rise with ξ and tend to the Gumbel values as $\xi \to 0$, so EV risk measures are **sensitive to the tail index**. GARP's Example 3.5 ($\mu = 2\%$, $\sigma = 0.7\%$, $\xi = 0.3$, $n = 100$) shows the size of the gap:

EV VaR	Frechet ($\xi = 0.3$)	Gumbel ($\xi = 0$)
99.5%	2.537%	2.483%
99.9%	4.322%	3.611%

The gap widens sharply as the confidence level rises. This is why the choice of case matters.

4. Choosing between Frechet and Gumbel

1. **Confidence in the parent.** If we are confident the parent is, say, a t, choose the EV distribution in whose domain of attraction it lies: the Frechet ($\xi > 0$).
2. **Statistical test.** Test the significance of ξ: Gumbel if insignificant, Frechet otherwise. The danger: wrongly concluding $\xi = 0$ **understates** extreme risk.
3. **Conservatism.** Given model risk, and since the estimated risk measure *increases* with the tail index, the safer option is **always to choose the Frechet**.

5. Using EVT in risk management

- **Risk measures.** A fitted EV distribution gives both **VaR and ES** for very high confidence levels, and can return values *beyond* the worst loss in the sample (historical simulation cannot).
- **Which distribution the probabilities refer to.** GEV probabilities refer to the distribution of *extreme losses*, not the parent. The 5th percentile of the GEV is a small extreme loss: on the left of the extreme-loss distribution but in the right tail of the parent. The link is $\Pr\{M_n < M_n^*\} = [\alpha]^n$.
- **Estimating the parameters.**
  - *Maximum likelihood*: well grounded; consistent and asymptotically normal if $\xi > -1/2$; needs numerical optimisation of a complex likelihood; small-sample problems.
  - *Regression* (Gumbel's method): order the extremes and fit by least squares. Easier than ML.
  - *Semi-parametric*: the **Hill estimator** of ξ. The hard part is choosing k, how many tail observations to use: a larger k means more data and lower variance but more bias. Hill plots may never settle ("Hill horror plots").
- **Horizon scaling.** The mean scales with h, the standard deviation with $\sqrt{h}$, the tail index not at all. Empirically $\text{VaR}(h) = \text{VaR}(1)\, h^{\kappa}$ with $\kappa \approx 0.45$, so the square-root rule is usually inappropriate for EV distributions.
- **Limitations.** Few extreme observations and wide confidence intervals; considerable model risk and **subjective judgement** (block size, threshold, k); sensitivity to small-sample effects, biases and non-linearities. "We shouldn't ask more of the theory than it can deliver."

The operational-risk framing (shape S6)

In operational risk, EVT complements the loss distribution approach (LDA) in the tail of the severity distribution. **Block maxima** (Fisher-Tippett) examines the maxima equally spaced in time: the maximum loss per period and per unit of measure. **POT** looks at losses above a high threshold. The caution: EVT is valid only if **a single mechanism** is responsible for all observed losses. Losses pooled across institutions, business lines and risk types are unlikely to meet this.

Formulas to recognise (Vardeez: not for heavy calculation)

GEV (Fisher-Tippett)	$\exp\left[-\left(1+\xi\frac{x-\mu}{\sigma}\right)^{-1/\xi}\right]$ for $\xi \neq 0$; $\exp[-\exp(-\frac{x-\mu}{\sigma})]$ for $\xi = 0$
GEV quantile	Frechet: $x_p = \mu - \frac{\sigma}{\xi}\left[1-(-\ln p)^{-\xi}\right]$; Gumbel: $x_p = \mu - \sigma \ln[-\ln p]$
EV VaR (block size n)	Replace p by α^n: Frechet $\mu_n - \frac{\sigma_n}{\xi_n}\left[1-(-n\ln\alpha)^{-\xi_n}\right]$; Gumbel $\mu_n - \sigma_n \ln[-n\ln\alpha]$
Standardised Gumbel	5% quantile -1.0972; 95% quantile 2.9702

Exam traps

- **Polarity flip on ξ.** $\xi > 0$ is Frechet, heavy tails, the case for financial and loss data. $\xi < 0$ is Weibull, lighter than normal. A statement attaching "lighter" or "Weibull" to a positive tail index is wrong.
- **Wrong limit theorem.** Fisher-Tippett gives the GEV for maxima; GPBdH gives the generalised Pareto for threshold excesses; the CLT gives the normal for averages. GARP swaps them.
- **"The parent must be known" (or normal).** EVT works whatever the parent, which is usually unknown. Normal and lognormal are only examples of the Gumbel domain.

- $\xi \to 0$ **gives the Gumbel, not the normal.** Neither $\xi \to \infty$ nor $\xi < 0$ gives a normal distribution.
- **Testing $\xi = 0$ is not the safe route.** Wrongly accepting $\xi = 0$ understates risk; the conservative choice is Frechet.
- **Absolute claims.** EVT is not free of judgement, does not use the whole distribution, and in operational risk is not valid for pooled losses unless one mechanism produced them.
- **Estimation methods swapped.** Regression orders the extremes and uses least squares; ML maximises a likelihood; Hill averages log-exceedances and is very sensitive to k.

Practice questions: MR-3 b

Question MR-3 b.1

Bank Q147 • Schweser Module Quiz 3.1 Q1

According to the Fisher-Tippett theorem, as the sample size n gets large, the distribution of (normalised) extremes converges to a:

A. normal distribution.

B. uniform distribution.

C. generalised Pareto distribution.

D. generalised extreme value distribution.

Question MR-3 b.2

Bank Q144 • AnalystPrep Q.2180

If $\xi < 0$, the GEV becomes the Weibull distribution, but this distribution is rarely used to model financial returns mainly because:

A. its cumulative distribution has heavier than normal tails and very few empirical financial returns are heavy-tailed.

B. its cumulative distribution has lighter than normal tails and very few empirical financial returns are light-tailed.

C. it is asymmetric.

D. it is symmetric.

Question MR-3 b.3 Bank Q140 • AP Mock 2 Q30

Salim Abdalla, FRM, recommends the use of extreme value theory. While discussing EVT, the following statements are made:

I. To apply EVT, the underlying loss distribution can be any of the commonly used distributions, e.g. normal or lognormal, and will usually be unknown.

II. If the tail parameter of the generalised extreme value (GEV) distribution goes to zero, then the GEV essentially becomes a Gumbel distribution.

III. If the tail parameter of the GEV distribution is less than zero, the GEV becomes the normal distribution.

IV. As the threshold value is increased, the distribution of exceedances converges to a generalised Gumbel distribution.

Which of the statements above are incorrect?

A. I and III.

B. II and IV.

C. III and IV.

D. I and II.

Question MR-3 b.4 Bank Q151 • BT answers P2.T7.804.2 (rebuilt)

Sandra is fitting a distribution to the extreme loss tail of historical returns using block maxima. Her situation includes:

I. The parent distribution $F(x)$ is actually unknown.

II. The parent (non-extreme) distribution is well described by a Student's t, i.e. non-normal.

III. End users might prefer the extreme tail to have exponential tails (Gumbel).

IV. End users want the extreme tail distribution to exhibit positive (right) skew.

Which of these issues, in theory, DISQUALIFIES the generalised extreme value (GEV) distribution?

A. Only I, because $F(x)$ must be specified.

B. Only II, because the parent must be normal.

C. Both III and IV, because the GEV cannot be a Gumbel or have skew.

D. None of these issues disqualifies the GEV.

Question MR-3 b.5 New • built from GARP MR Ch.3, 3.1 (Dowd) and Vardeez

A market risk analyst fits a GEV distribution to the monthly maxima of daily trading losses on an equity book. The estimated tail index is $\xi = 0.12$, but a test cannot reject $\xi = 0$ at the 5% level. The head of model validation asks how the choice between the Frechet and Gumbel cases should be made. Which of the following statements is most appropriate?

A. Because ξ is not significantly different from zero, the Gumbel case should be adopted, since accepting $\xi = 0$ can only overstate, never understate, the extreme risk measures.

B. If the parent loss distribution is believed to be a Student's t, the Gumbel case is appropriate, because the t lies in the domain of attraction of the Gumbel distribution.

C. Given model risk, and since EV risk measures rise with the tail index, the conservative choice is the Frechet case, because wrongly concluding $\xi = 0$ would understate extreme risk.

D. The Weibull case should be considered first, because a tail index estimate close to zero indicates tails lighter than those of the normal distribution.

Question MR-3 b.6 Bank Q153 • GARP PE2 Q66 • Mock 4 Q67

A risk analyst at an investment bank compares methods for estimating the parameters of an extreme value (EV) distribution for operational losses, covering maximum likelihood, regression and semi-parametric approaches. Which of the following statements is correct?

A. The regression method orders the extreme observations and obtains least squares estimates of the EV parameters.

B. The maximum likelihood method estimates the tail index by averaging a chosen number of the most extreme observations.

C. The Hill estimator is insensitive to how many tail observations are included.

D. Maximum likelihood is simple to apply because it needs no optimisation of a likelihood function.

Question MR-3 b.7 Bank Q148 • BT answers L2.T5.86.2

Which of the following is TRUE about extreme value theory (EVT), according to Dowd?

A. EVT can produce an estimate for expected shortfall (ES) but NOT value at risk (VaR).

B. EVT can produce an estimate for value at risk (VaR) but NOT expected shortfall (ES).

C. EVT makes use of all information in the entire distribution.

D. The application of EVT methods involves much subjectivity, and estimates are very uncertain and prone to small-sample effects, biases and non-linearities.

Question MR-3 b.8 Workbook A17.1 • S1, S2, S5, S6 • Aug-2026 • Map #25

The head of operational risk modelling at a European insurer is presenting to newly hired analysts on how extreme value theory can complement the loss distribution approach in the tail of the severity distribution. She compares the block maxima (GEV) approach, applied to the largest loss per quarter in each unit of measure, with the peaks-over-threshold approach. Which of the following statements should she include in the presentation?

A. The block maxima approach fits a two-parameter generalised Pareto distribution to the largest loss in each quarter, so it needs fewer parameters than the peaks-over-threshold approach.

B. The block maxima approach needs no threshold choice, but it can waste useful data because a quarter may contain more than one extreme loss; the peaks-over-threshold approach models all losses above a high threshold.

C. A positive tail index places the fitted GEV in the Weibull case with lighter than normal tails, so a positive estimate signals that extreme losses are less likely than under a normal parent.

D. Because both approaches rest on extreme-value theorems, they remain valid for losses pooled across business lines and risk types, whatever mechanism produced each loss.

Solutions • MR-3 b

Solution MR-3 b.1 — Answer: D

Bank Q147 • Schweser Module Quiz 3.1 Q1

Step 1 – What the theorem is about

Fisher-Tippett concerns the limiting distribution of the *maximum* M_n of n iid observations as n grows.

Step 2 – The limit

The only non-degenerate limit is the GEV family, which nests the Frechet ($\xi > 0$), Gumbel ($\xi = 0$) and Weibull ($\xi < 0$).

$\Rightarrow$ For extremes, the GEV plays the role the normal plays for averages.

Interpretation

The parent distribution does not need to be known: the limit law for maxima is the GEV whatever the parent.

Why the other options are wrong

A is incorrect. The normal is the central limit theorem's limit for sums and averages, not for extremes.

B is incorrect. The uniform distribution is not a limit law for extremes; no theorem supports it.

C is incorrect. Sibling swap. The generalised Pareto is the limit for excesses over a high threshold (GPBdH theorem, the POT approach), not for block maxima.

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1 Generalised Extreme-Value Theory, eq. 3.1

Solution MR-3 b.2 — Answer: B

Bank Q144 • AnalystPrep Q.2180

Step 1 – What the Weibull case describes

$\xi < 0$ corresponds to parents with tails *lighter* than the normal, and the distribution has a finite upper bound.

Step 2 – What financial returns look like

Empirical financial returns are typically heavy-tailed, with ξ estimates positive (but below 0.35).

$\Rightarrow$ A light-tailed limiting law rarely fits financial data.

Interpretation

This is why risk management works with the Frechet and Gumbel cases only (Figure 1).

Why the other options are wrong

A is incorrect. Polarity flip. The Weibull has lighter, not heavier, tails; heavy tails are the Frechet case.

C is incorrect. Asymmetry is not the reason: financial returns are often skewed too.

D is incorrect. The GEV family is not symmetric, and symmetry is not the stated reason.

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1

Solution MR-3 b.3 — Answer: C

Bank Q140 • AP Mock 2 Q30

Step 1 – Check each statement

I: EVT works whatever the parent (normal, lognormal, t, ...), which will usually be unknown. **Correct.**
II: $\xi \to 0$ gives the Gumbel distribution. **Correct.**
III: $\xi < 0$ gives the Weibull (tails lighter than normal), not the normal. **Incorrect.**
IV: By the GPBdH theorem, exceedances over a rising threshold converge to the generalised *Pareto*, not a "generalised Gumbel". **Incorrect.**
$\Rightarrow$ The incorrect statements are III and IV.

Interpretation

Two classic GARP swaps in one item: the wrong case for $\xi < 0$ and the wrong limit for threshold exceedances.

Why the other options are wrong

A is incorrect. Treats I as false: EVT does not require a known parent.
B is incorrect. Treats II as false: $\xi \to 0$ does give the Gumbel. It also misses that III is false.
D is incorrect. Marks the two correct statements as incorrect.

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1 and 3.2

Solution MR-3 b.4 — Answer: D

Bank Q151 • BT answers P2.T7.804.2

Step 1 – I and II: the parent

Fisher-Tippett gives the GEV as the limit of normalised maxima without knowing $F(x)$, and without needing a normal parent. A t parent simply places the extremes in the Frechet domain ($\xi > 0$).

Step 2 – III and IV: the form of the GEV

The GEV nests the Gumbel ($\xi = 0$, exponential tails), and GEV maxima distributions are right-skewed (the Frechet most of all).
$\Rightarrow$ None of the four issues disqualifies the GEV.

Interpretation

The GEV is the universal limit law for maxima, whatever the parent, and it contains all three cases.

Why the other options are wrong

A is incorrect. Confuses EVT with parametric VaR. Like the CLT, Fisher-Tippett does not need the parent to be specified.
B is incorrect. Wrongly assumes a normal parent is required; a t parent just implies a heavy tail.
C is incorrect. The Gumbel is the $\xi = 0$ special case of the GEV, and GEV maxima are positively skewed.

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1, Figure 3.1

Solution MR-3 b.5 — Answer: C

New • built from GARP MR Ch.3, 3.1

Step 1 – The three ways to choose

Dowd lists three: (1) choose the EV distribution in whose domain of attraction a known parent lies; (2) test the significance of ξ; (3) given model risk, always choose the Frechet.

Step 2 – Apply them to this case

The test cannot reject $\xi = 0$, but Dowd warns that relying on it "leaves us open to the danger that we might incorrectly conclude that ξ is 0, and this could lead us to underestimate our extreme risk measures". Since the estimated risk measure increases with the tail index, the Frechet is the safer option.

$\Rightarrow$ The conservative answer is the Frechet case.

Interpretation

With $\xi = 0.12$ the Frechet and Gumbel VaRs are close at 99.5% but diverge quickly at 99.9% and beyond (Section 3 table), which is where the model will be used.

Why the other options are wrong

A is incorrect. Polarity flip. Wrongly accepting $\xi = 0$ *understates* extreme risk, because EV risk measures rise with ξ.

B is incorrect. Sibling swap. The t lies in the domain of attraction of the Frechet, not the Gumbel.

D is incorrect. Wrong case. Tails lighter than normal correspond to $\xi < 0$ (Weibull); an estimate near zero points to the Gumbel boundary, and the Weibull is of little use for financial losses.

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1, choosing between the Frechet and the Gumbel

Solution MR-3 b.6 — Answer: A

Bank Q153 • GARP PE2 Q66

Step 1 – Match each method to its mechanics

Regression: order the extreme values and fit the EV parameters by least squares; easier than ML.
Maximum likelihood: maximise a complex likelihood or log-likelihood function, numerically.
Semi-parametric (Hill): estimate the tail index from the log-exceedances of the top k observations; choosing k is its hardest step.

$\Rightarrow$ Only A describes its method correctly.

Interpretation

GARP tests the methods by attaching each one's mechanics to another. Read every option as "is this the right method for that description?"

Why the other options are wrong

B is incorrect. Describes the Hill (semi-parametric) estimator, not ML.

C is incorrect. The Hill estimator is highly sensitive to the number of tail observations k; that is its main difficulty (bias against variance, "Hill horror plots").

D is incorrect. ML requires maximising a complex likelihood function; it is the harder method, not the simpler one.

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1 Estimation of EV Parameters; SchweserNotes 2026 Book 1, Reading 3, LO 3.b

Solution MR-3 b.7 Answer: D

Bank Q148 • BT answers L2.T5.86.2

Step 1 – What EVT delivers

A fitted EV tail gives both a tail quantile (VaR) and the conditional tail mean (ES). A and B are false.

Step 2 – What EVT uses

EVT deliberately uses only the extreme observations and ignores the body of the distribution. C is false.

Step 3 – The limitation

Extremes are rare, so estimates depend on subjective choices (block size, threshold, k) and few data points: very uncertain and prone to small-sample effects, biases and non-linearities.
$\Rightarrow$ D is Dowd's own conclusion.

Interpretation

EVT is a powerful but data-scarce tool. Dowd's advice: estimate confidence intervals, stress test, and do not ask more of the theory than it can deliver.

Why the other options are wrong

A is incorrect. EVT also gives VaR, which is a tail quantile.
B is incorrect. EVT also gives ES once the tail distribution is fitted.
C is incorrect. Describes historical simulation or a full parametric fit; EVT discards the body.

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.4 Conclusions

Solution MR-3 b.8 Answer: B

Workbook A17.1 • S1, S2, S5, S6 • Aug-2026

Step 1 – What block maxima is (GEV theory)

By the Fisher-Tippett theorem, the distribution of the maximum of a large iid sample converges to the GEV, with location μ, scale σ and tail index ξ. The operational-risk reading frames it as examining "the maxima equally spaced in time (e.g., the maximum operational loss per period of time and per UoM)".

Step 2 – What POT is

POT models excess losses over a high threshold u; by the GPBdH theorem they converge to the two-parameter generalised Pareto distribution (β, ξ) as u gets large.

Step 3 – Compare them

Dowd: the GEV "typically involves an additional parameter relative to the POT", and block maxima "can involve some loss of useful data relative to the POT approach, because some blocks might have more than one extreme in them"; on the other hand, POT "requires us to grapple with the problem of choosing the threshold, and this problem does not arise with the GEV". Statement B carries exactly these properties.

Interpretation

Neither approach dominates: "either approach is usually reasonable". With a short internal loss history, POT keeps more of the scarce tail observations; block maxima avoids the subjective threshold and handles clustered losses better, at the cost of efficiency.

Why the other options are wrong

A is incorrect. Sibling swap. The two-parameter GP distribution belongs to POT; the GEV fitted to block maxima has three parameters, typically one more than POT.

C is incorrect. Polarity flip. $\xi > 0$ is the Frechet case: power-law, heavy tails, more extreme losses than a normal parent. The Weibull case with lighter than normal tails is $\xi < 0$.
D is incorrect. Absolute claim. The ORR reading cautions that EVT "would only be valid if a single mechanism is responsible for all observed losses"; with losses from different business lines and risk types this is unlikely, which "raises concerns about the applicability of EVT in operational risk".

LO: MR-3 b; MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks; ORR-4 f. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1 and GEV vs POT; GARP, Operational Risk and Resilience, Ch.4, Extreme Value Theory and EVT: A Word of Caution

MR-3 • Parametric Approaches (II): Extreme Value • Vardeez: IMPORTANT

MR-3 c

P1 CONCEPT

Describe the peaks-over-threshold (POT) approach.

Workbook: A17 Extreme Value Theory: GEV vs POT • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A17. The map reports "peaks over threshold" and "POT threshold low consideration" from the sheets (Map #25).
Vardeez: no separate pointer for this LO; his chapter note applies: do not go too deep into calculations.

Reported exam shapes

- **S3** Peaks over threshold: the GPBdH theorem and the GP distribution of excess losses.
- **S4** POT threshold set too low: more data, more bias, less extreme.

What the questions look like: what POT models (the excess $X - u$, conditional on exceeding u), what it needs (two parameters plus a threshold), and the direction of the threshold trade-off. Distractors reverse the bias/variance polarity or claim the threshold can be chosen objectively.

Background on EVT, the GEV and the tail index is under MR-3 b. The GP distribution itself is developed under MR-3 e; this LO is about what POT models and how it is set up.

1. What POT models

The **peaks-over-threshold (POT)** approach applies EVT to the distribution of **excess losses over a high threshold** u. Let X be an iid loss with distribution function $F(x)$:

$$F_u(x) = \Pr\{X - u \le x \mid X > u\} = \frac{F(x+u) - F(u)}{1 - F(u)}, \qquad x > 0$$

Read it as: the probability that a loss exceeds the threshold by *at most* x, given that it does exceed the threshold. Two things to see in it:

- The variable modelled is the **excess** $X - u$, not the loss X itself.
- The conditioning event is $X > u$: the body of the distribution is thrown away.

The parent $F(x)$ can be any of the usual distributions (normal, lognormal, t) and **will usually be unknown**. That does not matter: as u gets large, the GPBdH theorem says $F_u(x)$ converges to a **generalised Pareto (GP)** distribution with only two parameters, a scale $\beta > 0$ and the tail index ξ (the same ξ as in GEV theory). See MR-3 e.

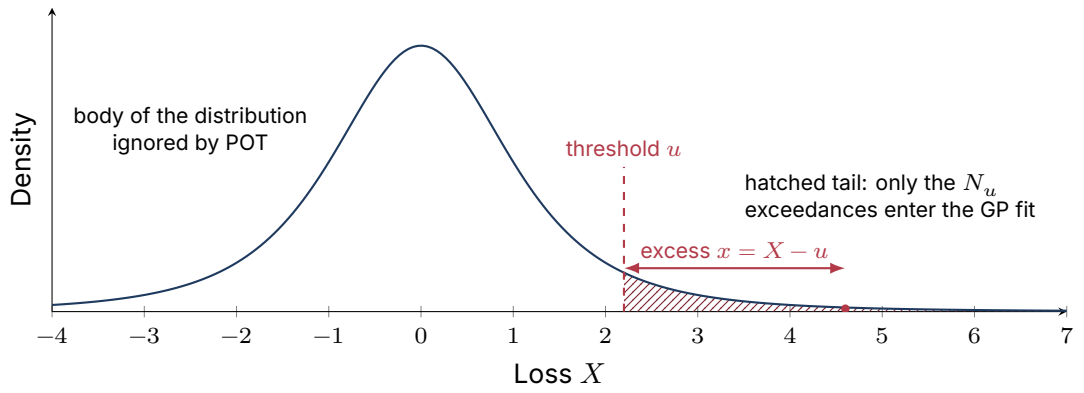


Figure 2. POT throws away the body of the loss distribution and models only what lies beyond the threshold u. The quantity modelled is the excess x, measured from u upward, not the loss itself.

2. From excess losses back to the loss tail

Rearranging $F_u(x)$ gives the tail of the loss distribution itself:

$$F(x) = (1 - F(u))\, G_{\xi,\beta}(x - u) + F(u), \qquad x > u$$

POT therefore needs an estimate of $F(u)$, the proportion of observations that do *not* exceed the threshold. The natural estimator is the observed proportion

$$\widehat{F(u)} = \frac{n - N_u}{n},$$

where N_u is the number of exceedances out of n observations. With β, ξ, u and N_u/n in hand, both **VaR and ES** follow in closed form (MR-3 e). ES exists only if $\xi < 1$.

3. The weak spot: choosing the threshold (shape S4)

The threshold u determines N_u. Dowd calls its choice "the weak spot of POT theory: it is inevitably arbitrary and therefore judgmental".

- u must be **high enough** for the GPBdH theorem to apply reasonably closely;
- but if u is **too high**, there are too few exceedances to estimate β and ξ reliably.

It is "very much akin to the problem of choosing k" for the Hill estimator: more data lowers variance but, by placing more weight on observations closer to the centre, raises bias.

	Lower threshold u	Higher threshold u
Exceedances N_u	More	Fewer
Variance of estimates	Lower (more data)	Higher (fewer data)
Bias	Higher: exceedances come from nearer the centre, GP approximation worse	Lower: deeper in the tail, GP approximation better
The risk	Sample of "extremes" is less extreme	Too few points for reliable estimates

A practical aid. Plot the mean-excess function (MEF) against u and choose a threshold where the MEF becomes horizontal. The choice still remains a judgement; maximum likelihood estimates β and ξ *given* u, it does not choose u.

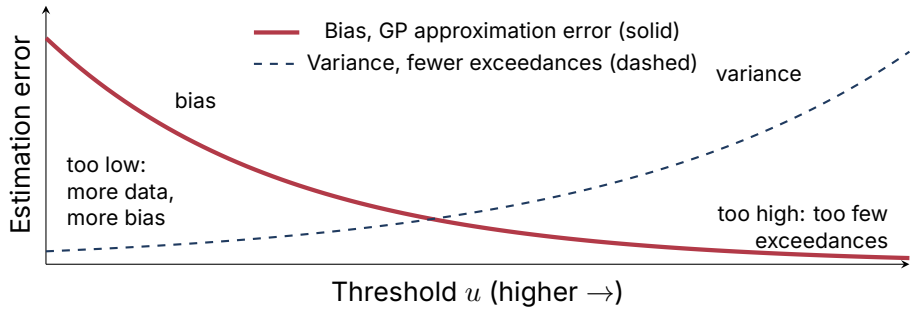


Figure 3. The threshold trade-off (a sketch, not data). Lowering u buys precision with bias; raising it buys GP accuracy with noise. There is no objectively optimal threshold.

Exam traps

- **Loss versus excess.** POT models $X - u$, conditional on $X > u$. Options that model X itself, or condition on $X = u$, are wrong.
- **Bias/variance polarity.** Lower u: more data, lower variance, *higher* bias. Higher u: fewer data, higher variance, *lower* bias. GARP reverses one half.
- **GPBdH is a large-threshold result.** It does not hold "for any threshold with enough exceedances". Lowering u weakens it.
- **The threshold is judgemental.** Maximum likelihood estimates β and ξ given u; it cannot make the threshold choice objective.
- $F(u)$ **versus** $1 - F(u)$**.** $F(u)$ is estimated by $(n - N_u)/n$, the share *below* the threshold; N_u/n is the exceedance share.
- **Parent.** POT does not require the parent to be known, or to be normal.

Practice questions: MR-3 c

Question MR-3 c.1

New • built from GARP MR Ch.3, 3.2 (Dowd) and Vardeez

A risk manager at an asset manager is replacing a normal-distribution VaR model with a peaks-over-threshold (POT) model for a commodity book whose daily loss distribution is unknown. Which of the following statements correctly describes the POT approach?

A. POT fits a distribution to the largest loss in each block of equal length, so it uses one observation per block and needs no threshold.

B. POT models the excess of each loss over a high threshold u, conditional on u being exceeded, and does not require the parent loss distribution to be known.

C. POT fits a distribution to all losses in the sample, including the body, and uses the threshold only to decide which losses to report.

D. POT requires the parent loss distribution to be normal or lognormal, because the excess distribution is derived from the parent's parameters.

Question MR-3 c.2 Bank Q154 • BT answers L2.T5.87.1

If X is a random iid loss with distribution function $F(x)$, and u is a threshold value of X, which expression defines the distribution of excess losses used in the peaks-over-threshold (POT) approach?

A. $\Pr\{X \leq x \mid X > u\}$

B. $\Pr\{X \leq x \mid X = u\}$

C. $\Pr\{X - u \leq x \mid X > u\}$

D. $\Pr\{X - u \leq x \mid X = u\}$

Question MR-3 c.3 New • built from GARP MR Ch.3, 3.2, eqs. 3.20 and 3.21

An analyst applying the POT approach to 2,000 daily losses sets a threshold $u = 2.5\%$, above which 60 losses fall, and fits a generalised Pareto distribution to the 60 excesses. To move from the excess-loss distribution back to the tail of the loss distribution itself, she needs an estimate of $F(u)$. Which of the following is the natural estimator, and why is it needed?

A. $60/2{,}000 = 3\%$, the share of exceedances, because $F(u)$ is the probability that a loss exceeds the threshold.

B. $1{,}940/2{,}000 = 97\%$, the share of losses not exceeding u, because the GP fit describes losses only conditional on exceeding u.

C. The fitted GP distribution evaluated at $x = u$, because once u is chosen the GP distribution describes the whole loss distribution.

D. The normal distribution function at u, using the sample mean and standard deviation, because POT assumes a normal parent below u.

Question MR-3 c.4 Bank Q158 • BT answers L2.T5.89.8 (rebuilt)

A risk manager has estimated the generalised Pareto parameters (scale β and tail index $\xi < 1$) from losses exceeding threshold u, along with the exceedance count N_u out of n observations. Which statement is correct?

A. Both VaR and expected shortfall can be computed from these estimates.

B. Only VaR can be computed; expected shortfall needs a separate non-parametric estimate.

C. Only expected shortfall can be computed; VaR needs a separate non-parametric estimate.

D. Neither can be computed, because POT only describes the distribution of excesses, not the tail of the loss itself.

Question MR-3 c.5 New • built from GARP MR Ch.3, 3.2 Estimation

A market risk team applying the POT approach to 1,500 daily losses must choose the threshold u. A junior analyst proposes several ways to make the choice. Which of the following is consistent with Dowd's treatment of threshold choice?

A. Plot the mean-excess function against u and choose a threshold around where it becomes horizontal, accepting that the choice remains judgemental.

B. Estimate u by maximum likelihood together with β and ξ, which makes the threshold an objective output of the fitting procedure.

C. Set u equal to the target 99.9% VaR, so that the GP distribution is fitted to exactly the 0.1% of losses that define the VaR.

D. Set u as low as possible, because the GPBdH theorem holds at any threshold once the number of exceedances is large.

Question MR-3 c.6 Workbook A17.2 • S3, S4 • Map #25

A market risk analyst at a Canadian bank fits a generalised Pareto distribution to 1,000 daily trading losses using the peaks-over-threshold approach. At the current threshold $u = 2.0\%$, there are 40 exceedances. To obtain more data for the estimation, a colleague proposes lowering the threshold to $u = 1.2\%$, which would give 150 exceedances. Which conclusion should the head of market risk reach about the proposal?

A. Lowering u to 1.2% raises the number of exceedances to 150 and also makes the GPBdH approximation more accurate, because the theorem holds for any threshold that leaves many exceedances.

B. Lowering u to 1.2% leaves the estimated tail index unchanged, because the generalised Pareto distribution describes excess losses whatever the parent distribution of the losses.

C. Lowering u to 1.2% raises the number of exceedances to 150 and lowers the variance of the estimates, but adds bias, because more exceedances come from closer to the centre of the distribution.

D. Lowering u to 1.2% reduces the bias of the estimates but raises their variance, because the additional exceedances come from closer to the centre of the distribution.

Solutions • MR-3 c

Solution MR-3 c.1 — Answer: B

New • built from GARP MR Ch.3, 3.2

Step 1 – What POT models

POT applies EVT to the distribution of excess losses over a high threshold: $F_u(x) = \Pr\{X - u \leq x \mid X > u\}$.

Step 2 – What it needs to know about the parent

The parent "can be any of the commonly used distributions ...and will usually be unknown"; the GPBdH theorem gives the limiting form of the excesses whatever it is.
$\Rightarrow$ B describes POT exactly.

Interpretation

This is the reason to move from a normal model to POT: the tail is modelled directly, without assuming a shape for the whole distribution.

Why the other options are wrong

A is incorrect. Sibling swap. One maximum per block and no threshold describes the block maxima (GEV) approach.
C is incorrect. POT discards the body; only the exceedances enter the fit (Figure 2).
D is incorrect. Absolute claim. The parent need not be normal or lognormal, and is usually unknown.

LO: MR-3 c: Describe the peaks-over-threshold (POT) approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2 The Peaks-over-Threshold Approach, eq. 3.18

Solution MR-3 c.2 — Answer: C

Bank Q154 • BT answers L2.T5.87.1

Step 1 – The conditioning event

POT studies losses that exceed the threshold, so the condition is $X > u$.

Step 2 – The variable modelled

The variable is the excess over the threshold, $X - u$, not the loss itself.
$\Rightarrow F_u(x) = \Pr\{X - u \leq x \mid X > u\}$.

Interpretation

Read the definition as "the probability that a loss exceeds u by at most x, given that it exceeds u".

Why the other options are wrong

A is incorrect. Right condition, wrong variable: models the loss X rather than the excess $X - u$.
B is incorrect. Conditions on $X = u$ (a zero-probability event for a continuous loss) and models X.
D is incorrect. Right variable, wrong condition: $X = u$ instead of $X > u$.

LO: MR-3 c: Describe the peaks-over-threshold (POT) approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, eq. 3.18

Solution MR-3 c.3 — Answer: B

New • built from GARP MR Ch.3, 3.2

Step 1 – Why $F(u)$ is needed

The GP fit describes $F_u(x)$, losses *conditional* on exceeding u. To get the unconditional tail, rearrange: $F(x) = (1 - F(u))\, G_{\xi,\beta}(x - u) + F(u)$.
$\Rightarrow$ The share of losses that never reach the threshold has to be supplied separately.

Step 2 – The natural estimator

$F(u)$ is the proportion of observations that do not exceed u:

$$\widehat{F(u)} = \frac{n - N_u}{n} = \frac{2{,}000 - 60}{2{,}000} = \frac{1{,}940}{2{,}000} = 97\%$$

Interpretation

The exceedance share $N_u/n = 3\%$ is what then appears in the POT VaR formula; it equals $1 - F(u)$, not $F(u)$.

Why the other options are wrong

A is incorrect. Wrong complement: 3% is $1 - F(u)$, the probability of exceeding u.
C is incorrect. The GP distribution describes only the excesses; it says nothing about losses below u.
D is incorrect. POT makes no normality assumption about the body; the empirical proportion is used instead.

LO: MR-3 c: Describe the peaks-over-threshold (POT) approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, eqs. 3.20 and 3.21

Solution MR-3 c.4 — Answer: A

Bank Q158 • BT answers L2.T5.89.8

Step 1 – From excesses to the loss tail

Adding u and using N_u/n converts the GP distribution of excesses into the tail of the loss distribution (MR-3 c Section 2).

Step 2 – The two risk measures

Inverting that tail gives VaR in closed form; ES is VaR plus the mean excess loss over VaR, also in closed form, and finite when $\xi < 1$.
$\Rightarrow$ Both VaR and ES come directly from β, ξ, u and N_u/n.

Interpretation

This is one of the goals of POT: a fully specified parametric tail beyond u that yields VaR and ES at very high confidence levels.

Why the other options are wrong

B is incorrect. ES has a closed-form expression from the same parameters.
C is incorrect. VaR is the starting point; it has the standard POT formula.
D is incorrect. Adding u and using N_u/n is exactly what turns the excess distribution into the loss tail.

LO: MR-3 c: Describe the peaks-over-threshold (POT) approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, eqs. 3.22 and 3.23

Solution MR-3 c.5 — Answer: A

New • built from GARP MR Ch.3, 3.2 Estimation

Step 1 – What Dowd says about the threshold

The choice of threshold is "inevitably arbitrary and therefore judgmental". u must be high enough for the GPBdH theorem to apply reasonably closely, but not so high that too few exceedances remain.

Step 2 – The aid he gives

As with the choice of k for the Hill estimator, plot the mean-excess function and choose a threshold where the MEF becomes horizontal.

$\Rightarrow$ A uses the recommended aid and keeps the judgement explicit.

Interpretation

A defensible process documents the threshold choice, shows the MEF plot, and reports confidence intervals for the resulting VaR and ES.

Why the other options are wrong

B is incorrect. Maximum likelihood estimates β and ξ *given* u; it does not remove the judgement in choosing u.

C is incorrect. A threshold at the 99.9% VaR leaves about 1.5 exceedances out of 1,500: far too few to estimate two parameters, and circular, since the VaR is the output.

D is incorrect. GPBdH holds as u gets large; lowering u to gather data adds bias.

LO: MR-3 c: Describe the peaks-over-threshold (POT) approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2 Estimation

Solution MR-3 c.6 — Answer: C

Workbook A17.2 • S3, S4

Step 1 – What the proposal changes

$$\frac{N_u}{n}: \frac{40}{1{,}000} = 4\% \longrightarrow \frac{150}{1{,}000} = 15\%$$

$\Rightarrow$ The threshold moves from roughly the 96th percentile of losses to roughly the 85th: the sample of "extremes" becomes larger and less extreme.

Step 2 – The trade-off in the reading

Dowd: "we want a threshold u to be sufficiently high for the GPBdH theorem to apply reasonably closely; but if u is too high, we won't have enough excess-threshold observations". The threshold problem "is very much akin to the problem of choosing k" for the Hill estimator, where more data "increases the precision of our estimator (and therefore reduces its variance), but also increases the bias of the tail estimator by placing relatively more weight on observations closer to the centre".

Interpretation

The proposal buys precision with bias: the GP limit is a large-threshold result, so the extra exceedances near the body of the distribution pull the fitted tail toward central behaviour. A defensible process documents the choice, checks where the mean-excess function becomes horizontal, and reports confidence intervals; the choice remains "inevitably arbitrary and therefore judgmental".

Why the other options are wrong

A is incorrect. Scope/condition error. GPBdH holds "as u gets large"; lowering u weakens the approximation. The number of exceedances is not the condition.

B is incorrect. Absolute claim. "Pretty much regardless" of the parent is a limiting result for a high threshold; the estimated ξ is sensitive to the threshold, which is why its choice is called the weak spot of POT.

D is incorrect. Polarity flip on the trade-off. More data near the centre lowers variance and raises bias, not the reverse.

LO: MR-3 c; MR-3 d; MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1 Semi-Parametric Estimation Methods; 3.2 Theory and Estimation, eqs. 3.18 to 3.24

MR-3 • Parametric Approaches (II): Extreme Value • Vardeez: IMPORTANT

MR-3 d

P1 CONCEPT

Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks.

Workbook: A17 Extreme Value Theory: GEV vs POT • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A17; "GEV vs POT distribution" is reported in the sheets (Map #25).

Vardeez: compare GEV vs POT.

Reported exam shape

- **S2** GEV versus POT: block maxima versus exceedances, data use, threshold.

What the questions look like: a statement set about the two approaches in which the attributes are swapped between them: "three parameters" and "wastes data in blocks" belong to GEV; "two parameters" and "needs a threshold" belong to POT. Workbook Question MR-3 b.8 (A17.1) also tests this shape.

1. Same theory, different targets

Both GEV and POT are manifestations of the **same underlying extreme value theory**, and both use the **tail index** ξ. GEV is geared to the distribution of extremes as such (the maxima of large samples or blocks); POT is geared to the distribution of exceedances over a high threshold. "In theory there is not too much to choose between them"; in practice there can be reasons to prefer one.

	GEV (block maxima)	POT
Models	The distribution of the maximum of each sample or block	The excess $X-u$ of losses over a high threshold u
Limit theorem	Fisher-Tippett	Gnedenko-Pickands-Balkema-deHaan (GPBdH)
Distribution	Generalised extreme value	Generalised Pareto
Parameters	Three: location μ, scale σ, tail index ξ	Two: scale β, tail index ξ (plus the chosen u)
Shared	Both rest on EVT, share ξ (a higher ξ means a heavier tail), need no knowledge of the parent, and can extrapolate beyond the worst loss in the sample.	
Data use	Can lose useful data: a block may contain more than one extreme, but only its maximum is kept	Uses every observation above the threshold
Judgement call	Block length	Threshold u, "the weak spot of POT"
Clustered losses	Block maxima are less clustered than the raw data (but longer blocks cost efficiency)	Exceedances cluster; needs extra work (declustering, conditional EVT)

2. Dowd's three points

1. One approach may be **more natural** in a given context, for example when limited data make one preferable.
2. The GEV typically involves **an additional parameter** relative to POT, and the block maxima approach can involve **some loss of useful data**, because some blocks may have more than one extreme in them. Both are disadvantages of the GEV.
3. On the other hand, POT requires us to grapple with **choosing the threshold**, a problem that does not arise with the GEV.

Conclusion: "either approach is usually reasonable", and one should choose the one that best suits the problem. Neither dominates in all circumstances.

Note on the source notes. Vardeez's table puts the extra parameter and the loss of useful data in one cell, which reads as if the extra parameter causes the data loss. They are two separate disadvantages: the data loss comes from keeping only one maximum per block.

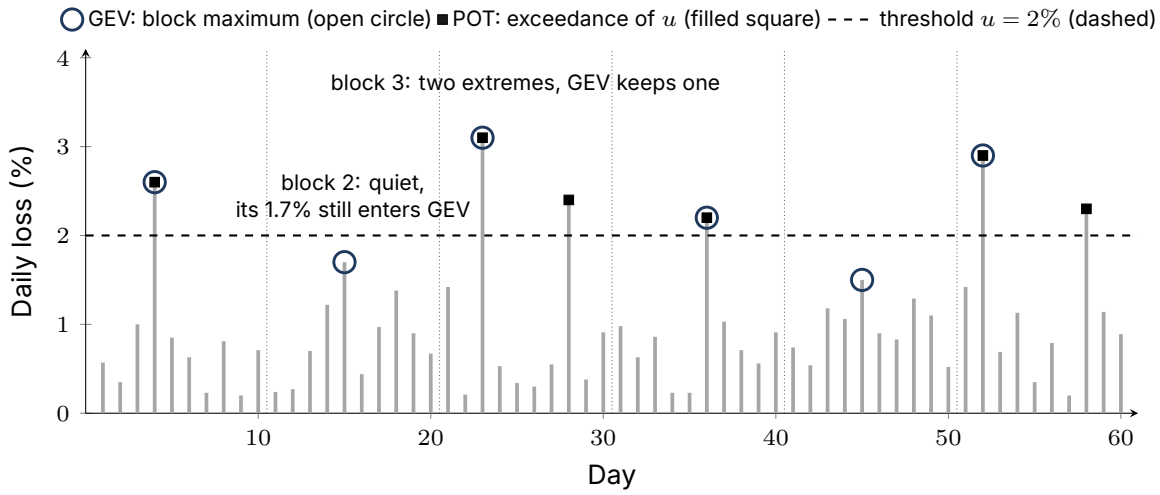


Figure 4. Sixty days in six blocks of ten. Block maxima keeps one point per block: it loses the second extremes in blocks 3 and 6 (2.4% and 2.3%) and admits the quiet maxima of blocks 2 and 5 (1.7% and 1.5%). POT keeps all six exceedances of u, but only after someone has chosen u.

3. The two judgement calls side by side

- **Block length (GEV).** Shorter blocks give more maxima (lower variance) but a worse GEV approximation (more bias). Longer blocks give fewer maxima (higher variance) but less bias, and the maxima are less affected by clustering.
- **Threshold (POT).** Lower u: more exceedances, lower variance, more bias. Higher u: fewer exceedances, higher variance, less bias (MR-3 c).

Both approaches therefore face a bias/variance trade-off; neither escapes judgement. Both are built on an iid assumption, so clustered (autocorrelated) losses are not handled automatically by either.

Exam traps

- **Sibling swap.** Three parameters, location parameter, block maxima, loss of data: GEV. Two parameters, threshold, all exceedances: POT. GARP moves one attribute across.

- **The shared parameter.** The two share ξ; options saying they share nothing are wrong.
- **"Superior in all circumstances".** Either approach is usually reasonable; the choice depends on the data and the problem.
- **Subjectivity on both sides.** POT's threshold and GEV's block length are both judgement calls; "avoiding a subjective choice" does not favour POT.
- **The parent.** Neither approach needs the parent distribution; an unknown parent does not favour either.
- **Clustering.** Neither approach handles autocorrelated clusters automatically.

Practice questions: MR-3 d

Question MR-3 d.1

Bank Q162 • Schweser Module Quiz 3.1 Q2

Compared with the generalised extreme value (GEV) approach, the peaks-over-threshold (POT) approach generally requires:

A. more estimated parameters than GEV and shares one parameter with GEV.

B. fewer estimated parameters than GEV and shares one parameter with GEV.

C. more estimated parameters than GEV and shares no parameters with GEV.

D. fewer estimated parameters than GEV and shares no parameters with GEV.

Question MR-3 d.2

Bank Q165 • BT answers L2.T5.89.1 (rebuilt)

Which statement about choosing between the GEV (block maxima) and POT approaches to extreme risk is most accurate?

A. GEV is superior in virtually all circumstances because it avoids choosing a threshold.

B. POT is superior in virtually all circumstances because it always uses more data.

C. Neither approach is superior in all circumstances; each has advantages and the choice depends on the data and the trade-offs involved.

D. They are unrelated methods and cannot be compared.

Question MR-3 d.3

Bank Q172 • BT answers P2.T7.805.3 (rebuilt)

Sally is comparing the generalised extreme value (GEV) block maxima method with the peaks-over-threshold (POT) method to model extreme losses in her firm's loss database. Consider the following considerations:

I. She wants to avoid discarding data by retaining only one extreme loss per block.

II. She prefers a model with fewer parameters to estimate.

III. She wants to avoid having to make a subjective tuning choice (block length for GEV, threshold for POT).

IV. She does not know the parent distribution of the losses.

Which of the considerations, taken individually, favour POT over GEV?

A. I only

B. I and II only

C. I, II and III only

D. II, III and IV only

Question MR-3 d.4

Bank Q170 • BT answers P2.T5.22.5.1

The board has asked Claire to develop risk estimates at very high confidence levels, e.g. 99.5% and 99.9%. Given the realities of her dataset (e.g. high time dependency), she may need to rely on extreme value theory (EVT). She recognises the two branches of univariate EVT are block maxima (GEV) and peaks-over-threshold (POT), and she is comparing them with non-parametric approaches such as historical simulation. With respect to the two univariate EVT approaches, each of the following statements is true of both approaches, EXCEPT which statement is false?

A. Each of the approaches automatically and easily fits distributions to a stochastic process with autocorrelated clusters.

B. Each of the fitted EV parametric distributions has a parameter that can be increased to model increasingly heavier tails.

C. She can fit a parametric distribution of extreme values to actual datasets without knowledge of the underlying distribution.

D. Unlike basic historical simulation, the EVT VaR or ES can return values that exceed the greatest loss in the actual historical dataset.

Question MR-3 d.5

Bank Q171 • BT answers P2.T5.22.6.1

In applying extreme value theory (EVT) to their investment firm's financial returns, Patricia and Peter are deciding between the classic block maxima (GEV) approach and the modern peaks-over-threshold (POT) approach. Patricia, who is more familiar with block maxima, makes the following points about that approach, but one is inaccurate. Each of the following is true EXCEPT which statement is false?

A. If the parent distribution is a Levy distribution, the extremes fall in the domain of attraction of the Frechet distribution.

B. For a given historical window, shorter blocks (and therefore more blocks) achieve lower variance but with more bias.

C. Block maxima is superior to POT because it uses the data more efficiently, and its Gumbel instance avoids the trade-off between bias and variance.

D. Time dependency in the loss data can be addressed with longer (and therefore fewer) blocks, giving less bias but at the cost of higher variance.

Question MR-3 d.6

Bank Q173 • GARP PE2 Q57 • Mock 4 Q58

The CRO of an investment bank is preparing to lead a panel discussion at a risk management conference on estimating extreme risk measures. The CRO plans to discuss the main branches of univariate extreme value theory as well as the characteristics of the generalised extreme value (GEV) approach and the peaks-over-threshold (POT) approach. Which of the following statements would be correct for the CRO to include in the discussion?

A. The block maxima approach can avoid the data loss issues that sometimes arise under the POT approach, which constitutes an advantage of GEV compared to POT.

B. The additional parameter that is included in the POT approach, but not the GEV approach, is a location parameter that measures the central tendency of the extreme loss distribution.

C. An advantage of the POT approach is that the distribution of extreme losses over a high threshold is assumed to always have the same form regardless of the distribution of all losses.

D. A drawback of the GEV approach is that the parameter estimates describing the extreme loss distribution suffer from the fact that the choice of the threshold over which losses are considered extreme is subjective.

Solutions • MR-3 d

Solution MR-3 d.1 — Answer: B

Bank Q162 • Schweser Module Quiz 3.1 Q2

Step 1 – Count the parameters

GEV: location μ, scale σ, tail index ξ (three). POT (generalised Pareto): scale β and tail index ξ (two), plus the chosen threshold u.

Step 2 – What they share

The tail index ξ is common to both.

$\Rightarrow$ POT needs fewer parameters and shares one with GEV.

Interpretation

The "extra" parameter in GEV is the location parameter; in POT the threshold plays a similar role but is chosen, not estimated.

Why the other options are wrong

A is incorrect. POT does not need more parameters; GEV carries the additional location parameter.

C is incorrect. Wrong on both counts.

D is incorrect. Right on the count, but ξ is shared.

LO: MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, GEV vs POT

Solution MR-3 d.2 — Answer: C

Bank Q165 • BT answers L2.T5.89.1

Step 1 – Advantages and drawbacks

POT needs fewer parameters and wastes less data, but requires choosing a threshold. GEV avoids the threshold but loses data and carries an extra parameter.

Step 2 – Dowd's conclusion

"Either approach is usually reasonable, and one should choose the one that seems to best suit the problem at hand."

$\Rightarrow$ Neither dominates.

Interpretation

GARP rewards the balanced statement and punishes "in virtually all circumstances".

Why the other options are wrong

A is incorrect. Absolute claim. GEV avoids a threshold but wastes data and needs an extra parameter.

B is incorrect. Absolute claim. POT's data efficiency is offset by the uncertainty of the threshold choice.

D is incorrect. Both come from EVT and share the tail index ξ; they are directly comparable.

LO: MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, GEV vs POT

Solution MR-3 d.3 — Answer: B

Bank Q172 • BT answers P2.T7.805.3

Step 1 – I and II

I: block maxima keeps only the maximum of each block, so it can lose data; POT uses every exceedance. **Favours POT.**
II: GEV estimates μ, σ and ξ; POT estimates only β and ξ. **Favours POT.**

Step 2 – III and IV

III: GEV requires a block length, POT a threshold; both are subjective. **Favours neither.**
IV: both the Fisher-Tippett and GPBdH theorems hold without knowing the parent. **Favours neither.**
$\Rightarrow$ I and II only.

Interpretation

These are exactly Dowd's two disadvantages of the GEV. The tuning choice and the unknown parent apply equally to both approaches.

Why the other options are wrong

A is incorrect. Omits II: forgets that GEV carries an extra location parameter.
C is incorrect. Adds III, but POT's threshold choice is just as subjective as GEV's block length.
D is incorrect. Drops I and adds III and IV, which apply equally to both methods.

LO: MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, GEV vs POT

Solution MR-3 d.4 — Answer: A

Bank Q170 • BT answers P2.T5.22.5.1

Step 1 – The iid assumption

Both GEV and POT are derived for iid losses. Time dependency (clustering) violates this, and neither method handles it automatically: it needs extra work (longer blocks for GEV, declustering or conditional EVT for POT).
$\Rightarrow$ A is false.

Step 2 – Check the shared features

B: ξ is the tail index in both the GEV and the GP; a higher ξ means a heavier tail. True.
C: the EV theorems hold whatever the parent. True.
D: a fitted parametric tail extrapolates, so EVT VaR or ES can exceed the worst observed loss; basic historical simulation cannot. True.

Interpretation

"Automatically and easily" is the absolute qualifier that gives A away. Clustered data are one of the refinements EVT needs, not something it handles for free.

Why the other options are wrong

B is incorrect. True statement: a higher shared ξ gives heavier tails in both.
C is incorrect. True statement: no knowledge of the parent is needed.
D is incorrect. True statement: parametric extrapolation goes beyond the sample maximum.

LO: MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2 and 3.3 Refinements to EV Approaches

Solution MR-3 d.5 — Answer: C

Bank Q171 • BT answers P2.T5.22.6.1

Step 1 – A: domain of attraction

A Levy parent is heavy-tailed, so its extremes fall in the Frechet domain ($\xi > 0$). True.

Step 2 – B and D: block length

Shorter blocks give more maxima (lower variance) but a worse approximation (more bias). Longer blocks give fewer maxima (higher variance) but less bias, and weaker dependence between the maxima. Both true.

Step 3 – C

Block maxima keeps one observation per block, so it uses data *less* efficiently than POT, and the bias/variance trade-off in block length exists in every GEV case, the Gumbel included.
$\Rightarrow$ C is the false statement.

Interpretation

Data efficiency is POT's advantage, not GEV's; GEV's advantage is that it needs no threshold.

Why the other options are wrong

A is incorrect. True: heavy-tailed parents such as the Levy, t or Pareto lie in the Frechet domain.
B is incorrect. True: the variance falls and the bias rises as blocks shorten.
D is incorrect. True: longer blocks reduce bias and dependence but raise variance.

LO: MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1, 3.2 and 3.3

Solution MR-3 d.6 — Answer: C

Bank Q173 • GARP PE2 Q57

Step 1 – C: the GPBdH result

By the GPBdH theorem, the distribution of excess losses over a sufficiently high threshold always has the same (generalised Pareto) form, "pretty much regardless of the distribution of the losses themselves". This is POT's advantage.

Step 2 – The swaps in A, B and D

A: data loss is a drawback of block maxima, not of POT. B: the extra (location) parameter belongs to GEV, not POT. D: the threshold is chosen in POT, not GEV.
$\Rightarrow$ Only C is correct.

Interpretation

A pure sibling-swap item: each wrong option takes a true property and attaches it to the other approach.

Why the other options are wrong

A is incorrect. Reversed: block maxima is the approach that loses data.
B is incorrect. Reversed: the location parameter is the extra parameter of GEV; POT has only β and ξ.
D is incorrect. Misattributed: the threshold choice is POT's weak spot; GEV's tuning choice is the block length.

LO: MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, Theory and GEV vs POT

MR-3 • Parametric Approaches (II): Extreme Value • Vardeez: IMPORTANT

MR-3 e

P1 CONCEPT

Discuss the application of the generalized Pareto (GP) distribution in the POT approach.

Workbook: A17 Extreme Value Theory: GEV vs POT • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A17 (shape S3) hits this LO's concept.
Vardeez: no separate pointer for this LO; his chapter note applies: do not go too deep into calculations.

Reported exam shape

- **S3** Peaks over threshold: the GPBdH theorem and the GP distribution of excess losses.

What the questions look like: which distribution the excesses converge to (GP, not GEV, normal or "generalised Gumbel"), what parameters the GP has (scale and tail index, plus a chosen threshold, no location), why it is the natural model, and the threshold trade-off. Workbook Question MR-3 c.6 (A17.2) also tags this LO.

1. The GPBdH theorem

As the threshold u gets large, the **Gnedenko-Pickands-Balkema-deHaan (GPBdH) theorem** says the distribution of excess losses $F_u(x)$ (MR-3 c) converges to a **generalised Pareto (GP)** distribution:

$$G_{\xi,\beta}(x) = \begin{cases} 1 - \left(1 + \dfrac{\xi x}{\beta}\right)^{-1/\xi} & \xi \neq 0 \\ 1 - \exp\left(-\dfrac{x}{\beta}\right) & \xi = 0 \end{cases} \qquad x \geq 0 \text{ for } \xi \geq 0; \quad 0 \leq x \leq -\beta/\xi \text{ for } \xi < 0.$$

- **Two parameters only:** a positive **scale** β and the **tail (shape) index** ξ, the *same* ξ as in GEV theory. No location parameter: the chosen threshold u plays that role.
- ξ can be positive, zero or negative; the cases of interest are $\xi \geq 0$, and above all $\xi > 0$ (heavy-tailed data).
- The result holds "pretty much regardless of the distribution of the losses themselves". Provided the threshold is high enough, the GP is therefore **the natural model for excess losses**.

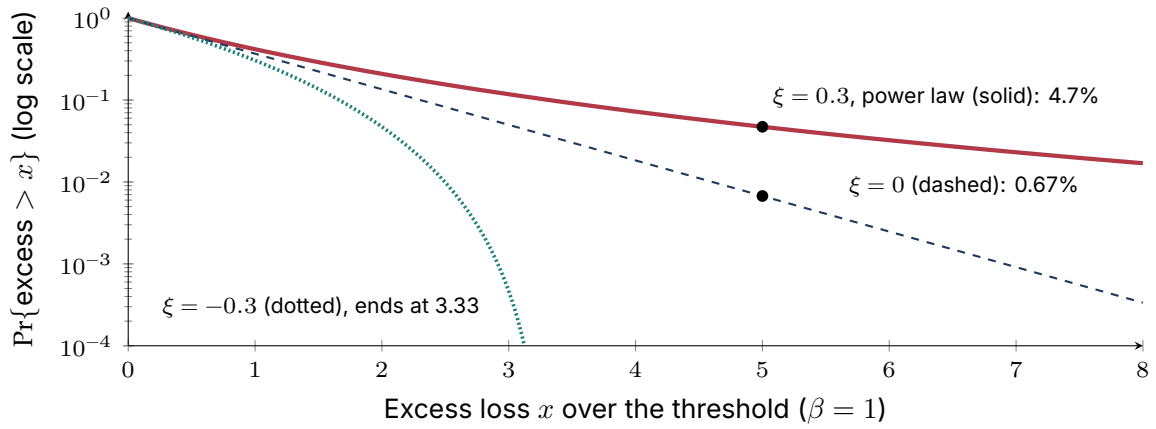


Figure 5. GP survival functions on a log scale. With $\xi = 0$ the tail is a straight line (exponential decay); with $\xi > 0$ it bends away above it, so large excesses stay far more likely (4.7% against 0.67% at $x = 5\beta$); with $\xi < 0$ the excess has a finite maximum. The tail index carries the same meaning as in the GEV (MR-3 b, Figure 1).

2. Why the GP fits loss data better than the normal

Relative to the normal, the GP density **dips below** the normal just before the tail and then **rises above** it through the tail, until the extreme tail. Empirical loss data behave this way: less mass in the shoulder, more mass in the tail. The GP then gives an approximately linear (in log terms) description of the tail that matches the data more closely than the normal.

3. Applying the GP in POT: the steps

1. **Choose the threshold** u, which fixes the number of exceedances N_u. The trade-off: high enough for GPBdH to apply reasonably closely, low enough to leave enough exceedances for estimation (MR-3 c, Figure 3). The choice is judgemental; the mean-excess plot helps.
2. **Estimate** β **and** ξ from the N_u excesses. Maximum likelihood is the most reliable (asymptotically normal, and so relatively well behaved, if $\xi > -0.5$); semi-parametric and moment-based methods also exist, but moment estimators can be unreliable.
3. **Estimate** $F(u)$ by the share of observations below the threshold, $(n - N_u)/n$.
4. **Compute the risk measures.** One of the goals of POT is VaR, and from VaR the expected shortfall: ES = VaR plus the mean excess loss over VaR, finite only if $\xi < 1$.

Formulas to recognise (Vardeez: not for heavy calculation)

GP (GPBdH limit)	$G_{\xi,\beta}(x) = 1-(1+\xi x/\beta)^{-1/\xi},\ \xi \neq 0; \quad 1-\exp(-x/\beta),\ \xi = 0$
Loss tail	$F(x) = 1 - \frac{N_u}{n}\left[1+\xi\frac{x-u}{\beta}\right]^{-1/\xi}, \quad x > u$
POT VaR	$\text{VaR} = u + \frac{\beta}{\xi}\left\{\left[\frac{n}{N_u}(1-\alpha)\right]^{-\xi} - 1\right\}$
POT ES ($\xi < 1$)	$\text{ES} = \frac{\text{VaR}}{1-\xi} + \frac{\beta - \xi u}{1-\xi}$
GARP Example 3.6	$\beta = 0.8, \xi = 0.15, u = 2\%, N_u/n = 4\%$: 99.5% VaR 3.952%, ES 5.238%; 99.9% VaR 5.942%, ES 7.578%

Correction to the source text. GARP's Example 3.6 (and the Tier A workbook, which repeats it) gives the 99.9% ES as 17.578. Recomputing with eq. 3.23: $\text{ES} = 5.942/0.85 + (0.8 - 0.15 \times 2)/0.85 = 6.991 + 0.588 = 7.578$. The "1" is a typo; the correct figure is 7.578%.

Exam traps

- **Wrong limit distribution.** Excesses over a rising threshold converge to the generalised *Pareto*. Not the GEV (that is for block maxima), not the normal, not a "generalised Gumbel".
- **Wrong parameter set.** GP: scale β and tail index ξ, plus a chosen u. A location parameter belongs to the GEV.
- **ξ is shared with the GEV.** It is not a GP-only parameter, and $\xi > 0$ means heavy tails in both.
- **Bounded tail.** A finite upper bound on the excess ($-\beta/\xi$) arises only when $\xi < 0$.
- **Threshold direction.** Higher u: better convergence to the GP, fewer data points. Lower u: more data, worse convergence (more bias).
- **Net effect on VaR of raising u.** With the same data, a higher u lowers N_u/n; u and N_u/n push VaR in opposite directions, so the net effect on VaR and ES is unclear.
- **ES needs $\xi < 1$.**

Practice questions: MR-3 e

Question MR-3 e.1

Bank Q175 • AnalystPrep Q.4010

As the threshold level, u, gets large, the Gnedenko-Pickands-Balkema-deHaan (GPBdH) theorem states that the distribution of above-threshold observations converges to:

A. a normal distribution.

B. a generalised extreme value distribution.

C. a generalised Pareto distribution.

D. a uniform distribution.

Question MR-3 e.2 New • built from GARP MR Ch.3, 3.2 and Vardeez

A member of the risk committee asks why the risk team models excess losses over a high threshold with a generalised Pareto (GP) distribution rather than with a normal distribution fitted to the same data. Which of the following is the best answer?

A. By the GPBdH theorem, excess losses over a high threshold converge to a GP distribution almost regardless of the parent, and the GP puts more mass in the tail than the normal does.

B. By the Fisher-Tippett theorem, excess losses converge to a GP distribution only when the parent is normal, so the GP and the normal give the same tail estimates.

C. The GP distribution has thinner tails than the normal, so it avoids the overstatement of extreme losses that empirical studies attribute to the normal.

D. The GP distribution has a location, a scale and a tail parameter, so unlike the normal it can be fitted to the whole loss distribution.

Question MR-3 e.3 Bank Q182 • BT answers L2.T5.88.1

The FRM reviews two EVT approaches: peaks-over-threshold (POT) and block maxima. POT is characterised by a generalised Pareto (GP) distribution, while block maxima is characterised by a generalised extreme-value (GEV) distribution. Which of the following describes the necessary parameters for the POT GP distribution?

A. Only scale and tail/shape index parameters.

B. Scale and tail/shape index parameters, plus the threshold (u) must be specified.

C. Only scale, location, and tail/shape index parameters.

D. Scale, location, and tail/shape index parameters, plus the threshold (u) must be specified.

Question MR-3 e.4 New • built from GARP MR Ch.3, 3.2, eqs. 3.19 and 3.23

A validation analyst reviews a POT model in which the fitted generalised Pareto parameters are $\beta = 0.8$ and $\xi = 0.15$ for a threshold of $u = 2\%$. Which of the following statements about these parameters is correct?

A. ξ is a tail index specific to the GP distribution with no counterpart in GEV theory, so it cannot be compared with block maxima estimates.

B. β is a location parameter that measures the central tendency of the excess losses, so it must be estimated together with u.

C. Because $\xi > 0$, the excess losses have a heavy, power-law tail, and the POT expected shortfall exists because $\xi < 1$.

D. Because ξ is close to zero, the excess-loss distribution has a finite upper bound, so no loss can exceed $u + \beta/\xi$.

Question MR-3 e.5

Bank Q174 • GARP PE1 Q57 • Mock 3 Q24

The CRO of a regional bank expresses concern that the bank's internal risk models do not adequately assess potential extreme losses. A risk analyst asks whether a model based on extreme value theory (EVT) would address the concern. Which of the following is correct when applying EVT and examining distributions of losses exceeding a threshold value?

A. As the threshold value is increased, the distribution of losses over a fixed threshold value converges to a generalised Pareto distribution.

B. If the tail parameter value of the generalised extreme-value (GEV) distribution goes to infinity, then the GEV essentially becomes a normal distribution.

C. To apply EVT, the underlying loss distribution must be either normal or lognormal.

D. The number of exceedances decreases as the threshold value decreases, which causes the reliability of the parameter estimates to increase.

Question MR-3 e.6

Bank Q183 • BT answers L2.T5.88.2

In the peaks-over-threshold (POT) approach, as the analyst selects a HIGHER threshold level, what is the trade-off encountered?

A. More data points but less help in selecting which distribution to employ.

B. More data points but less convergence to the generalised Pareto distribution.

C. Better indication of which distribution to employ but fewer data points on which to make estimates.

D. Better convergence to the generalised Pareto distribution but fewer data points on which to make estimates.

Question MR-3 e.7

Bank Q184 • BT answers P2.T7.805.2

Barbara the Risk Manager is selecting a threshold in her peaks-over-threshold (POT) extreme value theory model. She realises that the choice of threshold, in Dowd's words, "is the weak spot of POT theory: it is inevitably arbitrary and therefore judgmental." Her data set contains 1,000 observations. At current calibration the threshold implies 50 losses, so $N(u)/N = 50/1{,}000 = 5.0\%$. Regarding adjusting the threshold, which statement is TRUE?

A. If she increases the threshold, the bias increases.

B. If she increases the threshold, the variance decreases.

C. She should employ maximum likelihood estimation (MLE) to determine the objectively optimal threshold.

D. If she increases the threshold for the same dataset, she should expect $N(u)/N$ to decrease but with unclear effect on VaR and ES (they may increase or decrease).

Solutions • MR-3 e

Solution MR-3 e.1 — Answer: C

Bank Q175 • AnalystPrep Q.4010

Step 1 – What POT studies

The excess distribution $F_u(x) = \Pr\{X - u \leq x \mid X > u\}$.

Step 2 – The limit

GPBdH: as u gets large, $F_u(x)$ converges to the generalised Pareto distribution with scale β and tail index ξ.

$\Rightarrow$ The limit for threshold exceedances is the GP.

Interpretation

GPBdH is the POT counterpart of Fisher-Tippett: GP for excesses, GEV for maxima.

Why the other options are wrong

A is incorrect. The normal describes the body of many parents, not the extreme tail.

B is incorrect. Sibling swap: the GEV is the limit for block maxima (Fisher-Tippett).

D is incorrect. The uniform distribution has no role in extreme value theory.

LO: MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, eq. 3.19

Solution MR-3 e.2 — Answer: A

New • built from GARP MR Ch.3, 3.2

Step 1 – The theoretical reason

GPBdH: the distribution of excess losses always has the same form in the limit, "pretty much regardless of the distribution of the losses themselves". Provided u is high enough, the GP is the natural model for excess losses.

Step 2 – The empirical reason

Relative to the normal, the GP dips below it before the tail and rises above it in the tail, which is how empirical loss data behave.

$\Rightarrow$ A gives both reasons correctly.

Interpretation

A normal fitted to all the data is driven by the centre and understates the tail; the GP is fitted to the tail itself.

Why the other options are wrong

B is incorrect. Wrong theorem and wrong condition: Fisher-Tippett is about maxima, and GPBdH does not need a normal parent.

C is incorrect. Polarity flip: the GP (for $\xi > 0$) has heavier tails than the normal; the normal understates, not overstates, extreme losses.

D is incorrect. Wrong parameter set: the GP has only a scale and a tail index, and it is fitted to the excesses, not the whole distribution.

LO: MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2 Theory; SchweserNotes 2026 Book 1, Reading 3, LO 3.e

Solution MR-3 e.3 — Answer: B

Bank Q182 • BT answers L2.T5.88.1

Step 1 – The GP parameters

The GP distribution has two parameters: a positive scale β and the tail (shape) index ξ.

Step 2 – The threshold

Unlike the GEV there is no location parameter; the analyst must choose the threshold u, which also determines N_u and enters the VaR formula.

$\Rightarrow$ Scale and tail index, plus a specified threshold.

Interpretation

The threshold replaces the GEV's location parameter, but it is chosen by judgement, not estimated.

Why the other options are wrong

A is incorrect. Omits the threshold, which must be selected before anything can be estimated.

C is incorrect. The location parameter belongs to the GEV; the threshold is also missing.

D is incorrect. Mixes the GEV's location parameter into the GP model.

LO: MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, eq. 3.19

Solution MR-3 e.4 — Answer: C

New • built from GARP MR Ch.3, 3.2

Step 1 – Read each parameter

$\beta = 0.8$ is the (positive) *scale* parameter. $\xi = 0.15$ is the tail index, the same parameter as in GEV theory.

Step 2 – What $\xi = 0.15$ implies

$\xi > 0$: a heavy, power-law tail (Figure 5). The ES formula $\text{ES} = \text{VaR}/(1-\xi) + (\beta - \xi u)/(1-\xi)$ requires $\xi < 1$, which holds.

$\Rightarrow$ C is correct.

Interpretation

An estimate of 0.15 is typical for financial data (positive but below 0.35), and it can be compared directly with tail index estimates from a block maxima fit.

Why the other options are wrong

A is incorrect. ξ is the same tail index as in GEV theory.

B is incorrect. β is a scale parameter; the GP has no location parameter.

D is incorrect. A finite upper bound, at $u - \beta/\xi$, exists only when $\xi < 0$. With $\xi = 0.15$ the tail is unbounded; $u + \beta/\xi = 7.33\%$ has no meaning here.

LO: MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, eqs. 3.19 and 3.23

Solution MR-3 e.5 — Answer: A

Bank Q174 • GARP PE1 Q57

Step 1 – The GPBdH theorem

As the threshold u becomes large, the distribution of excesses converges to a generalised Pareto distribution, for a wide class of parents.

$\Rightarrow$ A is correct.

Step 2 – Check the others
$\xi = 0$ (not infinity) gives the light-tailed Gumbel; EVT needs no normality assumption; lowering the threshold *increases* the number of exceedances.

Interpretation
GARP mixes a GEV fact, a parent-distribution fact and a threshold fact into one statement set; only the GPBdH statement survives.

Why the other options are wrong
B is incorrect. $\xi = 0$, not infinity, gives the Gumbel, whose light tails resemble the normal or lognormal; no value of ξ gives a normal.
C is incorrect. EVT does not require a normal or lognormal parent; those are only examples of the $\xi = 0$ domain.
D is incorrect. Reversed: lowering the threshold raises the number of exceedances (lower variance) but worsens the GP approximation (more bias).

LO: MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.1 and 3.2

Solution MR-3 e.6 — Answer: D

Bank Q183 • BT answers L2.T5.88.2

Step 1 – Convergence
GPBdH: the excess distribution converges to the GP as u gets large, so a higher u improves the approximation.

Step 2 – Data
A higher u leaves fewer exceedances N_u, so β and ξ are estimated less precisely.
$\Rightarrow$ Better convergence, fewer data points.

Interpretation
This is the same trade-off as MR-3 c.6 seen from the other side: a high threshold buys lower bias at the cost of higher variance.

Why the other options are wrong
A is incorrect. A higher threshold reduces the data points, and distribution choice is not the issue.
B is incorrect. Reverses both effects.
C is incorrect. Fewer data points is right, but the benefit is convergence to the GP, not a better indication of which distribution to use: the GP is already given by theory.

LO: MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2 Estimation

Solution MR-3 e.7 — Answer: D

Bank Q184 • BT answers P2.T7.805.2

Step 1 – The mechanical effect
With the same 1,000 observations, raising u leaves fewer exceedances, so $N(u)/N$ falls below 5.0%.

Step 2 – The effect on VaR
In $\text{VaR} = u + \frac{\beta}{\xi}\left\{\left[\frac{n}{N_u}(1-\alpha)\right]^{-\xi} - 1\right\}$, a larger u pushes VaR up while a smaller N_u/n pushes it down (and β, ξ will be re-estimated).

$\Rightarrow$ The net effect on VaR and ES is unclear.

Step 3 – Bias, variance and MLE

A higher threshold improves GP convergence, so bias *falls*; fewer exceedances raise variance. MLE estimates β and ξ given u; it does not choose u.

Interpretation

Example from the bank's companion item: at $u = 4\%$ ($N_u/n = 2.8\%$) a 99.9% VaR of 8.670% becomes 8.630% at $u = 5\%$ ($N_u/n = 1.7\%$) with the same β and ξ: the two effects nearly cancel.

Why the other options are wrong

A is incorrect. Reverses the bias effect: a higher threshold reduces bias.
B is incorrect. Reverses the variance effect: fewer exceedances increase variance.
C is incorrect. MLE fits the parameters given u; the threshold choice stays judgemental.

LO: MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.7, 3.2, eq. 3.22 and Estimation

MR-8 • Correlation Basics: Definitions, Applications, and Terminology • Vardeez: MODERATE

MR-8 c

P1 CONCEPT

Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.

Workbook: A8 Correlation Swaps, Quanto Options and Concentration Ratio • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: A8 sits on recall map #8 (7 of 16 sittings). "Quanto options" is reported by name alongside the correlation swap.
Vardeez: quanto options (his chapter pointer: "Quanto Option & Correlation Swap").

Reported exam shape

- **S4** Quanto options: what correlation risk they carry, and in which direction correlation moves the price.

What the questions look like: conceptual "which statement is correct". The distractors flip the direction (higher correlation, higher price), swap the roles (positive correlation favours the *buyer*), or drop one of the seller's two unknowns. A second family asks which multi-asset option gets *cheaper* as correlation falls: the answer is always one of the two exceptions, the option on the worse of two or the basket option.

1. Background: correlation risk and correlation trading

- **Financial correlation risk** is the risk of financial loss due to adverse movements in correlation between two or more variables. The variables can be financial (default correlation between two debtors) or non-financial (political tension and an exchange rate). Correlation risk can be *non-monotonic*.
- **Correlation trading** means trading assets whose price is determined, at least in part, by the co-movement of one or more assets in time. The traders try to forecast changes in correlation and gain from them. Two instrument families: **multi-asset (correlation) options** and **correlation swaps** (MR-8 d).
- Multi-asset options are also called **rainbow options** or **mountain-range options**. They are priced analytically with extensions of Black-Scholes-Merton (1973).

2. Multi-asset options and the direction of correlation

S_1, S_2 are the two asset prices at maturity, K the strike, n_i the basket weights.

Option	Payoff	Lower correlation
Option on the better of two	$\max(S_1, S_2)$	**higher price**
Call on the maximum of two	$\max[0,\ \max(S_1, S_2) - K]$	**higher price**
Exchange option (e.g. a convertible bond)	$\max(0,\ S_2 - S_1)$	**higher price**
Spread call option	$\max[0,\ (S_2 - S_1) - K]$	**higher price**
Option on the better of two or cash	$\max(S_1, S_2, \text{cash})$	**higher price**
Dual strike call option	$\max(0,\ S_1 - K_1,\ S_2 - K_2)$	**higher price**
Option on the worse of two	$\min(S_1, S_2)$	**lower price**
Basket option	$\max\left[\sum n_i S_i - K,\ 0\right]$	**lower price**

Why most go up when correlation falls. With a low, preferably negative, correlation, if one asset falls (on average) the other rises, so one of the two is likely to end high and the payoff is likely to be large. The option pays on the *best* asset or on the *gap* between them, and dispersion helps both.

Why the two exceptions go down. The worse-of option pays on the *weaker* asset: the more the two diverge, the lower the minimum. A basket option pays on a weighted *sum*: lower correlation means more offsetting moves inside the basket, lower basket volatility, lower option value.

The exchange option. Rewrite $\max(0, S_2 - S_1) = S_1 \max(0,\ S_2/S_1 - 1)$ and price it with Black-Scholes-Merton on the ratio S_2/S_1, whose volatility carries the covariance:

$$\sigma_E = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2}$$

As $\rho \to +1$ the spread barely moves, σ_E shrinks and the price goes to almost zero. GARP's Figure 8.4 shows the price close to 0 at high correlation and USD 15.08 at a correlation of -1.

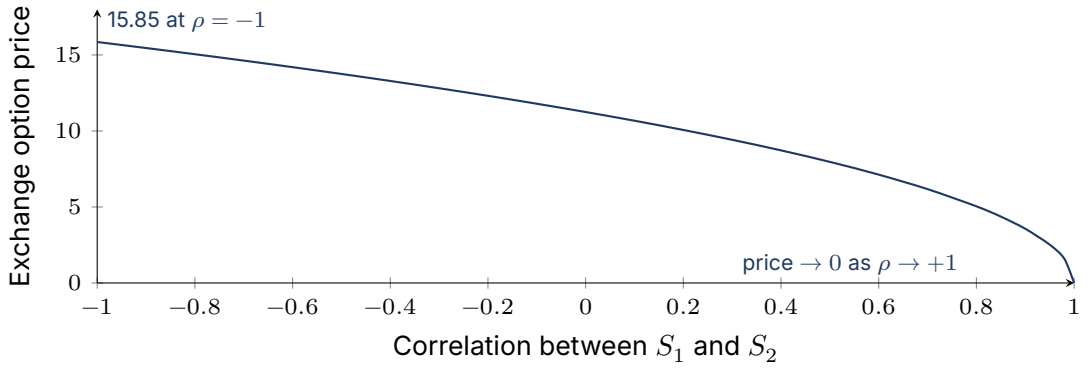


Figure 1. Exchange option $\max(0, S_2 - S_1)$ priced with Margrabe's formula, $S_1 = S_2 = 100$, both volatilities 20%, one year (illustrative parameters, not GARP's). The shape is GARP's Figure 8.4: a strong, monotonic negative relation between price and correlation.

3. Quanto options

A **quanto option** lets a domestic investor exchange a potential option payoff in a *foreign* currency back into the *home* currency at a **fixed** exchange rate (usually today's spot). It protects the investor against currency risk.

GARP's example. An American believes the Nikkei will rise but worries that a falling yen would eat the profit. She buys a quanto call on the Nikkei; the yen payoff is converted into dollars at a fixed rate. "Quanto" comes from *quantity*: the amount to be re-exchanged is unknown because it depends on the option payoff.

The seller does not know two things:

1. how deep the call will be in the money at maturity, i.e. **which yen amount** has to be converted into dollars;
2. the **exchange rate at maturity** at which that stochastic yen payoff will be converted.

The correlation between (1) and (2), the underlying S' and the exchange rate X (quoted as **domestic per unit of foreign**, USD per 1 yen), drives the price.

	Positive correlation (Nikkei, X)	Negative correlation (Nikkei, X)
Nikkei rises with	a *rising* yen	a *falling* yen
Seller's cost	Needs only a small yen amount to deliver the dollar payoff	Needs more yen to meet the dollar payment
Who it favours	The **seller**	The **buyer**
Quanto price	**Lower**	**Higher**

Rule: the more positive the correlation, the lower the quanto price; the lower the correlation, the more expensive the quanto. This is the same negative price-correlation relation as the exchange option.

	Most multi-asset options	Worse-of two & basket
Correlation falls	Price rises	Price falls
Holder is	Short correlation	Long correlation
Reason	Dispersion raises the best asset or the gap	Dispersion lowers the minimum; offsets cut basket volatility

Exam traps

- **Quanto polarity.** More positive correlation (underlying vs domestic-per-foreign rate) means a *cheaper* quanto. Higher correlation, higher price is the planted flip.
- **Role misassignment.** Positive correlation favours the *seller*. The buyer's conversion rate is fixed, so a rising foreign currency gives the buyer nothing extra.
- **One unknown dropped.** The seller faces two unknowns: the foreign-currency amount and the exchange rate at maturity. "Only the exchange rate is uncertain" is wrong.
- **Direction reversed.** The payoff is earned in foreign currency and converted *into* domestic currency, not the other way round.

- **"The fixed rate removes correlation."** The fixed rate removes the *buyer's* FX risk only; the seller still bears it, so correlation remains a pricing input.
- **The two exceptions.** Worse-of-two and basket options get *cheaper* as correlation falls. Every other multi-asset option in the list gets dearer.

Practice questions: MR-8 c

Question MR-8 c.1

Bank Q412 • AnalystPrep Q.1546

A quanto option allows a domestic investor to convert a potential option payoff, earned in a foreign currency, back into the domestic currency at a fixed exchange rate, protecting the investor against currency risk. From a risk management standpoint, the financial institution selling a quanto option does not know which two things at the time of sale?

A. The foreign-currency amount to be converted into domestic currency, and the exchange rate at option maturity at which that foreign-currency payoff will be converted.

B. The domestic-currency amount to be converted into foreign currency, and the exchange rate at option maturity at which that domestic-currency payoff will be converted.

C. The foreign currency that is correlated with the domestic currency, and the impact of the correlation on the buying and selling of quanto options.

D. The domestic-currency amount to be converted into foreign currency, and the impact of the correlation on the buying and selling of quanto options.

Question MR-8 c.2

Workbook A8.5 • S4 • Map #8

A structurer at a Singapore bank is presenting to a group of newly hired analysts. A US client has bought a quanto call on the Nikkei 225 whose yen payoff will be converted into US dollars at today's spot rate. The exchange rate X is quoted as US dollars per one yen. Which of the following statements in the presentation is correct?

A. The more positive the correlation between the Nikkei and X, the more expensive the quanto call, because the seller must convert a larger yen payoff.

B. The seller of the quanto call is uncertain only about the exchange rate at maturity, because the yen payoff it must convert is fixed at inception.

C. If the correlation between the Nikkei and X is negative, a rising Nikkei comes with a falling yen, so the seller needs more yen for the dollar payoff and the quanto is more expensive.

D. A positive correlation between the Nikkei and X favours the buyer of the quanto call, because a rising yen converts the Nikkei gain into more US dollars.

Question MR-8 c.3 Bank Q414 • BT P2.T5.24.1.1 (rebuilt)

A US investor buys a quanto call on the German DAX 30 index. The payoff is in euros but is converted into US dollars at a fixed exchange rate, so the investor is protected against currency risk. How does the correlation between the DAX 30 and the euro (relative to the dollar) affect the price of the quanto call?

A. Positive correlation increases the price, because the euro strengthens as the DAX rises and the seller needs more euros to cover the payoff.

B. Negative correlation decreases the price, because the euro strengthens as the DAX falls and the call becomes less likely to finish in the money.

C. Correlation has no effect on the price, because the conversion rate is fixed at inception.

D. Negative correlation increases the price, because the euro tends to weaken as the DAX rises, so the seller needs more euros to convert the profit into dollars.

Question MR-8 c.4 Bank Q415 • BT P2.T5.501.2

For which of the following multi-asset options does an increase in the correlation between S_1 and S_2 imply an increase in the option price? (S_1 and S_2 are the asset prices at maturity and K is the strike.)

A. Call on the maximum of two: payoff $= \max[0,\ \max(S_1, S_2) - K]$

B. Option on the worse of two: payoff $= \min(S_1, S_2)$

C. Exchange option: payoff $= \max(0,\ S_2 - S_1)$

D. Spread call option: payoff $= \max[0,\ (S_2 - S_1) - K]$

Question MR-8 c.5 New • built from GARP Ch.8, 8.5 (multi-asset options)

A structurer at an investment bank is repricing a book of two-asset options written on the same pair of stocks. A risk model update cuts the correlation between the two stocks from 0.6 to 0.1, with all other inputs unchanged. Which of the following options in the book will most likely become *cheaper*?

A. A call on the maximum of the two stocks

B. A spread call option on the difference between the two stocks

C. A basket call option on an equally weighted portfolio of the two stocks

D. An exchange option giving the right to swap one stock for the other

Question MR-8 c.6 Bank Q413 • BT Mock Q46 (rebuilt)

Bertha believes that correlation between assets is going to decrease. Which of the following trades best expresses her view, i.e., gains value if correlation falls?

A. Buy an option on the worse of two assets, payoff $= \min(S_1, S_2)$.

B. Buy an option on the better of two assets, payoff $= \max(S_1, S_2)$.

C. Buy correlation in a correlation swap (pay the fixed rate, receive realized correlation).

D. Sell (short) an exchange option, payoff $= \max(0,\ S_2 - S_1)$.

Question MR-8 c.7 Bank Q423 • BT L2.T5.21.3c (rebuilt)

Which of the following statements about the effect of correlation on the price of multi-asset correlation options is correct?

A. Lower correlation raises the price of most multi-asset options (such as an exchange option), but it lowers the price of a call on the worse of two assets.

B. An exchange option is worth the most when the two asset prices are almost perfectly positively correlated.

C. Correlation affects the price of rainbow options but is irrelevant to basket options.

D. A higher correlation increases the price of every multi-asset correlation option.

Solutions • MR-8 c

Solution MR-8 c.1 — Answer: A

Bank Q412 • AnalystPrep Q.1546

Step 1 – Where the payoff is earned

A quanto call pays off on a foreign underlying, so the payoff is earned in the *foreign* currency and converted *into* the domestic currency at the fixed rate.

Step 2 – The seller's two unknowns

(1) How deep in the money the call finishes, i.e. the foreign-currency amount to convert; (2) the exchange rate at maturity at which the seller must source that amount.

$\Rightarrow$ Only A names both unknowns in the right direction (foreign into domestic).

Interpretation

The correlation between these two unknowns is what makes the quanto a correlation product. The correlation is a pricing input, not an unknown quantity in itself.

Why the other options are wrong

B is incorrect. Reverses the conversion direction: the payoff is earned in foreign currency, not domestic.

C is incorrect. The currencies are known at sale; the correlation is an input to pricing, not one of the two unknown amounts.

D is incorrect. Reverses the direction and substitutes the correlation effect for the unknown exchange rate.

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.1 (GARP Ch.8), 8.5 Trading and Correlation (quanto option)

Solution MR-8 c.2 — Answer: C

Workbook A8.5 • S4

Step 1 – The seller's two unknowns

The quanto seller does not know how deep the call will be in the money (the yen amount to convert) nor the exchange rate at which it is converted. Their correlation drives the price.

Step 2 – Read the sign

With X in USD per yen, a negative correlation means that when the Nikkei rises the yen falls in value, so more yen are needed to meet the dollar payment.

$\Rightarrow$ "The lower the correlation coefficient, the more expensive is the quanto option."

Interpretation

A quanto carries correlation risk between the foreign underlying and the exchange rate. The seller is effectively short the low-correlation scenario, the same negative price-correlation relationship as the exchange option in Figure 1.

Why the other options are wrong

A is incorrect. Polarity flip: the more positive the correlation, the *lower* the quanto price, because a rising Nikkei comes with a rising yen and fewer yen are needed for the dollar payoff.

B is incorrect. Scope error: drops the first of the seller's two unknowns. The yen payoff is not fixed; it depends on how deep in the money the call finishes ("quanto" comes from "quantity").

D is incorrect. Role misassignment: a positive correlation favours the *seller*, not the buyer. The buyer's conversion rate is fixed by the contract, so a rising yen does not give the buyer more

dollars.

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.
• **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation (quanto option)

Solution MR-8 c.3 — Answer: D

Bank Q414 • BT P2.T5.24.1.1

Step 1 – Set up the seller's problem

The seller owes a dollar amount fixed by (euro payoff) × (fixed rate). It must source those euros' dollar value at the market rate at maturity.

Step 2 – Negative correlation

If the DAX and the euro are negatively correlated, a rising DAX (large euro payoff) tends to come with a weaker euro, which is the costly scenario for the seller.
⇒ Lower (negative) correlation raises the quanto price; positive correlation lowers it.

Interpretation

Same rule as the Nikkei case with a different market: the sign of the correlation between the underlying and the domestic-per-foreign rate sets the direction.

Why the other options are wrong

A is incorrect. Reverses the effect: positive correlation *lowers* the price, because the euro strengthens along with the DAX and the conversion is cheap for the seller.
B is incorrect. Wrong direction, and wrong mechanism: correlation does not change whether the call finishes in the money; it changes what the payoff costs the seller.
C is incorrect. The fixed rate removes only the buyer's FX risk; the seller still bears it, so correlation is a pricing input.

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.
• **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation (quanto option)

Solution MR-8 c.4 — Answer: B

Bank Q415 • BT P2.T5.501.2

Step 1 – The general rule

For multi-asset options the lower the correlation, the higher the price, because dispersion makes a large payoff more likely. Raising correlation therefore *lowers* most prices.

Step 2 – Find the exception

The option on the worse of two pays the minimum. With higher correlation the assets move together, so the weaker one does not lag far behind and the minimum is higher.
⇒ Only the worse-of option rises in price as correlation rises.

Interpretation

The worse-of holder is long correlation; the holders of the other three are short correlation.

Why the other options are wrong

A is incorrect. The call on the maximum gains from dispersion, so its price falls as correlation rises.
C is incorrect. The exchange option is close to zero as correlation approaches 1, because the spread $S_2 - S_1$ hardly moves.
D is incorrect. The spread call depends on the volatility of $S_2 - S_1$, which shrinks as correlation rises.

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.

• **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation (multi-asset options)

Solution MR-8 c.5 Answer: C

New • built from GARP Ch.8, 8.5

Step 1 – Classify each option

Call on the maximum, spread call and exchange option all pay on the *best* asset or on the *gap* between the two: lower correlation means more dispersion and a higher price.

Step 2 – The basket exception

A basket call pays on the weighted *sum* of the two stocks. Lower correlation means more offsetting moves inside the basket, so the basket's volatility falls and the option is worth less.
$\Rightarrow$ Of the four, only the basket call gets cheaper when correlation falls from 0.6 to 0.1.

Interpretation

GARP lists exactly two exceptions to "lower correlation, higher price": the option on the worse of two and the basket option. Questions on this LO usually test the first; this one tests the second.

Why the other options are wrong

A is incorrect. The call on the maximum benefits from dispersion: with lower correlation one stock is more likely to finish high, so the price rises.
B is incorrect. The spread call is priced off the volatility of the difference, $\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$, which rises as ρ falls.
D is incorrect. The exchange option is a spread option with zero strike; its price rises as correlation falls (Figure 1).

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.
• **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation (multi-asset options, Figure 8.4)

Solution MR-8 c.6 Answer: B

Bank Q413 • BT Mock Q46

Step 1 – Translate the view

Bertha expects correlation to fall, so she wants to be *short* correlation: a position whose value rises as correlation drops.

Step 2 – Test each trade

Better-of option: lower correlation spreads the assets apart, so $\max(S_1, S_2)$ is larger on average; owning it gains.
Worse-of option: lower correlation lowers $\min(S_1, S_2)$; owning it loses.
Paying fixed in a correlation swap gains when realised correlation exceeds the fixed rate: long correlation.
Short exchange option: the exchange option gains value as correlation falls, so the short position loses.
$\Rightarrow$ Only the long better-of option expresses a falling-correlation view.

Interpretation

Owning most multi-asset options is a short-correlation position; owning the worse-of option, buying correlation in a swap, or selling an exchange option are long-correlation positions.

Why the other options are wrong

A is incorrect. The worse-of option is the one multi-asset option that is long correlation; its price falls as correlation falls.
C is incorrect. The correlation buyer pays fixed and receives realised correlation, so it profits

when correlation rises, the opposite of her view.
D is incorrect. The exchange option's price rises as correlation falls, so a short position loses: it is long correlation.

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.
• **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation

Solution MR-8 c.7 Answer: A

Bank Q423 • BT L2.T5.21.3c

Step 1 – The general direction
With low correlation one asset is more likely to rise while the other falls, so most multi-asset options (exchange, spread, best-of) are worth more.

Step 2 – The exchange option at the extremes
$\sigma_E = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$ falls as ρ rises; at $\rho \to 1$ the price is close to zero (Figure 1).

Step 3 – The exception
The option on the worse of two pays on the minimum, which falls when correlation falls.
$\Rightarrow$ A states both the rule and the exception correctly.

Interpretation
The statement-style question rewards knowing both the direction and the exceptions. GARP's own list also makes the basket option an exception (Question MR-8 c.5).

Why the other options are wrong
B is incorrect. Reverses the relation: near-perfect positive correlation makes the exchange option almost worthless.
C is incorrect. A basket's value depends on the basket variance, which depends on the pairwise correlations, so correlation matters for baskets too.
D is incorrect. Absolute and reversed: higher correlation generally *lowers* the price, and the worse-of option is the counterexample in the other direction.

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options.
• **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation (multi-asset options, Figure 8.4)

MR-8 • Correlation Basics: Definitions, Applications, and Terminology • Vardeez: MODERATE

MR-8 d

P1 CONCEPT

Describe the structure, uses, and payoffs of a correlation swap.

Workbook: A8 Correlation Swaps, Quanto Options and Concentration Ratio • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic. Your August 2026 recall: "correlation swap payoff". Map #8 reports "1 on correlation swap return (numerical)" and "correlation swap" twice in separate sittings.
Vardeez: correlation swap.

Reported exam shapes

- **S1** Correlation swap payoff to the fixed-rate payer, plain numerical as seen in August.
- **S2** The same swap from the seller's (fixed-rate receiver's) side, with irrelevant volatilities.
- **S3** Correlation swap return per unit notional, reverse-solved for the strike.

What the questions look like: the LO verb is "describe", so expect structure-and-uses statements. The payoff calculation is in because it was *reported from the exam* (August recall), not because the wording asks for it. The traps are the averaging normaliser, the party flip and the dropped strike.

1. Structure

A **correlation swap** exchanges a **fixed** (known) correlation for the correlation that actually occurs over the life of the swap, the **realised** or **stochastic** (unknown) correlation. It is a fixed-for-floating swap with correlation in place of an interest rate.

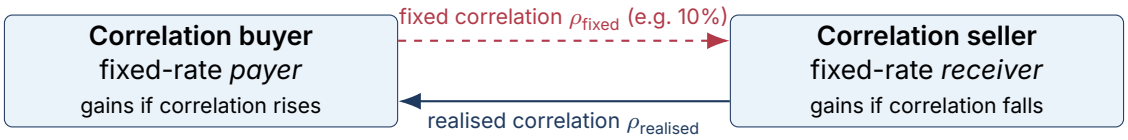


Figure 2. Dashed arrow: the fixed leg; solid arrow: the realised leg. Paying fixed is "buying correlation"; receiving fixed is "selling correlation". Net settlement at maturity: notional × (realised − fixed) to the buyer.

- **Buying correlation** = paying fixed. The present value of the swap *increases* for the correlation buyer if realised correlation increases.
- **Selling correlation** = receiving fixed. The mirror position.

2. Payoff

Realised correlation is the average of the pairwise Pearson correlations of the assets' daily log returns, $\ln(S_t/S_{t-1})$, over the swap's life, counting **each pair once** $(i > j)$:

$$\rho_{\text{realised}} = \frac{2}{n^2 - n} \sum_{i>j} \rho_{i,j}$$

$$\boxed{\text{Payoff}_{\text{payer}} = N\left(\rho_{\text{realised}} - \rho_{\text{fixed}}\right)} \qquad \text{Payoff}_{\text{receiver}} = N\left(\rho_{\text{fixed}} - \rho_{\text{realised}}\right)$$

$(n^2 - n)/2$ is the number of distinct pairs, so the formula is simply the average of the off-diagonal pairs: 3 pairs for $n = 3$, 6 pairs for $n = 4$.

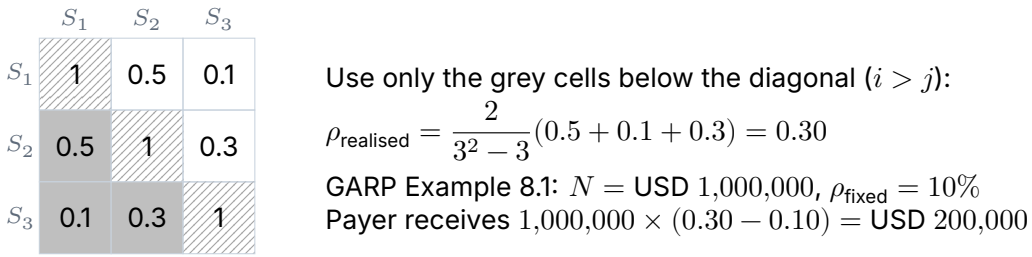


Figure 3. The realised correlation matrix at maturity (GARP Table 8.2). Grey cells ($i > j$) are used; the hatched diagonal 1s and the white upper triangle (duplicates) are ignored.

Vardeez's example. Fixed 0.20, USD 1 million, three assets with $\rho_{2,1} = 0.6$, $\rho_{3,1} = 0.2$, $\rho_{3,2} = 0.04$: $\rho_{\text{realised}} = \frac{1}{3}(0.84) = 0.28$; buyer's payoff $1{,}000{,}000 \times (0.28 - 0.20) =$ USD 80,000.

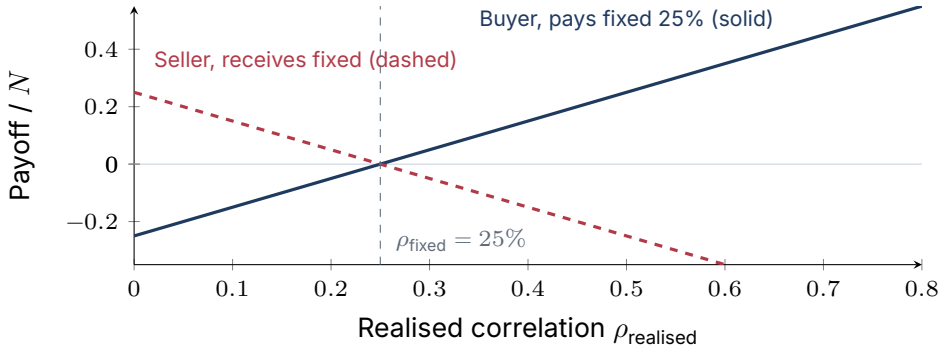


Figure 4. Solid line: buyer; dashed line: seller. Payoff per unit notional is linear in realised correlation and crosses zero at the fixed rate. Volatilities and maturity do not enter the payoff.

3. Uses

- **Indirect crash hedge.** When stock prices decrease, the correlation between the stocks typically *increases* (MR-8 g, MR-9). A fixed correlation payer therefore protects themselves indirectly against a stock market decline.
- **Buying correlation with options.** Another way of buying correlation is to **buy puts on an index** (e.g. the Dow) and **sell puts on its individual stocks**. Correlation and volatility are positively related, so if correlation rises in a downturn, the implied volatility of the index put rises and is expected to outperform the loss on the short single-stock puts.
- **The London Whale (2012)** ran the mirror image in credit. JP Morgan's Bruno Iksil **sold CDS on the CDX.NA.IG.9 index** and "hedged" by **buying single-name CDS**. In a recovering economy volatility and correlation typically decrease, so the sold index CDS should outperform. Instead a short squeeze cost JP Morgan over USD 2 billion.

4. Valuation caveat

At the time GARP's text was written there is **no industry-standard valuation model** for correlation swaps. To apply swap valuation techniques you need a **term structure of correlation**, and none exists. Traders use historical data to anticipate ρ_{realised}, or apply emerging **stochastic correlation models**.

	Buyer (fixed-rate payer)	Seller (fixed-rate receiver)
Also called	Buying correlation	Selling correlation
Payoff	$N(\rho_{\text{realised}} - \rho_{\text{fixed}})$	$N(\rho_{\text{fixed}} - \rho_{\text{realised}})$
Gains when	Realised correlation rises above fixed	Realised correlation stays below fixed
Equity sell-off	Hedged (correlations rise)	Exposed
Option replica	Long index puts, short single-stock puts	Short index puts, long single-stock puts

Formula card: MR-8 d

Realised correlation	$\rho_{\text{realised}} = \dfrac{2}{n^2 - n} \sum_{i>j} \rho_{i,j}$ (average of the $(n^2 - n)/2$ distinct pairs)
Payer (buyer)	$N(\rho_{\text{realised}} - \rho_{\text{fixed}})$
Receiver (seller)	$N(\rho_{\text{fixed}} - \rho_{\text{realised}})$
Return per unit notional	Payoff$/N$; reverse-solve: $\rho_{\text{fixed}} = \rho_{\text{realised}} \pm$ Payoff$/N$ (receiver $+$, payer $-$)

Exam traps

- **Wrong normaliser or wrong cells.** Averaging all n^2 cells (diagonal 1s included), using $2/n^2$ or $1/(n^2 - n)$, dividing by n instead of the number of pairs, or summing both triangles and still applying $2/(n^2 - n)$. Only the $i > j$ cells, once each.
- **Party flip.** The fixed-rate payer buys correlation and gains when realised exceeds fixed; the receiver has the mirror payoff. Read whose payoff the stem asks for.
- **Dropped strike.** $N \times \rho_{\text{realised}}$ is not the payoff; subtract the fixed rate.
- **Red herrings.** Volatilities, maturity and asset weights do not enter $N(\rho_{\text{realised}} - \rho_{\text{fixed}})$.
- **Option replica reversed.** Buying correlation is long *index* puts and short *single-stock* puts, not the other way round.
- **Variance, not correlation.** The payoff is on realised minus fixed *correlation*, not variance (that is a variance swap).

Practice questions: MR-8 d

Question MR-8 d.1

New • built from GARP Ch.8, 8.5 (correlation swaps)

The chief investment officer of a pension fund that holds a large, diversified US equity portfolio is worried about a broad market sell-off. A derivatives adviser proposes using correlation trades to protect the fund. Which of the following statements made by the adviser is correct?

A. Paying the fixed rate in a correlation swap on the fund's main holdings indirectly hedges a sell-off, because correlations between stocks typically rise when stock prices fall.

B. Receiving the fixed rate in a correlation swap indirectly hedges a sell-off, because the fund locks in a known correlation while realised correlation falls in a downturn.

C. Buying put options on the individual stocks and selling put options on the index is a way of buying correlation, because index volatility rises with correlation.

D. The swap can be valued with standard swap techniques, because traders can bootstrap a market-implied term structure of correlation.

Question MR-8 d.2

Bank Q430 • GARP PE2 Q33 (rebuilt)

The manager of an equity trading desk presents to newly hired analysts on derivatives used to take positions on equity correlation, including rainbow options, quanto options and correlation swaps. Which of the following statements could the manager correctly make?

A. The payoff for each of these products is highest when the price movements of the underlying assets are uncorrelated.

B. Investors can take a synthetic long position in equity correlation by buying put options on an equity index and selling put options on the components of that index.

C. The payoff for the correlation buyer in a correlation swap is negative if the realized correlation of the underlying assets is higher than the fixed correlation in the swap.

D. The payoff to the buyer of a correlation swap is the notional multiplied by the difference between realised and fixed variance of the index.

Question MR-8 d.3

Workbook A8.1 • S1 • Aug-2026 • Map #8

A junior risk analyst at a USD-based hedge fund is presenting the settlement of a 1-year correlation swap to a group of newly hired analysts. The fund pays a fixed correlation of 25% on a notional of USD 5,000,000 on a basket of three equities. The realised pairwise Pearson correlations of daily log-returns at maturity are:

	Asset 1	Asset 2	Asset 3
Asset 1	1	0.6	0.2
Asset 2	0.6	1	0.4
Asset 3	0.2	0.4	1

Which of the following is closest to the payoff to the fund at maturity?

A. USD 750,000

B. USD 1,750,000

C. USD 2,000,000

D. USD 2,750,000

Question MR-8 d.4

Bank Q431 • Mock 2 Q50 (rebuilt)

An investor buys a correlation swap with a fixed correlation rate of 0.3 and a notional of USD 750,000 for one year on a portfolio of four assets. The pairwise correlations of the daily log returns at maturity are:

	Asset 1	Asset 2	Asset 3	Asset 4
Asset 1	1	0.24	0.61	0.08
Asset 2	0.24	1	0.12	0.36
Asset 3	0.61	0.12	1	0.45
Asset 4	0.08	0.36	0.45	1

What is the payoff at maturity for the swap buyer?

A. USD 7,500

B. USD 123,750

C. USD 136,875

D. USD 232,500

Question MR-8 d.5

Workbook A8.2 • S2 • Map #8

A treasury risk analyst at a European bank is preparing a report on a 6-month correlation swap in which the bank receives a fixed correlation of 35% on a notional of EUR 8,000,000 on a three-stock basket. Annualised volatilities of the three stocks over the period were 22%, 31% and 18%. The realised pairwise correlations at maturity are:

	Stock X	Stock Y	Stock Z
Stock X	1	0.7	0.5
Stock Y	0.7	1	0.3
Stock Z	0.5	0.3	1

Which of the following is closest to the bank's payoff at maturity?

A. EUR $-1{,}200{,}000$

B. EUR 133,333

C. EUR 800,000

D. EUR 1,200,000

Question MR-8 d.6

Workbook A8.3 • S3 • Map #8

An internal auditor at a Japanese securities firm is validating a booked correlation swap whose term sheet is missing the fixed correlation. The firm was the fixed-rate receiver on a notional of JPY 600,000,000 on three Nikkei constituents and received JPY 90,000,000 at maturity. The realised pairwise correlations were:

	Stock 1	Stock 2	Stock 3
Stock 1	1	0.55	0.40
Stock 2	0.55	1	0.25
Stock 3	0.40	0.25	1

Which of the following is closest to the fixed correlation in the swap?

A. 15%

B. 25%

C. 42%

D. 55%

Question MR-8 d.7 New • built from GARP Ch.8, 8.5 (London Whale, valuation)

A risk analyst is preparing a case-study note on correlation positions in credit and on how correlation swaps are valued. Which of the following statements in the note is correct?

A. In 2012 the "London Whale" sold CDS on the CDX.NA.IG.9 index and bought CDS on individual names, a position expected to profit if volatility and correlation fell in a recovering economy.

B. In 2012 the "London Whale" bought CDS on the CDX.NA.IG.9 index and sold CDS on individual names, a position expected to profit if correlation rose in a recovering economy.

C. Correlation swaps are valued with an industry-standard model that discounts the fixed leg off a market term structure of correlation.

D. The present value of a correlation swap rises for the fixed-rate receiver when realised correlation increases during the life of the swap.

Solutions • MR-8 d

Solution MR-8 d.1 — Answer: A

New • built from GARP Ch.8, 8.5

Step 1 – What the fund fears

In a broad sell-off the correlation between stocks typically increases (Dow correlation rose from about 27% to over 50% in 2008 to 2009).

Step 2 – Which side gains from rising correlation

The fixed-rate payer (correlation buyer) receives $N(\rho_{\text{realised}} - \rho_{\text{fixed}})$, which is positive when realised correlation rises above the fixed rate.

$\Rightarrow$ Paying fixed is an indirect hedge against a stock market decline.

Interpretation

The hedge is indirect: the swap pays on correlation, not on price. It works only because correlation and falling markets go together.

Why the other options are wrong

B is incorrect. Party flip: the receiver sells correlation and loses when realised correlation rises, which is what happens in a downturn.

C is incorrect. Reversed legs: buying correlation with options is long *index* puts and short *single-stock* puts.

D is incorrect. GARP: there is no industry-standard valuation model and no term structure of correlation exists; traders use historical data or stochastic correlation models.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. • **Reference:** Meissner, Ch.1 (GARP Ch.8), 8.5 Trading and Correlation

Solution MR-8 d.2 — Answer: B

Bank Q430 • GARP PE2 Q33

Step 1 – Correlation and index volatility

Index volatility rises with the correlation among its constituents, while single-stock volatilities are largely unaffected.

Step 2 – The option replica

Long index puts plus short constituent puts therefore gains when correlation rises: a synthetic long correlation position.

$\Rightarrow$ B is GARP's "another way of buying correlation".

Interpretation

Each wrong option misstates the correlation swap payoff $N(\rho_{\text{realised}} - \rho_{\text{fixed}})$ or the products' dependence on correlation.

Why the other options are wrong

A is incorrect. Absolute and wrong: a correlation swap buyer gains when correlation is *high*, and the worse-of option is worth more when assets are correlated; no single rule makes all of them highest at zero correlation.

C is incorrect. Reversed: the buyer's payoff is *positive* when realised correlation exceeds the fixed rate.

D is incorrect. Sibling swap: realised minus fixed *variance* is a variance swap; a correlation swap pays on correlation.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. • **Reference:** Meissner, Ch.1, 8.5 Trading

and Correlation

Solution MR-8 d.3 Answer: A

Workbook A8.1 • S1 • Aug-2026

Step 1 – Identify the pairs with $i > j$

Only the three cells below the diagonal enter Equation 8.6: $\rho_{2,1} = 0.6$, $\rho_{3,1} = 0.2$, $\rho_{3,2} = 0.4$.

$$\sum_{i>j} \rho_{i,j} = 0.6 + 0.2 + 0.4 = 1.2$$

$\Rightarrow$ The diagonal 1s (each asset with itself) carry no information about co-movement and are excluded.

Step 2 – Normaliser

$$\frac{2}{n^2 - n} = \frac{2}{9 - 3} = \frac{1}{3}$$

$\Rightarrow$ With three assets there are three distinct pairs, so the formula is the simple average of the pairs.

Step 3 – Realised correlation

$$\rho_{\text{realised}} = \tfrac{1}{3} \times 1.2 = 0.40$$

$\Rightarrow$ The basket's average pairwise correlation over the year was 40%, above the 25% the fund locked in.

Step 4 – Payoff to the fixed-rate payer

$$\text{Payoff}_{\text{payer}} = N(\rho_{\text{realised}} - \rho_{\text{fixed}}) = 5{,}000{,}000 \times (0.40 - 0.25) = \text{USD } 750{,}000$$

$\Rightarrow$ The fund bought correlation at 25% and correlation came in at 40%, so it receives 15 correlation points times the notional.

Interpretation

A positive payoff confirms the fund was long correlation. Since correlations tend to rise in sell-offs, this position is the indirect equity-crash hedge described in MR-8. Had the fund received fixed, the same numbers would give USD $-750{,}000$.

Why the other options are wrong

B is incorrect. Diagonal included: averaged all nine cells, $(3 + 2 \times 1.2)/9 = 0.60$, giving $5{,}000{,}000 \times (0.60 - 0.25) = 1{,}750{,}000$.

C is incorrect. Dropped term: paid out the realised correlation times the notional, $5{,}000{,}000 \times 0.40 = 2{,}000{,}000$, forgetting to subtract the fixed strike.

D is incorrect. Double counting: summed both triangles (2.4) and still applied $2/(n^2 - n)$, so $\rho = 0.80$ and $5{,}000{,}000 \times (0.80 - 0.25) = 2{,}750{,}000$.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.1, 8.5 Trading and Correlation, Equations 8.6 and 8.7, Example 8.1

Solution MR-8 d.4

Answer: A

Bank Q431 • Mock 2 Q50

Step 1 – Count and sum the distinct pairs

With $n = 4$ there are $(16 - 4)/2 = 6$ pairs below the diagonal:

$$\sum_{i>j} \rho_{i,j} = 0.24 + 0.61 + 0.08 + 0.12 + 0.36 + 0.45 = 1.86$$

Step 2 – Realised correlation

$$\rho_{\text{realised}} = \frac{2}{4^2 - 4} \times 1.86 = \frac{1.86}{6} = 0.31$$

$\Rightarrow$ The average of the six pairs is 31%, just above the 30% fixed rate.

Step 3 – Buyer's payoff

$$750{,}000 \times (0.31 - 0.30) = \text{USD } 7{,}500$$

Interpretation

With four assets the normaliser is $2/12 = 1/6$, not $1/3$: the number of pairs grows as $n(n-1)/2$. The buyer's gain is small because realised correlation barely beat the strike.

Why the other options are wrong

B is incorrect. Divided the pair sum by $n = 4$ instead of by 6 pairs: $1.86/4 = 0.465$, $750{,}000 \times (0.465 - 0.30) = 123{,}750$.

C is incorrect. Averaged all 16 cells including the diagonal: $(4 + 2 \times 1.86)/16 = 0.4825$, $750{,}000 \times 0.1825 = 136{,}875$.

D is incorrect. Dropped the strike: $750{,}000 \times 0.31 = 232{,}500$.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. • **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation, Equations 8.6 and 8.7

Solution MR-8 d.5

Answer: A

Workbook A8.2 • S2

Step 1 – Sum of pairwise correlations ($i > j$)

$$\sum_{i>j} \rho_{i,j} = 0.7 + 0.5 + 0.3 = 1.5$$

$\Rightarrow$ Three distinct pairs; the 1s on the diagonal and the duplicate upper triangle are ignored.

Step 2 – Realised correlation

$$\rho_{\text{realised}} = \frac{2}{3^2 - 3} \times 1.5 = \frac{1.5}{3} = 0.50$$

$\Rightarrow$ The basket's average pairwise correlation was 50%.

Step 3 – Payer's payoff (Equation 8.7)

$$N(\rho_{\text{realised}} - \rho_{\text{fixed}}) = 8{,}000{,}000 \times (0.50 - 0.35) = \text{EUR } 1{,}200{,}000$$

$\Rightarrow$ This is what the fixed-rate payer (the correlation buyer, the bank's counterparty) receives.

Step 4 – Receiver's payoff

$$\text{Payoff}_{\text{receiver}} = N(\rho_{\text{fixed}} - \rho_{\text{realised}}) = \text{EUR } -1{,}200{,}000$$

$\Rightarrow$ The bank sold correlation at 35% and correlation rose to 50%, so the bank pays EUR 1.2 million.

Interpretation

Selling correlation is effectively short protection against a sell-off; the loss is the cost of correlation rising. The stock volatilities are irrelevant: Equation 8.7 depends only on realised versus fixed correlation.

Why the other options are wrong

B is incorrect. Wrong normaliser $2/n^2 = 2/9$: $\rho = (2/9)(1.5) = 0.3333$, so the receiver payoff becomes $-8{,}000{,}000 \times (0.3333 - 0.35) = +133{,}333$.

C is incorrect. Wrong normaliser $1/(n^2 - n)$ (forgot the 2): $\rho = 1.5/6 = 0.25$, receiver payoff $-8{,}000{,}000 \times (0.25 - 0.35) = +800{,}000$.

D is incorrect. Party flip: reported the payer's payoff $+1{,}200{,}000$, although the bank receives fixed.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. • **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation, Equations 8.6 and 8.7

Solution MR-8 d.6

Answer: D

Workbook A8.3 • S3

Step 1 – Realised correlation

$$\rho_{\text{realised}} = \frac{2}{3^2 - 3}(0.55 + 0.40 + 0.25) = \frac{1.20}{3} = 0.40$$

$\Rightarrow$ Average pairwise co-movement over the swap life was 40%.

Step 2 – Return per unit notional

$$\frac{\text{Payoff}_{\text{receiver}}}{N} = \frac{90{,}000{,}000}{600{,}000{,}000} = 0.15$$

$\Rightarrow$ The firm earned 15 correlation points; a positive receiver payoff means realised correlation came in below the fixed rate.

Step 3 – Reverse-solve the receiver's payoff

$$\frac{\text{Payoff}_{\text{receiver}}}{N} = \rho_{\text{fixed}} - \rho_{\text{realised}} \quad \Rightarrow \quad \rho_{\text{fixed}} = 0.40 + 0.15 = 0.55$$

$\Rightarrow$ The firm had sold correlation at 55%.

Interpretation

The booked receipt is consistent only with a 55% strike. A term sheet showing a lower fixed correlation alongside a positive JPY 90 million receipt would be an exception for the auditor to escalate, because a receiver with a strike below realised correlation pays rather than receives.

Why the other options are wrong

A is incorrect. Intermediate result: reported the return per unit notional, $90/600 = 0.15$, as if it were the strike.

B is incorrect. Party flip: treated the firm as the payer, $\rho_{\text{fixed}} = \rho_{\text{realised}} - 0.15 = 0.25$.

C is incorrect. Wrong normaliser $2/n^2 = 2/9$: $\rho = (2/9)(1.20) = 0.2667$, then $0.2667 + 0.15 = 0.42$.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. • **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation, Equations 8.6 and 8.7

Solution MR-8 d.7 **Answer: A**

New • built from GARP Ch.8, 8.5

Step 1 – The London Whale trade

Bruno Iksil sold CDS on the CDX.NA.IG.9 bond index and "hedged" it by buying CDS on individual bonds. In a recovering economy volatility and correlation typically decrease, so the sold index CDS should outperform the losses on the single-name CDS.
$\Rightarrow$ It was a short-correlation position in credit, the mirror of buying index puts and selling single-stock puts.

Step 2 – Valuation

GARP: no industry-standard valuation model exists and no term structure of correlation exists; traders use historical data or emerging stochastic correlation models.

Step 3 – Who gains from rising correlation

The present value rises for the correlation *buyer* (fixed-rate payer) when realised correlation increases.

Interpretation

The case is a reminder that a hedge built on index versus components is a correlation bet. The trade went wrong in a short squeeze that cost JP Morgan over USD 2 billion.

Why the other options are wrong

B is incorrect. Sibling swap: reverses both legs and the correlation view; Iksil *sold* index protection and expected correlation to *fall*.
C is incorrect. True-fact-wrong-claim: there is no industry-standard model and no market term structure of correlation.
D is incorrect. Party flip: rising realised correlation raises the value for the fixed-rate *payer*, not the receiver.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. • **Reference:** Meissner, Ch.1, 8.5 Trading and Correlation

MR-8 • Correlation Basics: Definitions, Applications, and Terminology • Vardeez: MODERATE

MR-8 g

P1 BOTH

Explain how correlation risk relates to systemic and concentration risk.

Workbook: A8 Correlation Swaps, Quanto Options and Concentration Ratio • Tier A

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A8, map #8 and #76). The handwritten exam sheet flags "concentration ratio concept and calculation" from this chapter.
Vardeez: concentration ratio: concept and calculation.

Reported exam shape

- **S5** Concentration ratio concept and calculation, tied to the joint default probability.

What the questions look like: a mix. Calculations: the ratio $1/n$, the joint default probability from the binomial correlation formula, and the worst-case loss built on it. Concepts: why correlations rise in a systemic crash, and why a lower concentration ratio helps only if default correlation is below 1.

1. Correlation risk and systemic risk

Systemic risk is the risk that a financial market or an entire financial system collapses.

- **Examples.** The credit market froze in September 2008 after Lehman filed for bankruptcy; lending resumed only slowly even with Fed guarantees. The Dow fell 53.54%, from 14,093 (October 2007) to 6,547 (March 2009). All but one of the Dow 30 fell (Walmart was up); 489 of the S&P 500 declined.
- **The link.** A systemic decline involves almost the entire market, so **correlations between stocks increase sharply**. Dow correlation rose from a non-crisis average of about **27%** to **over 50%** (August 2008 to March 2009), peaking at **96.97%** in February 2009. Correlations of US stocks with bonds and international equities rose too.
- **Consequence.** Portfolios considered well diversified in benign times suffered **unexpected losses**: diversification fails exactly when it is needed most. Systemic failures typically spread to the economy (lower GDP, higher unemployment).

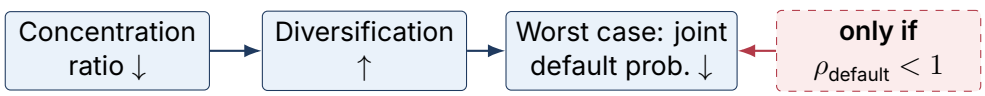
2. Correlation risk and concentration risk

Concentration risk: the risk of financial loss due to a concentrated exposure to a specific group of counterparties. It is quantified with the **concentration ratio**:

$$\text{Concentration ratio} = \frac{1}{n} \qquad (n \text{ loans of equal size})$$

Ten equal loans: 0.1. One loan: 1. **The lower the ratio, the more diversified the default risk, assuming the default correlation is smaller than 1.**

Sector concentration. Lending to more sectors raises sector diversification and reduces default risk, because *intra*-sector defaults are more highly correlated than defaults across sectors.



3. The joint default probability

Treat default of X and Y as two binomial variables (1 if default, 0 otherwise). Their standard deviations and the joint probability that both default are:

$$\sigma_X = \sqrt{P_X(1-P_X)} \qquad \boxed{P(X \cap Y) = \rho_{XY}\sqrt{P_X(1-P_X)}\,\sqrt{P_Y(1-P_Y)} + P_X P_Y}$$

At $\rho_{XY} = 0$ the joint probability collapses to $P_X P_Y$; at $\rho_{XY} = 1$ with equal PDs it equals the single PD.

GARP Example 8.3. Bank C has USD 10 million to lend.

Case	Concentration ratio	Worst case
(a) one loan of 10m to W, $P_W = 10\%$	1	EL $= 10{,}000{,}000 \times 0.10 =$ USD 1,000,000
(b) 5m each to X and Y, PDs 10%	1/2	Loss of USD 10m only if both default: $P(X \cap Y)$
(c) 3.33m each to X, Y, Z, PDs 10%	1/3	Worst case falls further

For case (b), $\sigma = \sqrt{0.1 \times 0.9} = 0.3$, so $P(X \cap Y) = 0.09\,\rho + 0.01$:

ρ_{XY}	1	0.5	0	-0.1
$P(X \cap Y)$	10%	5.5%	1%	0.1%

At $\rho = 1$ the two loans default together or survive together: no benefit from the lower ratio. Even a slightly **negative** correlation can leave a **positive** joint default probability (0.1% at -0.1) when the PDs are high and the binomial standard deviations low. Both a lower concentration ratio and a lower correlation reduce the creditor's worst case; GARP adds that a higher (copula) correlation gives a higher credit VaR.

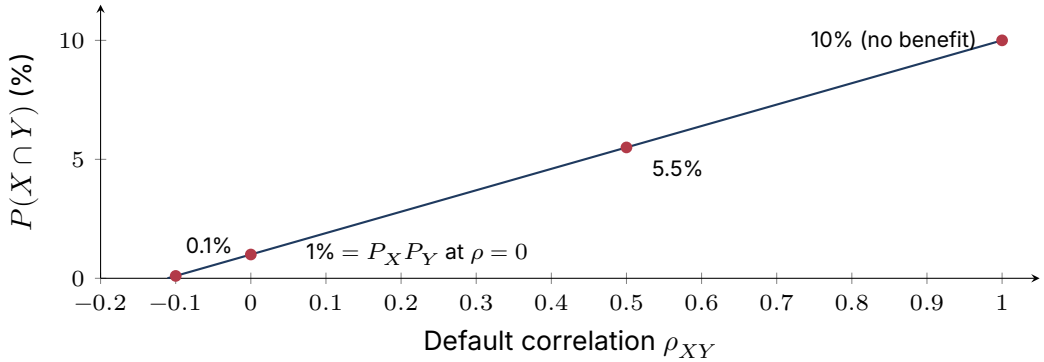


Figure 5. Joint default probability of two loans with 10% PD each (GARP Example 8.3). Linear in ρ; it reaches zero only at $\rho = -0.111$, so a mildly negative correlation still leaves a positive worst case.

Correction to the source notes (rounding). Vardeez's two-loan example (USD 2.5 million each, PD 5%) shows $P(AB) = 0.02627$ and EL $= \$131{,}350$ at $\rho = 0.5$, because $\sqrt{0.00226}$ was rounded. Exactly: $0.5 \times (0.05 \times 0.95) + 0.0025 = 0.02625$, so EL $= 0.02625 \times 5{,}000{,}000 = \$131{,}250$. The $\rho = 1$ ($250,000) and $\rho = 0$ ($12,500) figures are correct.

Formula card: MR-8 g

Concentration ratio	$1/n$ for n equal loans (lower is more diversified)
Binomial std. dev.	$\sigma_X = \sqrt{P_X(1-P_X)}$
Joint default probability	$P(X \cap Y) = \rho_{XY}\,\sigma_X\sigma_Y + P_X P_Y$
Special cases	$\rho = 0$: $P_X P_Y$; $\rho = 1$, equal PDs: P_X; zero joint PD at $\rho = -P_X P_Y/(\sigma_X\sigma_Y)$
Worst-case loss	$P(X \cap Y) \times$ total exposure $\times$ LGD

Exam traps

- **Ratio upside down.** The ratio is $1/n$, not n. A *lower* ratio means *more* diversification.
- **Scope of the benefit.** A lower ratio helps only if default correlation is below 1; at $\rho = 1$ the loans default together and the worst case is unchanged.
- **Dropped term.** Keeping only $\rho\sigma_X\sigma_Y$, or only $P_X P_Y$ (the independence case). Both terms are needed.
- **Variances for standard deviations.** $\sigma_X\sigma_Y$ is $\sqrt{P_X(1-P_X)}\sqrt{P_Y(1-P_Y)}$; for equal PDs this equals $P(1-P)$, not its square root.
- $\sqrt{\rho}$ **instead of** ρ**.** The correlation enters linearly.
- **Wrong subset for** n**.** Count *all* equal-sized loans in the book, not one sub-portfolio.
- **Systemic direction.** Correlations *rise* in a crisis; diversification benefits shrink, they do not grow.
- **Sector polarity.** Intra-sector default correlations are *higher*, so sector diversification lowers risk.

Practice questions: MR-8 g

Question MR-8 g.1

Bank Q460 • AnalystPrep Q.1556 (rebuilt)

Systemic risk is the risk of a collapse of the entire financial system. From August 2008 to March 2009, the average pairwise correlation among US stocks rose from about 27% to over 50%. Which of the following best describes the relationship between systemic risk and correlation risk?

A. They are unrelated; the rise in correlation reflected company-specific factors, so diversification benefits were unchanged.

B. In a systemic crisis correlations rise sharply, so correlation risk is greatest when diversification is most needed and diversification benefits shrink.

C. A systemic crisis lowers correlations as investors rotate across asset classes, which increases diversification benefits.

D. Correlations rose only among US stocks; correlations with bonds and international equities were unaffected.

Question MR-8 g.2

Bank Q464 • Mock 1 Q1 (rebuilt)

Bank Q's loan book consists of bullet loans of identical size, each to a separate counterparty, in two sub-portfolios:

Sub-portfolio	Number of loans	Size per loan (USD m)	Marginal PD
Prime	14	5	5.0%
Subprime	8	5	12.0%

Which of the following is correct regarding the concentration ratio of the combined portfolio and the benefit of reducing it?

A. The ratio is 0.0455; if the pairwise default correlation across all counterparties were 1, reducing the ratio would give no reduction in joint default risk.

B. The ratio is 0.0455; reducing the ratio lowers the worst-case joint default probability irrespective of the level of default correlation.

C. The ratio is 0.0714; if the pairwise default correlation across all counterparties were 1, reducing the ratio would give no reduction in joint default risk.

D. The ratio is 0.0714; reducing the ratio lowers the worst-case joint default probability irrespective of the level of default correlation.

Question MR-8 g.3

Bank Q458 • Mock 1 Q3

Two prime counterparties each have a marginal default probability of 5.0%. The risk committee instructs that a joint default correlation of 0.5 be applied. Which of the following is closest to their joint probability of default?

A. 2.13%

B. 2.38%

C. 2.63%

D. 3.61%

Question MR-8 g.4

Workbook A8.4 • S5 • Map #8 • Map #76

A credit risk manager at a German commercial bank is reviewing a proposal from the corporate lending desk. Today the bank has a single EUR 20 million loan to one borrower. The desk proposes to replace it with two loans of EUR 10 million each to two unrelated borrowers, X and Y, in different sectors. The manager's inputs are:

- One-year default probability of X: 8%; of Y: 12%
- Default correlation between X and Y (binomial Pearson correlation): 0.25
- Expected recovery on either loan: 40%

Using Meissner's framework, which of the following correctly states the bank's concentration ratio after the change and the probability of the worst-case scenario in which both X and Y default?

A. Concentration ratio 0.50; joint default probability 3.16%

B. Concentration ratio 0.50; joint default probability 0.96%

C. Concentration ratio 0.50; joint default probability 2.20%

D. Concentration ratio 2.00; joint default probability 3.16%

Question MR-8 g.5

Bank Q465 • Schweser Module Quiz 8.3 Q1 (rebuilt)

A creditor makes a $4 million loan to Company X and a $4 million loan to Company Y. Each company has a 7% default probability, and the default correlation coefficient is 0.6. The expected loss for this creditor under the worst-case scenario (both default), assuming LGD of 100%, is closest to:

A. $39,200

B. $312,480

C. $351,680

D. $560,000

Question MR-8 g.6

New • built from GARP Ch.8, 8.8 (Example 8.3)

A credit officer is reviewing two equal loans to borrowers X and Y. Each borrower has a one-year default probability of 10%, so the binomial standard deviation of each default indicator is 30%. Which of the following statements in the officer's memo is correct?

A. With a default correlation of -0.1 the joint default probability is still positive, at about 0.1%, so a slightly negative correlation does not eliminate the worst case.

B. With any negative default correlation the joint default probability is zero, so a negatively correlated pair removes the worst case entirely.

C. With a default correlation of 1, splitting one loan into these two equal loans halves the worst-case probability from 10% to 5%.

D. Lending to two borrowers in the same sector reduces concentration risk more than lending across sectors, because intra-sector default correlations are lower.

Solutions • MR-8 g

Solution MR-8 g.1

Answer: B

Bank Q460 • AnalystPrep Q.1556

Step 1 – What drives returns in a systemic crisis

Market-wide factors dominate asset-specific ones, so almost all assets fall together.

Step 2 – Read the evidence

Average Dow correlation rose from about 27% to over 50% (peak 96.97% in February 2009); correlations with bonds and international equities also rose.

$\Rightarrow$ Correlation risk is greatest, and diversification weakest, in a systemic crisis.

Interpretation

This is why "well diversified" portfolios suffered unexpected losses in 2008 to 2009: the diversification assumed in benign times disappeared in the tail.

Why the other options are wrong

A is incorrect. Treats a market-wide rise in correlation as idiosyncratic; it is driven by systematic factors.

C is incorrect. Polarity flip: correlations rise, not fall, in a crisis.

D is incorrect. Scope error: GARP notes that correlations of US equities with bonds and international equities also increased.

LO: MR-8 g: Explain how correlation risk relates to systemic and concentration risk. • **Reference:** Meissner, Ch.1 (GARP Ch.8), 8.7 Correlation Risk and Systemic Risk

Solution MR-8 g.2

Answer: A

Bank Q464 • Mock 1 Q1

Step 1 – Count all equal-sized loans

$n = 14 + 8 = 22$ loans of USD 5 million each.

$$\text{Concentration ratio} = \frac{1}{22} = 0.0455$$

$\Rightarrow$ The PDs differ across sub-portfolios, but the ratio only counts equal-sized exposures.

Step 2 – When does a lower ratio help?

$P(X \cap Y) = \rho\,\sigma_X\sigma_Y + P_X P_Y$. At $\rho = 1$ all counterparties default together or survive together, so spreading the book over more names gives no reduction in the worst case.

Interpretation

The benefit of a lower concentration ratio is conditional on default correlation being below 1. The PDs in the table are a red herring for the ratio itself.

Why the other options are wrong

B is incorrect. Right ratio, but overextends the rule: at $\rho = 1$ diversification gives nothing.

C is incorrect. Used the prime sub-portfolio only: $1/14 = 0.0714$.

D is incorrect. Wrong ratio $(1/14)$ and the same over-extension as B.

LO: MR-8 g: Explain how correlation risk relates to systemic and concentration risk. • **Reference:** Meissner, Ch.1, 8.8 Correlation Risk and Concentration Risk, Example 8.3

Solution MR-8 g.3 — Answer: C

Bank Q458 • Mock 1 Q3

Step 1 – Binomial standard deviations

$$\sigma_X\sigma_Y = \sqrt{0.05 \times 0.95} \times \sqrt{0.05 \times 0.95} = 0.05 \times 0.95 = 0.0475$$

Step 2 – Correlation term

$$\rho\,\sigma_X\sigma_Y = 0.5 \times 0.0475 = 0.02375$$

Step 3 – Add the independence term

$$P(X \cap Y) = 0.02375 + 0.05 \times 0.05 = 0.02375 + 0.0025 = 0.02625 = 2.63\%$$

$\Rightarrow$ More than ten times the 0.25% that independence would give.

Interpretation

Joint PD = correlation-driven part plus the independent part. With equal PDs, $\sigma_X\sigma_Y$ is simply $P(1-P)$.

Why the other options are wrong

A is incorrect. Subtracted P_XP_Y instead of adding: $0.02375 - 0.0025 = 2.13\%$.
B is incorrect. Dropped P_XP_Y: reported only the correlation term, 2.38%.
D is incorrect. Used $\sqrt{0.5} = 0.7071$ in place of ρ: $0.7071 \times 0.0475 + 0.0025 = 3.61\%$.

LO: MR-8 g: Explain how correlation risk relates to systemic and concentration risk. • **Reference:** Meissner, Ch.1, 8.8, Equations 8.11 and 8.12

Solution MR-8 g.4 — Answer: A

Workbook A8.4 • S5

Step 1 – Concentration ratio

$$\text{Concentration ratio} = \frac{1}{n} = \frac{1}{2} = 0.50$$

$\Rightarrow$ Two loans of equal size halve the ratio from 1 (a single loan) to 0.50: the default risk is more diversified, provided the default correlation is below 1.

Step 2 – Standard deviations of the binomial default variables (Equation 8.12)

$$\sigma_X = \sqrt{0.08 \times 0.92} = 0.2713, \qquad \sigma_Y = \sqrt{0.12 \times 0.88} = 0.3250$$

Step 3 – Joint default probability (Equation 8.11)

$$\begin{aligned} P(X \cap Y) &= \rho_{XY}\sigma_X\sigma_Y + P_XP_Y = 0.25 \times 0.2713 \times 0.3250 + 0.08 \times 0.12 \\ &= 0.0220 + 0.0096 = 3.16\% \end{aligned}$$

$\Rightarrow$ There is a 3.16% chance that both borrowers default within the year, the bank's worst case.

Interpretation

Under the old single loan the worst case (default of the one borrower) had the full PD of that borrower; after splitting, losing both loans requires a joint default, here 3.16%. The benefit comes from both levers Meissner links: a lower concentration ratio and a correlation below 1. If X and Y were in the same sector, intra-sector default correlation would be higher and the joint probability would rise. The 40% recovery is a red herring: it changes the size of the loss, not its probability.

Why the other options are wrong

B is incorrect. Dropped term: assumed independence and kept only $P_X P_Y = 0.08 \times 0.12 = 0.96\%$, ignoring the 0.25 default correlation.
C is incorrect. Dropped term: kept only the correlation term $0.25 \times 0.2713 \times 0.3250 = 2.20\%$ and forgot to add $P_X P_Y$.
D is incorrect. Polarity flip on the ratio: reported $n/1 = 2.00$ instead of $1/n$; a higher number would wrongly signal more concentration after diversifying.

LO: MR-8 g: Explain how correlation risk relates to systemic and concentration risk. • **Reference:** Meissner, Ch.1, 8.8 Correlation Risk and Concentration Risk, Example 8.3, Equations 8.11 and 8.12

Solution MR-8 g.5 Answer: C

Bank Q465 • Schweser Module Quiz 8.3 Q1

Step 1 – Joint default probability

$$\sigma_X \sigma_Y = 0.07 \times 0.93 = 0.0651$$

$$P(X \cap Y) = 0.6 \times 0.0651 + 0.07 \times 0.07 = 0.03906 + 0.0049 = 0.04396$$

Step 2 – Worst-case exposure

Both loans are lost: $4{,}000{,}000 + 4{,}000{,}000 = \$8{,}000{,}000$ at 100% LGD.

Step 3 – Expected loss under the worst case

$$0.04396 \times 8{,}000{,}000 = \$351{,}680$$

$\Rightarrow$ The worst-case expected loss is the joint PD times the total exposure.

Interpretation

This is GARP's framing of Example 8.3: the creditor's worst case is the joint default, and its probability (not just its size) depends on correlation.

Why the other options are wrong

A is incorrect. Set the correlation to zero: joint PD $= 0.0049$, $0.0049 \times 8m = \$39{,}200$.
B is incorrect. Omitted the $P_X P_Y$ term: $0.03906 \times 8m = \$312{,}480$.
D is incorrect. Assumed $\rho = 1$, so the joint PD equals the 7% PD: $0.07 \times 8m = \$560{,}000$.

LO: MR-8 g: Explain how correlation risk relates to systemic and concentration risk. • **Reference:** Meissner, Ch.1, 8.8 Correlation Risk and Concentration Risk, Equation 8.11

Solution MR-8 g.6 Answer: A

New • built from GARP Ch.8, 8.8

Step 1 – Joint PD as a function of correlation

$$P(X \cap Y) = \rho \times 0.3 \times 0.3 + 0.1 \times 0.1 = 0.09\,\rho + 0.01$$

Step 2 – Evaluate the statements

$\rho = -0.1$: $-0.009 + 0.01 = 0.001 = 0.1\%$, still positive. It reaches zero only at $\rho = -0.01/0.09 = -0.111$.
$\rho = 1$: $0.09 + 0.01 = 10\%$, the same as one 10m loan with a 10% PD.
$\Rightarrow$ A is GARP's own observation: even a slightly negative correlation can leave a positive joint default probability when the PDs are high.

Interpretation

The worst case shrinks as correlation falls, but it is not eliminated by a small negative correlation, and the concentration ratio buys nothing when $\rho = 1$ (Figure 5).

Why the other options are wrong

B is incorrect. Absolute qualifier: "any" negative correlation is wrong; the joint PD is positive down to $\rho = -0.111$ here.

C is incorrect. At $\rho = 1$ the loans default together or survive together: the worst case stays at 10%, there is no benefit from the lower ratio.

D is incorrect. Polarity flip: intra-sector default correlations are *higher*, so lending across sectors diversifies more.

LO: MR-8 g: Explain how correlation risk relates to systemic and concentration risk. • **Reference:** Meissner, Ch.1, 8.8 Correlation Risk and Concentration Risk, Example 8.3, Equation 8.11

MR-14 • The Art of Term Structure Models: Drift • Vardeez: VERY IMPORTANT

MR-14 a

P1 BOTH

Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift.

Workbook: A7 Ho-Lee, Time-Dependent Drift and Volatility • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A7: the short-rate trees of this reading came up in the August 2026 sitting (Ho-Lee plug values, time-dependent drift and volatility theory). Model 1 and Model 2 are the building blocks of every tree in MR-14 and MR-15.

Vardeez (chapter): "Vasicek model calc can come; Ho-Lee model; calc of expected rate in T years. Calculations are super important here."

Reported exam shape for this LO

- **A7 S6** Sibling swap against Vasicek and Models 1 and 2: a feature of one model (drift, mean reversion, parallel shifts, flat volatility) attributed to another.

What the questions look like: two kinds. (1) Plug-in tree nodes: $r_0 \pm \sigma\sqrt{dt}$ for Model 1, $r_0 + \lambda dt \pm \sigma\sqrt{dt}$ for Model 2, one or two steps out. (2) Effectiveness statements: term-structure shape, flat volatility, parallel shifts, recombining, negative rates, equilibrium versus arbitrage-free.

1. Notation for every short-rate model in this reading

Symbol	Meaning
dr	change in the (continuously compounded, annualised) short rate over a small interval dt
dt	time step in years: one month $= 1/12$, so $\sqrt{dt} = \sqrt{1/12} = 0.2887$
σ	**annual basis-point volatility** of rate changes (e.g. 113 bp $= 1.13\%$)
dw	normal random variable, mean 0, **standard deviation** $\sqrt{dt}$
λ	drift added to the rate (risk premium plus expected change), per year
$k,\ \theta$	speed of mean reversion and long-run (risk-neutral) level: Vasicek, MR-14 f, g

Every tree in this reading moves the rate by *drift* $\times\, dt$ plus or minus *volatility* $\times\sqrt{dt}$, with probability $\frac{1}{2}$ on each branch. The models differ only in the drift term.

2. Model 1: normally distributed rates, no drift

$$\boxed{dr = \sigma\, dw} \qquad E[dr] = 0, \qquad V[dr] = \sigma^2 dt, \qquad \text{SD of the rate change} = \sigma\sqrt{dt}$$

The tree. From r_0 the rate goes to $r_0 + \sigma\sqrt{dt}$ or $r_0 - \sigma\sqrt{dt}$. On date 2 the nodes are $r_0 + 2\sigma\sqrt{dt}$, r_0, $r_0 - 2\sigma\sqrt{dt}$: up-then-down and down-then-up land on the same rate, so the tree **recombines**.

GARP example. $r_0 = 6.18\%$, $\sigma = 1.13\%$, $dt = 1/12$: $\sigma\sqrt{dt} = 1.13\% \times 0.2887 = 0.326\%$, or 32.6 bp a month. Date-1 nodes 6.506% and 5.854%; date-2 nodes 6.832%, 6.180%, 5.528% (Figure 1, left).

Effectiveness of Model 1

- **Term structure slopes down, only because of convexity.** With no drift, expected future rates equal r_0; the convexity effect lowers longer par rates (GARP Table 14.1: -0.8 bp at 2 years, -5.1 at 5, -18.8 at 10, -135.3 at 30). It is *not* flat.
- **Flat term structure of volatility** at σ (113 bp at every term), while the market's is **humped**.

- **One factor, parallel shifts.** The only factor is the short rate: a 10 bp rise in r raises every rate by 10 bp.
- **Too inflexible.** Calibrating r_0 to the 10-year par rate leaves every other point off the market curve.
- **Negative rates.** Normal (Gaussian) model, so rates can go negative; the probability rises with the horizon (MR-14 b, c).
- **Equilibrium model:** no attempt is made to match the initial term structure.

3. Model 2: constant drift (risk premium)

$$\boxed{dr = \lambda\,dt + \sigma\,dw} \qquad E[dr] = \lambda\,dt, \qquad \text{SD} = \sigma\sqrt{dt}$$

The tree. Every node is shifted up by λdt per step, whatever the direction of the shock:

$$\text{date 1: } r_0 + \lambda dt \pm \sigma\sqrt{dt} \qquad \text{date 2: } r_0 + 2\lambda dt + \{2,\ 0,\ -2\}\,\sigma\sqrt{dt}$$

The middle node on date 2 is the single value $r_0 + 2\lambda dt$, so the tree still **recombines**.

GARP example. $r_0 = 5.138\%$, $\lambda = 0.229\%$, $\sigma = 1.10\%$. Drift $0.229\%/12 = 1.9$ bp a month; SD $1.10\% \times 0.2887 = 31.75$ bp a month. With a realised $dw = 0.15$: $dr = 0.229\%/12 + 1.10\% \times 0.15 = 0.1841\%$, new rate 5.322%.

Terminal distribution. After T years the rate is normal with mean $r_0 + \lambda T$ and SD $\sigma\sqrt{T}$. GARP: after 10 years, mean $5.138\% + 10 \times 0.229\% = 7.428\%$, SD $1.10\%\sqrt{10} = 347.9$ bp. The positive drift raises the mean, so negative rates become less likely (but not impossible).

What the drift is. λ is a *risk-neutral* drift: a mix of the true expected change in the rate and a risk premium. A 1.9 bp monthly drift could be 1.9 bp of expected rise, or 1 bp of expected rise plus 0.9 bp of premium, or a 0.1 bp expected fall with 2 bp of premium.

Effectiveness of Model 2

- **Upward-sloping term structure** from the drift: calibrating r_0 and λ to the 2- and 10-year par rates fits the market far better than Model 1.
- **But the drift is implausibly large.** Read as a risk premium alone, $\lambda = 0.229\%$ with 110 bp volatility is a Sharpe ratio of about 0.21; read as expected rate rises, it says rates in year 10 exceed rates in year 9, hard to defend in the long end. The fitted curve overshoots the 30-year par rate by about 25 bp.
- **Nothing else changes.** Moving from Model 1 to Model 2 does not change the *flat* term structure of volatility, the size of the convexity effect or the *parallel-shift* nature.
- Still an **equilibrium** model (Ho-Lee, MR-14 d, is the arbitrage-free generalisation).

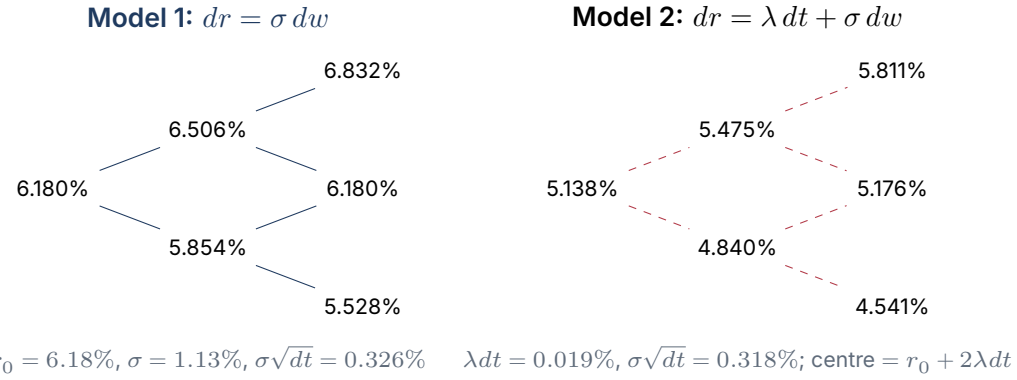


Figure 1. Two-step monthly trees, all branch probabilities $\frac{1}{2}$ (GARP's numbers). Both recombine. Model 1 (solid branches): the centre node stays at r_0. Model 2 (dashed branches): the whole tree climbs by λdt each step.

	Model 1 (no drift)	Model 2 (constant drift)
Process	$dr = \sigma\, dw$	$dr = \lambda\, dt + \sigma\, dw$
Date-2 nodes	$r_0 \pm 2\sigma\sqrt{dt}$, r_0	$r_0 + 2\lambda dt \pm 2\sigma\sqrt{dt}$, $r_0 + 2\lambda dt$
Recombines	Yes	Yes
Term structure	Downward sloping (convexity only)	Upward sloping (drift)
Volatility structure	Flat at σ (market: humped)	Flat at σ (unchanged)
Shifts	Parallel	Parallel
Negative rates	Possible, rising with horizon	Possible, less likely
Weakness	Too inflexible to fit curve shapes	Drift too large to justify (long end)
Class	Equilibrium	Equilibrium

Formula card: MR-14 a

Model 1 step	$r_{t+dt} = r_t \pm \sigma\sqrt{dt}$; realised: $dr = \sigma\, dw$
Model 2 step	$r_{t+dt} = r_t + \lambda dt \pm \sigma\sqrt{dt}$; realised: $dr = \lambda dt + \sigma\, dw$
Date-2 nodes	$r_0 + n\lambda dt + \{2, 0, -2\}\sigma\sqrt{dt}$ ($\lambda = 0$ for Model 1)
Terminal distribution	Normal, mean $r_0 + \lambda T$, SD $\sigma\sqrt{T}$
Scaling	drift $\times\, dt$; volatility $\times\sqrt{dt}$ ($\sqrt{1/12} = 0.2887$, $\sqrt{1/4} = 0.5$)

Exam traps

- dt **versus** $\sqrt{dt}$**.** Drift scales with dt, volatility with $\sqrt{dt}$. $\sigma/12$ shrinks the branches; $\lambda\sqrt{dt}$ or an unscaled λ inflates the drift.
- **Drift on every branch.** In Model 2 the drift is added on the down branches too; subtracting it, or dropping it, are the standard distractors.
- **One step versus two.** Date-2 outer nodes sit $2\sigma\sqrt{dt}$ from the centre; quoting the date-1 node is a planted option.
- **A realised** dw **is not a node.** The middle node on date 2 is $r_0 + 2\lambda dt$ whatever dw happened to be.
- **"Model 1 is flat."** False: its term structure slopes down through convexity; only its *volatility*

structure is flat.

- **Sibling swap.** Mean reversion and non-parallel shifts belong to Vasicek; a humped volatility structure to Gauss+; neither drift specification produces them.

Practice questions: MR-14 a

Question MR-14 a.1

Bank Q653 • BT P2.T5.302.1 (rebuilt)

Analyst Greg builds a monthly interest rate tree ($dt = 1/12$) with Tuckman's Model 1 (normally distributed rates, zero drift, $dr = \sigma\, dw$). The current short rate is 4.0% and annual basis-point volatility is 180 basis points. Part of the tree:

Node	Month 0	Month 1	Month 2
Top			[2,2] = ?
Up		4.520%	
Middle	$r(0) = 4.000\%$		4.000%
Down		3.480%	
Bottom			2.961%

What is the missing rate at node [2,2]?

A. 4.30%

B. 4.73%

C. 5.04%

D. 4.52%

Question MR-14 a.2

Bank Q654 • BT P2.T5.302.2 (rebuilt)

Analyst Barbara builds a monthly interest rate tree ($dt = 1/12$) with Tuckman's Model 2 (normally distributed rates with drift, $dr = \lambda\, dt + \sigma\, dw$). The current short rate is 3.0%, annual basis-point volatility is 200 basis points and the annual drift is $\lambda = 50$ basis points. Part of the tree:

Node	Month 0	Month 1	Month 2
Top			4.238%
Up		3.619%	
Middle	$r(0) = 3.000\%$		3.083%
Down		2.464%	
Bottom			[2,0] = ?

What is the missing rate at node [2,0]?

A. 1.85%

B. 1.93%

C. 2.46%

D. 1.76%

Question MR-14 a.3 Bank Q651 • AnalystPrep Q.4014 (rebuilt)

Under Model 2 the current short rate is 5.26%, annual basis-point volatility is 115 basis points ($\sigma = 1.15\%$), annual drift is $\lambda = 0.25\%$, and $dt = 1/12$. After one month the random variable dw takes the value 0.25. What is the short-term rate after one month?

A. 5.7975%

B. 5.5475%

C. 5.5683%

D. 5.6128%

Question MR-14 a.4 Bank Q658 • Schweser Module Quiz 14.1 Q2

Using Model 2, assume a current short-term rate of 8%, an annual drift of 50 bps, and a short-term rate standard deviation of 2%. In addition, assume the ex-post realization of the dw random variable is 0.3. After constructing a 2-period interest rate tree with annual periods, what is the interest rate in the middle node at the end of Year 2?

A. 8.0%

B. 8.8%

C. 9.0%

D. 9.6%

Question MR-14 a.5 Bank Q657 • Mock 4 Q57

An analyst is modelling the short-term interest rate using a model with constant drift and constant volatility: $dr = \lambda\,dt + \sigma\,dw$. The current short-term rate is 3.00%, the annual drift λ is 0.30%, and the annual basis-point volatility σ is 1.20%. What are the expected value and the standard deviation of the short-term rate two years from now?

A. Expected rate 3.30%; standard deviation 1.70%

B. Expected rate 3.60%; standard deviation 1.70%

C. Expected rate 3.60%; standard deviation 1.20%

D. Expected rate 3.60%; standard deviation 2.40%

Question MR-14 a.6 Bank Q650 • AnalystPrep Q.2859 (rebuilt)

In Model 2 ($dr = \lambda\,dt + \sigma\,dw$) the current short rate is zero, the drift λ is positive and the basis-point volatility σ is zero. Which statement is correct?

A. The short rate will be negative about half of the time.

B. The long-term spot rate will equal the short rate at all maturities.

C. The short rate rises deterministically and never goes negative, so long-term spot rates exceed the short rate.

D. The long-term spot rate will be below the short rate because of the convexity effect.

Question MR-14 a.7 Bank Q656 • BT P2.T5.306.3 (rebuilt)

Which of the following statements about Tuckman's Model 1 (no drift) and Model 2 (constant drift) is correct?

A. Model 1 predicts a hump-shaped term structure of volatility, consistent with observed volatility.

B. In Model 1, a change in the short-term rate leads to a parallel shift in the predicted yield curve.

C. The Model 2 tree is non-recombining because the drift is added only to the up-branches.

D. Model 2's constant drift is easy to justify as an expected rate increase over very long horizons.

Question MR-14 a.8 Bank Q655 • BT P2.T5.302.3

Tuckman's Model 1 assumes zero drift and is also called a normal model. His Model 2 adds a term for drift. Each of the following is true about these two models, according to Tuckman, EXCEPT for:

A. A weakness of Model 1 (normal model) is that the short-term rate can become negative.

B. Model 1 implies a term structure that is perfectly flat at the current rate for all maturities, including the long-term rates.

C. Model 2 is more capable of producing an upward-sloping term structure, which is often observed.

D. Model 2 is an equilibrium model rather than an arbitrage-free model because no attempt is made to match the term structure closely.

Solutions • MR-14 a

Solution MR-14 a.1

Answer: C

Bank Q653 • BT P2.T5.302.1

Step 1 – Size of one step

$$\sigma\sqrt{dt} = 1.80\% \times 0.28868 = 0.5196\%$$

$\Rightarrow$ This matches the month-1 nodes: $4.000\% \pm 0.520\% = 4.520\%$ and 3.480%.

Step 2 – Top node after two up moves

$$r_{uu} = r_0 + 2\sigma\sqrt{dt} = 4.000\% + 2 \times 0.5196\% = 5.039\% \approx 5.04\%$$

$\Rightarrow$ With zero drift the tree is symmetric: the bottom node $4.000\% - 1.039\% = 2.961\%$ confirms it.

Interpretation

Model 1 recombines, so the date-2 nodes are r_0 and $r_0 \pm 2\sigma\sqrt{dt}$. The given bottom node is a free check on your step size.

Why the other options are wrong

A is incorrect. Scaled volatility by dt instead of $\sqrt{dt}$: $4\% + 2 \times 1.8\%/12 = 4.30\%$.
B is incorrect. Used the two-month standard deviation $\sigma\sqrt{2dt} = 0.735\%$ instead of two up steps: 4.73%.
D is incorrect. Gave the month-1 up node (one step only): 4.52%.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.1 Model 1

Solution MR-14 a.2

Answer: B

Bank Q654 • BT P2.T5.302.2

Step 1 – Drift and shock per step

$$\lambda dt = \frac{0.50\%}{12} = 0.0417\%, \qquad \sigma\sqrt{dt} = 2.00\% \times 0.28868 = 0.5774\%$$

Step 2 – Bottom node: two drifts, two down shocks

$$r_{dd} = 3.000\% + 2(0.0417\%) - 2(0.5774\%) = 1.929\% \approx 1.93\%$$

$\Rightarrow$ The drift is added on every branch, down moves included.

Interpretation

Check against the given nodes: the middle node $3.000\% + 2 \times 0.0417\% = 3.083\%$ and the top node $3.083\% + 1.155\% = 4.238\%$ both match. The tree recombines with its centre lifted by $2\lambda dt$.

Why the other options are wrong

A is incorrect. Left out the drift (Model 1): $3.0\% - 1.155\% = 1.85\%$.
C is incorrect. Gave the month-1 down node (one step only): 2.46%.
D is incorrect. Subtracted the drift on the down branches: $3.0\% - 0.083\% - 1.155\% = 1.76\%$.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.2 Model 2, tree following eq. 14.6

Solution MR-14 a.3

Answer: C

Bank Q651 • AnalystPrep Q.4014

Step 1 – Drift over the month

$$\lambda dt = \frac{0.25\%}{12} = 0.0208\%$$

Step 2 – Shock from the realised dw

$$\sigma\, dw = 1.15\% \times 0.25 = 0.2875\%$$

$\Rightarrow$ dw is the realisation of a variable whose SD is already $\sqrt{dt}$, so no further $\sqrt{dt}$ is applied.

Step 3 – New rate

$$r_{1/12} = 5.26\% + 0.0208\% + 0.2875\% = 5.5683\%$$

Interpretation

Same structure as GARP's eq. 14.6 ($0.229\%/12 + 1.10\% \times 0.15$). The only judgement call is to treat dw as a realised value of dw, not as a standard normal Z.

Why the other options are wrong

A is incorrect. Added the full annual drift: $5.26\% + 0.25\% + 0.2875\% = 5.7975\%$.
B is incorrect. Omitted the drift (Model 1): $5.26\% + 0.2875\% = 5.5475\%$.
D is incorrect. Replaced the realised dw by $\sqrt{dt}$: $5.26\% + 0.0208\% + 1.15\% \times 0.2887 = 5.6128\%$.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.2, eq. 14.6

Solution MR-14 a.4

Answer: C

Bank Q658 • Schweser Module Quiz 14.1 Q2

Step 1 – What the middle node is

In the recombining Model 2 tree, the up-down and down-up paths cancel the volatility terms, so the date-2 middle node is $r_0 + 2\lambda dt$.

Step 2 – Plug in (annual steps, $dt = 1$)

$$r_{\text{mid}} = 8\% + 2 \times 0.50\% \times 1 = 9.0\%$$

$\Rightarrow$ The realised $dw = 0.3$ moves one simulated path; it does not move a node of the tree.

Interpretation

The dw and the 2% volatility are red herrings for this node. They would matter for the outer nodes ($9\% \pm 2 \times 2\%$) or for a single simulated path.

Why the other options are wrong

A is incorrect. Ignored the drift (the Model 1 answer): 8.0%.
B is incorrect. One drift plus the dw realisation: $8\% + 0.5\% + 0.3\% = 8.8\%$.
D is incorrect. Added the dw realisation in each period: $8\% + 2 \times (0.5\% + 0.3\%) = 9.6\%$.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.2; SchweserNotes 2026 Book 1, Reading 14

Solution MR-14 a.5

Answer: B

Bank Q657 • Mock 4 Q57

Step 1 – Expected rate: drift accumulates linearly

$$E[r_2] = r_0 + \lambda T = 3.00\% + 0.30\% \times 2 = 3.60\%$$

Step 2 – Standard deviation: grows with $\sqrt{T}$

$$\text{SD} = \sigma\sqrt{T} = 1.20\% \times \sqrt{2} = 1.70\%$$

$\Rightarrow$ The terminal distribution of Model 2 is normal with mean $r_0 + \lambda T$ and SD $\sigma\sqrt{T}$.

Interpretation

This is GARP's terminal-distribution result (10 years: 7.428% and 347.9 bp). Without mean reversion nothing dampens the spread, so the SD grows without limit as $\sqrt{T}$ (contrast Vasicek, MR-14 g).

Why the other options are wrong

A is incorrect. One year of drift instead of two: $3.00\% + 0.30\% = 3.30\%$.
C is incorrect. Used the one-year volatility without scaling by $\sqrt{2}$: 1.20%.
D is incorrect. Scaled the volatility linearly with time: $1.20\% \times 2 = 2.40\%$.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.2 (terminal distribution after one and 10 years)

Solution MR-14 a.6

Answer: C

Bank Q650 • AnalystPrep Q.2859

Step 1 – Zero volatility removes the random term

With $\sigma = 0$, $dr = \lambda dt$: the rate path is $r_t = 0 + \lambda t$, rising deterministically and never negative.

Step 2 – Spot rates

With no volatility there is no convexity effect, so spot rates are averages of the expected short rates. Each term's average of a rising path exceeds today's rate of zero, and grows with maturity.
$\Rightarrow$ The term structure slopes upward.

Interpretation

The question isolates what each building block does: volatility creates both negative-rate risk and the convexity effect; drift tilts the curve. Switch off volatility and only the drift's upward tilt remains.

Why the other options are wrong

A is incorrect. Negative rates come from random shocks; with $\sigma = 0$ there are none.
B is incorrect. A curve flat at the short rate needs zero drift as well as zero volatility.
D is incorrect. Convexity lowers long rates only when volatility is positive; here it is zero.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1 and 14.2

Solution MR-14 a.7 Answer: B

Bank Q656 • BT P2.T5.306.3

Step 1 – Factor structure of Model 1

The only factor is the short rate and volatility is constant, so a shock to r moves every point of the curve by the same amount. GARP: a 10 bp rise raises all rates by 10 bp.
$\Rightarrow$ Model 1 is a model of parallel shifts: B is correct.

Step 2 – Check the others

Model 1's volatility structure is flat, not humped. Model 2 adds λdt on every branch, so up-down and down-up still meet at $r_0 + 2\lambda dt$. A constant drift read as expected rate rises is hardest to justify in the long end.

Interpretation

Each wrong option is a feature borrowed from elsewhere: the hump is the market (and Gauss+); non-recombination is Vasicek before adjustment; the long-horizon defence of drift reverses GARP's caveat.

Why the other options are wrong

A is incorrect. Model 1's volatility structure is flat at σ; the hump is what the market shows and Model 1 fails to capture.
C is incorrect. The drift is added on every branch, so the Model 2 tree recombines.
D is incorrect. Reverses the caveat: rising expected rates are hardest to justify at very long horizons.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1 (Figure 14.3), 14.2

Solution MR-14 a.8 Answer: B

Bank Q655 • BT P2.T5.302.3

Step 1 – Test each statement

A: normal rates can go below zero (true). C: the positive drift lets Model 2 produce the usual upward slope (true). D: no attempt is made to match the term structure, so both are equilibrium models (true).

Step 2 – The false one

B: with zero drift, the convexity effect pulls longer rates down (-18.8 bp at 10 years, -135.3 bp at 30), so Model 1's term structure is *downward sloping*, not flat.
$\Rightarrow$ The EXCEPT answer is B.

Interpretation

"No drift" means expected future short rates equal today's rate. It does not mean a flat curve, because rates also embed convexity, which grows with maturity and volatility.

Why the other options are wrong

A is incorrect. True statement: the main drawback of a normal model.
C is incorrect. True statement: the drift is what produces the upward slope.
D is incorrect. True statement: Models 1 and 2 are equilibrium models; Ho-Lee is the first arbitrage-free step.

LO: MR-14 a: Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1 (Table 14.1, Figure 14.2), 14.2

MR-14 • The Art of Term Structure Models: Drift • Vardeez: VERY IMPORTANT

MR-14 b

P1 CALC

Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift.

Workbook: A7 Ho-Lee, Time-Dependent Drift and Volatility • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A7 (Model 1 is the base case of the Ho-Lee and Model 3 trees seen in August 2026).
Vardeez (chapter): "Calculations are super important here."

Reported exam shape for this LO

- **A7 S6** Sibling swap against Vasicek and Models 1 and 2 (the Model 1 node is the planted distractor in Ho-Lee questions).

What the questions look like: pure plug-ins of $dr = \sigma\, dw$ and $\sigma\sqrt{dt}$. The whole difficulty is *what number to multiply*: a realised dw (use it as is), a standard normal draw Z (multiply by $\sqrt{dt}$), or the standard deviation of dw (a distractor).

1. The rate change

$$\boxed{dr = \sigma\, dw} \qquad dw \sim N\left(0,\ \sqrt{dt}\right)$$

The change in the short rate is the annual basis-point volatility times the realised dw. Three ways the random term can be given:

The stem gives	What to do
A realised value of dw ("dw takes the value 0.15")	$dr = \sigma \times dw$. The $\sqrt{dt}$ is already inside dw; do *not* multiply by 0.2887 again.
A standard normal draw Z ("the inverse normal gives $Z = 1.24$")	$dw = Z\sqrt{dt}$, so $dr = \sigma Z\sqrt{dt}$.
"dw has standard deviation $\sqrt{1/12} = 0.2887$"	That is a description of dw, not a draw. It is the distractor.

GARP example. $r_0 = 6.18\%$, $\sigma = 1.13\%$, $dt = 1/12$, and dw takes the value 0.15:

$$dr = 1.13\% \times 0.15 = 0.17\% \quad (17\text{ bp}), \qquad r_{1/12} = 6.18\% + 0.17\% = 6.35\%.$$

2. The standard deviation of the rate change

Because $E[dw] = 0$, the expected change (the drift) is zero. Because the SD of dw is $\sqrt{dt}$:

$$\boxed{\text{SD}(dr) = \sigma\sqrt{dt}} \qquad \text{GARP: } 1.13\% \times \sqrt{1/12} = 0.326\% = 32.6\text{ bp per month.}$$

The tree reproduces exactly these two moments: with branches $\pm\sigma\sqrt{dt}$ at probability $\frac{1}{2}$,

$$E[dr] = \tfrac{1}{2}\sigma\sqrt{dt} + \tfrac{1}{2}(-\sigma\sqrt{dt}) = 0, \qquad V[dr] = \tfrac{1}{2}\sigma^2 dt + \tfrac{1}{2}\sigma^2 dt - 0 = \sigma^2 dt.$$

Over several steps the variances add: after n steps of length dt the SD is $\sigma\sqrt{n\,dt}$, and after T years it is $\sigma\sqrt{T}$. One step is $\pm\sigma\sqrt{dt}$; two steps put the outer nodes at $\pm 2\sigma\sqrt{dt}$, but the *standard deviation* after two steps is only $\sigma\sqrt{2dt}$.

Negative rates rise with the horizon. GARP: after 10 years the SD is $1.13\%\sqrt{10} = 3.573\%$; a shock of only $6.18/3.573 = 1.73$ standard deviations takes 6.18% below zero (Figure 2). Over one year the distribution almost certainly stays positive.

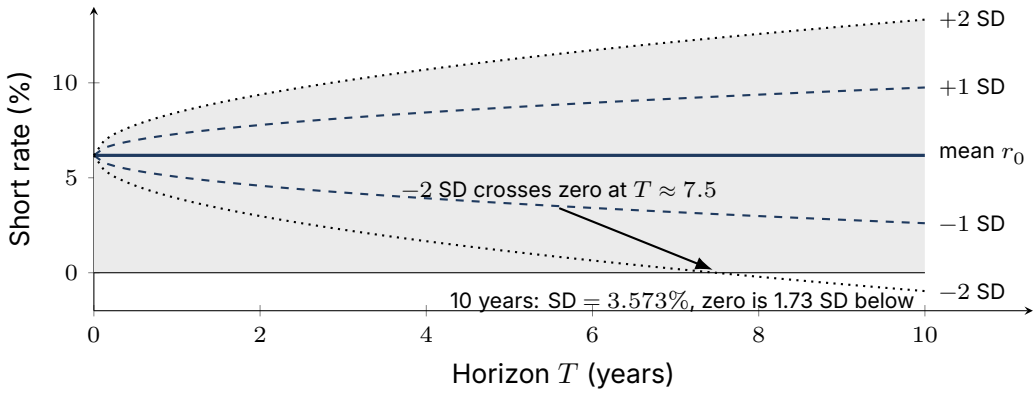


Figure 2. Model 1 with GARP's inputs ($r_0 = 6.18\%$, $\sigma = 1.13\%$). The SD of the terminal rate grows with $\sqrt{T}$ and nothing pulls the rate back, so the chance of a negative short rate grows with the horizon.

Formula card: MR-14 b

Rate change	$dr = \sigma\,dw$; new rate $= r_0 + \sigma\,dw$
From a standard normal Z	$dw = Z\sqrt{dt}$, $dr = \sigma Z\sqrt{dt}$
SD of the rate change	$\sigma\sqrt{dt}$ (monthly: $\sigma \times 0.2887$; quarterly: $\sigma \times 0.5$)
SD after n steps / T years	$\sigma\sqrt{n\,dt}$ / $\sigma\sqrt{T}$
Tree nodes	date 1: $r_0 \pm \sigma\sqrt{dt}$; date 2: $r_0 \pm 2\sigma\sqrt{dt}, r_0$
Moments of the tree	$E[dr] = 0$, $V[dr] = \sigma^2 dt$

Exam traps

- **Double scaling.** Multiplying a realised dw by $\sqrt{dt}$ again (the change shrinks by 0.2887).
- **Missing scaling.** Treating a standard normal Z as if it were dw (the change is 3.5 times too large for monthly steps).
- **SD of dw used as the draw.** $\sigma \times 0.2887$ is the SD of the change, not the realised change.
- **dt for $\sqrt{dt}$.** $\sigma/12$ instead of $\sigma/\sqrt{12}$.
- **Units.** 75 bp is 0.75%, not 7.5%. Keep the sign of a negative dw.
- **Linear SD.** SDs over several steps add in *variance*: $\sigma\sqrt{2dt}$, not $2\sigma\sqrt{dt}$ (that is the outer node).
- **Labels.** "Change in the rate" versus "new rate": GARP swaps them in the options.

Practice questions: MR-14 b

Question MR-14 b.1

Bank Q660 • AP Mock 1 Q13

Using a term structure model with no drift and normally distributed rates, assume that the short-term interest rate is 5%, annual volatility is 75 basis points, and after one month the realisation of dw (a normally distributed random variable with mean 0 and standard deviation $\sqrt{dt}$) is 0.2. What is the new spot rate?

A. 5.15%

B. 4.85%

C. 3.50%

D. 6.50%

Question MR-14 b.2

Bank Q662 • AnalystPrep Q.2854 (rebuilt)

The current short-term rate is 7.31%. Volatility is 167 basis points per year and the time interval is one month ($dt = 1/12$). Under the normally distributed, no-drift model (Model 1), after one month the random variable dw (mean zero, standard deviation 0.2887) takes the value 0.21. What is the change in the short-term rate?

A. 10 basis points

B. 35 basis points

C. 48 basis points

D. 3 basis points

Question MR-14 b.3

Bank Q661 • AnalystPrep Q.1651

An FRM candidate is analyzing the movement in short-term interest rates assuming the rates are normally distributed with no drift. The current short-term rate is 7.18%, the time interval is 3 months (3/12 of a year), and volatility is 118 basis points per year. After three months, the random variable dw takes the value 0.18. What are the change in the short-term rate and the short-term rate after 3 months?

A. Change 0.2124%; rate after 3 months 7.3924%

B. Change 7.3924%; rate after 3 months 0.2124%

C. Change 1.2924%; rate after 3 months 8.4724%

D. Change 8.4724%; rate after 3 months 1.2924%

Question MR-14 b.4 Bank Q665 • Schweser Module Quiz 14.1 Q1 (rebuilt)

Using Model 1, assume the current short-term rate is 5%, annual volatility is 80 basis points, and dw is a normally distributed random variable with mean 0 and standard deviation $\sqrt{dt}$. After one month the realisation of dw is -0.5. What are the change in the spot rate and the new spot rate?

A. Change +40 bp; new rate 5.40%

B. Change -40 bp; new rate 4.60%

C. Change -11.5 bp; new rate 4.885%

D. Change -50 bp; new rate 4.50%

Question MR-14 b.5 Bank Q666 • BT P2.T5.303.1 (rebuilt)

Analyst Barry simulates short-term rates with Tuckman's Model 1 (no drift): $dr = \sigma\, dw$. The time step is one month ($dt = 1/12$), the initial rate is 4.00%, and annual basis-point volatility is 200 bp. In the first step the random uniform draw is 0.8925, which maps via the inverse normal to a standard normal value $Z = 1.240$ (so $dw = Z\sqrt{dt}$). To what level does the rate evolve after the first month?

A. 3.284%

B. 4.207%

C. 4.716%

D. 6.480%

Question MR-14 b.6 New • built from GARP Ch.14, 14.1 (eqs. 14.3, 14.4)

A risk analyst at a Malaysian bank uses Model 1 ($dr = \sigma\, dw$) to describe the MYR short rate for the ALCO pack. The current short rate is 4.50% and the annual basis-point volatility is 150 bp. The pack needs two figures: the standard deviation of the change in the short rate over the next month, and the standard deviation of the short rate five years from now. Which pair is closest?

A. 0.125%; 3.354%

B. 0.433%; 3.354%

C. 0.433%; 7.500%

D. 1.500%; 3.354%

Question MR-14 b.7 New • built from GARP Ch.14, 14.1 (tree moments)

A quant builds a Model 1 short-rate tree with quarterly steps ($dt = 0.25$), starting from $r_0 = 3.00\%$ with annual basis-point volatility of 90 bp and probability $\frac{1}{2}$ on every branch. Which of the following is closest to the standard deviation of the short rate after two steps, and to the distance between the highest and the lowest node on date 2?

A. Standard deviation 0.636%; distance 0.900%

B. Standard deviation 0.636%; distance 1.800%

C. Standard deviation 0.900%; distance 1.800%

D. Standard deviation 0.450%; distance 0.900%

Solutions • MR-14 b

Solution MR-14 b.1

Answer: A

Bank Q660 • AP Mock 1 Q13

Step 1 – Rate change

$$dr = \sigma\, dw = 0.75\% \times 0.2 = 0.15\% = 15\text{ bp}$$

$\Rightarrow dw = 0.2$ is a realisation of dw; its $\sqrt{dt}$ scaling is already inside it.

Step 2 – New rate

$$r_{1/12} = 5.00\% + 0.15\% = 5.15\%$$

Interpretation

No drift, so the whole move is volatility times the realised shock, added with its sign.

Why the other options are wrong

B is incorrect. Sign error: subtracted the change although dw is positive: 4.85%.
C is incorrect. Read 75 bp as 7.5%: $dr = 1.5\%$, and subtracted it: 3.50%.
D is incorrect. Read 75 bp as 7.5% and added it: 6.50%.

LO: MR-14 b: Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1, eqs. 14.1 and 14.2

Solution MR-14 b.2

Answer: B

Bank Q662 • AnalystPrep Q.2854

Step 1 – Identify the draw

The stem gives the realised value of dw (0.21). The 0.2887 is the standard deviation of dw, not a multiplier.

Step 2 – Rate change

$$dr = \sigma\, dw = 167\text{ bp} \times 0.21 = 35.1\text{ bp} \approx 35\text{ bp}$$

$\Rightarrow$ The new rate would be $7.31\% + 0.35\% = 7.66\%$ (not asked).

Interpretation

Every wrong option uses the 0.2887 or the dt somewhere it does not belong. The current rate 7.31% is not needed for the change.

Why the other options are wrong

A is incorrect. Multiplied by the SD of dw as well: $167 \times 0.2887 \times 0.21 = 10.1$ bp.
C is incorrect. Used the SD of dw in place of the draw: $167 \times 0.2887 = 48.2$ bp (this is the SD of the monthly change).
D is incorrect. Scaled by dt: $167 \times 0.21/12 = 2.9$ bp.

LO: MR-14 b: Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1, eq. 14.2

Solution MR-14 b.3

Answer: A

Bank Q661 • AnalystPrep Q.1651

Step 1 – Rate change

$$dr = \sigma\, dw = 1.18\% \times 0.18 = 0.2124\%$$

Step 2 – Rate after three months

$$r = 7.18\% + 0.2124\% = 7.3924\%$$

$\Rightarrow$ The 3-month interval is already reflected in the realised dw; it does not enter separately.

Interpretation

The options test reading discipline: the right numbers appear twice, once with the labels swapped.

Why the other options are wrong

B is incorrect. Correct numbers with the labels swapped.
C is incorrect. Multiplied the current rate by dw instead of the volatility: $7.18\% \times 0.18 = 1.2924\%$, new rate 8.4724%.
D is incorrect. The C error with the labels swapped.

LO: MR-14 b: Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1, eq. 14.2

Solution MR-14 b.4

Answer: B

Bank Q665 • Schweser Module Quiz 14.1 Q1

Step 1 – Rate change, keeping the sign

$$dr = \sigma\, dw = 0.80\% \times (-0.5) = -0.40\% = -40 \text{ bp}$$

Step 2 – New rate

$$r = 5.00\% - 0.40\% = 4.60\%$$

Interpretation

A negative draw moves the rate down; with no drift there is nothing else.

Why the other options are wrong

A is incorrect. Dropped the negative sign of dw: $+40$ bp, 5.40%.
C is incorrect. Applied $\sqrt{dt}$ again: $0.80\% \times 0.5 \times 0.2887 = 11.5$ bp, 4.885%.
D is incorrect. Ignored the volatility and treated $dw = -0.5$ as -0.5%: 4.50%.

LO: MR-14 b: Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1; SchweserNotes 2026 Book 1, Reading 14

Solution MR-14 b.5

Answer: C

Bank Q666 • BT P2.T5.303.1

Step 1 – Turn the standard normal draw into dw

$$dw = Z\sqrt{dt} = 1.240 \times \sqrt{1/12} = 1.240 \times 0.28868 = 0.3580$$

$\Rightarrow$ Here the stem gives Z, not dw, so the $\sqrt{dt}$ scaling is needed.

Step 2 – Rate change and new rate

$$dr = 2.00\% \times 0.3580 = 0.716\%, \qquad r_{1/12} = 4.00\% + 0.716\% = 4.716\%$$

Interpretation

Compare MR-14 b.2: when the stem gives a realised dw you use it directly; when it gives a standard normal Z (a simulation), you scale by $\sqrt{dt}$ first.

Why the other options are wrong

A is incorrect. Sign error: subtracted the change: 3.284%.
B is incorrect. Scaled by dt instead of $\sqrt{dt}$: $2\% \times 1.24/12 = 0.207\%$, 4.207%.
D is incorrect. Omitted the $\sqrt{dt}$ scaling: $2\% \times 1.24 = 2.48\%$, 6.480%.

LO: MR-14 b: Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1, eq. 14.1

Solution MR-14 b.6

Answer: B

New • GARP Ch.14, 14.1

Step 1 – SD of the one-month change

$$\sigma\sqrt{dt} = 1.50\% \times \sqrt{1/12} = 1.50\% \times 0.28868 = 0.433\%$$

$\Rightarrow$ About 43 bp a month, the same calculation as GARP's 32.6 bp for $\sigma = 1.13\%$.

Step 2 – SD of the rate in five years

$$\sigma\sqrt{T} = 1.50\% \times \sqrt{5} = 3.354\%$$

$\Rightarrow$ Monthly variances add over 60 steps: $60 \times \sigma^2/12 = 5\sigma^2$.

Interpretation

The current rate is not needed: Model 1's dispersion depends only on σ and the horizon. With a 4.50% rate and a 3.354% SD, a shock of only 1.34 SD would take the rate below zero after five years.

Why the other options are wrong

A is incorrect. Scaled the monthly figure by dt instead of $\sqrt{dt}$: $1.50\%/12 = 0.125\%$.
C is incorrect. Scaled the five-year SD linearly with time: $1.50\% \times 5 = 7.50\%$.
D is incorrect. Reported the annual volatility as the monthly SD: 1.500%.

LO: MR-14 b: Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1, eq. 14.4 and the 10-year terminal SD example

Solution MR-14 b.7

Answer: B

New • GARP Ch.14, 14.1

Step 1 – One step

$$\sigma\sqrt{dt} = 0.90\% \times \sqrt{0.25} = 0.45\%$$

Step 2 – Standard deviation after two steps: variances add

$$\text{SD} = \sigma\sqrt{2dt} = 0.90\% \times \sqrt{0.5} = 0.636\%$$

$\Rightarrow$ Check on the tree: nodes $\pm 0.90\%$ with probability $\frac{1}{4}$ each and 0 with $\frac{1}{2}$ give variance $\frac{1}{2}(0.90\%)^2$, SD 0.636%.

Step 3 – Highest minus lowest node

$$(r_0 + 2\sigma\sqrt{dt}) - (r_0 - 2\sigma\sqrt{dt}) = 4\sigma\sqrt{dt} = 1.80\%$$

Interpretation

The outer nodes sit $2\sigma\sqrt{dt}$ from the centre, but the standard deviation grows only with $\sqrt{n}$ because half of the two-step paths (up-down, down-up) cancel. r_0 is not needed.

Why the other options are wrong

A is incorrect. Correct SD, but measured the distance on date 1 ($2\sigma\sqrt{dt} = 0.90\%$).

C is incorrect. Added the step SDs linearly ($2 \times 0.45\% = 0.90\%$) and called it the standard deviation.

D is incorrect. Gave the one-step SD (0.45%) and the date-1 distance.

LO: MR-14 b: Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.1, eqs. 14.3 and 14.4

MR-14 • The Art of Term Structure Models: Drift • Vardeez: VERY IMPORTANT

MR-14 d

P1 CALC

Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift.

Workbook: A7 Ho-Lee, Time-Dependent Drift and Volatility • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A7 and a confirmed August 2026 recall: "Ho-Lee numerical, plug values".
Vardeez: Ho-Lee model (chapter note: "calculations are super important here").

Reported exam shapes

- **A7 S1** Interest rate after 2 months in the top node of a monthly Ho-Lee tree.
- **A7 S2** Ho-Lee and time-dependent volatility models: rate in the upper nodes.
- **A7 S3** Ho-Lee and Model 3 two-step plug and play (Model 3 is MR-15 a, b).
- **A7 S4** No-arbitrage model, node-2 Δr and the drift read back from the nodes.
- **A7 S5** Two questions on the Ho-Lee short-rate tree (incl. a reverse-solve).
- **A7 S7** August plain form: Ho-Lee plug values.

What the questions look like: numbers in, node out. "Rate in the top (or lowest) node after two months", "change from today to node 2", or backwards: "which λ_1 (or σ) produced these nodes". Distractors are built from dt versus $\sqrt{dt}$, $2\lambda_1$ instead of $\lambda_1 + \lambda_2$, one $\sigma\sqrt{dt}$ instead of two, and the Model 1 node with no drift.

1. The Ho-Lee process

$$\boxed{dr = \lambda_t\, dt + \sigma\, dw}$$

Model 2 with one change: the drift may be different in every period. GARP's illustration: an annualised drift of -20 bp over the first month, $+20$ bp over the second month, and so on. Each λ_t is still a mix of risk premium and expected change in the short rate, and it may be negative. Volatility σ stays constant, so the tree still recombines.

2. Building the tree

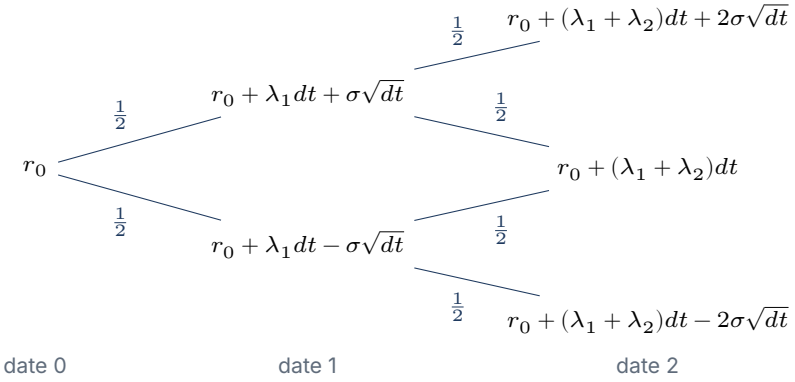


Figure 3. The Ho-Lee tree. Every node on date n carries the *cumulative* drift $(\lambda_1 + \cdots + \lambda_n)dt$; the volatility part is $\{+1, -1\}\sigma\sqrt{dt}$ on date 1 and $\{+2, 0, -2\}\sigma\sqrt{dt}$ on date 2. If $\lambda_1 = \lambda_2$ it is the Model 2 tree; if both are zero it is Model 1.

Worked example. Monthly tree, $r_0 = 2.00\%$, $\sigma = 1.20\%$, $\lambda_1 = 0.50\%$, $\lambda_2 = 1.00\%$.

- Per-step pieces: $\lambda_1 dt = 0.50\%/12 = 0.0417\%$; $\sigma\sqrt{dt} = 1.20\% \times 0.28868 = 0.3464\%$.
- Date 1: $2.00\% + 0.0417\% \pm 0.3464\% = 2.388\%$ and 1.695%.
- Date 2 centre: $r_0 + (\lambda_1 + \lambda_2)dt = 2.00\% + 1.50\%/12 = 2.125\%$.
- Date 2 outer nodes: $2.125\% \pm 2 \times 0.3464\% = 2.818\%$ and 1.432%.
- Node-2 Δr to the top node: $(\lambda_1 + \lambda_2)dt + 2\sigma\sqrt{dt} = 0.125\% + 0.693\% = +81.8$ bp.

A simulated path (rather than a tree node) uses standard normal draws $Z_1, Z_2, \ldots$:

$$r_{n\,dt} = r_0 + (\lambda_1 + \cdots + \lambda_n)\,dt + \sigma\sqrt{dt}\,(Z_1 + \cdots + Z_n).$$

3. Calibration: why Ho-Lee is arbitrage-free

The free parameters $\lambda_1, \lambda_2, \ldots$ are chosen one at a time to match the prices of securities with fixed cash flows:

1. With $dt = 1/12$, set r_0 equal to the one-month rate.
2. Find λ_1 so that the tree produces the market two-month spot rate.
3. Find λ_2 so that it produces the market three-month spot rate; continue until the tree ends.

The rate curves then **match all the rates input into the model**: Ho-Lee is arbitrage-free (MR-14 e).

What the time-dependent drift does *not* change. Just as adding a constant drift to Model 1 did not change the shape of the term structure of volatility or the parallel-shift nature, a time-dependent drift does not either. Ho-Lee is still a normal model (negative rates possible), with a flat volatility structure and parallel shifts. Calibration buys fit, not structure.

4. Reading the tree backwards (shape S4)

Because the $\pm\sigma\sqrt{dt}$ shocks cancel when you average two branches, the nodes reveal the parameters:

$$\lambda_1 = \frac{\frac{1}{2}(r_u + r_d) - r_0}{dt}, \qquad \sigma = \frac{r_u - r_d}{2\sqrt{dt}}, \qquad (\lambda_1 + \lambda_2)\,dt = r_{\text{mid},2} - r_0.$$

Example: $r_0 = 2.50\%$, $r_u = 2.8887\%$, $r_d = 2.3113\%$ (monthly): average 2.6000%, so $\lambda_1 dt = 0.10\%$ and $\lambda_1 = 1.20\%$; $\sigma = 0.5774\%/(2 \times 0.28868) = 1.00\%$. The monthly shift $\lambda_1 dt$ is an intermediate result; the question asks for the annualised λ_1.

Formula card: MR-14 d

Process	$dr = \lambda_t\,dt + \sigma\,dw$
Date-1 nodes	$r_0 + \lambda_1 dt \pm \sigma\sqrt{dt}$
Date-2 nodes	$r_0 + (\lambda_1 + \lambda_2)dt + \{2, 0, -2\}\,\sigma\sqrt{dt}$
Top node Δr (2 steps)	$(\lambda_1 + \lambda_2)dt + 2\sigma\sqrt{dt}$; date-2 range $= 4\sigma\sqrt{dt}$
Simulated path	$r_0 + \sum \lambda_i\,dt + \sigma\sqrt{dt}\sum Z_i$
Reverse-solve	$\lambda_1 = [\frac{1}{2}(r_u + r_d) - r_0]/dt$; $\sigma = (r_u - r_d)/(2\sqrt{dt})$
Special cases	$\lambda_1 = \lambda_2 = \lambda$: Model 2; all $\lambda_t = 0$: Model 1

Exam traps

- $\sqrt{dt}$ **versus** dt**.** Drift scales with dt, the shock with $\sqrt{dt}$. σdt shrinks the branches; $\lambda\sqrt{dt}$ inflates the drift.
- **Wrong drift sum at date 2.** The date-2 nodes carry $(\lambda_1 + \lambda_2)dt$, not $2\lambda_1 dt$ and not $\lambda_2 dt$ alone. Forgetting the drift altogether is the Model 1 node.
- **Drift on down moves.** The drift is *added* on every branch; subtracting it on the way down is a planted option.
- **Outer nodes.** Date-2 top and bottom sit $2\sigma\sqrt{dt}$ from the centre (range $4\sigma\sqrt{dt}$); a one-σ top node or a $2\sigma\sqrt{dt}$ range is the date-1 picture.
- λ_2 **is idle on date 1.** A date-1 node never uses λ_2.
- **Intermediate results.** In reverse-solves, $\lambda_1 dt$ (monthly) and σ are intermediates; annualise λ.
- **Fit is not structure.** Calibrating λ_t does not create mean reversion, a humped volatility structure or a floor at zero.

Practice questions: MR-14 d

Question MR-14 d.1

Bank Q670 • AnalystPrep Q.1656 (rebuilt)

In the Ho-Lee model, the drift of the short-term rate is allowed to differ from period to period (time-dependent drift). What does the drift in a given period represent?

A. A combination of the risk premium and the expected change in the short-term rate over that period

B. Only the expected change in the short-term rate over that period

C. The expected change in the volatility of the short-term rate over that period

D. Only the risk premium demanded by investors over that period

Question MR-14 d.2

Workbook A7.1 • S2, S5, S7 • Aug-2026 • Map #7

A junior risk analyst at an Australian bank is building a monthly Ho-Lee short-rate tree for AUD swaps. The current one-month rate is 3.20%, annual basis-point volatility is 120 basis points, the calibrated drift for the first month is $\lambda_1 = 0.60\%$ and for the second month is $\lambda_2 = 0.90\%$. Which of the following is closest to the short rate in the up state at the end of the first month?

A. 2.904%

B. 3.350%

C. 3.546%

D. 3.596%

Question MR-14 d.3

Bank Q671 • GARP PE1 Q48 • Mock 3 Q32

A risk analyst constructs a binomial interest rate tree by using the Ho-Lee model. The time step is monthly and the annualized drift is 80 bps in the first month and 120 bps in the second month. Assuming the current annualized short-term rate is 3.2% and the annual basis-point volatility is 2.1%, what is the interest rate at the lowest node after 2 months?

A. 1.82%

B. 2.15%

C. 2.76%

D. 3.03%

Question MR-14 d.4 Workbook A7.2 • S1, S2, S3, S4, S5 • Map #7

A rates quant at a Canadian pension fund is extending a monthly Ho-Lee tree for CAD to the end of the second month, to check how wide the range of short rates becomes. Inputs:

- Current one-month rate r_0: 1.50%; annual basis-point volatility σ: 160 basis points
- Calibrated drift: $\lambda_1 = 0.60\%$ for month 1 and $\lambda_2 = 1.20\%$ for month 2 (annualised)

Which of the following is closest to the short rate in the top node after two months, and to the gap between the top and bottom nodes on that date?

A. Top node 2.112%; gap 0.924%

B. Top node 2.524%; gap 1.848%

C. Top node 2.574%; gap 0.924%

D. Top node 2.574%; gap 1.848%

Question MR-14 d.5 Bank Q673 • BT P2.T5.304.2 (rebuilt, key corrected)

An analyst uses the Ho-Lee model, $dr = \lambda(t)dt + \sigma dw$, with monthly steps ($dt = 1/12$), current rate $r(0) = 4.000\%$, annual basis-point volatility of 300 bps, and annualized drifts of $\lambda(1) = 110$ bps in month one and $\lambda(2) = 70$ bps in month two. In the simulation, the standard normal draw is -0.52 in month one and $+0.84$ in month two. What is the simulated short rate after two months, $r(2/12)$?

A. 4.277%

B. 4.877%

C. 4.427%

D. 5.110%

Question MR-14 d.6 Workbook A7.3 • S4, S5 • Map #7

A model validator at a Singapore bank is reviewing a monthly Ho-Lee tree for SGD. The documentation is missing λ_1, but the tree shows $r_0 = 2.50\%$ and date-1 nodes of 2.8887% (up) and 2.3113% (down). The model's two-year par rate is 2.95%. Which of the following is closest to the annualised drift λ_1?

A. 0.10%

B. 1.00%

C. 1.20%

D. 4.66%

Question MR-14 d.7 New • built from GARP Ch.14, 14.3 and Workbook A7 S4

A rates analyst is checking a quarterly Ho-Lee tree ($dt = 0.25$) built by a colleague. The tree starts from $r_0 = 2.00\%$ with annual basis-point volatility of 100 bp. The calibrated drift for the first quarter is $\lambda_1 = 0.40\%$, and the middle node on date 2 shows 2.30%. Which of the following is closest to the annualised drift for the second quarter, λ_2, and to the rate in the top node on date 2?

A. $\lambda_2 = 0.20\%$; top node 3.30%

B. $\lambda_2 = 0.80\%$; top node 2.80%

C. $\lambda_2 = 0.80\%$; top node 3.30%

D. $\lambda_2 = 1.20\%$; top node 3.30%

Solutions • MR-14 d

Solution MR-14 d.1

Answer: A

Bank Q670 • AnalystPrep Q.1656

Step 1 – What the risk-neutral drift contains

As in Model 2, the drift of a risk-neutral process is a combination of the true expected change in the short rate and a risk premium. Ho-Lee only lets that combination differ from period to period.

Step 2 – Why it cannot be split

The λ_t are calibrated to market prices, and prices do not depend on how the drift divides between expectation and premium.

$\Rightarrow$ A drift of -20 bp in month 1 and $+20$ bp in month 2 is one number per period, of mixed origin.

Interpretation

This is why a fitted drift that jumps around from one year to the next is suspicious (MR-14 e): it implies implausible changes in expectations or in the risk premium.

Why the other options are wrong

B is incorrect. Ignores the risk premium: the drift is risk-neutral.

C is incorrect. Confuses drift with volatility; σ is a separate, constant parameter in Ho-Lee.

D is incorrect. Ignores the expected-change component.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.3

Solution MR-14 d.2

Answer: D

Workbook A7.1 • S2, S5, S7 • Aug-2026

Step 1 – Time step

$$dt = \frac{1}{12} = 0.08333, \qquad \sqrt{dt} = 0.28868$$

$\Rightarrow$ One monthly step, measured in years.

Step 2 – Drift over the first step

$$\lambda_1 dt = 0.60\% \times \frac{1}{12} = 0.050\%$$

$\Rightarrow$ The risk-neutral drift lifts the rate by 5 basis points over the month in both states.

Step 3 – Volatility perturbation

$$\sigma\sqrt{dt} = 1.20\% \times 0.28868 = 0.3464\%$$

$\Rightarrow$ One standard deviation of the monthly rate change is about 34.6 basis points.

Step 4 – Up-state node

$$r_u = r_0 + \lambda_1 dt + \sigma\sqrt{dt} = 3.20\% + 0.050\% + 0.3464\% = 3.596\%$$

$\Rightarrow$ If the shock is up, the one-month rate one month from now is about 3.60%.

Interpretation

λ_2 is irrelevant until date 2. The node is the date-1 building block from which λ_2 is later chosen so the model reprices the market three-month spot rate.

Why the other options are wrong

A is incorrect. Down-state node: $3.20\% + 0.050\% - 0.3464\% = 2.904\%$.
B is incorrect. dt instead of $\sqrt{dt}$ on volatility: $3.20\% + 0.050\% + 1.20\%/12 = 3.350\%$.
C is incorrect. Dropped the drift (used Model 1): $3.20\% + 0.3464\% = 3.546\%$.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3, eq. 14.7

Solution MR-14 d.3

Answer: B

Bank Q671 • GARP PE1 Q48

Step 1 – Cumulative drift to date 2

$$(\lambda_1 + \lambda_2)\, dt = \frac{0.80\% + 1.20\%}{12} = 0.1667\%$$

Step 2 – Volatility step

$$\sigma\sqrt{dt} = 2.10\% \times 0.28868 = 0.6062\%$$

Step 3 – Lowest node: two down moves

$$r_{dd} = 3.20\% + 0.1667\% - 2 \times 0.6062\% = 2.154\% \approx 2.15\%$$

$\Rightarrow$ The drift is added even on the way down; only the shock changes sign.

Interpretation

The centre node on date 2 is 3.367% and the range is $4\sigma\sqrt{dt} = 2.42\%$. Every distractor is one sign or one step away from the right path.

Why the other options are wrong

A is incorrect. Subtracted the drift: $3.20\% - 0.1667\% - 1.2124\% = 1.82\%$.
C is incorrect. Only one down move: $3.20\% + 0.1667\% - 0.6062\% = 2.76\%$.
D is incorrect. Subtracted the drift and applied no shock: $3.20\% - 0.1667\% = 3.03\%$.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.3, eq. 14.7 and tree

Solution MR-14 d.4

Answer: D

Workbook A7.2 • S1, S2, S3, S4, S5

Step 1 – Cumulative drift to date 2

$$(\lambda_1 + \lambda_2)\, dt = (0.60\% + 1.20\%) \times \frac{1}{12} = 0.150\%$$

$\Rightarrow$ The two monthly drifts together shift every date-2 node up by 15 basis points.

Step 2 – Volatility perturbation per step

$$\sigma\sqrt{dt} = 1.60\% \times 0.28868 = 0.4619\%$$

$\Rightarrow$ Each monthly up or down move is about 46 basis points.

Step 3 – Top node (two up moves) and node-2 Δr

$$r_{uu} = r_0 + (\lambda_1 + \lambda_2)dt + 2\sigma\sqrt{dt} = 1.50\% + 0.150\% + 0.9238\% = 2.574\%, \qquad \Delta r = +107.4 \text{ bp}$$

$\Rightarrow$ After two up shocks the one-month rate is about 2.57%.

Step 4 – Bottom node and the gap

$$r_{dd} = 1.650\% - 0.9238\% = 0.726\%, \qquad r_{uu} - r_{dd} = 4\sigma\sqrt{dt} = 1.848\%$$

$\Rightarrow$ The date-2 nodes are 2.574%, 1.650% and 0.726%. The drift cancels in the gap; only volatility sets the width.

Interpretation

The range still stays above zero at two months, but in a normal model the probability of negative rates rises with the horizon (MR-14 c): the width grows with $\sqrt{t}$ while the drift grows with t.

Why the other options are wrong

A is incorrect. Placed the date-2 top node only one $\sigma\sqrt{dt}$ above the centre (the date-1 picture): $1.650\% + 0.4619\% = 2.112\%$, gap $2\sigma\sqrt{dt} = 0.924\%$.

B is incorrect. Used λ_1 for both months ($2\lambda_1 dt = 0.10\%$) instead of $(\lambda_1 + \lambda_2)dt$: $1.50\% + 0.10\% + 0.9238\% = 2.524\%$.

C is incorrect. Correct top node, but measured the gap as $2\sigma\sqrt{dt} = 0.924\%$, the spread between the two date-1 nodes, not the date-2 range.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3

Solution MR-14 d.5 **Answer: C**

Bank Q673 • BT P2.T5.304.2

Step 1 – Cumulative drift

$$(\lambda_1 + \lambda_2)\, dt = \frac{1.10\% + 0.70\%}{12} = 0.150\%$$

Step 2 – Cumulative shock

$$\sigma\sqrt{dt}\,(Z_1 + Z_2) = 3.00\% \times 0.28868 \times (-0.52 + 0.84) = 0.8660\% \times 0.32 = 0.2771\%$$

$\Rightarrow$ Each standard normal draw is scaled by $\sigma\sqrt{dt}$; the two monthly shocks add.

Step 3 – Simulated rate

$$r(2/12) = 4.000\% + 0.150\% + 0.277\% = 4.427\%$$

Interpretation

A simulated path is not a tree node: the draws are arbitrary normals, not ± 1. The structure is the same as the tree, cumulative drift times dt plus $\sigma\sqrt{dt}$ times the sum of shocks. (The bank's original key was corrected to C.)

Why the other options are wrong

A is incorrect. Left out the drift: $4.000\% + 0.277\% = 4.277\%$.

B is incorrect. Used only the second month's shock: $4.000\% + 0.150\% + 0.84 \times 0.8660\% = 4.877\%$.

D is incorrect. Forgot the $\sqrt{dt}$ on the shocks: $4.000\% + 0.150\% + 0.32 \times 3.00\% = 5.110\%$.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.3, eq. 14.7

Solution MR-14 d.6

Answer: C

Workbook A7.3 • S4, S5

Step 1 – Average of the date-1 nodes

$$\tfrac{1}{2}(r_u + r_d) = \tfrac{1}{2}(2.8887\% + 2.3113\%) = 2.6000\%$$

$\Rightarrow$ The $\pm\sigma\sqrt{dt}$ shocks cancel in the average, leaving $r_0 + \lambda_1 dt$.

Step 2 – Drift over the step

$$\lambda_1 dt = 2.6000\% - 2.5000\% = 0.1000\%$$

$\Rightarrow$ The month's drift raises the expected short rate by 10 basis points.

Step 3 – Annualise

$$\lambda_1 = \frac{0.1000\%}{1/12} = 1.20\%$$

$\Rightarrow$ λ_1 is quoted per year, like every parameter in $dr = \lambda(t)dt + \sigma dw$.

Check – Implied volatility (not asked)

$$\sigma = \frac{r_u - r_d}{2\sqrt{dt}} = \frac{0.5774\%}{2 \times 0.28868} = 1.00\%$$

Interpretation

A 1.20% annualised drift for month 1 is what makes the model reprice the market two-month spot rate. The two-year par rate is irrelevant to this step.

Why the other options are wrong

A is incorrect. Intermediate result: the monthly shift $\lambda_1 dt = 0.10\%$, not annualised.
B is incorrect. Sibling quantity: computed $\sigma = 1.00\%$ from the node spread instead of the drift.
D is incorrect. Used the up node alone, treating its volatility shock as drift: $(2.8887\% - 2.50\%) \times 12 = 4.66\%$.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3

Solution MR-14 d.7

Answer: C

New • GARP Ch.14, 14.3

Step 1 – Read the cumulative drift off the middle node

$$(\lambda_1 + \lambda_2)\, dt = r_{\text{mid},2} - r_0 = 2.30\% - 2.00\% = 0.30\% \quad \Rightarrow \quad \lambda_1 + \lambda_2 = \frac{0.30\%}{0.25} = 1.20\%$$

$\Rightarrow$ The shocks cancel on the middle node, so it carries pure drift.

Step 2 – Second-quarter drift

$$\lambda_2 = 1.20\% - 0.40\% = 0.80\%$$

Step 3 – Top node on date 2

$$\sigma\sqrt{dt} = 1.00\% \times \sqrt{0.25} = 0.50\%, \qquad r_{uu} = 2.30\% + 2 \times 0.50\% = 3.30\%$$

Interpretation

This is the calibration logic run backwards: the middle node pins down the drift, the node spread pins down the volatility. Here $\lambda_2 > \lambda_1$, which Ho-Lee allows and Model 2 would not.

Why the other options are wrong

A is incorrect. Did not annualise the second-quarter shift: $0.30\% - 0.40\% \times 0.25 = 0.20\%$ is $\lambda_2 dt$, not λ_2.

B is incorrect. Correct λ_2, but the top node placed one $\sigma\sqrt{dt}$ above the centre: 2.80% (the date-1 spacing).

D is incorrect. Forgot to subtract λ_1: reported the cumulative $\lambda_1 + \lambda_2 = 1.20\%$ as λ_2.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.3 (tree and calibration procedure)

MR-14 • The Art of Term Structure Models: Drift • Vardeez: VERY IMPORTANT

MR-14 e

P1 CONCEPT

Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices.

Workbook: A7 Ho-Lee, Time-Dependent Drift and Volatility • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A7. August 2026 reported "no-arbitrage model" items and a "time-dependent drift and vol, theory question".
Vardeez (chapter): Ho-Lee and the arbitrage-free idea are part of his "very important" MR-14 list.

Reported exam shape for this LO

- **A7 S4** No-arbitrage model, node-2 Δr and the drift read back from the nodes. The numerical side is drilled in MR-14 d (Questions d.4, d.6, d.7); this block covers the theory around it.

What the questions look like: "which statement is correct" or roman-numeral sets on four ideas: what makes a model arbitrage-free, what arbitrage-free models are used for, why fitting to prices can mislead, and why relative value needs an equilibrium model.

1. Two ways to build a term structure model

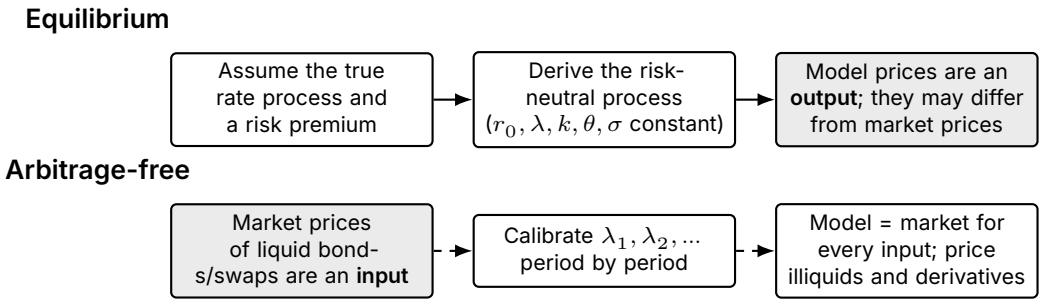


Figure 4. Direction of the logic. Equilibrium models (Models 1 and 2, Vasicek) start from assumptions and produce prices. Arbitrage-free models (Ho-Lee) start from prices and back out the drift that reproduces them. Shaded boxes: where market prices sit in each approach.

	Equilibrium model	Arbitrage-free model
Examples	Model 1, Model 2, Vasicek (and CIR)	Ho-Lee (time-dependent drift); Model 3 for options
Parameters	A few constants	One free drift per period (calibrated)
Fit to today's curve	Approximate, by design	Exact for every input price
Market prices are	Something to explain and compare with	Assumed fair, taken as given
Best used for	Relative value: rich/cheap analysis, choosing hedges, finding temporary mispricings	Quoting illiquid securities from liquid ones; valuing and hedging derivatives for market making or proprietary trading
Main risk	Misfits the observed curve	Inherits any distortion in the input prices; a bad model stays bad

2. Uses and benefits of arbitrage-free models

1. **Quoting securities that are not actively traded from more liquid ones.** A customer asks for a swap to 3 years and 4 months; only 3- and 4-year (or 2- and 5-year) swaps trade. The arbitrage-free model interpolates.
   - **Benefit over curve fitting:** the interpolation rests on economic reasoning (the expectations, risk premium and convexity built into neighbouring rates), not on an arbitrary functional form such as linear or quadratic interpolation.
   - **Caveat:** the advantage depends on the model's assumptions. A poor volatility assumption gives a poor convexity estimate and can make the model worse than a simple technique.
2. **Valuing and hedging derivatives** for market making or proprietary trading. Practitioners take the underlying as fairly priced (an option on a 10-year bond: assume the 10-year bond is fair; check that separately). Because the model matches the traded prices by construction, it is ideal for pricing derivatives *given* the underlyings.

3. The issue of fitting models to market prices

The case for fitting: market prices contain a lot of information about future rate behaviour, so a fitted model captures it. Two warnings:

1. **A mediocre or bad model cannot be rescued by calibration.** If parallel shifts are not good enough for the application, adding a fitted time-dependent drift to a parallel-shift model does not make it any more suitable.
2. **Fitting assumes the prices are fair in the context of the model.** Prices are often distorted by supply and demand imbalances, taxes, liquidity differences and other factors unrelated to rates. Fitting then attributes those effects to the rate process.

GARP's example. A large bank liquidates its 7-year bonds or swaps, depressing prices and raising rates around 7 years. It would be wrong to conclude that expected rates in seven years have risen. The 7-year securities are "fair" given the liquidity event, but not fair with respect to expected rates and the risk premium; investors might buy them cheap and hold through the event.

The drift test. A risk-neutral drift arises only from expectations and risk premium. A fitted drift that jumps between years 15–16 and 16–17 implies an implausible change in one or the other, so it is probably capturing non-interest-rate effects.

Relative value needs equilibrium models. An arbitrage-free model concludes, by construction, that all the bond or swap prices it was fitted to are fair relative to one another. Investors choosing among securities, market makers selecting hedges and traders looking for temporary mispricings must use equilibrium models.

In practice the line blurs. A model may use a deterministic drift for a few years (short-term forecasts) and a constant drift after; or take the 2-, 5-, 10- and 30-year rates as given (the most liquid points treated as fair) and value everything else. Blending the two approaches is part of the art of term structure modelling.

Exam traps

- **Direction swap.** "Start from the true process and a risk premium, derive prices" is *equilibrium*; "start from prices, back out the drift" is *arbitrage-free*.
- **Relative value.** Arbitrage-free models cannot say one fitted bond is rich or cheap to another: they assume both are fair.
- **Fit is not fairness.** Matching market prices does not imply fair values or accurate hedges (bad model; distorted prices).
- **Fit is not structure.** A fitted drift does not create mean reversion, a humped volatility structure or a zero floor.
- **Classification.** Models 1 and 2 and Vasicek are equilibrium models (constant parameters cannot match an arbitrary curve); Ho-Lee is arbitrage-free.
- **Forecasting.** A calibrated drift mixes expectation and premium: it is not a forecast of rates.

Practice questions: MR-14 e

Question MR-14 e.1

Bank Q674 • AnalystPrep Q.1636

You are asked to start with assumptions about the interest rate process and about the risk premium demanded by the market for bearing interest rate risk, and then derive the risk-neutral process. This approach results in:

A. An arbitrage-free model

B. An equilibrium model

C. The matching of the initial term structure

D. The pricing of all fixed-income securities by arbitrage

Question MR-14 e.2 Bank Q676 • AnalystPrep Q.1657 (rebuilt)

According to Tuckman, what is the most important use of arbitrage-free interest rate models?

A. Judging whether one security is rich or cheap relative to another

B. Forecasting the expected path of future short-term rates

C. Explaining the slope of the term structure through economic fundamentals

D. Quoting prices of illiquid or non-standard securities, and valuing derivatives, consistently with the prices of liquid securities

Question MR-14 e.3 Bank Q677 • AnalystPrep Q.1658 (rebuilt)

A portfolio manager wants to judge whether one bond is rich or cheap relative to another bond. According to Tuckman, which model choice is appropriate?

A. An equilibrium model, because arbitrage-free models assume both securities are fairly priced and so cannot reveal relative mispricing

B. An arbitrage-free model, because calibrating to both bond prices makes their values directly comparable

C. An arbitrage-free model, because time-dependent drift lets it fit more prices than an equilibrium model

D. Either type, because both give identical values for liquid securities

Question MR-14 e.4 Bank Q680 • Schweser Module Quiz 14.2 Q3 (rebuilt)

A risk analyst makes three statements about term structure models:

Statement 1: The Ho-Lee model is more general than Model 2, since constant drift (and no drift) are special cases of Ho-Lee.

Statement 2: Like Ho-Lee, the Vasicek model, with its constant parameters, can be calibrated to match any observed term structure.

Statement 3: Arbitrage-free models are well suited to relative value analysis of two securities, since both are matched to market prices.

How many of the statements are correct?

A. 0

B. 1

C. 2

D. 3

Question MR-14 e.5 Bank Q678 • BT P2.T5.304.3

Tuckman's short-term interest rate models classify into two categories: equilibrium models and arbitrage-free models. Consider the following statements about no-arbitrage interest rate models:

I. Model 1 (normally distributed rates and no drift) and Model 2 (constant drift and risk premium) are classified as arbitrage-free models.

II. The Ho-Lee Model, due to time-dependent drift, may be used to match the observed prices of securities and, therefore, may be classified as an arbitrage-free model.

III. Important uses of arbitrage-free models include quoting non-actively traded securities (based on prices of liquid securities) and valuing derivatives for purposes of market-making or proprietary trading.

IV. Because an arbitrage-free model matches market prices does not necessarily imply that it provides fair values and accurate hedges: the model itself could be bad, and/or market prices may not be "fair" in the context of the model.

Which of the above statements are TRUE, according to Tuckman?

A. None

B. I. and II. only

C. II., III., and IV.

D. All are true

Question MR-14 e.6 New • built from GARP Ch.14, 14.4 (seven-year liquidation example)

A swap dealer calibrates a Ho-Lee model to the EUR swap curve every morning. Overnight, a large bank liquidates its book of swaps with about seven years to maturity, pushing 7-year swap rates up by 12 bp while neighbouring maturities barely move. After recalibration, the fitted drift for years six to eight jumps. Which of the following is the most appropriate conclusion for the dealer's model risk manager?

A. The recalibrated drift correctly signals that the market now expects short rates around year seven to be higher, so hedges should be adjusted for that expectation.

B. Fitting to the distorted prices attributes a liquidity effect to the interest rate process; the 7-year swaps are fair given the liquidation, not fair with respect to expected rates and the risk premium.

C. Because Ho-Lee is arbitrage-free, its recalibrated prices are fair values and its hedges remain accurate by construction.

D. An equilibrium model would be forced to reproduce the higher 7-year rate as well, so the choice between model types does not matter here.

Question MR-14 e.7 Workbook A7.5 • S6, S7 • Aug-2026 • Map #7

A senior risk analyst at a UK insurer is reviewing a GBP rates desk that uses the Ho-Lee model to quote odd-maturity swaps and Model 3 (time-dependent volatility, $dr = \lambda(t)dt + \sigma e^{-\alpha t}dw$) to quote caplets. Which of the following statements is correct to include in the review?

A. Because its drift varies with time, the Ho-Lee model produces a humped term structure of volatility that matches market implied volatilities.

B. Because $\lambda(t)$ is calibrated to market spot rates, the Ho-Lee model rules out negative short-term rates.

C. The Ho-Lee drift can match all input spot rates, but the model remains a parallel-shift model with a flat term structure of volatility.

D. The exponential decay in Model 3's volatility introduces mean reversion, so Model 3 is no longer a parallel-shift model.

Solutions • MR-14 e

Solution MR-14 e.1 — Answer: B

Bank Q674 • AnalystPrep Q.1636

Step 1 – Identify the direction of the logic

The approach starts from the true rate process plus the market risk premium and derives the risk-neutral process; the term structure is an *output*.

Step 2 – Classify

That is the defining procedure of an equilibrium model (Models 1 and 2, Vasicek). An arbitrage-free model runs the other way: it takes prices as given and chooses the drift to reproduce them.
$\Rightarrow$ Equilibrium.

Interpretation

The question is the Figure 4 picture in words. Watch for the reverse wording, which describes an arbitrage-free model.

Why the other options are wrong

A is incorrect. Reverses the direction: arbitrage-free models start from observed prices.
C is incorrect. Matching the initial term structure is the property of arbitrage-free models, not of assumption-driven ones.
D is incorrect. Overstates: an assumption-driven model need not price every security consistently with the market.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.2 (Models 1 and 2 as equilibrium models), 14.4

Solution MR-14 e.2 — Answer: D

Bank Q676 • AnalystPrep Q.1657

Step 1 – What calibration buys

Arbitrage-free models are fitted so that model prices equal the prices of liquid securities (on-the-run bonds, standard swaps).

Step 2 – The two uses GARP names

(1) Quoting securities that are not actively traded (an odd-maturity swap) from the liquid ones; (2) valuing and hedging derivatives for market making or proprietary trading, taking the underlying as fairly priced.
$\Rightarrow$ Consistent pricing relative to the liquid market: D.

Interpretation

Each wrong option is a job for a different tool: relative value and economic explanation need equilibrium models; forecasting is impossible from a drift that mixes expectation and risk premium.

Why the other options are wrong

A is incorrect. Relative value is exactly what arbitrage-free models cannot do: they assume all fitted securities are fair.
B is incorrect. The calibrated drift mixes expected change and risk premium, so it is not a forecast.
C is incorrect. Explaining the curve from fundamentals is the domain of equilibrium models.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.4

Solution MR-14 e.3 — Answer: A

Bank Q677 • AnalystPrep Q.1658

Step 1 – What relative value needs

Judging one bond rich or cheap to another requires a model whose values can *disagree* with market prices.

Step 2 – Why arbitrage-free models fail here

They are calibrated to treat the input prices as correct, so the fitted securities show no mispricing by construction.

$\Rightarrow$ Use an equilibrium model.

Interpretation

GARP: "investors wanting to choose among securities, market makers looking to pick up value by strategically selecting hedging securities, or traders looking to profit from temporary mispricings must rely on equilibrium models".

Why the other options are wrong

B is incorrect. Calibration builds in the assumption that both prices are right, so the comparison reveals nothing.

C is incorrect. Fitting more prices is a strength for quoting illiquids, and the very reason it fails for relative value.

D is incorrect. The two model types agree on liquid securities only because the arbitrage-free one is forced to.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.4

Solution MR-14 e.4 — Answer: B

Bank Q680 • Schweser Module Quiz 14.2 Q3

Step 1 – Statement 1

Setting $\lambda_1 = \lambda_2 = \ldots$ gives Model 2, and setting them to zero gives Model 1, so Ho-Lee nests both. **True.**

Step 2 – Statement 2

Vasicek's constant k, θ, σ give no free parameter per maturity, so it cannot match an arbitrary curve (GARP: probably not flexible enough). **False.**

Step 3 – Statement 3

Arbitrage-free models assume both securities are fairly priced, so they cannot do relative value. **False.**

$\Rightarrow$ Exactly one statement is correct.

Interpretation

Statement 2 is a sibling swap (Ho-Lee's flexibility given to Vasicek); Statement 3 turns the main limitation of arbitrage-free models into a supposed strength.

Why the other options are wrong

A is incorrect. Misses that Statement 1 is true.

C is incorrect. Accepts one of the two false statements.

D is incorrect. Accepts both false statements.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.3–14.5

Solution MR-14 e.5 — Answer: C

Bank Q678 • BT P2.T5.304.3

Step 1 – Statement I

Models 1 and 2 have zero or constant drift and make no attempt to match the curve: they are *equilibrium* models. **False.**

Step 2 – Statements II and III

Ho-Lee's time-dependent drift lets it match observed prices, so it is arbitrage-free (II true). Quoting non-traded securities and valuing derivatives are GARP's two uses (III true).

Step 3 – Statement IV

GARP's two warnings: a bad model cannot be rescued by calibration, and market prices may not be fair in the context of the model (IV true).

$\Rightarrow$ II, III and IV.

Interpretation

Statement I is the classification trap; the rest are GARP's text almost verbatim.

Why the other options are wrong

A is incorrect. II, III and IV are all true.

B is incorrect. Accepts the misclassification in I and rejects the true III and IV.

D is incorrect. Fails to catch that I misclassifies Models 1 and 2.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.2–14.4

Solution MR-14 e.6 — Answer: B

New • GARP Ch.14, 14.4

Step 1 – What moved the 7-year rate

A forced liquidation: a supply and demand imbalance unrelated to expected rates or the risk premium.

Step 2 – What fitting does with it

An arbitrage-free model takes the new 7-year price as fair and changes the drift to reproduce it, attributing a liquidity effect to the rate process. The jump in the fitted drift is the warning sign: a risk-neutral drift arises only from expectations and risk premium, and neither plausibly jumped overnight at year seven.

$\Rightarrow$ The 7-year swaps are fair given the liquidity event, not fair with respect to the expected evolution of rates.

Interpretation

This is GARP's own example. Traders might buy the cheapened 7-year swaps and hold them past the liquidity event. An equilibrium model would flag them as cheap instead of absorbing the move.

Why the other options are wrong

A is incorrect. Reads the fitted drift as a forecast; it is the liquidity distortion fed into the rate process.

C is incorrect. Absolute claim: matching market prices does not imply fair values or accurate hedges.

D is incorrect. True fact, wrong model: an equilibrium model is *not* forced to fit the 7-year rate; that is why it can flag the mispricing.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market

prices. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.4 Desirability of Fitting to the Term Structure

Solution MR-14 e.7 Answer: C

Workbook A7.5 • S6, S7 • Aug-2026

Step 1 – What time-dependent drift buys

The free parameters $\lambda_1, \lambda_2, \ldots$ are chosen one at a time to match the two-, three-, … -period spot rates, so the rate curves match all the rates input into the model (MR-14 d, e).

Step 2 – What it does not change

Adding a constant drift (Model 2) or a time-dependent drift (Ho-Lee) does not affect the shape of the term structure of volatility nor the parallel-shift characteristic of the model. Volatility is σ at every term.

Step 3 – Model 3 contrast

Time-dependent volatility fits caplet prices across expiries, but as for any model without mean reversion, Model 3 is a parallel-shift model with a flat term structure of volatility, flat at levels that change over time (MR-15 a, c).

Interpretation

Calibration buys fit, not structure. If the desk's hedges rely on non-parallel moves, neither model fixes that: a mediocre model cannot be rescued by calibrating it to market prices.

Why the other options are wrong

A is incorrect. Sibling swap: a humped volatility structure is a Gauss+ feature (no shock on r); drift has no effect on the volatility shape in Ho-Lee.

B is incorrect. Scope/condition error: calibrating the drift fits prices but leaves the distribution normal, so negative rates remain possible; ruling them out needs a different distribution (CIR or lognormal) or setting negative shadow rates to zero.

D is incorrect. Sibling swap: Model 3 can mimic the Vasicek terminal distribution, but it has no mean reversion and stays a parallel-shift model.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices. MR-15 c: Assess the efficacy of time-dependent volatility models. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3 and 14.4; Ch.10, 15.1

MR-14 • The Art of Term Structure Models: Drift • Vardeez: VERY IMPORTANT

MR-14 f

P1 BOTH

Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion.

Workbook: A7 Ho-Lee, Time-Dependent Drift • A25 Gauss+ and Vasicek • D1 CIR and Lognormal

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A7 hits this LO (sibling-swap shape), and A25 (Gauss+ and Vasicek, August 2026 recall) builds on the same Vasicek process.
Vardeez: "Vasicek model": "Vasicek model calc can come"; "calculations are super important here".

Reported exam shape for this LO

- **A7 S6** Sibling swap against Vasicek and Models 1 and 2 (mean reversion, recombination and drift attributed to the wrong model).

What the questions look like: the LO verb is "describe", but Vardeez flags calculations, so expect both. Numbers: $\theta = r_\infty + \lambda/k$, the two date-1 nodes, or the averaged date-2 middle node. Concepts: why the simple tree does not recombine, how it is made to recombine, and what the risk premium does to θ.

1. The Vasicek process: mean reversion

$$\boxed{dr = k(\theta - r)\,dt + \sigma\,dw}$$

- $k > 0$ is the **speed of mean reversion**: higher k, faster pull. θ is the **long-run value** (central tendency) of the short rate in the *risk-neutral* process.
- **The drift changes sign:** $k(\theta - r)$ is positive when $r < \theta$ (rate pushed up) and negative when $r > \theta$ (pushed down). The further r is from θ, the larger the pull.
- **Economic logic.** The economy tends toward an equilibrium set by fundamentals (productivity of capital, long-run monetary policy). Mean reversion is reasonable but strong: it breaks down in hyperinflation and other structural breaks ("rates mean revert until they don't").

Splitting θ into expectations and risk premium. If the true process reverts to r_∞ and a constant risk premium λ enters as a drift,

$$dr = k(r_\infty - r)dt + \lambda dt + \sigma dw = k\left(\left[r_\infty + \frac{\lambda}{k}\right] - r\right)dt + \sigma dw \qquad \Longrightarrow \qquad \boxed{\theta = r_\infty + \frac{\lambda}{k}}$$

Very many (an infinite number of) combinations of r_∞ and λ give the same θ, hence the same prices: **expectations cannot be separated from the risk premium by observing prices**. GARP: $r_\infty = 6.179\%$, $\lambda = 0.229\%$, $k = 0.025$ give $\theta = 15.339\%$.

2. The simple tree, and why it does not recombine

Each node moves by the drift computed *at that node* plus or minus $\sigma\sqrt{dt}$:

$$r_{\text{up}} = r + k(\theta - r)dt + \sigma\sqrt{dt}, \qquad r_{\text{down}} = r + k(\theta - r)dt - \sigma\sqrt{dt}.$$

Worked example (Vardeez / Schweser). $k = 0.03$, $\sigma = 1.50\%$, $r_\infty = 6\%$, $\lambda = 0.36\%$, $r_0 = 6.20\%$, $dt = 1/12$.

- $\theta \approx 6\% + 0.36\%/0.03 = 18\%$.
- Drift at r_0: $0.03 \times (18\% - 6.20\%)/12 = 0.0295\%$; shock $\sigma\sqrt{dt} = 1.50\%/\sqrt{12} = 0.4330\%$ (43 bp).
- Date 1: $6.20\% + 0.0295\% \pm 0.4330\% = 6.663\%$ and 5.796%.
- From 6.663% the drift is $0.03(18\% - 6.663\%)/12 = 0.0283\%$: nodes 7.124% and 6.258%.
- From 5.796% the drift is $0.03(18\% - 5.796\%)/12 = 0.0305\%$: nodes 6.260% and 5.394%.

The up-down path ends at 6.258%, the down-up path at 6.260%. The volatility parts cancel exactly, but the down node is further from θ, so its drift is larger. **Mean reversion is what breaks recombination.**

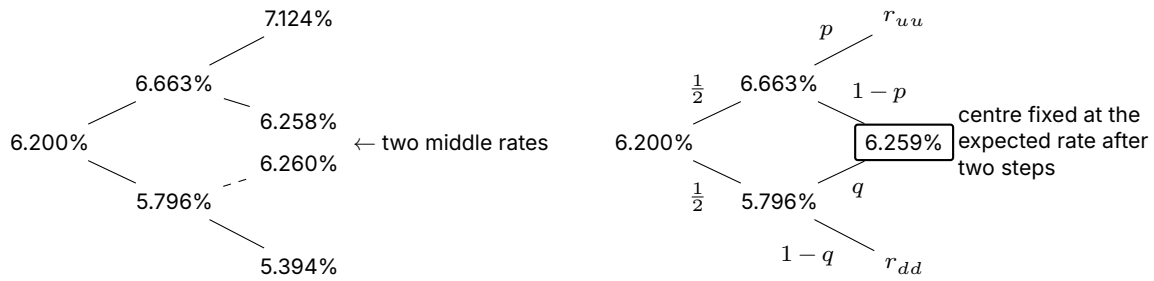


Figure 5. Vasicek with $k = 0.03$, $\theta = 18\%$, $\sigma = 1.50\%$, monthly steps. Left: the drift is recomputed at each node, so the up-down (6.258%) and down-up (6.260%, dashed branch) paths miss each other. Right: the centre node is fixed (boxed) and r_{uu}, p, r_{dd}, q are solved for; probabilities are no longer $\frac{1}{2}$ away from the centre.

3. Making the tree recombine

There are many ways; GARP presents one:

1. **Fix the centre node on date 2** at the *expected* value of the process after two steps (the shocks have zero expectation, so the drift alone determines it). Vardeez and Schweser describe this as averaging the two middle nodes: $(6.258\% + 6.260\%)/2 = 6.259\%$. The two are the same number, because the Vasicek drift is linear in r:

$$\tfrac{1}{2}(r_{ud} + r_{du}) = m_1 + k(\theta - m_1)dt, \qquad m_1 = r_0 + k(\theta - r_0)dt.$$

2. **Drop the 50/50 assumption** on the outer branches: use $(p, 1-p)$ from the up node and $(q, 1-q)$ from the down node.

3. **Solve for r_{uu} and p** so that the branches from the up node match the process's expected rate and standard deviation from that node:

$$p\, r_{uu} + (1-p)\, r_{\text{mid}} = r_u + k(\theta - r_u)dt, \qquad \sqrt{p\,(r_{uu} - m_u)^2 + (1-p)(r_{\text{mid}} - m_u)^2} = \sigma\sqrt{dt},$$

where $m_u = r_u + k(\theta - r_u)dt$ is the expected rate one step after the up node. Then do the same for r_{dd} and q from the down node. Repeat date by date. From the centre node itself, adding and subtracting $\sigma\sqrt{dt}$ around its expected value with 50/50 probabilities still works.

GARP's numbers ($k = 0.025$, $\theta = 15.339\%$, $\sigma = 1.26\%$, $r_0 = 5.121\%$): date 1: 5.5060% and 4.7786%; simple date-2 middles 5.1628% and 5.1643%; centre fixed at 5.1635%; solved $r_{uu} = 5.8909\%$, $p = 0.4990$; $r_{dd} = 4.4361\%$, $q = 0.5011$; SD per step $1.26\%\sqrt{1/12} = 0.3637\%$.

4. Where Vasicek sits among the drift choices

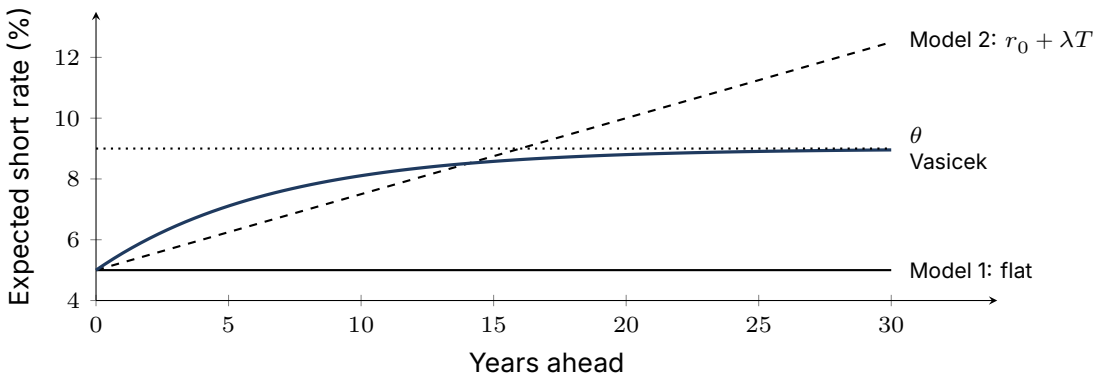


Figure 6. Illustrative expected paths from $r_0 = 5\%$: Model 1 (solid, flat), Model 2 with $\lambda = 0.25\%$ (dashed, rises without limit), Vasicek with $k = 0.15$, $\theta = 9\%$ (thick, bends toward the dotted θ line and never crosses it).

Model	Process	Tree	What it produces
Model 1	$dr = \sigma dw$	Recombining	Downward-sloping curve (convexity); flat volatility; parallel shifts; negative rates
Model 2	$dr = \lambda dt + \sigma dw$	Recombining	Upward slope from drift; drift hard to justify in the long end
Ho-Lee	$dr = \lambda_t dt + \sigma dw$	Recombining	Any shape (each λ_t calibrated, can be negative); still flat volatility, parallel shifts
Vasicek	$dr = k(\theta - r)dt + \sigma dw$	Not recombining until the centre is fixed	Rates pulled to θ; downward-sloping volatility; non-parallel shifts; still normal (negative rates possible)

The four models differ in one place only: the drift term. Model 1 has none, Model 2 a constant, Ho-Lee a time-varying constant, and Vasicek a drift that depends on where the rate currently is.

Formula card: MR-14 f

Process	$dr = k(\theta - r)dt + \sigma dw$
Risk-neutral long-run level	$\theta = r_\infty + \lambda/k$
Node step	$r\pm$: $r + k(\theta - r)dt \pm \sigma\sqrt{dt}$ (drift recomputed at each node)
Recombined centre	expected rate after two steps $= \frac{1}{2}(r_{ud} + r_{du})$
Outer nodes	solve r_{uu}, p (and r_{dd}, q) to match the node's expected rate and SD $\sigma\sqrt{dt}$

Exam traps

- **θ arithmetic.** $\theta = r_\infty + \lambda/k$: adding λ without dividing by k, multiplying by k, or subtracting are the standard distractors.
- **Drift sign.** The drift pulls *toward* θ; a node below θ moves up on average. Moving away from θ is a planted option.

- **Unscaled drift or volatility.** $k(\theta - r)$ must be multiplied by dt; σ by $\sqrt{dt}$.
- **Same drift everywhere.** Reusing the date-0 drift at the date-1 nodes gives a recombining (Model 2-like) tree and the wrong nodes.
- **"Impossible to recombine".** False: averaging the middle nodes and re-solving the probabilities and outer nodes produces a recombining tree.
- **Mean reversion is not a floor.** Vasicek rates are normal; mean reversion damps the spread but does not stop negative rates (that needs CIR or lognormal, MR-15).

Practice questions: MR-14 f

Question MR-14 f.1

Bank Q683 • Mock 1 Q40 (rebuilt)

A risk manager is reviewing short-term interest rate assumptions by applying the Vasicek model. The first task is to estimate the long-run value of the short-term interest rate. The current interest rate is 5% and you estimate the true long-term interest rate is 5.5%, annual drift (risk premium) of 0.32%, annual volatility of 1.8%, $dt = 1/12$, and a mean reversion adjustment (k) of 0.04. What is the long-run value of the short-term rate, assuming risk neutrality?

A. 5.82%

B. 5.51%

C. 13.50%

D. -2.50%

Question MR-14 f.2

Bank Q681 • AnalystPrep Q.6530

In the Vasicek model, a risk premium (λ) is introduced. What is the key implication of this risk premium for the observed term structure of interest rates?

A. The risk premium directly affects the volatility of the short-term rate.

B. The risk premium is indistinguishable from the long-run value of the short-term rate (θ) when observing security prices.

C. The risk premium only affects the short end of the yield curve and has no impact on long-term rates.

D. The risk premium causes the short-term rate to become negatively correlated with the long-run value (θ).

Question MR-14 f.3 Bank Q684 • Mock 2 Q67 (rebuilt)

Using the Vasicek model, assume a current short-term rate of 5.2% and an annual volatility of the interest rate process of 3.5%. Also assume that the long-run mean-reverting level is 14.2% with a speed of adjustment of 0.4. Within a binomial interest rate tree, what are the upper and lower node rates after the first month?

A. Upper 9.81%; lower 7.79%

B. Upper 5.91%; lower 3.89%

C. Upper 6.51%; lower 4.49%

D. Upper 9.00%; lower 2.00%

Question MR-14 f.4 Bank Q682 • BT P2.T5.305.1 (rebuilt)

Analyst Steven builds a monthly ($dt = 1/12$) short-rate tree under the Vasicek model $dr = k(\theta - r)dt + \sigma dw$ with $r(0) = 8.00\%$, $\theta = 3.00\%$, $k = 0.60$ and annual volatility $\sigma = 2.00\%$ (200 bps). After the first month the tree has $r(u) = 8.3274\%$ and $r(d) = 7.1726\%$. The unrecombined tree is then made recombining by averaging the two middle nodes of month 2 (up-down and down-up). What is the value of the recombined middle node at month 2?

A. 7.484%

B. 7.513%

C. 7.542%

D. 7.750%

Question MR-14 f.5 New • built from GARP Ch.14, 14.5 (recombining tree)

A junior quant's two-step monthly Vasicek tree does not recombine: the up-down and down-up paths end 2 bp apart. Which of the following describes the procedure GARP presents for turning it into a recombining tree?

A. Keep probabilities of $\frac{1}{2}$ on every branch and replace the two middle nodes by their average; the outer nodes on date 2 are left as first computed.

B. Fix the centre node on date 2 at the expected rate after two steps, then choose the outer nodes and the branch probabilities so that, from each date-1 node, the tree matches the process's expected rate and standard deviation.

C. Drop the mean-reversion drift in the second step so that the up-down and down-up paths cancel exactly, as they do in Model 2.

D. Set the branch probabilities so that the tree's expected rate equals θ on every date, keeping the move per step at $\sigma\sqrt{dt}$.

Question MR-14 f.6 Bank Q687 • BT P2.T5.305.3

Each of the following statements is true about the Vasicek Model EXCEPT:

A. The Vasicek Model is an equilibrium model like the Cox-Ingersoll-Ross (CIR) model and unlike the Ho-Lee model, which is an arbitrage-free (aka, no-arbitrage) model.

B. In the Vasicek Model, half-life is given by $\ln(2)/k$ and can be interpreted as the time it takes for the short-rate (the factor) to progress half the distance toward the long-run value (θ).

C. The mean reversion in Vasicek dampens volatility relative to the same process without mean reversion, particularly at long horizons.

D. Due to the mean-reversion, it is impossible to develop a rate tree that recombines under Vasicek.

Question MR-14 f.7 Workbook D1.2 • S2, S3, S4, S5 • Map #39

A model validation analyst at a Japanese bank is asked to recommend a one-factor short-rate model for the bank's interest rate desk. The desk sets three requirements: (i) the short rate cannot become negative; (ii) the basis-point volatility of the short rate rises with the level of the rate, but less than proportionally; and (iii) the short rate is pulled toward a long-run level. Which recommendation, with its justification, should the analyst make?

A. The Vasicek model, because mean reversion pulls the short rate toward its long-run level and therefore prevents the short rate from becoming negative.

B. The CIR model, because its basis-point volatility rises with the square root of the rate and falls to zero at a zero rate, where the drift is positive.

C. The Black-Karasinski model, because the logarithm of the rate mean-reverts and its basis-point volatility rises with the square root of the rate.

D. The Ho-Lee model, because fitting its time-dependent drift to market bond prices guarantees that no node of the rate tree becomes negative.

Solutions • MR-14 f

Solution MR-14 f.1

Answer: C

Bank Q683 • Mock 1 Q40

Step 1 – Risk-neutral long-run value

$$\theta = r_\infty + \frac{\lambda}{k} = 5.5\% + \frac{0.32\%}{0.04} = 5.5\% + 8.0\% = 13.50\%$$

$\Rightarrow$ The risk premium is capitalised into the long-run level by dividing by the speed of reversion.

Interpretation

A small premium becomes a large shift in θ when k is small. The current rate, volatility and dt are not needed for θ; they would be needed for the first tree step.

Why the other options are wrong

A is incorrect. Added λ without dividing by k: $5.5\% + 0.32\% = 5.82\%$.
B is incorrect. Multiplied instead of divided: $5.5\% + 0.32\% \times 0.04 = 5.51\%$.
D is incorrect. Subtracted the premium term: $5.5\% - 8.0\% = -2.50\%$.

LO: MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, eqs. 14.9 and 14.10

Solution MR-14 f.2

Answer: B

Bank Q681 • AnalystPrep Q.6530

Step 1 – Where λ enters

$dr = k(r_\infty - r)dt + \lambda dt + \sigma dw = k([r_\infty + \lambda/k] - r)dt + \sigma dw$: the premium only shifts the long-run level to $\theta = r_\infty + \lambda/k$.

Step 2 – What prices can see

Prices depend only on the risk-neutral process, i.e. on θ. Infinitely many pairs (r_∞, λ) give the same θ and the same prices.

$\Rightarrow$ From prices alone, the premium cannot be told apart from the long-run expectation.

Interpretation

This is why "expectations versus risk premium" needs extra assumptions (an economist's r_∞, say) and cannot be read off a calibrated model.

Why the other options are wrong

A is incorrect. λ changes the drift only; σ is untouched.
C is incorrect. A shift in θ feeds through the whole curve, most at long maturities.
D is incorrect. θ is a constant parameter; no correlation is implied.

LO: MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, eq. 14.10 and text

Solution MR-14 f.3

Answer: C

Bank Q684 • Mock 2 Q67

Step 1 – Monthly drift at r_0

$$k(\theta - r_0)dt = 0.4 \times (14.2\% - 5.2\%) \times \frac{1}{12} = 0.30\%$$

$\Rightarrow$ The rate is far below θ, so the pull is upward and strong.

Step 2 – Monthly shock

$$\sigma\sqrt{dt} = 3.5\% \times 0.28868 = 1.0104\%$$

Step 3 – Date-1 nodes

$$r_u = 5.2\% + 0.30\% + 1.0104\% = 6.51\%, \qquad r_d = 5.2\% + 0.30\% - 1.0104\% = 4.49\%$$

Interpretation

The same drift applies to both date-1 nodes because both start from r_0. Only from date 1 onward does each node get its own drift, which is what breaks recombination.

Why the other options are wrong

A is incorrect. Forgot to scale the drift by dt (3.6% instead of 0.30%): $5.2\% + 3.6\% \pm 1.01\%$.
B is incorrect. Drift with the wrong sign (moving away from θ): $5.2\% - 0.30\% \pm 1.01\%$.
D is incorrect. Used the annual volatility as the monthly shock: $5.2\% + 0.30\% \pm 3.5\%$.

LO: MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, first time step

Solution MR-14 f.4 **Answer: B**

Bank Q682 • BT P2.T5.305.1

Step 1 – Shock per step

$$\sigma\sqrt{dt} = 2.00\% \times 0.28868 = 0.5774\%$$

Step 2 – Up-down node (drift at 8.3274%)

$$0.60\times(3\%-8.3274\%)/12 = -0.2664\%, \qquad r_{ud} = 8.3274\%-0.2664\%-0.5774\% = 7.4837\%$$

Step 3 – Down-up node (drift at 7.1726%)

$$0.60\times(3\%-7.1726\%)/12 = -0.2086\%, \qquad r_{du} = 7.1726\%-0.2086\%+0.5774\% = 7.5414\%$$

$\Rightarrow$ The higher node is pulled down harder, so the paths miss by about 6 bp.

Step 4 – Recombined centre

$$\tfrac{1}{2}(7.4837\% + 7.5414\%) = 7.5125\% \approx 7.513\%$$

Interpretation

Check: the expected rate after two steps is $7.75\% + 0.60(3\% - 7.75\%)/12 = 7.5125\%$, the same number, as it must be with a drift linear in r.

Why the other options are wrong

A is incorrect. Used only the up-down node: 7.484%.
C is incorrect. Used only the down-up node: 7.542%.
D is incorrect. Applied one step of drift and stopped: $8\% - 0.25\% = 7.75\%$ (the expected rate after one step).

LO: MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5; SchweserNotes 2026 Book 1, Reading 14 (Figure 14.9)

Solution MR-14 f.5

Answer: B

New • GARP Ch.14, 14.5

Step 1 – Fix the centre

Since the expected shock is zero, the drift alone fixes the expected rate after two steps; GARP sets the date-2 centre node equal to it (5.1635% in its example; numerically the same as averaging the two middle nodes).

Step 2 – Free the probabilities and outer nodes

From the up node, choose r_{uu} and p so that $p\,r_{uu} + (1-p)r_{\text{mid}}$ equals the process's expected rate from that node and the branch SD equals $\sigma\sqrt{dt}$; likewise r_{dd} and q from the down node.
$\Rightarrow$ GARP's solution: $r_{uu} = 5.8909\%$, $p = 0.4990$; $r_{dd} = 4.4361\%$, $q = 0.5011$.

Interpretation

The tree recombines and still honours the Vasicek process at every node: mean reversion is kept, and the probabilities absorb the asymmetry.

Why the other options are wrong

A is incorrect. Averaging alone, with outer nodes and $\frac{1}{2}$ probabilities unchanged, no longer matches the expected rate and volatility from each date-1 node.
C is incorrect. Removing the mean-reverting drift turns the model into something else; recombination must be achieved with the drift kept.
D is incorrect. Wrong target: the expected rate is not θ on each date; it moves gradually toward θ at speed k.

LO: MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, eqs. 14.12 to 14.21

Solution MR-14 f.6

Answer: D

Bank Q687 • BT P2.T5.305.3

Step 1 – Test the true statements

A: Vasicek and CIR are equilibrium models; Ho-Lee is arbitrage-free (true). B: half-life $= \ln 2/k$, the time to cover half the distance to θ (true). C: mean reversion dampens dispersion, particularly at long horizons (353 versus 398 bp at 10 years in GARP; true).

Step 2 – The false one

D: the simple tree does not recombine, but fixing the centre node and re-solving the outer nodes and probabilities produces a recombining tree.
$\Rightarrow$ "Impossible" is false: D.

Interpretation

The absolute word "impossible" is the giveaway; GARP says there are many ways to represent Vasicek with a recombining tree.

Why the other options are wrong

A is incorrect. True statement.
B is incorrect. True statement (MR-14 g).
C is incorrect. True statement (MR-14 g).

LO: MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the

Vasicek Model with mean reversion. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5

Solution MR-14 f.7 Answer: B

Workbook D1.2 • S2, S3, S4, S5

Step 1 – Requirement (ii) sorts the volatility specifications

Normal models (Ho-Lee, Vasicek) have constant basis-point volatility; lognormal models (Courtadon, Model 4, Salomon Brothers, Black-Karasinski) have basis-point volatility σr, proportional to the rate; CIR has $\sigma\sqrt{r}$, rising less than proportionally.

Step 2 – Requirement (i)

CIR: basis-point volatility is zero at a zero rate and the drift $k(\theta - 0)$ is positive there, so the rate cannot go negative.

Step 3 – Requirement (iii)

CIR's drift $k(\theta - r)$ pulls the rate toward θ. Only CIR meets all three.

Interpretation

The Vasicek option is the one tied to this LO: mean reversion is a pull, not a floor. (CIR and lognormal models are MR-15 d, f.)

Why the other options are wrong

A is incorrect. Scope/condition error. Vasicek does mean-revert, but its basis-point volatility is constant and rates are normally distributed, so negative rates remain possible; mean reversion only damps the spread of future rates.

C is incorrect. Sibling swap. Black-Karasinski does mean-revert in the log rate and stays positive, but as a lognormal model its basis-point volatility is proportional to the rate, the property CIR does not have.

D is incorrect. Absolute claim. Ho-Lee rates are normal with constant volatility; fitting the drift matches bond prices but does not stop low nodes from going negative, and Ho-Lee has no long-run level at all.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. MR-15 f: Describe lognormal models with deterministic drift and mean reversion. MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models. MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion. • **Reference:** Tuckman and Serrat 3e, Ch.10, 15.2 to 15.4; Ch.9, 14.3 and 14.4

MR-14 • The Art of Term Structure Models: Drift • Vardeez: VERY IMPORTANT

MR-14 g

P1 CALC

Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life.

Workbook: A25 Gauss+ and Vasicek • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A25 (August 2026 recall: Gauss+; the sheet adds "Vasicek model calculation can come" and flags LO 14.g separately).
Vardeez: "Vasicek calculations, incl. expected rate in T (e.g. 7) years (his 'LO 14.g')"; chapter note "calculations are super important here".

Reported exam shapes

- **A25 S1** Closed-book $E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT})$, with θ built from r_∞ and the risk premium.
- **A25 S2** Closed-book half-life $\ln 2/k$.
- **A25 S3** Vasicek rate change and standard deviation (one-month drift; terminal standard deviation).
- **A25 S4** LO 14.g expected rate in T years.

What the questions look like: formula recall with no formula sheet. Two outputs per question are common (expected rate and half-life; drift and SD). Distractors: $1/k$ for the half-life, swapped weights, r_∞ in place of θ, λ not divided by k, $\sigma\sqrt{T}$ in place of the mean-reverting SD.

1. One step: the rate change and its standard deviation

$$dr = k(\theta - r)\,dt + \sigma\,dw, \qquad \boxed{E[dr] = k(\theta - r)\,dt}, \qquad \boxed{\text{SD}(dr) = \sigma\sqrt{dt}}$$

- The expected change is the drift only ($E[dw] = 0$). A *realised* change adds $\sigma \times$(realised dw).
- If the stem gives r_∞ and λ, build $\theta = r_\infty + \lambda/k$ first (MR-14 f).
- **GARP:** $k = 0.025$, $\theta = 15.339\%$, $r_0 = 5.121\%$: $E[dr] = 0.025 \times (15.339\% - 5.121\%)/12 = 0.0213\%$ (2.13 bp); SD $= 1.26\%\sqrt{1/12} = 36.4$ bp.

2. Expected rate in T years

The distance between the current rate and θ decays exponentially at the speed k:

$$\boxed{E[r_T] = \theta - (\theta - r_0)\,e^{-kT} = r_0\,e^{-kT} + \theta\,(1 - e^{-kT})}$$

A weighted average of today's rate and θ; the weight on today's rate, e^{-kT}, decays with the horizon.

Vardeez's example ($r_0 = 6.2\%$, $k = 0.03$, $\theta = 18\%$, $T = 10$):

1. Gap today: $18\% - 6.2\% = 11.8\%$.
2. Gap after 10 years: $11.8\% \times e^{-0.03\times 10} = 11.8\% \times 0.7408 = 8.74\%$.
3. Expected rate: $18\% - 8.74\% = 9.26\%$.

GARP's example ($r_0 = 5.121\%$, $\theta = 15.339\%$, $k = 0.025$): gap $10.218\% \times e^{-0.25} = 7.9578\%$, so $E[r_{10}] = 7.3812\%$.

3. Half-life

The half-life τ is the time for the rate to cover *half* the distance to θ:

$$(\theta - r_0)\, e^{-k\tau} = \tfrac{1}{2}(\theta - r_0) \quad \Longrightarrow \quad \boxed{\tau = \frac{\ln 2}{k}}$$

It depends only on k (not on r_0, θ or σ). Vardeez: $\ln 2/0.03 = 23.1$ years, when the expected rate reaches $18\% - 5.9\% = 12.1\%$. GARP: $\ln 2/0.025 = 27.73$ years. Larger k, shorter half-life. Do not confuse it with $1/k$, the time for the gap to fall to $e^{-1} = 37\%$.

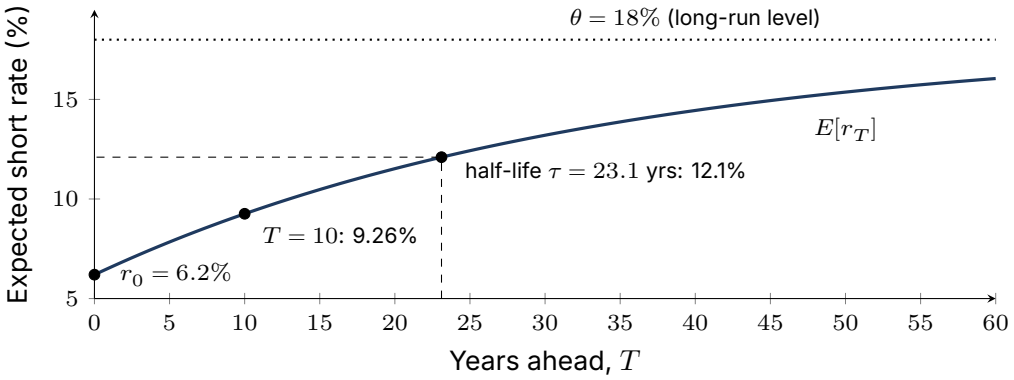


Figure 7. Vardeez's example. The expected rate approaches θ asymptotically and never reaches it. At the half-life exactly half of the 11.8-point gap has closed. Both worked answers sit on this one curve.

4. Standard deviation of the rate in T years

$$\boxed{\mathrm{SD}(r_T) = \sqrt{\frac{\sigma^2}{2k}\left(1 - e^{-2kT}\right)}} \qquad \text{versus} \qquad \sigma\sqrt{T} \text{ without mean reversion.}$$

GARP: $\sigma = 1.26\%$, $k = 0.025$, $T = 10$: $\sqrt{0.0126^2(1 - e^{-0.5})/0.05} = 353$ bp, against $1.26\%\sqrt{10} = 398$ bp. "126 bp per year" does not mean the terminal SD grows with $\sqrt{T}$: the pull toward θ dampens dispersion, particularly at long horizons, and caps the variance at $\sigma^2/(2k)$.

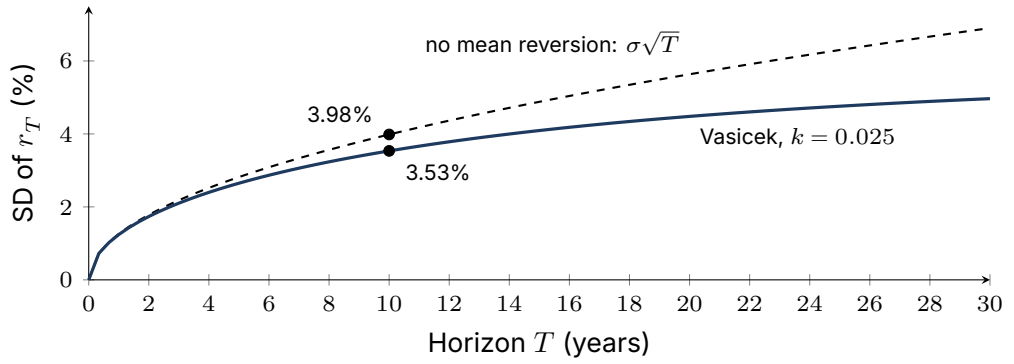


Figure 8. Terminal standard deviation with GARP's $\sigma = 1.26\%$. Solid: Vasicek. Dashed: the same σ without mean reversion. The gap widens with the horizon.

Effectiveness in one paragraph (MR-14 h, context). Mean reversion gives a downward-sloping term structure of volatility (better than Model 1's flat one, but it overstates short-term volatility), non-parallel shifts (a 10 bp rise in the short rate moves the 30-year spot rate only about 7 bp), a smaller convexity effect at long terms (about 75 bp at 30 years versus 135 bp), and reads k as the length of economic news: low k (long half-life) means long-lived news, high k short-lived news.

Formula card: MR-14 g

Long-run level	$\theta = r_\infty + \lambda/k$
Expected one-step change	$k(\theta - r)\,dt$; realised: $k(\theta - r)dt + \sigma\,dw$
SD of one-step change	$\sigma\sqrt{dt}$
Expected rate in T years	$E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT}) = \theta - (\theta - r_0)e^{-kT}$
Half-life	$\tau = \ln 2/k$ (inverse: $k = \ln 2/\tau$)
SD of r_T	$\sqrt{\sigma^2(1 - e^{-2kT})/(2k)}$; no mean reversion: $\sigma\sqrt{T}$; limit $\sigma/\sqrt{2k}$

Exam traps

- $1/k$ **instead of** $\ln 2/k$**.** $1/k$ is the time constant (63% of the gap closed), not the half-life. Also $\ln 2/(2k)$ and $0.5/k$ appear as options.
- **Swapped weights.** e^{-kT} goes on r_0, not on θ; swapping overstates how far the rate has reverted.
- r_∞ **instead of** θ**.** For pricing, revert to the risk-neutral $\theta = r_\infty + \lambda/k$, not to the economist's r_∞.
- λ **not divided by** k**.** $\theta = r_\infty + \lambda$ is the classic dropped-term error.
- **Drift not scaled.** $k(\theta - r)$ is per year; multiply by dt for one month.
- **Terminal SD.** $1 - e^{-kT}$ instead of $1 - e^{-2kT}$; σ^2/k instead of $\sigma^2/(2k)$; or $\sigma\sqrt{T}$ (ignores mean reversion).
- **Expected versus realised change.** The expected change is the drift alone; adding $\sigma\,dw$ gives one realised path.

Practice questions: MR-14 g

Question MR-14 g.1

Bank Q697 • AnalystPrep Q.1662 (rebuilt)

In a risk-neutral Vasicek model, $dr = k(\theta - r)dt + \sigma dw$ with $\theta = r_\infty + \lambda/k$. Assume $k = 0.020$, $\sigma = 125$ basis points per year, $r_\infty = 6.180\%$, $\lambda = 0.227\%$ per year, current short rate $r = 5.212\%$, a one-month interval ($dt = 1/12$), and a realized random shock of $dw = 0.14$ (use it directly in $\sigma \cdot dw$). What are the change in the short rate over the month and the standard deviation of the monthly rate change?

A. Change of 17.66 bp; standard deviation of 36.08 bp

B. Change of 19.55 bp; standard deviation of 125.00 bp

C. Change of 19.55 bp; standard deviation of 36.08 bp

D. Change of 2.05 bp; standard deviation of 36.08 bp

Question MR-14 g.2

Bank Q688 • AnalystPrep Q.1664 (rebuilt)

In a Vasicek model, the half-life of the short rate is 28.72 years. Which statement is correct?

A. The rate takes 28.72 years to cover half of the distance between its current value and its long-run mean-reverting level, which implies a speed of reversion $k \approx 0.0241$.

B. The rate takes 28.72 years to reach its long-run mean-reverting level, which implies $k \approx 0.0348$.

C. The rate takes 57.44 years to cover half of the distance to its long-run level, which implies $k \approx 0.0121$.

D. The rate takes 14.36 years to cover half of the distance to its long-run level, which implies $k \approx 0.0483$.

Question MR-14 g.3

Bank Q696 • GARP PE2 Q46 • Mock 4 Q47

A quantitative analyst on the fixed-income desk of an investment bank is applying the Vasicek model to estimate future short-term interest rates. The model is $dr = k(\theta - r)dt + \sigma dw$, where dr is the change in the short-term interest rate, θ is the estimated long-run value of the short-term interest rate, k is the mean reversion rate, r is the current level of the short-term interest rate, σ is the annual basis-point volatility of the short-term interest rate, dt is the time interval measured in years, and dw is a normally distributed random variable with mean zero and standard deviation equal to the square root of dt. The analyst gathers the following information:

Current short-term interest rate (r)	3.35%
Long-run value of short-term interest rate (θ)	4.55%
Mean reversion rate (k)	0.06
Annual basis-point volatility (σ)	120 bps

The analyst then creates an interest rate tree, determines the expected short-term interest rate after 8 years, and calculates how long it will take the short-term interest rate to revert halfway to the long-run value. Which of the following statements would be correct for the analyst to make?

A. The expected short-term interest rate is 3.81% and the half-life is 11.6 years.

B. The expected short-term interest rate is 3.81% and the half-life is 16.7 years.

C. The expected short-term interest rate is 4.09% and the half-life is 11.6 years.

D. The expected short-term interest rate is 4.09% and the half-life is 16.7 years.

Question MR-14 g.4

Workbook A25.4 • S1, S3, S4 • Map #40

A treasury analyst at a Brazilian bank prices BRL rate products with a Vasicek model. The economist's real-world view and the desk's risk premium are:

- Current short rate r_0: 2.00%; speed of mean reversion k: 0.15
- Real-world long-run value of the short rate r_∞: 4.00%
- Risk premium entering as a constant drift λ: 0.30% per year

For pricing, which of the following is closest to the expected short rate in 5 years under the risk-neutral process, and to the expected change in the short rate over the next month?

A. 3.06%; +2.5 bps

B. 3.21%; +2.9 bps

C. 3.89%; +5.0 bps

D. 4.11%; +5.0 bps

Question MR-14 g.5 Bank Q689 • AnalystPrep Q.1665 (rebuilt)

In a Vasicek model, the speed of mean reversion is $k = 0.025$ and the annual volatility is 126 basis points. The short rate in 10 years is normally distributed with an expected value of 7.4812%. What is the standard deviation of the short rate in 10 years?

A. 265 basis points

B. 353 basis points

C. 398 basis points

D. 500 basis points

Question MR-14 g.6 Workbook A25.5 • S2, S3 • Map #40

A risk consultant at an Australian bank is explaining to the board risk committee how long rate shocks persist in the bank's Vasicek model. Parameters: $r_0 = 3.50\%$, $\theta = 5.00\%$, $k = 0.08$, $\sigma = 110$ bps per year. Which of the following is closest to the half-life of the short rate, and to the standard deviation of the short rate 10 years from now?

A. Half-life 8.66 years; standard deviation 2.04%

B. Half-life 8.66 years; standard deviation 2.46%

C. Half-life 8.66 years; standard deviation 3.48%

D. Half-life 12.50 years; standard deviation 2.46%

Solutions • MR-14 g

Solution MR-14 g.1 — Answer: C

Bank Q697 • AnalystPrep Q.1662

Step 1 – Risk-neutral long-run level

$$\theta = r_\infty + \frac{\lambda}{k} = 6.180\% + \frac{0.227\%}{0.020} = 17.53\%$$

Step 2 – Drift over the month

$$k(\theta - r)\,dt = 0.020 \times (17.53\% - 5.212\%) \times \frac{1}{12} = 0.0205\% = 2.05 \text{ bp}$$

Step 3 – Shock and total change

$$\sigma\,dw = 125 \times 0.14 = 17.50 \text{ bp}, \qquad dr = 2.05 + 17.50 = 19.55 \text{ bp}$$

Step 4 – Standard deviation of the monthly change

$$\sigma\sqrt{dt} = 125 \times 0.28868 = 36.08 \text{ bp}$$

Interpretation

These are GARP's parameters (14.5): the drift is tiny because k is small, so almost all of the realised move is the shock. The question asks for the realised change (drift plus shock), not the expected change (drift alone).

Why the other options are wrong

A is incorrect. Used $r_\infty = 6.18\%$ in place of θ in the drift (ignored λ/k): $0.17 + 17.50 = 17.66$ bp.

B is incorrect. Correct change, but quoted the annual 125 bp as the monthly SD.

D is incorrect. Reported the drift alone (the expected change), omitting $\sigma\,dw$.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, eqs. 14.9 to 14.11

Solution MR-14 g.2 — Answer: A

Bank Q688 • AnalystPrep Q.1664

Step 1 – Definition

The half-life is the time for the gap between the rate and θ to shrink by half: $\tau = \ln 2/k$.

Step 2 – Invert for k

$$k = \frac{\ln 2}{\tau} = \frac{0.69315}{28.72} = 0.0241$$

$\Rightarrow$ After 28.72 years the gap is 50% of its starting size, not zero.

Interpretation

The rate never reaches θ; it approaches it asymptotically. Half-life is the intuitive way of quoting the speed of reversion.

Why the other options are wrong

B is incorrect. Treats the half-life as the time to reach θ and uses $k = 1/28.72 = 0.0348$.

C is incorrect. Doubles the half-life: $\ln 2/57.44 = 0.0121$.
D is incorrect. Halves it: $\ln 2/14.36 = 0.0483$.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, eqs. 14.24 and 14.25

Solution MR-14 g.3

Answer: A

Bank Q696 • GARP PE2 Q46

Step 1 – Decay factor

$$e^{-kT} = e^{-0.06\times 8} = e^{-0.48} = 0.6188$$

Step 2 – Expected rate in 8 years

$$E[r_8] = \theta + (r_0 - \theta)e^{-kT} = 4.55\% + (3.35\% - 4.55\%) \times 0.6188 = 3.807\% \approx 3.81\%$$

$\Rightarrow$ Equivalently $0.6188 \times 3.35\% + 0.3812 \times 4.55\%$: only about 38% of the gap has closed.

Step 3 – Half-life

$$\tau = \frac{\ln 2}{0.06} = 11.55 \approx 11.6 \text{ years}$$

Interpretation

The 120 bp volatility and the tree are red herrings: neither the expected rate nor the half-life depends on σ. Each wrong option combines one or both of the two classic slips.

Why the other options are wrong

B is incorrect. Right rate, but $1/k = 16.7$ years is the time constant (about 63% of the gap closed), not the half-life.
C is incorrect. Right half-life, but the weights swapped (0.6188 on θ): 4.09%, overstating the reversion.
D is incorrect. Both errors: swapped weights and $1/k$.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, eqs. 14.23 and 14.25

Solution MR-14 g.4

Answer: D

Workbook A25.4 • S1, S3, S4

Step 1 – Risk-neutral long-run value

$$\theta = r_\infty + \frac{\lambda}{k} = 4.00\% + \frac{0.30\%}{0.15} = 6.00\%$$

$\Rightarrow$ Prices reflect expectations plus risk premium; the risk-neutral process reverts to 6.00%, not to the economist's 4.00%.

Step 2 – Decay factor

$$e^{-kT} = e^{-0.15\times 5} = e^{-0.75} = 0.47237$$

$\Rightarrow$ After 5 years, 47.2% of today's 4-point gap to θ remains.

Step 3 – Expected rate in 5 years

$$E[r_5] = r_0 e^{-kT} + \theta(1 - e^{-kT}) = 2.00\%(0.47237) + 6.00\%(0.52763)$$
$$= 0.9447\% + 3.1658\% = 4.11\%$$

$\Rightarrow$ Equivalently, θ minus the remaining gap: $6.00\% - 4.00\% \times 0.47237 = 4.11\%$.

Step 4 – One-month expected change

$$E[dr] = k(\theta - r_0)\,dt = 0.15 \times (6.00\% - 2.00\%) \times \frac{1}{12} = 0.050\% = +5.0 \text{ bps}$$

Interpretation

The half-life is $\ln 2/0.15 = 4.6$ years, so after 5 years the expected rate has covered a little more than half the distance to θ. Because the drift shrinks as the gap closes, the 5-year move (211 bps) is far less than 60 months of today's 5 bp drift.

Why the other options are wrong

A is incorrect. Wrong-input twin: reverted to the real-world $r_\infty = 4.00\%$ instead of θ: $2.00\%(0.47237) + 4.00\%(0.52763) = 3.06\%$; drift $0.15 \times 2.00\%/12 = 2.5$ bps.

B is incorrect. Dropped the division by k: $\theta = 4.00\% + 0.30\% = 4.30\%$, giving 3.21% and $0.15 \times 2.30\%/12 = 2.9$ bps.

C is incorrect. Swapped the weights: $2.00\%(0.52763) + 6.00\%(0.47237) = 3.89\%$; the one-month drift is unaffected.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.5 eqs. 14.9, 14.10, 14.23; 4e, Ch.9, 16.1 eqs. 16.2 to 16.4

Solution MR-14 g.5 **Answer: B**

Bank Q689 • AnalystPrep Q.1665

Step 1 – Terminal SD formula

$$\text{SD}(r_T) = \sqrt{\frac{\sigma^2}{2k}(1 - e^{-2kT})}$$

Step 2 – Plug in

$$2kT = 0.5, \quad 1 - e^{-0.5} = 0.3935, \quad \frac{\sigma^2}{2k} = \frac{(1.26\%)^2}{0.05} = 31.752\ (\%^2)$$
$$\text{SD} = \sqrt{31.752 \times 0.3935} = \sqrt{12.49} = 3.53\% \approx 353 \text{ bp}$$

Interpretation

GARP's own example (eq. 14.27): mean reversion cuts the 10-year SD from 398 bp to 353 bp. The expected value of 7.4812% is a red herring for the SD.

Why the other options are wrong

A is incorrect. Used $1 - e^{-kT}$ in place of $1 - e^{-2kT}$: $\sqrt{31.752 \times 0.2212} = 2.65\%$.

C is incorrect. Ignored mean reversion: $1.26\%\sqrt{10} = 3.98\%$.

D is incorrect. Used σ^2/k instead of $\sigma^2/(2k)$, doubling the variance: 5.00%.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. • **Reference:** Tuckman & Serrat 3e, Ch.9, 14.5, eqs. 14.26 and 14.27

Solution MR-14 g.6 **Answer: B**

Workbook A25.5 • S2, S3

Step 1 – Half-life

$$h = \frac{\ln 2}{k} = \frac{0.69315}{0.08} = 8.66 \text{ years}$$

$\Rightarrow$ A shock to the short rate takes about 8.7 years to decay halfway toward θ. r_0 and θ do not enter.

Step 2 – Variance factor

$$\frac{1 - e^{-2kT}}{2k} = \frac{1 - e^{-1.6}}{0.16} = \frac{1 - 0.20190}{0.16} = 4.9881$$

Step 3 – Terminal standard deviation

$$\text{SD} = \sigma\sqrt{4.9881} = 1.10\% \times 2.2334 = 2.46\%$$

$\Rightarrow$ The short rate in 10 years is normal with a standard deviation of about 246 bps around its expectation.

Interpretation

Without mean reversion the standard deviation would be $1.10\%\sqrt{10} = 3.48\%$: the pull toward θ dampens dispersion, particularly at long horizons, and the same effect makes the volatility term structure slope downward. For the committee, an 8.7-year half-life means rate news is fairly long-lived in this calibration.

Why the other options are wrong

A is incorrect. Dropped the 2 in the exponent: $1.10\%\sqrt{(1 - e^{-0.8})/0.16} = 2.04\%$.
C is incorrect. Ignored mean reversion: $\sigma\sqrt{T} = 1.10\%\sqrt{10} = 3.48\%$.
D is incorrect. Used $1/k = 12.5$ years (the exponential time constant) instead of $\ln 2/k$.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. MR-14 h: Describe the effectiveness of the Vasicek Model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.5 eqs. 14.25 to 14.27, Figure 14.5

MR-15 • The Art of Term Structure Models: Volatility and Distribution • Vardeez: VERY IMPORTANT

MR-15 a

P1 CONCEPT

Describe the short-term rate process under a model with time-dependent volatility.

Workbook: A7 Ho-Lee, Time-Dependent Drift and Volatility • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: in August 2026 the term structure questions included "time-dependent drift and vol: theory question" next to the Ho-Lee plug-in calculation.
Vardeez: "Model 3" (time-dependent volatility). For the whole chapter he lists the CIR model, the concept of terminal distributions, Model 3 and the lognormal model, and asks you to understand the interpretation.

Reported exam shapes

- **A7·S2** Ho-Lee and time-dependent volatility models: the rate in the upper nodes (the calculation is in MR-15 b).
- **A7·S7** The August plain form: a theory question on time-dependent drift and volatility.

What the questions look like: "which statement is correct" about what $\sigma(t)$ depends on, what Model 3 does and does not change relative to Ho-Lee, and the sibling traps (mean reversion, CIR, lognormal, Gauss+). The August theory question is Workbook A7.5, filed under MR-14 e; read it with this block.

1. The general process

The previous reading (MR-14) let the *drift* depend on time (Ho-Lee). This reading lets the *volatility* depend on time as well:

$$\boxed{dr = \underbrace{\lambda(t)\,dt}_{\text{time-dependent drift}} + \underbrace{\sigma(t)\,dw}_{\text{time-dependent volatility}}} \qquad dw \sim N\left(0, \sqrt{dt}\right)$$

- $\sigma(t)$ is a **deterministic function of calendar time**. It is the annualised basis-point volatility of the short rate *at date* t: if $\sigma(1) = 1.26\%$ and $\sigma(2) = 1.20\%$, the short rate's volatility is 126 bp per year in one year and 120 bp per year in two years.
- $\sigma(t)$ contains **no** r. Volatility does not depend on the level of rates. (Rate-level dependence is the next part of the reading: CIR has $\sigma\sqrt{r}$, the lognormal model σr.)
- $\sigma(t)$ can rise, fall or stay flat from period to period. The general model imposes no shape.
- How it nests the earlier models: Model 1 is $dr = \sigma\,dw$; Model 2 adds a constant drift λ; Ho-Lee makes the drift $\lambda(t)$; the time-dependent volatility model also makes the volatility $\sigma(t)$.

2. Model 3: the special case

To make the general model concrete, Tuckman uses **Model 3**:

$$\boxed{dr = \lambda(t)\,dt + \sigma\,\mathrm{e}^{-\alpha t}\,dw}$$

- σ is the volatility at $t = 0$; with $\alpha > 0$ the volatility **declines exponentially toward zero**. (It could have been designed to decline to some other constant; Model 3 just uses zero.)
- α is the **decay rate**. $\alpha = 0$ gives back Ho-Lee (constant σ), not Vasicek.

- The distribution of rates stays **normal**: Model 3 is not a lognormal model and negative rates remain possible.

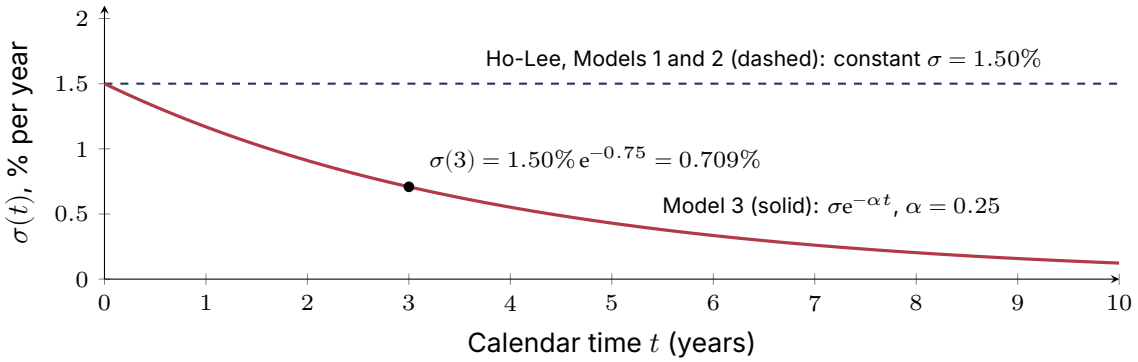


Figure 1. The volatility of the short rate as a function of calendar time. In Model 3 it starts at σ and decays toward zero; in Ho-Lee it is the same at every date. Each step in a tree uses the volatility in force at that date.

3. Why build it: fitting option prices

- Just as a time-dependent **drift** is used to fit many **bond or swap rates** (Ho-Lee), a time-dependent **volatility** is used to fit many **option prices**.
- **Caplets.** A caplet pays $\max(r - K, 0)$ on its expiry date, so its value depends on the distribution of the short rate at that date. Caplets expiring on different dates imply different volatilities; the flexibility of $\lambda(t)$ and $\sigma(t)$ lets the model match them all. Deterministic volatility functions are popular with market makers in interest rate options (caps and floors).

4. What Model 3 does *not* change

- **Still a parallel-shift model.** As for any model without mean reversion, a shock to the short rate moves all rates by the same amount.
- **Term structure of volatility is flat.** At any date the volatility is the same across maturities; because $\sigma(t)$ changes over time, the structure is "flat at levels that change over time, but it is still always flat".
- **No mean reversion.** The decay in volatility is not a pull toward a long-run rate; it only changes how wide future shocks are.

> **Correction to the source notes.** Vardeez's notes (MR-15 c) say that in Model 3 "the volatility term structure is declining". GARP (Tuckman, Section 15.1) says the opposite: Model 3's term structure of volatility is **flat** at every date, at a level that falls over time; it is the mean-reverting (Vasicek) models whose term structure of volatility slopes **downward**. What declines in Model 3 is $\sigma(t)$ *through time*, not volatility *across maturities*. Use GARP's version.

5. The Model 3 versus Vasicek trap

- Set Model 3's initial volatility and decay rate α equal to Vasicek's σ and mean-reversion speed k: the **standard deviations of the terminal distributions are identical**.
- Also make Model 3's drift match Vasicek's average rate path: the **terminal distributions are identical**, an even stronger result.

- But the models remain very different: Model 3 is a parallel-shift model with a flat volatility term structure; Vasicek has non-parallel shifts and a downward-sloping volatility term structure. Same terminal distribution, completely different shift behaviour.

	Ho-Lee	Model 3	Vasicek
Process	$\lambda(t)dt + \sigma dw$	$\lambda(t)dt + \sigma e^{-\alpha t} dw$	$k(\theta - r)dt + \sigma dw$
Volatility	Constant	Falls with time, not with r	Constant
Mean reversion	No	No	Yes
Shifts	Parallel	Parallel	Non-parallel
Vol. term structure	Flat	Flat, level changes over time	Downward sloping
Fits	Bond/swap rates	Option (caplet) prices	Economic intuition
Distribution	Normal	Normal	Normal

Exam traps

- **Time, not level.** $\sigma(t)$ depends on calendar time only. Rate-level dependence ($\sigma\sqrt{r}, \sigma r$) is CIR or lognormal.
- **Decay is not mean reversion.** Model 3 can reproduce the Vasicek terminal distribution but stays a parallel-shift model.
- **Flat, not declining, volatility term structure** at each date (see the correction box).
- $\alpha = 0$ **gives Ho-Lee**, not Vasicek, and not Model 1 (the drift is still $\lambda(t)$).
- **Drift versus volatility roles.** Time-dependent drift fits bond and swap rates; time-dependent volatility fits option prices. Swapping the two is a standard distractor.
- **No shape restriction.** The general $\sigma(t)$ can increase, decrease or stay constant; "must be increasing" is false.
- **Reading** $\sigma(t)$**.** The argument is the date in years: $\sigma(2)$ is the volatility at $t = 2$, not over the second half-year.

Practice questions: MR-15 a

Question MR-15 a.1 Bank Q705 • AnalystPrep Q.1670 (rebuilt)

A term structure model specifies the short-term rate process as $dr = \lambda(t)\,dt + \sigma(t)\,dw$. On which factor does the volatility of the short rate depend in this model?

A. The current level of the short-term rate, as in the CIR model.

B. Time only, through the function $\sigma(t)$.

C. Neither the rate level nor time; volatility is a constant as in the Ho-Lee model.

D. The drift $\lambda(t)$, since both are functions of time.

Question MR-15 a.2 Bank Q709 • AnalystPrep Q.1681 (rebuilt)

A model of the short rate has the form $dr = \lambda(t)\,dt + \sigma(t)\,dw$, where t is measured in years. If $\sigma(1) = 1.25\%$ and $\sigma(2) = 1.28\%$, which statement is correct?

A. The volatility of the short rate is 125 bps per year at $t = 1$ year and 128 bps per year at $t = 2$ years.

B. The volatility of the short rate is 128 bps per year at $t = 1$ year and 125 bps per year at $t = 2$ years.

C. The volatility of the short rate is 125 bps per year at $t = 6$ months and 128 bps per year at $t = 1$ year.

D. The volatility of the short rate is 1.25% in the first year and 1.28% in the second year, constant within each year, so volatility is declining over time.

Question MR-15 a.3 Bank Q706 • BT notes P2.T5.306.3 (rebuilt, key corrected)

Consider the following summary of the interest rate models in Tuckman:

I. Model 1 is a simple equilibrium model with normally distributed rates and no drift.

II. Model 2 adds a constant drift (risk premium) to Model 1 and is also an equilibrium model with normally distributed rates.

III. The Ho-Lee model has normally distributed rates and a time-dependent drift, and can be calibrated so that it is arbitrage-free.

IV. The Vasicek model has normally distributed rates with mean reversion toward a long-run value.

V. Model 3 replaces the normal distribution of rates with a lognormal distribution while keeping volatility constant.

Which of the statements are true?

A. I and II only

B. III and V only

C. I, II, III and IV only

D. All of the statements

Question MR-15 a.4 Bank Q714 • Schweser Module Quiz 15.1 Q1 (rebuilt)

Regarding the validity of time-dependent volatility models, which of the following statements is (are) correct?

I. They are flexible because volatility can change from period to period; however, volatility must be an increasing function of time.

II. Time-dependent volatility functions are useful for pricing interest rate caps and floors.

A. I only.

B. II only.

C. Both I and II.

D. Neither I nor II.

Question MR-15 a.5 Bank Q715 • AnalystPrep Q.1671 (rebuilt)

A dealer wants to price a 5-year cap on the short rate r, made up of caplets expiring each period. Market prices of caplets at different expiries imply different volatilities. A caplet pays $\max(r - K, 0)$ on its expiry date, so its value depends on the distribution of the short rate at that date. Which model feature best allows the dealer to match all the caplet prices?

A. A single constant volatility, since all caplets share the same strike.

B. A time-dependent drift only, with constant volatility.

C. Mean reversion with constant volatility.

D. A deterministic, time-dependent volatility function $\sigma(t)$ calibrated to the caplets.

Question MR-15 a.6 New • built from GARP Ch.15 (Tuckman 15.1) and Vardeez

A rates desk replaces its Ho-Lee model with Model 3, $dr = \lambda(t)\,dt + \sigma e^{-\alpha t}\,dw$ with $\alpha > 0$, so that it can quote caplets across expiries. The head of market risk asks how the short-rate process has changed. Which of the following statements is correct?

A. Because volatility decays over time, a shock to the short rate now moves short-term rates more than long-term rates, so Model 3 produces non-parallel shifts.

B. At any given date the term structure of volatility is flat, but its level changes over time as $\sigma e^{-\alpha t}$ declines.

C. Because volatility now depends on the level of the short rate, basis-point volatility falls when rates fall, which prevents negative rates.

D. Setting $\alpha = 0$ turns Model 3 into the Vasicek model with constant volatility.

Question MR-15 a.7 New • built from GARP Ch.15 (Tuckman 15.1) and Vardeez

A model validator compares a Vasicek model, $dr = k(\theta - r)\,dt + \sigma\,dw$, with a Model 3 implementation in which the initial volatility equals the Vasicek σ, the decay rate α equals the Vasicek mean-reversion speed k, and the time-dependent drift $\lambda(t)$ is set to match the average path of rates in the Vasicek model. Which of the following conclusions is correct?

A. The two models produce identical terminal distributions of the short rate, yet Model 3 remains a parallel-shift model while Vasicek does not.

B. Because the terminal distributions are identical, the two models imply identical yield curve shifts and therefore identical hedge ratios.

C. The terminal standard deviations would match only if α were set equal to half of k, so the two terminal distributions differ.

D. Model 3 has a wider terminal distribution than Vasicek at every horizon, because Model 3 has no mean reversion.

Solutions • MR-15 a

Solution MR-15 a.1

Answer: B

Bank Q705 • AnalystPrep Q.1670

Step 1 – Read the diffusion term

The shock is $\sigma(t)\,dw$; the coefficient $\sigma(t)$ is a deterministic function of calendar time.

Step 2 – Check for the rate level

Nothing in $\sigma(t)$ involves r, so basis-point volatility does not change with the level of rates.
$\Rightarrow$ Volatility depends on time only.

Interpretation

Letting σ vary with t is exactly what distinguishes the time-dependent volatility model (Model 3: $\sigma e^{-\alpha t}$) from Ho-Lee. Rate-level dependence arrives later in the reading with CIR ($\sigma\sqrt{r}$) and the lognormal model (σr).

Why the other options are wrong

A is incorrect. Rate-level dependence belongs to CIR and the lognormal model; $\sigma(t)$ has no r in it.

C is incorrect. Describes Ho-Lee or Model 1 with constant σ; the point of $\sigma(t)$ is that volatility changes over time.

D is incorrect. Both coefficients are functions of t, but they are separate functions; volatility is not driven by the drift.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eq. 15.1

Solution MR-15 a.2

Answer: A

Bank Q709 • AnalystPrep Q.1681

Step 1 – What the argument of $\sigma(\cdot)$ means

$\sigma(t)$ is the annualised basis-point volatility of the short rate evaluated at date t, in years.

Step 2 – Read the two values

$\sigma(1) = 1.25\%$: 125 bp per year at $t = 1$. $\sigma(2) = 1.28\%$: 128 bp per year at $t = 2$.
$\Rightarrow$ Volatility rises slightly with time here; the general model does not force it to decline.

Interpretation

This mirrors GARP's own illustration ($\sigma(1) = 1.26\%$, $\sigma(2) = 1.20\%$ means 126 bp per year in one year and 120 bp per year in two). The general $\sigma(t)$ can go either way; only the Model 3 special case forces a decline.

Why the other options are wrong

B is incorrect. Swaps the two values.

C is incorrect. Misreads the arguments as six months and one year; they are one and two years.

D is incorrect. Reads $\sigma(t)$ as a step function by year and then gets the direction wrong: $1.28\% > 1.25\%$, so volatility increases.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1

Solution MR-15 a.3 — Answer: C

Bank Q706 • BT notes P2.T5.306.3

Step 1 – Check I to IV

I: Model 1, $dr = \sigma dw$: normal, no drift. True. II: Model 2, $dr = \lambda dt + \sigma dw$: constant drift, normal. True. III: Ho-Lee, $dr = \lambda(t)dt + \sigma dw$, drift fitted to the market curve: arbitrage-free. True. IV: Vasicek, $dr = k(\theta - r)dt + \sigma dw$: normal with mean reversion. True.

Step 2 – Check V

Model 3 is $dr = \lambda(t)dt + \sigma e^{-\alpha t} dw$: rates stay *normal* and volatility is *time-dependent*, not constant. False on both counts.

$\Rightarrow$ I, II, III and IV only.

Interpretation

Model 3 differs from Ho-Lee in one place only, the volatility term. Lognormal rates are a separate model (Model 4) later in the same reading.

Why the other options are wrong

A is incorrect. Correct statements but incomplete: omits III and IV, which are also true.
B is incorrect. Includes the false statement V and omits I, II and IV.
D is incorrect. Accepts V; Model 3 is neither lognormal nor constant-volatility.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.9 and Ch.10, 15.1 eq. 15.2

Solution MR-15 a.4 — Answer: B

Bank Q714 • Schweser Module Quiz 15.1 Q1

Step 1 – Statement I

The general $\sigma(t)$ can be increasing, decreasing or constant between periods. No monotonicity is required. False.

Step 2 – Statement II

A cap or floor is a series of caplets or floorlets paying off on different dates, so its price depends on $\sigma(t)$ at several future dates. The flexibility of $\sigma(t)$ is exactly what helps. True.

$\Rightarrow$ Only II is correct.

Interpretation

Flexibility to set the volatility date by date is the selling point of these models; the "must be increasing" clause is the planted absolute qualifier.

Why the other options are wrong

A is incorrect. Accepts the false restriction in I and rejects the valid cap and floor use.
C is incorrect. Accepts the false restriction in I.
D is incorrect. Rejects II, though caps and floors are the main application.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1; SchweserNotes 2026 Book 1, Reading 15

Solution MR-15 a.5 — Answer: D

Bank Q715 • AnalystPrep Q.1671

Step 1 – What drives each caplet's value

Each caplet depends on the spread of the short-rate distribution at its own expiry, so a different effective volatility is needed for each expiry.

Step 2 – Which model input controls that spread date by date

A deterministic $\sigma(t)$ sets the volatility at each date, so it can be fitted to one caplet expiry at a time.

$\Rightarrow$ A time-dependent volatility function matches the whole strip of caplets.

Interpretation

GARP's parallel: a time-dependent drift fits many bond or swap rates, a time-dependent volatility fits many option prices. Market makers in rate options use deterministic volatility functions for exactly this reason.

Why the other options are wrong

A is incorrect. One constant volatility cannot reproduce caplet prices that imply different volatilities by expiry.

B is incorrect. Drift moves the mean of the rate, not its dispersion, so it cannot fit volatility differences (that is the Ho-Lee role).

C is incorrect. Mean reversion with constant volatility imposes a fixed volatility pattern with few parameters; it is the choice for valuing and hedging, not for fitting caplet prices exactly.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1

Solution MR-15 a.6 Answer: B

New • built from GARP Ch.15

Step 1 – What Model 3 adds

Only the volatility term changes: $\sigma e^{-\alpha t}$ instead of a constant σ. It depends on calendar time, not on the level of r.

Step 2 – What it keeps

There is still no mean reversion, so a shock to the short rate moves all rates equally: a parallel-shift model. At any date, volatility is the same across maturities, so the term structure of volatility is flat; because $\sigma(t)$ falls with time, the level of that flat structure falls with time.

$\Rightarrow$ "Flat at levels that change over time, but still always flat."

Interpretation

Decay in volatility is easy to mistake for mean reversion. It changes how wide future shocks are, not where rates are pulled to, so it cannot create non-parallel shifts.

Why the other options are wrong

A is incorrect. Polarity of the model class: without mean reversion, shocks move all rates equally. Non-parallel shifts need mean reversion (Vasicek).

C is incorrect. Sibling swap: rate-level dependence is CIR ($\sigma\sqrt{r}$) or lognormal (σr). Model 3 is normal, so negative rates remain possible.

D is incorrect. Wrong nesting: with $\alpha = 0$ the volatility is a constant σ and the drift is still $\lambda(t)$, which is Ho-Lee. Vasicek needs a drift $k(\theta - r)$.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eq. 15.2

Solution MR-15 a.7 Answer: A

New • built from GARP Ch.15

Step 1 – Terminal standard deviation

With Model 3's initial volatility and decay rate equal to Vasicek's σ and k, the standard deviations

of the terminal distributions are identical (Tuckman, Section 15.1).

Step 2 – Terminal mean

With Model 3's drift matched to Vasicek's average rate path, the means match too, so the terminal distributions are identical.

Step 3 – What still differs

Model 3 has no mean reversion: it is a parallel-shift model with a flat volatility term structure. Vasicek is a non-parallel shift model with a downward-sloping volatility term structure.
$\Rightarrow$ Same terminal distribution, different dynamics.

Interpretation

Matching terminal distributions makes the two models agree on anything that depends only on the distribution of the short rate at a date. Anything that depends on how the whole curve moves (hedge ratios across maturities, curve trades) still differs.

Why the other options are wrong

B is incorrect. Identical terminal distributions do not imply identical curve dynamics; Model 3 shifts in parallel, Vasicek does not.

C is incorrect. Wrong condition: GARP sets α equal to k, not $k/2$, and the terminal standard deviations then match exactly.

D is incorrect. With the parameters matched there is no difference in width; the decay in Model 3's volatility does the job mean reversion does in Vasicek.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1 and Figure 15.1

MR-15 • The Art of Term Structure Models: Volatility and Distribution • Vardeez: VERY IMPORTANT

MR-15 b

P1 CALC

Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility.

Workbook: A7 Ho-Lee, Time-Dependent Drift and Volatility • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: the August 2026 recall map lists "Ho-Lee and time-dependent volatility models: calculate value of rate in upper nodes" and "Ho-Lee & Model 3, 2-step plug and play calculation".
Vardeez: "Model 3" calculation; the concept of terminal distributions.

Reported exam shapes

- **A7·S2** Ho-Lee and time-dependent volatility models: the rate in the upper nodes.
- **A7·S3** Ho-Lee and Model 3 two-step plug and play.

What the questions look like: one-step dr with a given dw realisation; the rate at the upper node after two steps, each step with its own λ and $\sigma(t)$; the standard deviation of the rate change at a date or over several steps; and how that standard deviation behaves as the horizon grows. Expect red-herring inputs (r_0 when only a change is asked).

1. One step of Model 3

For a step of length dt starting at date t:

$$\boxed{dr = \lambda(t)\,dt + \sigma \mathrm{e}^{-\alpha t}\,dw} \qquad dw \sim N\left(0, \sqrt{dt}\right)$$

- **Drift** scales with dt (annual λ times the fraction of a year).
- **Shock** is the volatility in force at that date times dw. If you are given a *realised* dw it already contains $\sqrt{dt}$; if you are building a tree, the shock is $\pm\sigma(t)\sqrt{dt}$.
- **Standard deviation of the step** $= \sigma(t)\sqrt{dt} = \sigma \mathrm{e}^{-\alpha t}\sqrt{dt}$.

Worked example (Vardeez, Schweser). $r_0 = 5\%$, $\lambda = 0.24\%$, $\sigma = 1.50\%$, $\alpha = 0.30$, one month, realised $dw = 0.2$:

$$dr = 0.24\% \times \tfrac{1}{12} + 1.50\%\,\mathrm{e}^{-0.3/12} \times 0.2 = 0.02\% + 0.29\% = 0.31\% \;\Rightarrow\; r = 5.31\%$$

With constant volatility (Ho-Lee) the same draw gives $0.02\% + 0.30\% = 0.32\%$: slightly larger, because Model 3's volatility has already started to decay.

2. Two steps on a tree

Each step uses *its own* drift and *its own* volatility. With the volatility at the start of each step, $\sigma_1 = \sigma(0)$ and $\sigma_2 = \sigma(dt)$:

$$r_{uu} = r_0 + (\lambda_1 + \lambda_2)\,dt + [\sigma_1 + \sigma_2]\,\sqrt{dt} \qquad \mathrm{SD}_{2\text{ steps}} = \sqrt{dt}\,\sqrt{\sigma_1^2 + \sigma_2^2}$$

The two steps are independent, so **variances add**, not standard deviations. Because $\sigma_1 \neq \sigma_2$, the up-down and down-up nodes no longer coincide.

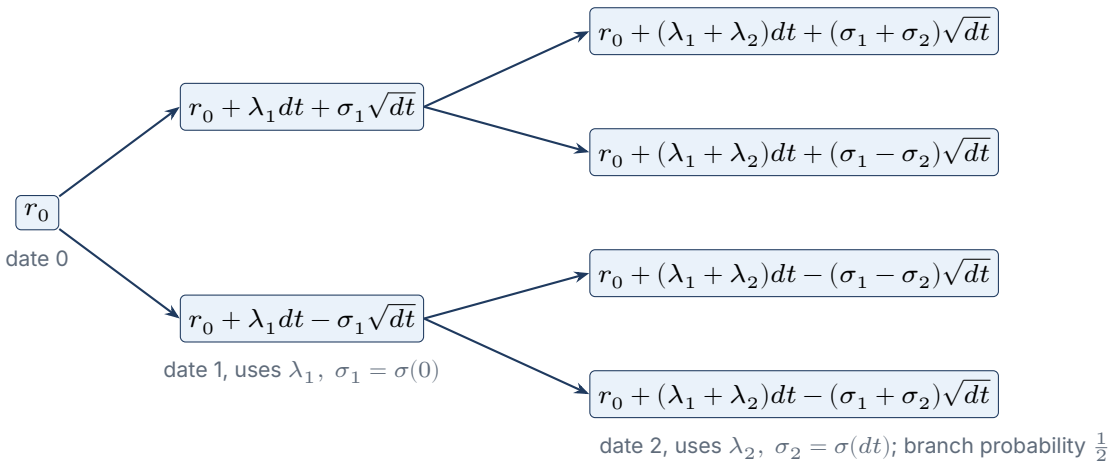


Figure 2. A two-step Model 3 tree. The top node collects both drifts and both volatilities. With $\sigma_2 < \sigma_1$ the two middle nodes differ by $2(\sigma_1 - \sigma_2)\sqrt{dt}$, so the tree does not recombine; with $\sigma_1 = \sigma_2$ it collapses to the Ho-Lee tree.

3. Behaviour of the standard deviation

Two different standard deviations appear in questions. Keep them apart:

- **Per-step (instantaneous) volatility** $\sigma(t) = \sigma e^{-\alpha t}$: starts at σ and **declines toward zero**. Example: $\sigma = 1.20\%$, $\alpha = 0.25$: $\sigma(2) = 0.73\%$, $\sigma(4) = 0.44\%$.
- **Standard deviation of the terminal distribution** (the short rate at horizon T, seen from today): accumulates all the steps, so it **rises with the horizon**, quickly at first and then more slowly, **levelling off** because later steps add less and less variance.

For reference (GARP presents the result without derivation; it is the same formula as the Vasicek variance with k replaced by α, which is why the two models' terminal standard deviations match):

$$\mathrm{SD}(r_T) = \sigma\sqrt{\frac{1 - e^{-2\alpha T}}{2\alpha}} \xrightarrow{T\to\infty} \frac{\sigma}{\sqrt{2\alpha}} \qquad \text{versus} \qquad \sigma\sqrt{T} \text{ with constant volatility}$$

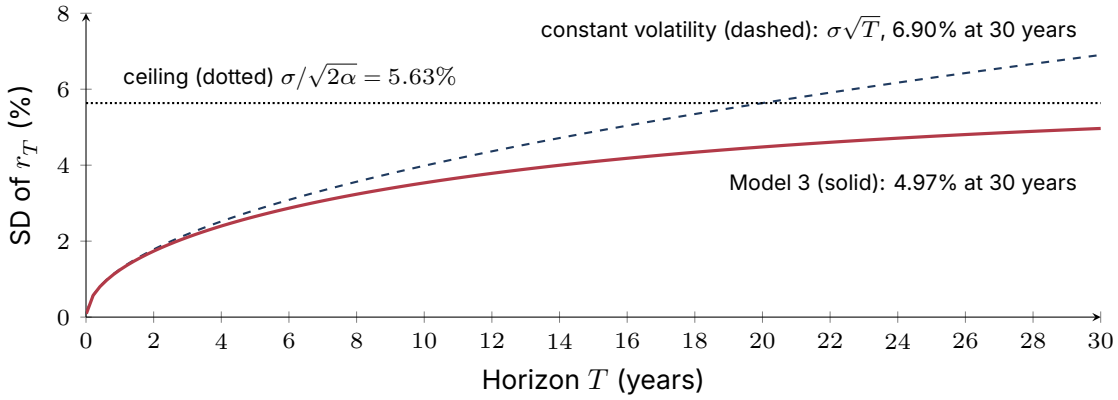


Figure 3. Standard deviation of the terminal distribution of the short rate in Model 3 (GARP's Figure 15.1 parameters) against a constant-volatility model. Model 3 rises rapidly at first, then more slowly, toward a ceiling instead of growing without bound.

Formula card: MR-15 b

Model 3	$dr = \lambda(t)\,dt + \sigma e^{-\alpha t}\,dw, \quad dw \sim N(0, \sqrt{dt})$
Volatility at date t	$\sigma(t) = \sigma e^{-\alpha t}$ (per year)
One-step change	$dr = \lambda\,dt + \sigma(t)\,dw$ (realised dw already includes $\sqrt{dt}$)
One-step SD	$\sigma(t)\sqrt{dt}$
Upper node, two steps	$r_0 + (\lambda_1 + \lambda_2)dt + (\sigma_1 + \sigma_2)\sqrt{dt}$
SD after n steps	$\sqrt{dt}\,\sqrt{\sigma_1^2 + \cdots + \sigma_n^2}$ (variances add)
Terminal SD (reference)	$\sigma\sqrt{(1 - e^{-2\alpha T})/(2\alpha)}$, ceiling $\sigma/\sqrt{2\alpha}$
Useful numbers	$\sqrt{1/12} = 0.28868, \quad \sqrt{0.25} = 0.5, \quad \sqrt{1/252} = 0.06299$

Exam traps

- **$\sqrt{dt}$ versus dt.** Drift scales with dt; the volatility perturbation scales with $\sqrt{dt}$. Using $\sigma\,dt$ shrinks the branches; $\lambda\sqrt{dt}$ inflates the drift.
- **Constant versus decaying volatility.** Use $\sigma e^{-\alpha t}$ at each step's date. Reusing the initial σ is Ho-Lee.
- **Which date.** The step that begins at $t = 3$ uses $\sigma(3)$; check the convention in the stem (start of step versus end of step).
- **Adding standard deviations.** Independent steps add in variance. $\sigma_1\sqrt{dt} + \sigma_2\sqrt{dt}$ is the upper node's distance from the centre, not a standard deviation.
- **Double $\sqrt{dt}$.** A realised dw "already scaled" must not be multiplied by $\sqrt{dt}$ again.
- **Sign of the exponent.** $e^{+\alpha t}$ makes volatility grow, which is not Model 3.
- **Instantaneous versus terminal.** $\sigma(t)$ declines to zero; the SD of the terminal distribution rises and levels off. Answers that say "the terminal SD falls to zero" confuse the two.
- **Intermediate result.** One step's move is not the two-step answer; the date-1 node is not the date-2 node.

Practice questions: MR-15 b

Question MR-15 b.1

Bank Q710 • AnalystPrep Q.1682 (rebuilt)

In Model 3, $dr = \lambda(t)\,dt + \sigma e^{-\alpha t}\,dw$ with $\sigma = 1.20\%$, $\alpha = 0.25$ and $\lambda(t)$ constant. Which statement best describes the standard deviation of the short-rate change per unit of dw?

A. It starts at 1.20% per year and declines exponentially toward zero, reaching about 0.44% at $t = 4$ years.

B. It starts at 1.20% per year and declines exponentially toward zero, reaching about 0.73% at $t = 4$ years.

C. It starts at 1.20% and rises exponentially with time, reaching about 3.26% at $t = 4$ years.

D. It is constant at 1.20% per year; α only affects the drift.

Question MR-15 b.2

New • built from GARP Ch.15 (approved sample)

A fixed-income analyst models the short rate with GARP's Model 3, $dr = \lambda(t)\,dt + \sigma e^{-\alpha t}\,dw$, using monthly steps. The parameters are $r_0 = 4.0\%$, $\sigma = 1.50\%$ per year, $\alpha = 0.25$, and drift $\lambda(t) = 0.20\%$ per year. What is the standard deviation of the rate change over the one-month step that begins at $t = 3$ years?

A. 5.9 bp

B. 20.5 bp

C. 43.3 bp

D. 70.9 bp

Question MR-15 b.3

Bank Q707 • BT P2.T5.306.1

An analyst applies Tuckman's Model 3 for the short-term rate: $dr = \lambda(t)\,dt + \sigma e^{-\alpha t}\,dw$. Assumptions:

- Time step is monthly, $dt = 1/12$ year
- Initial short rate $r(0) = 4.00\%$
- Basis-point volatility parameter $\sigma = 3.00\%$ per year
- Drift $\lambda(t)$ is constant at 1.30% per year
- $\alpha = 0.380$

In the 60th month ($t = 5.0$ years), the random uniform draw is 0.20, so the standard normal draw is NORM.S.INV$(0.20) = -0.8416$ and $dw = -0.8416 \times \sqrt{1/12} = -0.2430$ (already time-scaled). Which is nearest to the change in the rate, dr, in the 60th month?

A. -1.00%

B. 0.00%

C. $+1.00\%$

D. $+2.00\%$

Question MR-15 b.4

Bank Q708 • GARP PE1 Q54 (rebuilt)

A bank models the short rate with $dr = \lambda(t)\,dt + \sigma(t)\,dw$, where dw has mean zero and standard deviation $\sqrt{dt}$, using a monthly binomial tree ($dt = 1/12$) with up and down moves equally likely. Inputs:

Current short rate (date 0)	2.75%
Drift $\lambda(1)$	0.0018 per year
Drift $\lambda(2)$	0.0030 per year
Annualized volatility $\sigma(1)$	0.0050
Annualized volatility $\sigma(2)$	0.0080

What is the change in the rate from date 0 to the upper node (two up-moves) at date 2?

A. 16 bps

B. 42 bps

C. 26 bps

D. 86 bps

Question MR-15 b.5

Workbook A7.4 • S2, S3 • Map #7

An interest rate options market maker at a Japanese bank quotes caplets off Model 3, $dr = \lambda(t)dt + \sigma e^{-\alpha t}dw$, on a quarterly tree ($dt = 0.25$). For each quarter the desk uses the volatility at the start of that quarter, $\sigma(0)$ for the first step and $\sigma(0.25)$ for the second. Inputs (annualised):

- $r_0 = 2.40\%$; $\lambda_1 = 0.80\%$; $\lambda_2 = 1.00\%$
- Initial basis-point volatility $\sigma = 150$ bps; decay rate $\alpha = 0.60$

Which of the following is closest to the short rate in the upper node after two quarters, and to the standard deviation of the short rate after two quarters?

A. Upper node 4.246%; standard deviation 0.990%

B. Upper node 4.246%; standard deviation 1.396%

C. Upper node 4.350%; standard deviation 1.061%

D. Upper node 4.696%; standard deviation 0.990%

Question MR-15 b.6

New • built from GARP Ch.15 and Workbook A7

A risk analyst at a pension fund runs a coarse annual tree ($dt = 1$) for Model 3 to sense-check the dispersion of the short rate three years ahead. For each year the tree uses the volatility at the start of that year. The inputs are:

- Current short rate: 3.10%; drift $\lambda(t)$: 0.25% per year
- Initial basis-point volatility $\sigma = 120$ bps per year; decay rate $\alpha = 0.20$

Which of the following is closest to the standard deviation of the short rate after three years on this tree?

A. 1.43%

B. 1.75%

C. 2.08%

D. 2.99%

Question MR-15 b.7 New • built from GARP Ch.15, Figure 15.1

A market maker in interest rate options uses Model 3 with $\sigma = 126$ bps per year and $\alpha = 0.025$, the parameters of GARP's Figure 15.1. A junior analyst asks how the standard deviation of the terminal distribution of the short rate behaves as the horizon lengthens. Which of the following is the correct answer?

A. It declines toward zero, because the volatility $\sigma e^{-\alpha t}$ decays to zero.

B. It grows in proportion to $\sqrt{T}$, reaching about 6.90% at 30 years, as in a constant-volatility model.

C. It rises rapidly at first and then more slowly, levelling off toward about 5.63%, so it is about 4.97% at 30 years.

D. It rises rapidly at first and then levels off at the initial volatility of 1.26%.

Solutions • MR-15 b

Solution MR-15 b.1

Answer: A

Bank Q710 • AnalystPrep Q.1682

Step 1 – Volatility at $t = 0$

$\sigma e^{0} = 1.20\%$.

Step 2 – Volatility at $t = 4$

$$1.20\% \times e^{-0.25\times 4} = 1.20\% \times e^{-1} = 0.4415\% \approx 0.44\%$$

$\Rightarrow$ With $\alpha > 0$, $e^{-\alpha t}$ decays toward zero, so the per-unit-dw volatility falls exponentially.

Interpretation

This is the instantaneous volatility at a date, not the standard deviation of the terminal distribution, which keeps rising with the horizon (Question MR-15 b.7).

Why the other options are wrong

B is incorrect. 0.73% is the value at $t = 2$ ($1.20\% \times e^{-0.5}$), not $t = 4$.
C is incorrect. Dropped the minus sign in the exponent: $1.20\% \times e^{+1} = 3.26\%$.
D is incorrect. Confuses Model 3 with Ho-Lee or Model 1 (constant σ); α governs the decay of volatility, not the drift.

LO: MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eq. 15.2

Solution MR-15 b.2

Answer: B

New • built from GARP Ch.15

Step 1 – Volatility in force at $t = 3$

$$\sigma(3) = 1.50\% \times e^{-0.25\times 3} = 1.50\% \times 0.47237 = 0.70855\% \text{ per year}$$

$\Rightarrow$ Three years in, the volatility has decayed to less than half its starting value.

Step 2 – Scale to one month

$$\text{SD} = \sigma(3)\sqrt{dt} = 0.70855\% \times \sqrt{1/12} = 0.70855\% \times 0.28868 = 0.2045\% \approx 20.5 \text{ bp}$$

Interpretation

Only the volatility term matters for the standard deviation of a step. $r_0 = 4.0\%$ and $\lambda = 0.20\%$ are given but not needed: the drift shifts the mean of the change, not its dispersion.

Why the other options are wrong

A is incorrect. Scaled by dt instead of $\sqrt{dt}$: $0.70855\%/12 = 0.0590\% = 5.9$ bp.
C is incorrect. Left out the decay, using the initial σ: $1.50\% \times 0.28868 = 0.4330\% = 43.3$ bp (the Ho-Lee answer).
D is incorrect. Left out the time scaling: $\sigma(3) = 0.7086\% = 70.9$ bp is an annual figure.

LO: MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eq. 15.2

Solution MR-15 b.3

Answer: B

Bank Q707 • BT P2.T5.306.1

Step 1 – Drift over the month

$$\lambda\, dt = 1.30\% \times \tfrac{1}{12} = 0.1083\%$$

Step 2 – Volatility in force at $t = 5$

$$\sigma \mathrm{e}^{-\alpha t} = 3.00\% \times \mathrm{e}^{-0.380\times 5} = 3.00\% \times \mathrm{e}^{-1.9} = 3.00\% \times 0.14957 = 0.4487\%$$

$\Rightarrow$ By month 60 the volatility has decayed from 300 bp to about 45 bp per year.

Step 3 – Shock

$$0.4487\% \times (-0.2430) = -0.1090\%$$

Step 4 – Total change

$$dr = 0.1083\% - 0.1090\% = -0.0007\% \approx 0.00\%$$

Interpretation

The negative draw almost exactly cancels the monthly drift because the decayed volatility is so small. $r(0) = 4.00\%$ is not needed. The stem gives dw already scaled by $\sqrt{1/12}$; multiplying by $\sqrt{dt}$ again would be a double count.

Why the other options are wrong

A is incorrect. Ignored the decay: $0.1083\% + 3.00\% \times (-0.2430) = -0.62\%$, nearest -1.00%.
C is incorrect. Used the full annual drift without dt: $1.30\% - 0.109\% = +1.19\%$, nearest $+1.00\%$.
D is incorrect. Unscaled drift, no decay and the sign of the draw flipped: $1.30\% + 3.00\% \times 0.2430 = +2.03\%$.

LO: MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eq. 15.2

Solution MR-15 b.4

Answer: B

Bank Q708 • GARP PE1 Q54

Step 1 – Up-move from date 0 to date 1

$$\lambda(1)\, dt + \sigma(1)\sqrt{dt} = \tfrac{0.0018}{12} + 0.0050 \times 0.28868 = 0.000150 + 0.001443 = 0.001593\ (15.9\ \text{bp})$$

Step 2 – Up-move from date 1 to date 2

$$\lambda(2)\, dt + \sigma(2)\sqrt{dt} = \tfrac{0.0030}{12} + 0.0080 \times 0.28868 = 0.000250 + 0.002309 = 0.002559\ (25.6\ \text{bp})$$

Step 3 – Change to the upper node

$$15.9 + 25.6 = 41.5\ \text{bp} \approx 42\ \text{bp} \quad (\text{upper node } 2.75\% + 0.415\% = 3.165\%)$$

⇒ Each step uses that step's own drift and volatility, and the two up-moves add.

Interpretation

This is the plug-and-play shape reported from the exam (A7·S3): two steps, two drifts, two volatilities. The current rate of 2.75% is only needed if the question asks for the level of the node.

Why the other options are wrong

A is incorrect. 16 bp is only the first step.
C is incorrect. 26 bp is only the second step.
D is incorrect. Added the annual drifts without dt: $18 + 30 = 48$ bp, plus the volatility terms of 37.5 bp, gives 86 bp.

LO: MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eq. 15.1

Solution MR-15 b.5 **Answer: A**

Workbook A7.4 • S2, S3

Step 1 – Volatility for each step

$$\sigma(0) = 1.50\%, \qquad \sigma(0.25) = 1.50\% \times \mathrm{e}^{-0.60\times 0.25} = 1.50\% \times 0.8607 = 1.2911\%$$

⇒ Volatility decays from 150 to about 129 bp per year over the first quarter.

Step 2 – Perturbations per step

$$\sigma(0)\sqrt{dt} = 1.50\% \times 0.5 = 0.7500\%, \qquad \sigma(0.25)\sqrt{dt} = 1.2911\% \times 0.5 = 0.6455\%$$

Step 3 – Drift over two quarters

$$(\lambda_1 + \lambda_2)\, dt = (0.80\% + 1.00\%) \times 0.25 = 0.450\%$$

Step 4 – Upper node

$$r_{uu} = 2.40\% + 0.450\% + 0.7500\% + 0.6455\% = 4.246\%$$

⇒ Two up moves, each sized by the volatility in force during that quarter.

Step 5 – Standard deviation after two quarters

$$\mathrm{SD} = \sqrt{dt}\,\sqrt{\sigma(0)^2 + \sigma(0.25)^2} = 0.5 \times \sqrt{1.50^2 + 1.2911^2}\,\% = 0.5 \times 1.9791\% = 0.990\%$$

⇒ The two independent steps add in variance. Because volatility decays, the SD rises more slowly with horizon than $\sigma\sqrt{t}$.

Interpretation

A constant-volatility Ho-Lee tree with the same σ gives a wider distribution (1.061%); the decay is what lets the desk match caplets that expire later at lower volatilities. With σ changing between steps the up-down and down-up nodes no longer coincide (2.954% and 2.746%), and Model 3 remains a parallel-shift model with a flat (time-varying) volatility term structure.

Why the other options are wrong

B is incorrect. Correct node, but added the standard deviations linearly, $0.7500\% + 0.6455\% = 1.396\%$: this is the upper node's distance from r_0 plus drift, not a standard deviation.

C is incorrect. Ignored the decay and used $\sigma = 1.50\%$ in both steps (Ho-Lee): $2.85\% + 2 \times 0.75\% = 4.350\%$, SD $1.50\%\sqrt{0.5} = 1.061\%$.
D is incorrect. Scaled the drift by $\sqrt{dt}$ instead of dt: $2.40\% + 1.80\% \times 0.5 + 1.3955\% = 4.696\%$.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eqs. 15.1 and 15.2, Figure 15.1

Solution MR-15 b.6 — Answer: B

New • built from GARP Ch.15 and Workbook A7

Step 1 – Volatility at the start of each year

$$\sigma(0) = 1.20\%, \quad \sigma(1) = 1.20\%\,\mathrm{e}^{-0.20} = 0.9825\%, \quad \sigma(2) = 1.20\%\,\mathrm{e}^{-0.40} = 0.8044\%$$

Step 2 – Add the variances ($dt = 1$)

$$\text{SD} = \sqrt{1} \times \sqrt{1.20^2 + 0.9825^2 + 0.8044^2}\,\% = \sqrt{3.0523}\,\% = 1.747\%$$

$\Rightarrow$ About 1.75%: each later year adds less variance because its volatility has decayed.

Interpretation

A constant-volatility model would give $1.20\%\sqrt{3} = 2.08\%$; the decay narrows the three-year distribution by about a sixth. The current rate and the drift move the centre of the distribution, not its width, so they are not needed. (On a fine grid the same inputs approach $\sigma\sqrt{(1 - \mathrm{e}^{-2\alpha T})/(2\alpha)} = 1.59\%$; the coarse annual tree overstates it because it holds each year's volatility at its starting value.)

Why the other options are wrong

A is incorrect. Used the volatility at the end of each year, $\sigma(1)$, $\sigma(2)$, $\sigma(3)$: $\sqrt{0.9653 + 0.6470 + 0.4337} = 1.43\%$. The stem says start of year.
C is incorrect. Ignored the decay (Ho-Lee): $1.20\% \times \sqrt{3} = 2.08\%$.
D is incorrect. Added the standard deviations linearly: $1.20 + 0.98 + 0.80 = 2.99\%$.

LO: MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, eq. 15.2, Figure 15.1

Solution MR-15 b.7 — Answer: C

New • built from GARP Ch.15, Figure 15.1

Step 1 – Separate the two standard deviations

The instantaneous volatility $\sigma\mathrm{e}^{-\alpha t}$ falls toward zero. The terminal standard deviation adds up the variance of every step from today to T, so it can only rise with T.

Step 2 – How fast it rises

Early steps carry nearly the full σ, so the terminal SD rises rapidly at first; later steps add less and less, so it rises more slowly. GARP's Figure 15.1 shows exactly this shape.

Step 3 – Where it levels off

$$\frac{\sigma}{\sqrt{2\alpha}} = \frac{1.26\%}{\sqrt{0.05}} = 5.63\%, \qquad \text{SD}(r_{30}) = 1.26\%\sqrt{\frac{1-\mathrm{e}^{-1.5}}{0.05}} = 4.97\%$$

Interpretation

The shape mirrors the effect of mean reversion on horizon standard deviation in Vasicek, which is why matching α to k makes the two models' terminal standard deviations identical (MR-15 a).

Why the other options are wrong

A is incorrect. Confuses the instantaneous volatility at a date with the dispersion of the terminal distribution: the first declines, the second accumulates.

B is incorrect. Ignores the decay: $1.26\% \times \sqrt{30} = 6.90\%$ is the constant-volatility answer.

D is incorrect. Right shape, wrong ceiling: the initial annual volatility is not the limit of the accumulated standard deviation.

LO: MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. • **Reference:** Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1, Figure 15.1

MR-16 • The Vasicek and Gauss+ Models • Vardeez: VERY IMPORTANT

MR-16 a

P1 CONCEPT

Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics.

Workbook: A25 Gauss+ and Vasicek • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: "Gauss+" was reported from the August 2026 sitting, and the factor-change calculation came up in its plain form.
Vardeez: new chapter, highly testable.

Reported exam shapes

- **A25·S5** Gauss+ multi-factor structure: no shock on r, the $r \to m \to l \to \mu$ cascade, two sources of risk.

What the questions look like: "which statement is correct / which is false" about the cascade, what each factor represents, why the short rate has no random term, what the "+" means, and why the model can produce a hump-shaped volatility term structure. Workbook A25.1 tests the no-shock feature through a one-line calculation.

1. Starting point: what Vasicek cannot do

Vasicek, $dr = k(\theta - r)\,dt + \sigma\,dw$, has one factor (the short rate) that mean-reverts to a constant θ (full treatment in MR-14 f, g and MR-16 b). Because its only volatility is the shock to r, the volatility of forward rates is σe^{-kt}: **highest at the short end and declining with term**. Empirically, short-rate volatility is *low*, because central banks peg the policy rate, and the term structure of volatility **rises quickly with term and then flattens or declines** (a hump). One factor also cannot hedge intermediate- and shorter-term bonds well. Gauss+ is built to fix this.

2. The structure: a cascade of three factors

Risk-neutral dynamics (Tuckman eqs. 16.9 to 16.12):

$$
\begin{aligned}
dr_t &= -\alpha_r\,(r_t - m_t)\,dt && \text{no random term}\\
dm_t &= -\alpha_m\,(m_t - l_t)\,dt + \sigma_m\left(\rho\,dW_t^1 + \sqrt{1-\rho^2}\,dW_t^2\right)\\
dl_t &= -\alpha_l\,(l_t - \mu)\,dt + \sigma_l\,dW_t^1\\
\mathrm{E}[dW_t^1\,dW_t^2] &= 0
\end{aligned}
$$

Factor	Reverts to	What it represents
r_t **short rate**	m_t, fastest	The policy rate (fed funds target). Pegged by the central bank and moved gradually, so it gets no shock of its own.
m_t **medium-term**	l_t, slower	Business cycle and monetary policy: where the market thinks the short rate will be in two to three years (a "forward shadow rate" that can go deeply negative near the zero bound).
l_t **long-term**	μ, slowest	Long-term expectations of inflation and the real rate (demographics, production technology).
μ **constant**	–	Long-run target: a long-term expectation of the short rate plus a risk premium.

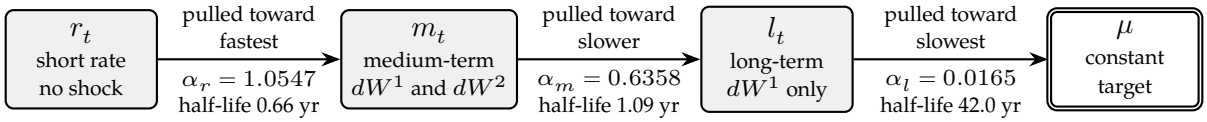


Figure 1. The cascade of convergence. Each factor is pulled toward the one on its right; only μ is constant. Half-lives $\ln 2/\alpha$ use GARP's Table 16.1 estimates: about eight months, 13 months and 42 years.

Drift signs. Each drift is $-\alpha \times (\text{factor} - \text{target})$: a factor above its target drifts down, below its target drifts up, at its target has no drift. If $r_t < m_t$ the short rate rises toward m_t; if $r_t = m_t$ it does not move.

3. The single most examinable feature: no shock on r

- Equation 16.9 has **no volatility term**. Over a short interval the short rate is fixed and only drifts toward m, mirroring a central bank that pegs the policy rate but moves the target over time.
- This keeps **short-rate volatility low**. Mean-reverting factors on their own generate a downward-sloping volatility term structure; suppressing the shock on r lets Gauss+ match the empirically observed **hump**: low volatility at the very short end, a peak at intermediate or longer terms, then a decline.
- **The name.** "Gauss": rates are normally (Gaussian) distributed. "+": the model has **three factors (state variables) but only two sources of risk**, dW^1 and dW^2. The short rate is a state variable that is not itself a source of risk.

4. How the shocks are wired

- l loads on dW^1 only: its fluctuation has volatility $\sigma_l\sqrt{dt}$.
- m loads ρ on dW^1 and $\sqrt{1-\rho^2}$ on dW^2. Since $\rho^2 + (1-\rho^2) = 1$, its fluctuation has volatility exactly $\sigma_m\sqrt{dt}$ (eq. 16.13).
- Covariance $\text{Cov}(dm, dl) = \rho\,\sigma_m\sigma_l\,dt$, so $\text{Corr}(dm, dl) = \rho$ (eqs. 16.14, 16.15).
- The medium and long factors move continuously (expectations move smoothly); the real short rate jumps on policy dates, which the model approximates by a continuous path.
- These are risk-neutral dynamics. In GARP's application only the long-term factor earns a risk premium.

5. What the mean-reversion speeds imply

- Expected ordering: $\alpha_r > \alpha_m > \alpha_l$ (Table 16.1: 1.0547, 0.6358, 0.0165).
- **Larger** α_r: changes in m and l make their way into the term structure of rates faster.
- **Larger** α_m: m converges to l faster, so m affects longer-term yields less.
- **Smaller** α_l: l converges to μ more slowly, so the effect of l on all longer-term yields is more similar, that is, **more parallel**.
- Result (GARP Figure 16.7): the short factor affects only the very short end; the medium factor drives the two- to three-year region; the long factor has its biggest impact at six to eight years and, beyond 10 years, is the **sole** factor driving forward rate changes.

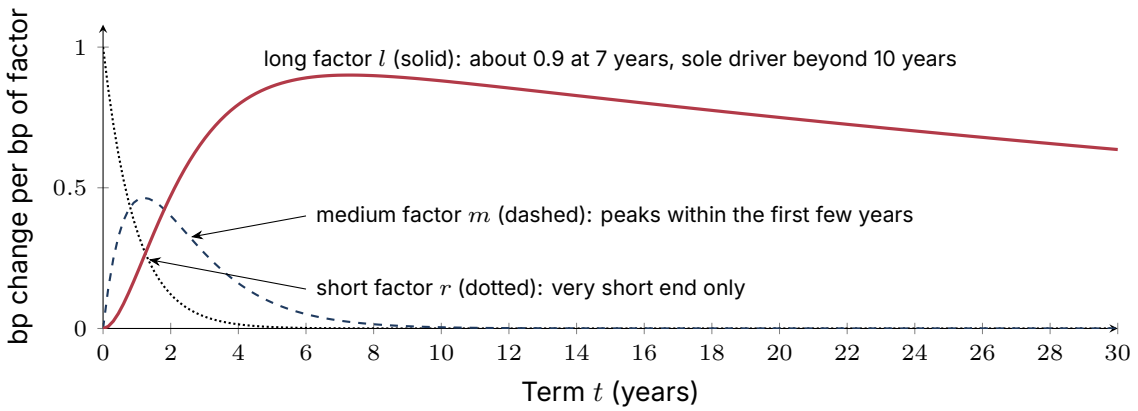


Figure 2. Change in the expected short rate at term t (an instantaneous forward rate, convexity ignored) for a one-unit change in each factor, computed from the Table 16.1 mean-reversion speeds. The pattern is the one in GARP's Figure 16.7.

Table 16.1: Gauss+ parameters, US Treasury zeros, January 2014 to January 2022

Mean reversion	$\alpha_r = 1.0547$ (half-life 0.66 yr), $\alpha_m = 0.6358$ (1.09 yr), $\alpha_l = 0.0165$ (42.01 yr)
Volatilities	$\sigma_m = 109.2$ bp, $\sigma_l = 96.4$ bp (no σ_r)
Correlation, target	$\rho = 0.212$, $\mu = 10.555\%$ (high, but l reverts to it very slowly and it includes a risk premium)

Exam traps

- **Shock on r.** Equation 16.9 has no dW term. Any statement giving the short rate its own volatility, or saying short-end volatility is the highest point of the curve, is wrong.
- **What the "+" means.** Three factors, two sources of risk. Not "an added factor to make volatility slope down", not "three sources of risk".
- **Hump, not decline.** Gauss+ matches a hump-shaped volatility term structure; a monotonically declining one is the Vasicek picture.
- **Moving targets.** r reverts to m, m to l, l to μ. Only μ is fixed; the short rate does not revert to μ directly.
- **No shock is not no drift.** With no random term, r still drifts toward m.
- **Speed polarity.** Larger α means faster convergence and a shorter half-life; smaller α_l means a more parallel effect of l on long yields.
- **Gaussian.** Rates are normal in Gauss+, so negative rates are possible; it is not a lognormal model.

Practice questions: MR-16 a

Question MR-16 a.1

Bank Q744 • Schweser Module Quiz 16.1 Q2

The role of the medium-term factor in the Gauss+ model is that the factor:

A. remains constant over time.

B. directly determines the pricing of long-term bonds.

C. reflects economic influences, such as business cycles.

D. represents the risk premium in the interest rate structure.

Question MR-16 a.2

Bank Q740 • AnalystPrep Q.6520 (rebuilt)

In the Gauss+ model, suppose at some date $r_t > m_t$ and $m_t > l_t > \mu$. Ignoring random shocks, which statement best describes the drift of the factors?

A. r_t drifts down toward m_t, m_t drifts down toward l_t, and l_t drifts down toward μ.

B. r_t drifts up toward m_t, while m_t and l_t drift down.

C. r_t drifts down toward μ directly, bypassing m_t.

D. r_t does not move because it has no random shock.

Question MR-16 a.3

New • built from GARP Ch.16 (Tuckman 16.2) and Vardeez

A risk manager is reviewing a rates desk's new Gauss+ implementation, in which $dr_t = -\alpha_r(r_t - m_t)dt$, $dm_t = -\alpha_m(m_t - l_t)dt + \sigma_m(\rho\, dW_t^1 + \sqrt{1-\rho^2}\, dW_t^2)$ and $dl_t = -\alpha_l(l_t - \mu)dt + \sigma_l\, dW_t^1$. Which of the following statements about the model's structure is correct?

A. The model has three state variables but only two sources of risk, because the short-term rate has no random shock of its own.

B. The model has three sources of risk, one for each of r, m and l, and the "+" in its name refers to this third source.

C. The short-term rate carries its own volatility term, which makes short-end volatility the highest point of the volatility term structure.

D. Rates are lognormally distributed in Gauss+, which keeps the short-term rate above zero at all horizons.

Question MR-16 a.4

Bank Q742 • BT notes P2.T5.23.5.3

Each of the following is true about the Gauss+ model EXCEPT which is false?

A. In the cascade form, the short-term rate mean reverts to the medium-term factor, which reverts to the long-term factor.

B. Gauss+ has three state variables r_t, m_t, l_t, denoting the short-term rate, a medium-term factor, and a long-term factor.

C. The long-term factor mean reverts to a long-term constant level and reflects long-term trends such as inflation expectations and productivity.

D. The "+" reflects an added factor that ensures a downward-sloping term structure of volatility.

Question MR-16 a.5 Bank Q738 • AnalystPrep Q.6518 (rebuilt)

Which volatility term structure can the Gauss+ model be calibrated to replicate, in line with empirically observed interest rate behavior?

A. Volatility that is highest at the short end and declines steadily with maturity.

B. Volatility that rises steadily with maturity.

C. Low volatility at the short end, higher volatility at medium maturities, and declining volatility at long maturities (hump-shaped).

D. Constant volatility across all maturities.

Question MR-16 a.6 New • built from GARP Ch.16 (Tuckman 16.3)

A quantitative analyst is explaining to a portfolio manager how the mean-reversion parameters of the Gauss+ model shape the way factor changes pass through to the term structure of rates. Which of the following statements is consistent with the model?

A. A larger α_r makes changes in the factors m and l make their way into the term structure of rates faster.

B. A larger α_m makes m converge to l more slowly and increases the effect of m on longer-term yields.

C. A smaller α_l makes the effect of l on longer-term yields less parallel across terms.

D. A larger α_l lengthens the half-life of the long-term factor, so shocks to l persist longer.

Question MR-16 a.7 Workbook A25.1 • S5, S6 • Aug-2026 • Map #40

A junior risk analyst at a US asset manager is presenting the Gauss+ model to a group of newly hired analysts, using the book's estimated parameters: $\alpha_r = 1.0547$, $\alpha_m = 0.6358$, $\alpha_l = 0.0165$, $\sigma_m = 109.2$ bps, $\sigma_l = 96.4$ bps. Today the short-term rate is $r = 5.25\%$ and the medium-term factor is $m = 4.10\%$. The desk's previous one-factor Vasicek implementation used a short-rate volatility of 95 bps, and this month's realised increment is $dw = +0.20$ (already scaled by $\sqrt{dt}$). Which of the following is closest to the short-term rate one month from now ($dt = 1/12$) under Gauss+?

A. 5.149%

B. 5.189%

C. 5.339%

D. 5.351%

Solutions • MR-16 a

Solution MR-16 a.1

Answer: C

Bank Q744 • Schweser Module Quiz 16.1 Q2

Step 1 – Place m in the cascade

m is the middle level: r reverts to it and it reverts to l.

Step 2 – Its economic meaning

It captures business-cycle and monetary-policy influences: where the market believes the short rate will be in two to three years.
$\Rightarrow$ It reflects economic influences such as business cycles.

Interpretation

Because r reverts to m quickly, m acts as a leading indicator of the short rate; GARP's Figure 16.8 shows it rising in 2014, a couple of years before the Fed began raising rates.

Why the other options are wrong

A is incorrect. m is dynamic, with its own drift toward l and its own shocks. Only μ is constant.
B is incorrect. Long-term yields are driven by l; beyond 10 years l is the sole driver of forward rate changes.
D is incorrect. The risk premium sits in μ (and, in GARP's application, is earned by the long-term factor), not in m.

LO: MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2

Solution MR-16 a.2

Answer: A

Bank Q740 • AnalystPrep Q.6520

Step 1 – Short rate

$-\alpha_r(r_t - m_t) < 0$ since $r_t > m_t$: r falls toward m.

Step 2 – Medium-term factor

$-\alpha_m(m_t - l_t) < 0$ since $m_t > l_t$: m falls toward l.

Step 3 – Long-term factor

$-\alpha_l(l_t - \mu) < 0$ since $l_t > \mu$: l falls toward μ, the only fixed target.
$\Rightarrow$ Every factor sits above its target, so all three drift down.

Interpretation

Each drift is proportional to the gap from the next factor in the cascade. The sign rule is the same for all three: above target, down; below target, up.

Why the other options are wrong

B is incorrect. Wrong sign on the short-rate drift: with $r_t > m_t$ the drift is negative.
C is incorrect. The short rate reverts to the moving target m_t, not directly to μ.
D is incorrect. Confuses having no shock with having no drift: r still drifts toward m.

LO: MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2, eqs. 16.9 to 16.11

Solution MR-16 a.3 — Answer: A

New • built from GARP Ch.16

Step 1 – Count the state variables

Describing the state of the world requires knowing r, m and l: three factors.

Step 2 – Count the sources of risk

Only dW^1 and dW^2 appear; dr has no random term.
$\Rightarrow$ Three factors, two sources of risk: that is the "+".

Interpretation

The missing shock on r is deliberate. It reflects a central bank pegging the policy rate, keeps short-rate volatility low and lets the model produce the hump-shaped volatility term structure seen in the data.

Why the other options are wrong

B is incorrect. Miscounts the sources of risk: there are only two. The "+" marks a state variable that is *not* a source of risk.
C is incorrect. Polarity: the short rate has no volatility term, which makes short-end volatility *low*, not the highest point.
D is incorrect. Wrong distribution: "Gauss" means rates are normally distributed, so negative rates are possible.

LO: MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2

Solution MR-16 a.4 — Answer: D

Bank Q742 • BT notes P2.T5.23.5.3

Step 1 – Check A to C

A restates the cascade, B the three state variables, and C the long-term factor reverting to the constant μ and reflecting long-run trends. All true.

Step 2 – Check D

The "+" denotes three factors with only two sources of risk. And the model is built to match a *hump-shaped* volatility term structure, not a downward-sloping one.
$\Rightarrow$ D is the false statement.

Interpretation

A monotonically declining volatility structure is what one-factor mean-reverting models (Vasicek) produce; Gauss+ removes the shock on r precisely to escape it.

Why the other options are wrong

A is incorrect. True: the cascade.
B is incorrect. True: the three state variables.
C is incorrect. True: l reverts to μ and represents long-term trends.

LO: MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2

Solution MR-16 a.5 — Answer: C

Bank Q738 • AnalystPrep Q.6518

Step 1 – Short end

The short rate has no shock and only follows m, so short-end volatility is low.

Step 2 – Intermediate terms

The medium and long factors carry their own shocks, so volatility rises toward intermediate maturities.

Step 3 – Long end

Mean reversion damps the effect of shocks at long terms, so volatility declines.
$\Rightarrow$ Low, then higher, then declining: a hump.

Interpretation

Empirical and implied volatility term structures have this hump largely because of central banks. A one-factor mean-reverting model can only produce a curve that is highest at the short end.

Why the other options are wrong

A is incorrect. A monotonic decline from the short end is the Vasicek shape.
B is incorrect. Volatility does not rise forever; it declines at long maturities.
D is incorrect. A flat volatility structure belongs to parallel-shift models without mean reversion (Model 1, Ho-Lee, Model 3).

LO: MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2, Figure 16.6

Solution MR-16 a.6 — Answer: A

New • built from GARP Ch.16

Step 1 – Role of α_r

r reverts to m at speed α_r. The faster it does so, the faster changes in m (and, through m, in l) show up in the short rate and hence in the term structure.

Step 2 – Check the other speeds

Larger α_m: m converges to l *faster* and affects longer-term yields *less*. Smaller α_l: l converges to μ more slowly, so its effect on all longer-term yields is *more* parallel. Half-life is $\ln 2/\alpha$, so a larger α_l *shortens* it.
$\Rightarrow$ Only A matches Tuckman.

Interpretation

This is why the estimated $\alpha_l = 0.0165$ (half-life 42 years) makes the long factor behave almost like a parallel shift of the long end, while $\alpha_r = 1.0547$ confines the short factor to the very short end.

Why the other options are wrong

B is incorrect. Polarity flip twice: a larger α_m means faster convergence and a smaller effect on longer yields.
C is incorrect. Polarity flip: a smaller α_l makes the effect of l *more* parallel.
D is incorrect. Half-life is inversely related to speed: $\ln 2/\alpha_l$ falls as α_l rises.

LO: MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.3, Figure 16.7

Solution MR-16 a.7 **Answer: A**

Workbook A25.1 • S5, S6 • Aug-2026

Step 1 – State the equation

$$dr_t = -\alpha_r(r_t - m_t)\,dt \qquad \text{(Equation 16.9, no random term)}$$

$\Rightarrow$ The short-term rate is pushed only toward the medium-term factor, the way a central bank moves its policy target gradually.

Step 2 – Gap to the target

$$r - m = 5.25\% - 4.10\% = 1.15\%$$

$\Rightarrow$ The short rate sits 115 bps above where the medium-term factor says it is heading.

Step 3 – Annualised drift

$$-\alpha_r(r - m) = -1.0547 \times 1.15\% = -1.2129\%$$

$\Rightarrow$ At the current gap, the short rate would fall about 121 bps over a full year.

Step 4 – One-month change and new level

$$dr = -1.2129\% \times \tfrac{1}{12} = -0.1011\%, \qquad r_{1/12} = 5.25\% - 0.1011\% = 5.149\%$$

$\Rightarrow$ With no shock in Equation 16.9, the short rate's path over dt is deterministic: the expected and actual moves coincide.

Interpretation

The short rate falls about 10 bps toward the medium-term factor, reflecting a market that expects easing. Neither σ_m, σ_l nor the Vasicek volatility and dw enter: the absence of a shock on r is what keeps short-rate volatility low and lets Gauss+ match the hump-shaped volatility term structure.

Why the other options are wrong

B is incorrect. Wrong speed (sibling swap): used $\alpha_m = 0.6358$: $5.25\% - 0.6358 \times 1.15\%/12 = 5.189\%$.

C is incorrect. Added a Vasicek-style shock to the short rate: $5.149\% + 0.95\% \times 0.20 = 5.339\%$. Equation 16.9 has no random term.

D is incorrect. Sign flip: wrote the drift as $+\alpha_r(r-m)dt$, so the rate moves away from its target: $5.25\% + 0.1011\% = 5.351\%$.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eq. 16.9, Table 16.1

MR-16 • The Vasicek and Gauss+ Models • Vardeez: VERY IMPORTANT

MR-16 b

P1 CONCEPT

Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model.

Workbook: A25 Gauss+ and Vasicek • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A25, Gauss+ and Vasicek); the Gauss+ structure shape (A25·S5) hits this LO's concept.
Vardeez: new chapter, highly testable.

Reported exam shapes

- **A25·S5** Gauss+ multi-factor structure (no shock on r, the cascade, two sources of risk) set against Vasicek's single factor.

What the questions look like: side-by-side comparisons (factors, targets, speeds, volatility term structure), matching a volatility curve to a model, and "which model for which task". The Vasicek calculations (expected rate, half-life, $\theta = r_\infty + \lambda/k$) sit under MR-14 g; Workbook A25.4 (filed there) is the one that also tags this LO.

1. Side by side

	Vasicek	Gauss+
Process	$dr = k(\theta - r)dt + \sigma dw$	Cascade $r \to m \to l \to \mu$ (eqs. 16.9 to 16.12)
Factors	One: the short rate	Three state variables r, m, l; two sources of risk
Short-rate shock	Present: $\sigma\, dw$	Absent: r only drifts toward m
Target	One constant θ	Moving targets m and l; only μ is constant
Speeds	One, k	Three, $\alpha_r > \alpha_m > \alpha_l$
Volatility parameters	One, σ (fits one point of the curve)	σ_m, σ_l and correlation ρ
Vol. term structure	Forward vol σe^{-kt}: declining from the short end	Hump: low short end, peak at intermediate terms, then declining
Distribution	Normal	Normal
Strength	Simple; analytic formulae; one factor explains much of the variability at long maturities	Flexible fit of curve shapes and volatilities; factors have economic meaning
Weakness	Too inflexible for most uses (see below)	More parameters to estimate; more complex
Typical use	Teaching; simple pricing and hedging of long bonds	Proprietary relative-value and macro trading, hedging, derivatives across the curve

2. Vasicek: features and where it works

- **Mean reversion** sets the expected path: below θ the rate is expected to rise, above θ to fall, with drift $k(\theta - r)$. GARP example: $r = 2\%$, $k = 0.0165$, $\theta = 11\%$ gives 14.85 bp a year, about 1.2 bp a month; with $\sigma = 95$ bp the monthly shock has SD $95\sqrt{1/12} = 27.4$ bp.
- **Risk premium is indistinguishable from expectations**: with a constant premium λ, the risk-neutral target is $\theta = r_\infty + \lambda/k$, and infinitely many (r_∞, λ) pairs give the same prices.
- **Analytic formulae** for $\mathrm{E}[r_t]$, $\mathrm{V}[r_t]$, forwards and spots; half-life $\ln 2/k$ (42 years for $k = 0.0165$).
- **Downward-sloping volatility**: forward-rate volatility (and sensitivity to r) is $\sigma \mathrm{e}^{-kt}$, capturing the empirical fact that volatility falls at longer terms.
- **Where it is reasonable**: pricing, comparing and hedging *long-term* bonds first callable after some intermediate number of years (it can match 10- to 30-year noncallables and the 10-year rate volatility), and trading long bonds against a model calibrated to the 10-year rate.

Why it is not widely used (three reasons): (1) not flexible enough to approximate the wide variety of observed term structures; calibrating to one point relies too heavily on the model for all the others; (2) a second factor captures significantly more variability, and more than one factor is needed to hedge intermediate- and shorter-term bonds; (3) it cannot capture a volatility term structure that rises quickly with term and then flattens or declines, so it cannot handle volatility products spread across the curve or hedge one segment with another.

3. Gauss+: what it adds

- Three cascading factors with economic meaning (policy rate, business cycle, long-run inflation and real rate).
- No shock on r, so it can match the **hump-shaped** volatility term structure (Figure 1).
- Several volatility parameters and a correlation: it fits the market's volatility term structure, not a single point.
- **Applications.** Gauss+ is popular with practitioners for **proprietary trading**. In **relative value** trading, positions are hedged against factor moves, for example receiving forwards that are high to the model and converging while paying ones that are low. In **macro** trading, the trader takes views on the factors and splits long forwards into expectations, risk premium and convexity. The model can also price bonds, swaps, caps and floors.
- **Estimation in stages** (MR-16 d): speeds from regression coefficients, then volatilities and ρ from the volatility term structure, then μ. Much easier than maximum likelihood.

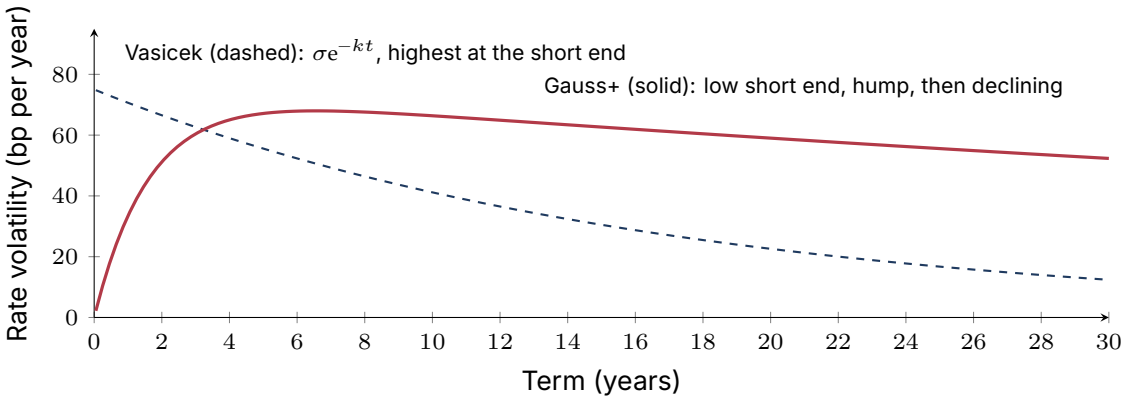


Figure 1. Illustrative shapes of the volatility term structure. A one-factor mean-reverting model can only decline from the short end; Gauss+, with no shock on the short rate, rises from a low short-end volatility to a hump before declining, as observed in markets (GARP Figure 16.6).

Exam traps

- **Reversed roles.** Vasicek is the single-factor model with a fixed target; Gauss+ is the multi-factor cascade with moving targets.
- **Stochastic volatility.** Neither model has it: both use constant volatility parameters. The difference is the number of factors and the missing shock on r.
- **Matching curves.** Monotonic decline from the short end: Vasicek. Hump (rise then decline): Gauss+. Flat: parallel-shift models. Steadily rising: neither.
- **One volatility point.** Calibrating Vasicek's single σ to one maturity leaves the others mismatched; it does not "automatically" fit the curve.
- **Vasicek is not useless.** It still works for long-term bonds callable after an intermediate period and for teaching; the weakness is at intermediate and short maturities and in volatility products.
- **Gauss+ in practice.** It is a practitioner tool for proprietary trading and hedging, not a teaching toy.

Practice questions: MR-16 b

Question MR-16 b.1

Bank Q752 • AnalystPrep Q.6524 (rebuilt)

Which statement correctly contrasts the mean reversion dynamics of the Vasicek and Gauss+ models?

A. Both models revert to a single fixed long-run rate; Gauss+ merely adds more noise.

B. In Vasicek the target moves dynamically, while in Gauss+ the target is fixed.

C. In Gauss+ all three factors revert to the fixed parameter μ at the same speed.

D. In Vasicek the short rate reverts to a fixed θ; in Gauss+ r reverts to m, m reverts to l, and l reverts to a fixed μ, with speeds fastest for r and slowest for l.

Question MR-16 b.2

Bank Q754 • BT P2.T5.25.7.2 (rebuilt)

The table shows annualized rate volatility (%) by maturity for three candidate volatility term structures.

Maturity (years)	1	3	5	7	10	14
Curve X	1.81	1.48	1.21	0.99	0.74	0.49
Curve Y	1.80	2.20	2.40	2.00	1.00	0.60
Curve Z	0.80	1.20	1.50	1.80	2.10	2.40

Which assignment correctly matches the curves to the volatility term structures produced by the Vasicek and Gauss+ models?

A. Curve X is Vasicek; Curve Y is Gauss+.

B. Curve Y is Vasicek; Curve X is Gauss+.

C. Curve Z is Vasicek; Curve Y is Gauss+.

D. Curve X is Vasicek; Curve Z is Gauss+.

Question MR-16 b.3

Bank Q749 • AnalystPrep Q.1667 (rebuilt)

A risk manager calibrates the Vasicek model's volatility parameter so that the model matches the market volatility at one point on the interest rate curve. What is the most likely consequence, and how does Gauss+ differ?

A. The model will automatically match market volatility at all other maturities; Gauss+ gives no improvement.

B. The model will match other maturities only if the curve is hump-shaped; Gauss+ is calibrated to a single maturity as well.

C. The model will generally mismatch market volatility at other maturities, impairing derivative pricing and hedging; Gauss+, with several volatility parameters and a correlation, can replicate a hump-shaped volatility term structure.

D. The mismatch affects zero-coupon bond prices but not derivative prices; Gauss+ is no different in this respect.

Question MR-16 b.4

Bank Q753 • AnalystPrep Q.6525 (rebuilt)

Which limitation of the Vasicek model most directly reduces its usefulness for hedging a fixed-income portfolio across the yield curve?

A. It cannot price zero-coupon bonds in closed form.

B. It offers little flexibility in the shape of the curve, so it cannot match market rates accurately enough for effective hedging.

C. It assumes volatility is stochastic, making hedge ratios unstable.

D. It has no mean reversion, so rates drift without anchor.

Question MR-16 b.5 Bank Q745 • AP Mock 1 Q14 (rebuilt)

A bank's quant team has two tasks: (i) a training exercise illustrating the basic effect of mean reversion on a zero-coupon bond price, and (ii) pricing interest rate caps and floors consistently with the market's hump-shaped volatility term structure. Which model assignment is most appropriate?

A. Vasicek for both tasks, because its single-factor closed-form structure is sufficient for any rate derivative.

B. Vasicek for task (i) and Gauss+ for task (ii).

C. Gauss+ for task (i) and Vasicek for task (ii), because the more complex model is needed to teach mean reversion.

D. Either model for both tasks, because both are calibrated to the same volatility structure.

Question MR-16 b.6 New • built from GARP Ch.16 (Tuckman 16.1, 16.4)

The head of rates at an insurer is deciding which applications of the Vasicek and Gauss+ models to approve. Which of the following proposed uses is consistent with Tuckman's assessment of the two models?

A. Using Vasicek to price, compare and hedge long-term bonds that are first callable after some intermediate number of years, calibrated to match 10- to 30-year noncallable bonds and the 10-year rate volatility.

B. Using Vasicek to hedge intermediate- and shorter-term bonds, because a single factor explains most term structure variability at every maturity.

C. Restricting Gauss+ to classroom use, because practitioners avoid it for proprietary trading since its factors cannot be recovered from market data.

D. Using Gauss+ for caps across the curve because, unlike Vasicek, it adds a random shock to the short-term rate.

Solutions • MR-16 b

Solution MR-16 b.1 — Answer: D

Bank Q752 • AnalystPrep Q.6524

Step 1 – Vasicek

$dr = k(\theta - r)dt + \sigma dw$: one rate, one fixed target θ, one speed k.

Step 2 – Gauss+

A cascade of moving targets: r seeks m, m seeks l, l seeks μ. Only μ is fixed. Speeds are ordered $\alpha_r > \alpha_m > \alpha_l$.
$\Rightarrow$ D describes both correctly.

Interpretation

Replacing a single fixed target with a cascade of moving targets at very different speeds is what gives Gauss+ its richer dynamics across short, medium and long horizons.

Why the other options are wrong

A is incorrect. Gauss+ changes the mean-reversion structure itself, not just the noise.
B is incorrect. Reverses the models: the moving targets are in Gauss+.
C is incorrect. The factors revert to each other, not all to μ, and at different speeds.

LO: MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.1 and 16.2

Solution MR-16 b.2 — Answer: A

Bank Q754 • BT P2.T5.25.7.2

Step 1 – Vasicek's shape

One mean-reverting factor and one volatility parameter: volatility σe^{-kt} declines monotonically with maturity. Curve X is exactly $2\%\, e^{-0.1t}$ (1.81, 1.48, 1.21, 0.99, 0.74, 0.49).

Step 2 – Gauss+'s shape

No shock on r and separate medium and long factors give a hump: Curve Y rises to 2.40 at five years, then falls.

Step 3 – Curve Z

Volatility rising with maturity throughout is not a shape either model is described as producing.
$\Rightarrow$ X is Vasicek, Y is Gauss+.

Interpretation

In a "match the curve" question, look first for monotonic decline from the short end (Vasicek), then for a rise followed by a fall (Gauss+). A flat curve would point to a parallel-shift model without mean reversion.

Why the other options are wrong

B is incorrect. Reverses the models: Vasicek cannot produce a hump.
C is incorrect. Vasicek's volatility decays with maturity; it does not rise.
D is incorrect. Right on Vasicek, but Gauss+ gives a hump (rise then decline), not a perpetual rise.

LO: MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, Figures 16.3 and 16.6

Solution MR-16 b.3 — Answer: C

Bank Q749 • AnalystPrep Q.1667

Step 1 – One parameter, one point

Vasicek has a single volatility parameter σ, so it can be calibrated to the volatility at one maturity only (GARP: "the parameter σ can be used to approximate an implied option volatility at one point of the term structure").

Step 2 – Consequence elsewhere

Volatilities at other maturities are then dictated by the model's σe^{-kt} shape and generally miss the market's hump, impairing the pricing of volatility products and hedging across segments.

Step 3 – Gauss+

With σ_m, σ_l and ρ (and no shock on r), Gauss+ is calibrated to the whole volatility term structure.

Interpretation

This is Vasicek's third weakness in GARP: it cannot capture a volatility structure that rises quickly with term and then flattens or declines, so it cannot handle options spread across the term structure.

Why the other options are wrong

A is incorrect. Claims one parameter fits everything and denies the Gauss+ improvement.
B is incorrect. The fit depends on the number of volatility parameters, not on the curve being hump-shaped; and Gauss+ is not limited to one maturity.
D is incorrect. Volatility errors matter most for derivatives; Gauss+ was designed to fix exactly this.

LO: MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.1 and 16.3

Solution MR-16 b.4 — Answer: B

Bank Q753 • AnalystPrep Q.6525

Step 1 – What hedging across the curve needs

The model must match market rates (and sensitivities) at many maturities, so that bonds in one segment can hedge bonds in another.

Step 2 – Vasicek's limitation

One factor gives little flexibility in curve shapes; calibrating to one point relies too heavily on the model for the rest. More than one factor is needed to hedge intermediate- and shorter-term bonds.

$\Rightarrow$ Limited curve flexibility is the binding limitation.

Interpretation

Gauss+'s three factors address this; GARP lists inflexibility as the first reason Vasicek "is not very widely used".

Why the other options are wrong

A is incorrect. Vasicek does give closed-form zero-coupon prices; that is one of its advantages.
C is incorrect. Vasicek's volatility is constant, not stochastic.
D is incorrect. Mean reversion is Vasicek's defining feature.

LO: MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.1

Solution MR-16 b.5 Answer: B

Bank Q745 • AP Mock 1 Q14

Step 1 – Task (i)

Illustrating mean reversion on a zero-coupon bond is a simple application: Vasicek's single factor and analytic formulae make it the natural teaching tool.

Step 2 – Task (ii)

Caps and floors priced consistently with a hump-shaped volatility term structure need a model that fits volatilities across maturities: Gauss+.

$\Rightarrow$ Vasicek for (i), Gauss+ for (ii).

Interpretation

The choice follows the purpose: simplicity where it suffices, flexibility where volatility across the curve matters.

Why the other options are wrong

A is incorrect. Vasicek's single volatility parameter cannot reproduce the market's volatility term structure.

C is incorrect. Reverses the roles: mean reversion is taught most easily in the simpler model, and Vasicek cannot price the caps consistently.

D is incorrect. The models differ precisely in their volatility structure.

LO: MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.1 and 16.2

Solution MR-16 b.6 Answer: A

New • built from GARP Ch.16

Step 1 – Where Vasicek is reasonable

GARP: the model "can reasonably be used, for example, to price, compare values, and hedge long-term bonds that are first callable after some intermediate number of years", because it can match noncallables from 10 to 30 years and the volatility of 10-year rates.

Step 2 – Check the other proposals

Vasicek needs more than one factor to hedge intermediate and shorter bonds. Gauss+ is well known among practitioners for proprietary trading, and its factors are fitted daily to the fed funds rate and the two- and 10-year forwards. Gauss+ *removes* the shock on the short rate.

$\Rightarrow$ Only A is consistent with Tuckman.

Interpretation

A good comparison answer names what each model is *for*. Vasicek: simple long-end pricing and hedging. Gauss+: relative-value and macro trading and products spread across the curve.

Why the other options are wrong

B is incorrect. One factor explains a lot of variability mainly at longer maturities; more than one factor is needed for intermediate and shorter bonds.

C is incorrect. Reversed: Gauss+ is popular for proprietary trading, and its factors are recovered from market data every day.

D is incorrect. Sibling swap: Vasicek has the short-rate shock; Gauss+ is defined by its absence.

LO: MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.1, 16.3 and 16.4

MR-16 • The Vasicek and Gauss+ Models • Vardeez: VERY IMPORTANT

MR-16 c

P1 CALC

Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model.

Workbook: A25 Gauss+ and Vasicek • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: the Gauss+ factor changes were reported from August 2026 in their plain form.
Vardeez: new chapter, highly testable. (His notes skip this LO; the notes below are written from the GARP text.)

Reported exam shapes

- **A25·S6** August plain form: Gauss+ factor changes (short rate, medium-term factor, long-term factor).
- **A25·S5** The structure behind the bookkeeping: no shock on r, the cascade, two sources of risk.

What the questions look like: given today's r, m, l, the parameters and realised shocks, compute dr, dm or dl over dt. Distractors come from the wrong speed, the wrong target, a sign flip, a shock added to r, a missing dt, a double $\sqrt{dt}$ or the wrong loading on dW^2. Workbook A25.1 (short rate) is worked under MR-16 a.

1. Read each equation before plugging in

$$\begin{aligned} dr &= -\alpha_r\,(r-m)\,dt \\ dm &= -\alpha_m\,(m-l)\,dt + \sigma_m\left(\rho\,dW^1 + \sqrt{1-\rho^2}\,dW^2\right) \\ dl &= -\alpha_l\,(l-\mu)\,dt + \sigma_l\,dW^1 \end{aligned}$$

- dr has **no random term**: pure drift toward m, at speed α_r.
- dm is driven by **both** shocks, weighted ρ and $\sqrt{1-\rho^2}$; it reverts to l at α_m.
- dl is driven by dW^1 **only**; it reverts to μ at α_l.
- Drifts are annual, so multiply by dt. A *realised* dW already includes $\sqrt{dt}$; if the stem gives a standard normal draw z instead, use $dW = z\sqrt{dt}$.
- With all shocks zero (or for the *expected* change), only the drift terms remain.

2. Worked example: one month of factor changes

Today $r = 2.00\%$, $m = 3.00\%$, $l = 4.00\%$; Table 16.1 parameters ($\alpha_r = 1.0547$, $\alpha_m = 0.6358$, $\alpha_l = 0.0165$, $\sigma_m = 109.2$ bp, $\sigma_l = 96.4$ bp, $\rho = 0.212$, $\mu = 10.555\%$); $dt = 1/12$; realised $dW^1 = 0.10$, $dW^2 = -0.05$.

Short-term factor, drift only:

$$dr = -1.0547\,(2.00\% - 3.00\%)\,\tfrac{1}{12} = +0.0879\%$$

The short rate is below the medium-term factor, so it is pulled up.

Medium-term factor, drift plus both shocks, with $\sqrt{1-0.212^2} = 0.9773$:

$$\text{drift} = -0.6358\,(3.00\% - 4.00\%)\,\tfrac{1}{12} = +0.0530\%$$

$$\text{shock} = 1.092\%\,[0.212 \times 0.10 + 0.9773 \times (-0.05)] = -0.0302\%$$

$$dm = 0.0530\% - 0.0302\% = +0.0228\%$$

Long-term factor, drift plus dW^1 only:

$$\text{drift} = -0.0165\,(4.00\% - 10.555\%)\,\tfrac{1}{12} = +0.0090\%, \qquad \text{shock} = 0.964\% \times 0.10 = +0.0964\%$$

$$dl = 0.0090\% + 0.0964\% = +0.1054\%$$

What the answers show. The long-term factor moves the most (10.5 bp) despite the slowest mean reversion, because almost all of its move is the shock; its drift is under 1 bp. The short rate moves 8.8 bp with no randomness at all. Slow mean reversion does not mean a small move; it means the factor is not pulled back quickly once it has moved.

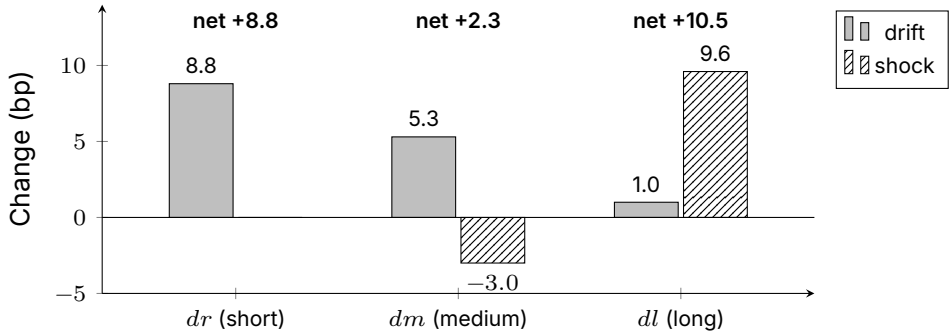


Figure 1. The worked example split into drift (grey fill) and shock (hatched). The short rate is all drift; the medium factor's drift of +5.3 bp is partly offset by a shock of -3.0 bp (net +2.3 bp); the long factor is almost all shock (net +10.5 bp).

Formula card: MR-16 c

Short rate	$dr = -\alpha_r(r - m)\,dt$ (no shock; expected = actual)
Medium factor	$dm = -\alpha_m(m - l)\,dt + \sigma_m\left(\rho\,dW^1 + \sqrt{1-\rho^2}\,dW^2\right)$
Long factor	$dl = -\alpha_l(l - \mu)\,dt + \sigma_l\,dW^1$
Shock size	$\text{SD}(dm) = \sigma_m\sqrt{dt}$, $\text{SD}(dl) = \sigma_l\sqrt{dt}$, $\text{Corr}(dm, dl) = \rho$
Half-life	$\ln 2/\alpha$ (0.66, 1.09 and 42.01 years in Table 16.1)
Useful numbers	$\sqrt{1-0.212^2} = 0.9773$; $\sqrt{1/12} = 0.28868$; $\sqrt{1/252} = 0.06299$

Exam traps

- **Shock on r.** Adding a Vasicek-style $\sigma\,dw$ to the short rate is the defining error.
- **Wrong speed or wrong target.** r uses α_r and reverts to m; m uses α_m and reverts to l; l uses α_l and reverts to μ.
- **Sign.** The drift is $-\alpha(\text{factor} - \text{target})$. A factor below its target drifts *up*.
- **Loadings.** m loads ρ on dW^1 and $\sqrt{1-\rho^2}$ on dW^2; swapping them, using $1-\rho^2$, or adding

the raw shocks is wrong. l loads on dW^1 only.

- **Scaling.** Drift needs dt. A realised dW already contains $\sqrt{dt}$; multiplying again is a double count.
- **Intermediate results.** The drift alone, or the shock alone, is not the change in m or l.
- **Red herrings.** Vasicek volatilities, σ_m in a short-rate question, r in a medium-factor question, α_m in a long-factor question.

Practice questions: MR-16 c

Question MR-16 c.1

Bank Q761 • Schweser Module Quiz 16.1 Q4

Assume the short-term rate is 5.0% and the medium-term rate is 6.0%. By using the Gauss+ model, if the corresponding mean reversion speed is 1.3 and the time step is three months, what is the change in the short-term rate?

A. -0.325%.

B. $+0.325\%$.

C. -3.250%.

D. $+3.250\%$.

Question MR-16 c.2

Bank Q760 • BT notes P2.T5.25.7.1 (rebuilt)

In the Gauss+ model, $dr_t = -a_r(r_t - m_t)dt$, $dm_t = -a_m(m_t - l_t)dt + \ldots$, $dl_t = -a_l(l_t - \mu)dt + \ldots$, with $a_r = 0.5$, $a_m = 0.2$, $a_l = 0.05$ and $\mu = 5\%$. Currently $r = 2\%$, $m = 3\%$, $l = 4\%$. Setting all random shocks to zero and using $dt = 0.25$ years, which factor has the smallest change over the step, and what is it?

A. Short-term factor, $+0.125\%$

B. Medium-term factor, $+0.050\%$

C. Long-term factor, $+0.0125\%$

D. None; each gap is 1% so all three factors change by the same amount

Question MR-16 c.3

Bank Q757 • AnalystPrep Q.6529 (rebuilt, key corrected)

In the Gauss+ model the long-term factor follows $dl_t = a_l(\mu - l_t)\,dt + \sigma_l\,dW_l$. Given $a_l = 0.3$, $l_t = 0.045$, $\mu = 0.04$, $\sigma_l = 0.025$, a realized shock $dW_l = 0.03$ and $dt = 0.01$, what is the change in the long-term factor over the step?

A. 0.000735

B. -0.000015

C. 0.000750

D. 0.000765

Question MR-16 c.4 Bank Q759 • BT P2.T5.25.8.1 (rebuilt, key corrected)

The medium-term factor in the Gauss+ model follows $dm_t = -a_m(m_t - l_t)dt + \sigma_m(\rho\, dW_1 + \sqrt{1-\rho^2}\, dW_2)$. Given $\sigma_m = 0.01$ per year, $\rho = 0.6$, $dt = 1/252$, and a day in which $dW_1 = \sqrt{dt}$ and $dW_2 = 0$, what is the size of the stochastic (diffusion) contribution to the daily change in m_t?

A. 0.0378%

B. 0.0630%

C. 0.0504%

D. 0.6000%

Question MR-16 c.5 Workbook A25.2 • S5, S6 • Aug-2026 • Map #40

A rates strategist at a hedge fund is simulating one monthly step ($dt = 1/12$) of the Gauss+ medium-term factor with the book's parameters: $\alpha_m = 0.6358$, $\sigma_m = 109.2$ bps (1.092% per year), $\rho = 0.212$. Current factor values are $r = 1.75\%$, $m = 3.00\%$ and $l = 4.20\%$. The realised increments for the month are $dW^1 = -0.2309$ and $dW^2 = -0.3464$. Which of the following is closest to the change in the medium-term factor?

A. -56.7 bps

B. -48.9 bps

C. -36.0 bps

D. $+6.4$ bps

Question MR-16 c.6 Workbook A25.3 • S2, S5, S6 • Aug-2026 • Map #40

A model validator at a European bank is checking a monthly ($dt = 1/12$) simulation step of the long-term factor in a EUR implementation of the Gauss+ model that reuses the book's parameters: $\alpha_l = 0.0165$, $\alpha_m = 0.6358$, $\sigma_l = 96.4$ bps (0.964% per year), $\mu = 10.555\%$ and $\rho = 0.212$. The long-term factor is $l = 3.50\%$ and the medium-term factor is $m = 2.80\%$. The realised increment for the month is $dW^1 = +0.15$ (already scaled, so it includes $\sqrt{dt}$). Which of the following is closest to the change in the long-term factor?

A. $+5.1$ bps

B. $+15.4$ bps

C. $+26.1$ bps

D. $+51.8$ bps

Question MR-16 c.7 New • built from GARP Ch.16, Table 16.1 and Vardeez

A rates strategist steps the Gauss+ model forward one month ($dt = 1/12$) using GARP's Table 16.1 parameters:

α_r	1.0547	σ_m	109.2 bps
α_m	0.6358	σ_l	96.4 bps
α_l	0.0165	ρ	0.212
μ	10.555%		

Today $r = 2.00\%$, $m = 3.00\%$ and $l = 4.00\%$. The realised increments for the month, already scaled by $\sqrt{dt}$, are $dW^1 = +0.10$ and $dW^2 = -0.05$. Which of the following is closest to the changes in the three factors $(dr,\ dm,\ dl)$, in basis points?

A. $(+5.3,\ +2.3,\ +10.5)$

B. $(+8.8,\ +2.3,\ +9.5)$

C. $(+8.8,\ +2.3,\ +10.5)$

D. $(+8.8,\ +14.8,\ +10.5)$

Solutions • MR-16 c

Solution MR-16 c.1

Answer: B

Bank Q761 • Schweser Module Quiz 16.1 Q4

Step 1 – State the equation

$$dr = -\alpha_r(r - m)\,dt, \qquad dt = 0.25$$

Step 2 – Gap and change

$$r - m = 5.0\% - 6.0\% = -1.0\%, \qquad dr = -1.3 \times (-1.0\%) \times 0.25 = +0.325\%$$

$\Rightarrow$ The short rate is below the medium-term factor, so it drifts up.

Interpretation

No volatility enters: the short rate has no shock in Gauss+, so this is both the expected and the actual change.

Why the other options are wrong

A is incorrect. Sign error: the rate moves away from the medium-term factor.
C is incorrect. Wrong sign and a factor-of-ten slip.
D is incorrect. Right direction, ten times too large (the dt applied as 2.5 or the gap read as 10%).

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eq. 16.9

Solution MR-16 c.2

Answer: C

Bank Q760 • BT notes P2.T5.25.7.1

Step 1 – The three gaps

$m - r = 1\%$, $l - m = 1\%$, $\mu - l = 1\%$: every factor sits 1% below its target, so all drift up.

Step 2 – The three changes

$$dr = 0.5 \times 1\% \times 0.25 = +0.125\%, \qquad dm = 0.2 \times 1\% \times 0.25 = +0.050\%$$

$$dl = 0.05 \times 1\% \times 0.25 = +0.0125\%$$

$\Rightarrow$ With equal gaps the change is proportional to the speed; the long-term factor, with the smallest α, moves least.

Interpretation

This is the cascade in numbers: fastest response at the short end, slowest at the long end. With shocks switched on, the ranking of actual moves can reverse (MR-16 c.7).

Why the other options are wrong

A is incorrect. The largest change, not the smallest.
B is incorrect. The medium factor's change, larger than the long factor's.
D is incorrect. Equal gaps do not give equal changes; the speeds differ.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eqs. 16.9 to 16.11

Solution MR-16 c.3

Answer: A

Bank Q757 • AnalystPrep Q.6529

Step 1 – Drift

$$a_l(\mu - l_t)\,dt = 0.3 \times (0.04 - 0.045) \times 0.01 = -0.000015$$

$\Rightarrow$ l is above μ, so the drift is negative.

Step 2 – Shock

$$\sigma_l\,dW_l = 0.025 \times 0.03 = 0.000750$$

$\Rightarrow$ dW_l is the realised shock, so no extra $\sqrt{dt}$.

Step 3 – Total

$$dl = -0.000015 + 0.000750 = 0.000735$$

Interpretation

The stem writes the drift as $a_l(\mu - l)$, which is the same as GARP's $-\alpha_l(l - \mu)$. As in the worked example, the long factor's move is dominated by the shock.

Why the other options are wrong

B is incorrect. Drift only: omitted the shock.
C is incorrect. Shock only: omitted the drift.
D is incorrect. Sign error on the drift: $+0.000015 + 0.000750$.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eq. 16.11

Solution MR-16 c.4

Answer: A

Bank Q759 • BT P2.T5.25.8.1

Step 1 – Size of dW_1

$$\sqrt{dt} = \sqrt{1/252} = 0.06299$$

Step 2 – Diffusion term

$$\sigma_m\left(\rho\,dW_1 + \sqrt{1-\rho^2} \times 0\right) = 0.01 \times 0.6 \times 0.06299 = 0.000378 = 0.0378\%$$

$\Rightarrow$ With $dW_2 = 0$ only the ρ-weighted dW_1 term remains.

Interpretation

The ρ loading on dW_1 is what makes m correlated with l; the $\sqrt{1-\rho^2}$ loading on dW_2 is m's own, independent risk.

Why the other options are wrong

B is incorrect. Dropped ρ: $0.01 \times 0.06299 = 0.0630\%$.
C is incorrect. Used $\sqrt{1-\rho^2} = 0.8$ (the dW_2 weight) instead of $\rho = 0.6$: 0.0504%.
D is incorrect. Forgot the $\sqrt{dt}$: $0.01 \times 0.6 = 0.6\%$, an annual-size shock.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eqs. 16.10 and 16.13

Solution MR-16 c.5 **Answer: C**

Workbook A25.2 • S5, S6 • Aug-2026

Step 1 – State the equation

$$dm_t = -\alpha_m(m_t - l_t)\,dt + \sigma_m\left(\rho\,dW^1 + \sqrt{1-\rho^2}\,dW^2\right) \quad \text{(Equation 16.10)}$$

$\Rightarrow$ m reverts to the long-term factor l, not to r and not to μ; the short rate $r = 1.75\%$ is irrelevant here.

Step 2 – Drift over the month

$$-\alpha_m(m-l)\,dt = -0.6358 \times (3.00\% - 4.20\%) \times \tfrac{1}{12} = +0.0636\%$$

$\Rightarrow$ m is 120 bps below l, so the trend pulls it up by about 6.4 bps this month.

Step 3 – Loading on the second shock

$$\sqrt{1-\rho^2} = \sqrt{1-0.212^2} = 0.97727$$

$\Rightarrow$ This weight gives the medium-term fluctuation volatility $\sigma_m\sqrt{dt}$ and correlation exactly ρ with the long-term factor (Equations 16.13 to 16.15).

Step 4 – Combined standardised shock

$$\begin{aligned}\rho\,dW^1 + \sqrt{1-\rho^2}\,dW^2 &= 0.212 \times (-0.2309) + 0.97727 \times (-0.3464)\\ &= -0.04895 - 0.33853 = -0.38748\end{aligned}$$

$\Rightarrow$ Most of the medium-term move comes from the factor's own independent shock dW^2.

Step 5 – Scale by volatility

$$\sigma_m \times (-0.38748) = 1.092\% \times (-0.38748) = -0.4231\%$$

Step 6 – Total change

$$dm = 0.0636\% - 0.4231\% = -0.3595\% \approx -36.0 \text{ bps}$$

$\Rightarrow$ The negative shock overwhelms the upward drift toward l.

Interpretation

The medium-term factor, the market's "forward shadow rate" for where the short rate will be in two to three years, fell about 36 bps. Since r reverts quickly to m ($\alpha_r = 1.0547$), this lowers the expected path of the short rate over the next year or so.

Why the other options are wrong

A is incorrect. Dropped the correlation loadings and added the raw increments: $0.0636\% + 1.092\% \times (-0.2309 - 0.3464) = -56.7$ bps.

B is incorrect. Wrong target (sibling swap): reverted m to r instead of l: $-0.6358 \times (3.00\% - 1.75\%)/12 - 0.4231\% = -48.9$ bps.

D is incorrect. Intermediate result: the drift alone, $+6.4$ bps, treating m like the shock-free short rate.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eqs. 16.10 and 16.13 to 16.15, Table 16.1

Solution MR-16 c.6

Answer: B

Workbook A25.3 • S2, S5, S6 • Aug-2026

Step 1 – State the equation

$$dl_t = -\alpha_l(l_t - \mu)\,dt + \sigma_l\,dW^1 \quad \text{(Equation 16.11)}$$

$\Rightarrow$ l reverts to the constant μ at speed α_l and loads only on dW^1; m, α_m and ρ do not enter.

Step 2 – Gap to the target

$$l - \mu = 3.50\% - 10.555\% = -7.055\%$$

Step 3 – Drift over the month

$$-\alpha_l(l-\mu)\,dt = -0.0165 \times (-7.055\%) \times \tfrac{1}{12} = +0.0097\%$$

$\Rightarrow$ Despite the 7-point gap, the slowest speed in the cascade (half-life $\ln 2/0.0165 = 42.0$ years) moves l up by only about 1 bp a month.

Step 4 – Random fluctuation

$$\sigma_l\,dW^1 = 0.964\% \times 0.15 = 0.1446\%$$

$\Rightarrow$ The shock adds about 14.5 bps; the monthly standard deviation of this term is $0.964\% \times \sqrt{1/12} = 27.8$ bps.

Step 5 – Total change

$$dl = 0.0097\% + 0.1446\% = 0.1543\% \approx +15.4 \text{ bps}$$

Interpretation

Because α_l is so small, l behaves almost like a random walk: its effect on longer-term forwards is nearly parallel and, beyond 10 years, it is the sole factor driving forward-rate changes. A 15 bp rise in l lifts the long end almost one-for-one.

Why the other options are wrong

A is incorrect. Double $\sqrt{dt}$: multiplied the already-scaled increment by $\sqrt{1/12}$ again: $0.97 + 14.46 \times 0.28868 = 5.1$ bps.

C is incorrect. Dropped dt on the drift: $0.0165 \times 7.055\% = 11.64$ bps plus 14.46 bps $= 26.1$ bps.

D is incorrect. Wrong speed (sibling swap): used α_m: $0.6358 \times 7.055\%/12 = 37.38$ bps plus 14.46 bps $= 51.8$ bps.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eq. 16.11, Table 16.1

Solution MR-16 c.7

Answer: C

New • built from GARP Ch.16, Table 16.1

Step 1 – Short rate, drift only

$$dr = -1.0547 \times (2.00\% - 3.00\%) \times \tfrac{1}{12} = +0.0879\% = +8.8 \text{ bp}$$

Step 2 – Medium factor, drift plus both shocks

$$\text{drift} = -0.6358 \times (3.00\% - 4.00\%) \times \tfrac{1}{12} = +0.0530\%, \qquad \sqrt{1 - 0.212^2} = 0.9773$$

$$\text{shock} = 1.092\% \times [0.212 \times 0.10 + 0.9773 \times (-0.05)] = 1.092\% \times (-0.02767) = -0.0302\%$$

$$dm = 0.0530\% - 0.0302\% = +0.0228\% = +2.3 \text{ bp}$$

Step 3 – Long factor, drift plus dW^1

$$\text{drift} = -0.0165 \times (4.00\% - 10.555\%) \times \tfrac{1}{12} = +0.0090\%, \quad \text{shock} = 0.964\% \times 0.10 = 0.0964\%$$

$$dl = +0.1054\% = +10.5 \text{ bp}$$

$\Rightarrow (+8.8,\ +2.3,\ +10.5)$ bp.

Interpretation

The long factor moves most even though it has by far the slowest mean reversion, because its move is almost all shock. The short rate moves 8.8 bp with no randomness at all. Compare MR-16 c.2: without shocks the ranking is the reverse.

Why the other options are wrong

A is incorrect. Used α_m for the short rate: $0.6358 \times 1\%/12 = 5.3$ bp.
B is incorrect. Reverted l to m instead of μ: drift $-0.0165 \times 1\%/12 = -0.14$ bp, so $dl = 9.5$ bp.
D is incorrect. Swapped the loadings in dm, putting ρ on dW^2 and $\sqrt{1-\rho^2}$ on dW^1: shock $+9.5$ bp, so $dm = 14.8$ bp.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. • **Reference:** Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eqs. 16.9 to 16.11, Table 16.1

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 a

P1 CONCEPT

Describe a volatility smile and volatility skew.

Workbook: A3 Volatility Smiles, Skews and Put-Call Parity • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A3 (Map #4, 9 of 16 sources). Your August 2026 recall: "volatility skew, 1Q: effect of changing maturity and strike (equity option)".
Vardeez: full chapter is important.

Reported exam shape

- **A3·S1** Equity skew: the effect of changing strike and maturity on implied volatility, plain form.

What the questions look like: definition and shape questions. Which way does implied volatility move as the strike rises? Which option has the lowest implied volatility? What is on each axis? The trap is swapping the FX smile and the equity skew.

1. Implied volatility: the input that is backed out

The BSM call price depends on six inputs, $c = f\{S,\ K,\ T,\ r,\ q,\ \sigma\}$. Five are observable; σ is not. Set the BSM price equal to the **market** price and solve backwards for the σ that makes them equal: that σ is the **implied volatility**.

- It is a **market quote, not a property of the asset**: it depends on the pricing model used to invert the price. Two traders using different models get different implied volatilities from the same market price.
- It is not historical volatility: no data window is involved.
- If BSM held exactly (constant volatility, lognormal prices), every strike would back out the **same** implied volatility and the plot would be a **flat line**.

Traders do use BSM, but not as originally intended: they let the volatility depend on strike and maturity (Hull).

2. Smile, skew and surface

- **Volatility smile:** implied volatility of options with a *given maturity* plotted against strike price, usually as the ratio K/S_0 (moneyness).
- **Volatility skew ("smirk"):** the term used for **equity** options, where implied volatility *falls* as the strike rises. Essentially an inverse relationship between implied volatility and K/S_0.
- **Volatility surface:** implied volatility as a function of both strike *and* time to maturity (a three-dimensional plot). Smiles are its slices at fixed maturity.

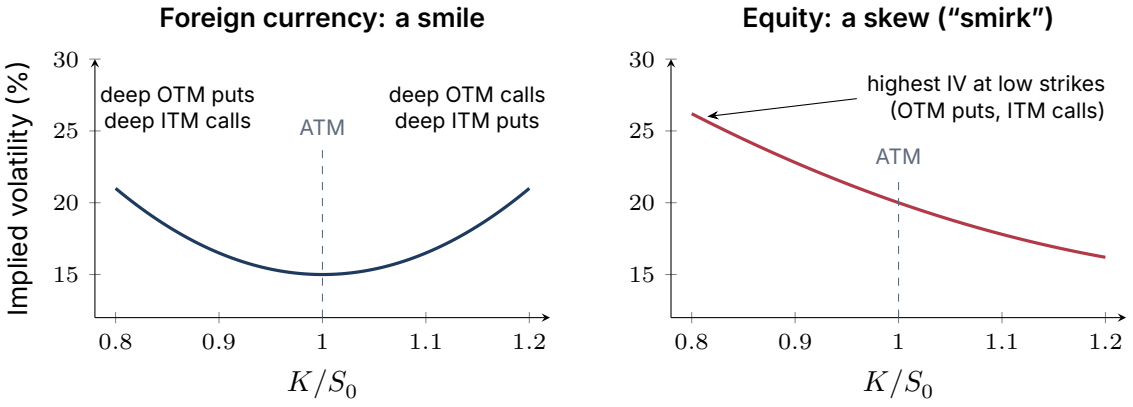


Figure 1. Both curves plot implied volatility against moneyness, read left to right as the strike rises. The currency smile is roughly symmetric and lowest at the money. The equity skew falls steadily: low-strike options carry the highest implied volatility and the curve does not turn back up.

3. Reading the curves

- **Low strike** ($K/S_0 < 1$) means an OTM *put* or an ITM *call*. **High strike** means an OTM *call* or an ITM *put*. For a given strike the call and the put share one implied volatility (MR-17 b), so "which option has the lowest IV?" is really "which option has the highest strike?" on an equity skew.
- **FX:** implied volatility is relatively low at the money and rises progressively as the option moves either into or out of the money.
- **Equity:** since the October 1987 crash the curve slopes down. Before 1987 there was no marked smile for equity options.
- **Maturity (the August shape).** Two separate effects, both from Hull 17.5:
  - The *level*: at-the-money implied volatility tends to **increase** with maturity when short-dated volatilities are historically **low** (they are expected to rise), and to **decrease** with maturity when they are historically high.
  - The *shape*: the smile or skew becomes **less pronounced** as maturity increases (jumps average out over long horizons).

	FX smile	Equity skew
Shape	U-shaped, lowest at the money	Downward sloping in K/S_0
Highest IV	Both wings (deep ITM and deep OTM)	Low strikes (OTM puts, ITM calls)
Lowest IV	At the money	High strikes (ITM puts, OTM calls)
History	Priced in by the late 1980s	Only since October 1987
Longer maturity	Smile flattens	Skew flattens

Exam traps

- **Polarity flip on strike.** For equities implied volatility *falls* as the strike rises.
- **Sibling swap.** "Lowest at the money, higher at both ends" is the FX smile, not the equity skew.
- **ITM call vs ITM put.** An ITM call has a low strike (high IV on an equity skew); an ITM put has a high strike (low IV).
- **Condition on maturity.** The term structure slopes up only when short-dated volatilities are historically low.
- **Historical vs implied.** The smile uses implied volatility at a fixed maturity, not historical volatility across dates.
- **Vega is not the smile.** A hump-shaped vega curve is a model sensitivity; it exists even when the smile is flat.

Practice questions: MR-17 a

Question MR-17 a.1

Bank Q768 • AnalystPrep Q.1706

Option traders in equity and foreign currency markets often refer to a "volatility smile." Which of the following statements gives the correct definition of a volatility smile?

A. The plot of the implied volatility of financial markets that determines option prices.

B. The plot of the implied volatility of an option with a given maturity as a function of its strike price.

C. The plot of the implied volatility of an option with an infinite life as a function of its current price.

D. The pattern of changes in historical volatility across trading days for options with different expiration dates.

Question MR-17 a.2

Bank Q770 • BT notes P2.T5.01.3 (rebuilt)

Two traders look at the same European option on the same underlying, with the same market price, strike and maturity. Trader 1 uses the Black-Scholes-Merton model and Trader 2 uses a different option pricing model. Which statement about the implied volatility each trader calculates, and hence the volatility smile each plots, is most accurate?

A. The implied volatilities must be identical, because they are backed out of the same market price.

B. The implied volatilities can differ, because implied volatility depends on the pricing model used to invert the market price.

C. The implied volatilities must be identical, because volatility is a property of the underlying asset alone.

D. The implied volatilities can differ only if the traders use different historical data windows.

Question MR-17 a.3 Bank Q771 • BT L2.T5.06.3 (rebuilt, key corrected)

An analyst notes that a plot of option vega against the underlying stock price is hump-shaped: vega is highest for at-the-money options and lower for out-of-the-money options. Which statement about whether this plot is consistent with the volatility smile is most accurate?

A. It is consistent with the smile, because the smile also shows the largest values for at-the-money options.

B. It is not the smile: vega is a model-based sensitivity of option value to volatility, whereas the smile plots implied volatility against strike (or K/S_0).

C. It is consistent with the smile, because high vega at the money means at-the-money options must have the highest implied volatility.

D. It is not the smile, because under Black-Scholes-Merton vega would be constant across strikes, while implied volatility would be hump-shaped.

Question MR-17 a.4 Bank Q820 • Mock 1 Q16 • Schweser Module Quiz 17.1 Q3

Which of the following is true of currency option volatility smiles?

A. Currency options generate implied volatilities that are higher for deep-in-the-money options and lower for deep-out-of-the-money options.

B. Currency options generate implied volatilities that are higher for deep-out-of-the-money options and lower for deep-in-the-money options.

C. Currency options generate implied volatilities that are higher for at-the-money options and lower for deep-out-of-the-money and deep-in-the-money options.

D. Currency options generate implied volatilities that are higher for deep-in-the-money options and deep-out-of-the-money options and lower for at-the-money options.

Question MR-17 a.5 Bank Q822 • Mock 1 Q9

Assuming an option in the equity options portfolio has an at-the-money strike of 50, which of the following options is most likely to have the lowest level of implied volatility in real life?

A. At-the-money put.

B. In-the-money call.

C. Out-of-the-money put.

D. In-the-money put.

Question MR-17 a.6 Workbook A3.1 • S1, S5 • Aug-2026 • Map #4

A senior equity derivatives analyst at a Hong Kong bank is presenting implied volatilities of European options on a stock index to a group of newly hired analysts. Short-dated implied volatilities on the index are currently at historically low levels. Which of the following statements should the analyst include in the presentation?

A. Implied volatility rises as the strike price rises, and at-the-money implied volatility is expected to increase with option maturity.

B. Implied volatility is lowest at the money and rises toward both lower and higher strikes, and at-the-money implied volatility is expected to increase with option maturity.

C. Implied volatility falls as the strike price rises, and at-the-money implied volatility is expected to decrease with option maturity.

D. Implied volatility falls as the strike price rises, and at-the-money implied volatility is expected to increase with option maturity.

Solutions • MR-17 a

Solution MR-17 a.1

Answer: B

Bank Q768 • AnalystPrep Q.1706

Step 1 – What goes into the curve

Implied volatilities are backed out of observed option prices with BSM, holding the maturity fixed.

Step 2 – What goes on the axes

Implied volatility is plotted against the strike price (or K/S_0). A non-flat curve is the smile or skew.

$\Rightarrow$ A smile is implied volatility as a function of strike for options of one maturity.

Interpretation

The two fixed elements, "implied" and "given maturity", are what the wrong options remove or replace.

Why the other options are wrong

A is incorrect. Too vague: it makes implied volatility a market-wide quantity, with no strike axis and no fixed maturity, so there is no curve.

C is incorrect. A smile needs a finite, fixed maturity, and the horizontal axis is the strike, not the option's current price.

D is incorrect. Confuses implied with historical volatility and plots against time rather than strike.

LO: MR-17 a: Describe a volatility smile and volatility skew. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), introduction

Solution MR-17 a.2

Answer: B

Bank Q770 • BT notes P2.T5.01.3

Step 1 – Definition

Implied volatility is the volatility input that makes a *model* price equal the market price.

Step 2 – Model dependence

Different models map volatility to price differently, so the same market price inverts to different volatilities.

$\Rightarrow$ The smile each trader plots is model-specific; they agree only if they use the same model.

Interpretation

Implied volatility is a quote in the language of a model. Under BSM it is the quote traders use, which is why a smile in BSM terms says the market does not price with a lognormal distribution.

Why the other options are wrong

A is incorrect. A common market price is not enough: the inversion also depends on the model.

C is incorrect. Confuses implied volatility with an objective property of the asset.

D is incorrect. Confuses implied with historical volatility; implied volatility uses no data window.

LO: MR-17 a: Describe a volatility smile and volatility skew. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), introduction and 17.6

Solution MR-17 a.3 Answer: B

Bank Q771 • BT L2.T5.06.3

Step 1 – What the smile plots

Market implied volatility (vertical axis) against strike or K/S_0 (horizontal axis).

Step 2 – What the vega curve plots

$\partial V/\partial\sigma$, computed from the model, against the stock price.

Step 3 – Test the link

If BSM held exactly, implied volatility would be flat across strikes (no smile), yet vega would still be hump-shaped.

$\Rightarrow$ The hump in vega says nothing about the smile.

Interpretation

Two different objects can both peak near the money. Only the smile is a statement about market prices.

Why the other options are wrong

A is incorrect. Matches only the "highest at the money" feature; the FX smile is lowest at the money, and neither smile is a vega plot.

C is incorrect. Large vega means high sensitivity to volatility, not a high level of implied volatility.

D is incorrect. Reverses the facts: under BSM implied volatility is flat while vega stays hump-shaped.

LO: MR-17 a: Describe a volatility smile and volatility skew. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), introduction; Ch.19 vega

Solution MR-17 a.4 Answer: D

Bank Q820 • Mock 1 Q16

Step 1 – Recall the shape

The FX smile is roughly symmetric and lowest at the money.

Step 2 – Read both wings

Deep ITM and deep OTM options both carry higher implied volatility than ATM options.

$\Rightarrow$ Only D has both wings high and the centre low.

Interpretation

The reason is fat tails on both sides of the exchange rate distribution (MR-17 c, d): both wings are dearer than BSM at the ATM volatility.

Why the other options are wrong

A is incorrect. A one-sided pattern (only deep ITM high): this is an equity-style skew read from the call side.

B is incorrect. Another one-sided pattern: it ignores the high implied volatility of deep ITM options.

C is incorrect. Describes a frown (highest at the money), the shape of a single anticipated jump or a central-bank band, not the usual currency smile.

LO: MR-17 a: Describe a volatility smile and volatility skew. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2, Figure 17.1

Solution MR-17 a.5 — Answer: D

Bank Q822 • Mock 1 Q9

Step 1 – Shape

Equity implied volatility falls as the strike rises (skew).

Step 2 – Locate each strike relative to 50

ATM put: 50. ITM call: below 50. OTM put: below 50. ITM put: **above** 50.
⇒ The highest strike, the ITM put, sits furthest right on the skew and has the lowest implied volatility.

Interpretation

Put or call does not matter at a given strike (put-call parity); only the strike does. Translate "ITM/OTM call/put" into "low or high strike" first.

Why the other options are wrong

A is incorrect. Strike 50 sits in the middle of the skew, above the implied volatility at strikes above 50.
B is incorrect. An ITM call has a strike below 50, the high-volatility side; it confuses ITM with a high strike, which is true only for puts.
C is incorrect. An OTM put has a strike below 50, where implied volatility is highest (demand for crash protection).

LO: MR-17 a: Describe a volatility smile and volatility skew. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.3, Figure 17.3

Solution MR-17 a.6 — Answer: D

Workbook A3.1 • S1, S5 • Aug-2026

Step 1 – Strike: the equity skew

For equity options (stocks and indices) the volatility decreases as the strike price increases (Hull 17.3). Low-strike options, deep OTM puts and deep ITM calls, carry the highest implied volatilities.

Step 2 – Maturity: the term structure

Implied volatility tends to be an increasing function of maturity when short-dated volatilities are historically low, because volatilities are then expected to increase (Hull 17.5). The stem sets that condition.

Step 3 – The second maturity effect

Across a surface, the skew itself becomes less pronounced as maturity increases: jumps are averaged out over long horizons and the impact of non-constant volatility on implied volatility usually shrinks.

Interpretation

D is the only statement with both directions right. If short-dated volatilities were historically high instead, the same skew would sit on a term structure that decreases with maturity.

Why the other options are wrong

A is incorrect. Polarity flip on strike: for equities implied volatility falls, not rises, as the strike increases.
B is incorrect. Sibling swap: a U-shape lowest at the money is the foreign currency smile, not the equity skew.
C is incorrect. Scope/condition error: a term structure that decreases with maturity applies when short-dated volatilities are historically high; the stem says they are low.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. • **Reference:** Hull, OFOD 11e, Ch.20, 17.3 and 17.5, Table 17.2

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 b

P1 CONCEPT

Explain the implications of put-call parity on the implied volatility of call and put options.

Workbook: A3 Volatility Smiles, Skews and Put-Call Parity • Tier A • Map #4, #71

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A3. Map #71 reports "put call parity" as a standalone theory question, tested in its smile context, not as a Part I recap.
Vardeez: full chapter is important.

Reported exam shape

- **A3·S7** A European call and put with the same strike and maturity share one implied volatility; tested in its smile context (and as a distractor elsewhere: "parity equalises implied volatility across strikes").

What the questions look like: "what does parity imply?", "the call IV is 24%, the put IV is 27%: what does it mean and which trade?", and the one applied form, "the call trades \$0.28 above BSM: where should the put trade?". No option-pricing calculation is needed.

1. Parity holds for market prices and for BSM prices

$$p + S_0 e^{-qT} = c + K e^{-rT} \qquad \Longleftrightarrow \qquad c - p = S_0 e^{-qT} - K e^{-rT}$$

(For currency options q is the foreign risk-free rate.) Parity rests on a simple no-arbitrage argument with **no assumption about the distribution** of the asset price: it is true whether or not prices are lognormal. So it holds for market prices, and it holds for BSM prices at *any* volatility:

$$p_{mkt} + S_0 e^{-qT} = c_{mkt} + K e^{-rT}, \qquad p_{BSM} + S_0 e^{-qT} = c_{BSM} + K e^{-rT}$$

Subtracting:

$$\boxed{p_{BSM} - p_{mkt} = c_{BSM} - c_{mkt}} \qquad \text{(Hull eq. 17.2)}$$

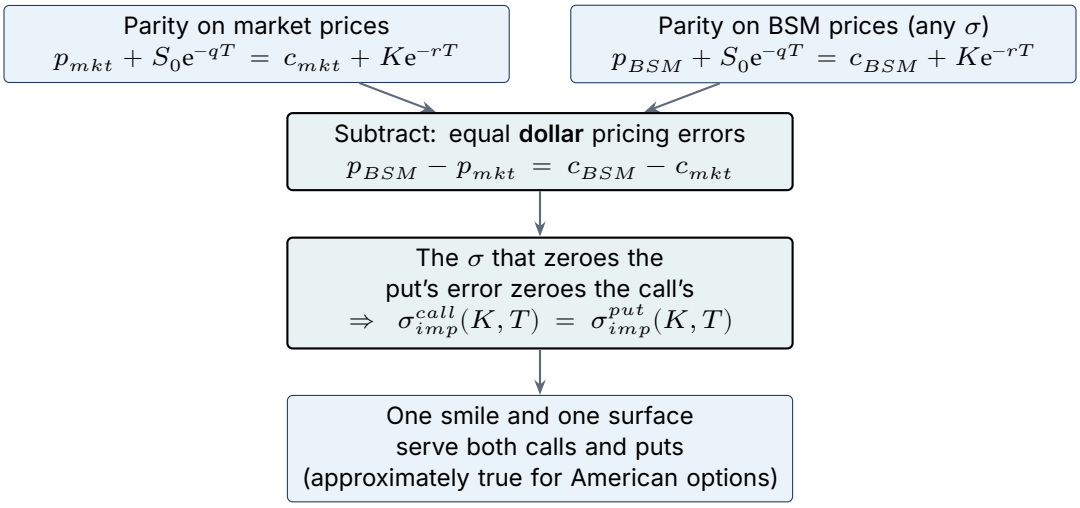


Figure 2. The chain from parity to a single implied volatility. Nothing in it assumes lognormal prices.

2. The implications

1. **Equal dollar errors.** For the same strike and maturity, the market price of a European put deviates from its BSM price by *exactly* the same dollar amount as the call. Not proportionally: the same dollars.
2. **Equal implied volatility.** If the put's implied volatility is 22%, then $p_{BSM} = p_{mkt}$ at 22%, so $c_{BSM} = c_{mkt}$ at 22%: the call's implied volatility is also 22%.
3. **One smile.** The smile (and the surface) is the same for European calls and puts, so we never need to say which one we are quoting. Approximately true for American options.
4. **Any distribution.** The result does not need lognormal prices. A gap between call and put implied volatility at the same K and T is an arbitrage whatever the true distribution.
5. **Same strike only.** Parity says nothing about implied volatilities at *different* strikes. Those differ, and that difference is the smile.
6. **Equal vegas.** $c - p = S_0 e^{-qT} - Ke^{-rT}$ contains no volatility, so when volatility moves from σ_1 to σ_2 (all else unchanged) the call and the put change in value by the same amount.

Example (Hull 17.1). Spot \$0.60, $r = 5\%$, $r_f = 10\%$, 1-year call at $K = 0.59$ trades at 0.0236 (implied volatility 14.5%). Parity: $p = 0.0236 + 0.59e^{-0.05} - 0.60e^{-0.10} = 0.0419$, and the put's implied volatility is also 14.5%.

3. When the two implied volatilities differ

A higher implied volatility means a higher price relative to BSM, so the option with the **higher** implied volatility is the **rich** leg.

	Put IV > call IV (put rich)	Call IV > put IV (call rich)
Options	Sell the put, buy the call	Sell the call, buy the put
Stock	Short the stock	Buy the stock
Cash	Invest Ke^{-rT} at the risk-free rate	Borrow Ke^{-rT}
At T	Long call + short put buys the stock for K, closing the short	Short call + long put sells the stock for K, repaying the loan

The profit is locked in whether or not prices are lognormal, because the payoffs cancel state by state.

Exam traps

- **Scope error.** Parity equalises call and put implied volatility at the *same* strike and maturity, not across strikes.
- **Condition error.** "The arbitrage exists only if prices are lognormal": wrong. Parity is distribution-free.
- **Sibling swap.** "The skew gives puts higher IV than calls": the skew is across strikes, not between a call and a put at one strike.
- **Rich leg polarity.** Higher implied volatility means more expensive: sell that leg.
- **Equal dollars, not equal percentages.** Scaling the put by c_{mkt}/c_{BSM} is wrong.
- **Delta vs vega.** Calls and puts have opposite deltas but the same (positive) vega.

Practice questions: MR-17 b

Question MR-17 b.1

Bank Q786 • Schweser Module Quiz 17.1 Q1 (rebuilt)

A European call and a European put have the same underlying, strike price and time to expiration. Market prices of both deviate from their Black-Scholes-Merton prices (calculated with a single volatility). Put-call parity implies that the implied volatilities of the call and the put are:

A. equal, because parity holds for market prices and BSM prices, so the pricing deviations must be the same for the put and the call.

B. unequal, because puts and calls respond to volatility with opposite signs.

C. unequal, because the put has a higher probability of finishing in the money when the call is out of the money.

D. equal only if the options are at-the-money.

Question MR-17 b.2

Bank Q776 • AnalystPrep Q.1707 (rebuilt)

Which of the following statements about the implied volatility of European options is correct?

A. For the same strike and maturity, the implied volatility of a European call equals that of a European put, so their volatility smiles are identical.

B. For the same strike and maturity, the implied volatility of a European put is higher than that of a European call whenever the underlying price distribution is not lognormal.

C. A European call with a shorter maturity has the same implied volatility as a European put with a longer maturity at the same strike.

D. For the same strike and maturity, the implied volatility of a European call is higher than that of a European put because calls have unlimited upside.

Question MR-17 b.3

Bank Q778 • AnalystPrep Q.1710 (rebuilt)

The implied volatility of a European put is found to be 25%, meaning that the BSM put price equals the market put price at that volatility. A European call has the same strike price and maturity as the put. Using put-call parity, what can be concluded about the call's implied volatility?

A. It is also 25%, because parity forces the call's pricing error to equal the put's pricing error, which is zero at 25%.

B. It is greater than 25%, because calls carry more upside risk than puts.

C. It is less than 25%, because the put premium includes a crash premium that the call lacks.

D. It cannot be determined without knowing the current underlying price.

Question MR-17 b.4 New • built from GARP Ch.17 Q17.26

A volatility trader at a Dutch bank holds a European call and a European put on a non-dividend-paying index, with the same strike price and the same time to maturity. Over a short period the index level and interest rates are unchanged, but implied volatility rises from σ_1 to σ_2. A junior colleague argues that the two options must change in value by different amounts. Which of the following statements is correct?

A. The call gains and the put loses value, because the two options have deltas of opposite sign.

B. Both options gain by the same dollar amount, because $c - p = S_0 - K\mathrm{e}^{-rT}$ does not depend on volatility.

C. Both options gain value, but the call gains more because its payoff is unlimited on the upside.

D. Both options gain by the same dollar amount only if they are at the money.

Question MR-17 b.5 Bank Q782 • BT P2.T5.409.2 (rebuilt)

An analyst estimates a stock's volatility at 34.0% per annum. The Black-Scholes-Merton (BSM) price of a European call is USD 6.95, but the call trades in the market at USD 7.23. The BSM price of a European put on the same stock with the same strike and maturity is USD 5.12. Which is NEAREST to the likely market price of the put?

A. USD 4.84

B. USD 5.12

C. USD 5.33

D. USD 5.40

Question MR-17 b.6 Bank Q780 • GARP book Ch.17 Q17.20 (rebuilt)

A stock trades at \$40. A 6-month European call with strike \$30 has an implied volatility of 35%, and a 6-month European call with strike \$50 has an implied volatility of 28%. The risk-free rate is 5% (continuous) and no dividends are expected. Using BSM, the call prices are \$11.155 and \$0.725, so put-call parity gives put prices of \$0.414 (strike 30) and \$9.491 (strike 50). What are the implied volatilities of the 6-month European puts with strikes of \$30 and \$50?

A. 35% for the $K = 30$ put and 28% for the $K = 50$ put.

B. 28% for the $K = 30$ put and 35% for the $K = 50$ put.

C. 31.5% for both puts (the average of the two call volatilities).

D. Both puts have a lower implied volatility than the matching calls, since the puts are priced by parity from lower call values.

Question MR-17 b.7 Workbook A3.4 • S7 • Map #4, #71

An options trader at a Swiss bank observes European call and put options on a non-dividend-paying stock with the same strike price and the same maturity. Using BSM, the call's implied volatility is 24% and the put's implied volatility is 27%. The head of market risk asks what the discrepancy means. Which of the following is the most appropriate response?

A. No action is needed, because the equity skew assigns higher implied volatilities to puts than to calls.

B. An arbitrage exists only if the stock price is lognormally distributed, because the implied volatilities come from BSM.

C. The call is expensive relative to the put, so the trader should sell the call, buy the put and buy the stock, borrowing at the risk-free rate.

D. The put is expensive relative to the call, so the trader should sell the put, buy the call and short the stock, investing the proceeds at the risk-free rate.

Solutions • MR-17 b

Solution MR-17 b.1

Answer: A

Bank Q786 • Schweser Module Quiz 17.1 Q1

Step 1 – Parity on both price sets

$c - p = S - PV(K)$ holds for market prices and for BSM prices at any volatility.

Step 2 – Subtract

$c_{mkt} - c_{BSM} = p_{mkt} - p_{BSM}$: the dollar deviation is identical for the call and the put.

Step 3 – From prices to volatility

The volatility that removes the call's deviation removes the put's as well.
$\Rightarrow$ Equal implied volatilities, at every strike.

Interpretation

The result needs only no-arbitrage. It is why the smile is the same for calls and puts.

Why the other options are wrong

B is incorrect. Confuses delta with vega: calls and puts have opposite deltas, but both have positive (and equal) vega.
C is incorrect. Mixes moneyness with implied volatility; complementary moneyness does not change the parity result.
D is incorrect. Wrongly restricts the result to at-the-money options; parity applies at every strike.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.1, eqs. 17.1 and 17.2

Solution MR-17 b.2

Answer: A

Bank Q776 • AnalystPrep Q.1707

Step 1 – Parity is a no-arbitrage identity

$c_{mkt} - c_{BSM} = p_{mkt} - p_{BSM}$ for the same K and T.

Step 2 – Consequence

The volatility that sets $c_{BSM} = c_{mkt}$ also sets $p_{BSM} = p_{mkt}$.
$\Rightarrow$ Call and put implied volatilities coincide, so the smiles are identical.

Interpretation

Two conditions sit in the correct statement: same strike *and* same maturity. Each distractor breaks one of them or adds a payoff story.

Why the other options are wrong

B is incorrect. Condition error: parity is distribution-free, so it holds when prices are not log-normal too; a gap would be an arbitrage.
C is incorrect. Scope error: parity links options with the same maturity; different maturities generally have different implied volatilities (term structure).
D is incorrect. Confuses payoff asymmetry with implied volatility.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.1

Solution MR-17 b.3 — Answer: A

Bank Q778 • AnalystPrep Q.1710

Step 1 – Equal errors

$p_{BSM} - p_{mkt} = c_{BSM} - c_{mkt}$ at any volatility.

Step 2 – Evaluate at 25%

At 25% the put's error is zero, so the call's error is zero too.
$\Rightarrow$ The call's implied volatility is 25%.

Interpretation

This is Hull's own illustration (with 22%). No prices or underlying level are needed.

Why the other options are wrong

B is incorrect. Treats implied volatility as reflecting payoff asymmetry; parity rules out a different call volatility.
C is incorrect. Applies the equity-skew (crash premium) intuition, which works across strikes; here the strike is the same.
D is incorrect. The result follows from the parity difference alone; S_0 cancels.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.1

Solution MR-17 b.4 — Answer: B

New • built from GARP Ch.17 Q17.26

Step 1 – Parity before and after

At σ_1: $c_1 - p_1 = S_0 - K\mathrm{e}^{-rT}$. At σ_2: $c_2 - p_2 = S_0 - K\mathrm{e}^{-rT}$ (same right-hand side, since S_0, K, r, T are unchanged and volatility does not appear).

Step 2 – Subtract

$c_2 - c_1 = p_2 - p_1$.
$\Rightarrow$ Both options change by the same dollar amount: call vega equals put vega. Both gain, since vega is positive.

Interpretation

This is the price-change version of "one implied volatility for the call and the put". Holding the parity right-hand side fixed is what makes it work: over a short period with the index unchanged, only volatility moves.

Why the other options are wrong

A is incorrect. Sibling swap: opposite signs belong to delta (sensitivity to the index), not to vega.
C is incorrect. Payoff asymmetry does not change the parity difference, which contains no volatility term.
D is incorrect. Scope error: the result holds at every strike, not only at the money.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.1; Practice Question 17.26

Solution MR-17 b.5 — Answer: D

Bank Q782 • BT P2.T5.409.2

Step 1 – Equal dollar deviations

$c_{mkt} - c_{BSM} = p_{mkt} - p_{BSM}$.

Step 2 – Call deviation

$7.23 - 6.95 = +0.28$.

$\Rightarrow$ The market prices the call 0.28 above BSM, so it must price the put 0.28 above BSM as well.

Step 3 – Put market price

$p_{mkt} = 5.12 + 0.28 = \text{USD } 5.40$.

Interpretation

The analyst's 34% is below the market's implied volatility for this strike and maturity. Both options imply the same, higher volatility.

Why the other options are wrong

A is incorrect. Subtracted the deviation: $5.12 - 0.28 = 4.84$. The call is above its BSM price, so the put is too.

B is incorrect. Assumed the put trades at its BSM price, ignoring that parity transmits the call's mispricing.

C is incorrect. Scaled proportionally: $5.12 \times 7.23/6.95 = 5.33$. Parity requires equal dollar, not equal percentage, deviations.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.1, eq. 17.2; Practice Question 17.10

Solution MR-17 b.6 — Answer: A

Bank Q780 • GARP book Ch.17 Q17.20

Step 1 – The puts are priced by parity from the calls

$K = 30$: $11.155 + 30\text{e}^{-0.025} - 40 = 0.414$. $K = 50$: $0.725 + 50\text{e}^{-0.025} - 40 = 9.491$.

Step 2 – Parity holds for BSM prices at any volatility

A put priced by parity from a call valued at volatility σ has a BSM-implied volatility of σ.

$\Rightarrow$ 35% for the $K = 30$ put and 28% for the $K = 50$ put (checked by inverting BSM: 35.0% and 28.0%).

Interpretation

The skew (35% at the low strike, 28% at the high strike) is the same for puts and calls. Parity works strike by strike.

Why the other options are wrong

B is incorrect. Swapped the strikes' volatilities.

C is incorrect. Averaged across strikes; implied volatility is strike-specific and parity applies at each strike separately.

D is incorrect. Puts priced by parity are consistent with the calls, so their implied volatilities are equal, not lower.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.1; Practice Question 17.20

Solution MR-17 b.7 — Answer: D

Workbook A3.4 • S7

Step 1 – What parity implies

Put-call parity $p + S_0 = c + K\text{e}^{-rT}$ holds for BSM prices and for market prices, so $p_{BS} - p_{mkt} = c_{BS} - c_{mkt}$ (17.2). At any volatility the dollar pricing errors are equal, so a European call and put with the same strike and maturity must have the same implied volatility.

Step 2 – Which leg is rich

A higher implied volatility means a higher price relative to BSM. At 27% versus 24%, the put is priced too high relative to the call: $p_{mkt} + S_0 > c_{mkt} + K\mathrm{e}^{-rT}$.

Step 3 – The arbitrage

Sell the put, buy the call and short the stock: the cash received $p - c + S_0$ exceeds $K\mathrm{e}^{-rT}$. Invest it at the risk-free rate. At T the long call and short put together buy the stock for K whatever happens, which closes the short; the invested proceeds exceed K.

Interpretation

Parity is a no-arbitrage relation with no assumption about the asset price distribution, so the profit is locked in whether or not prices are lognormal. This is why the smile is the same for calls and puts.

Why the other options are wrong

A is incorrect. Sibling swap: the skew relates implied volatility to the strike, across strikes. At the same strike and maturity the call and put must share one implied volatility.

B is incorrect. Scope/condition error: put-call parity does not require any assumption about the probability distribution; it is true both when the distribution is lognormal and when it is not.

C is incorrect. Polarity flip: the put, with the higher implied volatility, is the expensive leg. Selling the call and buying the put would buy the overpriced side.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. • **Reference:** Hull, OFOD 11e, Ch.20, 17.1, eqs. 17.1 and 17.2, Example 17.1; Practice Questions 17.8 and 17.17

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 c

P1 CONCEPT

Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset.

Workbook: A3 Volatility Smiles, Skews and Put-Call Parity • Tier A • Map #4

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A3. Map #4 reports "lognormal vols vs implied vols for FX and index options".
Vardeez: full chapter is important. His notes label this LO "Distributions and Fat Tails".

Reported exam shape

- **A3·S4** Lognormal vols versus implied vols for FX and index options: with one at-the-money volatility, which options are priced above or below market?

What the questions look like: translate in either direction between three things: the tail shape of the implied distribution, the shape of the implied volatility curve, and whether deep OTM options are dear or cheap relative to a lognormal (single-volatility) price.

1. One rule connects all three

The **implied distribution** is the risk-neutral distribution of the future asset price that is consistent with the smile (Hull's appendix recovers it from butterfly prices). Compare it with a lognormal of the same mean and standard deviation, then read the tails through deep OTM options:

The rule. Where the implied distribution has *more* probability than the lognormal, the option that pays off in that region is priced *relatively high*, so its implied volatility is *relatively high*. Less probability means a relatively low price and a relatively low implied volatility.

- A deep OTM **call** at K_2 ($K_2/S_0 \gg 1$) pays off only in the **right** tail.
- A deep OTM **put** at K_1 ($K_1/S_0 \ll 1$) pays off only in the **left** tail.
- A **flat** implied volatility curve corresponds exactly to a lognormal implied distribution (BSM). Any slope or curvature means the market's distribution is not lognormal.

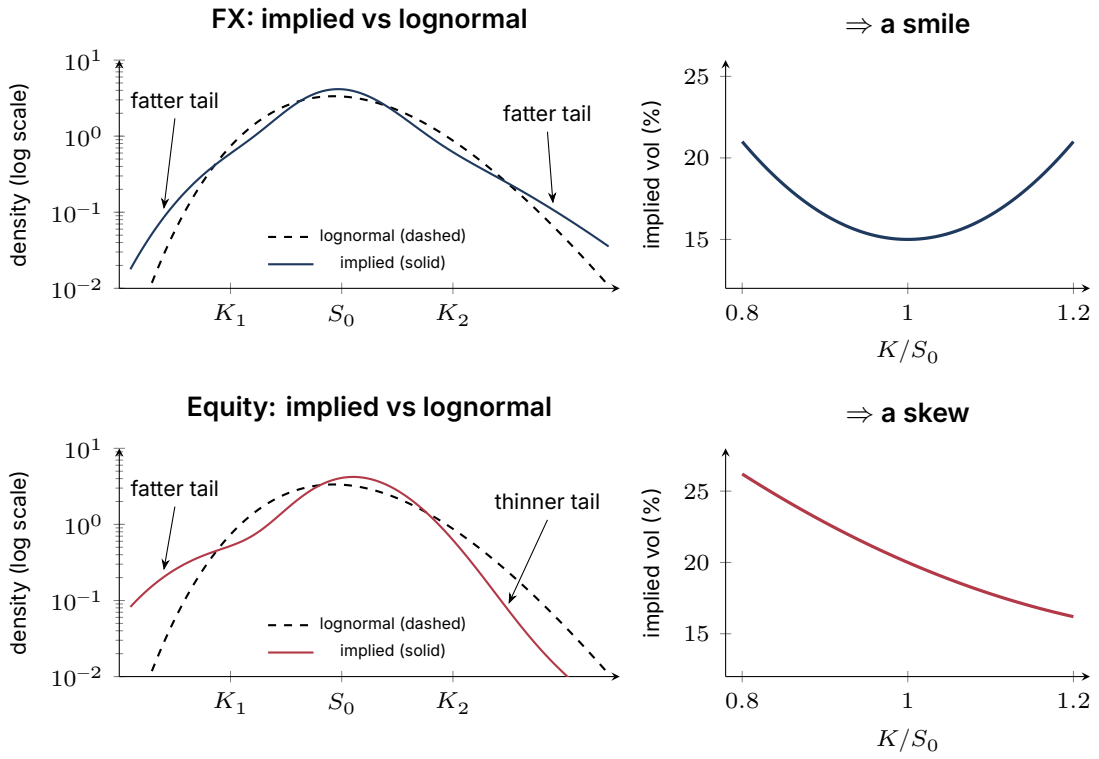


Figure 3. The distribution on the left is the curve on the right, read a different way. The densities use a log scale so the tails are visible (both implied curves have the same standard deviation as the lognormal). Top: the currency implied distribution is fatter than lognormal in both tails (and more peaked), so options struck well below and well above the money are dearer than BSM says and implied volatility rises at both ends. Bottom: the equity implied distribution is fatter in the left tail and thinner in the right, so only low-strike options are dear and implied volatility falls across the range.

2. FX and equity, side by side

	Foreign currency	Equity (stocks, indices)
Implied vs lognormal	Heavier left *and* right tail; more peaked	Heavier left tail, *less* heavy right tail
Deep OTM put (K_1)	Relatively high price, high IV	Relatively high price, high IV
Deep OTM call (K_2)	Relatively high price, high IV	Relatively *low* price, low IV
Curve	Smile (U-shape)	Skew (falls with strike)
One ATM vol (lognormal) prices	*Both* wings below market	OTM puts below market, OTM calls *above* market

The last row is the August shape (A3·S4): a single at-the-money volatility is too low for every option whose implied volatility sits above the ATM level, so those options are underpriced relative to the market, and vice versa.

3. Other shapes, same rule

Implied distribution vs lognormal	Implied volatility curve
Lognormal (BSM holds)	Flat
Both tails heavier (kurtosis > 3), e.g. stochastic volatility, normal mixture	Smile
Heavier left, thinner right (negative price-volatility correlation)	Downward skew
Heavier right, thinner left (*positive* price-volatility correlation)	Upward skew
Both tails thinner (platykurtic; exchange rate held in a narrow band)	Frown (highest at the money)
Bimodal: one large anticipated jump (MR-17 i)	Frown

Risk reversal. Traders summarise the tilt as RR $= \sigma_{imp}(25\Delta \text{ call}) - \sigma_{imp}(-25\Delta \text{ put})$. Both are OTM options: a heavier right tail and lighter left tail give RR > 0; the usual equity skew gives RR < 0; a symmetric smile gives RR ≈ 0.

Deep OTM options as "options on volatility". Their value comes almost entirely from the tails, so they are the instruments most sensitive to the shape of the distribution (Hull Q17.16).

Exam traps

- **Sibling swap.** A heavier *right* tail is the FX property, not the equity one.
- **Polarity flip.** The FX implied distribution has a *heavier* (not thinner) left tail, so deep OTM FX puts are underpriced by a single ATM vol.
- **Pricing direction.** "Priced below market" with a lognormal means the market (implied) price is higher, i.e. the implied distribution puts more mass in that tail.
- **Scope error.** Put-call parity does not equalise implied volatilities across strikes.
- **Correlation sign.** Negative price-volatility correlation gives the downward skew; positive gives an upward skew.
- **Flat means lognormal.** A smile or skew that "still has a lognormal implied distribution" is internally inconsistent.

Practice questions: MR-17 c

Question MR-17 c.1

Bank Q793 • AnalystPrep Q.1527 (rebuilt)

In the Black-Scholes-Merton model the underlying price follows geometric Brownian motion with constant volatility. If every option price in the market were consistent with this model, which of the following would be observed?

A. Implied volatility flat across strikes, and a lognormal implied distribution of the terminal price.

B. A U-shaped smile in implied volatility, and a lognormal implied distribution with fatter tails.

C. A downward-sloping skew in implied volatility, and a left-skewed implied distribution.

D. Implied volatility that varies with strike but a lognormal implied distribution.

Question MR-17 c.2

Bank Q802 • Schweser Module Quiz 17.1 Q2 (key corrected)

An empirical distribution that exhibits a fatter right tail than that of a lognormal distribution would indicate:

A. equal implied volatilities across low and high strike prices.

B. greater implied volatilities for low strike prices.

C. greater implied volatilities for high strike prices.

D. higher implied volatilities for mid-range strike prices.

Question MR-17 c.3

Bank Q800 • BT notes P2.T5.409.3 (rebuilt)

Implied volatilities computed from Google call options slope downward: implied volatility falls as the strike price rises, as is typical of equity options. An analyst draws these conclusions:

I. The implied distribution has a heavier left tail than a lognormal distribution.

II. The implied distribution has a lighter right tail than a lognormal distribution.

III. An out-of-the-money put with strike 90% of spot is priced at a higher implied volatility than an out-of-the-money call with strike 110% of spot.

IV. The implied distribution has a heavier right tail than a lognormal distribution.

Which conclusions are correct?

A. I only

B. II and III only

C. I, II and III only

D. I, II, III and IV

Question MR-17 c.4 Bank Q795 • BT P2.T5.410.3

Suppose the implied distribution of a certain option asset class exhibits, relative to the lognormal distribution, a lighter left tail and a heavier right tail. Define risk reversal (RR) as the difference in implied volatility between a 25-delta call and a -25-delta put: RR $=$ implied volatility [25-delta call] $-$ implied volatility [-25-delta put]. Which of the following is most likely TRUE of RR?

A. Negative risk reversal

B. Zero risk reversal

C. Positive risk reversal

D. Risk reversal approximately equal to the implied volatility of the 50-delta (ATM) call

Question MR-17 c.5 Bank Q794 • BT P2.T5.410.2

Suppose the volatility is uncertain and positively correlated with the underlying asset price. Relative to prices implied by constant volatility, which of the following scenarios is MOST PLAUSIBLE?

A. Lower price for out-of-the-money (OTM) put options and higher price for in-the-money (ITM) put options

B. Lower price for out-of-the-money (OTM) call options and higher price for in-the-money (ITM) call options

C. Lower price for at-the-money (ATM) call options and higher price for at-the-money (ATM) put options

D. Lower price for in-the-money (ITM) put options and higher price for out-of-the-money (OTM) put options

Question MR-17 c.6 Bank Q799 • BT notes P2.T5.23.7.3 (rebuilt)

Four archetypal shapes of implied volatility plotted against K/S are defined as follows. *Smile*: implied volatility is lowest at the money and higher at both low and high strikes. *Frown*: implied volatility is highest at the money and lower at both low and high strikes. *Downward skew*: implied volatility falls steadily as strike rises. *Upward skew*: implied volatility rises steadily as strike rises. Consider the scenarios:

I. Asset volatility is positively correlated with the asset price.

II. The asset price distribution has a heavy left tail and a light right tail.

III. The underlying is an exchange rate that trades in a very narrow band because of central bank policy.

IV. An option expires in one week and, because of an expected announcement, the stock will jump either up or down by more than 16% within the week.

Which shapes correspond to scenarios I to IV?

A. I. Upward skew, II. Downward skew, III. Frown, IV. Frown

B. I. Downward skew, II. Upward skew, III. Frown, IV. Frown

C. I. Upward skew, II. Downward skew, III. Smile, IV. Smile

D. I. Downward skew, II. Downward skew, III. Frown, IV. Smile

Question MR-17 c.7

Bank Q803 • GARP PE1 Q56 • Mock 3 Q25

A risk committee of the board of company ABC is discussing the difference between pricing deep out-of-the-money call options on ABC stock and pricing deep out-of-the-money call options on the USD/GBP foreign exchange (FX) rate using the Black-Scholes-Merton model. The committee considers pricing each of these two options based on two distinct probability distributions of underlying asset prices at the option expiration date: a lognormal probability distribution, and an implied risk-neutral probability distribution obtained from the volatility smile for each aforementioned option of the same maturity and the same moneyness. If the implied risk-neutral probability distribution is used instead of the lognormal distribution, which of the following is correct?

A. The price of the option on ABC stock would be relatively high and the price of the option on USD/GBP FX rate would be relatively low compared to those computed from the lognormal counterparts.

B. The price of the option on ABC stock would be relatively low and the price of the option on USD/GBP FX rate would be relatively high compared to those computed from the lognormal counterparts.

C. The price of the option on ABC stock would be relatively low and the price of the option on USD/GBP FX rate would be relatively low compared to those computed from the lognormal counterparts.

D. The price of the option on ABC stock would be relatively high and the price of the option on USD/GBP FX rate would be relatively high compared to those computed from the lognormal counterparts.

Question MR-17 c.8

Workbook A3.3 • S2, S3, S4, S5, S7 • Map #4

A junior trader at a Canadian dealer prices every European option on a stock index, and every European option on USD/CAD, with a single lognormal volatility per underlying equal to that underlying's at-the-money implied volatility. A market risk analyst compares the resulting model prices with market prices of options with the same maturity. Which of the following conclusions should the analyst reach?

A. Deep out-of-the-money index puts are priced below market, because the implied distribution has a heavier left tail than the lognormal.

B. Deep out-of-the-money index calls are priced below market, because the implied distribution has a heavier right tail than the lognormal.

C. Deep out-of-the-money USD/CAD puts are priced above market, because the implied distribution has a thinner left tail than the lognormal.

D. Options at every strike are priced at market, because put-call parity forces one implied volatility on every option with the same maturity.

Solutions • MR-17 c

Solution MR-17 c.1

Answer: A

Bank Q793 • AnalystPrep Q.1527

Step 1 – What BSM assumes

Constant-volatility GBM means $\ln S_T$ is normal, so S_T is lognormal.

Step 2 – What that does to implied volatility

One volatility prices every strike, so every strike backs out the same implied volatility.
$\Rightarrow$ Flat implied volatility and a lognormal implied distribution are two descriptions of the same assumption.

Interpretation

Any smile or skew is therefore evidence that the market's implied distribution is not lognormal.

Why the other options are wrong

B is incorrect. A U-shaped smile means fatter tails than lognormal, which contradicts "lognormal".

C is incorrect. Describes the equity skew, whose implied distribution is not lognormal; it violates BSM rather than reflecting it.

D is incorrect. Internally inconsistent: strike-dependent implied volatility means the implied distribution differs from the lognormal.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 and Appendix

Solution MR-17 c.2

Answer: C

Bank Q802 • Schweser Module Quiz 17.1 Q2

Step 1 – Where the extra probability sits

A fatter right tail means high underlying prices are more likely than under the lognormal.

Step 2 – Which options pay off there

High-strike options (OTM calls, ITM puts) pay off in that region, so their market prices exceed BSM prices at a single volatility.
$\Rightarrow$ Higher prices mean higher implied volatility at **high** strikes.

Interpretation

This is the upward-skew case (the mirror image of equities). The source key was corrected to C.

Why the other options are wrong

A is incorrect. Equal implied volatilities is the flat line of a lognormal distribution, which a fatter tail contradicts.

B is incorrect. Describes a fatter *left* tail (the equity skew); the strike side is reversed.

D is incorrect. Elevated mid-range volatility is a frown, which comes from thin tails, not a fat right tail.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 and 17.3; SchweserNotes Book 1, Reading 17

Solution MR-17 c.3 — Answer: C

Bank Q800 • BT notes P2.T5.409.3

Step 1 – Left side of the skew

High implied volatility at low strikes means options that pay off on large falls are dear: the left tail is heavier than lognormal. I is true.

Step 2 – Right side of the skew

Low implied volatility at high strikes means options that pay off on large rises are cheap: the right tail is lighter. II is true and IV is false.

Step 3 – Compare the two OTM options

The 90% put sits on the high-volatility side; the 110% call on the low-volatility side. III is true. $\Rightarrow$ I, II and III.

Interpretation

A downward skew carries three linked facts: heavy left tail, thin right tail, expensive downside protection.

Why the other options are wrong

A is incorrect. Recognises only the heavy left tail and misses the two other consequences of the same slope.

B is incorrect. Omits I, the core implication of high implied volatility at low strikes.

D is incorrect. Includes IV: a heavier right tail belongs to a smile or an upward skew, not to the equity skew.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.3, Figures 17.3 and 17.4

Solution MR-17 c.4 — Answer: C

Bank Q795 • BT P2.T5.410.3

Step 1 – Lighter left tail

Low-strike options are cheap, so the -25-delta put (OTM, low strike) has a *low* implied volatility.

Step 2 – Heavier right tail

High-strike options are dear, so the 25-delta call (OTM, high strike) has a *high* implied volatility. $\Rightarrow$ RR $=$ high $-$ low > 0: a positive risk reversal.

Interpretation

The 25-delta call sits in the right wing and the -25-delta put in the left wing, so RR measures the tilt of the curve.

Why the other options are wrong

A is incorrect. Negative RR is the usual equity result (heavy left, light right); the tails here are the other way round.

B is incorrect. Zero RR needs a symmetric smile with equally heavy tails.

D is incorrect. RR is a difference between two volatilities, not a level; it has no link to the ATM level.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 to 17.4

Solution MR-17 c.5 — Answer: A

Bank Q794 • BT P2.T5.410.2

Step 1 – Translate the correlation into tails

Positive correlation: volatility is high when the price is high and low when it is low. The implied distribution has a heavier right tail and a lighter left tail (upward skew).

Step 2 – Map strikes to tails

Low-strike options (OTM puts, ITM calls) depend on the left tail: cheaper than constant-volatility prices. High-strike options (OTM calls, ITM puts) depend on the right tail: dearer.
$\Rightarrow$ OTM put cheaper and ITM put dearer.

Interpretation

This is the mirror image of the equity case (negative correlation). Translate "ITM/OTM put/call" into low or high strike before choosing.

Why the other options are wrong

B is incorrect. Reversed for calls: OTM calls (high strike) are dearer here. This is the negative-correlation (equity) answer.
C is incorrect. ATM options are barely affected by the tails, and parity ties the ATM call and put together, so they cannot move in opposite directions.
D is incorrect. Reversed for puts: ITM puts (high strike) are dearer and OTM puts (low strike) cheaper under positive correlation.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.3; Practice Question 17.13

Solution MR-17 c.6 — Answer: A

Bank Q799 • BT notes P2.T5.23.7.3

Step 1 – I: positive price-volatility correlation

High-price outcomes carry more variance, so the right tail is heavier: high-strike options have higher implied volatility. Upward skew.

Step 2 – II: heavy left, light right

Low-strike options are dear: implied volatility falls with strike. Downward skew.

Step 3 – III: a narrow central-bank band

Extreme moves are truncated, so both tails are light and wing options are cheap. Frown.

Step 4 – IV: one large anticipated jump

The terminal distribution is bimodal; away-from-the-money options are cheap relative to ATM. Frown (MR-17 i).
$\Rightarrow$ Only A matches all four.

Interpretation

Every case uses the same rule: implied volatility is high where the implied distribution has more weight than the lognormal.

Why the other options are wrong

B is incorrect. Reverses both skews: a heavy left tail gives high implied volatility at *low* strikes.
C is incorrect. Gets the skews right but treats light tails (III) and a single jump (IV) as heavy-tailed smiles.
D is incorrect. Treats positive correlation as a downward skew (the equity, negative-correlation case) and calls the jump a smile.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2, 17.3 and 17.8; Practice Questions 17.13 and 17.15

Solution MR-17 c.7 Answer: B

Bank Q803 • GARP PE1 Q56

Step 1 – What a deep OTM call depends on

It pays off only in the right tail, so its price depends on right-tail probability.

Step 2 – Equity (ABC)

The equity skew implies a *thinner* right tail than lognormal: the call is worth relatively less.

Step 3 – FX (USD/GBP)

The FX smile implies heavier tails on both sides: the right tail is fatter, so the call is worth relatively more.

$\Rightarrow$ ABC relatively low, USD/GBP relatively high.

Interpretation

The same instrument (deep OTM call) moves in opposite directions on the two underlyings because only the right tail matters, and that is where equity and FX differ.

Why the other options are wrong

A is incorrect. Reverses both: treats equity as fat-right-tailed and FX as thin.

C is incorrect. Treats FX like equity, ignoring that the FX smile raises the right tail.

D is incorrect. Treats equity like FX, ignoring the skew that thins the right tail.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 and 17.3, Figures 17.2 and 17.4

Solution MR-17 c.8 Answer: A

Workbook A3.3 • S2, S3, S4, S5, S7

Step 1 – What a single lognormal volatility assumes

BSM with one volatility assumes a lognormal distribution. The market prices with a smile, which corresponds to the implied distribution (Hull 17.2, 17.3 and Appendix).

Step 2 – Index options

The equity implied distribution has a heavier left tail and a less heavy right tail than the lognormal. A deep OTM put pays off only in the left tail, so the market prices it relatively high, at an implied volatility above the at-the-money level. The trader's at-the-money volatility therefore prices it below market. A deep OTM call pays off only in the thin right tail and is priced relatively low by the market, so the trader prices it above market.

Step 3 – FX options

The FX implied distribution has heavier tails on both sides. Both deep OTM puts and deep OTM calls carry implied volatilities above the at-the-money level, so the trader prices both wings below market. Table 17.1 supports the market: 3 SD daily moves occur on 1.30% of days versus 0.27% under the lognormal model.

Interpretation

The trader is selling tail protection too cheaply on index puts and on both FX wings: the trade Business Snapshot 17.1 describes, from the wrong side.

Why the other options are wrong

B is incorrect. Sibling swap: a heavier right tail is the FX property, not the equity one. The index implied distribution has a less heavy right tail, so deep OTM index calls are priced above market by the lognormal.
C is incorrect. Polarity flip: the FX implied distribution has a heavier, not thinner, left tail, so deep OTM USD/CAD puts are priced below market.
D is incorrect. Scope error: put-call parity makes a call and a put with the same strike and maturity share one implied volatility; it does not equalise implied volatilities across strikes.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. • **Reference:** Hull, OFOD 11e, Ch.20, 17.1 to 17.3, Table 17.1, Business Snapshot 17.1

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 d

P1 CONCEPT

Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility.

Workbook: A3 Volatility Smiles, Skews and Put-Call Parity • Tier A • Map #4

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A3. Map #4: "2 questions from volatility smiles on FX and equity options" and "implied volatility: foreign exchange options shape".
Vardeez: full chapter is important.

Reported exam shapes

- **A3·S2** Two questions on FX and equity smiles.
- **A3·S3** Implied volatility shape for FX options.
- **A3·S4** Lognormal vols versus implied vols for FX and index options (worked in MR-17 c, Question MR-17 c.8).

What the questions look like: what the FX distribution looks like relative to the lognormal (heavier tails, more peaked), why (non-constant volatility, jumps), what that does to option prices and implied volatility, and how maturity changes the picture.

1. What FX distributions look like

- **Heavier tails than the lognormal** in both directions (kurtosis), and **more peaked**: both small and large moves are more likely than under the lognormal; intermediate moves are less likely.
- **Evidence (Hull Table 17.1).** Daily moves of 10 currencies against the USD, 2005 to 2015, measured in standard deviations of the daily change:

Move	Real world	Lognormal
>1 SD	23.32%	31.73%
>2 SD	4.67%	4.55%
>3 SD	1.30%	0.27%
>4 SD	0.49%	0.01%
>5 SD	0.24%	0.00%
>6 SD	0.13%	0.00%

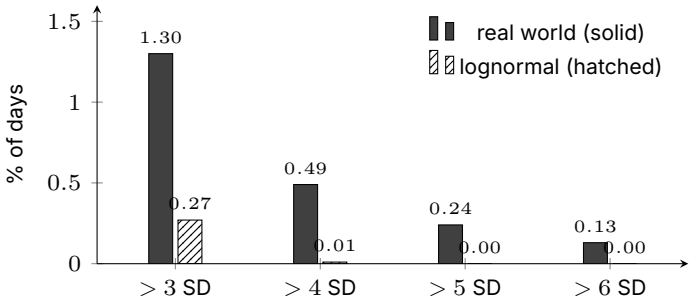


Figure 4. Hull Table 17.1. Moves beyond 3 SD happen about five times as often as the lognormal predicts, and moves beyond 4 to 6 SD, which the lognormal says should hardly ever happen, still occur. Note the first row: moves beyond 1 SD are *less* frequent than lognormal, which is the "more peaked" centre, not a sign of thin tails.

2. Why exchange rates are not lognormal

Two of the conditions for a lognormal price are:

1. The volatility of the asset is **constant**.
2. The price changes **smoothly, with no jumps**.

Neither holds for exchange rates: volatility is far from constant, and rates frequently **jump**, sometimes in response to central bank actions. Both make extreme outcomes more likely, in either direction (the volatility of an exchange rate has little correlation with its level, so the effect is roughly symmetric).

Maturity. As option maturity increases:

- the percentage impact of non-constant volatility on *prices* becomes **more** pronounced, but its impact on *implied volatility* usually becomes **less** pronounced;
- the percentage impact of jumps on both prices and implied volatility becomes **less** pronounced (jumps are averaged out).

⇒ Net result: the FX smile **flattens** as maturity increases. Short-dated options on jump-prone currencies show the most pronounced smile.

3. Implications for option prices and implied volatility

- Deep OTM calls (K_2) and deep OTM puts (K_1) are both worth **more** than BSM with the ATM volatility says, so both carry **higher implied volatilities**: the U-shaped smile, lowest at the money.
- **The trade (Business Snapshot 17.1).** If the market prices with one lognormal volatility and you know the tails are heavy, buy deep OTM calls and puts on many currencies and wait (a long strangle). The mid-1980s traders who knew this made money; by the late 1980s the smile was priced in and the opportunity disappeared.
- **A credible band** (a central bank holding the rate between 0.97 and 1.03) does the opposite: the tails are truncated, wing options rarely pay off, and implied volatility is highest at the money: a frown.

Exam traps

- **Sibling swap.** FX reasons are non-constant volatility and jumps; leverage and crashophobia belong to equities.
- **Tails vs centre.** The FX implied distribution has *both* tails heavier; moves of 1 to 2 SD are *less* likely, not more.
- **Maturity polarity.** The smile becomes *less* pronounced for longer maturities. The price impact of non-constant volatility grows with maturity; its implied-volatility impact shrinks.
- **Band vs free float.** A tight central-bank band thins the tails and gives a frown, not a smile.
- **Trade direction.** Heavy tails mean be *long* the wings (long strangle), not short volatility (short straddle) or long a butterfly.

Practice questions: MR-17 d

Question MR-17 d.1

Bank Q801 • GARP PE1 Q51 • Mock 3 Q29

A risk analyst at an investment bank is examining the assumptions the bank uses in its foreign exchange (FX) option pricing model. Currently, the implied volatility in a particular FX pair is relatively low if an option is at-the-money and becomes progressively higher as the option moves either more in-the-money or more out-of-the-money. How does the distribution of option prices on this FX pair implied by the Black-Scholes-Merton model compare to a lognormal distribution with the same mean and standard deviation?

A. The implied distribution has a heavier left tail and a less heavy right tail.

B. The implied distribution has a heavier left tail and a heavier right tail.

C. The implied distribution has a less heavy left tail and a heavier right tail.

D. The implied distribution has a less heavy left tail and a less heavy right tail.

Question MR-17 d.2

New • built from GARP Ch.17 Table 17.1 and Business Snapshot 17.1

A currency options strategist at a Japanese bank studies daily moves in 10 exchange rates against the USD over ten years and compares the frequencies with those predicted by a lognormal model:

Daily move	Real world	Lognormal model
$>$ 1 SD	23.32%	31.73%
$>$ 3 SD	1.30%	0.27%
$>$ 5 SD	0.24%	0.00%

Which of the following conclusions is most consistent with the data?

A. Exchange rate distributions are more peaked and have heavier tails than the lognormal, so deep out-of-the-money calls and puts should trade at higher implied volatilities than at-the-money options.

B. Exchange rate distributions have heavier tails than the lognormal only on the downside, so only deep out-of-the-money puts should trade at higher implied volatilities than at-the-money options.

C. Moves larger than 1 SD occur less often than the lognormal predicts, so exchange rate distributions have thinner tails and deep out-of-the-money options should trade at lower implied volatilities than at-the-money options.

D. Moves larger than 3 SD occur about five times as often as the lognormal predicts, so the at-the-money implied volatility should be raised while implied volatility stays flat across strikes.

Question MR-17 d.3 Bank Q807 • AnalystPrep Q.1713

In practice, exchange rates do not follow a lognormal distribution, as the exchange rate's volatility is far from constant and there are frequent jumps. Which of the following statements is true for this scenario?

A. As the maturity of the option increases, the percentage impact of non-constant volatility on prices becomes less pronounced.

B. The effects of variable volatility and jumps are independent of the maturity of options.

C. When the maturity of options increases, the volatility smile becomes less pronounced.

D. Variable volatility and jump risks have a uniform influence on options, regardless of strike or term structure.

Question MR-17 d.4 Bank Q808 • BT notes P2.T5.07.02 (rebuilt)

A risk manager compares four currency options and wants the one whose implied volatilities will deviate most from a flat line across strikes (the most pronounced volatility smile), because the BSM lognormal assumption fits it worst. Which option is it?

A. A 5-year option on a major currency pair with stable volatility

B. A 1-month option on a currency pair whose volatility frequently jumps because of central bank interventions and policy surprises

C. A 2-year option on a major currency pair with stable volatility

D. A 5-year option on an emerging-market currency

Question MR-17 d.5 New • built from GARP Ch.17 Q17.15

A central bank has credibly committed to keep an exchange rate between 0.97 and 1.03, and the rate currently trades at 1.00. An FX options trader at a Brazilian bank is asked what pattern of implied volatilities to expect for 3-month options on the rate. Which of the following is the most appropriate answer?

A. Implied volatility is highest at the money and lower for strikes near or beyond 0.97 and 1.03, because the band makes both tails thinner than the lognormal.

B. Implied volatility is lowest at the money and higher for strikes near or beyond 0.97 and 1.03, because central bank actions make the rate jump.

C. Implied volatility falls steadily as the strike rises, because the central bank defends the lower edge of the band more strongly.

D. Implied volatility is flat across strikes, because the band reduces the volatility of the rate by the same amount at every strike.

Question MR-17 d.6

Bank Q810 • BT notes P2.T5.410.1

Suppose most market participants think that foreign currency exchange rates are lognormally distributed, such that they are comfortable using the same (constant) volatility to value options with the Black-Scholes option pricing model (BSM OPM). Risk Analyst Sally has performed an analysis, however, and she knows that the lognormal assumption is not good: rather, she expects actual distributions with heavier tails, both left and right, than the lognormal. Which trade most takes advantage of this situation?

A. Long butterfly

B. Short straddle

C. Long strangle

D. Long bull spread

Question MR-17 d.7

Workbook A3.2 • S2, S3, S6 • Map #4

A risk consultant is reviewing the options desk of a German bank that trades EUR/USD options and options on a European equity index. The desk prices the two books with differently shaped volatility smiles, and the consultant is preparing a report explaining why. Which of the following statements is correct to include in the report?

A. The FX smile is U-shaped because exchange rate volatility is not constant and rates jump, while the equity skew slopes downward because equity prices and volatility are negatively correlated.

B. The FX smile is U-shaped because of the leverage effect and crashophobia, while the equity skew slopes downward because volatility is not constant and prices jump on central bank actions.

C. The FX smile is U-shaped because exchange rate volatility is not constant and rates jump, while the equity skew slopes downward because equity prices and volatility are positively correlated.

D. The FX smile is U-shaped because exchange rate volatility is not constant and rates jump, while the equity skew slopes downward in a pattern that was equally marked before the October 1987 crash.

Solutions • MR-17 d

Solution MR-17 d.1 — Answer: B

Bank Q801 • GARP PE1 Q51

Step 1 – Name the shape

Lowest at the money and rising in both directions: a symmetric smile.

Step 2 – Read each wing

High implied volatility at low strikes: the market prices large falls above the lognormal, so the left tail is heavier. High implied volatility at high strikes: large rises are priced above the lognormal, so the right tail is heavier.

$\Rightarrow$ Both tails heavier.

Interpretation

This is the textbook FX picture (Hull Figures 17.1 and 17.2). The four options are the four combinations of heavier or lighter on each side.

Why the other options are wrong

A is incorrect. The equity skew (only low strikes elevated); here the right wing is elevated too.

C is incorrect. An upward skew (only high strikes elevated); the left wing is elevated too.

D is incorrect. A frown (lowest implied volatility in the wings), the opposite of the smile described.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2, Figures 17.1 and 17.2

Solution MR-17 d.2 — Answer: A

New • built from GARP Ch.17 Table 17.1

Step 1 – Read the tails

Beyond 3 SD: 1.30% versus 0.27% (about $4.8\times$). Beyond 5 SD: 0.24% versus essentially zero. Large moves are far more frequent than lognormal: heavy tails.

Step 2 – Read the centre

Beyond 1 SD: 23.32% versus 31.73%. More days than predicted fall *inside* 1 SD: the distribution is more peaked.

Step 3 – Implication for options

Heavy tails on both sides make deep OTM calls and puts worth more than BSM at the ATM volatility, so their implied volatilities are higher: a smile.

Interpretation

Hull's point is that the table supports the smile traders use. The table counts moves of either sign, which is why the conclusion is about both wings.

Why the other options are wrong

B is incorrect. Sibling swap: a heavy downside only is the equity picture; the data measure moves in both directions.

C is incorrect. Misreads the first row: fewer moves beyond 1 SD reflects a more peaked centre, while the 3 to 5 SD rows show heavier tails.

D is incorrect. True fact, wrong conclusion: raising one volatility for all strikes keeps the lognormal shape and still underprices the wings relative to the ATM option.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2, Table 17.1, Business Snapshot 17.1

Solution MR-17 d.3 — Answer: C

Bank Q807 • AnalystPrep Q.1713

Step 1 – Non-constant volatility

With longer maturity its percentage effect on *prices* grows, but its effect on *implied volatility* usually shrinks.

Step 2 – Jumps

Their percentage effect on both prices and implied volatility shrinks with maturity (they average out).

$\Rightarrow$ Net: the smile becomes less pronounced as maturity increases.

Interpretation

Hull's sentence has two halves pointing in different directions for non-constant volatility. The exam uses the price half as a trap (option A).

Why the other options are wrong

A is incorrect. Polarity flip on the price effect: it becomes *more* pronounced with maturity.
B is incorrect. Both effects depend on maturity; they are strongest for short-dated options.
D is incorrect. The effects vary with strike (the smile) and with maturity, so they are not uniform.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2, Reasons for the Smile in Foreign Currency Options

Solution MR-17 d.4 — Answer: B

Bank Q808 • BT notes P2.T5.07.02

Step 1 – Sources of the smile

Non-constant volatility and jumps create heavier tails than lognormal.

Step 2 – Effect of maturity

Both effects are strongest, in implied-volatility terms, for short-dated options; long maturities flatten the smile.

Step 3 – Rank the four

Stable-volatility major pairs (A, C) deviate least. A 5-year maturity (A, D) mitigates the smile. The 1-month option on a jump-prone currency (B) combines the strongest source with the shortest maturity.

$\Rightarrow$ B.

Interpretation

Two dials, source strength and maturity, both point to B.

Why the other options are wrong

A is incorrect. Long maturity and stable volatility both reduce the smile: among the flattest.
C is incorrect. Stable volatility on a major pair fits the lognormal best.
D is incorrect. Emerging-market currencies can jump, but the 5-year maturity averages the jumps out.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2

Solution MR-17 d.5 — Answer: A

New • built from GARP Ch.17 Q17.15

Step 1 – What the band does to the distribution

A credible band truncates the distribution at 0.97 and 1.03: both tails are much thinner than a lognormal with the same ATM volatility.

Step 2 – Implication for prices

Options struck near or beyond the band edges have far less chance of paying off than BSM implies, so they are cheap and their implied volatilities are low.

$\Rightarrow$ Implied volatility is highest at the money and falls toward the wings: a frown, the inverse of the usual FX smile.

Interpretation

The same rule as always: less probability in a tail than the lognormal means lower implied volatility for options that pay off there.

Why the other options are wrong

B is incorrect. Applies the free-floating FX story (jumps on central bank actions) to a rate the central bank is holding inside a band; that gives a smile only when tails are heavy.

C is incorrect. Sibling swap: a downward skew needs a heavier left tail and thinner right tail; a symmetric band thins both.

D is incorrect. A band changes the *shape* of the distribution, not just its scale, so implied volatility cannot stay flat.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), Practice Question 17.15

Solution MR-17 d.6 — Answer: C

Bank Q810 • BT notes P2.T5.410.1

Step 1 – What the market misprices

With one constant volatility the market underprices deep OTM puts and calls relative to Sally's heavy-tailed view.

Step 2 – Choose the position

Be long both tails: buy an OTM put and an OTM call, which is a long strangle.

$\Rightarrow$ Long strangle.

Interpretation

This is Business Snapshot 17.1: buy deep OTM calls and puts on many currencies and wait.

Why the other options are wrong

A is incorrect. A long butterfly is short the wings and profits from small moves: the opposite of a fat-tail view.

B is incorrect. A short straddle is short volatility and loses on large moves.

D is incorrect. A bull spread is a directional bet; Sally's view is symmetric and about the tails.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), Business Snapshot 17.1

Solution MR-17 d.7 **Answer: A**

Workbook A3.2 • S2, S3, S6

Step 1 – FX shape and reason

Implied volatility is relatively low at the money and progressively higher into or out of the money (Hull 17.2). Two lognormal conditions fail: volatility is far from constant and exchange rates frequently jump, sometimes on central bank actions. Both make extreme outcomes, in either direction, more likely.

Step 2 – Equity shape and reason

Volatility decreases as the strike increases (Hull 17.3). There is a negative correlation between equity prices and volatility, explained by leverage, the volatility feedback effect and crashophobia, so declines bring higher volatility (heavy left tail) and rises bring lower volatility (thin right tail).

Interpretation

The consultant should explain each smile by its own market's mechanism; the two shapes are not interchangeable even though both depart from the lognormal.

Why the other options are wrong

B is incorrect. Sibling swap: the explanations are exchanged. Leverage and crashophobia explain the equity skew; non-constant volatility and jumps (including central bank actions) explain the FX smile.

C is incorrect. Polarity flip: the equity correlation between prices and volatility is negative; a positive correlation would thicken the right tail instead.

D is incorrect. Scope error: there was no marked volatility smile for equity options before October 1987; the skew has existed only since the crash, which is why crashophobia is one explanation.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20, 17.2 and 17.3, Business Snapshot 17.2

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 e

P1 CONCEPT

Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape.

Workbook: A3 Volatility Smiles, Skews and Put-Call Parity • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic A3, the most heavily tagged LO in it. Map #4: "equity options volatility smile property"; August 2026 recall on the equity skew.
Vardeez: full chapter is important.

Reported exam shapes

- **A3·S1** Equity skew: effect of changing strike and maturity (Question MR-17 a.6).
- **A3·S2, S3** FX and equity smiles; FX shape (Questions MR-17 a.4, d.1, d.7).
- **A3·S5** Equity options volatility smile property.
- **A3·S6** Sibling swap: FX explanation attributed to equities or vice versa (Question MR-17 d.7). The map's flagged trap.

What the questions look like: describe each shape, and match each shape to its explanation. The workbook questions for this LO sit under MR-17 a (A3.1) and MR-17 d (A3.2); this block adds the explanation questions.

1. The two shapes, recapped

- **Foreign currency: a smile.** Implied volatility is higher for deep ITM and deep OTM options than for ATM options. Implied distribution: both tails heavier than lognormal. (Shape detail in MR-17 a; distribution in MR-17 c and d.)
- **Equity: a skew or smirk.** Higher implied volatility for low strikes (OTM puts, ITM calls) than for high strikes (ITM puts, OTM calls): an inverse relationship with K/S_0. Implied distribution: heavier left tail, thinner right tail. Equity traders believe large *down* moves are more likely than large up moves, compared with the lognormal.
- **History.** Before the October 1987 crash there was no marked equity smile; implied volatilities were much less dependent on the strike.

2. Why equities skew

The core fact is a **negative correlation between equity prices and volatility**: as prices fall, volatility rises, making even larger declines possible; as prices rise, volatility falls, making further rises less likely. That produces the heavy left tail and thin right tail. Three explanations (Hull 17.3; Vardeez lists the first and third):

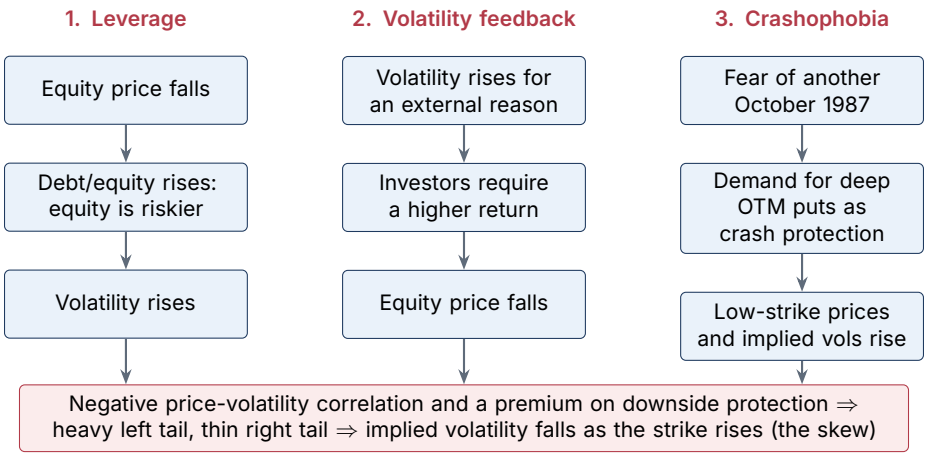


Figure 5. The three explanations of the equity skew. Note the opposite causal directions of the first two: leverage runs price → volatility; volatility feedback runs volatility → price.

Evidence for crashophobia (Rubinstein). Declines in the S&P 500 tend to be accompanied by a *steepening* of the skew; when the S&P rises the skew becomes less steep. Demand for downside protection (e.g. during a geopolitical crisis) bids up OTM put prices and steepens the skew.

3. Why currencies smile

Exchange rate volatility is not constant and rates jump (sometimes on central bank actions), which makes extreme moves in *either* direction more likely than lognormal (MR-17 d). Long-dated options show less of a smile than short-dated ones.

	Foreign currency	Equity
Shape	Smile: low ATM, high in both wings	Skew: high at low strikes, falling with strike
Implied distribution	Both tails heavier, more peaked	Heavier left, thinner right
Explanation	Non-constant volatility; jumps (central banks)	Negative price-volatility correlation: leverage, volatility feedback, crashophobia
Since	Priced in by late 1980s	October 1987 crash
Longer maturity	Less pronounced	Less pronounced

Exam traps

- **Sibling swap (the map's flagged trap).** Leverage and crashophobia explain equities; non-constant volatility and jumps explain FX.
- **Correlation sign.** Equity price and volatility are *negatively* correlated.
- **Crashophobia is a fear priced into options**, not a statement that realised volatility rises when

prices fall (that is leverage).

- **Causal direction.** Leverage: price falls, then volatility rises. Volatility feedback: volatility rises, then price falls.
- **1987.** The equity skew has existed only since the crash; it was not "equally marked" before.
- **Demand at low strikes** raises implied volatility at low strikes and steepens the downward skew; it does not create an upward smirk.

Practice questions: MR-17 e

Workbook questions for this LO: A3.1 (worked as Question MR-17 a.6) and A3.2 (worked as Question MR-17 d.7). Attempt them again here before the questions below.

Question MR-17 e.1

Bank Q823 • Schweser Module Quiz 17.1 Q4

Which of the following statements regarding equity option volatility is true?

A. There is higher implied price volatility for away-from-the-money equity options.

B. "Crashophobia" suggests actual equity volatility increases when stock prices decline.

C. Compared to the lognormal distribution, traders believe the probability of large down movements in price is similar to large up movements.

D. Increasing leverage at lower equity prices suggests increasing volatility.

Question MR-17 e.2

New • built from GARP Ch.17, 17.3 and Business Snapshot 17.2

A strategist at a UK asset manager is explaining to clients why implied volatility on FTSE 100 options falls as the strike price rises. Which of the following explanations is correctly described?

A. Leverage effect: an external rise in volatility makes investors require a higher return, so the equity price falls.

B. Volatility feedback effect: a fall in the equity price raises the firm's debt-to-equity ratio, so equity volatility rises.

C. Crashophobia: traders fear a repeat of the October 1987 crash, which is supported by the skew steepening when the S&P 500 falls and flattening when it rises.

D. Crashophobia: traders fear sharp equity declines, which is supported by the skew having been equally marked before and after October 1987.

Question MR-17 e.3

Bank Q815 • BT notes P2.T5.02.4 (rebuilt)

A stock trades at 60 with a risk-free rate of 5% (continuously compounded). A 6-month European put with strike 54 is priced by the Black-Scholes-Merton model using a volatility of 30%, giving a price of 1.96. The put actually trades at 2.49, which corresponds to an implied volatility of about 34%. Which explanation is most consistent with this higher implied volatility?

A. Crashophobia: investors pay a premium for out-of-the-money puts as protection against a sharp fall in equity prices.

B. A violation of put-call parity, which makes puts systematically more expensive than calls at the same strike.

C. Investors believe the left tail of the stock price distribution is thinner than lognormal.

D. Speculative demand for out-of-the-money calls raising implied volatility at low strikes.

Question MR-17 e.4

Bank Q821 • Mock 1 Q51 (rebuilt)

During a geopolitical crisis, investors increase their demand for protection against extreme losses, and the prices of deep out-of-the-money equity put options rise significantly relative to at-the-money options. Which of the following best explains this behavior and its effect on the equity volatility skew?

A. Investors expect lower volatility during crises, so the skew flattens as deep OTM put prices rise.

B. Demand for downside tail-risk protection increases, raising implied volatility at low strikes and steepening the skew.

C. Higher demand for OTM puts raises implied volatility at high strikes, so the skew becomes an upward-sloping smirk.

D. Put-call parity forces deep OTM put prices to rise whenever call prices rise, leaving the skew unchanged.

Question MR-17 e.5

Bank Q818 • BT P2.T5.23.7.1 (rebuilt)

Peter asserts that equity prices and volatility are negatively correlated. A stock trades at 60, the risk-free rate is 5% and all options have 6 months to expiry. The table gives BSM prices using a flat 30% volatility and observed market prices.

Strike	BSM call (30%)	BSM put (30%)	Market call	Market put
45	16.45	0.34	16.80	0.69
60	5.78	4.30	5.78	4.30
75	1.32	14.47	0.87	14.02

What is the shape of the implied volatility smile, and is it consistent with Peter's assertion?

A. Upward-sloping skew; consistent with Peter's assertion.

B. Downward-sloping skew; consistent with Peter's assertion.

C. Downward-sloping skew; not consistent with Peter's assertion.

D. Symmetric smile; not consistent with Peter's assertion.

Question MR-17 e.6

Bank Q814 • GARP book Ch.17 Q17.25 (rebuilt)

An analyst studies 5,000 daily returns of an equity index. Each return is standardized by the sample mean and standard deviation. She finds 11 returns below -3 standard deviations and 5 returns above $+3$ standard deviations. Under a normal distribution, $P(Z < -3) = P(Z > 3) \approx 0.00135$. Which statement is the most appropriate conclusion?

A. Under normality about 6.75 observations are expected in each tail; the excess of down moves over up moves implies a left-skewed distribution, consistent with a downward-sloping implied volatility skew.

B. Both tail counts exceed what a normal distribution implies, so implied volatility should form a symmetric smile as for currency options.

C. Up moves exceed what is expected under normality, so implied volatility should rise with strike (upward-sloping skew).

D. The counts are close to the normal expectation of 6.75 per tail, so the implied volatility curve should be flat.

Question MR-17 e.7

Bank Q819 • GARP PE2 Q19 • Mock 4 Q20

An option pricing analyst at an investment bank has been asked to write a report examining the relationship between option prices and implied volatility curves. The analyst notes that the implied volatility curves of different underlying assets often have different shapes and explains the reasons why this occurs. Which of the following statements can correctly be included in the report?

A. The implied volatility smile commonly seen in equity options is due to the higher probability of a greater than three standard deviation price change than would be expected if prices are lognormally distributed.

B. The implied volatility smile commonly seen in foreign exchange rate options is due to the higher probability of a price change of between one and two standard deviations from the mean than would be expected if prices are lognormally distributed.

C. Demand for option protection against steep drops in equity prices leads to higher prices in out-of-the-money puts relative to out-of-the-money calls, which creates a downward-sloping implied volatility skew in these options.

D. Demand for option protection against the impact of unexpected central bank announcements on foreign exchange rates leads to higher prices, and higher implied volatilities, for at-the-money options relative to out-of-the-money options.

Solutions • MR-17 e

Solution MR-17 e.1

Answer: D

Bank Q823 • Schweser Module Quiz 17.1 Q4

Step 1 – The equity shape

A smirk: higher implied volatility for low strikes, lower for high strikes.

Step 2 – The leverage explanation

When equity value falls, debt-to-equity rises, making equity riskier: volatility rises as the price falls.

$\Rightarrow$ D states the leverage effect correctly.

Interpretation

Leverage is about *actual* volatility rising; crashophobia is about *fear* priced into options. Keep them apart.

Why the other options are wrong

A is incorrect. "Higher for away-from-the-money options" (both wings) is the currency smile; for equities only low strikes are elevated.

B is incorrect. Crashophobia is market fear priced into low-strike options, not a claim about realised volatility; that claim is the leverage effect.

C is incorrect. Traders see large down moves as *more* probable than large up moves, which is exactly why the skew exists.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.3; SchweserNotes Book 1, Reading 17

Solution MR-17 e.2

Answer: C

New • built from GARP Ch.17, 17.3

Step 1 – Match each name to its mechanism

Leverage: price falls $\rightarrow$ leverage rises $\rightarrow$ volatility rises. Volatility feedback: volatility rises (external) $\rightarrow$ required return rises $\rightarrow$ price falls.

Step 2 – Crashophobia and its evidence

Rubinstein's explanation: traders fear another crash like October 1987 and price options accordingly. Support: the skew steepens when the S&P 500 falls and flattens when it rises.

$\Rightarrow$ C is the only correctly described pairing.

Interpretation

A and B are the classic name swap between the two correlation mechanisms. D attaches a false history to the right name.

Why the other options are wrong

A is incorrect. Sibling swap: this mechanism (volatility first, then price) is the volatility feedback effect.

B is incorrect. Sibling swap: this mechanism (price first, then leverage and volatility) is the leverage effect.

D is incorrect. Scope error: the equity skew has existed only since October 1987, which is why crashophobia is a plausible explanation.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.3, Business Snapshot 17.2

Solution MR-17 e.3 — Answer: A

Bank Q815 • BT notes P2.T5.02.4

Step 1 – Locate the option

$K/S_0 = 54/60 = 0.90$: an OTM put, on the low-strike side of the skew.

Step 2 – Read the price

Market 2.49 versus BSM 1.96 at 30% (verified: 1.958); the implied volatility is about 34% (verified: 33.99%).

$\Rightarrow$ Extra premium on a low-strike put is the equity skew; the standard explanation is crashophobia (alongside leverage).

Interpretation

The numbers only confirm that the low-strike put is dear; the question asks which explanation fits.

Why the other options are wrong

B is incorrect. Parity equalises call and put implied volatility at the same strike; it cannot explain a skew across strikes, and it is not violated here.

C is incorrect. Polarity flip: a thinner left tail would make OTM puts cheaper and lower their implied volatility.

D is incorrect. Demand for OTM calls affects high strikes, not a low-strike put.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.3, Business Snapshot 17.2

Solution MR-17 e.4 — Answer: B

Bank Q821 • Mock 1 Q51

Step 1 – What the demand does

Buyers bid up deep OTM puts: crashophobia in action.

Step 2 – From price to implied volatility

Higher price means higher implied volatility at low strikes.

$\Rightarrow$ Low-strike implied volatility rises relative to ATM and high strikes: the skew steepens.

Interpretation

This matches the empirical support for crashophobia: the skew steepens in sell-offs.

Why the other options are wrong

A is incorrect. Higher put prices mean higher, not lower, implied volatility; the skew steepens rather than flattens.

C is incorrect. Direction error: the demand is at low strikes (puts), so the downward slope steepens; it does not flip upward.

D is incorrect. Parity links a call and a put at the same strike; it does not hold the skew fixed when demand shifts across strikes.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), Business Snapshot 17.2

Solution MR-17 e.5 — Answer: B

Bank Q818 • BT P2.T5.23.7.1

Step 1 – $K = 45$

Market call $16.80 > 16.45$ and market put $0.69 > 0.34$: both options are dearer than BSM at

30%, so implied volatility is above 30% (about 36%).

Step 2 – $K = 60$

Market equals BSM: implied volatility 30%.

Step 3 – $K = 75$

Market call $0.87 < 1.32$ and market put $14.02 < 14.47$: implied volatility below 30% (about 26%).

$\Rightarrow$ Implied volatility falls as the strike rises (36%, 30%, 26%, verified by inverting BSM): a downward skew.

Step 4 – Link to the assertion

Negative price-volatility correlation fattens the left tail and thins the right tail, which is exactly what produces this skew.

Interpretation

Note that at each strike the call and put deviate from BSM by the same dollar amount (0.35 at 45, 0.45 at 75): parity at work, so one implied volatility per strike.

Why the other options are wrong

A is incorrect. Reverses the direction: market prices exceed BSM at the low strike and fall below it at the high strike.

C is incorrect. Correct shape, wrong inference: a downward skew is the signature of negative correlation.

D is incorrect. A symmetric smile needs implied volatility above 30% at both 45 and 75; at 75 it is below.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.1 and 17.3

Solution MR-17 e.6 **Answer: A**

Bank Q814 • GARP book Ch.17 Q17.25

Step 1 – Expected count per tail under normality

$5{,}000 \times 0.00135 = 6.75$.

Step 2 – Compare

Down tail: 11 observed, about 63% above 6.75: heavier than normal. Up tail: 5 observed, about 26% below 6.75: thinner than normal.

$\Rightarrow$ Heavy left tail and thin right tail: a left-skewed distribution, which gives high implied volatility at low strikes (the downward skew).

Interpretation

The arithmetic is only the expected count; the question is about reading tail asymmetry into the shape of the curve.

Why the other options are wrong

B is incorrect. Only the down tail exceeds 6.75; a symmetric smile (the currency pattern) needs both tails heavy.

C is incorrect. Reverses the asymmetry: the up tail is the thin one.

D is incorrect. Treats 11 and 5 as close to 6.75, ignoring the clear asymmetry.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), Practice Question 17.25

Solution MR-17 e.7 **Answer: C**

Bank Q819 • GARP PE2 Q19

Step 1 – Equity

Equity options show a downward skew, not a symmetric smile. Demand for crash protection raises OTM put prices relative to OTM calls (crashophobia): statement C.

Step 2 – FX

The FX smile is lowest at the money and higher away from it, because the far tails (beyond about 3 SD) are more likely than lognormal; moves of 1 to 2 SD are *less* likely.
$\Rightarrow$ Only C survives.

Interpretation

Each wrong option attaches a real feature to the wrong market or flips the shape. This is the map's sibling-swap trap.

Why the other options are wrong

A is incorrect. Sibling swap: a symmetric fat-tail story ("greater than 3 SD" in both directions) is the currency explanation; equities have a heavy left tail only.
B is incorrect. The extra FX probability is in the far tails; 1 to 2 SD moves are less likely than lognormal (Table 17.1).
D is incorrect. Polarity flip: the FX smile has the *lowest* implied volatility at the money; ATM highest would be a frown.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 and 17.3, Table 17.1, Business Snapshot 17.2

MR-18 • Fundamental Review of the Trading Book • Vardeez: VERY IMPORTANT

MR-18 a

P1 CONCEPT

Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes.

Workbook: A4 FRTB • Tier A • August 2026 recall B2 ES and coherence • Tier B

WHAT IS TESTED HERE

Why P1. Tier A workbook topic: FRTB came up in the August 2026 sitting ("FRTB, 2Q"). **Vardeez:** the full chapter is important; refer to the GARP book for curvature risk.

Reported exam shapes

- **A4 S1** Stressed ES at 97.5% replaces 10-day VaR at 99%, and why.
- **A4 S5** FRTB vs Basel I (1996 Amendment), Basel II, Basel 2.5: overall changes, each change attached to the right accord.
- **A4 S6** Curvature risk charge, from the GARP text.
- **B2 S2** ES as the average of tail VaRs read off a confidence-level table, against the 99% VaR it replaces.

What the questions look like: "which of the following is correct" (or "EXCEPT") statements about what changed and why. Distractors move a feature to the wrong accord, flip the fat-tail polarity, or claim near-equality of ES and VaR for *any* distribution. The one numerical shape (B2 S2) is a tail-average read-off, not a formula.

1. Timeline and the three regimes

- May 2012: Basel Committee consultative document, the "Fundamental Review of the Trading Book". Comments, revisions and Quantitative Impact Studies (QISs: banks estimate the effect of proposed rules) followed.
- January 2019: final rules ("Minimum Capital Requirements for Market Risk"). Agreed implementation January 2023, with many jurisdictions a year or two later.

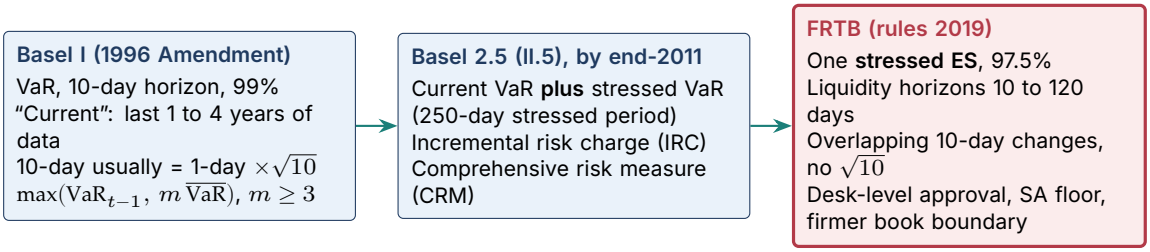


Figure 1. How market risk capital evolved. Basel II (2004) is missing on purpose: it kept market risk on the 1996 Amendment basis; its innovations were IRB credit risk weights, operational risk capital and the three pillars.

Two changes happen at the FRTB step and are easy to conflate: the **measure** changes (VaR to ES) and the **confidence level** changes (99% to 97.5%). The stressed-period requirement carries over from Basel 2.5, so the FRTB measure is a *stressed* ES, not a current one.

2. Why ES replaces VaR (shape S1)

VaR only asks "how bad can things get?": it reports the threshold and says nothing about losses beyond it. ES asks how bad things are *on average* once the threshold is breached.

Normal losses: the two nearly coincide. With mean μ and standard deviation σ,

$$\text{VaR}_{99\%} = \mu + 2.326\,\sigma \qquad \text{ES}_{97.5\%} = \mu + 2.338\,\sigma$$

The 97.5% level was chosen precisely so that, for a normal distribution, the switch is roughly capital-neutral (a difference of 0.012σ).

Fat-tailed losses: they separate. When the loss distribution has a heavier tail than the normal, the 97.5% ES can be *considerably greater* than the 99% VaR. The switch bites exactly where tail risk is fat.

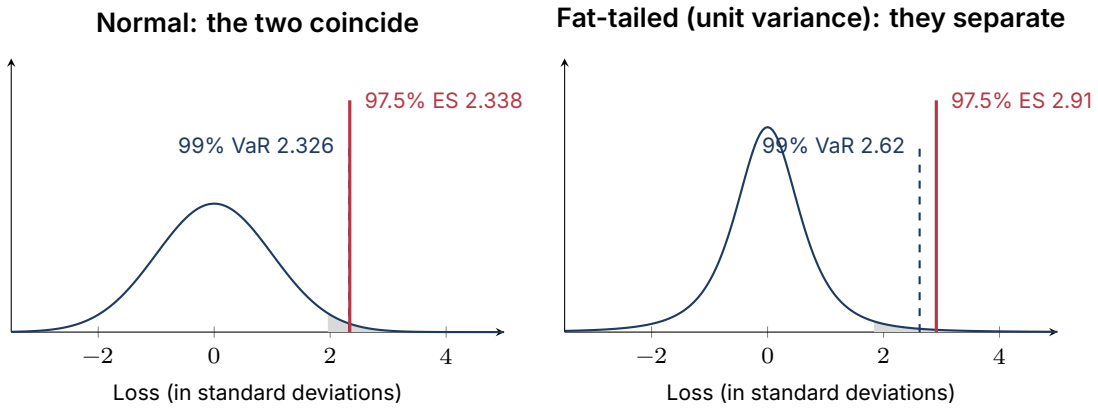


Figure 2. Dashed line: 99% VaR; solid line: 97.5% ES; grey shading: the 2.5% tail that ES averages over. Left: normal losses, the average lands almost exactly on the 99% VaR cut. Right: a Student-t with 3 degrees of freedom rescaled to unit variance; the same tail holds much larger losses, so ES moves well beyond VaR.

The bond example: VaR reports nothing

A \$950 million bond portfolio has a 2% probability of default, in which case (in this example) the whole \$950 million is lost; otherwise nothing is lost.

- **95% VaR = \$0.** Only 2% of outcomes involve a loss, so the 95th percentile outcome is "no default". VaR says the portfolio is riskless.
- **95% ES = \$380 million.** The 5% tail contains the 2% default outcome and 3% of no-loss outcomes, so $2/5 = 40\%$ of the tail is a total loss: $0.40 \times \$950\text{m} = \$380\text{m}$.

> **Correction to the source notes.** Vardeez's "precision point" reads the 40% as a loss rate on default and says the true ES should be $0.02/0.05 \times \$380\text{m} = \$152\text{m}$. That misreads the example. In the Schweser source the whole \$950m is lost on default, and 40% is the share of the 5% tail that is a default ($2\%/5\%$). So $0.40 \times \$950\text{m} = \$380\text{m}$ *is* the probability-weighted tail average, i.e. the ES. Same logic in Schweser's quiz: \$825m, 3% default probability, 95% ES $= 0.60 \times 825 = \$495\text{m}$.

Reading ES off a table of tail VaRs (shape B2 S2)

ES can be estimated by slicing the tail beyond the confidence level into n equal-probability slices and averaging the $n - 1$ VaRs at the slice boundaries. With 10 slices beyond 97.5% there are 9 tail VaRs (97.75% to 99.75%); the 97.5% VaR itself marks the start of the tail and is *not* in the average. For a heavy-tailed desk the result exceeds the 99% VaR (Question MR-18 a.5).

3. The new architecture

- **Two approaches, both always computed.** FRTB has a standardized approach (SA) and an internal models approach (IMA). Even a bank approved for IMA must also calculate SA capital: the SA is a floor. December 2017: total capital at least 72.5% of standardized, reached by 2028 after a five-year phase-in. The trend since 2008 is *less* reliance on internal models.
- **Desk level.** Most calculations are done per trading desk, and IMA permission is granted desk by desk: the FX desk can be on IMA while the equity desk is not.
- **Liquidity horizons** replace the single 10-day horizon (MR-18 b), and ES is computed from actual overlapping 10-day changes in a stressed period, not 1-day changes $\times\sqrt{10}$.
- **Firmer trading book / banking book boundary**, plus desk-level backtesting and P&L attribution tests (MR-18 c).

The standardized approach in three components

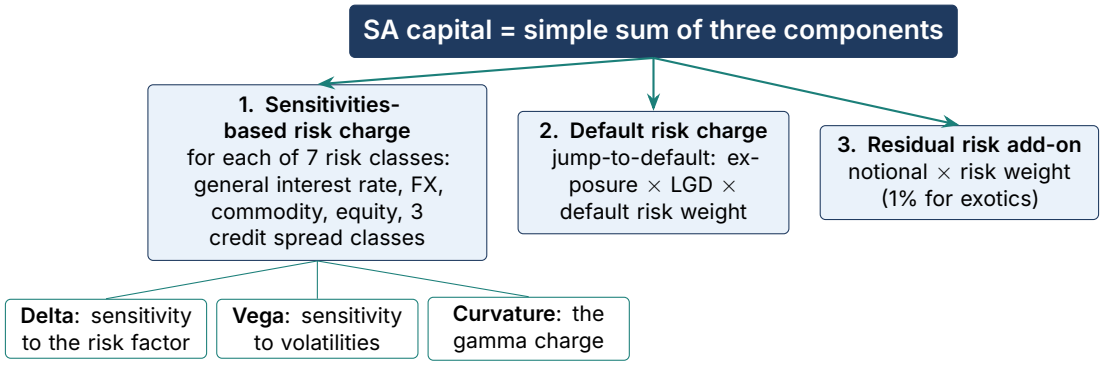


Figure 3. The FRTB standardized approach. No diversification is allowed across risk classes, nor between delta and vega within a class.

Delta and vega charges. Within a risk class, Delta charge $= \sqrt{\sum_i \sum_j \rho_{ij}\,\delta_i\delta_j\,W_iW_j}$ with risk weights W_i and correlations ρ_{ij} set by the Basel Committee and sensitivities δ_i from the bank. Vega is the same with volatilities as risk factors.

- Equity, FX and commodity deltas are per *percentage* change: a 1% rise adding \$3,000 gives $\delta = 3{,}000/0.01 = 300{,}000$. Rate and spread deltas are per *absolute* change: 1 bp costing \$200 gives $\delta = -200/0.0001 = -2{,}000{,}000$.
- **Risk weights embed the liquidity horizon and the ES confidence level:** W = stressed daily volatility $\times\sqrt{LH} \times 2.338$, e.g. $0.02 \times \sqrt{20} \times 2.338 = 0.209$.
- Each vertex of a term structure (e.g. 10 rate vertices from 3 months to 30 years) is a separate risk factor.
- Correlations are tested at 1.25, 1.00 and 0.75 times the Basel values; the *greatest* charge is used.

Curvature risk charge (shape S6, from GARP)

The SA's charge for gamma risk. Shock risk factor i up and down by its risk weight W_i and remove the delta effect:

$$\text{Curvature charge}_i = \max\Big(\underbrace{W_i\delta_i - \Delta V_i^{+}}_{\text{up shock}},\ \underbrace{-W_i\delta_i - \Delta V_i^{-}}_{\text{down shock}},\ 0\Big)$$

Each term is the extra loss from curvature after the linear (delta) effect has been removed. A negative result counts as zero. For a term structure, all points shift in parallel by the *largest* W_i on the curve, and each point's own δ_i removes delta.

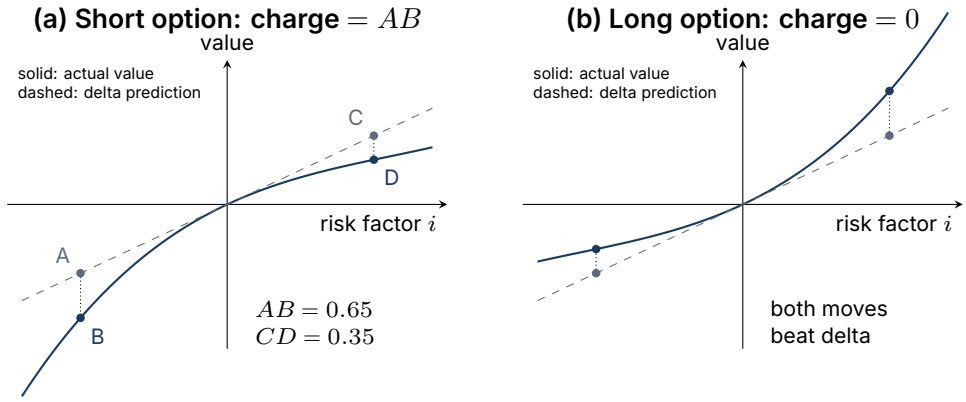


Figure 4. GARP Figure 18.1 redrawn. Dashed line: value predicted by delta; solid curve: actual value. Shocks of $\pm W_i$. In (a) both moves are worse than delta predicts, the charge is the larger shortfall, AB. In (b) curvature raises value relative to delta for both moves, so both terms are negative and the charge is zero.

Numerical illustration. $\delta = 300{,}000$, $W = 0.209$, so $W\delta = 62{,}700$. Short book: $\Delta V^+ = 41{,}200$, $\Delta V^- = -88{,}900$; terms $62{,}700 - 41{,}200 = 21{,}500$ and $-62{,}700 + 88{,}900 = 26{,}200$; charge \$26,200. Long book: $\Delta V^+ = 80{,}000$, $\Delta V^- = -48{,}000$; terms $-17{,}300$ and $-14{,}700$; charge 0.

Default risk charge and residual add-on. Jump-to-default is charged separately from credit spread risk: exposure × LGD × default risk weight (senior debt LGD 75%; A-rated weight 3%; equity LGD 100%). The residual add-on is notional × risk weight (1% for exotic options that cannot be built from vanillas). A simplified SA for smaller banks (June 2017, included in the 2019 rules) drops vega and gamma.

	Basel I (1996 Amd.)	Basel 2.5	FRTB
Measure	VaR	VaR + stressed VaR	Stressed ES
Confidence	99%	99%	97.5%
Horizon	10 days ($\sqrt{10}$ scaling allowed)	10 days	10 to 120 days by liquidity; overlapping 10-day changes
Calibration	Current (1 to 4 years)	Current + 250-day stress	250-day stress only
Credit in trading book	Specific risk	IRC (99.9%, 1 yr, defaults + downgrades)	Default risk charge (jump-to-default)
Securitizations	Specific risk	CRM (own models)	Standardized only
Model approval	Bank-wide	Bank-wide	Desk by desk; SA always computed

Exam traps

- **Polarity flip on tails.** Heavy tails make 97.5% ES *greater* than 99% VaR, not smaller. Near-equality (2.338 vs 2.326) holds for the *normal* only, not "for any distribution".
- **Current vs stressed.** "Current, calibrated to the last one to four years" is Basel I VaR. The FRTB measure is a stressed ES.
- **Wrong accord.** Stressed VaR, IRC and CRM are Basel 2.5, not Basel II. Basel II's innovations were credit risk (IRB), operational risk capital and the three pillars.
- **FRTB adds stressed VaR to ES**: false. It replaces VaR plus stressed VaR with one stressed ES.

- $\sqrt{10}$ **scaling** of 1-day changes is Basel I / 2.5 practice. FRTB uses actual overlapping 10-day changes.
- **More reliance on internal models**: false. FRTB reduces it (SA floor, desk-level approval). Calculations move *to* the desk level, not away from it.
- **Curvature polarity.** Short options attract the curvature charge; long options give zero.
- **FRTB drops the default charge**: false. The IRC is replaced by a default risk charge.
- **Tail VaR average.** Average the $n - 1$ VaRs *beyond* the start of the tail; do not include the start VaR and do not divide by n.

Practice questions: MR-18 a

Question MR-18 a.1

Bank Q856 • GARP PE1 Q13 • Mock 3 Q67

A regulatory analyst at a large multinational bank is examining regulatory requirements the bank must comply with under the Basel Committee's FRTB guidelines. The analyst explores how the FRTB guidelines evolved from the Basel I and Basel II.5 frameworks. Which of the following is correct regarding the FRTB?

A. While Basel I and Basel II.5 allowed market risk to be calculated at the trading desk level, FRTB requires that market risk be calculated on a firm-wide basis.

B. While Basel I and Basel II.5 emphasized the use of a standardized approach to calculating market risk, FRTB encourages each bank to develop and rely on an internal models approach.

C. FRTB standardizes the liquidity horizon for all risk factors in the market risk capital calculation as 10 days, rather than the different horizons used in Basel I and Basel II.5.

D. FRTB requires that the stressed ES measure be used in determining market risk capital, rather than the VaR and stressed VaR that were used in Basel I and Basel II.5.

Question MR-18 a.2

Bank Q855 • BT P2.T7.608.1 (rebuilt)

The Basel Committee's Fundamental Review of the Trading Book (FRTB) revises the market risk framework. Compared to Basel I and Basel II.5, each of the following is true EXCEPT which statement is false?

A. The Basel II.5 internal models approach required the sum of current VaR and stressed VaR, whereas the FRTB requires a single stressed expected shortfall.

B. Basel I and Basel II.5 used a 10-day horizon for all risk factors, whereas the FRTB uses five liquidity horizons (10, 20, 40, 60 and 120 days) depending on the liquidity of the risk factor.

C. Basel II.5 required a default-related charge for credit-dependent positions, but the FRTB eliminates any such charge and relies on VaR and ES alone.

D. Basel I and Basel II.5 based capital on VaR at a 99% confidence level, whereas the FRTB bases capital on expected shortfall at a 97.5% confidence level.

Question MR-18 a.3

Workbook A4.2 • S1 • Aug-2026 • Map #5

The CRO of a Japanese bank is preparing a board risk committee briefing on why FRTB replaces 10-day 99% VaR with 97.5% expected shortfall for market risk capital under the internal models approach. One desk head argues the change is cosmetic, since for his desk, whose losses are close to normally distributed, the two measures give almost the same JPY figure. Which of the following statements should the CRO include as correct?

A. With loss distributions whose tails are heavier than the normal, 97.5% ES can be considerably lower than 99% VaR, so the change mainly reduces capital for fat-tailed desks.

B. Like the Basel I VaR measure, the 97.5% ES is a current measure, calibrated to market movements over the one to four years immediately preceding the calculation date.

C. Because 97.5% ES and 99% VaR are almost exactly equal for any loss distribution, the switch affects capital only through the new liquidity horizons.

D. For a normal loss distribution the two measures nearly coincide, but with heavier tails 97.5% ES can be considerably greater, and FRTB calibrates the ES to stressed market conditions.

Question MR-18 a.4

Bank Q852 • BT P2.T5.405.2 (rebuilt)

Which of the following statements about the move from 99% VaR to 97.5% expected shortfall (ES) under the FRTB is INCORRECT?

A. VaR gives a loss threshold but says nothing about the size of losses beyond it, whereas ES measures the expected loss given that the threshold is exceeded.

B. For a normal distribution, 97.5% ES (mean plus 2.338 standard deviations) is almost identical to 99% VaR (mean plus 2.326 standard deviations), but the two can differ considerably when tails are fat.

C. ES is easier to backtest than VaR, which is why the FRTB requires the stressed ES itself to be backtested.

D. The FRTB bases capital on a single ES calibrated to a stressed period, rather than adding a current 10-day VaR to a stressed VaR.

Question MR-18 a.5

Workbook B2.2 • S2 • Map #12

A market risk manager at an Australian bank is preparing a trading desk for the switch from a 99% VaR to a 97.5% ES under FRTB. The desk's P/L distribution is heavy-tailed. The desk's model slices the tail beyond the 97.5% confidence level into 10 equal-probability slices and reports the VaR at each confidence level (AUD million):

Confidence	97.50%	97.75%	98.00%	98.25%	98.50%	98.75%	99.00%	99.25%	99.50%	99.75%
VaR	4.29	4.50	4.75	5.04	5.39	5.84	6.42	7.26	8.60	11.43

Which of the following is closest to the desk's 97.5% ES and the change in its headline risk number when the 97.5% ES replaces the 99% VaR?

A. ES AUD 5.92 million; change −AUD 0.50 million

B. ES AUD 6.35 million; change −AUD 0.07 million

C. ES AUD 6.42 million; change AUD 0.00 million

D. ES AUD 6.58 million; change +AUD 0.16 million

Question MR-18 a.6

New • built from GARP Ch.18, Section 18.2 and Vardeez

A market risk manager at a mid-sized European bank is preparing a briefing note on the FRTB sensitivities-based standardized approach for the bank's risk committee. Which of the following statements should the manager include as correct?

A. Risk weights are set by the Basel Committee as multiples of stressed daily volatility that reflect each risk factor's liquidity horizon and the 97.5% ES confidence level.

B. Diversification is recognized between the delta and vega charges within a risk class, and across the seven risk classes, through correlations set by the Basel Committee.

C. Banks set the risk weights and correlations from their own data, and the regulator then applies a multiplier of at least 1.5 to the resulting charge.

D. The correlations specified by the Basel Committee are applied exactly as given, so the bank reports a single charge calculated at the prescribed correlation levels.

Question MR-18 a.7 Workbook A4.4 • S3, S5, S6 • Map #5

A risk consultant is preparing a one-page comparison of market risk capital under the 1996 Amendment to Basel I, Basel II, Basel 2.5 and FRTB for the chief compliance officer of a UK bank. The FRTB section discusses two GBP options desks under the standardized approach: one is short a book of at-the-money options and the other is long a similar book. Which of the following statements is correct to include?

A. FRTB moves from 99% to 97.5% confidence and from one 10-day horizon to liquidity horizons, but still lets banks scale one-day changes to 10 days with a square-root-of-10 multiplier.

B. Basel II overhauled market risk capital by adding a stressed VaR term to the current VaR term, alongside the introduction of its three pillars.

C. Under FRTB's standardized approach, the long-option desk attracts the curvature risk charge, because its gamma raises value relative to delta for both up and down moves.

D. Under FRTB's standardized approach, the short-option desk's curvature charge is the larger of its two shocked losses net of the delta effect, with negative results set to zero.

Solutions • MR-18 a

Solution MR-18 a.1

Answer: D

Bank Q856 • GARP PE1 Q13

Step 1 – Place the measures in each accord

Basel I: 10-day 99% VaR. Basel II.5: current VaR *plus* stressed VaR. FRTB: a single stressed ES at 97.5%.

⇒ D states the defining FRTB change.

Step 2 – Check the other claims against the FRTB design

FRTB moves calculations *to* the desk level and grants IMA permission desk by desk; it reduces reliance on internal models (SA always computed, SA floor); and it replaces the single 10-day horizon with five liquidity horizons.

Interpretation

Three of the four options are reversals of what FRTB did. GARP's favourite pattern for this LO is to state the old regime's feature as the new one's.

Why the other options are wrong

A is incorrect. Reversed: FRTB introduces desk-level calculation and approval; Basel I and II.5 were firm-wide.

B is incorrect. Reversed: FRTB reduces reliance on internal models; the standardized approach is always computed and acts as a floor.

C is incorrect. Reversed: the uniform 10-day horizon is the old regime; FRTB uses 10, 20, 40, 60 and 120 days.

LO: MR-18 a: Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes. • **Reference:** Hull, RMFI 6e, Ch.18 (GARP Ch.18), Section 18.1

Solution MR-18 a.2

Answer: C

Bank Q855 • BT P2.T7.608.1

Step 1 – Test each statement

A: true, VaR plus stressed VaR is replaced by one stressed ES. B: true, the five horizons are 10, 20, 40, 60 and 120 days. D: true, 99% VaR becomes 97.5% ES.

Step 2 – Find the false one

C claims FRTB eliminates any default-related charge. It does not: jump-to-default risk keeps its own charge (the default risk charge in the SA; a 99.9%, one-year VaR under the IMA), replacing the Basel 2.5 IRC.

⇒ C is the false statement.

Interpretation

"Eliminates" is an absolute claim: GARP often hides the false option in a word like this. The default charge is *replaced*, not removed.

Why the other options are wrong

A is incorrect. True statement: the single stressed ES replaces the VaR plus stressed VaR sum.

B is incorrect. True statement. (Picking it usually comes from remembering the superseded list with a 250-day horizon; see the correction in MR-18 b.)

D is incorrect. True statement: the confidence level moves from 99% (VaR) to 97.5% (ES).

LO: MR-18 a • **Reference:** Hull, RMFI 6e, Ch.18, Sections 18.1, 18.2.3, 18.3.3

Solution MR-18 a.3 Answer: D

Workbook A4.2 • S1 • Aug-2026

Step 1 – Compare the two measures under normality

For normal losses with mean μ and standard deviation σ: 99% VaR $= \mu + 2.326\sigma$ and 97.5% ES $= \mu + 2.338\sigma$.

$\Rightarrow$ The desk head is right for his desk: the multipliers differ by only 0.012σ.

Step 2 – Drop the normality condition

For non-normal losses the measures are not equivalent; when the loss distribution has a heavier tail than the normal, 97.5% ES can be considerably greater than 99% VaR.

$\Rightarrow$ ES averages the losses beyond the quantile, so it sees the shape of the tail that VaR ignores. The change is not cosmetic for fat-tailed desks.

Step 3 – The calibration

The measure is stressed ES at 97.5%: calculated from how market variables moved during stressed market conditions, as for Basel 2.5 stressed VaR.

Interpretation

The CRO's answer to the desk head: near-equality is a property of the normal case only; the capital consequences come from fat tails, stressed calibration and (separately) liquidity horizons.

Why the other options are wrong

A is incorrect. Polarity flip: heavier tails make 97.5% ES considerably *greater* than 99% VaR, raising capital for fat-tailed desks.

B is incorrect. Sibling swap: "current" calibration over the preceding one to four years describes the Basel I VaR; the FRTB measure is stressed ES.

C is incorrect. Scope/condition error: near-equality holds for normal distributions only; extending it to "any" distribution drops the condition.

LO: MR-18 a • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.1

Solution MR-18 a.4 Answer: C

Bank Q852 • BT P2.T5.405.2

Step 1 – Check the motivation statements

A is the core motivation: VaR is a threshold, ES is the expected loss beyond it. B is the normal benchmark (2.338 vs 2.326) with the correct fat-tail caveat. D is the FRTB design.

Step 2 – Check the backtesting claim

GARP gives two reasons FRTB does *not* backtest the stressed ES: ES is more difficult to backtest than VaR, and a stressed measure cannot be backtested at all, since stressed data are not expected to recur with the same frequency. FRTB backtests a 1-day VaR instead.

$\Rightarrow$ C gets both halves wrong: ES is harder, not easier, and the stressed ES is never backtested.

Interpretation

This is a sibling swap across LOs: the capital measure (stressed ES) and the backtested measure (1-day VaR at 99% and 97.5%) are different numbers. Details of the backtest are in MR-18 c.

Why the other options are wrong

A is incorrect. True statement: it is the central motivation for ES.

B is incorrect. True statement: the near-equality under normality is why 97.5% was chosen.

D is incorrect. True statement: one stressed ES replaces VaR plus stressed VaR.

LO: MR-18 a • **Reference:** Hull, RMFI 6e, Ch.18, Sections 18.1 and 18.3.1

Solution MR-18 a.5 — Answer: D

Workbook B2.2 • S2

Step 1 – Identify the tail VaRs

With $n = 10$ slices beyond 97.5% there are $n - 1 = 9$ tail VaRs, at 97.75% through 99.75%. The 97.5% VaR (4.29) marks the start of the tail and is excluded.

Step 2 – Average the tail VaRs

$$\text{ES}_{97.5\%} \approx \frac{4.50 + 4.75 + 5.04 + 5.39 + 5.84 + 6.42 + 7.26 + 8.60 + 11.43}{9} = \frac{59.23}{9} = 6.58$$

$\Rightarrow$ On the worst 2.5% of days the desk loses about AUD 6.58 million on average.

Step 3 – Compare with the 99% VaR

$$\Delta = \text{ES}_{97.5\%} - \text{VaR}_{99\%} = 6.58 - 6.42 = +0.16$$

Interpretation

For a normal distribution the two numbers would almost coincide (2.338σ against 2.326σ); because this desk's tail is heavy, the 97.5% ES exceeds the 99% VaR, which is the FRTB motivation for capturing tail risk. With only 10 slices the estimate is, if anything, too low: the average-tail-VaR estimate rises towards the true ES as n grows.

Why the other options are wrong

A is incorrect. Divided the nine tail VaRs by the number of slices (10) instead of the number of tail VaRs (9): $59.23/10 = 5.92$, change $5.92 - 6.42 = -0.50$.

B is incorrect. Included the 97.5% VaR in the average (the off-by-one on tail membership): $(59.23 + 4.29)/10 = 6.35$, change -0.07.

C is incorrect. Applied the normal-distribution equivalence to a heavy-tailed desk and set the 97.5% ES equal to the 99% VaR, so the change is zero.

LO: MR-1 c; MR-1 d; MR-18 a • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.4, Tables 1.1 and 1.2; Hull, RMFI 6e, Ch.18, 18.1

Solution MR-18 a.6 — Answer: A

New • built from GARP Ch.18, Section 18.2

Step 1 – Who sets what

In the sensitivities-based method the bank supplies the sensitivities δ_i; the Basel Committee sets the risk weights W_i and the correlations ρ_{ij}.

Step 2 – How the risk weights are built

W_i = stressed daily volatility $\times\sqrt{LH} \times 2.338$: the $\sqrt{LH}$ brings in the liquidity horizon and 2.338 is the normal multiplier for 97.5% ES. Example: $0.02 \times \sqrt{20} \times 2.338 = 0.209$.

$\Rightarrow$ A is correct.

Step 3 – Diversification and correlation scenarios

There is no diversification across risk classes, nor between delta and vega within a class. Correlations are tested at 1.25, 1.00 and 0.75 times the Basel values and the greatest charge is used.

Interpretation

The SA is "standardized" because the regulator fixes the weights and correlations; the bank's only input is its own sensitivities. This is also how liquidity horizons enter the SA (see Question MR-18 b.3).

Why the other options are wrong

B is incorrect. Overstates diversification: none is allowed across risk classes or between delta and vega.

C is incorrect. Sibling swap: banks do not set SA weights or correlations; the multiplier $m_c \geq 1.5$ belongs to the IMA capital formula.

D is incorrect. Drops the correlation scenarios: the charge is computed at 1.25, 1.00 and 0.75 times the Basel correlations and the greatest is used.

LO: MR-18 a • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.2 and footnote 5

Solution MR-18 a.7

Answer: D

Workbook A4.4 • S3, S5, S6

Step 1 – Curvature mechanics (Section 18.2.2)

Shock risk factor i up and down by its risk weight W_i. Compute $W_i\delta_i - \Delta V_i^+$ and $-W_i\delta_i - \Delta V_i^-$; the charge is the greater, and a negative result counts as zero.

$\Rightarrow$ Each term is the extra loss from curvature after the linear (delta) effect has been removed.

Step 2 – Apply to the two desks

Short options (Figure 4a): both moves are worse than delta predicts, so both terms are positive and the charge is the larger (AB in the figure). Long options (Figure 4b): curvature increases value relative to delta for both moves, both terms are negative, so the charge is zero.

$\Rightarrow$ Illustration: short book terms 21,500 and 26,200 give 26,200; long book terms $-17{,}300$ and $-14{,}700$ give 0.

Step 3 – Place the accords

1996 Amendment: 10-day 99% VaR, usually 1-day VaR $\times\sqrt{10}$, $m \geq 3$. Basel II: IRB credit risk, operational risk capital, three pillars; market risk retained. Basel 2.5: stressed VaR, IRC, CRM. FRTB: stressed 97.5% ES from overlapping 10-day changes, liquidity horizons, desk-level approval, SA with delta, vega and curvature.

Interpretation

The curvature charge is the standardized approach's gamma charge, and it falls on the short-gamma desk.

Why the other options are wrong

A is incorrect. Sibling swap: $\sqrt{10}$ scaling of one-day changes is the Basel I / Basel 2.5 practice; FRTB requires actual overlapping 10-day changes from the stressed period.

B is incorrect. Wrong accord: stressed VaR is a Basel 2.5 change (end-2011); Basel II's innovations were credit risk weights, operational risk capital and the three pillars.

C is incorrect. Polarity flip: the reason given is why a long option book receives a zero curvature charge; it is the short book that attracts it.

LO: MR-18 a; ORR-21 c; ORR-22 a • **Reference:** Hull, RMFI 6e, Ch.18, Sections 18.1, 18.2.2, 18.3; Carey (GARP), Ch.21 and Ch.22

MR-18 • Fundamental Review of the Trading Book • Vardeez: VERY IMPORTANT

MR-18 b

P1 BOTH

Compare the various liquidity horizons proposed by the FRTB for different asset classes and explain how a bank can calculate its expected shortfall using the various horizons.

Workbook: A4 FRTB • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A4, August 2026 recall).
Vardeez: the full chapter is important.

Reported exam shape

- **A4 S2** Liquidity horizons by asset class and the cascade ES.

What the questions look like: two halves, matching the two verbs. "Compare" gives concept items: which risk factor sits in which horizon, how horizons enter the SA and the IMA. "Explain how a bank can calculate" gives the cascade calculation: five nested ES numbers, the $\sqrt{(LH_j - LH_{j-1})/10}$ weights, root-sum-of-squares. Twists: a missing ES run, a red-herring 1-day VaR, or the subset-of-risk-factors scaling.

1. The five liquidity horizons

A liquidity horizon is the time needed to exit or hedge a risk factor exposure *in stressed markets* without materially moving prices. FRTB replaces the single 10-day horizon with five, and allocates every risk factor to one of them (GARP Table 18.1). The changes over the horizon in stressed markets are called *shocks*.

Correction to the source notes. Vardeez's notes list the horizons as 10, 20, 60, 120 and 250 days, with "structured products" at 250 days. That list is from the superseded 2012 consultation. The GARP 2026 text (Hull, Table 18.1) and Schweser both give **10, 20, 40, 60 and 120 days**. The 250 days in FRTB is the length of the *stressed period*, not a horizon. Use the table below, and the cascade weights 1, 1, 2, 2, 6 that follow from it. GARP tests this list directly (Question MR-18 b.1).

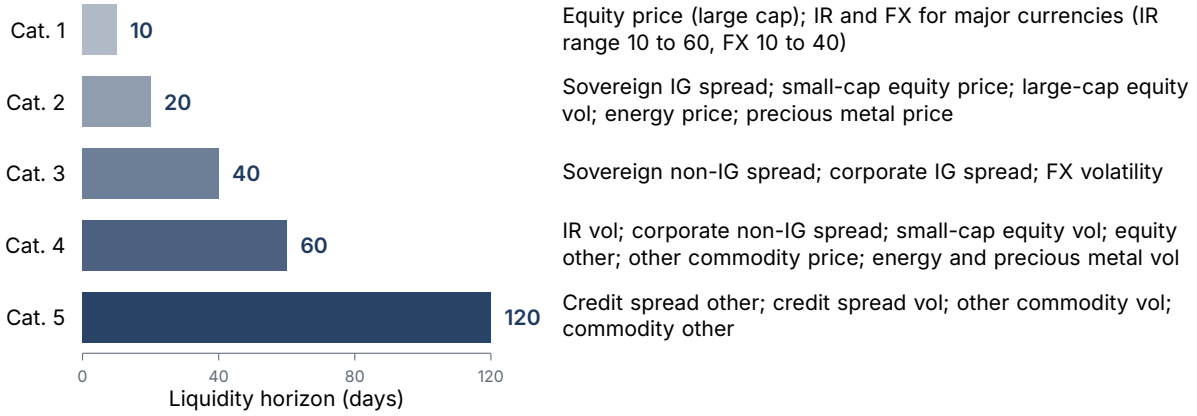


Figure 5. FRTB liquidity horizons and the risk factors assigned to each (Hull, Table 18.1). Interest rates (10 to 60 days) and FX rates (10 to 40 days) depend on the currency. Credit spreads span 20 to 120 days; credit spread volatility is 120 days.

Patterns that make the table easier to remember

- **Volatilities sit at or above their level.** Large-cap equity 10, its volatility 20; FX up to 40, FX vol 40; energy and precious metal price 20, their vols 60; other commodity price 60, its vol 120; credit spread up to 120, credit spread vol 120.
- **Credit spreads walk up with credit quality falling**: sovereign IG 20, sovereign non-IG 40, corporate IG 40, corporate non-IG 60, other 120.
- **Large cap is more liquid than small cap**: equity price 10 vs 20; equity vol 20 vs 60.

How the horizons enter each approach

- **Standardized approach:** through the Basel risk weights, $W =$ stressed daily vol $\times\sqrt{LH} \times 2.338$ (MR-18 a). Diversification enters through prescribed correlations, not through the weights.
- **Internal models approach:** through the cascade of nested ES calculations below.

2. Computing ES under the IMA

Overlapping 10-day changes from a stressed period

The IMA requires stressed ES at 97.5%; FRTB does not prescribe the method, but historical simulation is likely. All calculations use **10-day changes** observed in the stressed period:

- Trial 1 uses the changes from Day 0 to Day 10, trial 2 from Day 1 to Day 11, and so on: **overlapping** periods.
- Successive trials share 9 of their 10 daily changes, so they are **not independent**. This does *not bias* the result but *reduces the effective sample size*, making the estimate noisier. Non-overlapping periods would need a very long history.
- Basel I and Basel 2.5 allowed 10-day changes to be deduced from 1-day changes with a $\sqrt{10}$ multiplier. FRTB does not: it judges 10-day changes more relevant, since successive days may not be independent while successive 10-day periods reasonably can be.

Initial idea (2012) versus the adopted method: the first proposal measured each factor over its own full horizon in each trial (e.g. Day 0 to 10 for large-cap equity, Day 0 to 60 for a non-IG spread). It was revised (December 2014) to the five-category cascade to reduce implementation cost.

The cascade

Define category j as the risk factors with horizon LH_j ($LH_1, \ldots, LH_5 = 10, 20, 40, 60, 120$). Compute five ES numbers, each from a 10-day shock:

	Cat. 1 (10 days)	Cat. 2 (20 days)	Cat. 3 (40 days)	Cat. 4 (60 days)	Cat. 5 (120 days)	weight $(LH_j - LH_{j-1})/10$
ES_1	shock	shock	shock	shock	shock	1 (base)
ES_2	held	shock	shock	shock	shock	1
ES_3	held	held	shock	shock	shock	2
ES_4	held	held	held	shock	shock	2
ES_5	held	held	held	held	shock	6

Figure 6. The cascade (dark cells: shocked; white cells: held constant). Each row freezes one more of the most liquid categories, so only progressively less liquid exposures keep moving. The weight is the number of *extra* independent 10-day blocks for which that row's factors are still exposed.

$$\mathrm{ES} = \sqrt{\mathrm{ES}_1^2 + \sum_{j=2}^{5} \left(\mathrm{ES}_j \sqrt{\frac{LH_j - LH_{j-1}}{10}} \right)^2} = \sqrt{\mathrm{ES}_1^2 + \mathrm{ES}_2^2 + 2\,\mathrm{ES}_3^2 + 2\,\mathrm{ES}_4^2 + 6\,\mathrm{ES}_5^2}$$

Why it looks like this (an extension of the square-root rule). Suppose only categories 1 and 2 exist. All factors move in the first 10 days; category 2 factors move again over a further 10 days, independently, so $\mathrm{ES} = \sqrt{\mathrm{ES}_1^2 + \mathrm{ES}_2^2}$. Add category 3 (40 days): that would be right if category 3 had a 20-day horizon; it needs a further 20 days, independent of what came before, with ES over 20 days $= \sqrt{2}\times$ its 10-day ES, which adds $2\,\mathrm{ES}_3^2$. And so on. The cascade can be used for VaR as well.

Worked example (GARP Problem 18.7 style). $\mathrm{ES}_1, \ldots, \mathrm{ES}_5 = 10, 8, 6, 4, 2$:

$$\mathrm{ES} = \sqrt{10^2 + 8^2 + 2(6^2) + 2(4^2) + 6(2^2)} = \sqrt{100 + 64 + 72 + 32 + 24} = \sqrt{292} = 17.09$$

Missing category. If a desk has no category 1 factors, holding category 1 constant changes nothing, so $\mathrm{ES}_2 = \mathrm{ES}_1$. The term is not zero (Question MR-18 b.7).

Around the cascade

- **Run count.** Five ES calculations per desk: six desks need $5 \times 6 = 30$.
- **Stressed period search** goes back as far as 2007. If history is missing for some risk factors, the search may use a *subset* covering at least 75% of current factors, and the stressed ES is scaled *up*:

  $$\mathrm{ES}_{\text{stressed, full}} = \mathrm{ES}_{\text{stressed, subset}} \times \frac{\mathrm{ES}_{\text{last 12 months, all factors}}}{\mathrm{ES}_{\text{last 12 months, subset}}}$$

  This can double the runs from 30 to 60.
- **Partial ES and weighted ES.** A desk's partial ES shocks only that desk's factors. The sum of partial ES always exceeds the total-portfolio ES. $\mathrm{WES} = \lambda\,\mathrm{ES}_T + (1-\lambda)\sum_j \mathrm{ES}_{Pj}$ with $\lambda = 0.5$.
- **Non-modelable risk factors** (fewer than 24 observations a year, or more than one month between observations) are handled by stress tests, giving a charge NMC.
- **IMA capital on day t:** $\max\left(\mathrm{WES}_{t-1} + \mathrm{NMC}_{t-1},\ m_c\,\mathrm{WES}_{\text{avg}} + \mathrm{NMC}_{\text{avg}}\right)$, averages over the previous 60 days, $m_c \geq 1.5$.

Formula card: MR-18 b

Horizons	10, 20, 40, 60, 120 days (categories 1 to 5)
Cascade weights	$(LH_j - LH_{j-1})/10 = 1,\ 1,\ 2,\ 2,\ 6$
Liquidity-adjusted ES	$\sqrt{\mathrm{ES}_1^2 + \mathrm{ES}_2^2 + 2\,\mathrm{ES}_3^2 + 2\,\mathrm{ES}_4^2 + 6\,\mathrm{ES}_5^2}$
No category-1 factors	$\mathrm{ES}_2 = \mathrm{ES}_1$
Subset scaling	stressed ES $\times\ \mathrm{ES}_{12m,\text{all}}/\mathrm{ES}_{12m,\text{subset}}$ (subset $\geq 75\%$ of factors)
Weighted ES	$\mathrm{WES} = 0.5\,\mathrm{ES}_T + 0.5\sum_j \mathrm{ES}_{Pj}$
IMA capital	$\max(\mathrm{WES}_{t-1} + \mathrm{NMC}_{t-1},\ m_c\,\mathrm{WES}_{\text{avg}} + \mathrm{NMC}_{\text{avg}})$, $m_c \geq 1.5$, 60-day averages
SA risk weight	stressed daily vol $\times\sqrt{LH} \times 2.338$

Exam traps

- **Full-horizon weights.** Scaling by $\sqrt{LH_j/10}$ (weights 1, 2, 4, 6, 12) instead of the increment $\sqrt{(LH_j - LH_{j-1})/10}$ (1, 1, 2, 2, 6). It double counts time already covered by earlier rows.
- **Linear sum.** Adding the scaled ES terms (perfect correlation) instead of root-sum-of-squares (independence across blocks).
- **Unscaled quadrature.** Root-sum-of-squares with all weights equal to 1 ignores the extra horizon.
- **Dropped term.** A missing ES_2 for a desk with no 10-day factors equals ES_1, not zero.
- **The superseded list.** 10, 20, 60, 120, 250 is the 2012 proposal; 250 days is the stressed-period length.
- $\sqrt{10}$ **on 1-day changes.** Basel I / 2.5 practice. FRTB uses overlapping 10-day changes; a 1-day VaR in the stem is a red herring (it is for backtesting).
- **Overlap means bias**: false. Overlap means no independence, smaller effective sample, noisier estimate; no bias.
- **Inverted subset ratio.** The subset understates risk, so the ratio (all / subset) is above 1 and scales ES *up*.

Practice questions: MR-18 b

Question MR-18 b.1

Bank Q865 • AnalystPrep Q.2364 (rebuilt)

Under the FRTB, each risk factor is assigned to one of five categories according to the time needed to exit or hedge the exposure in stressed markets. Which of the following lists the liquidity horizons (in days) of the five categories?

A. 10, 20, 40, 60, 120

B. 10, 20, 60, 120, 250

C. 1, 5, 10, 20, 60

D. 5, 10, 20, 40, 120

Question MR-18 b.2

New • built from GARP Ch.18, Table 18.1

A market risk analyst at a universal bank is mapping the risk factors of a new multi-asset trading desk to the FRTB liquidity horizon categories before the desk applies for internal models approval. Which of the following assignments is correct?

A. Large-cap equity price volatility: 10 days.

B. Corporate investment-grade credit spread: 40 days.

C. Foreign exchange volatility: 60 days.

D. Credit spread volatility: 60 days.

Question MR-18 b.3

Bank Q868 • GARP PE2 Q7 • Mock 4 Q8

A regulatory capital analyst at a large European bank is studying the liquidity horizons specified for different risk factors in the Fundamental Review of the Trading Book (FRTB). The analyst examines how these liquidity horizons are included in and impact the bank's calculations of capital requirements. Which of the following statements is correct?

A. Under the standardized approach, the liquidity horizons of risk factors are incorporated into market risk capital calculations through the risk weights determined by the Basel Committee.

B. When a bank uses the historical simulation approach, the impact of 1-day changes in risk factors are scaled to the liquidity horizon for the risk factor by multiplying by the square root of time.

C. The standardized approach incorporates the diversification benefits between risk factors with different liquidity horizons through the risk weights set by the Basel Committee.

D. The historical simulation approach estimates a liquidity-adjusted ES by assuming a 10-day liquidity horizon for all risk factors, since their behavior will most likely be highly correlated during volatile market conditions.

Question MR-18 b.4

New • built from GARP Ch.18, Section 18.3

A model validation analyst is reviewing a bank's FRTB internal models implementation, which estimates stressed 97.5% ES by historical simulation over a 250-day stressed period. The first simulation trial uses the changes in all risk factors between Day 0 and Day 10 of the stressed period, the second uses the changes between Day 1 and Day 11, and so on. Which of the following statements about this design is correct?

A. Successive trials share most of their daily changes and are not independent; this does not bias the ES estimate, but it reduces the effective sample size and makes the estimate noisier.

B. Successive trials share most of their daily changes, so the ES estimate is biased upward, and the bank must divide the result by $\sqrt{10}$ before using it for capital.

C. The design breaches FRTB, which requires non-overlapping 10-day periods so that the observations on losses are independent.

D. The design is unnecessary under FRTB, which allows the bank to compute 1-day changes and scale them by $\sqrt{10}$, avoiding overlapping periods altogether.

Question MR-18 b.5

Bank Q867 • BT P2.T7.609.1 (rebuilt)

Under the FRTB internal models approach, risk factors are placed in five liquidity-horizon categories (10, 20, 40, 60 and 120 days). Which of the following best summarizes how the liquidity-adjusted expected shortfall is estimated?

A. ES_j is the ES from a 10-day shock to categories j through 5 only, holding categories 1 to $j-1$ constant. Each ES_j $(j \geq 2)$ is scaled by $\sqrt{(LH_j - LH_{j-1})/10}$, and total ES is the square root of the sum of the squared terms, reflecting independence across horizons.

B. ES_j is the ES from a 10-day shock to categories j through 5 only, scaled by $\sqrt{(LH_j - LH_{j-1})/10}$, but total ES is the simple sum of the five terms because variable changes are assumed perfectly correlated, the most conservative assumption.

C. ES_j is the ES from a 10-day shock to categories j through 5 only, but each is scaled by $\sqrt{LH_j/10}$ (the full horizon), and total ES is the square root of the sum of the squared terms.

D. ES is computed once from a 10-day shock to all risk factors and then multiplied by $\sqrt{120/10}$, so that every risk factor is treated as if it had the longest liquidity horizon.

Question MR-18 b.6

Bank Q864 • GARP book Ch.18 Q18.7 (rebuilt)

Under the FRTB internal models approach, a bank's risk factors fall into five liquidity-horizon categories: Category 1 (10 days), 2 (20 days), 3 (40 days), 4 (60 days) and 5 (120 days). ES_j denotes the 97.5% expected shortfall from a 10-day shock to risk factors in categories j through 5 only, holding categories 1 to $j-1$ constant. The bank estimates (USD million):

j	1	2	3	4	5
ES_j	10	8	6	4	2

Assuming independence of shocks across horizon buckets, what is the liquidity-adjusted ES?

A. USD 14.83 million

B. USD 17.09 million

C. USD 22.72 million

D. USD 30.00 million

Question MR-18 b.7

Workbook A4.1 • S2 • Aug-2026 • Map #5

A market risk analyst at a Canadian bank is producing the FRTB internal models approach figures for the bank's credit trading desk. The desk's only risk factors are:

- sovereign investment-grade credit spreads
- corporate investment-grade credit spreads
- corporate non-investment-grade credit spreads
- credit spread volatilities

Using overlapping 10-day changes from the stressed period, the analyst has computed stressed 97.5% ES (CAD million): ES_1 (all risk factors shocked) $= 18.0$; $\text{ES}_3 = 12.0$; $\text{ES}_4 = 7.5$; $\text{ES}_5 = 3.0$. The ES_2 run was dropped from the overnight batch. The desk's 1-day 99% VaR used for backtesting is CAD 4.6 million. Which of the following is closest to the desk's liquidity-adjusted ES?

A. CAD 27.9 million

B. CAD 33.2 million

C. CAD 44.6 million

D. CAD 70.9 million

Question MR-18 b.8

Bank Q862 • AnalystPrep Q.2365 / Q.2998 (rebuilt)

A bank cannot find a stressed 250-day period using all market variables because of missing history for some of them. Under the FRTB it computes stressed ES on a subset of variables, giving USD 80 million for the stressed period. Over the most recent 12 months, ES using all variables is USD 40 million and ES using only the subset is USD 32 million. What stressed ES should the bank use?

A. USD 100 million

B. USD 64 million

C. USD 80 million

D. USD 88 million

Solutions • MR-18 b

Solution MR-18 b.1

Answer: A

Bank Q865 • AnalystPrep Q.2364

Step 1 – Recall Table 18.1

Category 1 = 10 days, 2 = 20, 3 = 40, 4 = 60, 5 = 120.
⇒ List A.

Interpretation

This is the question Vardeez's notes would get wrong: his list (B) is the superseded 2012 proposal. In FRTB, 250 days is the length of the stressed period used to calibrate ES, not a liquidity horizon.

Why the other options are wrong

B is incorrect. The superseded list; 250 days is the stressed-period length, and the longest horizon is 120 days.
C is incorrect. FRTB has no 1-day or 5-day horizon; the shortest is 10 days (the old Basel I and II.5 horizon).
D is incorrect. Starts at 5 days; no FRTB category is shorter than 10 days.

LO: MR-18 b: Compare the various liquidity horizons proposed by the FRTB for different asset classes and explain how a bank can calculate its expected shortfall using the various horizons. • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.1 and Table 18.1

Solution MR-18 b.2

Answer: B

New • built from GARP Table 18.1

Step 1 – Look up each factor in Table 18.1

Large-cap equity price volatility: 20. Corporate IG credit spread: 40. FX volatility: 40. Credit spread volatility: 120.
⇒ Only B matches.

Step 2 – Use the patterns

Volatilities sit at or above their level (large-cap equity 10, its vol 20; FX up to 40, FX vol 40). Credit spreads lengthen as quality falls (sovereign IG 20, corporate IG 40, corporate non-IG 60), and credit spread volatility is the longest, 120.

Interpretation

Each wrong option moves one factor into a neighbouring bucket: the typical GARP distractor for a lookup table.

Why the other options are wrong

A is incorrect. Large-cap equity price is 10 days; its volatility is 20.
C is incorrect. FX volatility is 40 days; 60 is interest rate volatility.
D is incorrect. Credit spread volatility is 120 days; 60 is the corporate non-IG spread.

LO: MR-18 b • **Reference:** Hull, RMFI 6e, Ch.18, Table 18.1

Solution MR-18 b.3

Answer: A

Bank Q868 • GARP PE2 Q7

Step 1 – Standardized approach

The Basel Committee sets risk weights as multiples of stressed daily volatility that reflect the liquidity horizon and the confidence level: $W = \sigma_{\text{stressed}} \times \sqrt{LH} \times 2.338$ (e.g. $0.02\sqrt{20} \times 2.338 =$

0.209).
⇒ A is correct: liquidity horizons enter the SA through the risk weights.

Step 2 – Internal models approach

The base shock is an actual 10-day change from the stressed period; longer horizons enter through the cascade, scaled by $\sqrt{(LH_j - LH_{j-1})/10}$. 1-day changes are not scaled up.

Interpretation

Two different channels: weights in the SA, the cascade in the IMA. Diversification in the SA comes from the prescribed correlations, not from the weights.

Why the other options are wrong

B is incorrect. Describes Basel I / II.5 $\sqrt{\text{time}}$ scaling of 1-day changes; FRTB uses 10-day changes and the cascade.
C is incorrect. Risk weights capture liquidity horizons, not diversification; diversification comes through the Basel correlations.
D is incorrect. The liquidity-adjusted ES uses five horizons (10 to 120 days), not a uniform 10 days.

LO: MR-18 b • **Reference:** Hull, RMFI 6e, Ch.18, Sections 18.2 and 18.3

Solution MR-18 b.4 Answer: A

New • built from GARP Ch.18, Section 18.3

Step 1 – What overlap does

Adjacent trials (Day 0 to 10, Day 1 to 11) share 9 of their 10 daily changes, so successive trials are not independent.

Step 2 – What overlap does not do

GARP: this "does not bias the results, but it reduces the effective sample size, making results more noisy".
⇒ A is correct.

Step 3 – Why FRTB accepts it

Non-overlapping 10-day periods would require a very long history. The Basel Committee judges 10-day changes more relevant than 1-day changes, because successive days may not be independent while successive 10-day periods reasonably can be.

Interpretation

The FRTB design trades statistical efficiency (overlap) for relevance (actual 10-day moves in a stressed period).

Why the other options are wrong

B is incorrect. Overlap does not bias the estimate, and there is no $\sqrt{10}$ correction.
C is incorrect. Reversed: FRTB *requires* overlapping periods; non-overlapping periods are what econometricians would prefer but are infeasible.
D is incorrect. $\sqrt{10}$ scaling of 1-day changes is the Basel I / II.5 practice that FRTB abandons.

LO: MR-18 b • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3 and Problem 18.3

Solution MR-18 b.5 Answer: A

Bank Q867 • BT P2.T7.609.1

Step 1 – The nesting

ES_1 shocks all categories; ES_2 shocks categories 2 to 5 with category 1 held constant; ...; ES_5

shocks category 5 only.

Step 2 – The scaling

Each ES_j ($j \geq 2$) is scaled by $\sqrt{(LH_j - LH_{j-1})/10}$: squared weights 1, 2, 2, 6. ES_1 is unscaled.

Step 3 – The aggregation

Successive blocks are assumed independent, so the terms combine in quadrature (root-sum-of-squares).

$\Rightarrow$ Only A has all three features.

Interpretation

The three distractors are the three classic cascade errors: linear addition, full-horizon scaling, and one blanket horizon.

Why the other options are wrong

B is incorrect. A simple sum assumes perfect correlation across blocks; the formula assumes independence.

C is incorrect. Uses the full horizon LH_j instead of the increment $LH_j - LH_{j-1}$, double counting time already covered by earlier rows.

D is incorrect. Gives every factor the longest horizon; FRTB gives each factor its own horizon through the cascade.

LO: MR-18 b • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3, eq. 18.2

Solution MR-18 b.6 **Answer: B**

Bank Q864 • GARP book Ch.18 Q18.7

Step 1 – Cascade weights

$(LH_j - LH_{j-1})/10$: $j = 2$: $10/10 = 1$; $j = 3$: $20/10 = 2$; $j = 4$: $20/10 = 2$; $j = 5$: $60/10 = 6$.

Step 2 – Squared, weighted terms

$10^2 = 100$; $1 \times 8^2 = 64$; $2 \times 6^2 = 72$; $2 \times 4^2 = 32$; $6 \times 2^2 = 24$.

Step 3 – Liquidity-adjusted ES

$$\text{ES} = \sqrt{100 + 64 + 72 + 32 + 24} = \sqrt{292} = 17.09$$

$\Rightarrow$ USD 17.09 million, about 1.7 times the 10-day all-factor ES.

Interpretation

The longer-horizon categories add capital even though their own ES numbers are small, because they stay exposed for more independent 10-day blocks (category 5 for six extra blocks).

Why the other options are wrong

A is incorrect. Root-sum-of-squares with no horizon scaling: $\sqrt{100 + 64 + 36 + 16 + 4} = 14.83$.

C is incorrect. Full-horizon weights $LH_j/10 = 1, 2, 4, 6, 12$: $\sqrt{100 + 128 + 144 + 96 + 48} = \sqrt{516} = 22.72$.

D is incorrect. Simple sum $10 + 8 + 6 + 4 + 2 = 30$: perfect correlation and no scaling.

LO: MR-18 b • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3, eq. 18.2

Solution MR-18 b.7 — Answer: B

Workbook A4.1 • S2 • Aug-2026

Step 1 – Map each risk factor to its liquidity horizon (Table 18.1)

Sovereign IG spread: 20 days (category 2). Corporate IG spread: 40 days (category 3). Corporate non-IG spread: 60 days (category 4). Credit spread volatility: 120 days (category 5).
⇒ The desk has no category 1 (10-day) risk factors at all.

Step 2 – Infer the missing ES_2

ES_2 shocks categories 2 to 5 and holds category 1 constant. With no category 1 factors, holding them constant changes nothing, so $\text{ES}_2 = \text{ES}_1 = 18.0$.
⇒ The missing run is not a zero: every factor on this desk still moves in the second 10-day block.

Step 3 – Cascade weights

$(LH_j - LH_{j-1})/10$: 1, 1, 2, 2, 6.

Step 4 – Squared, weighted terms

$1 \times 18.0^2 = 324.0$; $1 \times 18.0^2 = 324.0$; $2 \times 12.0^2 = 288.0$; $2 \times 7.5^2 = 112.5$; $6 \times 3.0^2 = 54.0$.

Step 5 – Liquidity-adjusted ES

$$\text{ES} = \sqrt{324.0 + 324.0 + 288.0 + 112.5 + 54.0} = \sqrt{1102.5} = 33.20$$

⇒ CAD 33.2 million, about 1.84 times the 10-day all-factor ES: the long horizons of spreads and spread volatility add capital that a single 10-day horizon would miss.

Interpretation

The 1-day 99% VaR is a red herring: it is used only for the desk-level backtest, and scaling it by $\sqrt{10}$ is the Basel I / 2.5 practice that FRTB abandons.

Why the other options are wrong

A is incorrect. Dropped term: treats the missing ES_2 as zero, $\sqrt{324.0 + 288.0 + 112.5 + 54.0} = \sqrt{778.5} = 27.90$.

C is incorrect. Full-horizon weights $\sqrt{LH_j/10}$ (1, 2, 4, 6, 12): $\sqrt{324 + 648 + 576 + 337.5 + 108} = \sqrt{1993.5} = 44.65$.

D is incorrect. Adds the scaled ES linearly (perfect correlation): $18.0 + 18.0 + 12.0\sqrt{2} + 7.5\sqrt{2} + 3.0\sqrt{6} = 70.93$.

LO: MR-18 b • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3 and Table 18.1

Solution MR-18 b.8 — Answer: A

Bank Q862 • AnalystPrep Q.2365 / Q.2998

Step 1 – The FRTB scaling rule

When the stressed-period search uses a subset of risk factors (at least 75% of current factors), the stressed ES is scaled by $\text{ES}_{12m,\text{all}}/\text{ES}_{12m,\text{subset}}$.

Step 2 – The ratio

$$\frac{40}{32} = 1.25$$

⇒ The subset captures only 80% of the recent risk, so the stressed figure must be grossed up.

Step 3 – Scaled stressed ES

$$80 \times 1.25 = \text{USD } 100 \text{ million}$$

Interpretation

The ratio is measured on recent data, where both versions can be computed, and then applied to the stressed period, where only the subset can.

Why the other options are wrong

B is incorrect. Inverted the ratio: $80 \times 32/40 = 64$, scaling the figure down.

C is incorrect. Used the subset stressed ES unscaled, ignoring the missing variables.

D is incorrect. Added the difference $40 - 32 = 8$ instead of multiplying by the ratio.

LO: MR-18 b • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3

MR-18 • Fundamental Review of the Trading Book • Vardeez: VERY IMPORTANT

MR-18 c

P1 CONCEPT

Explain the FRTB revisions to Basel regulations in the following areas: - Classification of positions in the trading book compared to the banking book. - Backtesting, profit and loss attribution, credit risk, and securitizations.

Workbook: A4 FRTB • Tier A • August 2026 recall

WHAT IS TESTED HERE

Why P1. Tier A workbook topic (A4, August 2026 recall).
Vardeez: the full chapter is important.

Reported exam shapes

- **A4 S3** Standardized vs internal models approach, with the desk-level backtesting and P&L attribution tests that decide which applies.
- **A4 S4** Trading book / banking book boundary.

What the questions look like: concept statements, plus "eligibility pack" vignettes where you apply the thresholds (more than 12 / more than 30 exceptions; $\pm 10\%$ and 20%; four or more breaches) to desk data. No formula beyond reading those thresholds.

1. Trading book versus banking book

	Trading book	Banking book
Holds	Instruments the bank intends to trade	Instruments expected to be held to maturity
Valuation	Marked to market	Held at cost
Capital	Market risk capital	Credit risk capital

The problem. The two capitals are calculated quite differently, which invited regulatory arbitrage: banks held credit-dependent instruments in the trading book because they then attracted less capital. (Basel 2.5's IRC made this less attractive.)

FRTB's fix: a clearer, less subjective boundary.

- **"Intent to trade" is no longer sufficient.** The bank must be *able to trade* the instrument *and* manage its risks on a trading desk. Day-to-day value changes should affect equity and pose risks to solvency.
- **Assignment at initiation.** Instruments are assigned to a book when they are initiated, with strict rules against moving them.
- **Transfers only in extraordinary circumstances** (examples: closing a trading desk; a change in accounting standards on fair value recognition).
- **Any capital benefit from moving is disallowed**: the original capital requirement is retained.

2. Backtesting

FRTB does **not** backtest the stressed ES used for capital, for two reasons: (1) ES is more difficult to backtest than VaR; (2) a stressed measure cannot be backtested at all, because the stressed data it uses are not expected to recur with the same frequency.

Instead **each trading desk** backtests a **1-day VaR** on the **most recent 12 months** of data, at **both** 99% and 97.5%:

Confidence	Expected exceptions (≈250 days)	Fails if exceptions are
99%	2.5	more than 12
97.5%	6.25	more than 30

Breaching *either* limit sends the desk to the standardized approach until neither condition holds. The lower confidence level carries the higher allowance because it is expected to be breached more often. Regulators may also ask for other tests, e.g. that daily p-values (probability of a worse outcome than the one observed) are uniformly distributed.

3. Profit and loss attribution

Compare each day's actual P&L with the model's prediction. With U = actual minus model P&L and V = actual P&L ("actual" means hypothetical: the P&L if there had been no trading that day):

$$\text{Measure 1} = \frac{\text{Mean}(U)}{\text{SD}(V)} \in [-10\%, +10\%] \qquad \text{Measure 2} = \frac{\text{Var}(U)}{\text{Var}(V)} < 20\%$$

Measure 1 tests for *bias* (is the model systematically off?), so it is two-sided. Measure 2 tests for *noise* (how much variation the model fails to explain), so it is one-sided. **Four or more** situations in 12 months outside these ranges: the desk must use the standardized approach.

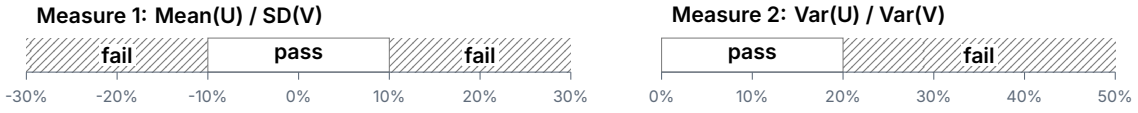


Figure 7. P&L attribution bands (hatched: fail zone). Measure 1 is two-sided (bias either way fails); Measure 2 is one-sided (there is no such thing as too little unexplained variance).

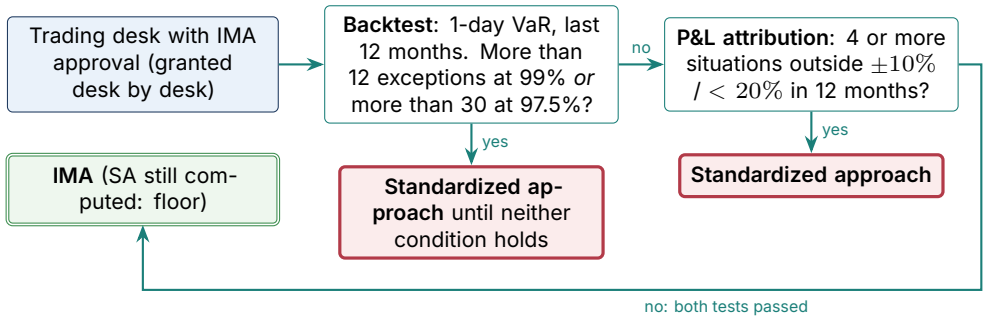


Figure 8. Desk-level eligibility for the internal models approach. Failing one desk does not affect other desks.

4. Credit risk

FRTB separates two credit risks of exposure to a company:

	Credit spread risk	Jump-to-default risk
What it is	The spread changes, so mark-to-market value changes	The company defaults: immediate, potentially large loss
IMA treatment	Like market risk: inside the ES process; horizons 20 to 120 days (spread vol 120)	Like banking-book default risk: VaR with **1-year horizon, 99.9%** (incremental default risk, IDR)
SA treatment	Delta / vega / curvature	Default risk charge: exposure × LGD × risk weight
Scope		All instruments subject to default, **including equities** (LGD 100% in the SA)

What changed from Basel 2.5. The IRC (99.9%, one year, defaults and downgrades) assumed a *constant level of risk*: a position that deteriorated was assumed replaced by a similar one at the end of its liquidity horizon (e.g. an A-rated bond with a three-month horizon replaced by another A-rated bond). FRTB replaces the IRC with the default risk charge and drops the replacement assumption in favour of incremental marking to market.

5. Securitizations

Basel 2.5's comprehensive risk measure (CRM) let banks, with approval, use their own models for securitization products (ABS, CDOs). The Basel Committee found too much variation in the capital calculated by different banks *for the same portfolio*. FRTB therefore requires the **standardized approach** for securitizations. The motivation is consistency, not conservatism.

> **The theme across all five areas:** FRTB narrows bank discretion. The book boundary is fixed and reclassification barred; desks that fail backtesting or P&L attribution lose their internal model; securitizations lose internal models outright.

Area	Before FRTB	FRTB
Book boundary	Intent to trade; moves possible	Able to trade and manage on a desk; fixed at initiation; moves only in extraordinary cases, no capital benefit
Backtesting	1996 rule: 1-day 99% VaR, 250 days; 10 or more exceptions sets $m = 4$	Per desk, 1-day VaR at 99% and 97.5%, 12 months; > 12 or > 30: SA
P&L attribution	None	Mean(U)/SD(V) in $\pm 10\%$, Var(U)/Var(V) $< 20\%$; 4+ breaches: SA
Credit	IRC with constant-risk replacement	Spread risk in ES; jump-to-default at 99.9%, 1 year, equities included
Securitizations	CRM with own models	Standardized approach only

Exam traps

- **Backtesting the stressed ES**: never. The backtest is on 1-day VaR, not the 10-day stressed ES.
- **One level or both?** GARP: both 99% and 97.5% are used (Schweser loosely says "either"); failing *either* limit sends the desk to SA.
- **Threshold swaps.** More than 12 at 99%, more than 30 at 97.5%; not the 1996 rule (10 or more in 250 days, multiplier 4). FRTB switches the approach, it does not raise a multiplier.

- **P&L attribution counts.** Four or more situations; three does not trigger. Measure 1 is $\pm 10\%$; Measure 2 is $< 20\%$ (do not apply $\pm 10\%$ to the variance ratio).
- **Intent to trade alone**: no longer sufficient.
- **Reclassification**: not "never" and not "at the bank's discretion": only in extraordinary circumstances, and any capital benefit is disallowed.
- **Jump-to-default parameters**: 99.9%, one year (not 97.5%, 10 days); equities included.
- **Securitizations**: standardized because of inconsistency across banks, not because models were too lenient.

Practice questions: MR-18 c

Question MR-18 c.1

Bank Q886 • BT notes P2.T5.23.8.2 (rebuilt)

Under the FRTB, a bank initially assigns a position to the banking book and later wants to move it to the trading book because it expects the move to lower its capital requirement. Which statement best describes the FRTB treatment?

A. Reclassification is permitted at the bank's discretion provided it is disclosed to the supervisor.

B. Reclassification is permitted only in extraordinary circumstances, and any capital benefit from the move is disallowed.

C. Reclassification is permitted only in extraordinary circumstances, and the bank keeps the full capital benefit of the new category.

D. Reclassification is never permitted under any circumstances.

Question MR-18 c.2

Bank Q884 • BT P2.T7.609.2 (rebuilt)

Each of the following statements about FRTB backtesting is true EXCEPT which is false?

A. Stressed expected shortfall measures are not backtested because extreme stressed values may not recur with the same frequency.

B. Backtesting uses a one-day horizon and the latest 12 months of data.

C. At the 99% level, more than 12 exceptions in the period requires the standardized approach.

D. The backtest must be performed on the 97.5% stressed ES used for the capital charge, with a 10-day horizon.

Question MR-18 c.3

Bank Q879 • AnalystPrep Q.6536 (rebuilt)

Under the FRTB profit-and-loss attribution test, U is the difference between the actual (hypothetical) and model-predicted P&L on a day and V is the actual (hypothetical) P&L. The test uses Mean(U)/SD(V), which must lie between -10% and $+10\%$, and Var(U)/Var(V), which must be less than 20%. Which of the following readings falls outside the requirement and counts toward the four-or-more failures in 12 months that force a desk onto the standardized approach?

A. Mean(U)/SD$(V) = -8\%$

B. Var(U)/Var$(V) = 12\%$

C. Mean(U)/SD$(V) = +9\%$

D. Var(U)/Var$(V) = 22\%$

Question MR-18 c.4

Workbook A4.3 • S3, S4 • Map #5

The head of market risk at an Australian bank is reviewing the FRTB eligibility pack for four trading desks, all of which currently hold internal models approval. Results for the most recent 12 months:

Desk	1-day 99% VaR exceptions	1-day 97.5% VaR exceptions	P&L attribution situations outside ranges
Rates	11	34	1
FX	7	22	2
Equity	10	27	3

The credit desk holds a portfolio of illiquid loans that it has documented an intent to trade but is not able to trade or hedge on the desk. Which of the following conclusions is correct?

A. The rates desk must calculate capital using the standardized approach, even though its 99% VaR exceptions are within the permitted number.

B. The equity desk must calculate capital using the standardized approach, because its P&L attribution ratios fell outside the ranges in three situations.

C. The credit desk may keep the loans in the trading book, because its documented intent to trade them meets the FRTB trading book test.

D. The FX desk's internal models approval fully replaces the standardized approach, so no standardized capital needs to be calculated for it.

Question MR-18 c.5 Bank Q870 • BT notes P2.T5.23.8.1 (rebuilt)

Which of the following statements about the FRTB's treatment of credit-dependent positions is FALSE?

A. Credit spread risk, the risk that a change in the credit spread alters mark-to-market value, can be addressed within the expected shortfall calculation.

B. Jump-to-default risk is subject to the incremental default risk (IDR) charge, calculated as a 99.9% VaR with a one-year horizon.

C. Under the internal models approach, the liquidity horizon for credit spreads ranges from 20 to 120 days, and the liquidity horizon for credit spread volatility is 120 days.

D. The FRTB requires no market risk capital charge for jump-to-default risk, because default exposures are charged only under the credit risk rules of the banking book.

Question MR-18 c.6 Bank Q885 • BT P2.T7.609.3 (rebuilt)

Basel II.5 introduced the incremental risk charge (IRC) to stop banks reducing capital by choosing the trading book over the banking book for credit-dependent instruments. How does the FRTB address credit trades?

A. The FRTB makes no changes to the IRC.

B. The FRTB edits the IRC to produce a capital charge based on a 10-day, 97.5% expected shortfall (ES).

C. The FRTB replaces the IRC with a default risk charge (DRC), which recognizes jump-to-default risk and mandates the inclusion of equity products.

D. The FRTB replaces the IRC with a default risk charge (DRC), which does not recognize jump-to-default risk and explicitly excludes equity products.

Question MR-18 c.7 New • built from GARP Ch.18, Section 18.3.4 and Vardeez

The head of regulatory capital at a bank with a large book of asset-backed securities and collateralized debt obligations is briefing the CFO on how FRTB changes the capital treatment of these positions. Which of the following statements is correct?

A. Basel II.5's comprehensive risk measure let banks use approved internal models; because capital for the same portfolio varied too much across banks, FRTB requires the standardized approach for securitizations.

B. Basel II.5's comprehensive risk measure let banks use approved internal models; because those models were found to be systematically too lenient, FRTB applies a higher multiplier to securitization ES.

C. FRTB allows securitizations to remain under the internal models approach, provided the desk holding them passes both the backtesting and the P&L attribution tests.

D. FRTB moves securitizations out of the trading book into the banking book, where they attract credit risk capital under the IRB approach.

Solutions • MR-18 c

Solution MR-18 c.1

Answer: B

Bank Q886 • BT notes P2.T5.23.8.2

Step 1 – The FRTB rule

Instruments are assigned to a book at initiation. Transfers are allowed only in extraordinary circumstances (desk closure, a change in accounting standards on fair value), and any capital benefit from the move is disallowed.

⇒ B states both halves.

Step 2 – Why

The lower capital on one side of the boundary is exactly what created the incentive to game it; disallowing the benefit removes the incentive.

Interpretation

The three distractors are the three ways to get this wrong: too loose (A), right condition but wrong consequence (C), too absolute (D).

Why the other options are wrong

A is incorrect. Discretionary reclassification is the pre-FRTB arbitrage the rule is designed to stop.

C is incorrect. Right condition, wrong consequence: the original capital requirement is retained.

D is incorrect. Absolute claim: transfers are allowed in extraordinary circumstances.

LO: MR-18 c: Explain the FRTB revisions to Basel regulations in the following areas: classification of positions in the trading book compared to the banking book; backtesting, profit and loss attribution, credit risk, and securitizations. • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.4

Solution MR-18 c.2

Answer: D

Bank Q884 • BT P2.T7.609.2

Step 1 – What is backtested

Each desk backtests a 1-day VaR on the most recent 12 months, at 99% and 97.5%. Limits: more than 12 exceptions at 99%, more than 30 at 97.5%.

⇒ B and C are true.

Step 2 – What is not backtested

The stressed ES is not backtested: ES is harder to backtest than VaR, and a stressed measure cannot be backtested because stressed data are not expected to recur with the same frequency.

⇒ A is true; D is false on every count (measure, stress, horizon).

Interpretation

Keep the two numbers apart: capital comes from 10-day-based stressed ES; eligibility comes from 1-day VaR backtests.

Why the other options are wrong

A is incorrect. True: it is GARP's second reason for not backtesting stressed ES.

B is incorrect. True: one-day horizon, latest 12 months.

C is incorrect. True: more than 12 exceptions at 99% triggers the standardized approach.

LO: MR-18 c • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3.1

Solution MR-18 c.3 — Answer: D

Bank Q879 • AnalystPrep Q.6536

Step 1 – Measure 1 (bias), two-sided band $[-10\%, +10\%]$
-8% and $+9\%$ are both inside the band.

Step 2 – Measure 2 (noise), one-sided limit $< 20\%$
12% passes; 22% fails.
$\Rightarrow$ Only D counts towards the four-or-more breaches.

Interpretation
A sign on Measure 1 is not a breach by itself, and a "small-looking" variance ratio can still be inside its limit because that limit is 20%, not 10%.

Why the other options are wrong
A is incorrect. Treats a negative ratio as a breach; -8% is inside $\pm 10\%$.
B is incorrect. Applies the $\pm 10\%$ band to the variance ratio; its limit is 20%.
C is incorrect. $+9\%$ is inside the band, close to but not beyond the limit.

LO: MR-18 c • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3.2

Solution MR-18 c.4 — Answer: A

Workbook A4.3 • S3, S4

Step 1 – Backtesting test, desk by desk
Rule: more than 12 exceptions at 99% or more than 30 at 97.5% sends the desk to the standardized approach. Rates: $11 \leq 12$ passes at 99%, but $34 > 30$ fails at 97.5%. FX: 7, 22: passes. Equity: 10, 27: passes.
$\Rightarrow$ Either condition alone is enough. The rates desk stays on the standardized approach until neither condition holds.

Step 2 – P&L attribution test
Rule: four or more situations in 12 months outside the ranges. Rates 1, FX 2, Equity 3: all below four.
$\Rightarrow$ No desk loses approval on P&L attribution this year; the equity desk is one breach from the threshold.

Step 3 – Trading book boundary
Under FRTB, intent to trade is no longer sufficient: the bank must be able to trade and manage the risks on a trading desk. The credit desk's loans fail this test.

Interpretation
The recommendation: move the rates desk to standardized capital, keep FX and equity on internal models while watching equity's attribution, and review the credit desk's loans for banking book treatment. Because approval is desk by desk, the rates result does not affect other desks. (The 1996 Amendment backtest, 10 or more exceptions in 250 days forcing a multiplier of 4, would have penalised the equity desk; FRTB's desk-level thresholds do not.)

Why the other options are wrong
B is incorrect. Scope/condition error: the trigger is four or more situations in 12 months; three does not trigger the standardized approach.
C is incorrect. Sibling swap: substitutes the pre-FRTB "intent to trade" test for the FRTB test, which requires the ability to trade and manage the risks on a desk.
D is incorrect. Absolute claim: even with internal models approval, banks must calculate capital under both approaches; the standardized approach provides a floor.

LO: MR-18 c; MR-18 a • **Reference:** Hull, RMFI 6e, Ch.18, Sections 18.1, 18.3.1, 18.3.2, 18.4

Solution MR-18 c.5 Answer: D

Bank Q870 • BT notes P2.T5.23.8.1

Step 1 – Two credit risks

Credit spread risk is handled like market risk, inside the ES process (A true), with horizons of 20 to 120 days for spreads and 120 days for spread volatility (C true).

Step 2 – Jump-to-default

Jump-to-default risk in the trading book has its own charge: under the IMA a VaR with a one-year horizon and 99.9% confidence (B true), mirroring banking-book default risk; under the SA a default risk charge.

$\Rightarrow$ D is false: FRTB does charge trading-book jump-to-default risk.

Interpretation

"Handled in the same way as default risks in the banking book" describes the *method* (1-year, 99.9% VaR), not a transfer of the charge to the banking book.

Why the other options are wrong

A is incorrect. True statement: spread risk sits in the ES process.
B is incorrect. True statement: 99.9%, one year (numbers often confused with 97.5% ES).
C is incorrect. True statement, straight from Table 18.1.

LO: MR-18 c • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3.3 and Table 18.1

Solution MR-18 c.6 Answer: C

Bank Q885 • BT P2.T7.609.3

Step 1 – What happens to the IRC

FRTB replaces the IRC with a default risk charge that targets jump-to-default risk.

Step 2 – Its parameters and scope

Under the IMA it is a 99.9%, one-year VaR, not a 10-day 97.5% ES. It applies to all instruments subject to default, including equities (LGD 100% in the SA).

$\Rightarrow$ C is correct.

Interpretation

Spread risk moves into the ES; default risk keeps a separate, long-horizon, very-high-confidence charge. The IRC's constant-level-of-risk replacement assumption is dropped.

Why the other options are wrong

A is incorrect. The IRC is replaced, not retained.
B is incorrect. Mixes the default charge with the ES capital measure; default risk uses 99.9% over one year.
D is incorrect. Reverses both features: jump-to-default is the point of the charge, and equities are included.

LO: MR-18 c • **Reference:** Hull, RMFI 6e, Ch.18, Sections 18.2.3 and 18.3.3

Solution MR-18 c.7 Answer: A

New • built from GARP Ch.18, Section 18.3.4

Step 1 – The Basel 2.5 position

The CRM charge covered securitization products (ABS, CDOs) and allowed banks, with regulatory approval, to use their own models.

Step 2 – The problem the Basel Committee found

Too much variation in the capital charges calculated by different banks for the same portfolio.

Step 3 – The FRTB response

The standardized approach must be used for securitizations.
$\Rightarrow$ A is correct.

Interpretation

The motivation is consistency, not conservatism: models were judged too variable, not too lenient. The positions stay in the trading book; they simply lose internal models.

Why the other options are wrong

B is incorrect. Wrong motivation and wrong mechanism: variability, not leniency; and the m_c multiplier applies to WES under the IMA, not to securitizations.
C is incorrect. Securitizations lose internal models outright, whatever the desk's test results.
D is incorrect. Sibling swap with the book boundary: FRTB does not move securitizations to the banking book.

LO: MR-18 c • **Reference:** Hull, RMFI 6e, Ch.18, Section 18.3.4

MARKET RISK • PART

P2

Second priority

Tier B topics, or LOs Vardeez flags inside a Very Important chapter. 12 LOs from 5 readings.

MR-1 • Estimating Market Risk Measures: An Introduction and Overview • Vardeez: VERY IMPORTANT

MR-1 c

P2 CALC

Estimate the expected shortfall given profit and loss (P&L) or return data.

Workbook: B2 Expected Shortfall and Coherence • Tier B

WHAT IS TESTED HERE

Why P2. A Tier B workbook topic (B2, Expected Shortfall and Coherence) hits this LO, inside a chapter Vardeez rates Very Important.
Vardeez: calculation of ES.

Reported exam shapes

- **B2·S1** ES from a sample (and its bootstrap resamples): exactly which losses enter the tail average, 4, 5 or 6. The off-by-one on tail membership is the intended distractor.
- **B2·S2** ES as the average of tail VaRs read off a confidence-level table, against the 99% VaR it replaces under FRTB.

What the questions look like: numerical. Either a sorted list of the worst losses (count the tail correctly), a table of VaRs at rising confidence levels (average the right rows), or μ and σ of a normal (use the closed form). The two workbook questions on these shapes are worked elsewhere in this book: B2.1 as Question MR-2 a.4 and B2.2 as Question MR-18 a.5.

1. What expected shortfall measures

VaR answers "how bad can it get with α confidence?". It says nothing about how bad the losses are once VaR is breached. **Expected shortfall (ES)** answers "if we are in the tail, what is the average loss?": it is the *probability-weighted average of the losses beyond VaR*.

- ES $\geq$ VaR at the same confidence level, always. If an answer gives ES below VaR, it is wrong.
- ES is a **coherent** risk measure (it is subadditive); VaR is not in general. This is why FRTB moves from 99% VaR to 97.5% ES.
- For a standard normal: 95% VaR $= 1.645$, 95% ES $= 2.063$.

2. Route 1: historical P/L data (count the tail)

With n equally likely observations and confidence level α:

$$\begin{aligned} \text{tail size} &= (1-\alpha)\, n \text{ observations} \\ \text{VaR} &= \text{the } \big[(1-\alpha)n + 1\big]\text{th largest loss} \\ \text{ES} &= \text{average of the } (1-\alpha)n \text{ largest losses} \end{aligned}$$

Dowd's convention: VaR is "the highest loss observation outside the tail", so the VaR observation is *not* averaged into ES.

- 1,000 observations, 95%: 50 tail losses. VaR is the 51st largest loss (1.704 in Dowd's example); ES is the average of the 50 largest (2.196).
- 100 observations, 95%: 5 tail losses. VaR is the 6th largest, ES the average of the 5 largest.
- 500 observations, 99%: 5 tail losses. 300 observations, 99%: 3 tail losses.

When the stem lists the exceedances instead. If you are told the VaR and given the losses that exceed it, ES is the plain average of *all* the listed exceedances (including any outlier), divided by how many there are.

3. Route 2: average the tail VaRs (slicing)

When you can read off quantiles but have no formula for ES, slice the tail:

1. Divide the tail mass $1-\alpha$ into N equal-probability slices of size $L=(1-\alpha)/N$.
2. Take the VaRs at the $N-1$ *interior* boundaries: $\alpha+L,\ \alpha+2L,\ \ldots,\ \alpha+(N-1)L$.
3. ES $\approx$ the simple average of those $N-1$ VaRs.

The VaR at α itself is the tail threshold, not a slice boundary inside the tail, so it is left out. The 100% quantile is not used either.

Standard normal, 95% ES	Slices N	VaRs averaged	Estimate	True ES
96%, 97%, 98%, 99%	5	4	2.003	2.063
95.5%, 96%, ..., 99.5%	10	9	2.025	2.063
...	100	99	2.056	2.063
...	10,000	9,999	2.063	2.063

The estimate *rises* with N and converges to the true value from below: coarse slicing understates ES because it misses the far tail.

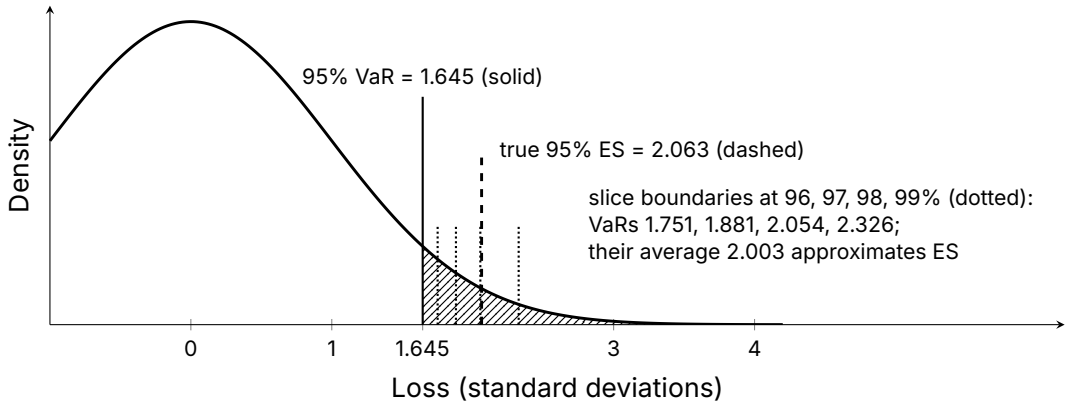


Figure 1. The 5% loss tail (hatched) is cut into five equal-probability slices. The four interior slice VaRs (dotted) are averaged: 2.003, just under the true ES of 2.063 (dashed). More slices push the estimate up towards 2.063.

4. Route 3: closed form for a normal

In loss-positive (L/P) units, with loss mean μ_L:

$$\boxed{\mathrm{ES}_\alpha = \mu_L + \sigma\,\frac{\phi(z_\alpha)}{1-\alpha}} \qquad \text{where} \qquad \phi(z)=\frac{\mathrm{e}^{-z^2/2}}{\sqrt{2\pi}}$$

If the stem gives a *return* mean μ_r (profit positive), the sign flips: $\mathrm{ES}=-\mu_r+\sigma\,\phi(z_\alpha)/(1-\alpha)$. Hull writes the same thing as $\sigma\,\mathrm{e}^{-Y^2/2}/(\sqrt{2\pi}(1-X))$ with X the confidence level and Y its quantile: X and Y are two different inputs.

Confidence α	z_α (VaR multiple)	$\phi(z_\alpha)/(1-\alpha)$ (ES multiple)	Gap
95%	1.645	2.063	0.418
97.5%	1.960	2.338	0.378
99%	2.326	2.665	0.339

FRTB link (shape B2·S2). For a normal, 97.5% ES $= \mu + 2.338\sigma$ is almost exactly the old 99% VaR $= \mu + 2.326\sigma$. With a fat tail, 97.5% ES can be considerably larger than 99% VaR, which is the point of the switch (see MR-18 a).

5. Route 4: discrete loss distributions

When losses come in lumps (defaults of a few bonds, a discrete P/L table), fill *exactly* $1-\alpha$ of probability with the worst outcomes, splitting the outcome that straddles the boundary:

$$\text{ES}_\alpha = \frac{1}{1-\alpha} \sum_{\text{worst outcomes}} (\text{probability used}) \times (\text{loss})$$

Example: three independent bonds, \$100 each, PD 10%, no recovery. $P(3 \text{ defaults}) = 0.1\%$, $P(2) = 2.7\%$, $P(1) = 24.3\%$. The 5% tail is 0.1% at \$300, 2.7% at \$200 and the remaining 2.2% at \$100: ES $= (0.3 + 5.4 + 2.2)/0.05 =$ \$158, while 95% VaR is only \$100. Dividing by the probability of the default states alone (2.8%) instead of the full 5% is the classic error.

Formula card: MR-1 c

HS tail size	$(1-\alpha)\,n$ observations; VaR is the next one, $(1-\alpha)n+1$
HS ES	average of the $(1-\alpha)n$ largest losses (VaR observation excluded)
Slicing	$\text{ES} \approx \frac{1}{N-1}\sum_{k=1}^{N-1} \text{VaR}_{\alpha+kL}$, $\quad L = (1-\alpha)/N$
Normal ES (L/P)	$\text{ES} = \mu_L + \sigma\,\phi(z_\alpha)/(1-\alpha)$; returns: $-\mu_r + \sigma\,\phi(z_\alpha)/(1-\alpha)$
Multiples	95%: 2.063 97.5%: 2.338 99%: 2.665
Discrete	fill exactly $1-\alpha$ of mass with the worst outcomes, then divide by $1-\alpha$

Exam traps

- **Off-by-one on the tail.** Averaging the VaR observation in (one too many), or dropping the last tail loss (one too few). With 100 observations at 95% you average exactly 5 losses.
- **Slicing off-by-one.** Including the VaR at α itself, or dividing by N instead of $N-1$.
- **Reporting a single quantile.** The VaR, the largest VaR in the table, or the single worst loss is not ES.
- **Sign of the mean.** Loss mean is *added*; return mean is *subtracted*. Ignoring the mean altogether is the other slip.
- **VaR multiple for ES.** Using 1.96 at 97.5% gives VaR; ES needs 2.338.
- **Dropping the outlier.** ES exists to capture the large losses; every exceedance enters the average.
- **Discrete tails.** Dividing by the probability of the loss states only, or taking the worst outcome as ES.

Practice questions: MR-1 c

Question MR-1 c.1

Bank Q57 • Mock 2 Q27 (rebuilt)

A risk manager has 100 daily P&L observations for a portfolio. The six largest daily losses (USD million) are 14, 11, 9, 8, 7 and 6; every other observation is a smaller loss or a profit. Using the historical simulation convention that expected shortfall is the average of the losses in the tail beyond the VaR cut-off, the 95% one-day expected shortfall is closest to:

A. USD 7.0 million

B. USD 9.2 million

C. USD 9.8 million

D. USD 10.5 million

Question MR-1 c.2

Bank Q54 • Mock 2 Q63 (rebuilt)

A risk manager at Bank ABC uses profit/loss data from the previous 1,000 trading days and estimates a 1-day 99% VaR of USD 10 million. Exactly nine historical losses exceed this VaR; in USD millions they are: 12, 15, 17, 18, 21, 22, 24, 27 and 51. Estimating expected shortfall as the average of the losses beyond VaR, what is the portfolio's 99% expected shortfall?

A. USD 19.5 million

B. USD 20.7 million

C. USD 21.7 million

D. USD 23.0 million

Question MR-1 c.3

Bank Q44 • BT P2.T5.22.2.1 (rebuilt)

Peter has fitted a parametric distribution to his portfolio's returns and can compute any quantile (the inverse CDF), but has no closed-form formula for the expected shortfall. Let C be the confidence level, $S = 1 - C$ the significance level, N the number of equal-probability slices of the tail, and $L = S/N$ the size of each slice. How can Peter approximate the C expected shortfall?

A. Average the N tail VaRs at confidence levels $C + L,\ C + 2L,\ \ldots,\ C + NL$.

B. Average the $N - 1$ tail VaRs at confidence levels $C + L,\ C + 2L,\ \ldots,\ C + (N - 1)L$.

C. Average the N tail VaRs at confidence levels $C,\ C + L,\ \ldots,\ C + (N - 1)L$.

D. It is not possible to approximate ES from a set of tail VaRs; ES must be derived from the density directly.

Question MR-1 c.4

Bank Q51 • GARP PE1 Q52 • Mock 3 Q28 • B2·S2

A wealth management firm has JPY 72 billion in assets under management. The portfolio manager estimates the daily VaR at various confidence levels as follows:

Confidence level	VaR (JPY)	Confidence level	VaR (JPY)
95.0%	332,760,000	97.5%	367,882,500
95.5%	336,292,500	98.0%	378,412,500
96.0%	340,095,000	98.5%	392,452,500
96.5%	350,332,500	99.0%	410,880,000
97.0%	359,107,500	99.5%	439,252,500

Using the approach of dividing the tail into equal-probability slices and averaging the VaRs, what is the closest estimate of the daily expected shortfall at the 97.5% confidence level?

A. JPY 394 million

B. JPY 398 million

C. JPY 405 million

D. JPY 414 million

Question MR-1 c.5

Bank Q41 • BT Mock Q1 (rebuilt)

An exposure has a value of USD 30.0 million. In L(+)/P(−) format its expected loss is 4.0% with volatility 15.0%, i.e. $\mu_{L/P} = +4.0\%$ and $\sigma = 15.0\%$. The exposure is normally distributed. Which is nearest to the single-period 97.5% expected shortfall (ES)?

A. USD 9.32 million

B. USD 10.02 million

C. USD 10.52 million

D. USD 11.72 million

Question MR-1 c.6

Bank Q50 • BT P2.T5.22.1.3

Consider a portfolio of three independent, identical junk bonds. Each has a default probability of 10.0% over a single period, a face value of $100.00 and is priced at par; the loss given default is 100% (no recovery). The portfolio value is $300.00. What is the portfolio's 95.0% expected shortfall (ES)?

A. $100.00

B. $158.00

C. $300.00

D. Ambiguous

Question MR-1 c.7 Bank Q42 • BT L2.T5.70.2 (rebuilt)

A process has a 5% probability of producing a material loss and otherwise produces no loss (Bernoulli). If the loss occurs, its severity is normally distributed with mean USD 4 million and standard deviation USD 2 million. What is the 95% expected shortfall of the loss?

A. USD 0.20 million

B. USD 4.00 million

C. USD 7.29 million

D. USD 8.13 million

Solutions • MR-1 c

Solution MR-1 c.1

Answer: C

Bank Q57 • Mock 2 Q27

Step 1 – Size the tail

$$(1 - 0.95) \times 100 = 5 \text{ observations}$$

$\Rightarrow$ The tail holds the 5 largest losses; the 6th largest (6) is the 95% VaR, the cut-off.

Step 2 – Average the tail

$$\text{ES}_{95} = \frac{14 + 11 + 9 + 8 + 7}{5} = \frac{49}{5} = 9.8$$

$\Rightarrow$ On the worst 5% of days the average loss is USD 9.8 million.

Interpretation

This is shape B2·S1: the whole question is which losses enter the average. The sixth observation is listed precisely so that you can include it by mistake.

Why the other options are wrong

A is incorrect. 7.0 is the 5th largest loss, a single order statistic, not an average of the tail.
B is incorrect. Averaged six losses, including the VaR observation: $55/6 = 9.17$.
D is incorrect. Averaged only four losses: $42/4 = 10.5$.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.4 Estimating Coherent Risk Measures

Solution MR-1 c.2

Answer: D

Bank Q54 • Mock 2 Q63

Step 1 – Identify the tail losses

The stem gives the nine losses that exceed the 99% VaR. ES is the average loss given that VaR is exceeded, so all nine enter, including the 51.

Step 2 – Average them

$$\text{ES}_{99} = \frac{12 + 15 + 17 + 18 + 21 + 22 + 24 + 27 + 51}{9} = \frac{207}{9} = 23.0$$

$\Rightarrow$ ES (23.0) is well above VaR (10), as it must be.

Interpretation

The 51 is not an outlier to be trimmed: capturing exactly that kind of loss is why ES exists. With 1,000 days one would expect 10 exceedances at 99%; here there are 9, and ES averages the ones observed.

Why the other options are wrong

A is incorrect. Dropped the 51 as an outlier: $156/8 = 19.5$.
B is incorrect. Divided by 10 (1% of 1,000) instead of the nine observed exceedances: $207/10 = 20.7$.
C is incorrect. Added the VaR itself as a tenth tail observation: $(207 + 10)/10 = 21.7$.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.4

Solution MR-1 c.3 — Answer: B

Bank Q44 • BT P2.T5.22.2.1

Step 1 – Where the slice boundaries sit

N slices of size L have boundaries at $C, C+L, \ldots, C+NL = 1$. The first is the tail threshold, the last is the 100% quantile. The $N-1$ boundaries strictly inside the tail are $C+L, \ldots, C+(N-1)L$.

Step 2 – Average those VaRs

ES is the average of VaR over the tail, so it is approximated by the simple average of the $N-1$ interior VaRs. It converges to the true ES as N grows.

$\Rightarrow$ With $N = 10$ at $C = 95\%$: average the VaRs at 95.5%, 96%, ..., 99.5%.

Interpretation

This is the general recipe behind every "table of VaRs" question (shape B2·S2). Counting the boundaries correctly is the whole question.

Why the other options are wrong

A is incorrect. Includes $C+NL = 100\%$, where VaR is unbounded (or the worst possible loss); the count is $N-1$, not N.

C is incorrect. Starts at C, so it includes the VaR at the threshold, which pulls the average down.

D is incorrect. False: averaging tail VaRs at rising confidence levels is exactly Dowd's method.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.4, Table 1.2

Solution MR-1 c.4 — Answer: C

Bank Q51 • GARP PE1 Q52

Step 1 – Identify the slices

The 2.5% tail beyond 97.5% is cut into five 0.5% slices, so there are four interior VaRs: 98.0%, 98.5%, 99.0%, 99.5%.

Step 2 – Average them

$$\text{ES}_{97.5} \approx \frac{378.4125 + 392.4525 + 410.88 + 439.2525}{4} = \frac{1{,}620.9975}{4} = 405.25 \text{ million}$$

$\Rightarrow$ About JPY 405 million.

Interpretation

The AUM (JPY 72 billion) and the rows below 97.5% are not needed. GARP builds the distractors from counting errors, so check the count before you add.

Why the other options are wrong

A is incorrect. Dropped the 99.5% VaR and averaged 98.0% to 99.0%: 393.9.

B is incorrect. Included the 97.5% boundary row and divided by 5: $1{,}988.88/5 = 397.8$.

D is incorrect. Dropped the 98.0% VaR and averaged 98.5% to 99.5%: 414.2.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.4

Solution MR-1 c.5 — Answer: D

Bank Q41 • BT Mock Q1

Step 1 – ES multiple at 97.5%

$$z_{0.975} = 1.96, \qquad \phi(1.96) = \frac{e^{-1.96^2/2}}{\sqrt{2\pi}} = 0.05845, \qquad \frac{0.05845}{0.025} = 2.338$$

Step 2 – ES in loss-positive units

$$\text{ES} = \mu_L + 2.338\,\sigma = 4.0\% + 2.338 \times 15.0\% = 39.07\%$$

$\Rightarrow$ In L/P format a positive mean is an expected *loss*, so it adds to the tail term.

Step 3 – Convert to dollars

$$0.3907 \times 30.0 = \text{USD } 11.72 \text{ million}$$

Interpretation

The 97.5% VaR is $4\% + 1.96 \times 15\% = 33.4\%$ (USD 10.02 million); ES is higher, as it must be.

Why the other options are wrong

A is incorrect. Subtracted the expected loss as if it were a return: $(2.338 \times 15\% - 4\%) \times 30 = 9.32$.
B is incorrect. Computed the 97.5% VaR: $(4\% + 1.96 \times 15\%) \times 30 = 10.02$.
C is incorrect. Ignored the mean: $2.338 \times 15\% \times 30 = 10.52$.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. • **Reference:** Hull, Risk Management and Financial Institutions, ES of a normal; Dowd, Ch.3, 1.4

Solution MR-1 c.6 — Answer: B

Bank Q50 • BT P2.T5.22.1.3

Step 1 – Loss distribution

Number of defaults $\sim$ Binomial(3, 0.10): $P(0) = 72.9\%$, $P(1) = 24.3\%$, $P(2) = 2.7\%$, $P(3) = 0.1\%$, with losses \$0, \$100, \$200, \$300.

Step 2 – Fill exactly 5% with the worst outcomes

0.1% at \$300, 2.7% at \$200 (2.8% so far), and the remaining 2.2% from the \$100 outcome.

Step 3 – Average over the 5%

$$\text{ES}_{95} = \frac{0.001(300) + 0.027(200) + 0.022(100)}{0.05} = \frac{0.3 + 5.4 + 2.2}{0.05} = \$158.00$$

$\Rightarrow$ The 95% VaR is \$100 (the outcome that straddles the 5% line); ES is \$158.

Interpretation

Discrete distributions rarely put an outcome exactly on the boundary, so ES splits the straddling outcome. The tail always contains exactly $1 - \alpha$ of probability.

Why the other options are wrong

A is incorrect. \$100 is the 95% VaR, not the tail average.
C is incorrect. \$300 is the worst outcome, which has only 0.1% probability.
D is incorrect. ES is well defined once the \$100 outcome is split; the tempting $E(L \mid L > \text{VaR}) = \$203.57$ conditions on 2.8% instead of 5%.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.4

Solution MR-1 c.7 **Answer: B**

Bank Q42 • BT L2.T5.70.2

Step 1 – What the 5% tail contains

The loss event has probability exactly 5%, so the worst 5% of outcomes *is* the loss event: every tail outcome is a draw from the severity distribution.

Step 2 – Average over the tail

$$\text{ES}_{95} = E[\text{severity}] = \text{USD } 4.00 \text{ million}$$

$\Rightarrow$ No second tail cut is applied inside the severity distribution.

Interpretation

The question tests whether you condition on the tail correctly. The unconditional expected loss averages over the whole distribution; ES averages over the tail only, and here the tail is the event itself.

Why the other options are wrong

A is incorrect. Unconditional expected loss, $5\% \times 4 = 0.20$; ES conditions on being in the tail.
C is incorrect. Applied the 95% VaR inside the severity distribution: $4 + 1.645 \times 2 = 7.29$.
D is incorrect. Applied a 95% ES inside the severity distribution: $4 + 2.063 \times 2 = 8.13$, a second tail cut.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.4

MR-1 • Estimating Market Risk Measures: An Introduction and Overview • Vardeez: VERY IMPORTANT

MR-1 e

P2 BOTH

Evaluate estimators of risk measures by estimating their standard errors.

Workbook: No workbook section • Vardeez focus • see A13 and B2 revision notes

WHAT IS TESTED HERE

Why P2. Vardeez flags this LO inside a chapter he rates Very Important. No workbook shape is reported.

Vardeez: interpretation of standard errors.

What the questions look like: mostly interpretation: what goes into the standard error, which way it moves with n, the bin width and the tail, and what a confidence interval around VaR means. One plug-in calculation of a confidence interval is possible (Dowd's Example 1.6 template). Four questions below are new, written from GARP's text, because the bank has only four on this LO.

1. Why a VaR estimate needs a standard error

A VaR or ES figure is an *estimate* from a finite sample. The standard error measures its **precision**: how much the estimate would move if we drew another sample. An estimator with a large standard error (a wide confidence interval) has limited practical value. Dowd's advice: get into the habit of estimating the standard error *whenever* you estimate the risk measure itself.

2. The standard error of a quantile

For a quantile q (VaR) estimated from n observations, with left-tail probability p:

$$\boxed{\text{se}(q) = \frac{\sqrt{p(1-p)/n}}{f(q)}} \qquad \text{90\% CI:} \quad \left[\, q - 1.645\,\text{se}(q),\;\; q + 1.645\,\text{se}(q) \,\right]$$

- $f(q)$ is the **probability mass in a bin of width h around** q: put q in the bin $[q - h/2,\, q + h/2]$, then $f(q) = 1 - P(\text{loss} > q + h/2) - P(\text{loss} < q - h/2)$.
- p is the probability of a loss beyond the top of that bin.
- The only inputs are p, n and $f(q)$. Return volatility and ES are not inputs.

Dowd's Example 1.6 (95% VaR of a standard normal, $q = 1.645$, $n = 1{,}000$):

Bin width h	p	$f(q)$	se(q)	90% CI
0.1: bin [1.595, 1.695]	0.0450	0.0104	0.630	[0.608, 2.682]
0.2: bin [1.545, 1.745]	0.0405	0.0207	0.301	[1.150, 2.140]

Both intervals are wide: even with 1,000 observations the 95% VaR is not pinned down well. Doubling h roughly doubles $f(q)$ and halves the standard error, but h is *arbitrary*, so the precision you report partly depends on a choice you made.

3. What moves the standard error

Change	Effect on se	Why
Larger sample n	Falls	se $\propto 1/\sqrt{n}$
Wider bin h	Falls	More mass $f(q)$ in the bin (and a slightly smaller p)
More extreme quantile (95% → 99%)	**Rises**	Moving into the tail, $p(1-p)$ falls but $f(q)$ falls much faster: data are sparse in the tail

Correction to the source notes: the tail direction. Vardeez lists "reduce tail probability $p \Rightarrow$ SE down", and Schweser's comparative statics say that a higher p raises the standard error. Both read only the numerator $p(1-p)$ while holding $f(q)$ fixed. GARP's own text (Dowd) is explicit: "the standard error rises as the probabilities become more extreme and we move further into the tail; hence, the more extreme the quantile, the less precise its estimator." With $n = 1{,}000$ and $h = 0.2$, se is 0.30 at the 95% quantile and 0.51 at the 99% quantile. In the exam, follow GARP: **more extreme quantile, larger standard error**. Only if a stem explicitly holds $f(q)$ constant does the formula-only reading (p up, se up) apply.

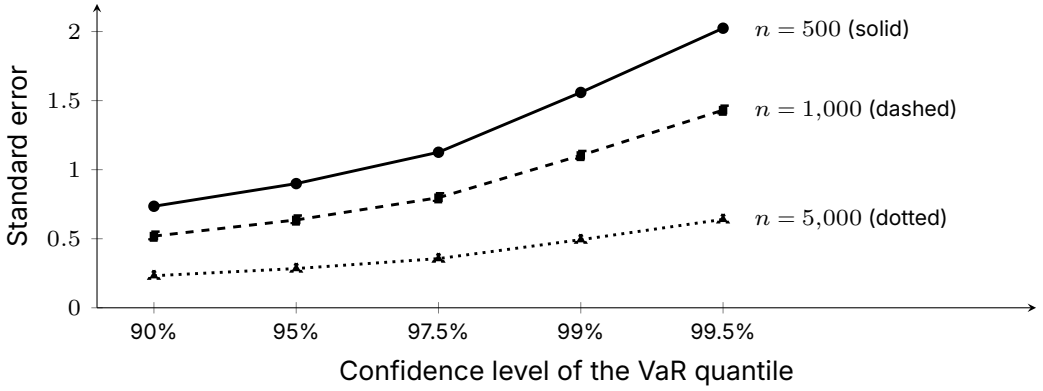


Figure 2. Quantile standard errors for a standard normal with bin width $h = 0.1$. Every curve rises as the quantile moves into the tail, and the whole curve drops as n grows.

4. Limits of the approach, and ES versus VaR

- The formula relies on asymptotic theory, so it needs large samples; it depends on the arbitrary bin width; and it gives **symmetric** intervals, which mislead for extreme quantiles whose true intervals are asymmetric (sparser data in the far tail). Order statistics and the bootstrap (MR-2) give better intervals.
- **Bootstrapping** estimates standard errors by resampling the observed data with replacement. No parametric distribution is needed.
- **ES versus VaR precision.** Yamai and Yoshiba: comparable standard errors for near-normal distributions, but ES estimators have much bigger standard errors when tails are heavy ("prone to the effects of large but infrequent losses"). Acerbi: tail heaviness raises ES standard errors relative to VaR, but in empirically realistic ranges the two are roughly comparable. Dowd's conclusion: estimate the standard error every time.

Formula card: MR-1 e

Standard error	$\text{se}(q) = \sqrt{p(1-p)/n} \,/\, f(q)$
Bin	$q \in [q-h/2,\ q+h/2];\ f(q) = 1 - P(L > q+h/2) - P(L < q-h/2)$
90% interval	$q \pm 1.645\,\text{se}(q)$ (95%: ± 1.96)
Direction	$n \uparrow$: se $\downarrow$ $h \uparrow$: se $\downarrow$ quantile more extreme: se $\uparrow$

Exam traps

- **Tail direction.** The more extreme the quantile, the *less* precise its estimator (GARP). Do not read the $p(1-p)$ term alone.
- **Bin width direction.** A wider bin *narrows* the interval, because $f(q)$ rises.
- **$f(q)$ is a probability mass, not the density value.** Compute it from the two tail probabilities around the bin.
- **Variance versus standard error.** Forgetting the square root gives an absurdly narrow interval.
- **Interval multiplier.** A 90% two-sided interval uses 1.645, not the VaR's 2.326.
- **Bootstrap needs no distribution.** It resamples the empirical data.

Practice questions: MR-1 e

Question MR-1 e.1

Bank Q72 • AnalystPrep Q.1481 (rebuilt)

An analyst constructs a confidence interval around a VaR estimate (a quantile of the loss distribution) using the standard error of the quantile estimator. Which set of inputs determines this standard error?

A. The left-tail probability p, the sample size n, and the probability mass $f(q)$ around the quantile

B. The sample size n and the volatility of returns only

C. The left-tail probability p and the probability mass $f(q)$ only, independent of sample size

D. The sample size n, the probability mass $f(q)$, and the expected shortfall at the quantile

Question MR-1 e.2

Bank Q74 • BT P2.T5.22.2.3 (rebuilt)

Dowd notes that the standard errors of risk-measure estimators are almost as important as the estimates themselves, and that bootstrapping can be used to estimate them. Each of the following statements about standard errors of risk measures is true EXCEPT:

A. The standard error is a measure of the precision of an estimate.

B. A larger standard error implies a wider confidence interval around the risk-measure estimate.

C. The standard error decreases as the sample size increases.

D. The standard error of VaR can be bootstrapped, but this requires a parametric distribution to be assumed.

Question MR-1 e.3 Bank Q73 • BT L2.T5.73.1 (rebuilt)

Each of the following is TRUE of the standard error of a quantile (VaR) estimator EXCEPT, holding all other factors constant:

A. The standard error falls as the sample size n rises.

B. The standard error rises as the bin width h around the quantile is widened.

C. The standard error rises as the left-tail probability p rises toward 0.5.

D. The standard error depends on the probability mass $f(q)$ around the quantile.

Question MR-1 e.4 New • built from GARP Ch.1 (Dowd 1.5, Example 1.6)

A market risk analyst estimates the 99% VaR of a standardised P/L series at $q = 2.326$ from $n = 1{,}000$ observations and wants a 90% confidence interval using the quantile standard error with a bin width of 0.2. For the fitted distribution, the probability of a loss greater than 2.426 is 0.0076 and the probability of a profit or a loss smaller than 2.226 is 0.9870. Which of the following is closest to the 90% confidence interval for the VaR?

A. $[1.14, 3.51]$

B. $[1.37, 3.28]$

C. $[1.49, 3.16]$

D. $[2.32, 2.33]$

Question MR-1 e.5 New • built from GARP Ch.1 (Dowd 1.5)

A risk manager re-runs Dowd's confidence interval for a 95% VaR estimated from 1,000 observations, widening the bin around the quantile from 0.1 to 0.2. The interval moves from about $[0.61, 2.68]$ to about $[1.15, 2.14]$. Which of the following statements is correct?

A. The interval narrows because the probability mass in the wider bin roughly doubles; since the bin width is arbitrary, the reported precision partly reflects a modelling choice.

B. The interval narrows because the left-tail probability p falls from 0.045 to 0.0405, and p is the only input that changes when the bin is widened.

C. The interval narrows, and because the quantile standard error gives symmetric intervals it is the most reliable approach for extreme quantiles such as the 99.9% VaR.

D. The interval should have widened, because a wider bin admits more observations and therefore more sampling noise around the quantile.

Question MR-1 e.6 New • built from GARP Ch.1 (Dowd 1.5)

A validation team compares the precision of several risk-measure estimators computed from the same 1,000-day history. Which of the following conclusions is consistent with GARP's reading on standard errors?

A. The 99% VaR estimate is less precise than the 95% VaR estimate, because the probability mass around the quantile falls faster than $p(1-p)$ as the quantile moves into the tail.

B. The 99% VaR estimate is more precise than the 95% VaR estimate, because $p(1-p)$ is smaller at $p = 0.01$ than at $p = 0.05$.

C. ES estimators always have smaller standard errors than VaR estimators, because averaging tail quantiles removes the effect of large but infrequent losses.

D. Once the sample exceeds 1,000 observations, the standard error no longer needs to be reported alongside the risk measure.

Solutions • MR-1 e

Solution MR-1 e.1

Answer: A

Bank Q72 • AnalystPrep Q.1481

Step 1 – Write the formula

$$\text{se}(q) = \frac{\sqrt{p(1-p)/n}}{f(q)}$$

Step 2 – Read off the inputs

Tail probability p, sample size n, and probability mass $f(q)$ in the bin around q. Nothing else enters.

Interpretation

Return volatility affects the quantile itself, not the sampling precision in this formula; ES is a different risk measure.

Why the other options are wrong

B is incorrect. Omits p and $f(q)$; volatility is not an input.
C is incorrect. Drops n, yet se shrinks with $\sqrt{n}$.
D is incorrect. Adds ES, which is not an input, and omits p.

LO: MR-1 e: Evaluate estimators of risk measures by estimating their standard errors. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.5, eq. 1.26

Solution MR-1 e.2

Answer: D

Bank Q74 • BT P2.T5.22.2.3

Step 1 – Check A to C

A: the standard error measures precision. True. B: the interval is the estimate $\pm$ a multiple of se, so a larger se means a wider interval. True. C: se $\propto 1/\sqrt{n}$. True.

Step 2 – Check D

Bootstrapping resamples the observed data with replacement. It needs no parametric distribution.
$\Rightarrow$ D is the false statement.

Interpretation

The bootstrap is the non-parametric route to standard errors and confidence intervals, and generally better than the asymptotic quantile formula.

Why the other options are wrong

A is incorrect. True: precision is exactly what the standard error describes.
B is incorrect. True: interval width is proportional to the standard error.
C is incorrect. True: more data, smaller standard error.

LO: MR-1 e: Evaluate estimators of risk measures by estimating their standard errors. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.5; Ch.4, 2.2

Solution MR-1 e.3 — Answer: B

Bank Q73 • BT L2.T5.73.1

Step 1 – Hold everything else fixed and read the formula

$\text{se}(q) = \sqrt{p(1-p)/n} \,/\, f(q)$.

Step 2 – Test each statement

A: se $\propto 1/\sqrt{n}$, so it falls as n rises. True. B: a wider bin *raises* $f(q)$ (and slightly lowers p), so se *falls*. False. C: with $f(q)$ held constant, $p(1-p)$ rises as p moves toward 0.5, so se rises. True under the stem's "all else constant". D: $f(q)$ is in the denominator. True.

$\Rightarrow$ B is the exception.

Interpretation

C is true only because the stem holds $f(q)$ fixed. In a real distribution, moving the quantile deeper into the tail also shrinks $f(q)$, and GARP's conclusion is that the more extreme quantile is *less* precise (see the correction box).

Why the other options are wrong

A is incorrect. True statement: larger samples are more precise.

C is incorrect. True statement under "holding all other factors constant": $p(1-p)$ peaks at $p = 0.5$.

D is incorrect. True statement: $f(q)$ is the denominator.

LO: MR-1 e: Evaluate estimators of risk measures by estimating their standard errors. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.5, eq. 1.26 and Example 1.6

Solution MR-1 e.4 — Answer: C

New • built from GARP Ch.1, Example 1.6

Step 1 – Probability mass in the bin

$$f(q) = 1 - 0.0076 - 0.9870 = 0.0054$$

$\Rightarrow$ The bin is $2.326 \pm 0.1 = [2.226,\ 2.426]$, and $p = 0.0076$ is the probability beyond its top.

Step 2 – Standard error

$$\text{se}(q) = \frac{\sqrt{0.0076 \times 0.9924/1000}}{0.0054} = \frac{0.002747}{0.0054} = 0.5086$$

Step 3 – 90% interval

$$2.326 \pm 1.645 \times 0.5086 = 2.326 \pm 0.837 = [1.49,\ 3.16]$$

$\Rightarrow$ Even with 1,000 observations, the 99% VaR could plausibly lie anywhere from about 1.5 to 3.2.

Interpretation

Compare Dowd's 95% interval with the same bin width, [1.15, 2.14] (se 0.30): moving out to 99% makes the estimate markedly less precise (se 0.51).

Why the other options are wrong

A is incorrect. Used the VaR multiplier 2.326 instead of 1.645 for a 90% interval: $2.326 \pm 2.326 \times 0.5086 = [1.14,\ 3.51]$.

B is incorrect. Used $p = 0.01$ (the VaR's tail probability) instead of the 0.0076 beyond the bin: se $= 0.5827$, $[1.37,\ 3.28]$.

D is incorrect. Forgot the square root, using the variance 0.0014 as if it were the standard error: $[2.324,\ 2.328]$.

LO: MR-1 e: Evaluate estimators of risk measures by estimating their standard errors. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.5, eqs. 1.26 and 1.27, Example 1.6

Solution MR-1 e.5 — Answer: A

New • built from GARP Ch.1, Example 1.6

Step 1 – What changes when h doubles

$f(q)$ goes from 0.0104 to 0.0207 (it roughly doubles), and p falls slightly from 0.045 to 0.0405.

Step 2 – Effect on the standard error

se falls from 0.630 to 0.301, driven mainly by the larger $f(q)$ in the denominator.
$\Rightarrow$ The interval narrows. But h is "essentially arbitrary", so the narrower interval is partly a product of the analyst's choice.

Interpretation

Dowd's own comment: even with $h = 0.2$ ("probably as wide as we can reasonably go") the interval is still rather wide.

Why the other options are wrong

B is incorrect. Both p and $f(q)$ change, and the $f(q)$ effect dominates; the fall in p alone would barely move the interval.
C is incorrect. True fact attached to the wrong conclusion: the intervals are symmetric, which is a *weakness* for extreme quantiles whose true intervals are asymmetric.
D is incorrect. Polarity flip: a wider bin raises $f(q)$ and narrows the interval.

LO: MR-1 e: Evaluate estimators of risk measures by estimating their standard errors. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.5, Example 1.6

Solution MR-1 e.6 — Answer: A

New • built from GARP Ch.1, Section 1.5

Step 1 – Tail direction

Moving from the 95% to the 99% quantile lowers $p(1-p)$ but lowers $f(q)$ much more, because observations are sparse in the tail. With $n = 1{,}000$ and $h = 0.2$: se 0.30 at 95%, 0.51 at 99%.
$\Rightarrow$ GARP: "the more extreme the quantile, the less precise its estimator".

Step 2 – ES versus VaR

Yamai and Yoshiba found ES standard errors much *bigger* than VaR's for heavy-tailed distributions; Acerbi found them roughly comparable in realistic ranges. Neither says ES is always more precise.

Interpretation

Dowd's practical advice is to report the standard error every time, whatever the sample size.

Why the other options are wrong

B is incorrect. Reads only the numerator $p(1-p)$; the shrinking $f(q)$ dominates.
C is incorrect. Polarity flip: heavy tails make ES estimators *more* prone to large infrequent losses.
D is incorrect. Absolute claim: there is no sample size beyond which the standard error stops mattering.

LO: MR-1 e: Evaluate estimators of risk measures by estimating their standard errors. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.5, Figure 1.9; Standard Errors in Estimators of Coherent Risk Measures

MR-1 • Estimating Market Risk Measures: An Introduction and Overview • Vardeez: VERY IMPORTANT

MR-1 f

P2 CONCEPT

Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution.

Workbook: No workbook section • Vardeez focus • see A13 revision notes

WHAT IS TESTED HERE

Why P2. Vardeez flags this LO inside a chapter he rates Very Important. No workbook shape is reported.

Vardeez: interpretation of QQ plots.

What the questions look like: a verbal or tabulated description of a QQ plot (where the points bend, which way, whether symmetric), and you name the distributional feature: normal fit, fat tails, skew, a pure location or scale difference, or an outlier. Plus a few "what is a QQ plot for" statements.

1. What a QQ plot is

Plot the quantiles of the **empirical** distribution (vertical axis) against the matching quantiles of a **reference** distribution, usually the normal (horizontal axis), at regular confidence levels. It is a *visual* check of fit: there is no test statistic, no p-value and no minimum sample size.

- **Linear plot:** the reference distribution fits. It also fits if the line is straight but has a different intercept or slope: that is a *linear transformation* (different location and scale), and the intercept and slope give a rough idea of the location and scale parameters.
- **Middle matches, tails do not:** the empirical distribution is symmetric in its body but its tails differ from the reference.

2. Reading the shapes

What the plot does	What it means
On the 45° line throughout	The empirical data follow the reference (e.g. normal)
Straight, but slope $\neq 1$ or shifted	Same shape, different scale or location (linear transformation)
Steeper than 1 in *both* tails, symmetric (S-shape)	Symmetric **fat tails**: leptokurtic, kurtosis above 3 (e.g. Student-t data against a normal)
Flatter than 1 in both tails	Thinner tails than the reference
Lower tail far below the line, upper tail below the line too	Long left tail, short right tail: **negative skew**
Upper tail far above the line, lower tail above the line	Long right tail: **positive skew**
One isolated point far from an otherwise good line	An **outlier**

Rule for the tails, with empirical quantiles on the vertical axis: in the lower tail, points *below* the line mean the empirical losses are more extreme (heavier left tail); in the upper tail, points *above* the line mean the gains are more extreme (heavier right tail). Steeper slope in a tail means heavier tail.

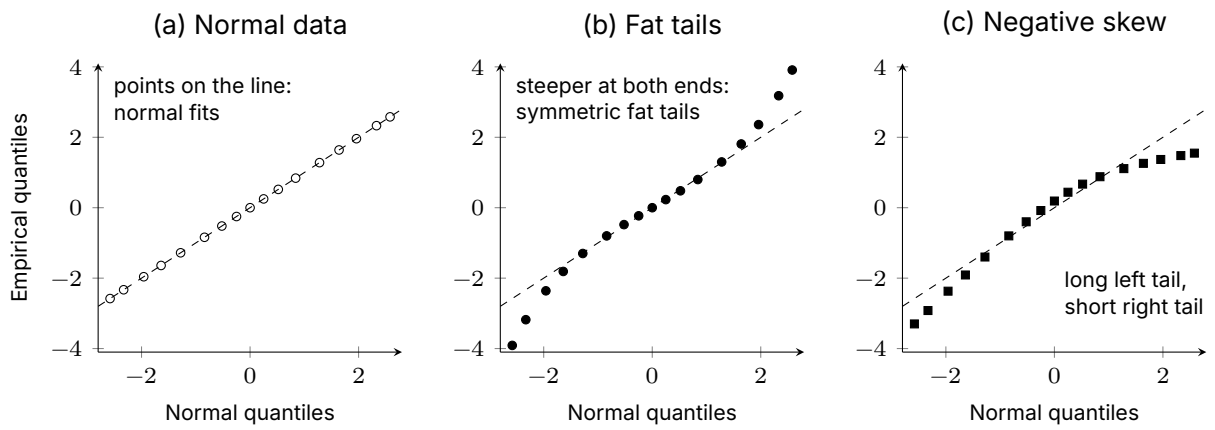


Figure 3. QQ plots against a normal reference (dashed 45° line). (a) Normal sample: points on the line. (b) Student-t type sample: the centre follows the line, both tails bend away and steepen. (c) Negatively skewed sample: the lower tail falls below the line, the upper tail flattens below it.

QQ plot is	QQ plot is not
A visual comparison of empirical and reference quantiles	A formal test that accepts or rejects at a confidence level
A way to spot fat tails, skew and outliers	A way to remove or smooth outliers
Usable with any reference distribution	A measure of an estimator's precision (that is the standard error, MR-1 e)
Informative on location and scale via intercept and slope	A direct estimate of skewness and kurtosis statistics

Exam traps

- **Scale versus tails.** A straight line with slope above 1 is a larger variance (same shape); only a bend in the tails means fat tails.
- **Symmetric bend versus one-sided bend.** Both tails bending away symmetrically is kurtosis; one tail heavy and the other light is skew.
- **Direction in each tail.** Lower tail below the line is a heavy left tail; upper tail *below* the line is a light right tail.
- **Purpose.** Visual fit diagnostic only: no p-value, no standard errors, no outlier removal.
- **QQ plots do reveal skew**; a statement that they show only kurtosis is false.

Practice questions: MR-1 f

Question MR-1 f.1

Bank Q83 • Schweser Module Quiz 1.2 Q3 (key corrected)

Which of the following best describes the main purpose of a quantile-quantile (QQ) plot?

A. To perform a formal statistical test that rejects or accepts a reference distribution at a given confidence level.

B. To visually compare the quantiles of an empirical sample with those of a reference distribution to judge whether the reference is a plausible fit.

C. To estimate the standard error of a quantile-based risk measure.

D. To compute VaR directly from the empirical distribution without assuming a model.

Question MR-1 f.2

Bank Q82 • GARP PE2 Q43 • Mock 4 Q44

A risk analyst is studying a series of graphs plotting the empirical distribution of a portfolio's profit and loss quantiles against the quantiles of different specified reference distributions. The analyst plans to use these QQ plots to gain insights into the properties of the empirical distribution. Which of the following statements is correct regarding QQ plots?

A. The tails of the empirical distribution are heavier than those of the reference distribution if the QQ plot has a steeper slope at its tails.

B. QQ plots are used to smooth the empirical distribution by removing outliers through the use of quantiles.

C. A distribution is a good reference distribution if data drawn from it produce a QQ plot that is non-linear in the central mass of the distribution.

D. The primary purpose of a QQ plot is to measure the skewness and kurtosis of the empirical data.

Question MR-1 f.3

Bank Q76 • AP Mock 2 Q12 (rebuilt)

Richard Wayne, FRM, plots the empirical quantiles of a return sample (vertical axis) against the quantiles of a standard normal distribution (horizontal axis). The points lie close to the 45-degree reference line through the central portion of the distribution, but at both ends they curve away from the line: below it in the lower tail and above it in the upper tail, forming an S-shape. Based on this QQ plot, which of the following is most likely correct?

A. The returns are normally distributed.

B. The returns have fatter tails than the normal distribution, with a roughly symmetric center.

C. The returns are negatively skewed relative to the normal distribution.

D. The returns are normal, but with a larger standard deviation than the sample estimate.

Question MR-1 f.4 Bank Q78 • BT L2.T5.74.1 (rebuilt)

An analyst plots the quantiles of an empirical return sample (vertical axis) against standard normal quantiles (horizontal axis). The plot is approximately a straight line with slope near 1 through the central region, but the slope becomes noticeably steeper than 1 in both the left and right tails, and the plot is symmetric about the origin. Which statement is most likely true?

A. The empirical distribution is normal with a larger variance than the reference.

B. The empirical distribution has positive skewness.

C. The empirical distribution is leptokurtic (kurtosis greater than 3) with symmetric, heavy tails.

D. A linear transformation of location and scale would make the empirical distribution approximately normal.

Question MR-1 f.5 New • built from GARP Ch.1 (Dowd 1.6, QQ plots)

A risk analyst plots the quantiles of a daily P/L series in USD thousands (vertical axis) against standard normal quantiles (horizontal axis). The points lie very close to a straight line along the whole range, including both tails; the line has an intercept of about 5 and a slope of about 40. Which of the following is the most appropriate conclusion?

A. The P/L is not normal, because the plot does not follow the 45-degree line.

B. The P/L has fat tails, because its quantiles are about 40 times larger than the normal quantiles.

C. The P/L is consistent with a normal distribution with a mean of about USD 5,000 and a standard deviation of about USD 40,000.

D. The P/L is positively skewed, because the intercept of the line is above zero.

Question MR-1 f.6 Bank Q80 • BT P2.T5.709.3 (rebuilt)

Peter, a risk analyst, wants to know whether a dataset is approximately normal. He plots empirical (actual) quantiles on the vertical axis against standard normal (predicted) quantiles on the horizontal axis and records these points:

Predicted quantile	−3.0	−2.0	0.0	2.0	3.0
Actual quantile	−6.0	−3.5	0.0	1.0	1.5

Based on the QQ plot, which statement is most likely true about this distribution?

A. The distribution is approximately normal.

B. It has light tails (left and right) with positive skew.

C. It has a very heavy left tail, a light right tail, and negative (left) skew.

D. It has a very heavy right tail, a light left tail, and positive (right) skew.

Question MR-1 f.7 Bank Q79 • BT L2.T5.74.2

Each of the following statements about the QQ plot is true EXCEPT:

A. Any distribution that produces a linear QQ plot is a good candidate distribution for the data.

B. The QQ plot is ideal for evaluating kurtosis but cannot effectively reveal skew.

C. The intercept and slope of a linear QQ plot give a rough idea of the location and scale parameters of the sample data.

D. A QQ plot can identify outliers even when the other observations are broadly consistent with the reference distribution.

Solutions • MR-1 f

Solution MR-1 f.1 — Answer: B

Bank Q83 • Schweser Module Quiz 1.2 Q3

Step 1 – What the plot does

It maps empirical quantiles one-to-one against the quantiles of a hypothesised distribution. If the plot is approximately linear, the reference is a plausible match; bends show where it fails.

Step 2 – What it does not do

It is visual inspection only: there is no test statistic, no standard error and no VaR output.

Interpretation

The QQ plot answers "which distribution do my data look like?", which is why it sits next to parametric VaR in Dowd's chapter.

Why the other options are wrong

A is incorrect. There is no formal test or confidence level in a QQ plot.
C is incorrect. Standard errors come from sampling theory or the bootstrap (MR-1 e).
D is incorrect. Computing VaR from the empirical distribution is historical simulation, not the purpose of the plot.

LO: MR-1 f: Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.6 The QQ Plot

Solution MR-1 f.2 — Answer: A

Bank Q82 • GARP PE2 Q43

Step 1 – Read a steep tail

With empirical quantiles on the vertical axis, a slope steeper than 1 in a tail means the empirical extreme quantiles lie further out than the reference's.
$\Rightarrow$ That is the definition of heavier tails.

Interpretation

This is GARP's standard framing: the shape of the plot, read tail by tail, tells you how the empirical distribution differs from the reference.

Why the other options are wrong

B is incorrect. QQ plots make outliers visible; they do not smooth data or remove them.
C is incorrect. Reversed: a good reference gives an approximately *linear* plot, including the central mass.
D is incorrect. The QQ plot is a visual fit diagnostic; skewness and kurtosis are measured with moment statistics.

LO: MR-1 f: Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.6

Solution MR-1 f.3 — Answer: B

Bank Q76 • AP Mock 2 Q12

Step 1 – The centre

Points close to the line in the middle: the body of the distribution resembles the normal.

Step 2 – The tails

Below the line in the lower tail: losses more extreme than normal. Above the line in the upper

tail: gains more extreme than normal. The departures are on both sides.
⇒ Symmetric fat tails, as in Figure 3(b).

Interpretation

An S-shape with a matching centre is the textbook signature of a leptokurtic distribution such as the Student-t.

Why the other options are wrong

A is incorrect. A normal sample would also follow the line in the tails.
C is incorrect. Skew would show as an asymmetric departure (one tail heavy, the other light), not departures on both sides.
D is incorrect. A larger standard deviation alone gives a straight line with a different slope everywhere, not a straight centre with bending tails.

LO: MR-1 f: Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.6

Solution MR-1 f.4 **Answer: C**

Bank Q78 • BT L2.T5.74.1

Step 1 – Centre linear, slope about 1

The body of the distribution matches the normal.

Step 2 – Steeper in both tails, symmetric

Extreme empirical quantiles lie further out than the normal's on both sides, and equally so.
⇒ Symmetric heavy tails: excess kurtosis (kurtosis above 3).

Interpretation

Steepness confined to the tails is what separates fat tails from a simple change in scale.

Why the other options are wrong

A is incorrect. A pure variance difference gives a straight line steeper than 1 everywhere, not only in the tails.
B is incorrect. Positive skew would show a heavy right tail only, not both.
D is incorrect. A location and scale change only shifts and tilts a straight line; it cannot straighten bending tails.

LO: MR-1 f: Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.6

Solution MR-1 f.5 **Answer: C**

New • built from GARP Ch.1, Section 1.6

Step 1 – A straight line means the shape fits

The points lie on a straight line throughout, including the tails, so the empirical distribution has the same shape as the normal reference.

Step 2 – Read location and scale from the line

If empirical $= a + b\times$ normal quantile, then a estimates the location and b the scale: mean about 5 and standard deviation about 40, in USD thousands.
⇒ A normal with mean about USD 5,000 and standard deviation about USD 40,000.

Interpretation

Dowd: a linear QQ plot means the reference fits "up to a linear transformation", and the intercept and slope give a rough idea of the location and scale parameters.

Why the other options are wrong

A is incorrect. A different slope or intercept does not reject normality; only bending does.
B is incorrect. Larger quantiles everywhere are a scale effect; fat tails need a bend in the tails.
D is incorrect. The intercept is the location; skew needs an asymmetric bend.

LO: MR-1 f: Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.6

Solution MR-1 f.6 — Answer: C

Bank Q80 • BT P2.T5.709.3

Step 1 – Left tail

At -3.0 the actual quantile is -6.0, far below the line: losses much more extreme than normal. A very heavy left tail.

Step 2 – Right tail

At 2.0 the actual is 1.0, at 3.0 it is 1.5: below the line, so gains are *less* extreme than normal. A light right tail.
$\Rightarrow$ Long left tail, short right tail: negative skew, as in Figure 3(c).

Interpretation

Read each tail separately: below the line is "heavier" on the left but "lighter" on the right.

Why the other options are wrong

A is incorrect. The points depart sharply from the line in both tails.
B is incorrect. The left tail is not light; the skew is negative.
D is incorrect. Reverses the reading: points below the line on the right mean a lighter right tail.

LO: MR-1 f: Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.6

Solution MR-1 f.7 — Answer: B

Bank Q79 • BT L2.T5.74.2

Step 1 – Check A, C, D

A linear plot makes the reference a good candidate (A); the intercept and slope approximate location and scale (C); an outlier stands out as an isolated point (D). All true.

Step 2 – Check B

Skew shows up as asymmetric tail behaviour (one tail bending away more than the other), so QQ plots do reveal skew.
$\Rightarrow$ B is false.

Interpretation

QQ plots reveal both kurtosis (symmetric bends) and skew (asymmetric bends).

Why the other options are wrong

A is incorrect. True statement: linearity indicates a good reference.
C is incorrect. True statement: intercept and slope map to location and scale.
D is incorrect. True statement: outliers stand out from the line.

LO: MR-1 f: Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution. • **Reference:** Dowd, Measuring Market Risk 2e, Ch.3, 1.6

MR-10 • Financial Correlation Modeling – Bottom-Up Approaches • Vardeez: NOT SO IMP

MR-10 b

P2 BOTH

Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets.

Workbook: B3 Default Correlation, Joint PD and Copulas • Tier B

WHAT IS TESTED HERE

Why P2. A Tier B workbook topic (B3, Default Correlation, Joint PD and Copulas) hits this LO. **Vardeez:** not likely to be tested.

Reported exam shape

- **B3·S4** Gaussian copula and implied correlation.

What the questions look like: mostly descriptive (what is mapped to what, what is preserved, which formula applies to two assets, strengths and weaknesses). A calculation can only be a joint default probability where the bivariate normal value is supplied or read from software; the skill tested is setting it up correctly (cumulative PDs, the right horizons, N^{-1} first, correlation last).

LO lettering. Vardeez's notes label this objective "MR-10 c" and the default-time procedure "MR-10 b". The official 2026 wording in the Master LO Map has the Gaussian copula and joint PD as **MR-10 b** (and the correlated default time as MR-10 c, a P4 LO). This book follows the map.

1. What a copula does

A copula joins several univariate (marginal) distributions into one multivariate distribution, **while preserving each marginal**. It does so by mapping every marginal onto a known, well-behaved distribution and defining the correlation there. Sklar (1959) showed that any joint distribution can be written this way. Vasicek brought one-factor copulas to finance (1987); Li introduced the multivariate default-time copula (2000). Copulas became the standard tool for CDOs, then fell out of favour in the 2007–2009 crisis.

2. The Gaussian copula

1. Take each marginal cumulative distribution $G_i(u_i)$ (any shape at all).
2. Map it **percentile to percentile** onto the univariate standard normal: $x_i = N^{-1}(G_i(u_i))$. The 5th percentile of G_i goes to the 5th percentile of $N(0,1)$, which is -1.645.
3. Join the mapped variables with the **multivariate standard normal** M_n and its correlation matrix ρ_M.

$$C_G[G_1(u_1), \dots, G_n(u_n)] = M_n[N^{-1}(G_1(u_1)), \dots, N^{-1}(G_n(u_n));\ \rho_M]$$

For default risk the marginals are the cumulative default probability curves $Q_i(t)$, which gives the **Gaussian default-time copula**:

$$C_{GD}[Q_1(t), \dots, Q_n(t)] = M_n[N^{-1}(Q_1(t)), \dots, N^{-1}(Q_n(t));\ \rho_M]$$

Key property: the original marginals are preserved; only the transformed variables x_i are (multivariate) normal. The times to default themselves are not normal. The standard normal target has mean 0 and standard deviation 1. With two assets, M_2 needs a single correlation ρ.

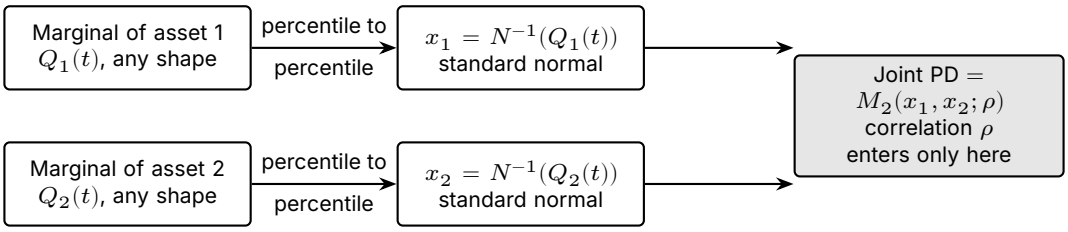


Figure 1. The three moves of a Gaussian copula for two assets. Each marginal is mapped percentile to percentile onto a standard normal (the marginal itself is untouched), and the correlation is imposed only between the two normals (shaded box).

3. Joint probability of default of two assets

Meissner's example: companies B and Caa, Gaussian default correlation 0.4.

Year t	$Q_B(t)$ cumulative	$N^{-1}(Q_B(t))$	$Q_{Caa}(t)$ cumulative	$N^{-1}(Q_{Caa}(t))$
1	6.51%	-1.5133	23.83%	-0.7118
2	14.16%	-1.0732	37.12%	-0.3287
3	21.03%	-0.8054	47.43%	-0.0645
4	27.04%	-0.6116	55.05%	0.1269
5	32.31%	-0.4590	60.09%	0.2557

- **Both default within one year:** $M_2(-1.5133,\ -0.7118;\ 0.4) = 3.44\%$, against $6.51\% \times 23.83\% = 1.55\%$ under independence.
- **B by year 3 and Caa by year 5:** $M_2(-0.8054,\ 0.2557;\ 0.4) = 16.93\%$. The horizons do not have to match.

Two things to watch: the inputs are **cumulative** default probabilities, so the answer is "both default *by* those dates", not "in those particular years"; and the copula correlation is *not* a default correlation. Plugging the copula ρ into the linear default-correlation formula

$$\pi_{12} = \pi_1\pi_2 + \rho\sqrt{\pi_1(1-\pi_1)\,\pi_2(1-\pi_2)}$$

is a classic distractor.

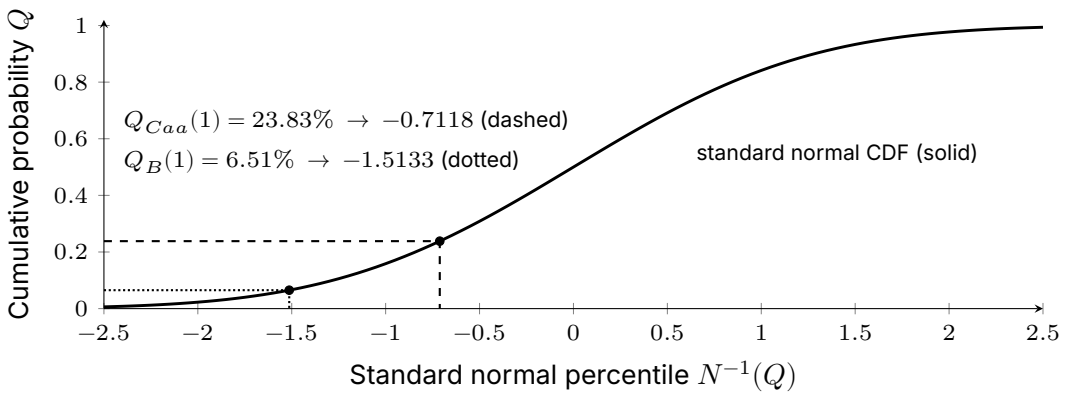


Figure 2. The percentile mapping for year 1: enter on the vertical axis with each cumulative default probability, meet the standard normal CDF, drop to the horizontal axis. The two percentiles then go into the bivariate normal with $\rho = 0.4$.

4. Strengths and weaknesses

Why it was popular	Why it is criticised
Joins any marginals while preserving them	**Zero (low) tail dependence**: joint extremes are understated, yet dependence rises in a crisis
One multivariate function handles 100+ assets (CDOs)	Hard to calibrate: one correlation cannot fit all CDO tranches (correlation skew)
Works for any horizon: joint PD by any dates	Static: no stochastic process for default intensity or correlation
Accepts a full correlation matrix ρ_M	Fell into disgrace in 2007–2009; still used, with caution

Implied correlation (B3·S4). When the copula correlation is calibrated so that model tranche values match market prices, it is an *implied* correlation: risk-neutral and market-based, not a historical estimate. In the one-factor version, once the common factor is realised the defaults are independent (conditional independence).

Formula card: MR-10 b

Mapping	$x_i = N^{-1}(Q_i(t))$, percentile to percentile
Two assets	joint PD $= M_2[N^{-1}(Q_1(t_1)),\ N^{-1}(Q_2(t_2));\ \rho]$
n assets	$M_n[\ldots;\ \rho_M]$ with an $n \times n$ correlation matrix
Bounds	$Q_1 Q_2 \leq$ joint PD $\leq \min(Q_1, Q_2)$ for $\rho \in [0, 1]$

Exam traps

- **Mean of the target.** Standard normal: mean 0, standard deviation 1 (not "mean and standard deviation of one").
- **What is normal.** Only the mapped x_i; the marginals (and default times) keep their own shapes.
- **Cumulative, not annual, PDs.** Use $Q(t)$, the probability of default *by* t.
- **Horizon per asset.** Each asset is mapped at its own horizon.
- **Copula ρ is not default correlation.** The linear formula with ρ gives a different (wrong) number.
- **M_2, not M_1 or M_n.** Two assets need the bivariate normal with one ρ.
- **Tail dependence.** The Gaussian copula's is low (zero asymptotically), a weakness, not a strength.

Practice questions: MR-10 b

Question MR-10 b.1

Bank Q515 • Mock 2 Q35

A portfolio strategist is training a junior analyst recently hired by her fund. The strategist would like the analyst's help with applying copulas, but she realizes this is not a concept that the analyst has worked with in the past. To educate the analyst on both correlation copulas and Gaussian copulas, the strategist is least likely to use which of the following descriptions?

A. A copula correlation model is designed to preserve the original marginal distributions while defining a correlation between them.

B. A Gaussian copula maps the marginal distribution of each variable to a normal distribution with a mean and standard deviation equal to one.

C. The purpose of a copula is to build a joint probability distribution between two or more variables and maintain those variables' individual marginal distributions.

D. A correlation copula will convert two or more unknown distributions that have distinct shapes and map them to a known distribution with well-defined properties.

Question MR-10 b.2

Bank Q518 • Schweser Module Quiz 10.1 Q5

Suppose a risk manager owns two non-investment-grade assets B and C and has determined their individual default probabilities for the next five years. Which of the following equations best defines how a Gaussian copula is constructed by the risk manager to estimate the joint probability of these two companies defaulting within the next year, assuming a Gaussian default correlation of 0.35?

A. $C_{GD}[Q_B(t), Q_C(t)] = M_2[N^{-1}(Q_B(t)), N^{-1}(Q_C(t)); \rho]$

B. $C[G_1(u_1), \ldots, G_n(u_n)] = F_n[F_1^{-1}(G_1(u_1)), \ldots, F_n^{-1}(G_n(u_n)); \rho_F]$

C. $C_{GD}[Q_1(t), \ldots, Q_n(t)] = M_n[N^{-1}(Q_1(t)), \ldots, N^{-1}(Q_n(t)); \rho_M]$

D. $M_n(\cdot) = Q_i(\tau_i)$

Question MR-10 b.3

Bank Q513 • BT P2.T5.505.3 (rebuilt)

The marginal one-year default probabilities of asset A and asset B are 5.0% and 7.0%. The Gaussian copula correlation is 0.30. Using a bivariate standard normal CDF (software), which value is closest to the joint one-year default probability?

A. 0.105%

B. 0.350%

C. 0.930%

D. 2.018%

Question MR-10 b.4 New • built from GARP Ch.10 (Meissner, Example 10.1) and Vardeez

A credit analyst models two borrowers, B and Caa, with a Gaussian default-time copula and a copula correlation of 0.4. The default probability term structures are:

Year	B: default prob. in year	B: cumulative	Caa: default prob. in year	Caa: cumulative
1	6.51%	6.51%	23.83%	23.83%
2	7.65%	14.16%	13.29%	37.12%
3	6.87%	21.03%	10.31%	47.43%
4	6.01%	27.04%	7.62%	55.05%
5	5.27%	32.31%	5.04%	60.09%

Software returns these values of the bivariate standard normal CDF with correlation 0.4: $M_2(-1.4855, -1.6410) = 1.20\%$; $M_2(-0.8054, -0.0645) = 14.66\%$; $M_2(-0.8054, 0.2557) = 16.93\%$; $M_2(-0.4590, 0.2557) = 24.93\%$. Which of the following is closest to the probability that B defaults by the end of year 3 and Caa defaults by the end of year 5?

A. 1.20%

B. 12.64%

C. 14.66%

D. 16.93%

Question MR-10 b.5 Bank Q514 • BT P2.T5.22.13.2

According to Meissner, the Gaussian copula is popular due to its convenient properties. In addition to convenience, what is true about the Gaussian copula?

A. The asymptotic tail dependence is high.

B. The univariate marginal distributions joined are normal.

C. It requires a single scalar (e.g., average) correlation because it cannot accept a correlation matrix.

D. It can be utilized in the default-time copula to model the joint probability of default within various (cumulative) time periods.

Question MR-10 b.6 Bank Q512 • BT P2.T5.505.2

About the Gaussian copula used in financial applications, each of the following is true EXCEPT which is not?

A. The Gaussian copula has low tail dependence, which is a weakness because dependencies (including correlations) increase in a crisis.

B. The Gaussian copula is difficult to calibrate to market prices; for example, it is difficult to calibrate CDO tranches with a single correlation model.

C. The Gaussian copula is principally static and allows only limited risk management; there is no stochastic process for the default intensity and default correlation.

D. The Gaussian copula is limited to market risk applications because it requires $n \times n$ pairwise correlation parameters, which is natural for a covariance matrix, but in credit risk there is no way to assume pairwise default correlations.

Question MR-10 b.7 Workbook B3.3 • S4, S6 • Map #16 • Map #60

A quantitative analyst at a Canadian bank is reviewing the documentation of the bank's portfolio credit model. The model joins each obligor's default-time distribution with a one-factor Gaussian copula, and its correlation parameter is calibrated so that modelled values of standard index tranches match their market prices. Which of the following statements in the documentation is correct?

A. The model assumes that the obligors' times to default are jointly normally distributed, so each obligor's marginal default-time distribution must itself be normal.

B. Conditional on a negative realisation of the common factor, an obligor with a higher factor loading has a lower conditional default probability than an otherwise identical obligor.

C. The correlation calibrated to the tranche prices is a real-world parameter, estimated from the historical default experience of the obligors in the index.

D. Once the common factor is realised, the obligors' defaults are independent, so all default dependence in the model enters through the obligors' loadings on that factor.

Solutions • MR-10 b

Solution MR-10 b.1 — Answer: B

Bank Q515 • Mock 2 Q35

Step 1 – Check each description against the definition

A copula preserves the marginals and defines a correlation (A); builds a joint distribution from marginals (C); maps unknown marginals onto a known distribution (D). All accurate.

Step 2 – Check the Gaussian target

The Gaussian copula maps to the *standard* normal: mean 0, standard deviation 1. B says mean and standard deviation both equal one.

$\Rightarrow$ B is the misdescription.

Interpretation

A one-word slip (mean "one" instead of zero) is enough to make a definition wrong; read the parameters.

Why the other options are wrong

A is incorrect. Accurate: preserving the marginals is the key property.

C is incorrect. Accurate: Sklar's construction of a joint distribution from marginals.

D is incorrect. Accurate: unknown shapes mapped to a known distribution.

LO: MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. • **Reference:** Meissner, Correlation Risk Modeling and Management, Ch.5, The Gaussian Copula

Solution MR-10 b.2 — Answer: A

Bank Q518 • Schweser Module Quiz 10.1 Q5

Step 1 – Two assets, one correlation

Two companies need the bivariate standard normal M_2 with a single correlation (0.35).

Step 2 – Map the one-year cumulative PDs

Each $Q(1)$ is mapped by N^{-1}, then M_2 is evaluated: $M_2[N^{-1}(Q_B(1)), N^{-1}(Q_C(1)); 0.35]$.

$\Rightarrow$ Option A.

Interpretation

B, C and D are all real formulas from the reading, just the wrong ones for this task.

Why the other options are wrong

B is incorrect. The generic copula with a generic distribution F, not the Gaussian default copula.

C is incorrect. The n-asset version with a correlation matrix, not the two-asset case.

D is incorrect. The default-time draw used to simulate a default time (MR-10 c), not a joint-probability equation.

LO: MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. • **Reference:** Meissner, Ch.5, The Gaussian Default Time Copula

Solution MR-10 b.3 — Answer: C

Bank Q513 • BT P2.T5.505.3

Step 1 – Map each PD

$$N^{-1}(0.05) = -1.6449, \qquad N^{-1}(0.07) = -1.4758$$

Step 2 – Evaluate the bivariate normal

$$\text{Joint PD} = M_2(-1.6449,\ -1.4758;\ 0.30) = 0.930\%$$

$\Rightarrow$ Above the independence value $0.05 \times 0.07 = 0.350\%$, as positive correlation requires, and below $\min(5\%, 7\%)$.

Interpretation

A quick sanity band: with ρ between 0 and 1 the joint PD must lie between 0.35% and 5%.

Why the other options are wrong

A is incorrect. Scaled the independence product by the correlation: $0.30 \times 0.35\% = 0.105\%$.
B is incorrect. Independence, $0.05 \times 0.07 = 0.350\%$: ignores ρ.
D is incorrect. Treated the copula correlation as a default correlation: $0.35\% + 0.30\sqrt{0.05(0.95)(0.07)(0.93)} = 2.018\%$.

LO: MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. • **Reference:** Meissner, Ch.5, Example 10.1

Solution MR-10 b.4 **Answer: D**

New • built from Meissner Example 10.1

Step 1 – Pick the cumulative PDs at each borrower's own horizon

B by year 3: $Q_B(3) = 21.03\%$. Caa by year 5: $Q_{Caa}(5) = 60.09\%$.

Step 2 – Map to standard normal percentiles

$$N^{-1}(0.2103) = -0.8054, \qquad N^{-1}(0.6009) = 0.2557$$

Step 3 – Read the bivariate normal

$$M_2(-0.8054,\ 0.2557;\ 0.4) = 16.93\%$$

$\Rightarrow$ A 16.93% chance that both have defaulted by those dates.

Interpretation

The horizons differ, and the method does not change: each borrower is mapped at its own date, and the correlation enters only in the last step. The 24.93% value in the stem is "both by year 5", a further trap.

Why the other options are wrong

A is incorrect. Used the in-year default probabilities (6.87% and 5.04%) instead of the cumulative ones: $M_2(-1.4855, -1.6410) = 1.20\%$.
B is incorrect. Ignored the copula and multiplied the marginals: $21.03\% \times 60.09\% = 12.64\%$.
C is incorrect. Used year 3 for both borrowers: $M_2(-0.8054, -0.0645) = 14.66\%$.

LO: MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. • **Reference:** Meissner, Ch.5, Example 10.1 (Question 2)

Solution MR-10 b.5 **Answer: D**

Bank Q514 • BT P2.T5.22.13.2

Step 1 – Recall the properties

The Gaussian copula maps arbitrary marginals, such as cumulative default probability curves $Q_i(t)$, onto a multivariate standard normal with correlation matrix ρ_M.

Step 2 – Horizon flexibility

Because $Q_i(t)$ exists for any t, the copula gives joint default probabilities over any cumulative period: this is the default-time copula.

$\Rightarrow$ D is correct.

Interpretation

Its tail dependence is the known weakness; its flexibility across horizons is a genuine strength.

Why the other options are wrong

A is incorrect. Polarity flip: Gaussian tail dependence is zero asymptotically (the t-copula has positive tail dependence).

B is incorrect. The marginals are arbitrary; only the transformed variables are normal.

C is incorrect. It accepts a full correlation matrix; a single average correlation is a simplification, not a requirement.

LO: MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. • **Reference:** Meissner, Ch.5, The Gaussian Copula

Solution MR-10 b.6 — Answer: D

Bank Q512 • BT P2.T5.505.2

Step 1 – Recognised weaknesses

Low tail dependence (A), difficult calibration with a single correlation across CDO tranches (B), and a static structure with no stochastic default intensity or correlation (C) are all genuine criticisms.

Step 2 – The false claim

D says the copula is limited to market risk. The Gaussian default-time copula was built precisely for credit risk.

$\Rightarrow$ D is the exception.

Interpretation

Remember both sides: designed for credit portfolios (CDOs), criticised for tail dependence and calibration.

Why the other options are wrong

A is incorrect. True: dependence rises in a crisis, which the Gaussian copula understates.

B is incorrect. True: the correlation skew across tranches is the calibration problem.

C is incorrect. True: the model is static.

LO: MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. • **Reference:** Meissner, Ch.5, The Gaussian Copula

Solution MR-10 b.7 — Answer: D

Workbook B3.3 • S4, S6

Step 1 – Conditional independence

In the single-factor model $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\epsilon_i$ (Hull: $x_i = a_i F + \sqrt{1-a_i^2}\,Z_i$) the idiosyncratic shocks are independent of one another and of the factor. Fixing m leaves only ϵ_i random, so defaults are conditionally independent and the pairwise correlation $\beta_i\beta_j$ comes entirely from the loadings.

Step 2 – Conditional default probability check

$p(m) = \Phi((k_i - \beta_i m)/\sqrt{1-\beta_i^2})$. With $\pi = 1\%$ ($k = -2.33$) and $m = -1$: $\beta = 0.4$ gives 1.78% (Malz Example 11.4); a higher β raises the numerator and shrinks the denominator, so the

conditional PD is higher.

Interpretation

D is correct. Conditional independence is what makes the factor model tractable and underlies the Vasicek large-portfolio result used for Credit VaR and IRB capital.

Why the other options are wrong

A is incorrect. Sibling swap (t_i for x_i). Hull: a company's time to default "is not even approximately normal"; the copula maps each t_i percentile-to-percentile into $x_i = N^{-1}[Q_i(t_i)]$ and assumes only the x_i are multivariate normal, so the marginals can take any shape.

B is incorrect. Polarity flip. When $m < 0$, a higher loading pushes the conditional mean $\beta_i m$ further down and the conditional PD up: cyclical firms suffer more in a downturn.

C is incorrect. Sibling swap. Implied correlation is inferred from market prices using risk-neutral default probabilities from single-name CDS spreads (Malz 22.4); it is a risk-neutral, market rather than credit risk, parameter, not a historical estimate.

LO: MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. • **Reference:** Malz, Ch.8, 8.3 Conditional Default Distributions; Hull, OFOD 11e, Ch.24, 24.2; Meissner, Ch.5, The Gaussian Copula; Malz, Ch.9, 22.4 Implied Correlation

MR-15 • The Art of Term Structure Models: Volatility and Distribution • Vardeez: VERY IMPORTANT

MR-15 d

P2 CONCEPT

Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models.

Workbook: D1 CIR and Lognormal Short-Rate Models • Tier D • Map #39

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter, and the Tier D workbook section D1 covers it (Map #39, "CIR model solving").
Vardeez: CIR model; concept of terminal distributions.

Reported exam shapes

- **D1·S3** Non-negativity: CIR and lognormal versus normal models.
- **D1·S5** The sibling set against Vasicek and Ho-Lee: which model has which drift and which volatility.

What the questions look like: "which statement correctly describes" items. The distractors swap $\sigma\sqrt{r}$ (CIR) with σr (lognormal), give Ho-Lee a mean reversion it does not have, or claim CIR's terminal distribution is normal.

1. Why let volatility depend on the rate

Models 1 and 2, Ho-Lee, Vasicek and Model 3 all set the **basis-point volatility** of the short rate independently of the level of the rate. GARP calls this "almost certainly not true at extreme levels":

- **High inflation, high rates.** These periods are unstable, so the expected size of the next rate change (basis-point volatility) is large.
- **Ultra-low rates.** Rates cannot fall much below zero, so zero acts as a barrier that *damps* volatility.

The fix is to make basis-point volatility an **increasing function of the rate**. The functional form decides how fast it increases: CIR uses $\sqrt{r}$, the lognormal family uses r.

2. The Cox-Ingersoll-Ross (CIR) model

$$\boxed{dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw}$$

- **Drift** $k(\theta - r)$: the same mean reversion as Vasicek. The rate is pulled toward the long-run level θ at speed k.
- **Volatility** $\sigma\sqrt{r}$: the drift term is not random and dw has standard deviation $\sqrt{dt}$, so the annualised standard deviation of dr (the **basis-point volatility**) is $\sigma\sqrt{r}$. The **parameter σ is constant; the basis-point volatility is not**. It rises with the rate, at a decreasing rate.
- **No negative rates.** Basis-point volatility is zero when $r = 0$, and the drift $k\theta$ is positive there. A rate at zero can only move up.

3. The lognormal (proportional volatility) family

Basis-point volatility proportional to the rate. Two examples in the reading:

Courtadon: $dr = k(\theta - r)\,dt + \sigma r\,dw$ Model 4 (simplest lognormal): $dr = ar\,dt + \sigma r\,dw$

Here σ is called the **yield volatility**: volatility as a *percentage of the rate*. Yield volatility is constant; basis-point volatility σr rises in a straight line with the rate. Like CIR, the basis-point volatility is zero at $r = 0$, so with a positive drift the rate cannot become negative.

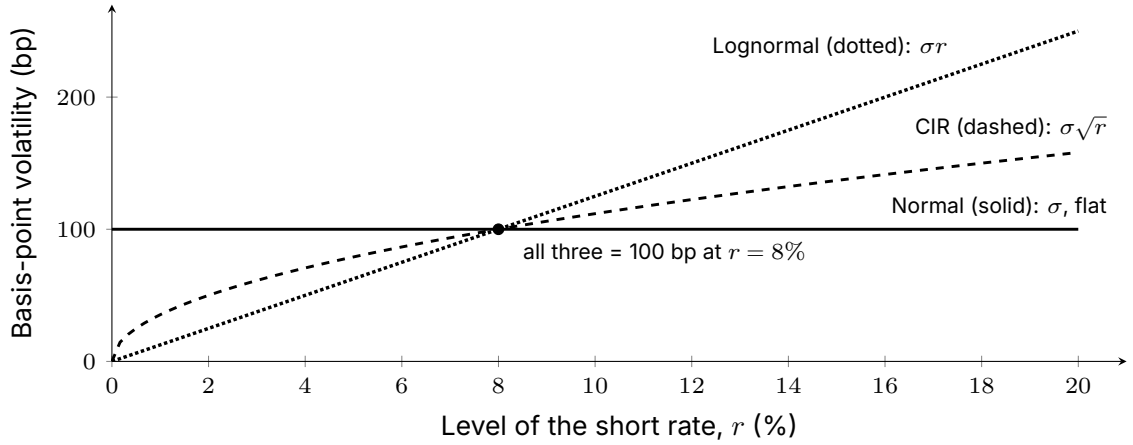


Figure 1. The three volatility specifications, each calibrated to 100 bp at an 8% rate (GARP Figure 15.2: $\sigma_{bp} = 0.01$, $\sigma_{CIR} = 0.0354$, $\sigma_Y = 12.5\%$). Normal is flat. CIR bends over because $\sqrt{r}$ grows more slowly than r. Lognormal is a straight line through the origin. CIR and lognormal both give zero volatility at a zero rate.

4. Terminal distributions (Vardeez's focus)

GARP sets the parameters so that, after ten years, the normal, CIR and lognormal distributions of the short rate all have a mean of 5% and a standard deviation of 2.32%. Even then they are not alike:

- **Normal:** symmetric, with a positive probability of negative rates (here about 1.6%). That probability *grows with the forecast horizon*.
- **CIR and lognormal:** non-negative and **right-skewed**. They stop at zero and lean right.
- **Consequence:** the probability of outcomes away from the mean differs enough to give **significantly different out-of-the-money option prices**. "The shape of the distribution used in an interest rate model is an important determinant of that model's performance."

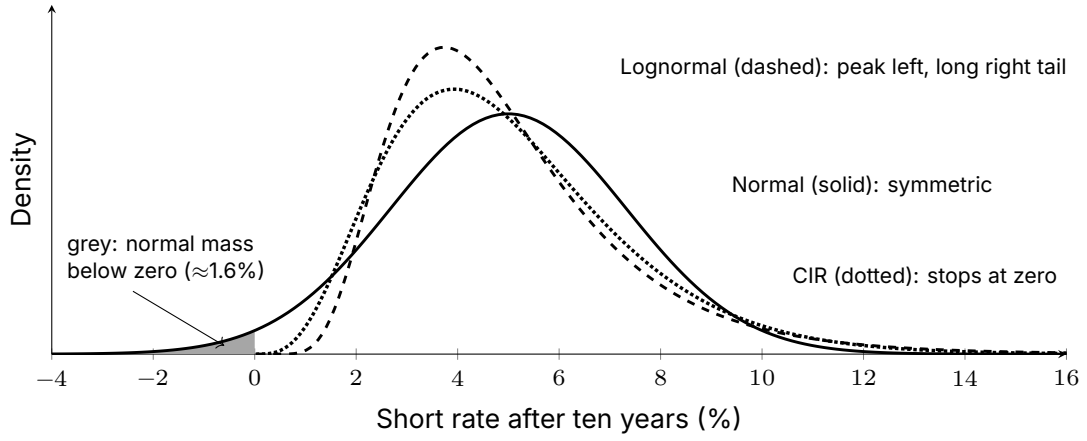


Figure 2. Terminal distributions with the same mean (5%) and standard deviation (2.32%), after GARP Figure 15.3. The CIR curve is drawn as a gamma distribution with the same mean and standard deviation (illustrative shape). Only the normal model puts probability below zero.

5. Is CIR always the better model? No

Non-negativity is "in some respects" an improvement, but model choice depends on the purpose. GARP's example: a trader who believes (i) constant volatility fits the current environment, (ii) the possibility of negative rates has a small impact on the securities being priced, and (iii) the computational simplicity of constant volatility is valuable "might very well opt for a model that allows some probability of negative rates".

	Normal (Vasicek, Ho-Lee)	CIR	Lognormal (Model 4, Courtadon)
Diffusion term	$\sigma\, dw$	$\sigma\sqrt{r}\, dw$	$\sigma r\, dw$
Basis-point vol	Constant σ	$\sigma\sqrt{r}$: rises, at a decreasing rate	σr: rises linearly
Yield vol	σ/r: falls with r	$\sigma/\sqrt{r}$: falls with r	σ: constant
Vol at $r = 0$	$\sigma > 0$	0	0
Negative rates	Possible	Impossible (positive drift at 0)	Impossible
Terminal distribution	Normal, symmetric	Non-negative, right-skewed	Non-negative, right-skewed
Mean reversion	Vasicek yes, Ho-Lee no	Yes, toward θ	Courtadon yes, Model 4 no

Correction to the source notes. Vardeez's comparison says that "in both cases [CIR and lognormal] yield-based volatility is constant; only basis point volatility changes." That holds for the lognormal family only. GARP: in the proportional specifications "yield volatility is constant"; in CIR the *parameter* σ is constant, while yield volatility $= \sigma\sqrt{r}/r = \sigma/\sqrt{r}$ **falls** as the rate rises. This is exactly what Question MR-15 e.2 tests.

Exam traps

- **Sibling swap $\sqrt{r}$ versus r.** CIR: $\sigma\sqrt{r}$. Lognormal: σr. Options often give each model the other's term.
- **Mean reversion is not what prevents negative rates.** Vasicek mean-reverts and can still go negative. Non-negativity comes from volatility vanishing at zero (CIR) or from multiplicative shocks (lognormal).
- **Low rates mean low basis-point volatility in CIR**, not high. "Low rates bring high volatility and so negative rates" reverses the mechanism.
- **CIR's terminal distribution is never normal**: it is non-negative and right-skewed.
- **Ho-Lee has no long-run level.** A negative drift "when the rate is above its long-run level" is mean reversion: Vasicek or CIR.
- **Absolute claims.** "CIR is always the better model" is false; the trader in GARP's example may rationally choose constant volatility.

Practice questions: MR-15 d

Question MR-15 d.1

Bank Q718 • AnalystPrep Q.1685 (rebuilt)

A short-rate model assumes that basis-point volatility increases with the level of the short rate, in proportion to the square root of the rate, and that the rate reverts to a long-run level θ at speed k. Which equation represents the dynamics of this model?

A. $dr = k(\theta - r)\,dt + \sigma\,dw$

B. $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$

C. $dr = k(\theta - r)\,dt + \sigma r\,dw$

D. $dr = k(r - \theta)\,dt + \sigma\sqrt{r}\,dw$

Question MR-15 d.2

Bank Q717 • AnalystPrep Q.1676 (rebuilt)

In the CIR model, basis-point volatility equals $\sigma\sqrt{r}$, which is zero when the short rate is zero; together with a positive drift at $r = 0$, this prevents negative rates. What is the main weakness of models with constant basis-point volatility (such as Vasicek or Ho-Lee) that this CIR feature addresses?

A. Shocks have the same size near zero as at high rate levels, so the short rate can become negative.

B. Constant basis-point volatility makes the short rate deterministic, so no interest rate risk is modelled.

C. Constant basis-point volatility prevents the model from including mean reversion toward a long-run rate.

D. Basis-point volatility rises roughly in proportion to the rate level, overstating risk when rates are high.

Question MR-15 d.3

Bank Q716 • GARP PE1 Q14 • AP Mock 3 Q14

A hedge fund's CRO asks the risk team to develop a term-structure model for fitting interest rates for the fund's options pricing. The team is comparing the Ho-Lee model (normal distribution with time-dependent drift) and the Cox-Ingersoll-Ross (CIR) model. Which of the following statements correctly describes one of these models?

A. In the Ho-Lee model, the drift of the short rate is a constant.

B. In the Ho-Lee model, when the short rate is above its long-run equilibrium level, the drift is negative.

C. In the CIR model, the volatility of the short rate declines exponentially to a constant long-run level.

D. In the CIR model, the basis-point volatility of the short rate is proportional to the square root of the rate.

Question MR-15 d.4

Bank Q719 • BT Mock Q11

For his term structure model, Peter considered the Vasicek model but, on reflection, prefers the Cox-Ingersoll-Ross (CIR) model. Which of the following statements about the CIR model is TRUE?

A. A drawback of the CIR model is that it does not mean revert to a long-term rate.

B. In the CIR model, the drift is time-dependent so that the model is an arbitrage-free fit to the curve.

C. An advantage of the CIR model is that the terminal distribution of the short rate is eventually normally distributed.

D. Unlike the Vasicek model's constant volatility, the CIR model's basis-point volatility is not constant, which (with positive drift) prevents the short rate from becoming negative.

Question MR-15 d.5

Bank Q720 • Schweser Module Quiz 15.2 Q2

Which of the following statements is most likely a disadvantage of the CIR model?

A. Interest rates are always non-negative.

B. Option prices from the CIR distribution may differ significantly from those under lognormal or normal distributions.

C. Basis-point volatility increases during periods of high inflation.

D. Long-run interest rates hover around a mean-reverting level.

Question MR-15 d.6

New • built from GARP MR Ch.15, 15.2 and Figure 15.3

A model validator compares three short-rate models whose parameters have been chosen so that, ten years ahead, the distribution of the short rate under each model has a mean of 5% and a standard deviation of 2.32%. The models are a normal model, the CIR model and a lognormal model. Which of the following conclusions is correct?

A. Because the three distributions share the same mean and standard deviation, the models will give the same prices for out-of-the-money interest rate options.

B. The normal model assigns a positive probability to negative rates, and this probability shrinks as the forecast horizon lengthens.

C. The CIR and lognormal distributions are non-negative and right-skewed, so out-of-the-money option prices can differ significantly from the normal model's.

D. The CIR distribution is left-skewed with a floor at the long-run level θ, while the lognormal distribution is symmetric around 5%.

Question MR-15 d.7 New • built from GARP MR Ch.15, 15.2

A dealer's rates trader must choose a short-rate model for a book of near-the-money swaptions. She believes that constant volatility describes the current environment well, that the possibility of negative rates has only a small impact on the prices of the securities she trades, and that the computational simplicity of a constant-volatility model is valuable. According to the reading, which choice is most defensible?

A. A constant basis-point volatility model, accepting that it allows some probability of negative rates.

B. The CIR model, because any model that allows negative rates must be rejected regardless of purpose.

C. A lognormal model, because its basis-point volatility is constant and it cannot produce negative rates.

D. The CIR model, because its terminal distribution is normal and therefore simpler to compute with.

Solutions • MR-15 d

Solution MR-15 d.1

Answer: B

Bank Q718 • AnalystPrep Q.1685

Step 1 – Translate each requirement into a term

Mean reversion toward θ at speed k: drift $k(\theta - r)\,dt$, which pulls r up when $r < \theta$ and down when $r > \theta$. Basis-point volatility proportional to $\sqrt{r}$: diffusion $\sigma\sqrt{r}\,dw$.

Step 2 – Combine

$dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, which is the CIR model.
$\Rightarrow$ CIR is Vasicek's drift with a $\sqrt{r}$-scaled shock.

Interpretation

Read the drift and the diffusion separately. The drift decides mean reversion (and its direction); the diffusion decides how volatility depends on the rate.

Why the other options are wrong

A is incorrect. Constant diffusion $\sigma\,dw$ is Vasicek: volatility does not depend on r.
C is incorrect. Diffusion $\sigma r\,dw$ is the proportional (lognormal, Courtadon) specification: linear in r, not square root.
D is incorrect. The drift sign is reversed. $k(r - \theta)$ pushes the rate away from θ.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. •
Reference: GARP MR Ch.15, 15.2, eq. 15.3

Solution MR-15 d.2

Answer: A

Bank Q717 • AnalystPrep Q.1676

Step 1 – What constant basis-point volatility implies

With diffusion $\sigma\,dw$, the standard deviation of dr is $\sigma\sqrt{dt}$ whatever the level of r. Near zero, a normal shock of that size can push the rate below zero, and a normal terminal distribution always has mass below zero, more so at long horizons.

Step 2 – What CIR changes

Scaling the shock by $\sqrt{r}$ makes it vanish at $r = 0$, where the drift $k\theta$ is positive.
$\Rightarrow$ The weakness addressed is shocks that do not shrink as rates approach zero.

Interpretation

The economic argument in the reading is the same: at ultra-low rates the zero bound damps volatility, which a constant-volatility model cannot capture.

Why the other options are wrong

B is incorrect. Constant does not mean zero. These models are stochastic, with volatility σ.
C is incorrect. Constant volatility and mean reversion coexist in Vasicek.
D is incorrect. Volatility proportional to the rate describes the lognormal model, not constant-volatility models.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. •
Reference: GARP MR Ch.15, 15.2

Solution MR-15 d.3 — Answer: D

Bank Q716 • GARP PE1 Q14

Step 1 – Recall both processes

Ho-Lee: $dr = \lambda(t)\,dt + \sigma\,dw$, a time-dependent drift calibrated to the curve, no mean reversion. CIR: $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, constant σ, basis-point volatility $\sigma\sqrt{r}$.

Step 2 – Test each statement

Only D matches: in CIR the basis-point volatility is $\sigma\sqrt{r}$, proportional to the square root of the rate.

$\Rightarrow$ Volatility shrinks toward zero as rates approach zero.

Interpretation

GARP builds the wrong options from neighbouring models: a constant drift (Model 2), mean reversion (Vasicek), and exponentially decaying volatility (Model 3).

Why the other options are wrong

A is incorrect. Ho-Lee's drift $\lambda(t)$ is time-dependent, not constant; a constant drift is Model 2.

B is incorrect. A negative drift above a long-run level is mean reversion, which Ho-Lee does not have (Vasicek and CIR do).

C is incorrect. Volatility that declines exponentially over time is Model 3's time-dependent volatility, not CIR's rate-dependent volatility.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. • **Reference:** GARP MR Ch.14, 14.3; Ch.15, 15.1 and 15.2

Solution MR-15 d.4 — Answer: D

Bank Q719 • BT Mock Q11

Step 1 – Compare the two processes

Vasicek: $dr = k(\theta - r)\,dt + \sigma\,dw$. CIR: $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$. Same drift, different volatility.

Step 2 – Why CIR rates stay non-negative

As $r \to 0$, basis-point volatility $\sigma\sqrt{r} \to 0$ while the drift $k\theta > 0$.

$\Rightarrow$ The rate cannot turn negative.

Interpretation

D is the only statement consistent with the CIR specification. The others contradict its mean reversion, its constant parameters, or the shape of its terminal distribution.

Why the other options are wrong

A is incorrect. CIR does mean revert to θ, exactly like Vasicek.

B is incorrect. CIR has constant parameters (an equilibrium model). A time-dependent drift fitted to the curve is the Ho-Lee feature.

C is incorrect. The CIR terminal distribution is non-negative and right-skewed, never normal; a normal distribution would allow negative rates.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. • **Reference:** GARP MR Ch.15, 15.2

Solution MR-15 d.5 Answer: B

Bank Q720 • Schweser Module Quiz 15.2 Q2

Step 1 – Sort the statements into advantages and drawbacks

Non-negative rates (A), basis-point volatility that rises in high-inflation periods (C) and mean reversion (D) are all reasons the reading gives *for* CIR.

Step 2 – Find the drawback

CIR's terminal distribution is right-skewed and non-negative, so its tail probabilities differ from those of the normal and lognormal models.

$\Rightarrow$ Out-of-the-money option prices may differ significantly across models: a model-choice risk.

Interpretation

The point comes from GARP's Figure 15.3: identical means and standard deviations, but different tails, hence different option prices.

Why the other options are wrong

A is incorrect. Non-negativity is a benefit.

C is incorrect. Higher basis-point volatility when rates are high is the realistic feature CIR was designed to capture.

D is incorrect. Mean reversion around a long-run level is a desirable, empirically sensible feature.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, Figure 15.3

Solution MR-15 d.6 Answer: C

New • GARP MR Ch.15, Figure 15.3

Step 1 – What matching two moments does and does not fix

Equal mean and standard deviation fix the centre and spread, not the shape. The normal is symmetric; CIR and lognormal are bounded at zero and right-skewed (Figure 2).

Step 2 – Consequence for options

Out-of-the-money options depend on the probability of outcomes far from the mean, which differs across the three shapes.

$\Rightarrow$ Prices of out-of-the-money options can differ significantly even with identical means and standard deviations.

Interpretation

This is the "concept of terminal distributions" that Vardeez points to: the shape of the distribution is "an important determinant of that model's performance".

Why the other options are wrong

A is incorrect. Absolute claim. Two matched moments do not make the tails equal, and out-of-the-money prices live in the tails.

B is incorrect. Polarity flip. The normal model's probability of negative rates *grows* with the horizon.

D is incorrect. Sibling swap of shapes: CIR is right-skewed with a floor at zero (not at θ), and the lognormal is right-skewed, not symmetric.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, Figure 15.3

Solution MR-15 d.7 **Answer: A**

New • GARP MR Ch.15, 15.2

Step 1 – Match the trader's three beliefs to the reading

GARP's example uses exactly these three beliefs (constant volatility fits now; negative rates matter little for the securities; simplicity has great value) and concludes that such a trader "might very well opt for a model that allows some probability of negative rates."

Step 2 – Check the alternatives against the model facts

Lognormal basis-point volatility is σr, not constant; CIR's terminal distribution is not normal.
$\Rightarrow$ The defensible choice is a constant-volatility model.

Interpretation

Non-negativity is "in some respects" an improvement, but "choosing a model depends on the purpose at hand."

Why the other options are wrong

B is incorrect. Absolute claim. The reading explicitly allows models with some probability of negative rates when that probability barely affects prices.

C is incorrect. Sibling swap. The lognormal model has constant *yield* volatility; its basis-point volatility rises with the rate.

D is incorrect. CIR's terminal distribution is non-negative and right-skewed, and CIR is the *more* complex model.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. • **Reference:** GARP MR Ch.15, 15.2

MR-15 • The Art of Term Structure Models: Volatility and Distribution • Vardeez: VERY IMPORTANT

MR-15 e

P2 CALC

Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models.

Workbook: D1 CIR and Lognormal Short-Rate Models • Tier D • Map #39

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter; Tier D workbook section D1 reports "CIR model solving".
Vardeez: CIR and lognormal rate change.

Reported exam shapes

- **D1·S1** CIR model solving: one step of the short-rate change with drift, volatility and a random draw.
- **D1·S2** CIR basis-point volatility $\sigma\sqrt{r}$ versus the constant parameter σ.
- **D1·S4** Lognormal models: basis-point volatility proportional to r, with deterministic drift or with mean reversion.

What the questions look like: a one-step CIR calculation with a table of r, θ, k, σ and a draw, where the distractors drop dt, drop $\sqrt{dt}$, drop $\sqrt{r}$ or flip the drift sign; plus statement questions on how basis-point and yield volatility move with the rate.

1. One step of the CIR model

Background: see MR-15 d. Discretise $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$ with $dw = z\sqrt{dt}$, where z is a standard normal draw:

$$\Delta r = \underbrace{k(\theta - r)\,dt}_{\text{drift, scales with } dt} + \underbrace{\sigma\sqrt{r}\,z\sqrt{dt}}_{\text{shock, scales with } \sqrt{dt}}$$

- **Read the stem for dw.** If it gives a standard normal draw z, multiply by $\sqrt{dt}$. If it gives "the realisation of dw" (already scaled), use it as given.
- **$\sqrt{r}$ uses the rate as a decimal.** $\sqrt{5\%} = \sqrt{0.05} = 0.2236$. Using $\sqrt{5} = 2.236$ gives a shock ten times too large.
- **The drift sign does the mean reversion.** Below θ the drift is positive; above θ it is negative.

Worked example (Vardeez, Schweser). $r = 5\%$, $\theta = 24\%$, $k = 0.04$, $\sigma = 1.50\%$, one month, $dw = 0.2$ (given as the realisation):

$$\Delta r = 0.04\,(0.24 - 0.05)\tfrac{1}{12} + 0.015\sqrt{0.05} \times 0.2 = 0.063\% + 0.067\% = 0.13\% \quad \Rightarrow \quad r_1 = 5.13\%$$

2. Basis-point volatility versus yield volatility

- **Basis-point volatility** is in rate units: the annualised standard deviation of dr (for example 100 bp).
- **Yield volatility** is a percentage of the rate: basis-point volatility divided by r (for example 12.5%).

Model	Basis-point vol	Yield vol	As r rises
Normal (Vasicek, Ho-Lee)	σ	σ/r	bp vol flat, yield vol falls
CIR	$\sigma\sqrt{r}$	$\sigma/\sqrt{r}$	bp vol rises, yield vol **falls**
Lognormal (Model 4, Courtadon)	σr	σ	bp vol rises, yield vol **constant**

Calibrating to a common basis-point volatility (GARP eqs. 15.6 to 15.8). For 100 bp at an 8% rate:

$$\sigma_{bp} = 0.01 \qquad \sigma_{CIR}\sqrt{0.08} = 0.01 \Rightarrow \sigma_{CIR} = 0.0354 \qquad \sigma_Y \times 0.08 = 0.01 \Rightarrow \sigma_Y = 12.5\%$$

From there, CIR basis-point volatility scales with $\sqrt{r/8\%}$ and lognormal with $r/8\%$. At a 2% rate: CIR 50 bp, lognormal 25 bp, normal still 100 bp.

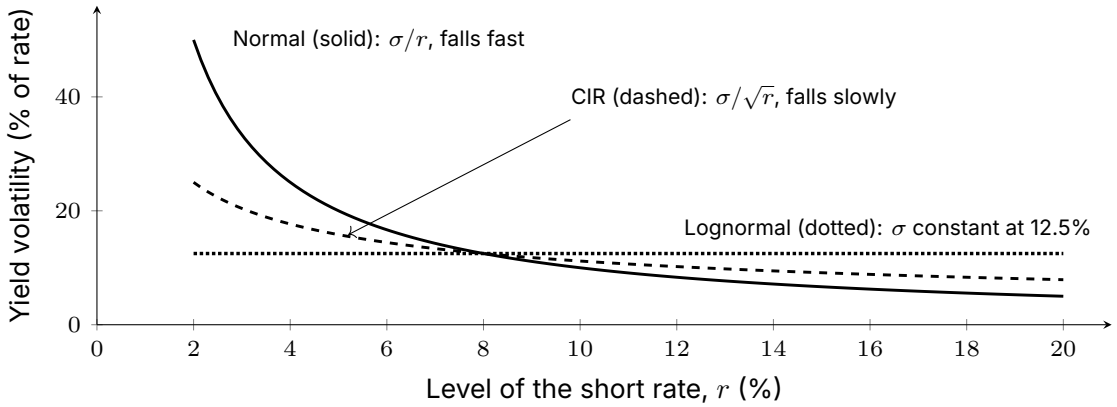


Figure 3. The same three calibrated models as Figure 1, now in yield-volatility units. Only the lognormal model keeps yield volatility constant; in CIR it falls as the rate rises.

3. One step of a lognormal model

For the lognormal model with drift the tree is built in $\ln r$ and exponentiated, so the step is **multiplicative**:

$$r_{\text{up}} = r_0\, \mathrm{e}^{a\,dt+\sigma\sqrt{dt}}, \qquad r_{\text{down}} = r_0\, \mathrm{e}^{a\,dt-\sigma\sqrt{dt}}$$

Example: $r_0 = 2.00\%$, $a = 0.50\%$ a year, $\sigma = 9\%$, monthly: $r_{\text{up}} = 2\% \times \mathrm{e}^{0.000417+0.025981} = 2.0535\%$. The 9% is a yield volatility, so it never enters as an additive basis-point shock. (Details of the lognormal models: MR-15 f.)

Formula card: MR-15 e

CIR one step	$\Delta r = k(\theta - r)\,dt + \sigma\sqrt{r}\,z\sqrt{dt}$ $(dw = z\sqrt{dt}$ unless given already scaled)
CIR volatilities	bp vol $= \sigma\sqrt{r}$ yield vol $= \sigma/\sqrt{r}$
Lognormal volatilities	bp vol $= \sigma r$ yield vol $= \sigma$
Lognormal tree step	$r_{\pm} = r_0\, \mathrm{e}^{a\,dt\pm\sigma\sqrt{dt}}$
Calibration at r^*	$\sigma_{CIR} = \sigma_{bp}/\sqrt{r^*}$, $\sigma_Y = \sigma_{bp}/r^*$ (0.0354 and 12.5% for 100 bp at 8%)

Exam traps

- σ **is not the basis-point volatility in CIR.** A parameter of 0.05 at $r = 3.6\%$ is $0.05\sqrt{0.036} = 94.9$ bp, not 500 bp.
- **Missing dt on the drift** or **missing $\sqrt{dt}$ on the shock**: two separate named mistakes. Check whether dw is already scaled.
- **Dropped $\sqrt{r}$**: treats CIR as Vasicek and inflates the shock.
- **Drift sign**: $k(\theta - r)$, not $k(r - \theta)$. Above θ the drift is negative.
- $\sqrt{5}$ **instead of** $\sqrt{0.05}$: a factor of 10.
- **CIR yield volatility falls** with the rate; only the lognormal keeps it constant.
- **Lognormal shocks are multiplicative**: $r_0 e^{\pm\sigma\sqrt{dt}}$, not $r_0 \pm \sigma\sqrt{dt}$.

Practice questions: MR-15 e

Question MR-15 e.1

Bank Q730 • Schweser Module Quiz 15.2 Q1 (rebuilt)

Which of the following choices correctly characterizes basis-point volatility and yield volatility as a function of the level of the short-term rate in the lognormal model $dr = ar\,dt + \sigma r\,dw$?

A. Basis-point volatility increases with the rate; yield volatility is constant.

B. Basis-point volatility is constant; yield volatility decreases with the rate.

C. Basis-point volatility increases with the rate; yield volatility decreases with the rate.

D. Basis-point volatility is constant; yield volatility increases with the rate.

Question MR-15 e.2

Bank Q729 • Mock 4 Q37

A fixed-income analyst is comparing the volatility specifications of different term structure models. The analyst focuses on the Cox-Ingersoll-Ross (CIR) model, in which the short-rate process is specified as $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$. Which of the following statements about volatility in the CIR model is correct?

A. Both basis-point volatility and yield volatility increase as the level of the short-term rate increases.

B. Both basis-point volatility and yield volatility decrease as the level of the short-term rate increases.

C. Basis-point volatility decreases and yield volatility increases as the level of the short-term rate increases.

D. Basis-point volatility increases and yield volatility decreases as the level of the short-term rate increases.

Question MR-15 e.3

Bank Q728 • GARP PE2 Q36

An analyst studying term structure models focuses on the Cox-Ingersoll-Ross (CIR) model and its treatment of the volatility of the short-term rate. Which of the following statements is correct regarding yield volatility and basis-point volatility in the CIR model?

A. Periods of extremely low short-term interest rates are accompanied by high basis-point volatility, which increases the possibility of negative interest rates in the CIR model.

B. In the CIR model, basis-point volatility is specified as a decreasing function of the mean reversion factor.

C. In the CIR model, yield volatility is specified as being constant while basis-point volatility is allowed to vary.

D. Basis-point volatility and yield volatility are used interchangeably to measure the same volatility in the CIR model.

Question MR-15 e.4

Bank Q722 • AP Mock 1 Q16 (rebuilt)

Use the Cox-Ingersoll-Ross model, $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, with a monthly step ($dt = 1/12$), to find the short-term rate after one month.

Current short-term rate	5%
Long-run value of the short-term rate, θ	7.5%
Speed of mean reversion, k	0.05
Volatility, σ	0.8%
Realized dw (already scaled for the time step)	0.3

A. 5.04%

B. 5.06%

C. 5.18%

D. 5.25%

Question MR-15 e.5

Workbook D1.1 • S1, S2 • Map #39

A risk analyst at an Australian bank is simulating monthly paths of the short rate under the risk-neutral Cox-Ingersoll-Ross (CIR) process $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, where dw is normal with mean zero and standard deviation $\sqrt{dt}$. For the first step of one path the inputs are:

- Current short rate: 3.60%; long-run value θ: 5.00%; speed of mean reversion k: 0.40
- Volatility parameter σ: 0.05
- Time step: one month ($dt = 1/12$); the standard normal draw for this step is $+1.2$

Which of the following is closest to the change in the short rate over the month and to the annualized basis-point volatility of the short rate at the start of the month?

A. $+37.5$ bp; 94.9 bp

B. $+88.9$ bp; 94.9 bp

C. $+118.5$ bp; 94.9 bp

D. $+177.9$ bp; 500.0 bp

Question MR-15 e.6

Bank Q727 • BT notes P2.T5.23.5.1 (rebuilt)

A CIR model, $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, is implemented with monthly steps ($dt = 1/12$). The current rate is 7.00%, $\sigma = 1.30\%$, $\theta = 5.00\%$ and $k = 0.30$. The random variable dw has mean zero and standard deviation $\sqrt{dt}$. If next month's shock is $+2.0$ standard deviations ($dw = 2.0\sqrt{1/12}$), which of the following is closest to the change in the rate?

A. -0.401%

B. $+0.149\%$

C. $+0.249\%$

D. $+0.701\%$

Question MR-15 e.7

Bank Q721 • BT P2.T5.307.2 (rebuilt)

An analyst builds a binomial tree for the lognormal model without mean reversion, $dr = \lambda r\,dt + \sigma r\,dw$, with time step $dt = 1/12$ (monthly), initial rate $r(0) = 2.00\%$, constant annual drift $\lambda = 0.50\%$, and annual yield volatility $\sigma = 9.00\%$. In the tree, the up-node after one month is $r(0)\,\mathrm{e}^{\lambda dt + \sigma\sqrt{dt}}$ and the down-node is $r(0)\,\mathrm{e}^{\lambda dt - \sigma\sqrt{dt}}$. What is the rate at the up-node after one month?

A. 1.9495%

B. 2.0535%

C. 2.0629%

D. 4.6397%

Question MR-15 e.8

New • built from GARP MR Ch.15, eqs. 15.6 to 15.8

A rates desk calibrates three short-rate models so that each gives a basis-point volatility of 100 bp when the short rate is 8%: a normal model (constant basis-point volatility), the CIR model (basis-point volatility $\sigma\sqrt{r}$) and a lognormal model (basis-point volatility σr). The parameters are then held fixed while the short rate falls to 2%. Which of the following is closest to the basis-point volatilities of the CIR model and of the lognormal model at the 2% rate?

A. CIR 25 bp; lognormal 50 bp

B. CIR 50 bp; lognormal 25 bp

C. CIR 100 bp; lognormal 100 bp

D. CIR 200 bp; lognormal 400 bp

Solutions • MR-15 e

Solution MR-15 e.1 — Answer: A

Bank Q730 • Schweser Module Quiz 15.2 Q1

Step 1 – Basis-point volatility

The coefficient on dw is σr, so basis-point volatility $= \sigma r$, which rises linearly with r.

Step 2 – Yield volatility

Yield volatility $= \sigma r / r = \sigma$, a constant.

$\Rightarrow$ With $\sigma = 20\%$: at $r = 2\%$ the basis-point volatility is 40 bp, at $r = 5\%$ it is 100 bp; yield volatility is 20% in both cases.

Interpretation

That is why σ in a lognormal model is called the yield volatility: volatility is specified as a fixed percentage of the rate.

Why the other options are wrong

B is incorrect. Describes a constant basis-point volatility model (Model 1, Vasicek), whose yield volatility σ/r falls with the rate.

C is incorrect. Describes CIR: basis-point volatility $\sigma\sqrt{r}$ rises, yield volatility $\sigma/\sqrt{r}$ falls.

D is incorrect. No model in the reading pairs constant basis-point volatility with rising yield volatility; constant basis-point volatility implies yield volatility falls with r.

LO: MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, eqs. 15.4 and 15.5

Solution MR-15 e.2 — Answer: D

Bank Q729 • Mock 4 Q37

Step 1 – Basis-point volatility

The coefficient on dw is $\sigma\sqrt{r}$, which rises with r (at a decreasing rate).

Step 2 – Yield volatility

Yield volatility $= \sigma\sqrt{r}/r = \sigma/\sqrt{r}$, which falls as r rises.

$\Rightarrow$ With $\sigma = 0.08$: $r = 1\%$ gives 0.80% and 80%; $r = 4\%$ gives 1.60% and 40%; $r = 9\%$ gives 2.40% and 26.7%.

Interpretation

Because $\sqrt{r}$ grows more slowly than r, volatility measured as a percentage of the rate declines. This is why the statement "yield volatility is constant in CIR" is wrong (see the correction box in MR-15 d).

Why the other options are wrong

A is incorrect. Yield volatility would rise only if basis-point volatility grew faster than proportionally with r; here it grows like $\sqrt{r}$.

B is incorrect. Basis-point volatility $\sigma\sqrt{r}$ clearly increases with r.

C is incorrect. Reverses both relationships.

LO: MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, eq. 15.3 and Figure 15.2

Solution MR-15 e.3 — Answer: C

Bank Q728 • GARP PE2 Q36

Step 1 – What CIR holds constant

In $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$ the parameter σ is constant.

Step 2 – What varies

The basis-point volatility $\sigma\sqrt{r}$ changes with the level of the short rate.
$\Rightarrow$ "The parameter σ is constant, but basis-point volatility is not" (GARP).

Interpretation

GARP's own key is C. In this item GARP uses "yield volatility" loosely for the constant parameter σ; strictly, CIR yield volatility $\sigma\sqrt{r}/r = \sigma/\sqrt{r}$ falls as the rate rises (Question MR-15 e.2). On the exam, pick the option that pairs a constant σ with a varying basis-point volatility; the other three are wrong on any reading.

Why the other options are wrong

A is incorrect. Reverses the mechanism: very low rates give *low* basis-point volatility, which is what keeps CIR rates non-negative.
B is incorrect. Basis-point volatility depends on the level of r, not on the mean reversion speed k, which governs the drift.
D is incorrect. The two are different measures: basis-point volatility is in rate units ($\sigma\sqrt{r}$), yield volatility is a percentage of the rate.

LO: MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, eq. 15.3

Solution MR-15 e.4 — Answer: B

Bank Q722 • AP Mock 1 Q16

Step 1 – Drift over one month

$$k(\theta - r)\,dt = 0.05 \times (7.5\% - 5\%) \times \tfrac{1}{12} = 0.0104\%$$

Step 2 – Shock (dw already scaled)

$$\sigma\sqrt{r}\,dw = 0.8\% \times \sqrt{0.05} \times 0.3 = 0.8\% \times 0.22361 \times 0.3 = 0.0537\%$$

$\Rightarrow$ No further $\sqrt{dt}$: the stem says dw is already scaled.

Step 3 – New rate

$$r_1 = 5\% + 0.0104\% + 0.0537\% = 5.064\% \approx 5.06\%$$

Interpretation

The rate is below θ, so the drift is positive; the shock is small because $\sqrt{0.05}$ shrinks the 0.8% parameter.

Why the other options are wrong

A is incorrect. Reversed the drift sign: $5\% - 0.0104\% + 0.0537\% = 5.043\%$.
C is incorrect. Did not multiply the drift by dt: $0.05 \times 2.5\% = 0.125\%$, giving 5.179%.
D is incorrect. Dropped $\sqrt{r}$ (Vasicek-style shock): $0.0104\% + 0.8\% \times 0.3 = 0.2504\%$, giving 5.25%.

LO: MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, eq. 15.3

Solution MR-15 e.5 **Answer: A**

Workbook D1.1 • S1, S2

Step 1 – Basis-point volatility

$$\sigma\sqrt{r} = 0.05 \times \sqrt{0.036} = 0.05 \times 0.18974 = 0.0094868 = 94.9 \text{ bp}$$

$\Rightarrow$ At a 3.60% short rate, one standard deviation of annual rate change is about 95 bp. The parameter 0.05 is not itself a basis-point volatility.

Step 2 – Drift over the month

$$k(\theta - r)\,dt = 0.40 \times (0.0500 - 0.0360) \times \tfrac{1}{12} = 0.000467 = 4.67 \text{ bp}$$

$\Rightarrow$ The rate is below θ, so mean reversion pulls it up.

Step 3 – Random term

$$dw = 1.2\sqrt{1/12} = 0.34641, \qquad \sigma\sqrt{r}\,dw = 0.0094868 \times 0.34641 = 0.003286 = 32.86 \text{ bp}$$

Step 4 – Rate change

$$\Delta r = 4.67 + 32.86 = 37.53 \text{ bp}, \qquad r_1 = 3.975\%$$

Interpretation

At the new rate the basis-point volatility is $0.05\sqrt{0.03975} = 99.7$ bp: CIR volatility rises with the square root of the rate. Had the draw driven the rate toward zero, volatility would have shrunk toward zero while the drift $k\theta > 0$ stayed positive, so the rate cannot become negative.

Why the other options are wrong

B is incorrect. Did not scale the drift by dt: $0.40 \times 0.014 = 0.0056$, plus 0.003286, gives 88.9 bp.

C is incorrect. Did not scale the shock by $\sqrt{dt}$: $0.000467 + 0.0094868 \times 1.2 = 0.011851$, or 118.5 bp.

D is incorrect. Treated the constant parameter $\sigma = 0.05$ as the basis-point volatility (500 bp) and omitted $\sqrt{r}$: $0.000467 + 0.05 \times 0.34641 = 0.017787$, or 177.9 bp.

LO: MR-15 d; MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** Tuckman and Serrat 3e, Ch.10, 15.2, eqs. 15.3 and 15.6 to 15.8 (GARP MR Ch.15)

Solution MR-15 e.6 **Answer: B**

Bank Q727 • BT notes P2.T5.23.5.1

Step 1 – Scale the shock

$$dw = 2.0 \times \sqrt{1/12} = 0.57735$$

Step 2 – Drift

$$k(\theta - r)\,dt = 0.30 \times (5\% - 7\%) \times \tfrac{1}{12} = -0.050\%$$

$\Rightarrow$ The rate is above θ, so mean reversion pulls it down.

Step 3 – Shock

$$\sigma\sqrt{r}\,dw = 1.30\% \times \sqrt{0.07} \times 0.57735 = 1.30\% \times 0.26458 \times 0.57735 = 0.1986\%$$

Step 4 – Change

$$\Delta r = -0.050\% + 0.1986\% = +0.149\%$$

Interpretation

Mean reversion pulls the 7% rate down by 5 bp, but the $+2\sigma$ shock adds almost 20 bp and dominates.

Why the other options are wrong

A is incorrect. Forgot to scale the drift by dt: $0.30 \times (-2\%) = -0.60\%$, so $-0.60\% + 0.1986\% = -0.401\%$.

C is incorrect. Wrong sign on the drift (used $r - \theta$): $+0.050\% + 0.1986\% = +0.249\%$.

D is incorrect. Omitted $\sqrt{r}$ from the shock: $1.30\% \times 0.57735 = 0.7506\%$, minus 0.050% gives $+0.701\%$.

LO: MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, eq. 15.3

Solution MR-15 e.7 **Answer: B**

Bank Q721 • BT P2.T5.307.2

Step 1 – Drift term

$$\lambda\,dt = 0.005/12 = 0.000417$$

Step 2 – Volatility term

$$\sigma\sqrt{dt} = 0.09 \times 0.288675 = 0.025981$$

Step 3 – Up-node

$$r_{\text{up}} = 2.00\% \times \mathrm{e}^{0.000417+0.025981} = 2.00\% \times 1.026749 = 2.0535\%$$

$\Rightarrow$ In a lognormal model the shock is multiplicative: the 9% moves the rate by about 2.6% *of the rate*, about 5 bp.

Interpretation

The drift is scaled by dt and the volatility by $\sqrt{dt}$, exactly as in the normal models; the difference is that both enter through the exponential.

Why the other options are wrong

A is incorrect. The down-node: $2.00\% \times \mathrm{e}^{0.000417-0.025981} = 1.9495\%$ (wrong sign on the volatility term).

C is incorrect. Used the full annual drift instead of $\lambda/12$: $2.00\% \times \mathrm{e}^{0.005+0.025981} = 2.0629\%$.

D is incorrect. Treated 9% as an additive basis-point volatility: $2.00\% + 0.0417\% + 9\% \times 0.2887 = 4.6397\%$, ignoring that volatility is proportional to the rate.

LO: MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** GARP MR Ch.15, 15.3 (tree for the Salomon Brothers model)

Solution MR-15 e.8 **Answer: B**

New • GARP MR Ch.15, eqs. 15.6 to 15.8

Step 1 – Calibrate at 8%

$$\sigma_{CIR}\sqrt{0.08} = 0.01 \Rightarrow \sigma_{CIR} = 0.035355 \qquad \sigma_Y \times 0.08 = 0.01 \Rightarrow \sigma_Y = 0.125$$

Step 2 – Re-evaluate at 2%

CIR: $0.035355 \times \sqrt{0.02} = 0.0050 = 50$ bp Lognormal: $0.125 \times 0.02 = 0.0025 = 25$ bp

$\Rightarrow$ Equivalently, CIR scales by $\sqrt{2/8} = 0.5$ and the lognormal by $2/8 = 0.25$.

Interpretation

When rates fall, the lognormal model's basis-point volatility shrinks fastest, CIR's more slowly, and the normal model's not at all (Figure 1). In yield-volatility terms CIR is now at $50/200 = 25\%$ while the lognormal stays at 12.5%.

Why the other options are wrong

A is incorrect. Sibling swap: gave CIR the proportional scaling (r) and the lognormal the square-root scaling.

C is incorrect. Treated both as constant basis-point volatility models, which is only true of the normal model.

D is incorrect. Scaled in the wrong direction, multiplying by $\sqrt{8/2} = 2$ and $8/2 = 4$: a falling rate *lowers* basis-point volatility in both models.

LO: MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. • **Reference:** GARP MR Ch.15, 15.2, eqs. 15.6 to 15.8 and Figure 15.2

MR-15 • The Art of Term Structure Models: Volatility and Distribution • Vardeez: VERY IMPORTANT

MR-15 f

P2 CONCEPT

Describe lognormal models with deterministic drift and mean reversion.

Workbook: D1 CIR and Lognormal Short-Rate Models • Tier D • Map #39

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter; Tier D workbook section D1 covers it.
Vardeez: lognormal model with deterministic drift: understand interpretation.

Reported exam shapes

- **D1·S4** Lognormal models: basis-point volatility proportional to r, with deterministic drift (Salomon Brothers) or with mean reversion (Black-Karasinski).

What the questions look like: statement questions. Additive versus multiplicative drift (Ho-Lee versus lognormal), why rates stay positive, why volatility is a percentage of the rate, and why the Black-Karasinski tree needs variable time steps to recombine. Workbook question D1.2, which tests this LO against CIR and Vasicek, is worked under MR-14 f.

1. Lognormal with deterministic drift: the Salomon Brothers model

$$
\begin{aligned}
dr &= a(t)\, r\, dt + \sigma r\, dw && \text{the model} \\
d[\ln r] &= \left[a(t) - \tfrac{1}{2}\sigma^2\right] dt + \sigma\, dw && \text{by Ito's lemma} \\
\boxed{d[\ln r] &= \tilde{a}(t)\, dt + \sigma\, dw} && \text{redefining } \tilde{a}(t) = a(t) - \tfrac{1}{2}\sigma^2
\end{aligned}
$$

- The **natural logarithm of the short rate is normal**, so the short rate is **lognormal**.
- It is "the Ho-Lee model based on the natural logarithm of the short rate": the drift $\tilde{a}(t)$ varies with time and is fitted to market bond prices, as in Ho-Lee. There is no mean reversion and no long-run level.
- **Build the tree in $\ln r$, then exponentiate.** In logs the steps are additive; in rates they become **multiplicative** (Figure 4).

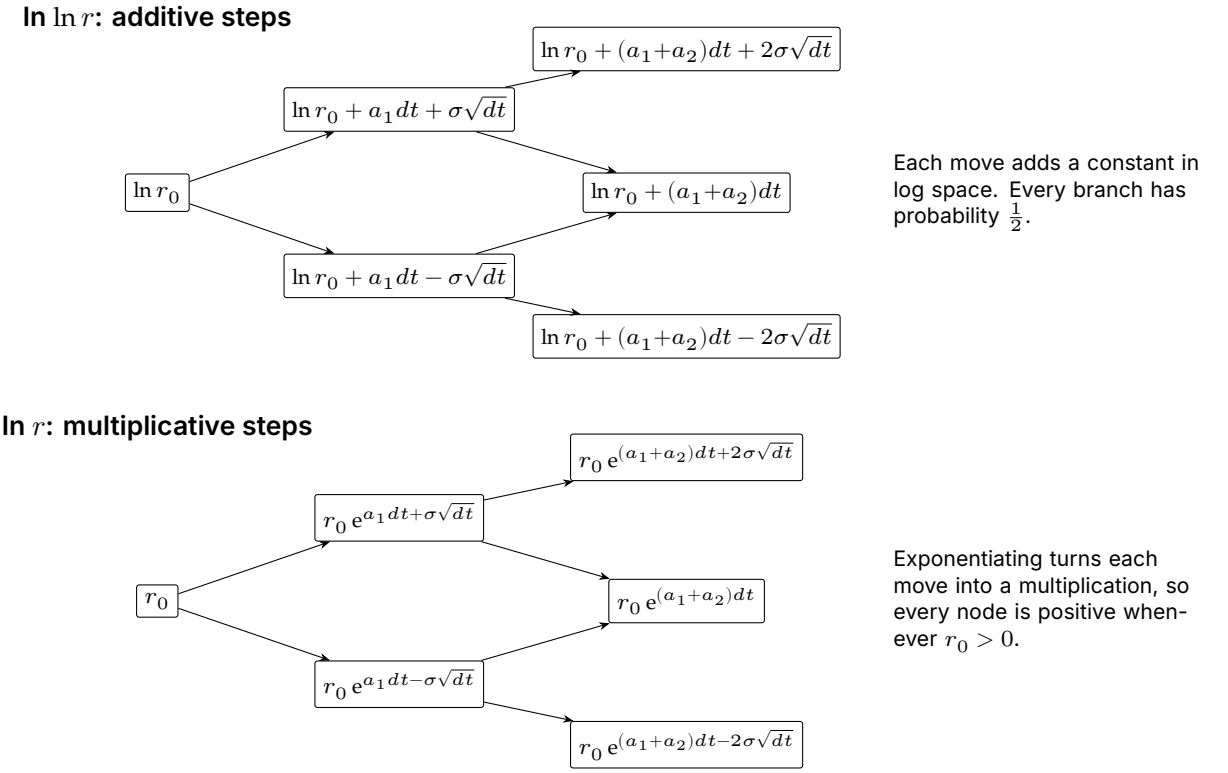


Figure 4. The Salomon Brothers tree, twice. Top: built in $\ln r$, with Ho-Lee-style time-dependent drifts a_1, a_2. Bottom: the same nodes exponentiated back into rates.

2. Why rates stay positive and why volatility is a percentage

- **Positive rates.** $\mathrm{e}^x > 0$ for every x, so as long as $r_0 > 0$ every node is positive. Ho-Lee's drifts are *additive* and its low nodes can go negative; the lognormal model's are *multiplicative*.
- **Volatility as a percentage of the rate.** For small x, $\mathrm{e}^x \approx 1 + x$. With $a_1 = 0$ and $dt = 1$, the up node is $r_0\mathrm{e}^{\sigma} \approx r_0(1 + \sigma)$. With $\sigma = 12.5\%$, the up-state rate is about 12.5% *above* the initial rate (not 12.5 percentage points). Hence lognormal volatility is quoted as yield volatility.

3. Lognormal with mean reversion: the Black-Karasinski model

$$\boxed{d[\ln r] = k(t)\,[\ln\theta(t) - \ln r]\,dt + \sigma(t)\,dw}$$

- The log of the short rate follows a **time-dependent Vasicek** process: it reverts to $\ln\theta(t)$ at speed $k(t)$ with volatility $\sigma(t)$. The rate is lognormal and stays positive.
- **Very flexible**: volatility, mean reversion speed and central tendency can all depend on time, "firmly placing the model in the arbitrage-free class." The user may keep or remove as much time dependence as desired.
- **The tree does not recombine naturally.** With a fixed time step it would recombine only if

$$k(2) = \frac{\sigma(1) - \sigma(2)}{\sigma(1)\,dt},$$

which would make the mean reversion speed completely determined by the volatility function. But the two do different jobs: **mean reversion controls the term structure of volatility**, while

time-dependent volatility controls the future volatility of the short rate (the prices of options expiring at different times).

- **Fix: vary the length of the time step.** Choose dt_1 freely, then set dt_2 so that

$$k(2) = \frac{1}{dt_2}\left[1 - \frac{\sigma(2)\sqrt{dt_2}}{\sigma(1)\sqrt{dt_1}}\right],$$

leaving mean reversion and volatility free to be chosen independently.

	Ho-Lee	Salomon Brothers	Black-Karasinski
Process	$dr = \lambda(t)dt + \sigma dw$	$d\ln r = \tilde{a}(t)dt + \sigma dw$	$d\ln r = k(t)[\ln\theta(t) - \ln r]dt + \sigma(t)dw$
Distribution of r	Normal	Lognormal	Lognormal
Drift	Time-dependent, additive	Time-dependent, multiplicative	Mean-reverting in $\ln r$
Mean reversion	None	None	Yes, toward $\ln\theta(t)$
Basis-point vol	Constant σ	σr	$\sigma(t)\,r$
Negative rates	Possible	Impossible	Impossible
Tree	Recombines	Recombines	Recombines only with variable dt

Exam traps

- **Additive versus multiplicative.** Ho-Lee drifts are additive; lognormal drifts are multiplicative. Options that swap them are the standard trap.
- **Salomon Brothers versus Black-Karasinski.** Deterministic drift only versus mean reversion in the log rate. Ho-Lee has no mean reversion and no long-run level.
- **Black-Karasinski is not an equilibrium model with constant drift**; it is time-dependent and arbitrage-free.
- **Black-Karasinski does not recombine "by definition"**: recombination is forced by choosing dt_2, not by adjusting probabilities.
- **The speed of mean reversion is a free parameter** $k(t)$: the model is neither "slow" nor "fast" by construction.
- **12.5% yield volatility** moves the rate by about 12.5% *of* the rate, not 12.5 percentage points.

Practice questions: MR-15 f

Question MR-15 f.1

Bank Q735 • Schweser Module Quiz 15.2 Q3

Which of the following statements best characterizes the differences between the Ho-Lee model with drift and the lognormal model with drift?

A. In the Ho-Lee model and the lognormal model, the drift terms are multiplicative.

B. In the Ho-Lee model and the lognormal model, the drift terms are additive.

C. In the Ho-Lee model the drift terms are multiplicative, but in the lognormal model the drift terms are additive.

D. In the Ho-Lee model the drift terms are additive, but in the lognormal model the drift terms are multiplicative.

Question MR-15 f.2

New • built from GARP MR Ch.15, 15.3

A junior quant describes the lognormal model with deterministic drift (the "Salomon Brothers" model), $dr = a(t)\,r\,dt + \sigma r\,dw$, to a model validation committee. Which of the following descriptions is correct?

A. The natural logarithm of the short rate follows a Ho-Lee-type process with a time-dependent drift, so the short rate is lognormal and every node of its tree is positive.

B. The natural logarithm of the short rate reverts toward a time-dependent central tendency $\ln \theta(t)$, so the short rate is lognormal and every node of its tree is positive.

C. The short rate follows a Ho-Lee-type process with a time-dependent drift, so the short rate is normal and low nodes of its tree may become negative.

D. The basis-point volatility of the short rate is proportional to the square root of the rate, so the short rate is non-negative but not lognormal.

Question MR-15 f.3

New • built from GARP MR Ch.15, 15.3

In a lognormal short-rate tree with $a_1 = 0$ and $dt = 1$, the up node after one period is $r_0\,\mathrm{e}^{\sigma}$. The model's volatility parameter is $\sigma = 12.5\%$ and the current short rate is 4.00%. Which statement best explains why volatility in this model is quoted as a percentage of the rate?

A. Because $\mathrm{e}^{\sigma} \approx 1 + \sigma$, the up-state rate is about 12.5% above the current rate, about 4.5%, so the size of the move scales with the level of the rate.

B. Because $\mathrm{e}^{\sigma} \approx 1 + \sigma$, the up-state rate is about 12.5 percentage points above the current rate, about 16.5%, so the size of the move is fixed in basis points.

C. Because the drift is additive, the up-state rate is the current rate plus 12.5% of the long-run level, so the size of the move scales with the long-run level.

D. Because the tree is built in rates rather than in logs, the up-state rate is the current rate times 12.5%, so the size of the move scales with the level of the rate.

Question MR-15 f.4 Bank Q731 • AnalystPrep Q.1679 (rebuilt)

Which of the following statements best describes the lognormal model with mean reversion (the Black-Karasinski model)?

A. The natural log of the short rate is normally distributed and mean-reverting, rates are always positive, and the mean-reversion speed, central tendency and volatility may all be time-dependent.

B. The short rate itself is normally distributed and can become negative, with constant mean-reversion speed, central tendency and volatility.

C. Basis point volatility is proportional to the square root of the short rate, and the short rate follows a non-central chi-square distribution.

D. The model produces a naturally recombining interest rate tree, so no adjustment to time steps is required.

Question MR-15 f.5 Bank Q736 • Schweser Module Quiz 15.2 Q4

Which of the following statements is true regarding the Black-Karasinski model?

A. The model produces an interest rate tree that is recombining by definition.

B. The model produces an interest rate tree that is recombining when the dt variable is manipulated.

C. The model is time-varying and mean-reverting with a slow speed of adjustment.

D. The model is time-varying and mean-reverting with a fast speed of adjustment.

Question MR-15 f.6 New • built from GARP MR Ch.15, 15.4

A Black-Karasinski tree with a fixed time step dt recombines only if $k(2) = [\sigma(1) - \sigma(2)] / [\sigma(1)\, dt]$. Instead of imposing this restriction, the model lets the length of the time step vary. Which of the following best explains why?

A. Mean reversion controls the term structure of volatility while time-dependent volatility controls the future volatility of the short rate, so the two must be chosen independently.

B. Mean reversion controls the future volatility of the short rate while time-dependent volatility controls the term structure of volatility, so the two must be chosen independently.

C. Imposing the restriction would make the model an equilibrium model, while varying the time step keeps the drift deterministic and removes mean reversion.

D. Varying the time step always makes the tree recombine, whatever mean reversion and volatility functions are chosen, so no restriction on dt_2 is needed.

Question MR-15 f.7 Bank Q734 • BT P2.T5.307.3

Each of the following is true about lognormal models of the short-term interest rate process EXCEPT for:

A. In the lognormal models, the natural logarithm of the short rate is normally distributed.

B. Like the Cox-Ingersoll-Ross (CIR), the lognormal models offer the advantage of not allowing negative rates.

C. The lognormal model without mean reversion is similar to the Ho-Lee model but based on the natural logarithm of the short rate instead of on the short rate itself.

D. The Black-Karasinski model is similar to the Ho-Lee model but instead is an equilibrium model with constant drift.

Solutions • MR-15 f

Solution MR-15 f.1 — Answer: D

Bank Q735 • Schweser Module Quiz 15.2 Q3

Step 1 – Ho-Lee

$dr = \lambda(t)\,dt + \sigma\,dw$: each period's drift is *added* to the rate.

Step 2 – Lognormal with drift

$d\ln r = \tilde{a}(t)\,dt + \sigma\,dw$: drifts add in logs, so after exponentiating they *multiply* the rate.

$\Rightarrow$ Additive for Ho-Lee, multiplicative for lognormal.

Interpretation

The multiplicative structure is also what keeps every lognormal rate positive (Figure 4).

Why the other options are wrong

A is incorrect. Makes Ho-Lee multiplicative; its drift adds to the rate.

B is incorrect. Ignores that exponentiation turns additive log drifts into multiplicative rate effects.

C is incorrect. Reverses the two models.

LO: MR-15 f: Describe lognormal models with deterministic drift and mean reversion. • **Reference:** GARP MR Ch.15, 15.3

Solution MR-15 f.2 — Answer: A

New • GARP MR Ch.15, 15.3

Step 1 – Apply Ito's lemma

$d[\ln r] = [a(t) - \frac{1}{2}\sigma^2]\,dt + \sigma\,dw = \tilde{a}(t)\,dt + \sigma\,dw$: a Ho-Lee process written on $\ln r$.

Step 2 – Consequences

$\ln r$ is normal, so r is lognormal; nodes are $r_0 e^{(\cdot)} > 0$.

$\Rightarrow$ "The Ho-Lee model based on the natural logarithm of the short rate."

Interpretation

The drift is deterministic and time-dependent (fitted to bond prices), with no mean reversion. That is what separates it from Black-Karasinski.

Why the other options are wrong

B is incorrect. Sibling swap: reversion toward $\ln\theta(t)$ is Black-Karasinski, the lognormal model *with* mean reversion.

C is incorrect. Describes Ho-Lee itself, on the rate rather than its logarithm, with normal rates that can go negative.

D is incorrect. Square-root basis-point volatility is CIR; the lognormal model's basis-point volatility is σr.

LO: MR-15 f: Describe lognormal models with deterministic drift and mean reversion. • **Reference:** GARP MR Ch.15, 15.3, eqs. 15.9 to 15.12

Solution MR-15 f.3 — Answer: A

New • GARP MR Ch.15, 15.3

Step 1 – Use the small-x approximation

$r_0 e^{\sigma} \approx r_0(1+\sigma) = 4.00\% \times 1.125 = 4.50\%$ (exactly $4.00\% \times e^{0.125} = 4.53\%$).

Step 2 – Interpret

The move is 12.5% *of* the rate, about 50 bp here. At a 2% rate the same σ would move it by about 25 bp.
$\Rightarrow$ Volatility is a percentage of the rate: yield volatility.

Interpretation

GARP: "Volatility is clearly a percentage of the rate...If, for example, $\sigma = 12.5\%$, then the short rate in the up-state is 12.5% above the initial short rate."

Why the other options are wrong

B is incorrect. Reads 12.5% as percentage points (an additive basis-point move), which is the normal-model reading.
C is incorrect. The lognormal drift is multiplicative, and there is no long-run level in the deterministic-drift model.
D is incorrect. The tree is built in logs and then exponentiated, and the up node is $r_0 e^{\sigma}$, not $r_0 \times \sigma$ (which would give 0.5%).

LO: MR-15 f: Describe lognormal models with deterministic drift and mean reversion. • **Reference:** GARP MR Ch.15, 15.3, eq. 15.13

Solution MR-15 f.4 — Answer: A

Bank Q731 • AnalystPrep Q.1679

Step 1 – The process

$d[\ln r] = k(t)[\ln \theta(t) - \ln r]\,dt + \sigma(t)\,dw$: a time-dependent Vasicek process on $\ln r$.

Step 2 – Consequences

$\ln r$ is normal and mean-reverting, $r = e^{\ln r} > 0$, and $k(t)$, $\theta(t)$, $\sigma(t)$ may all depend on time.
$\Rightarrow$ Positive, mean-reverting, flexible.

Interpretation

This flexibility is what places Black-Karasinski in the arbitrage-free class.

Why the other options are wrong

B is incorrect. Describes Vasicek: normal rates and constant parameters.
C is incorrect. Describes CIR: square-root volatility and a non-central chi-square distribution.
D is incorrect. The tree does not recombine naturally; the time steps must be recalibrated.

LO: MR-15 f: Describe lognormal models with deterministic drift and mean reversion. • **Reference:** GARP MR Ch.15, 15.4, eqs. 15.14 and 15.15

Solution MR-15 f.5 — Answer: B

Bank Q736 • Schweser Module Quiz 15.2 Q4

Step 1 – Why it does not recombine

With mean reversion in $\ln r$, an up move followed by a down move generally does not land where a down move followed by an up move does.

Step 2 – How it is forced to recombine

Choose dt_1 freely, then set dt_2 from $k(2) = \frac{1}{dt_2}\left[1 - \frac{\sigma(2)\sqrt{dt_2}}{\sigma(1)\sqrt{dt_1}}\right]$.
$\Rightarrow$ Recombination comes from manipulating dt.

Interpretation

The speed of adjustment is the free parameter $k(t)$; the model itself does not make it slow or fast.

Why the other options are wrong

A is incorrect. The tree is not recombining by definition.

C is incorrect. Nothing in the model makes the speed of adjustment slow.
D is incorrect. Nothing in the model makes the speed of adjustment fast.

LO: MR-15 f: Describe lognormal models with deterministic drift and mean reversion. • **Reference:** GARP MR Ch.15, 15.4, eqs. 15.20 to 15.23

Solution MR-15 f.6 Answer: A

New • GARP MR Ch.15, 15.4

Step 1 – What the restriction would do

Imposing $k(2) = [\sigma(1) - \sigma(2)]/[\sigma(1)dt]$ "would require that the mean reversion speed be completely determined by the time-dependent volatility function."

Step 2 – Why that is unacceptable

The two serve distinct purposes: mean reversion controls the term structure of volatility; time-dependent volatility controls the future volatility of the short rate, and hence the prices of options expiring at different times.
$\Rightarrow$ Letting dt vary keeps both free.

Interpretation

The length of the first step is arbitrary; the second is set to satisfy eq. 15.23.

Why the other options are wrong

B is incorrect. Sibling swap of the two roles.
C is incorrect. Black-Karasinski is arbitrage-free and keeps mean reversion; varying dt removes neither.
D is incorrect. Absolute claim: dt_2 is not free; it must be chosen to satisfy eq. 15.23.

LO: MR-15 f: Describe lognormal models with deterministic drift and mean reversion. • **Reference:** GARP MR Ch.15, 15.4

Solution MR-15 f.7 Answer: D

Bank Q734 • BT P2.T5.307.3

Step 1 – Check each statement

A: $\ln r$ is normal, so r is lognormal: true. B: $r = e^{\ln r} > 0$, as in CIR: true. C: the lognormal model with drift is Ho-Lee applied to $\ln r$: true.

Step 2 – Find the false one

D: Black-Karasinski is the lognormal model *with mean reversion*, a time-dependent Vasicek on $\ln r$; it is arbitrage-free, not an equilibrium model with constant drift.
$\Rightarrow$ D is the exception.

Interpretation

Pair the lognormal models with their normal cousins: Salomon Brothers is to Ho-Lee as Black-Karasinski is to (time-dependent) Vasicek.

Why the other options are wrong

A is incorrect. True statement, not the exception.
B is incorrect. True statement: positivity is shared by CIR and the lognormal models.
C is incorrect. True statement: the drift model is the Ho-Lee analogue in logs.

LO: MR-15 f: Describe lognormal models with deterministic drift and mean reversion. • **Reference:** GARP MR Ch.15, 15.3 and 15.4

MR-16 • The Vasicek and Gauss+ Models • Vardeez: VERY IMPORTANT (new chapter, highly testable)

MR-16 d

P2 CONCEPT

Explain how the parameters of the Gauss+ model can be estimated from empirical data.

Workbook: A25 Gauss+ and Vasicek • Tier A (no question on this LO)

WHAT IS TESTED HERE

Why P2. Vardeez flags the whole of MR-16 as a new, highly testable chapter, so this LO sits inside a Very Important chapter. No workbook section reports a shape on it; the Tier A section A25 covers MR-16 a to c.

Vardeez: the six-step simplified estimation procedure, and why the staging works.

What to know

- Which parameter is estimated at which stage, from which data, by which method.
- Why the mean reversion speeds can be found before the volatilities.
- How to read the estimated parameters (Table 16.1).

What the questions look like: "which parameter is calibrated by…" and "which pairing of parameter and method is correct" items, where the distractors swap the methods between the parameter groups.

1. The parameters to estimate

Structure and dynamics are covered in MR-16 a to c. In brief, the cascade is

$$
\begin{aligned}
dr_t &= -\alpha_r(r_t - m_t)\,dt && \text{no random shock: the "+"}\\
dm_t &= -\alpha_m(m_t - l_t)\,dt + \sigma_m\left(\rho\,dW_t^2 + \sqrt{1-\rho^2}\,dW_t^1\right) && \text{medium-term factor}\\
dl_t &= -\alpha_l(l_t - \mu)\,dt + \sigma_l\,dW_t^2 && \text{long-term factor, } dW^1 \perp dW^2
\end{aligned}
$$

There are **seven parameters**: three mean reversion speeds $\alpha_r, \alpha_m, \alpha_l$; two volatilities σ_m, σ_l; the correlation ρ between the medium- and long-term factors; and the long-run level μ. They could be estimated by **maximum likelihood**, the standard method in the term structure literature. The reading instead uses a **simplified, staged approach** that is "significantly easier to implement": each stage estimates a subset of the parameters.

2. The staged procedure

Step	What you do	Parameters
1	Choose a proxy for the short-term rate: the **fed funds target rate**.	Defines r
2	Run regressions of rate changes on changes in a **medium-term** (2-year) and a **long-term** (10-year) rate.	
3	Find the mean reversion speeds that **replicate the regression sensitivities** of Step 2.	$\alpha_r, \alpha_m, \alpha_l$
4	From market data, determine the **term structure of volatility**: observed volatilities of rates, or the volatilities the market uses to price interest rate derivatives.	
5	Find the volatility and correlation parameters that **replicate the volatility term structure** of Step 4.	σ_m, σ_l, ρ
6	Find the long-term reversion level by **minimising the sum of squared errors** of observed yields versus model yields across the whole sample.	μ

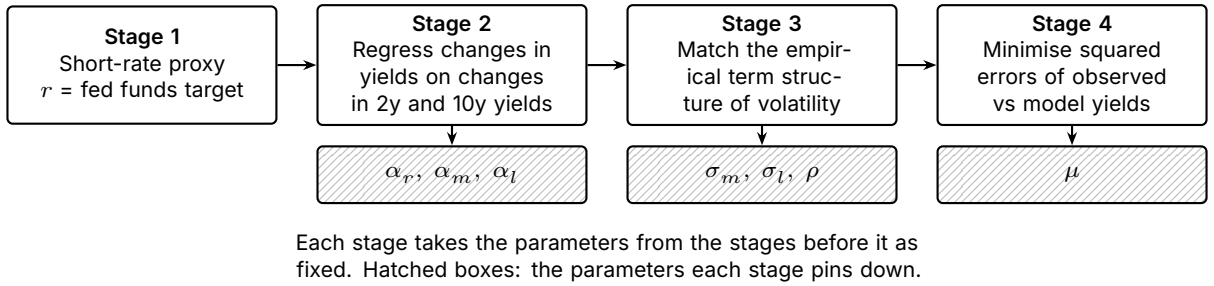


Figure 1. The simplified estimation of the Gauss+ model as four stages (the six steps of the table, with Steps 2 and 3, and 4 and 5, combined).

3. Why the staging works

- **The regression coefficients depend only on the mean reversion parameters, not on the volatility parameters.** So the α's can be pinned down first, from the regressions alone.
- With the α's fixed, the volatility and correlation parameters are chosen to match the **level and shape** of the volatility term structure.
- μ shifts the **level** of yields, not the sensitivities of their changes, so it comes last, from a least-squares fit of yield levels.

What the α's do to the term structure: the larger α_r, the faster changes in m and l reach the term structure; the larger α_m, the faster m converges to l and the less m affects longer-term yields; the smaller α_l, the slower l converges to μ and the more parallel the effect of l on all longer-term yields.

Wording of Step 2: Vardeez and Schweser versus GARP. Vardeez (following Schweser) writes Step 2 as "regress the changes in the short-term rate against changes in a medium-term rate (2-year) and a longer-term rate (10-year)". GARP's actual procedure regresses the changes in the **zero yields of various terms** on changes in the 2-year and the 10-year zero yields (Figure 16.5). For example, the 3-year yield loads 0.91 on the 2-year and 0.22 on the 10-year; the 15-year yield loads -0.20 and 1.10. The point both versions share, and the one exams test, is unchanged: these regression sensitivities identify only the mean reversion speeds. Expect either wording in a question stem.

4. Data choices and what the model is fitted to

- **Data:** daily, January 2014 to January 2022: the fed funds target rate and zero coupon bond prices derived from US Treasury prices (published by the Federal Reserve).
- **Why fed funds and not a repo rate?** A general collateral repo rate is theoretically more consistent with a Treasury term structure, but repo rates show occasional idiosyncratic jumps that complicate estimation without significant offsetting advantages.
- **Daily fit.** Once estimated, the factors are chosen each day to match the fed funds target and the **one-year rates two and ten years forward**. Those fitted points are fair by construction, which turns the model into a relative value tool for the rest of the curve.
- The factors m and l are **not** those forward rates themselves, in the way that r equals fed funds. How every forward rate relates to m and l depends on the estimated parameters, especially the α's.

5. Reading the estimates (GARP Table 16.1)

Parameter	Estimate	Half-life $= \ln 2/\alpha$ (years)
α_r	1.0547	0.66 (about 8 months)
α_m	0.6358	1.09 (about 13 months)
α_l	0.0165	42.01
σ_m	109.2 bp	
σ_l	96.4 bp	
ρ	0.212	
μ	10.555%	

- **Order of the speeds** is the one the economics predicts: central bank reaction (r to m) fastest, then m to l, then l to μ slowest by far.
- μ **above 10%** may look high, but l reverts to it very slowly and μ includes a **risk premium**.
- **Volatility parameters of 109 and 96 bp** translate into *lower* zero-yield volatilities, because the factors mean revert.



Figure 2. Half-lives of the three mean reversion processes (log scale; on a linear scale the first two would overlap). The long-term factor reverts about 64 times more slowly than the short rate: the "cascade of convergence" in numbers.

Parameter group	Matched to	NOT matched to
$\alpha_r, \alpha_m, \alpha_l$	Regression sensitivities of yield changes to 2y and 10y yield changes	Volatility term structure; yield levels
σ_m, σ_l, ρ	Empirical (or market-implied) term structure of volatility	Regression coefficients
μ	Least squares fit of observed versus model yield levels	Regressions; volatilities

Exam traps

- **Method swaps.** Speeds come from regressions, volatilities and ρ from the volatility term structure, μ from least squares on yields. Distractors move a parameter group to another group's method.
- **Reversed dependence.** The regression coefficients depend on the α's only, not on the volatilities. "Volatilities are set to one first" is invented.
- μ **is estimated last**, not first, and it shifts levels, not sensitivities.
- **The short-rate proxy is fed funds**, not a GC repo rate (repo has idiosyncratic jumps).
- m **and** l **are not the 2- and 10-year forward rates**; the model is fitted to those forwards each day.
- **Maximum likelihood** is the standard alternative, not the method used in the reading.

Practice questions: MR-16 d

Question MR-16 d.1 New • built from GARP MR Ch.16, 16.3

A quant is estimating the Gauss+ model from daily US data and must choose the empirical proxy for the short-term rate r. Which choice, with its justification, follows the reading?

A. The fed funds target rate, because general collateral repo rates show occasional idiosyncratic jumps that complicate the estimation without significant offsetting advantages.

B. A general collateral repo rate, because it is theoretically more consistent with a Treasury term structure and its jumps make the estimated volatility more realistic.

C. The two-year zero yield, because the short-term rate in the Gauss+ model is fitted each day to the one-year rate two years forward.

D. The fed funds target rate, because the short-term rate in the Gauss+ model carries its own random shock that must match fed funds volatility.

Question MR-16 d.2 Bank Q762 • Schweser Module Quiz 16.1 Q5

When using empirical data to estimate parameters for the Gauss+ model, which of the following parameters is calibrated by minimizing the squared errors of observed yields relative to model-predicted yields?

A. Long-term target.

B. Volatility parameter.

C. Correlation parameter.

D. Mean reversion speeds.

Question MR-16 d.3 Bank Q763 • AnalystPrep Q.6526 (rebuilt)

In the simplified procedure for estimating the Gauss+ model from empirical data, which of the following correctly matches a parameter with its estimation method?

A. Mean reversion speeds are found by minimizing squared errors between observed and model-implied yields.

B. The volatility and correlation parameters are found by regressing changes in the short-term rate on changes in medium- and long-term rates.

C. The long-term reversion level μ is found by minimizing squared errors between observed and model-implied yields across maturities.

D. Mean reversion speeds are found by matching the observed volatility term structure.

Question MR-16 d.4 Bank Q764 • AnalystPrep Q.6527 (rebuilt to GARP's procedure)

In the simplified approach to estimating the Gauss+ model, the mean reversion speeds are obtained first, from regressions of changes in zero yields of various terms on changes in the two-year and ten-year zero yields. Why can the speeds be estimated at this stage without first knowing the volatility parameters?

A. The regression coefficients depend only on the volatilities and correlation, which are observed directly.

B. The regression coefficients depend only on the mean reversion parameters, not on the volatility parameters.

C. The regression coefficients depend only on the long-term reversion level μ.

D. The regression coefficients depend on both the speeds and the volatilities, so the volatilities are first set to one.

Question MR-16 d.5 New • built from GARP MR Ch.16, 16.3 and Figure 16.5

In the estimation data, a regression of changes in the 3-year zero yield gives a coefficient of 0.91 on changes in the 2-year yield and 0.22 on changes in the 10-year yield; for the 15-year yield the coefficients are -0.20 and 1.10. Which of the following statements about this stage of the estimation is correct?

A. These coefficients are used to choose α_r, α_m and α_l so that the model's implied coefficients match them as closely as possible.

B. These coefficients are used to choose σ_m, σ_l and ρ so that the model's implied coefficients match them as closely as possible.

C. These coefficients are used to choose μ, because the coefficients shift with the level of long-term yields.

D. These coefficients cannot be matched by the model, so the speeds are instead taken from maximum likelihood estimates.

Question MR-16 d.6 New • built from GARP MR Ch.16, 16.3

After estimation, a trading desk uses the Gauss+ model for relative value. Each day the factors are chosen so that the model matches the fed funds target rate and the one-year rates two and ten years forward. Which statement is correct?

A. The medium- and long-term factors m and l are, by construction, equal to the one-year rates two and ten years forward.

B. The fitted forwards are fair by construction, and the relation of every forward rate to m and l depends on the estimated parameters, especially the mean reversion speeds.

C. Because the fitted forwards are fair by construction, every other point on the curve is also fair, so the model cannot identify rich or cheap securities.

D. The short-term rate r is fitted to the one-year rate two years forward, and m and l are both fitted to the ten-year forward.

Question MR-16 d.7 New • built from GARP MR Ch.16, Table 16.1

The Gauss+ estimates on US Treasury data are $\alpha_r = 1.0547$, $\alpha_m = 0.6358$, $\alpha_l = 0.0165$, $\sigma_m = 109.2$ bp, $\sigma_l = 96.4$ bp, $\rho = 0.212$ and $\mu = 10.555\%$. Which interpretation is correct?

A. The half-lives are about 0.66, 1.09 and 42 years, in the order the economics predicts, and μ seems high because it includes a risk premium and the long-term factor reverts to it very slowly.

B. The half-lives are about 0.66, 1.09 and 42 years, and the 109 and 96 bp volatility parameters translate into higher zero-yield volatilities because the factors mean revert.

C. The half-lives are about 1.05, 0.64 and 0.02 years, so the long-term factor reverts fastest of all, toward a μ that is a pure expectation of the short rate.

D. The half-lives are about 0.66, 1.09 and 42 years, and the long-term factor's slow reversion shows that μ was estimated first, from the regression stage.

Solutions • MR-16 d

Solution MR-16 d.1 — Answer: A

New • GARP MR Ch.16, 16.3

Step 1 – The reading's choice

"The short-term rate, r, is taken each day as equal to the fed funds target rate on that day."

Step 2 – The reason

A GC repo rate is "theoretically more consistent" with a Treasury term structure, but repo rates "exhibit occasional idiosyncratic jumps that complicate the estimation without significant offsetting advantages."

$\Rightarrow$ Fed funds target, for practical reasons.

Interpretation

The choice also fits the model's story: r reflects central bank policy, pegged at a target that moves with the business cycle.

Why the other options are wrong

B is incorrect. Takes the theoretical argument for repo but inverts the conclusion: the jumps are the reason *not* to use it.

C is incorrect. Confuses the proxy for r with the daily fitting targets; the two-year forward is fitted through the factors, not used as r.

D is incorrect. The short rate has *no* random shock in Gauss+ (that is the "+"); fed funds is chosen as a proxy, not to match a volatility.

LO: MR-16 d: Explain how the parameters of the Gauss+ model can be estimated from empirical data. • **Reference:** GARP MR Ch.16, 16.3

Solution MR-16 d.2 — Answer: A

Bank Q762 • Schweser Module Quiz 16.1 Q5

Step 1 – Map each parameter group to its stage

Speeds: regression sensitivities. Volatilities and correlation: the volatility term structure. Long-term level μ: least squares on yields.

Step 2 – Pick the least-squares parameter

Only μ is pinned down by fitting model yields to observed yields across the sample.

$\Rightarrow$ The long-term target.

Interpretation

μ sets the level to which the cascade ultimately reverts, so it is identified by yield levels, after the speeds and volatilities are fixed.

Why the other options are wrong

B is incorrect. Volatility parameters are fitted to the volatility term structure, not to yield levels.

C is incorrect. The correlation ρ is fitted together with the volatilities, to the volatility term structure.

D is incorrect. Mean reversion speeds come from the regression sensitivities of yield changes.

LO: MR-16 d: Explain how the parameters of the Gauss+ model can be estimated from empirical data. • **Reference:** GARP MR Ch.16, 16.3; SchweserNotes 2026 Book 1, LO 16.d

Solution MR-16 d.3 — Answer: C

Bank Q763 • AnalystPrep Q.6526

Step 1 – The correct pairings

Speeds $\alpha_r, \alpha_m, \alpha_l$: regression sensitivities (Steps 2 and 3). σ_m, σ_l, ρ: replicate the volatility term structure (Steps 4 and 5). μ: minimise squared yield errors (Step 6).

Step 2 – Test the options

Only C pairs a parameter with its own method.
$\Rightarrow$ μ by least squares across maturities.

Interpretation

Every distractor takes a correct method and attaches it to the wrong parameter group: the classic swap pattern for this LO.

Why the other options are wrong

A is incorrect. Swaps methods: speeds come from regressions, not from least squares on yields.
B is incorrect. Swaps methods: the regressions give the speeds, while volatility and correlation are fitted to the volatility term structure.
D is incorrect. The speeds do not depend on the volatilities, so matching the volatility term structure cannot identify them.

LO: MR-16 d: Explain how the parameters of the Gauss+ model can be estimated from empirical data. • **Reference:** GARP MR Ch.16, 16.3

Solution MR-16 d.4 — Answer: B

Bank Q764 • AnalystPrep Q.6527

Step 1 – What the regressions measure

The sensitivities of yield changes to changes in the 2-year and 10-year yields are functions of the reversion speeds and the maturity.

Step 2 – What the volatilities do

The volatilities scale the size of the factor moves but do not change these sensitivities.
$\Rightarrow$ "This staging is possible because the model regression coefficients depend only on the mean reversion parameters, not on the volatility parameters" (GARP).

Interpretation

The bank's original stem described the regression as "changes in a short-term rate proxy on changes in a medium-term and a longer-term rate" (the Schweser wording). The stem here follows GARP's procedure; the answer is the same either way.

Why the other options are wrong

A is incorrect. Reverses the dependence: the volatilities do not affect the coefficients.
C is incorrect. μ shifts yield levels and is calibrated last; it does not drive the sensitivities of changes.
D is incorrect. The coefficients do not depend on the volatilities, and no normalisation to one is used.

LO: MR-16 d: Explain how the parameters of the Gauss+ model can be estimated from empirical data. • **Reference:** GARP MR Ch.16, 16.3

Solution MR-16 d.5 — Answer: A

New • GARP MR Ch.16, Figure 16.5

Step 1 – What the coefficients are

They are the empirical sensitivities of each zero yield's changes to changes in the 2-year and 10-year zero yields (Figure 16.5).

Step 2 – What they identify

"All these regression coefficients implicitly describe the mean reversion parameters of the model." This stage chooses α_r, α_m, α_l so the model's implied coefficients match the empirical ones "as closely as possible."

$\Rightarrow$ The match is very close and within statistical confidence intervals.

Interpretation

Short maturities load mostly on the 2-year yield; long maturities load mostly on the 10-year yield (even negatively on the 2-year), which is the term-structure footprint of fast and slow mean reversion.

Why the other options are wrong

B is incorrect. Sibling swap: volatilities and ρ are matched to the volatility term structure, not to regression coefficients.

C is incorrect. μ is found last, from yield levels; the coefficients do not identify it.

D is incorrect. Polarity: the model matches the coefficients very closely; maximum likelihood is the standard alternative the reading avoids.

LO: MR-16 d: Explain how the parameters of the Gauss+ model can be estimated from empirical data. • **Reference:** GARP MR Ch.16, 16.3, Figure 16.5

Solution MR-16 d.6 — Answer: B

New • GARP MR Ch.16, 16.3

Step 1 – What is fitted each day

The factors are chosen to match the fed funds target and the one-year rates two and ten years forward, so "these two- and 10-year forward rates [are] fair by construction."

Step 2 – What m and l are

They are "in no way those forward rates themselves." The relation of every forward rate to m and l depends on the estimated parameters, in particular the mean reversion speeds.

$\Rightarrow$ B combines both facts.

Interpretation

Fitting a few points and reading the rest of the curve off the model is exactly what makes Gauss+ a relative value tool: securities away from the fitted points can be rich or cheap to the model.

Why the other options are wrong

A is incorrect. The factors are not the forward rates, unlike r, which does equal fed funds.

C is incorrect. Absolute claim: only the fitted points are fair by construction; the model is used to trade value elsewhere on the curve.

D is incorrect. r is set to the fed funds target, not to a forward rate.

LO: MR-16 d: Explain how the parameters of the Gauss+ model can be estimated from empirical data. • **Reference:** GARP MR Ch.16, 16.3

Solution MR-16 d.7 **Answer: A**

New • GARP MR Ch.16, Table 16.1

Step 1 – Half-lives

$$\frac{\ln 2}{1.0547} = 0.66, \qquad \frac{\ln 2}{0.6358} = 1.09, \qquad \frac{\ln 2}{0.0165} = 42.0 \text{ years}$$

$\Rightarrow$ About 8 months, 13 months and 42 years: central bank reaction fastest, then m to l, then l to μ.

Step 2 – Interpret μ and the volatilities

μ "might seem high at over 10%, but the long-term factor reverts very slowly to this target, and that target includes a risk premium." The 109 and 96 bp volatility parameters translate into *lower* zero-yield volatilities because of mean reversion.

Interpretation

The estimates come out in exactly the order the model's economic story predicts, which is a check on the estimation.

Why the other options are wrong

B is incorrect. Polarity flip: mean reversion makes zero-yield volatilities *lower* than the factor volatility parameters.

C is incorrect. Reports the speeds themselves as half-lives and reverses the order; μ includes a risk premium.

D is incorrect. μ is estimated last, by least squares on yield levels, not at the regression stage.

LO: MR-16 d: Explain how the parameters of the Gauss+ model can be estimated from empirical data. • **Reference:** GARP MR Ch.16, 16.3, Table 16.1

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 f

P2 CONCEPT

Describe alternative ways of characterizing the volatility smile.

Workbook: A3 Volatility Smiles section lists LOs a to i • no workbook question lands on this LO

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter ("full chapter is important"). The Tier A section A3 covers MR-17 a to i, but none of its reported exam shapes lands on this LO.
Vardeez: full chapter is important; no specific pointer for this LO.

Reported exam shapes: none from the recall workbooks.

What the questions look like: identify the alternative x-axis (strike, K/F_0, delta, a maturity-scaled measure), say why it is used, and know the at-the-money convention each one implies. Delta-axis questions test that a call's delta *falls* as the strike rises.

1. One y-axis, several possible x-axes

A volatility smile always has **implied volatility on the vertical axis**. What changes between the alternatives is the variable on the horizontal axis. The P1 blocks (MR-17 a to e) used the ratio K/S_0; every alternative below *replaces* that variable.

x-axis	What it does	At-the-money means
Strike K	Simplest, but the curve depends on the asset price: as S rises, the ATM strike rises and the whole curve moves **right** (left when S falls). A less stable picture.	$K = S_0$
K/S_0 (**moneyness**)	The standard choice (Figures 17.1 and 17.3 in GARP). Removes the horizontal drift of the strike axis.	$K/S_0 = 1$
K/F_0	A refinement: F_0 is the forward price for a contract maturing with the option.	$K = F_0$, because F_0, not S_0, is the expected price at maturity in a risk-neutral world
Option delta	Smile plotted against delta. Sometimes lets smiles be applied to options **other than European and American calls and puts**.	Call delta 0.5 or put delta -0.5 ("50-delta options")
$\frac{1}{\sqrt{T}} \ln\left(\frac{K}{F_0}\right)$	Scales moneyness by maturity. The smile is then **much less dependent on time to maturity**.	$K = F_0$ (value 0)

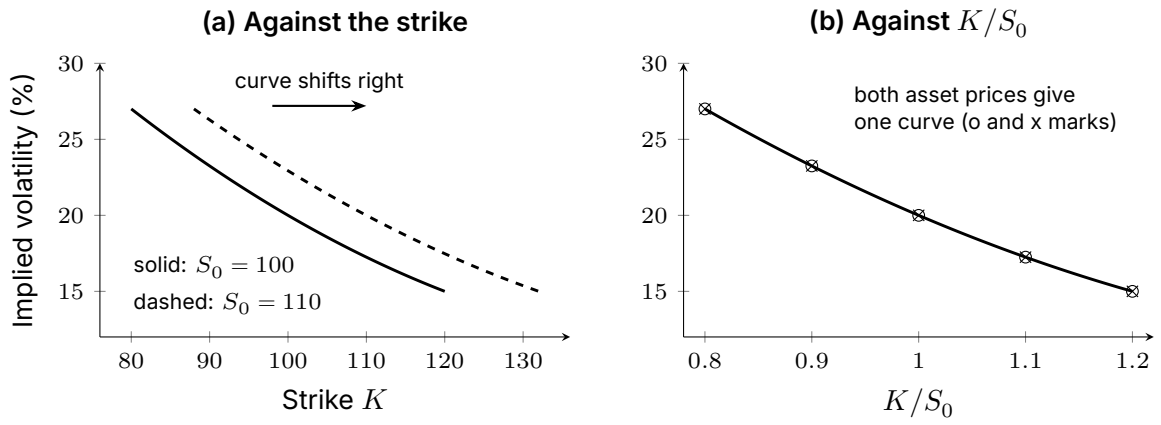


Figure 1. The same equity skew seen two ways (illustrative numbers). Plotted against the strike, the curve moves right when the asset price rises from 100 (solid) to 110 (dashed). Plotted against K/S_0, both asset prices give a single curve. This is why K/S_0 is the standard axis.

2. The delta axis, read correctly

- A call's delta **falls** as the strike rises: deep ITM calls (low strike) have delta near 1, deep OTM calls (high strike) near 0. So a delta axis running from low strike to high strike reads, for calls, something like $0.90,\ 0.75,\ 0.50,\ 0.25,\ 0.10$: monotone, not symmetric.
- A European put's delta lies between -1 and 0. The familiar "25–50–25" labels are shorthand (-0.25 put, 0.50 call, 0.25 call), not a list of actual call deltas.
- Illustration ($S_0 = 100$, $\sigma = 25\%$, $T = 1$, $r = 0$): call deltas at $K/S_0 = 0.8, 0.9, 1.0, 1.1, 1.2$ are $0.85, 0.71, 0.55, 0.40, 0.27$. Note that the 50-delta call is not exactly the $K = S_0$ call.

Precision point on the source notes. Vardeez (and Schweser) say plotting against K alone gives a "less stable" smile, and Vardeez glosses this as implied volatilities being "more inconsistent or less smooth". GARP's actual reason is narrower: the strike-axis curve **moves right or left as the asset price moves**, because the at-the-money strike moves with it (Figure 1). The curve is not less smooth; it drifts.

Exam traps

- **Delta direction.** Call delta decreases as strike increases. A delta sequence that rises with the strike, or that is symmetric (0.10, 0.25, 0.50, 0.25, 0.10), is wrong.
- **Put delta sign.** Put deltas are negative; the 50-delta put has delta -0.5, not $+0.5$.
- **ATM under K/F_0.** The argument is that F_0 is the *risk-neutral expected price at maturity*. It is not that $F_0 = S_0$, nor that delta is exactly 0.5 there.
- **What removes maturity dependence.** Only the scaled measure $\frac{1}{\sqrt{T}}\ln(K/F_0)$; K/F_0 alone does not.
- **Strike axis drift.** When S rises the strike-axis curve moves *right*, not up.

Practice questions: MR-17 f

Question MR-17 f.1

Bank Q825 • BT L2.T5.05.1

A trader plots the volatility smile for vanilla European options with option delta (instead of strike) on the x-axis. Moving from left to right along the axis (low strikes to high strikes), which is the most plausible sequence of actual deltas?

A. Call deltas $= [0.10,\ 0.25,\ 0.50,\ 0.75,\ 0.90]$

B. Call deltas $= [0.90,\ 0.75,\ 0.50,\ 0.25,\ 0.10]$

C. Call deltas $= [0.10,\ 0.25,\ 0.50,\ 0.25,\ 0.10]$

D. Put deltas $= [0.10,\ 0.25,\ 0.50,\ 0.25,\ 0.10]$

Question MR-17 f.2

Bank Q826 • BT notes P2.T5.05.01 (converted to multiple choice)

A volatility analyst proposes plotting the desk's smiles against option delta rather than against strike. Which of the following is a recognised advantage of characterizing the smile in terms of delta?

A. It sometimes makes it possible to apply volatility smiles to options other than European and American calls and puts.

B. It removes the need to back out implied volatilities, because delta can be observed directly in the market.

C. It makes the implied volatilities of calls and puts equal at the same strike, which a strike-based smile cannot do.

D. It defines an at-the-money option as one whose strike equals the current asset price, as on a strike axis.

Question MR-17 f.3

New • built from GARP Ch.17, 17.4

A risk analyst plots implied volatility against the strike price for a stock index on two dates. Between the dates the index rose about 10%, and the whole curve now sits further to the right. Which of the following statements is correct?

A. The curve moved right because the at-the-money strike rises with the asset price; this is why implied volatility is usually plotted against K/S_0 instead.

B. The curve moved right because implied volatility rose at every strike when the index rose; the strike axis shows this rise correctly.

C. Plotting against the strike is preferred, because the strike is fixed by contract while K/S_0 changes every day.

D. The shift shows that index prices are lognormal, because a lognormal smile moves one-for-one with the asset price.

Question MR-17 f.4 New • built from GARP Ch.17, 17.4

An FX options desk plots its smiles against K/F_0, where F_0 is the forward rate for the options' maturity, and defines an at-the-money option as one with $K = F_0$. Which of the following is the justification for this convention?

A. F_0, not S_0, is the expected value of the underlying on the option's maturity date in a risk-neutral world.

B. F_0 equals S_0 whenever interest rates are positive, so the two at-the-money definitions always coincide.

C. $K = F_0$ is the only strike at which a call has a delta of exactly 0.5 and a put a delta of exactly $+0.5$.

D. Using K/F_0 makes the smile independent of the option's maturity without any further adjustment.

Question MR-17 f.5 New • built from GARP Ch.17, 17.4

When the volatility smile is defined as the relationship between implied volatility and the delta of the option, how is an at-the-money option defined?

A. As a call with a delta of 0.5 or a put with a delta of -0.5.

B. As a call with a delta of 0.5 or a put with a delta of $+0.5$.

C. As any option whose strike equals the current asset price, whatever its delta.

D. As the call or put whose implied volatility is the lowest point on the smile.

Question MR-17 f.6 New • built from GARP Ch.17, 17.5

A financial engineer wants the smiles for 1-month, 1-year and 5-year options on the same asset to lie as close to one another as possible, so that one curve can serve all maturities. Defining T as time to maturity and F_0 as the forward price for the option's maturity, which horizontal axis does GARP's reading suggest?

A. $\dfrac{1}{\sqrt{T}} \ln\left(\dfrac{K}{F_0}\right)$

B. $\sqrt{T} \ln\left(\dfrac{K}{F_0}\right)$

C. K/S_0

D. The strike price K

Solutions • MR-17 f

Solution MR-17 f.1 — Answer: B

Bank Q825 • BT L2.T5.05.1

Step 1 – Direction of the axis

Left to right means the strike rises: from low-strike (in-the-money) calls to high-strike (out-of-the-money) calls.

Step 2 – How call delta moves with the strike

A call's delta falls as the strike rises, from near 1 (deep ITM) to near 0 (deep OTM).
$\Rightarrow$ The sequence must decrease monotonically: $0.90, 0.75, 0.50, 0.25, 0.10$.

Interpretation

The common "25–50–25" label is shorthand mixing puts and calls; the actual call deltas are monotone in strike.

Why the other options are wrong

A is incorrect. Reversed: call delta rises only as the strike falls, so this runs from high strike to low strike.

C is incorrect. Treats the symmetric shorthand labels as actual call deltas; delta is monotone in strike, not symmetric.

D is incorrect. European put deltas lie between -1 and 0, so a list of positive put deltas is impossible, and it is not monotone either.

LO: MR-17 f: Describe alternative ways of characterizing the volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.4

Solution MR-17 f.2 — Answer: A

Bank Q826 • BT notes P2.T5.05.01

Step 1 – What the delta axis buys

Hull: defining the smile against delta "sometimes makes it possible to apply volatility smiles to options other than European and American calls and puts". Delta is a standardised measure of moneyness that also carries maturity and volatility.
$\Rightarrow$ A states the recognised advantage.

Interpretation

The delta axis still plots *implied* volatility; it only changes the horizontal variable.

Why the other options are wrong

B is incorrect. The vertical axis is still implied volatility, which must be backed out of option prices; delta does not replace it.

C is incorrect. Put-call parity already makes call and put implied volatilities equal at the same strike and maturity, whatever the axis.

D is incorrect. Under the delta approach, at-the-money means a call delta of 0.5 or a put delta of -0.5, not $K = S_0$.

LO: MR-17 f: Describe alternative ways of characterizing the volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.4

Solution MR-17 f.3

Answer: A

New • GARP Ch.17, 17.4

Step 1 – Why the strike-axis curve drifts

The relationship between implied volatility and K depends on the asset price. As the asset price rises, the central at-the-money strike rises, so the curve moves right (left when the price falls).

Step 2 – The fix

Plot against K/S_0: the curve no longer drifts with the asset price (Figure 1).
$\Rightarrow$ A is GARP's argument.

Interpretation

The movement is horizontal, not vertical. Reading it as a change in the level of volatility is the trap in B.

Why the other options are wrong

B is incorrect. Polarity of the axis: the curve moved sideways because the ATM strike moved, not because volatility rose at every strike.
C is incorrect. Reversed preference: the drift of the strike axis is exactly why K/S_0 is preferred.
D is incorrect. A lognormal world gives a flat smile; the drift says nothing about lognormality.

LO: MR-17 f: Describe alternative ways of characterizing the volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.4

Solution MR-17 f.4

Answer: A

New • GARP Ch.17, 17.4

Step 1 – The refinement

Hull calculates the smile against K/F_0, with F_0 the forward price for the option's maturity, and traders often define at-the-money as $K = F_0$.

Step 2 – The argument

F_0, not S_0, is the expected asset price on the maturity date in a risk-neutral world.
$\Rightarrow$ A is the stated justification.

Interpretation

The convention centres the smile on where the asset is expected to be at expiry, not where it is today.

Why the other options are wrong

B is incorrect. $F_0 = S_0 e^{(r-q)T}$ differs from S_0 whenever the carry is not zero, so the two definitions generally differ.
C is incorrect. Put deltas are negative; the 50-delta put has delta -0.5. The delta definition of ATM is a different convention.
D is incorrect. Removing the maturity dependence needs the scaled measure $\frac{1}{\sqrt{T}} \ln(K/F_0)$, not K/F_0 alone.

LO: MR-17 f: Describe alternative ways of characterizing the volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.4

Solution MR-17 f.5

Answer: A

New • GARP Ch.17, 17.4

Step 1 – The delta convention

When the smile is defined against delta, an at-the-money option is a call with delta 0.5 or a put

with delta -0.5. These are "50-delta options".
$\Rightarrow$ A.

Interpretation

Because put delta = call delta -1 (no dividends), a 0.5 call and a -0.5 put at the same strike are consistent with each other.

Why the other options are wrong

B is incorrect. Sign error: a European put's delta is negative.
C is incorrect. That is the strike-based definition ($K = S_0$); under the delta approach the 50-delta option is generally not the $K = S_0$ option.
D is incorrect. The lowest point of a smile is a market outcome, not a definition of at-the-money; for an equity skew there is no interior minimum at all.

LO: MR-17 f: Describe alternative ways of characterizing the volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.4

Solution MR-17 f.6 Answer: A

New • GARP Ch.17, 17.5

Step 1 – The maturity problem

Smiles become less pronounced as maturity increases, so curves for different maturities differ when plotted against K or K/S_0.

Step 2 – The scaled measure

Some financial engineers define the smile against $\frac{1}{\sqrt{T}} \ln(K/F_0)$; the smile is then usually much less dependent on maturity.
$\Rightarrow$ A.

Interpretation

Dividing by $\sqrt{T}$ puts moneyness in units that grow with the square root of time, the same way the standard deviation of returns grows.

Why the other options are wrong

B is incorrect. Multiplies by $\sqrt{T}$ instead of dividing: it stretches long maturities further apart rather than bringing them together.
C is incorrect. K/S_0 removes the drift with the asset price but not the dependence on maturity.
D is incorrect. The strike axis also drifts with the asset price, so it is the least stable choice.

LO: MR-17 f: Describe alternative ways of characterizing the volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.5

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 g

P2 CONCEPT

Describe volatility term structures and volatility surfaces and how they may be used to price options.

Workbook: A3 Volatility Smiles section lists LOs a to i • no workbook question lands on this LO

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter ("full chapter is important"). No Tier A or B workbook shape lands on it.
Vardeez: full chapter is important; no specific pointer for this LO.

Reported exam shapes: none from the recall workbooks.

What the questions look like: which way the term structure slopes and why; what the axes of a surface are; which maturity has the more pronounced smile; how a trader uses the surface to price an option. GARP's own practice exam also tests *reading* a surface by linear interpolation (Question MR-17 g.4): the one applied form of this LO.

1. The volatility term structure

The **volatility term structure** lists implied volatilities of **at-the-money** options as a function of **time to maturity**.

- **Short-dated volatility historically low** ⇒ implied volatility tends to be an **increasing** function of maturity, because volatilities are expected to rise.
- **Short-dated volatility historically high** ⇒ a **decreasing** function of maturity, because volatilities are expected to fall.
- This is related to, but slightly different in meaning from, the mean reversion often seen in implied volatility. A GARCH(1,1) world gives the same picture: the slope depends on whether current volatility sits below or above its long-run level.

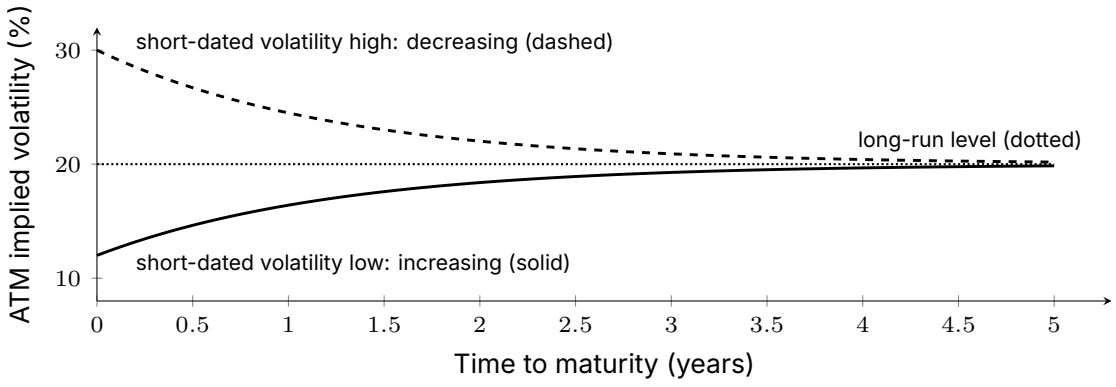


Figure 2. Two volatility term structures (illustrative). Both converge on the same long-run level: rising when short-dated volatility is low, falling when it is high.

2. The volatility surface

A **volatility surface** combines volatility smiles with the term structure: it tabulates the implied volatility appropriate for pricing an option with **any strike and any maturity**. One dimension is K/S_0, the other is time to maturity; the body holds BSM implied volatilities.

- **How it is filled.** Entries for options with reliable market data are calculated directly from their market prices. The rest of the table is typically determined by **interpolation**.
- **Shape.** The smile becomes **less pronounced as maturity increases**, for currency options and for most other assets. The reason (GARP 17.2): the percentage impact of jumps on implied volatility fades with maturity (over long horizons jumps get "averaged out"), and the percentage impact of nonconstant volatility on implied volatility usually becomes less pronounced too.
- **Axes.** Term structure: implied volatility against maturity (ATM). Smile: implied volatility against strike (one maturity). Surface: implied volatility against both.

	K/S_0				
Maturity	0.90	0.95	1.00	1.05	1.10
1 month	14.2	13.0	12.0	13.1	14.5
3 month	14.0	13.0	12.0	13.1	14.2
6 month	14.1	13.3	12.5	**13.4**	14.3
1 year	**14.7**	**14.0**	13.5	**14.0**	14.8
2 year	**15.0**	**14.4**	14.0	14.5	15.1
5 year	14.8	14.6	14.4	14.7	15.0

GARP Table 17.2: a volatility surface for foreign currency options (%). Bold cells are used in the two worked look-ups below.

Pricing a new option with the surface. Look up, or interpolate, the implied volatility for the option's maturity and K/S_0, then put that volatility into BSM or a binomial tree.

- 9-month option, $K/S_0 = 1.05$: halfway between 6 months (13.4) and 1 year (14.0), so $\sigma = 13.7\%$.
- 1.5-year option, $K/S_0 = 0.925$ (bilinear): 1 year $\to (14.7 + 14.0)/2 = 14.35$; 2 years $\to (15.0 + 14.4)/2 = 14.70$; halfway in maturity $\to \sigma = 14.525\%$.

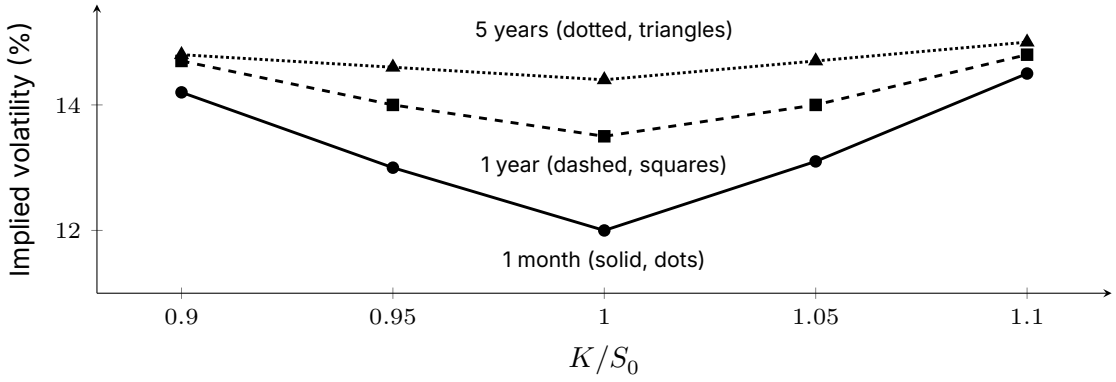


Figure 3. Three rows of GARP Table 17.2 drawn as smiles. Reading along a curve gives a smile; reading across the curves at $K/S_0 = 1$ gives the term structure. The 1-month smile spans 2.5 points, the 5-year smile only 0.6: the smile flattens with maturity.

3. What the model is doing (GARP 17.7)

- If traders use a different volatility for every option, BSM is arguably **no more than a sophisticated interpolation tool**: it keeps a new option's price consistent with the market prices of actively traded options.
- If traders switched to another plausible model, the surface and the smile would change, but arguably the **dollar prices quoted in the market would not change appreciably**.
- **Greek letters, and therefore hedging, do depend on the model.** An unrealistic model is liable to lead to poor hedging (see MR-17 h).
- Models matter most for derivatives that **do not trade actively**: many nonstandard exotics are model-dependent.
- Schweser's framing: the term structure and surface let traders confirm or reject a model's accuracy and consistency against market prices.

	Smile	Term structure	Surface
Plots	Implied vol vs strike (K/S_0)	ATM implied vol vs maturity	Implied vol vs strike *and* maturity
Holds fixed	Maturity	Moneyness (ATM)	Nothing
Shape facts	FX: smile; equity: skew	Up if short vol low, down if high	Smile flattens as maturity rises
Use	Price a strike	Price a maturity	Price any strike and maturity, by interpolation

Exam traps

- **Slope polarity.** High short-dated volatility gives a *downward*-sloping term structure; low gives upward.
- **Which smile is sharper.** Short-dated. Long-dated smiles are flatter because jumps average out.
- **Prices versus implied volatility.** The percentage impact of nonconstant volatility on *prices* grows with maturity, but its impact on *implied volatility* usually shrinks. Do not carry the first to the second.
- **Interpolation.** Interpolate in strike at both bracketing maturities, then in maturity. Typical errors: one dimension only, weights reversed, nearest grid point.
- **One volatility for all.** Marking every option at the ATM or an average implied volatility ignores the smile.
- **Model dependence.** Prices of liquid options are roughly model-free once the surface is fitted; Greeks and hedges are not.

Practice questions: MR-17 g

Question MR-17 g.1

Bank Q836 • BT notes P2.T5.01.2 (converted to multiple choice)

A junior analyst is labelling two charts in a volatility report. Which of the following correctly describes the axes of (i) a volatility term structure and (ii) a volatility surface?

A. (i) Implied volatility against time to maturity for at-the-money options; (ii) implied volatility against both strike (or K/S_0) and time to maturity.

B. (i) Implied volatility against strike for a fixed maturity; (ii) implied volatility against strike and the current asset price.

C. (i) Historical volatility against time to maturity; (ii) historical volatility against strike and time to maturity.

D. (i) At-the-money implied volatility against time to maturity; (ii) at-the-money implied volatility against maturity and the risk-free rate.

Question MR-17 g.2

Bank Q832 • BT L2.T5.06.1 (rebuilt)

Which of the following is the most plausible explanation for a downward-sloping implied volatility term structure?

A. Short-dated volatility is currently high relative to its long-run level, and volatility is mean reverting.

B. Short-dated volatility is currently low relative to its long-run level, and volatility is mean reverting.

C. Returns have lighter tails than the lognormal distribution (platykurtosis).

D. Returns are independent over time, so volatility is the same at all maturities.

Question MR-17 g.3

Bank Q837 • BT notes P2.T5.07.02 (converted to multiple choice)

A currency options trader compares the volatility smile for 3-month options with the smile for 2-year options on the same exchange rate. Which of the following is most likely?

A. The 3-month smile is more pronounced, because the percentage impact of jumps on implied volatility becomes less pronounced as maturity increases.

B. The 2-year smile is more pronounced, because the percentage impact of nonconstant volatility on option prices grows with maturity, and implied volatility follows prices.

C. The two smiles are equally pronounced, because implied volatility depends on K/S_0 and not on maturity.

D. The 2-year smile is more pronounced, because jumps accumulate over a longer life and make both tails heavier.

Question MR-17 g.4

Bank Q838 • GARP PE2 Q3 • Mock 4 Q4

A market-maker on the FX desk must quote an FX call option expiring in 7 months with $K/S_0 = 1.075$. The implied volatility surface (%) is:

Time to expiration	$K/S_0 = 0.90$	0.95	1.00	1.05	1.10
1 month	9.25	8.55	8.05	8.70	9.45
3 months	9.10	8.70	8.30	8.75	9.15
6 months	9.45	9.05	8.70	9.10	9.45
1 year	9.65	9.50	9.35	9.55	9.75

What implied volatility should the market-maker use?

A. 9.18%

B. 9.28%

C. 9.34%

D. 9.65%

Question MR-17 g.5

Bank Q834 • AP Mock 2 Q46

A trader wishes to mark to market a book of plain-vanilla stock options. He is long at-the-money calls and short deep out-of-the-money calls. The trader's bonus increases as the value of his book increases, and there is a pronounced smile on these options. Which approach should the trader use to mark the book?

A. Use historical volatility so as to correct for pricing mistakes inherent in the option market.

B. For each option, use the implied volatility of the most similar option tradable on the market.

C. Use the average of the implied volatilities for the traded options, since all options should have the same implied volatility under Black-Scholes.

D. Use the implied volatility of at-the-money options, since volatility estimation is more reliable there.

Question MR-17 g.6

New • built from GARP Ch.17, 17.5

A model validation team asks how the desk's FX volatility surface is populated before it is used to price new options. Which of the following describes standard practice?

A. Entries for options with reliable market data are implied volatilities calculated directly from their market prices; the rest of the table is typically filled by interpolation.

B. All entries are historical volatilities measured over a window equal to each option's maturity, then adjusted for the strike.

C. All entries are derived from the at-the-money implied volatility, scaled by $\sqrt{T}$ for maturity and held constant across strikes.

D. Entries are BSM option prices rather than volatilities, and a trader reads the price of a new option straight from the table.

Question MR-17 g.7 Bank Q829 • GARP book Ch.17 Q17.21 (converted to multiple choice)

A trader argues that "the Black-Scholes-Merton model is used by traders as an interpolation tool". Which of the following statements about the role of the pricing model is correct?

A. If traders switched from BSM to another plausible model, the volatility surface would change, but the dollar prices quoted for actively traded options would arguably not change appreciably.

B. If traders switched model, the dollar prices of actively traded options would change materially, while the Greek letters and hedges would stay the same.

C. The choice of model matters most for actively traded plain-vanilla options and least for non-standard exotic derivatives.

D. Because BSM is only an interpolation tool, the hedge ratios it produces do not depend on which model is used.

Solutions • MR-17 g

Solution MR-17 g.1 — Answer: A

Bank Q836 • BT notes P2.T5.01.2

Step 1 – Term structure

Implied volatility (vertical) against time to maturity (horizontal), for at-the-money options.

Step 2 – Surface

Three-dimensional: implied volatility against both strike (usually K/S_0) and time to maturity. It combines smiles with the term structure.

$\Rightarrow$ A.

Interpretation

The surface's slices are the smile (fixed maturity) and the term structure (fixed moneyness).

Why the other options are wrong

B is incorrect. Sibling swap: (i) describes a smile, and the surface's second dimension is maturity, not the asset price.

C is incorrect. Both objects are built from *implied* volatilities backed out of market prices, not historical volatilities.

D is incorrect. Part (i) is right, but the surface's dimensions are strike and maturity; the risk-free rate is an input, not an axis.

LO: MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.5

Solution MR-17 g.2 — Answer: A

Bank Q832 • BT L2.T5.06.1

Step 1 – What a longer-dated implied volatility represents

Roughly the expected average volatility over the option's life.

Step 2 – Apply mean reversion

If volatility is currently high, it is expected to fall toward its long-run level, so the average over longer lives is lower.

$\Rightarrow$ A high current level plus mean reversion gives a downward-sloping term structure (Figure 2, dashed).

Interpretation

GARP states the same pattern: decreasing in maturity when short-dated volatilities are historically high, because volatilities are expected to decrease.

Why the other options are wrong

B is incorrect. Polarity flip: low short-dated volatility gives an *upward*-sloping term structure.

C is incorrect. Tail shape changes the smile across strikes, not the slope across maturities.

D is incorrect. Constant volatility with independent returns gives a flat term structure, not a downward slope.

LO: MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.5

Solution MR-17 g.3 — Answer: A

Bank Q837 • BT notes P2.T5.07.02

Step 1 – What GARP says about maturity

The percentage impact of jumps on both prices and implied volatility becomes less pronounced as maturity increases; over long horizons jumps are "averaged out".

Step 2 – Nonconstant volatility

Its percentage impact on prices grows with maturity, but its impact on *implied volatility* usually becomes less pronounced.

$\Rightarrow$ Net result: the smile flattens as maturity rises, so the 3-month smile is the sharper one (Figure 3).

Interpretation

This holds for currency options and for options on most other assets.

Why the other options are wrong

B is incorrect. Transfers the price effect to implied volatility: the impact on prices grows with maturity, but the impact on implied volatility usually shrinks.

C is incorrect. Implied volatility depends on maturity as well as moneyness; that is the whole point of the surface.

D is incorrect. Reversed: jumps average out over long horizons, so long-dated distributions look closer to lognormal.

LO: MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 and 17.5

Solution MR-17 g.4 — Answer: C

Bank Q838 • GARP PE2 Q3

Step 1 – Bracket the option

Maturities: 6 months and 1 year. Strikes: 1.05 and 1.10; $K/S_0 = 1.075$ is their midpoint. 7 months is $1/6$ of the way from 6 to 12 months.

Step 2 – Interpolate in strike at each maturity

$$6 \text{ months}: \frac{9.10+9.45}{2} = 9.275 \qquad 1 \text{ year}: \frac{9.55+9.75}{2} = 9.65$$

Step 3 – Interpolate in maturity

$$\sigma = 9.275 + \tfrac{1}{6}(9.65 - 9.275) = 9.3375\% \approx 9.34\%$$

$\Rightarrow$ This is the volatility the market-maker puts into BSM to quote the option.

Interpretation

This is exactly how a surface is used to price an option whose strike and maturity are not on the grid.

Why the other options are wrong

A is incorrect. Interpolated in time only, at the 1.05 column: $9.10 + \frac{1}{6}(9.55 - 9.10) = 9.175$, ignoring the move to 1.075.

B is incorrect. Interpolated in strike only, at 6 months: 9.275, ignoring the extra month.

D is incorrect. Took the 1-year strike midpoint, 9.65, ignoring that 7 months is close to 6 months.

LO: MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.5, Table 17.2

Solution MR-17 g.5 — Answer: B

Bank Q834 • AP Mock 2 Q46

Step 1 – What marking to market requires

Values consistent with market prices for each strike and maturity.

Step 2 – Apply the smile

With a pronounced smile, each strike has its own implied volatility. Use the implied volatility of the most similar traded option (equivalently, read it off the surface).

$\Rightarrow$ B.

Interpretation

The incentive in the stem matters: a single volatility for every strike would let the trader bias the marks on the short OTM calls in his favour.

Why the other options are wrong

A is incorrect. Historical volatility is not a market price, so the marks would not be market-consistent.

C is incorrect. Averaging relies on the constant-volatility assumption that the smile contradicts.

D is incorrect. One ATM volatility ignores the smile and misprices the deep OTM calls he is short.

LO: MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.5

Solution MR-17 g.6 — Answer: A

New • GARP Ch.17, 17.5

Step 1 – Where the numbers come from

At any time some entries correspond to options with reliable market data; their implied volatilities are calculated directly from market prices.

Step 2 – The rest

The remaining entries are typically determined by interpolation.

$\Rightarrow$ A.

Interpretation

The surface is therefore anchored to traded prices; interpolation fills the gaps so any strike and maturity can be priced.

Why the other options are wrong

B is incorrect. The surface holds *implied* volatilities, not historical ones.

C is incorrect. Holding volatility constant across strikes would erase the smile the surface exists to capture.

D is incorrect. The table holds volatilities; the price comes from putting the looked-up volatility into BSM or a binomial tree.

LO: MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.5

Solution MR-17 g.7 — Answer: A

Bank Q829 • GARP book Ch.17 Q17.21

Step 1 – Why BSM is an interpolation tool

Traders use a different volatility for every option, so BSM serves to translate between a market

price and an implied volatility, and to price new options consistently with traded ones.

Step 2 – What changes if the model changes

The surface and smile would change, but quoted dollar prices of actively traded options arguably would not change appreciably. Greek letters and hedging strategies *do* depend on the model.

$\Rightarrow$ A.

Interpretation

Model risk sits in hedging and in instruments that do not trade actively, not in the prices of liquid vanillas.

Why the other options are wrong

B is incorrect. Reversed on both counts: prices of traded options are roughly model-free once fitted; Greeks are model-dependent.

C is incorrect. Reversed: models matter most for derivatives that do not trade actively, such as many nonstandard exotics.

D is incorrect. Hedge ratios are model-dependent; an unrealistic model is liable to lead to poor hedging.

LO: MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.7

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 h

P2 CONCEPT

Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures.

Workbook: A3 Volatility Smiles section lists LOs a to i • no workbook question lands on this LO

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter ("full chapter is important"). No Tier A or B workbook shape lands on it.
Vardeez: full chapter is important. His one result to memorise: $\Delta_{\text{BSM}} > \Delta_{\text{minimum variance}}$ (for equity options).

Reported exam shapes: none from the recall workbooks.

What the questions look like: the two effects of an equity price move (slide along the skew versus shift of the whole curve), which one dominates, and the direction of the minimum variance delta relative to the BSM delta, for calls and for puts.

1. The assumption the Greeks make

The formulas for delta and the other Greeks assume that **implied volatility stays the same when the asset price changes**. With a smile, that is not what is usually expected to happen, so the textbook Greeks can mislead.

2. Two effects when an equity price rises

1. **Slide along the skew.** S rises, so K/S_0 falls, and the option moves *left* along the downward-sloping skew to a **higher** implied volatility.
2. **Shift of the whole curve.** Equity prices and volatility are **negatively correlated**, so the entire curve moves **down** when the price rises (and up when it falls).

In practice **effect 2 dominates**: implied volatility tends to fall when the equity price rises, and rise when it falls.

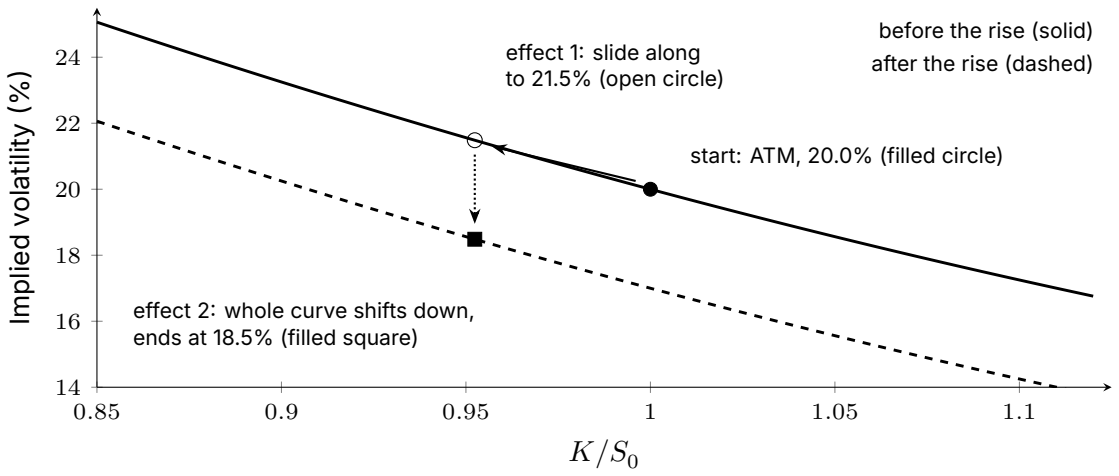


Figure 4. An equity price rise of 5% (illustrative). Effect 1 alone would raise implied volatility from 20.0% to 21.5%. The downward shift of the whole curve (effect 2) dominates, so implied volatility ends lower, at 18.5%.

3. The minimum variance delta

The delta that takes the relationship between implied volatility and the equity price into account is the **minimum variance delta**:

$$\Delta_{MV} = \Delta_{BSM} + \nu_{BSM} \frac{\partial E(\sigma_{imp})}{\partial S}$$

where Δ_{BSM} and ν_{BSM} (vega) come from constant-volatility BSM, and $E(\sigma_{imp})$ is the expected implied volatility as a function of S.

- $\nu_{BSM} > 0$ for calls and puts alike.
- For equities $\partial E(\sigma_{imp})/\partial S < 0$ (effect 2 dominates).
- So $\Delta_{MV} < \Delta_{BSM}$: **the BSM delta is higher than the minimum variance delta**. For a put this means the true delta is *more negative* than the BSM delta (e.g. below -0.40 when BSM gives -0.40).
- A hedger who uses the BSM delta on an equity option holds more of the underlying than the minimum variance hedge requires.

Precision point on vega. Vega is the sensitivity of the option's price to volatility; the smile is market implied volatility plotted against strike. One does not dictate the other: BSM vega is positive and peaks near the money whatever shape the smile takes (Question MR-17 h.5). A flat smile does not make vega zero.

Formula to recognise: MR-17 h

Minimum variance delta	$\Delta_{MV} = \Delta_{BSM} + \nu_{BSM}\,\partial E(\sigma_{imp})/\partial S$
Sign for equity options	$\nu_{BSM} > 0, \partial E(\sigma_{imp})/\partial S < 0 \Rightarrow \Delta_{MV} < \Delta_{BSM}$ (calls and puts)
Along a fixed skew only	$S \uparrow \Rightarrow K/S_0 \downarrow \Rightarrow \sigma_{imp} \uparrow$: sensitivity *above* BSM delta (effect 1 alone)

Exam traps

- **Which effect dominates.** The shift of the whole curve, not the slide along it.
- **Direction of the comparison.** BSM delta $>$ minimum variance delta for equity options. "Minimum variance delta is higher" is the standard distractor.
- **Correlation sign.** The curve falls when prices rise *because* the correlation is negative; a positive correlation would push it the other way.
- **Puts.** The adjustment is negative for puts too, so the true put delta is more negative than BSM's.
- **Along-the-curve-only questions.** If a stem says "ignore shifts of the whole curve", only effect 1 applies, and on a downward skew the sensitivity is *above* the BSM delta.
- **Vega versus the smile.** Vega is $\partial(\text{price})/\partial\sigma$, not the slope of the smile.

Practice questions: MR-17 h

Question MR-17 h.1

Bank Q839 • GARP book Ch.17 Q17.4 (converted to multiple choice)

A derivatives risk manager asks what is meant by a "minimum variance delta" for an equity index option. Which of the following is correct?

A. It is the delta that allows for the expected change in implied volatility when the asset price changes; for equity options it is lower than the BSM delta.

B. It is the delta that holds implied volatility constant when the asset price changes; for equity options it is higher than the BSM delta.

C. It is the delta that minimises vega exposure; for equity options it equals the BSM delta because vega is always positive.

D. It is the delta that allows for the expected change in implied volatility; for equity options it is higher than the BSM delta because the option moves up the skew.

Question MR-17 h.2

Bank Q846 • Schweser Module Quiz 17.2 Q1

When evaluating the impact of a volatility smile on the calculation of the option Greeks, which of the following statements is most likely correct?

A. If equity prices and volatilities are negatively correlated, equity price increases will move the option up the existing curve.

B. If equity prices and volatilities are positively correlated, equity price increases will move the entire option curve down.

C. The minimum variance delta will be higher than the Black-Scholes-Merton model delta.

D. When equity prices change, the effect from movements in the implied volatility curve up or down tends to dominate movements along the existing curve.

Question MR-17 h.3 Bank Q841 • BT L2.T5.08.2 (rebuilt, key corrected)

An at-the-money call ($S = K = \$20$) is priced in an equity market with "crashophobia": implied volatility is a decreasing function of K/S (a downward-sloping skew). Consider only the movement along this fixed skew (ignore any shift of the whole curve). When the stock rises to \$21, how does the sensitivity of the option price to the stock price compare with the BSM delta computed with constant volatility?

A. It is the same, because delta depends only on $N(d_1)$.

B. It is higher, because the rise in S lowers K/S and moves the option to a point with higher implied volatility.

C. It is lower, because the rise in S lowers K/S and moves the option to a point with lower implied volatility.

D. It is indeterminate, because the sign of the vega term depends on the strike.

Question MR-17 h.4 Bank Q845 • BT P2.T5.411.2 (rebuilt, key corrected)

A one-year at-the-money equity call with strike 100 trades at 13.28, implying a volatility of 30.0% (risk-free rate 3%). The Black-Scholes-Merton (BSM) delta is $N(d_1) = 0.60$ for the call and $0.60 - 1.0 = -0.40$ for the put with the same strike and maturity. The equity option exhibits a steep implied volatility skew (heavy left tail, light right tail). Empirically, equity prices and volatility are negatively correlated, so when the stock price rises the whole implied volatility curve shifts down, and this shift dominates the move along the existing curve. Which statement best describes the true (minimum variance) deltas relative to the BSM deltas of 0.60 and -0.40?

A. True call delta is higher than 0.60 and true put delta is higher than -0.40.

B. True call delta is lower than 0.60 and true put delta is lower than -0.40.

C. True call delta is equal to 0.60 and true put delta is equal to -0.40.

D. The relationship cannot be determined because delta is only a first-order approximation.

Question MR-17 h.5 Bank Q844 • BT P2.T5.409.1

In regard to the relationship, if any, between Greek vega and implied volatility, which of the following is TRUE?

A. If the implied volatility smile is a perfectly flat, horizontal line, then an increase in strike price has no effect on the option's price.

B. If the implied volatility smile is a perfectly flat, horizontal line, then Greek vega is zero.

C. Vega is the first derivative of the implied volatility smile; i.e., it plots the slopes of lines tangent to the implied volatility plot.

D. Under the Black-Scholes-Merton model, vega is always positive and highest for an at-the-money option, but these properties do not predict any particular shape of the implied volatility smile; a skew, smirk, frown or other shape is allowed.

Question MR-17 h.6 New • built from GARP Ch.17, 17.6

A hedger is short a large position in at-the-money calls on a stock index that shows the usual equity skew, and delta-hedges by buying the index using the Black-Scholes-Merton delta. Implied volatilities on the index tend to fall when the index rises. Relative to the minimum variance hedge, which of the following best describes her position?

A. She holds more of the index than the minimum variance hedge requires, because the BSM delta is higher than the minimum variance delta.

B. She holds less of the index than the minimum variance hedge requires, because the BSM delta ignores the rise in implied volatility along the skew.

C. Her hedge equals the minimum variance hedge, because positive vega offsets any change in implied volatility.

D. She holds more of the index than required when it rises and less when it falls, because the curve shifts in both directions.

Solutions • MR-17 h

Solution MR-17 h.1

Answer: A

Bank Q839 • GARP book Ch.17 Q17.4

Step 1 – Definition

The minimum variance delta takes into account the relationship between implied volatility and the asset price: $\Delta_{MV} = \Delta_{BSM} + \nu_{BSM}\,\partial E(\sigma_{imp})/\partial S$.

Step 2 – Sign for equities

$\nu_{BSM} > 0$ and, because of the negative price-volatility correlation, $\partial E(\sigma_{imp})/\partial S < 0$.
$\Rightarrow \Delta_{MV} < \Delta_{BSM}$.

Interpretation

It is the delta that minimises the variance of the hedged position once the expected move in implied volatility is allowed for.

Why the other options are wrong

B is incorrect. Holding implied volatility constant is what the BSM delta does; and the comparison is reversed.

C is incorrect. It is not about minimising vega, and it differs from the BSM delta precisely because vega is positive and volatility moves.

D is incorrect. Right definition, wrong direction: the downward shift of the whole curve dominates the move up the skew.

LO: MR-17 h: Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.6

Solution MR-17 h.2

Answer: D

Bank Q846 • Schweser Module Quiz 17.2 Q1

Step 1 – The two effects

A price change moves the option along the existing curve (K/S changes) and shifts the whole curve (negative correlation).

Step 2 – Which dominates

In practice the shift of the whole curve dominates movement along it.
$\Rightarrow$ D.

Interpretation

That dominance is why the minimum variance delta sits below the BSM delta.

Why the other options are wrong

A is incorrect. Moving up the existing curve is the along-the-curve effect, which comes from the skew's slope, not from the negative correlation; the correlation moves the whole curve down when prices rise.

B is incorrect. Reverses the correlation: the curve falls when prices rise *because* the correlation is negative.

C is incorrect. Reversed: the BSM delta is higher than the minimum variance delta.

LO: MR-17 h: Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.6; SchweserNotes 2026 Book 1, Reading 17

Solution MR-17 h.3

Answer: B

Bank Q841 • BT L2.T5.08.2

Step 1 – Total sensitivity

$$\frac{dC}{dS} = \Delta_{BSM} + \nu \frac{d\sigma}{dS}$$

Step 2 – Along a fixed downward skew

S from 20 to 21 lowers K/S from 1.00 to 0.952. On a downward-sloping skew, lower K/S means higher implied volatility, so $d\sigma/dS > 0$.
$\Rightarrow$ With $\nu > 0$, the extra term is positive: the sensitivity exceeds the BSM delta.

Interpretation

The stem switches off effect 2. With only effect 1 at work, the answer is the opposite of the minimum variance result; read the stem's scope carefully.

Why the other options are wrong

A is incorrect. Ignores that implied volatility changes as the option moves along the skew.
C is incorrect. Gets the skew backwards: lower K/S sits on the high-volatility side of an equity skew.
D is incorrect. A call's vega is always positive, so the sign is determinate.

LO: MR-17 h: Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.6

Solution MR-17 h.4

Answer: B

Bank Q845 • BT P2.T5.411.2

Step 1 – Check the BSM inputs

$d_1 = \frac{0.03+0.30^2/2}{0.30} = 0.25$, $N(d_1) = 0.599 \approx 0.60$; BSM call value 13.28 (verified). The data are consistent.

Step 2 – Sign of the adjustment

True delta $= \Delta_{BSM} + \nu\, d\sigma/dS$. The dominant downward shift of the curve makes net $d\sigma/dS < 0$; vega is positive for calls and puts.
$\Rightarrow$ Both adjustments are negative: true call delta < 0.60, true put delta < -0.40 (more negative).

Interpretation

Put delta $=$ call delta -1, so the same negative adjustment moves the put delta further below zero.

Why the other options are wrong

A is incorrect. Uses only the move along the existing curve, ignoring the dominant downward shift.
C is incorrect. Assumes implied volatility is unchanged when S moves, which is the BSM assumption the smile contradicts.
D is incorrect. Being a first-order measure does not stop a directional comparison; the sign of the vega term is known.

LO: MR-17 h: Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.6

Solution MR-17 h.5 — Answer: D

Bank Q844 • BT P2.T5.409.1

Step 1 – Two different objects

Vega $= \partial(\text{price})/\partial\sigma$; under BSM it is $S_0\phi(d_1)\sqrt{T} > 0$ and peaks near the money. The smile is the implied volatilities backed out of market prices across strikes.

Step 2 – Link between them

Vega's sign and its ATM peak hold for any smile shape; the smile's shape is set by the market's view of the distribution.

$\Rightarrow$ D is true.

Interpretation

This is the background to the minimum variance delta: vega converts a change in implied volatility into a change in price, whatever the smile looks like.

Why the other options are wrong

A is incorrect. A flat smile means one volatility for all strikes; a higher strike still lowers a call's price because moneyness changes.

B is incorrect. A flat smile is the BSM world, where vega is still positive.

C is incorrect. Confuses vega (derivative of price with respect to volatility) with the slope of the smile (derivative of implied volatility with respect to strike).

LO: MR-17 h: Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.6

Solution MR-17 h.6 — Answer: A

New • GARP Ch.17, 17.6

Step 1 – Which delta is larger

For equity options $\Delta_{MV} = \Delta_{BSM} + \nu\,\partial E(\sigma_{imp})/\partial S < \Delta_{BSM}$, since $\nu > 0$ and implied volatility falls when the index rises.

Step 2 – Translate into the hedge

Short calls are hedged by buying Δ units of the index. Using the larger BSM delta means buying more than the minimum variance hedge.

$\Rightarrow$ A: she is over-hedged relative to the minimum variance hedge.

Interpretation

When the index rises, the calls gain less than BSM predicts because their implied volatility falls, so fewer units of the index are needed to offset them.

Why the other options are wrong

B is incorrect. Uses effect 1 alone; the dominant effect is the downward shift of the whole curve.

C is incorrect. Positive vega is the reason the hedges differ, not a reason they coincide.

D is incorrect. The comparison has one sign: the curve moves against the price in both directions, so the BSM delta is too high whichever way the index moves.

LO: MR-17 h: Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures. •
Reference: Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.6

MR-17 • Volatility Smiles • Vardeez: VERY IMPORTANT

MR-17 i

P2 CONCEPT

Explain the impact of a single asset price jump on a volatility smile.

Workbook: A3 Volatility Smiles section lists LOs a to i • no workbook question lands on this LO

WHAT IS TESTED HERE

Why P2. Vardeez flags it inside a Very Important chapter ("full chapter is important"). No Tier A or B workbook shape lands on it.

Vardeez: full chapter is important. He treats smile versus frown as a polarity pair (his material for this LO sits stranded in the Gauss+ chapter as "LO 15.i"; it belongs here).

Reported exam shapes: none from the recall workbooks.

What the questions look like: what a single anticipated jump does to the distribution (bimodal), the resulting pattern (a frown: ATM implied volatility highest), why BSM struggles, and the contrast with the smile produced by random jumps.

1. A single anticipated jump makes the distribution bimodal

Suppose a stock is at \$50 and a news announcement due in a few days (a takeover outcome, a lawsuit verdict) will move it either up by \$8 or down by \$8. The distribution of the price in one month is then a **mixture of two lognormal distributions**, one for good news and one for bad news.

- The true distribution is **bimodal**, certainly not lognormal.
- It can have the **same mean and standard deviation** as a unimodal lognormal; what changes is the shape: probability piles up around the two post-news prices instead of around \$50.

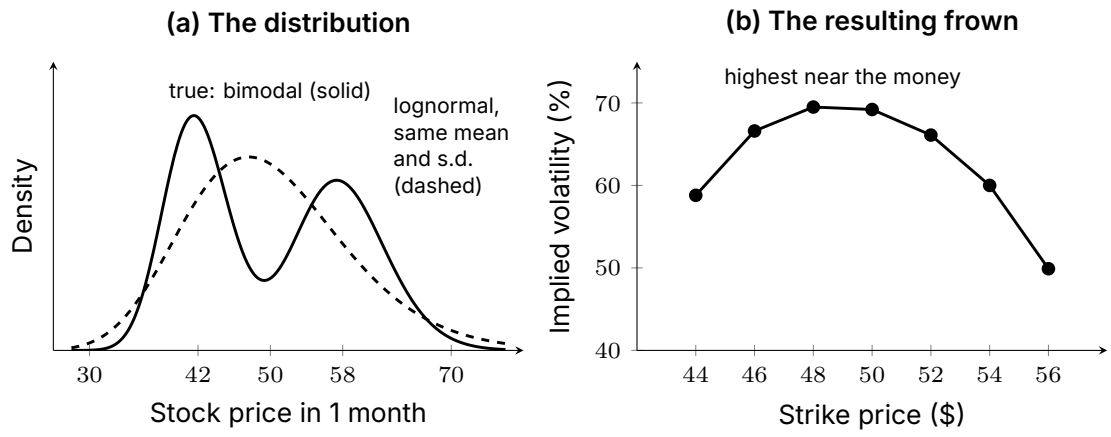


Figure 5. (a) Price in one month when news will move a \$50 stock to about \$42 or \$58 (illustrative mixture, 30% volatility after the news): the true distribution (solid) has two humps; the lognormal with the same mean and standard deviation (dashed) has one. (b) GARP's extreme case, where the price will be exactly \$42 or \$58: implied volatility from Table 17.3, highest near the money and falling into and out of the money.

2. GARP's extreme case: only two possible prices

Stock at \$50; in one month it will be either \$42 or \$58; $r = 12\%$. Then $u = 1.16$, $d = 0.84$, $a = \mathrm{e}^{0.12/12} = 1.0101$ and the risk-neutral up probability is $p = \frac{a-d}{u-d} = 0.5314$. Valuing 1-month European options on this two-point tree and backing out BSM implied volatilities gives:

Strike (\$)	42	44	46	48	50	52	54	56	58
Call (\$)	8.42	7.37	6.31	5.26	4.21	3.16	2.10	1.05	0.00
Put (\$)	0.00	0.93	1.86	2.78	3.71	4.64	5.57	6.50	7.42
Implied vol (%)	0.0	58.8	66.6	69.5	69.2	66.1	60.0	49.9	0.0

GARP Table 17.3, recomputed. Calls and puts at the same strike share one implied volatility (put-call parity, MR-17 b).

- The pattern is a **frown**: the opposite of the currency smile, with volatilities **declining as we move out of or into the money**.
- The implied volatility from the \$50 option would **overprice** options with strikes of \$44 or \$56.
- **Pricing in practice** (GARP Q17.12, Q17.18): value each option as a probability-weighted average of BSM values starting from each post-news price, with the post-news volatility. The risk-neutral probability comes from $S_0 = p\,S_u + (1-p)\,S_d$ (ignoring discounting over a day or so).

> **Correction to the source table.** GARP Table 17.3 prints the implied volatility at the \$56 strike as 49.0%. Inverting BSM on the table's own price (\$1.05) gives **49.9%**; the other entries agree with recomputation to within 0.1 point. The figure and table above use 49.9%. The shape, a frown, is unaffected.

3. Smile versus frown: a polarity pair

	Smile	Frown
Cause	Random jumps and nonconstant volatility: extreme moves in both directions more likely than lognormal (currencies, MR-17 d)	A single *anticipated* jump of known size: the distribution splits into two humps
Distribution	Unimodal, both tails heavier	Bimodal, same mean and s.d. possible
ATM implied vol	Lowest	Highest
Deep ITM / OTM	Highest	Lower
Maturity	Fades as maturity rises (jumps average out)	Fades as maturity rises (one jump is diluted by ordinary diffusion)

> **Correction to a bank key.** The bank's model answer to GARP Q17.7 ("What volatility smile is likely to be caused by jumps in the underlying asset price? Is it more pronounced for a 2-year or a 3-month option?") says a *frown*. GARP's own text says random jumps, like nonconstant volatility, make extreme outcomes more likely and are a reason for the currency *smile*, and that the impact of jumps fades as maturity increases. The answer is a smile, more pronounced for the 3-month option. A frown arises only from a single anticipated jump. Worked as Question MR-17 i.5.

Exam traps

- **Smile or frown.** A single anticipated jump gives a frown (ATM highest). Random jumps and fat tails give a smile (ATM lowest).
- **Bimodal, not fat-tailed.** The anticipated-jump distribution moves probability to two humps; it is not "heavier tails on both sides".
- **Using the ATM volatility for the wings.** With a frown it *overprices* away-from-the-money options.
- **Risk-neutral probability.** Solve $S_0 = pS_u + (1-p)S_d$ for the *favourable* outcome; swapping in $1-p$ is the distractor.
- **Maturity.** Jump effects fade with maturity; a "more pronounced for the 2-year option" option is wrong.

Practice questions: MR-17 i

Question MR-17 i.1

New • built from GARP Ch.17, 17.8

A biotech stock trades at \$50. A regulatory decision due in two days will move it to about \$58 if approval is granted or to about \$42 if it is refused. An analyst compares the distribution of the stock price in one month with a lognormal distribution that has the same mean and standard deviation. Which of the following statements is correct?

A. The true distribution is bimodal, with probability concentrated around the two post-decision prices rather than around \$50.

B. The true distribution is unimodal with heavier tails on both sides than the lognormal, as for an exchange rate.

C. The true distribution is unimodal with a heavier left tail and a thinner right tail, as for an equity index.

D. The true distribution is bimodal, and its standard deviation must exceed that of any lognormal with the same mean.

Question MR-17 i.2

New • built from GARP Ch.17, 17.8 and Table 17.3

A stock trades at \$50 and will be worth either \$42 or \$58 in one month, depending on a court ruling. A trader values one-month European options on a two-point tree and backs out BSM implied volatilities:

Strike (\$)	44	46	48	50	52	54	56
Implied volatility (%)	58.8	66.6	69.5	69.2	66.1	60.0	49.9

Which of the following statements is correct?

A. The pattern is a frown, so using the implied volatility of the \$50-strike option would overprice the \$44 and \$56 options.

B. The pattern is a smile, so using the implied volatility of the \$50-strike option would underprice the \$44 and \$56 options.

C. The pattern is a skew, with implied volatility falling as the strike rises, as for equity index options.

D. The pattern is a frown, so using the implied volatility of the \$50-strike option would underprice the \$44 and \$56 options.

Question MR-17 i.3 Bank Q848 • GARP book Ch.17 Q17.12 (converted to multiple choice)

A stock price is currently \$20. Tomorrow, news is expected to be announced that will either increase the price by \$5 or decrease the price by \$5. Which of the following best describes the problem with using Black-Scholes-Merton to value 1-month options on the stock, and a better approach?

A. The 1-month price distribution is bimodal rather than lognormal, so no single volatility prices all strikes; value each option as a probability-weighted average of BSM values starting from \$25 and from \$15.

B. The 1-month price distribution has heavier tails on both sides than the lognormal, so BSM should be used with a single higher volatility for all strikes.

C. There is no problem, because the expected stock price after the announcement is still about \$20, which is all BSM requires.

D. The 1-month price distribution is bimodal, so BSM should be applied with the at-the-money implied volatility, which then prices all strikes correctly.

Question MR-17 i.4 Bank Q849 • GARP book Ch.17 Q17.18 (rebuilt)

A major lawsuit ruling will be announced tomorrow. The stock price is \$60. If the ruling is favorable it jumps to \$75 and afterwards volatility is 25%; if unfavorable it jumps to \$50 and volatility is 40%. Ignore discounting over the one day. The 6-month risk-free rate is 6% (continuously compounded), no dividends. The risk-neutral probability of a favorable ruling is p, and 6-month European call prices today are the p-weighted average of BSM prices from each post-jump price. Which statement is correct?

A. $p = 0.40$; implied volatility rises steadily with strike (about 40% at $K = \$30$ up to about 48% at $K = \$80$).

B. $p = 0.40$; implied volatility is hump-shaped (a frown), about 47.8% at $K = \$40$, falling to about 40.4% at $K = \$80$.

C. $p = 0.60$; implied volatility is hump-shaped (a frown), about 47.8% at $K = \$40$, falling to about 40.4% at $K = \$80$.

D. $p = 0.40$; implied volatility is U-shaped (a smile) with its minimum near the \$60 strike.

Question MR-17 i.5 Bank Q847 • GARP book Ch.17 Q17.7 (converted to multiple choice, key corrected)

An options researcher asks what volatility smile is likely to be caused by jumps in the underlying asset price (occurring at random, as for exchange rates), and whether the pattern is likely to be more pronounced for a 2-year option than for a 3-month option. Which answer is correct?

A. A smile, with both tails heavier than lognormal, more pronounced for the 3-month option.

B. A smile, with both tails heavier than lognormal, more pronounced for the 2-year option.

C. A frown, with at-the-money implied volatility highest, more pronounced for the 3-month option.

D. A frown, with at-the-money implied volatility highest, more pronounced for the 2-year option.

Question MR-17 i.6 New • built from GARP Ch.17, 17.2 and 17.8

A volatility strategist is matching causes to implied volatility patterns. Which pairing is correct?

A. A single anticipated price jump produces a frown, with at-the-money implied volatility the highest; heavy tails on both sides produce a smile, with at-the-money implied volatility the lowest.

B. A single anticipated price jump produces a smile, with at-the-money implied volatility the lowest; heavy tails on both sides produce a frown, with at-the-money implied volatility the highest.

C. A single anticipated price jump produces a downward skew; a heavier left tail alone produces a frown.

D. A single anticipated price jump produces a frown, with at-the-money implied volatility the lowest; heavy tails on both sides produce a smile, with at-the-money implied volatility the highest.

Solutions • MR-17 i

Solution MR-17 i.1 — Answer: A

New • GARP Ch.17, 17.8

Step 1 – Model the news

The 1-month price is a mixture of two lognormals: one around \$58 (approval), one around \$42 (refusal).

Step 2 – Compare with the lognormal

The mixture is bimodal, and it can share its mean and standard deviation with a unimodal lognormal; only the shape differs.

$\Rightarrow$ A (Figure 5a).

Interpretation

This bimodal shape, not fat tails, is what drives the frown.

Why the other options are wrong

B is incorrect. Heavier tails on both sides is the random-jump, exchange-rate picture, which gives a smile.

C is incorrect. Heavier left tail and thinner right tail is the equity-index picture, which gives a skew.

D is incorrect. The mixture can have the same standard deviation as a lognormal; the difference is shape, not spread.

LO: MR-17 i: Explain the impact of a single asset price jump on a volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.8, Figure 17.5

Solution MR-17 i.2 — Answer: A

New • GARP Ch.17, 17.8, Table 17.3

Step 1 – Read the shape

Implied volatility peaks near \$48–\$50 (69.5%, 69.2%) and falls both ways, to 58.8% at \$44 and 49.9% at \$56.

$\Rightarrow$ A frown: the opposite of the currency smile.

Step 2 – Consequence for pricing

The \$44 and \$56 options have lower implied volatilities than the \$50 option. Pricing them at the \$50 option's 69.2% would use too high a volatility.

$\Rightarrow$ Overpricing, as GARP states.

Interpretation

This is the polarity of the frown: the wings are cheap relative to the at-the-money option.

Why the other options are wrong

B is incorrect. Wrong shape and wrong direction: the wings have *lower* implied volatility here.

C is incorrect. Volatility rises from \$44 to \$48 before falling, so this is a hump, not a monotone skew.

D is incorrect. Right shape, wrong consequence: too high a volatility overprices, it does not underprice.

LO: MR-17 i: Explain the impact of a single asset price jump on a volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.8, Table 17.3 (recomputed)

Solution MR-17 i.3 — Answer: A

Bank Q848 • GARP book Ch.17 Q17.12

Step 1 – What BSM assumes

A continuous lognormal diffusion with constant volatility, so a lognormal price in 1 month.

Step 2 – What the news does

The price jumps to about \$25 or \$15 tomorrow and then diffuses: a bimodal 1-month distribution with a mean near \$20.

$\Rightarrow$ No single volatility reproduces all strikes; implied volatilities form a frown.

Step 3 – A better approach

Value each option as a probability-weighted average of BSM values starting from \$25 and from \$15, each with the post-news volatility.

Interpretation

The expected price being unchanged does not rescue BSM; the shape of the distribution is wrong.

Why the other options are wrong

B is incorrect. Heavier tails on both sides is the random-jump picture; a single higher volatility still cannot fit a bimodal distribution.

C is incorrect. BSM needs a lognormal distribution, not just the right mean.

D is incorrect. With a frown, the ATM implied volatility overprices the away-from-the-money options.

LO: MR-17 i: Explain the impact of a single asset price jump on a volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.8, Q17.12

Solution MR-17 i.4 — Answer: B

Bank Q849 • GARP book Ch.17 Q17.18

Step 1 – Risk-neutral probability

$$60 = 75p + 50(1-p) \quad \Rightarrow \quad p = \frac{10}{25} = 0.40$$

Step 2 – Option prices

$C(K) = 0.40\,\text{BSM}(75, K, 25\%) + 0.60\,\text{BSM}(50, K, 40\%)$ with $T = 0.5$, $r = 6\%$. Verified: $K = 30$: 30.955; 40: 21.935; 50: 14.387; 60: 8.564; 70: 4.430; 80: 1.934.

Step 3 – Implied volatilities (inverting BSM at $S = 60$)

46.7, 47.8, 47.8, 46.1, 43.2, 40.4% for $K = 30$ to 80.

$\Rightarrow$ Highest at the middle strikes, lower at both ends: a frown.

Interpretation

The jump splits the distribution into two humps, giving the hump-shaped pattern. The different post-ruling volatilities tilt it, so it falls more on the high-strike side.

Why the other options are wrong

A is incorrect. Right p, but implied volatility does not rise with strike; it falls above about \$50.

C is incorrect. Uses 0.60, the probability of the unfavourable ruling; the expected price would then be \$65, not \$60.

D is incorrect. A U-shaped smile is the fat-tail pattern; a single jump produces the opposite, a hump.

LO: MR-17 i: Explain the impact of a single asset price jump on a volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.8, Q17.18

Solution MR-17 i.5 — Answer: A

Bank Q847 • GARP book Ch.17 Q17.7 (key corrected)

Step 1 – Random jumps and the tails

GARP 17.2: nonconstant volatility and jumps both make extreme outcomes more likely, which is why exchange rates have heavier tails on both sides and a U-shaped smile.

Step 2 – Maturity

The percentage impact of jumps on prices and implied volatility becomes less pronounced as maturity increases; over long horizons jumps are averaged out.
$\Rightarrow$ A smile, more pronounced for the 3-month option.

Interpretation

The bank's model answer gave a frown, which confuses random jumps with a single anticipated jump (see the correction box). Keep the two cases apart: random jumps $\rightarrow$ smile; one known event $\rightarrow$ frown.

Why the other options are wrong

B is incorrect. Right shape, wrong maturity: jump effects fade over long horizons.
C is incorrect. A frown comes from a single anticipated jump, not from random jumps.
D is incorrect. Wrong shape and wrong maturity.

LO: MR-17 i: Explain the impact of a single asset price jump on a volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 and footnote 2, Q17.7

Solution MR-17 i.6 — Answer: A

New • GARP Ch.17, 17.2 and 17.8

Step 1 – Single anticipated jump

Bimodal distribution $\Rightarrow$ ATM options are the dear ones $\Rightarrow$ frown, ATM implied volatility highest.

Step 2 – Heavy tails on both sides

Extreme moves more likely in both directions $\Rightarrow$ deep ITM and OTM options are dear $\Rightarrow$ smile, ATM implied volatility lowest.
$\Rightarrow$ A.

Interpretation

This is the polarity pair in Section 3. Option D keeps the labels but flips what each label means.

Why the other options are wrong

B is incorrect. Swaps the two causes.
C is incorrect. A downward skew comes from a heavy left tail and thin right tail (equities), not from a single jump; a heavy left tail alone does not give a frown.
D is incorrect. Right labels, wrong definitions: in a frown ATM implied volatility is highest, in a smile it is lowest.

LO: MR-17 i: Explain the impact of a single asset price jump on a volatility smile. • **Reference:** Hull, OFOD 11e, Ch.20 (GARP MR-17), 17.2 and 17.8

MARKET RISK • PART

P3

Third priority

Tier C and D topics, or LOs Vardeez flags in other chapters. 30 LOs from 9 readings.

MR-4 • Backtesting VaR • Vardeez: IMPORTANT (refer to crash-course notes)

MR-4 c

P3 BOTH

Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test.

Workbook: C2 Backtesting VaR: Exceedance vs PIT • Tier C • Map #21 (4/16)

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C2) hits this LO.
Vardeez: understand the calculation of failure rates and the model verification calculation "using the 2 tables" (GARP Table 4.1, decision errors, and Table 4.2, Kupiec's nonrejection regions).

Reported exam shapes

- **C2 S1** Backtesting VaR: counting exceptions against the model.
- **C2 S3** Kupiec's proportion-of-failures test (unconditional coverage) versus Christoffersen (independence, conditional coverage). The Christoffersen side is worked under MR-4 e.

What the questions look like: expected number of exceptions, the z-statistic, the largest number of exceptions that still passes, reading the nonrejection-region table, and "which statement about the failure-rate test is true".

1. Exceptions and the failure rate

Backtesting verifies that actual losses are in line with projected losses: the history of VaR forecasts is compared with realised P&L. An **exception** (exceedance) is a day on which the loss exceeds the previous day's VaR.

- **Too many exceptions:** the model *underestimates* risk (too little capital, regulatory penalties).
- **Too few exceptions:** the model *overstates* risk, an excessive and inefficient allocation of capital.

$$\text{Failure rate} = \frac{N}{T} \qquad N = \text{number of exceptions},\ T = \text{number of observations}$$

For an accurate model the failure rate is an *unbiased* estimator of $p = 1 - c$, where c is the VaR confidence level, and it converges to p as T grows. Example: a 99% VaR over 252 days with 6 exceptions has a failure rate of $6/252 = 2.38\%$ against the 1% expected, so the model appears to underestimate risk.

2. The model verification test

Each day is a Bernoulli trial (exception or not) with probability p, so the number of exceptions x is **binomial**:

$$f(x) = \binom{T}{x} p^x (1-p)^{T-x}, \qquad E(x) = pT, \qquad V(x) = p(1-p)T$$

The test is **nonparametric**: it makes no assumption about the return distribution, only that each day is an independent trial. For large T the binomial is approximated by the normal:

$$\boxed{z = \frac{x - pT}{\sqrt{p(1-p)T}} \sim N(0,1)}$$

At the two-tailed 95% *test* confidence level, reject the model if $|z| > 1.96$. The normal is only a convenient shortcut; the exact binomial can always be used instead.

JPMorgan, 1998. Daily revenue fell below the 95% VaR band on $x = 20$ of $T = 252$ days:

$$z = \frac{20 - 0.05 \times 252}{\sqrt{0.05 \times 0.95 \times 252}} = \frac{7.4}{3.46} = 2.14 > 1.96 \quad \Rightarrow \quad \text{reject the model}$$

Largest number of exceptions that still passes

Upper bound $= pT + z^* \sqrt{p(1-p)T}$, then **round down**. Example: 98% VaR, $T = 252$: $5.04 + 1.96 \times 2.222 = 9.40$, so 9 exceptions pass and 10 fail.

Kupiec's proportion-of-failures (POF) test

Kupiec defines the 95% nonrejection region with a log-likelihood ratio:

$$LR_{uc} = -2\ln\left[(1-p)^{T-N}p^N\right] + 2\ln\left\{\left[1 - \frac{N}{T}\right]^{T-N}\left(\frac{N}{T}\right)^N\right\} \sim \chi^2(1)$$

Reject if $LR_{uc} > 3.841$. This is equivalent to the z-test, because a $\chi^2(1)$ variable is the square of a standard normal ($1.96^2 = 3.84$). JPMorgan: $LR_{uc} = 3.91 > 3.841$, so reject, as with the z-test. The test is called *unconditional* because it looks only at the count, not at when the exceptions happen (MR-4 e).

Table 4.2: nonrejection regions (95% test confidence)

p	VaR confidence c	$T = 252$ days	$T = 510$ days	$T = 1{,}000$ days
0.01	99%	$N < 7$	$1 < N < 11$	$4 < N < 17$
0.025	97.5%	$2 < N < 12$	$6 < N < 21$	$15 < N < 36$
0.05	95%	$6 < N < 20$	$16 < N < 36$	$37 < N < 65$
0.075	92.5%	$11 < N < 28$	$27 < N < 51$	$59 < N < 92$
0.10	90%	$16 < N < 36$	$38 < N < 65$	$81 < N < 120$

How to read it:

- N *at or above* the upper limit: VaR is too low, the model understates large losses. N *at or below* the lower limit: the model is overly conservative.
- As a proportion N/T the interval **shrinks as** T **grows** (95% row: $[6/252,\ 20/252] = [2.4\%,\ 7.9\%]$ at $T = 252$; $[3.7\%,\ 6.5\%]$ at $T = 1{,}000$). More data makes a false model easier to reject.
- For small p the test is weak: at $p = 0.01$, $T = 252$ the region is $N < 7$, with *no lower limit*. There is no way to tell whether a very low count is luck or a model that systematically overstates risk. This is why some banks backtest at 95% and scale the capital with a larger multiplier.
- The **test** confidence level (95% here) is *not* the **VaR** confidence level c. Basel backtests a 99% VaR with its own decision rule.

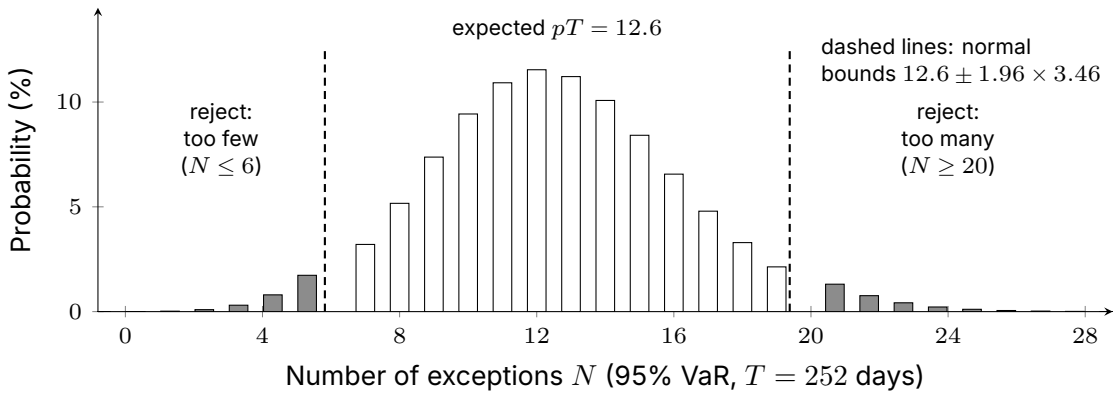


Figure 1. Binomial distribution of exceptions for a correct 95% VaR model over 252 days. White bars: Kupiec's nonrejection region $6 < N < 20$. Grey bars: rejection regions. The dashed lines are the normal-approximation bounds 5.8 and 19.4, which give the same integer region.

Formula card: MR-4 c

Failure rate	N/T, unbiased for $p = 1 - c$
Exceptions	$x \sim \text{Binomial}(T, p)$, $\quad E(x) = pT$, $\quad V(x) = p(1-p)T$
z-test	$z = (x - pT)/\sqrt{p(1-p)T}$; reject if $\lvert z \rvert > 1.96$ (two-tailed 95%)
Acceptance limit	$pT \pm z^* \sqrt{p(1-p)T}$, round the upper limit *down*
Kupiec POF	$LR_{uc} \sim \chi^2(1)$, reject if $LR_{uc} > 3.841$ $(= 1.96^2)$
Critical values	two-tailed: 90% 1.645, 95% 1.96, 99% 2.576; one-tailed 95% 1.645

Exam traps

- **Variance for standard deviation.** Dividing $x - pT$ by $p(1-p)T$ instead of its square root.
- **Dropped $(1-p)$.** Using $\sqrt{pT}$ as the standard deviation.
- **One-tailed for two-tailed.** 1.645 instead of 1.96 for a two-tailed 95% test (or 1.96 for a 90% test).
- **Rounding the limit up.** 9.40 means 9 exceptions pass; 10 is already rejected.
- **Expected count as the limit.** pT is the centre of the acceptance region, not its edge.
- **Confusing the two confidence levels.** The VaR level sets p; the test level sets the critical value.
- **Wrong column of Table 4.2.** Check T before reading the region.
- **Size of losses.** The exception test counts breaches; the size of a loss beyond VaR is not what it evaluates.

Practice questions: MR-4 c

Question MR-4 c.1

Bank Q211 • Schweser Module Quiz 4.1 Q1 (rebuilt)

A VaR model is built at a 97.5% confidence level and backtested over a 252-day period. How many exceptions are expected?

A. 2.5

B. 6.3

C. 12.6

D. 245.7

Question MR-4 c.2

Bank Q213 • Schweser Module Quiz 4.1 Q4 (rebuilt)

A risk manager is backtesting a VaR model at the 95% confidence level using 252 daily returns and observes 17 exceptions. Using the model verification z-score $z = (x - pT)/\sqrt{p(1-p)T}$, where $p = 1 - c$, what is the z-score for this sample?

A. 0.37

B. 1.24

C. 1.27

D. 1.96

Question MR-4 c.3

Bank Q203 • BT L2.T5.59.1 (rebuilt)

A bank backtests its 95% daily VaR and observes 19 exceptions over 250 trading days. Using the normal approximation to the binomial, what is the decision and test statistic for a 90% two-tailed test of the model?

A. Accept, with a test statistic of 0.55

B. Accept, with a test statistic of 1.89

C. Reject, with a test statistic of 1.89

D. Reject, with a test statistic of 1.84

Question MR-4 c.4

Bank Q198 • AP Mock 1 Q21 (rebuilt, key corrected)

A risk analyst is backtesting a bank's daily 98% VaR model over one year of 252 trading days. The backtest uses the normal approximation to the binomial and a two-tailed test at the 95% confidence level (critical value 1.96). What is the maximum number of exceptions (daily losses exceeding VaR) for which the analyst would still conclude that the model is correctly calibrated?

A. 5

B. 8

C. 9

D. 10

Question MR-4 c.5

Bank Q210 • Mock 2 Q31 (rebuilt)

A risk analyst backtests a bank's one-day 99% VaR model over eight years (250 trading days per year) at a 95% two-tailed confidence level. Assume i.i.d. daily returns and use the normal approximation to the binomial. What is the highest number of exceptions that is acceptable to conclude the model is correctly calibrated?

A. 20 exceptions

B. 28 exceptions

C. 29 exceptions

D. 31 exceptions

Question MR-4 c.6

New • built from GARP MR Ch.4, Table 4.2 (Kupiec, 1995)

A model validator at a regional bank reviews a one-day 97.5% VaR model. Over the last 510 trading days the desk recorded 6 exceptions. The validator uses the following excerpt of Kupiec's nonrejection regions, at the 95% test confidence level:

VaR confidence	$T = 252$	$T = 510$	$T = 1{,}000$
99%	$N < 7$	$1 < N < 11$	$4 < N < 17$
97.5%	$2 < N < 12$	$6 < N < 21$	$15 < N < 36$
95%	$6 < N < 20$	$16 < N < 36$	$37 < N < 65$

Which of the following conclusions is correct?

A. Do not reject the model: 6 exceptions lies inside the nonrejection region $2 < N < 12$.

B. Reject the model: 6 exceptions is at the lower limit of $6 < N < 21$, so the VaR model is overly conservative.

C. Reject the model: 6 exceptions exceeds the Basel green-zone limit of 4, so the VaR understates risk.

D. Do not reject the model: 6 exceptions is close to the expected count of $0.025 \times 252 = 6.3$.

Question MR-4 c.7

Bank Q199 • AP Mock 2 Q14

A large investment firm has a trading book whose size changes with perceived trading opportunities. The firm's chief risk analyst reports a one-day VaR at the 95% confidence level of GBP 110 million. Assuming losses are independently and identically distributed, which of the following would be convincing evidence that the analyst is estimating VaR poorly?

A. Over the past 250 days, there are no exceptions.

B. Over the past 250 days, the mean loss is GBP 132 million.

C. Over the past 250 days, the largest loss is GBP 1.1 billion.

D. Over the past 250 days, there are seven exceptions.

Question MR-4 c.8 Bank Q202 • BT L2.T5.58.1

In regard to Jorion's value at risk (VaR) backtest based on exceptions (failure rates), EACH of the following is true EXCEPT:

A. It is nonparametric.

B. If the bank's VaR model is accurate, the sample failure rate is an unbiased estimator of (1 − confidence).

C. The significance test of the backtest requires the assumption of a normal distribution in order to avoid a numerical/simulation approach.

D. The hypothesis test of the model's accuracy can assume a binomial distribution, which is easy to use.

Solutions • MR-4 c

Solution MR-4 c.1

Answer: B

Bank Q211 • Schweser Module Quiz 4.1 Q1

Step 1 – Exception probability

$p = 1 - c = 1 - 0.975 = 0.025$

Step 2 – Expected exceptions

$$E(x) = pT = 0.025 \times 252 = 6.3$$

$\Rightarrow$ A correct 97.5% model should breach on about 6 of 252 days.

Interpretation

The expected count is the tail probability times the number of observations; it is the centre of the acceptance region in every later test.

Why the other options are wrong

A is incorrect. Reported the 2.5% tail probability as a count, without multiplying by 252.
C is incorrect. Used 5% (a 95% VaR): $0.05 \times 252 = 12.6$.
D is incorrect. Multiplied the confidence level instead of the tail probability: $0.975 \times 252 = 245.7$, the expected number of *non*-exceptions.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 Model Verification Based on Failure Rates

Solution MR-4 c.2

Answer: C

Bank Q213 • Schweser Module Quiz 4.1 Q4

Step 1 – Expected count and excess

$p = 0.05$, $T = 252$: $pT = 12.6$, so $x - pT = 17 - 12.6 = 4.4$.

Step 2 – Standard deviation of the count

$$\sqrt{p(1-p)T} = \sqrt{0.05 \times 0.95 \times 252} = \sqrt{11.97} = 3.460$$

Step 3 – z-score

$$z = \frac{4.4}{3.460} = 1.27$$

$\Rightarrow$ Below 1.96, so the model is not rejected at the 95% test level.

Interpretation

The excess of 4.4 exceptions is only 1.27 standard deviations: well within normal sampling variation for a correct model.

Why the other options are wrong

A is incorrect. Divided by the variance instead of the standard deviation: $4.4/11.97 = 0.37$.
B is incorrect. Dropped the $(1-p)$ factor: $4.4/\sqrt{0.05 \times 252} = 4.4/3.550 = 1.24$.
D is incorrect. Quoted the 95% two-tailed critical value, which is the threshold, not the statistic.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, eq. 4.2

Solution MR-4 c.3

Answer: C

Bank Q203 • BT L2.T5.59.1

Step 1 – Expected count and standard deviation

$pT = 0.05 \times 250 = 12.5$; $\sqrt{0.05 \times 0.95 \times 250} = 3.446$.

Step 2 – Test statistic

$$z = \frac{19 - 12.5}{3.446} = 1.886 \approx 1.89$$

Step 3 – Decision at the 90% two-tailed level

The critical value is 1.645. Since $1.89 > 1.645$, reject.

$\Rightarrow$ The model produces too many exceptions to be accepted at 90% test confidence (it would pass at 95%, where the cut-off is 1.96).

Interpretation

The same data can pass or fail depending on the *test* confidence level chosen; a lower test confidence widens the rejection region.

Why the other options are wrong

A is incorrect. Divided by the variance (11.875): $6.5/11.875 = 0.55$, and accepted.

B is incorrect. Right statistic, wrong cut-off: compared 1.89 with the 95% value 1.96 instead of 1.645.

D is incorrect. Right decision, but omitted $(1 - p)$: $6.5/\sqrt{250 \times 0.05} = 1.84$.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, eq. 4.2

Solution MR-4 c.4

Answer: C

Bank Q198 • AP Mock 1 Q21

Step 1 – Expected count and standard deviation

$p = 0.02$, $T = 252$: $pT = 5.04$; $\sqrt{252 \times 0.02 \times 0.98} = 2.222$.

Step 2 – Upper limit of the acceptance region

$$5.04 + 1.96 \times 2.222 = 9.40$$

Step 3 – Round down and check

$x = 9$: $z = (9 - 5.04)/2.222 = 1.78 < 1.96$, accept. $x = 10$: $z = 2.23 > 1.96$, reject.

$\Rightarrow$ The largest acceptable count is 9.

Interpretation

The limit is a fractional number of exceptions; only whole counts inside it pass, so always round down and check the next integer.

Why the other options are wrong

A is incorrect. Gave the expected count $0.02 \times 252 \approx 5$, the centre of the region, not its edge.

B is incorrect. Used the one-tailed 1.645: $5.04 + 1.645 \times 2.222 = 8.70$.

D is incorrect. Rounded 9.40 up; 10 exceptions is already rejected ($z = 2.23$).

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, eq. 4.2

Solution MR-4 c.5

Answer: B

Bank Q210 • Mock 2 Q31

Step 1 – Sample and expected count

$T = 8 \times 250 = 2{,}000$, $p = 0.01$: $pT = 20$.

Step 2 – Standard deviation

$$\sqrt{2{,}000 \times 0.01 \times 0.99} = \sqrt{19.8} = 4.450$$

Step 3 – Upper limit

$$20 + 1.96 \times 4.450 = 28.72$$

$\Rightarrow$ 28 exceptions pass ($z = 1.80$); 29 fail ($z = 2.02$).

Interpretation

With eight years of data the expected count (20) is large enough for the normal approximation to work well, unlike a single year at 99% (2.5 expected).

Why the other options are wrong

A is incorrect. Gave the expected count, 20.
C is incorrect. Rounded 28.72 to the nearest integer; 29 lies outside the region.
D is incorrect. Used the 99% two-tailed value 2.576: $20 + 2.576 \times 4.450 = 31.5$.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, eq. 4.2

Solution MR-4 c.6

Answer: B

New • built from GARP MR Ch.4, Table 4.2

Step 1 – Pick the right cell

VaR confidence 97.5% ($p = 0.025$) and $T = 510$: the nonrejection region is $6 < N < 21$, so $N = 7, \ldots, 20$ pass.

Step 2 – Locate the observation

$N = 6$ is not strictly greater than 6, so it falls in the *lower* rejection region. (Check: Kupiec's $LR_{uc} = 4.55 > 3.841$.)
$\Rightarrow$ Too few exceptions: the expected count is $0.025 \times 510 = 12.75$, and 6 is too far below it.

Step 3 – Interpret

A count below the region means VaR is set too high: the model is overly conservative and ties up excess capital.

Interpretation

The table rejects in both directions. Reading the correct column matters: the same 6 exceptions would pass at $T = 252$, where 6.3 are expected.

Why the other options are wrong

A is incorrect. Read the $T = 252$ column ($2 < N < 12$) instead of $T = 510$.
C is incorrect. Applied the Basel green-zone rule (0 to 4 exceptions of a 99% VaR over 250 days) to a 97.5% VaR over 510 days, and read the direction wrongly.
D is incorrect. Computed the expected count with $T = 252$ instead of 510; over 510 days 12.75 are expected.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, Table 4.2 (adapted from Kupiec, 1995), eq. 4.3

Solution MR-4 c.7 — Answer: A

Bank Q199 • AP Mock 2 Q14

Step 1 – What an exception test can see

$p = 0.05$, $T = 250$: expected 12.5 exceptions, standard deviation $\sqrt{250 \times 0.05 \times 0.95} = 3.446$.

Step 2 – Test each count

Zero exceptions: $z = (0 - 12.5)/3.446 = -3.63$, far outside ± 1.96 (probability $0.95^{250} \approx 0.0003\%$).

Seven exceptions: $z = (7 - 12.5)/3.446 = -1.60$, inside ± 1.96.

$\Rightarrow$ Zero breaches is convincing evidence that VaR is grossly overstated.

Interpretation

A backtest rejects in both tails: too few exceptions is as much a sign of a poor model as too many.

Why the other options are wrong

B is incorrect. A mean loss is not an exception count; with a book that changes size it says nothing about breach frequency.

C is incorrect. The test counts breaches, not their size: one very large loss can occur in a well-calibrated model, since the tail beyond VaR is unbounded.

D is incorrect. Seven exceptions gives $z = -1.60$, inside the acceptance region.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.1 and 4.2

Solution MR-4 c.8 — Answer: C

Bank Q202 • BT L2.T5.58.1

Step 1 – Test each statement

A True: counting exceptions needs no assumption about the return distribution. **B** True: N/T is an unbiased measure whose expectation is $p = 1 - c$ for an accurate model. **D** True: each day is a Bernoulli trial, so the count is binomial.

Step 2 – Find the false one

C False: the normal is only a large-sample *approximation* to the binomial, a convenient shortcut. The exact binomial can be used directly, without simulation.

Interpretation

Jorion's framework is binomial and nonparametric; the z-statistic is an approximation, not a requirement.

Why the other options are wrong

A is incorrect. True statement: the test is nonparametric.

B is incorrect. True statement: the failure rate is an unbiased estimator of $1 - c$.

D is incorrect. True statement: the binomial is the natural distribution of the count.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, eqs. 4.1 and 4.2

MR-4 • Backtesting VaR • Vardeez: IMPORTANT (refer to crash-course notes)

MR-4 d

P3 CONCEPT

Identify and describe Type I and Type II errors in the context of a backtesting process.

Workbook: C2 Backtesting VaR: Exceedance vs PIT • Tier C • Map #21 (4/16)

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C2) hits this LO.
Vardeez: part of "failure rates and model validation using the 2 tables"; Table 4.1 is the decision-error grid.

Reported exam shape

- **C2 S4** Type I versus Type II errors, including the trade-off. The map lists this pair as "uncleared in your Confusables Registry".

What the questions look like: name the error in a scenario, swap-the-label distractors, and "how would you reduce the Type II error" (power). Probabilities are read from a given table, not computed.

1. The decision grid (Table 4.1)

The null hypothesis is that the VaR model is accurate.

Decision	Model is correct	Model is incorrect
Accept	OK	**Type II error**
Reject	**Type I error**	OK (power of the test)

- **Type I:** *rejecting an accurate model*. The bank is penalised for bad luck. Its probability is the significance level of the test, computed under the null.
- **Type II:** *accepting an inaccurate model*. Risk and capital stay understated: the regulator's bigger worry, because a bank that understates VaR on purpose gets a pass.
- **Power** $= 1 - P(\text{Type II})$: the probability of rejecting an incorrect model. A test is *powerful* when, for a low Type I rate, its Type II rate is also low.
- Type II needs a **specific alternative** (true coverage 98%? 97%? 92%?), so it has no single value. It is the harder error to quantify.
- One decision can produce only one kind of error: an acceptance risks Type II, a rejection risks Type I.

2. The trade-off, in numbers (Basel set-up)

Basel records exceptions of a 99% VaR over $T = 250$ days and rejects (yellow or red zone) from 5 exceptions. GARP's Table 4.4 gives both error rates:

	Correct model, $p = 1\%$		Incorrect model, $p = 3\%$	
N	$P(X = N)$	$P(X \geq N)$ (Type I)	$P(X = N)$	$P(X < N)$ (Type II)
3	21.5	45.7	3.8	1.9
4	13.4	24.2	7.2	5.7
5	6.7	**10.8**	10.9	**12.8**
6	2.7	4.1	13.8	23.7
7	1.0	1.4	14.9	37.5

- With a cut-off of 5, a correct model is rejected **10.8%** of the time (one bank in ten penalised for bad luck), and a model with only 97% coverage passes **12.8%** of the time.
- Moving the cut-off trades one error for the other: a lower cut-off (smaller green zone) lowers Type II and raises Type I; a higher cut-off does the reverse.

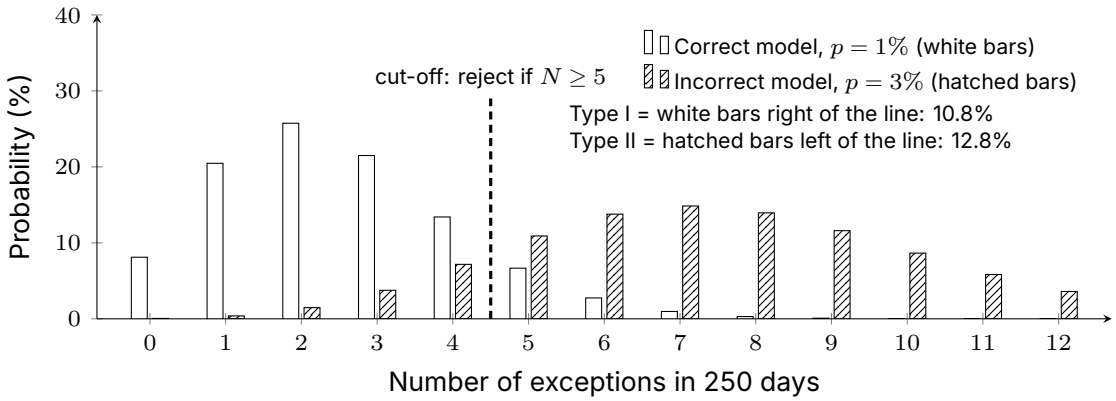


Figure 2. Both errors live on the same cut-off. Sliding the dashed line right shrinks the Type I area and grows the Type II area. Only moving the two distributions further apart (more data, or a lower VaR confidence level) shrinks both.

3. Raising the power of the backtest

- **Lower the VaR confidence level.** At 99% only 2.5 exceptions are expected, too few for a reliable test. At 95% (about 13 a year) with a cut-off of 17, Type I is 12.5% (close to Basel's 10.8%) and Type II falls to **7.4%**. Capital can be kept unchanged with a larger multiplier.
- **More observations.** With $T = 1{,}000$ and a cut-off of 14, Type I is 13.4% and Type II only **0.03%**. Use a horizon as short as possible (daily) to get more observations and limit the effect of changes in portfolio composition.
- **A lower test confidence level** (e.g. a 90% instead of 95% two-tailed test) narrows the acceptance region: Type II falls, Type I rises.
- What does *not* help: a higher VaR confidence level, a smaller sample, or a wider acceptance region.

Exam traps

- **Label swap.** Type I = reject a *correct* model; Type II = accept an *incorrect* one. Distractors simply swap the two definitions.
- **Both errors at once.** One accept or reject decision can produce only one error.
- **"A higher VaR confidence level makes the backtest more reliable."** The reverse: fewer exceptions, less power.
- **Power without Type I.** A test that rejects every model has zero Type II error and is useless; power is judged at a given Type I level.
- **Type I from the wrong distribution.** Type I is computed under the null ($p = 1\%$); Type II under a stated alternative ($p = 3\%$). Read the right column.
- **PDF for CDF.** An error rate is a cumulative probability ($P(X \geq N)$ or $P(X < N)$), not the probability of exactly one count.

Practice questions: MR-4 d

Question MR-4 d.1

Bank Q218 • AnalystPrep Q.6473 (rebuilt)

A regulator backtests a bank's internal VaR model using a null hypothesis that the model is accurate. Which statement correctly describes the consequences of the two possible errors?

A. A Type I error means an inaccurate model is accepted, leaving risk and capital understated; a Type II error means an accurate model is rejected, causing unnecessary penalties.

B. A Type I error means an accurate model is rejected, causing unnecessary penalties or recalibration; a Type II error means an inaccurate model is accepted, leaving risk and capital understated.

C. Both errors lead to understated risk, so the regulator should minimise both by increasing the confidence level of the VaR.

D. Type I and Type II errors always have the same probability, so their consequences cancel out in the backtest.

Question MR-4 d.2

Bank Q230 • GARP PE2 Q70 • Mock 4 Q71

A model validation team at a bank is backtesting the bank's VaR model. One team member is concerned that the validation could result in a Type I or a Type II error. Which of the following is correct regarding Type I and Type II errors?

A. The probability of committing a Type II error decreases when the sample size increases and the level of significance is held constant.

B. A backtest is considered statistically powerful if it minimizes the probability of committing a Type II error regardless of the probability of committing a Type I error.

C. A Type I error is committed when an incorrectly specified model is accepted.

D. A Type II error is committed when a correctly specified model is rejected.

Question MR-4 d.3

Bank Q215 • AP Mock 1 Q10

A model gives a VaR value of USD 5 million for a portfolio at a 99% confidence level. A one-year backtest, with statistical significance judged at the 95% level, reveals that losses exceeded USD 5 million on 8 occasions. The model is accepted as accurate. Assuming 252 days in a year, which of these statements is most likely true?

A. A Type I error has occurred.

B. A Type II error has occurred.

C. Both Type I and Type II errors have occurred.

D. The model has been accepted correctly.

Question MR-4 d.4

Bank Q214 • Mock 3 Q27

A new risk analyst assists in backtesting the bank's VaR model. The bank uses a 1-day VaR at 95% confidence but is considering a change to 99% confidence, as in the Basel framework. Which statement about this change is correct?

A. The decision to accept or reject a VaR model from a two-tailed backtest at the 95% test confidence level is less reliable for a 99% VaR model than for a 95% VaR model.

B. The 95% VaR model is less likely to be rejected by a backtest than the 99% VaR model.

C. When using a two-tailed 90% confidence test in a backtest, there is a smaller probability of incorrectly rejecting a 95% VaR model than a 99% VaR model.

D. Using a 99% VaR model will lower the probability of committing both Type I error and Type II error.

Question MR-4 d.5

Bank Q222 • BT L2.T5.60.1

Basel's traffic-light market risk backtest (green/yellow/red zones) tests the null hypothesis that the bank's 99% VaR model is accurate, with the yellow zone beginning at 5 exceptions in 250 days. Which of the following is MOST difficult to quantify?

A. Probability of Type I error

B. Probability of Type II error

C. Probability of accepting an accurate VaR model

D. Significance level of backtest

Question MR-4 d.6

Bank Q227 • BT P2.T5.22.7.2

Peter backtests his firm's one-day 99.0% VaR model over 500 daily observations. He rejects the model if he observes nine or more exceptions ($X \geq 9$) and otherwise accepts it. Below are the binomial pdf and CDF ($n = 500$) for a model whose true coverage is inaccurate, at 98.0% ($p = 0.02$), 97.0% and 96.0%.

Exceptions	98% pdf	98% CDF	97% pdf	97% CDF	96% pdf	96% CDF
4	1.83%	2.81%	0.06%	0.08%	0.00%	0.00%
5	3.71%	6.52%	0.18%	0.25%	0.00%	0.01%
6	6.24%	12.76%	0.45%	0.70%	0.02%	0.02%
7	8.99%	21.75%	0.98%	1.68%	0.04%	0.07%
8	11.30%	33.05%	1.86%	3.54%	0.11%	0.18%
9	12.61%	45.67%	3.15%	6.69%	0.26%	0.44%
10	12.64%	58.30%	4.79%	11.48%	0.53%	0.97%

If the firm's true coverage level is 98.0%, what is the implied probability of a Type II error?

A. 2.81%

B. 11.30%

C. 33.05%

D. 66.95%

Question MR-4 d.7

Bank Q229 • BT P2.T5.713.2

Barbara simulates the backtest of a proposed VaR model. The Risk Committee asked her to reduce the probability of a Type II error. The previous version assumed a 97.5% VaR model, a two-tailed backtest confidence level of 95.0%, and a sample of two years of daily P/L (500 trading days). Which of the following adjustments is the MOST EFFECTIVE way to reduce the Type II error rate?

A. Decrease the sample size from 500 days to 250 days

B. Increase the cutoff (the number of acceptable exceptions)

C. Increase the VaR confidence level from 97.5% to 99.0%

D. Decrease the two-tailed backtest confidence level from 95.0% to 90.0%

Solutions • MR-4 d

Solution MR-4 d.1

Answer: B

Bank Q218 • AnalystPrep Q.6473

Step 1 – Define the errors against the null "model is accurate"

Type I = rejecting a true null = rejecting an *accurate* model. Type II = failing to reject a false null = accepting an *inaccurate* model.

Step 2 – Attach the consequences

Type I: unnecessary penalties or recalibration (the bank pays for bad luck). Type II: risk and capital stay understated.

$\Rightarrow$ Only B matches both definitions and both consequences.

Interpretation

Regulators worry most about Type II, which lets a bank that understates its VaR through; banks worry most about Type I.

Why the other options are wrong

A is incorrect. Swaps the two definitions (the most common distractor on this LO).

C is incorrect. Rejecting an accurate model does not understate risk, and a higher VaR confidence level reduces test power rather than minimising both errors.

D is incorrect. The two probabilities are generally different; lowering one tends to raise the other.

LO: MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, Table 4.1

Solution MR-4 d.2

Answer: A

Bank Q230 • GARP PE2 Q70

Step 1 – Definitions

Type I = rejecting a correct model; Type II = accepting an incorrect model. So C and D are the definitions with the labels swapped.

Step 2 – Sample size

With the significance level (Type I rate) held fixed, a larger sample tightens the distribution of the failure rate. The test separates good from bad models better, so power rises and Type II falls.

$\Rightarrow$ A is correct.

Interpretation

More data is the one change that improves the test without paying for it with a higher Type I rate (GARP: $T = 1{,}000$ cuts Type II to 0.03%).

Why the other options are wrong

B is incorrect. Absolute qualifier: a test that rejects every model has zero Type II error but is useless. Power is judged at a given Type I level.

C is incorrect. Describes a Type II error.

D is incorrect. Describes a Type I error.

LO: MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 and The Basel Rules

Solution MR-4 d.3 — Answer: B

Bank Q215 • AP Mock 1 Q10

Step 1 – Was acceptance justified?

Expected exceptions $0.01 \times 252 = 2.52$; standard deviation $\sqrt{252 \times 0.01 \times 0.99} = 1.58$.

$$z = \frac{8 - 2.52}{1.58} = 3.47 > 1.96$$

$\Rightarrow$ The count is inconsistent with a correct 99% model: it should have been rejected.

Step 2 – Name the error

The model was *accepted* although the evidence says it is inaccurate: a Type II error.

Interpretation

Only one decision was made (accept), so only the error linked to acceptance, Type II, can have occurred.

Why the other options are wrong

A is incorrect. A Type I error requires rejecting an accurate model; this model was accepted.
C is incorrect. One decision cannot produce both errors.
D is incorrect. With $z = 3.47$ the acceptance was not correct. (Using the 5% test level as if it were the exception rate makes 8 look normal: that is the trap.)

LO: MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, Table 4.1, eq. 4.2

Solution MR-4 d.4 — Answer: A

Bank Q214 • Mock 3 Q27

Step 1 – Compare the information in the two samples

Over 250 days: a 95% VaR expects 12.5 exceptions (s.d. 3.45, 28% of the mean); a 99% VaR expects 2.5 (s.d. 1.57, 63% of the mean).
$\Rightarrow$ At 99% the count is far noisier relative to its mean.

Step 2 – Consequence for the decision

With few expected exceptions the test has little power, so accept/reject decisions about a 99% model are less reliable.

Interpretation

This is GARP's central point on Basel backtesting: the 99% level "generates too few exceptions for a reliable test".

Why the other options are wrong

B is incorrect. The rejection probability of a correct model is the test's significance level at either VaR level; for an incorrect model the 95% test is *more* likely to reject.
C is incorrect. The Type I rate is set by the test (10% here) whatever the VaR level.
D is incorrect. Changing the VaR level cannot lower both errors; a higher level actually lowers power.

LO: MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules

Solution MR-4 d.5 — Answer: B

Bank Q222 • BT L2.T5.60.1

Step 1 – What is computed under the null

Under $p = 1\%$, $T = 250$: $P(N \geq 5) = 10.8\%$ is the Type I error and the significance level; $P(\text{accept} \mid \text{accurate}) = 89.2\%$. All three are single, known numbers.

Step 2 – What needs an alternative

Type II requires the *true* coverage of a bad model. At 97% it is 12.8%, at 98% it is different, and so on. There is no single number.

$\Rightarrow$ Type II is the hardest to quantify.

Interpretation

This is why Basel's design is framed around a chosen Type I rate, with Type II quoted only for illustrative alternatives.

Why the other options are wrong

A is incorrect. Directly computable: 10.8%.

C is incorrect. Computable: $1 - 10.8\% = 89.2\%$.

D is incorrect. The significance level equals the Type I probability, 10.8%.

LO: MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules, Table 4.4

Solution MR-4 d.6 — Answer: C

Bank Q227 • BT P2.T5.22.7.2

Step 1 – Which outcomes are acceptances?

Peter accepts when $X \leq 8$.

Step 2 – Probability that the bad model is accepted

True coverage 98%: $P(X \leq 8) = \text{CDF}(8) = 33.05\%$ from the 98% column.

$\Rightarrow$ A third of the time the inaccurate model passes. Power $= 1 - 33.05\% = 66.95\%$.

Interpretation

Type II is a cumulative probability under the stated alternative, read at the last accepted count.

Why the other options are wrong

A is incorrect. CDF at 4 exceptions: the wrong cut-off (borrowed from the Basel green zone).

B is incorrect. PDF at exactly 8: one point, not the whole acceptance region.

D is incorrect. The power of the test, $1 - 33.05\%$, not the Type II error.

LO: MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 and Table 4.4

Solution MR-4 d.7 — Answer: D

Bank Q229 • BT P2.T5.713.2

Step 1 – What lowers Type II

Type II falls when the acceptance region narrows or the test gains information.

Step 2 – Test each option (illustration: $T = 500$, $p_0 = 2.5\%$, true $p = 3.5\%$)

95% two-tailed: Type I 3.0%, Type II 77.2%. 90% two-tailed: Type I 8.3%, Type II **60.9%**.

$T = 250$ at 95%: Type II 82.9% (worse). A higher cut-off widens acceptance (worse). A 99% VaR gives fewer exceptions (worse).

$\Rightarrow$ Lowering the backtest confidence level is the only change that cuts Type II.

Interpretation

The price is a higher Type I rate (3.0% to 8.3%): the trade-off of Figure 2 in action.

Why the other options are wrong

A is incorrect. Less data lowers power: Type II rises to about 83%.

B is incorrect. A higher cut-off accepts more bad models.

C is incorrect. A 99% VaR produces fewer expected exceptions, so deviations are harder to detect.

LO: MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 and The Basel Rules

MR-4 • Backtesting VaR • Vardeez: IMPORTANT (refer to crash-course notes)

MR-4 e

P3 CONCEPT

Explain the need to consider conditional coverage in the backtesting framework.

Workbook: C2 Backtesting VaR: Exceedance vs PIT • Tier C • Map #21 (4/16)

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C2) hits this LO.
Vardeez: conditional coverage $LR_{cc} = LR_{uc} + LR_{ind}$, and the two different critical values (3.84 and 5.99).

Reported exam shape

- **C2 S3** Kupiec's POF test (unconditional coverage) versus Christoffersen (independence, conditional coverage).

What the questions look like: what clustering means, which test detects it, and reading LR values against the right cut-off.

1. Why the count is not enough

Kupiec's test (MR-4 c) is **unconditional**: it ignores conditioning, or time variation, in the data. It checks only whether the *number* of exceptions is right. But exceptions can **cluster or "bunch"** closely in time, which should also invalidate the model.

- At 95% about 13 exceptions are expected a year, spread evenly. If 10 of them fall in the last two weeks, that is a red flag even though the count is fine.
- Bunching says the VaR is *not responding to current conditions*: volatility has risen and the model has not caught up, or traders have moved into unusual positions or risk "holes".
- A verification system should therefore measure **conditional coverage**: correct coverage *conditional on current conditions*. The remedy for a failure is a model that lets risk vary through time (e.g. GARCH or filtered historical simulation, MR-2 c).

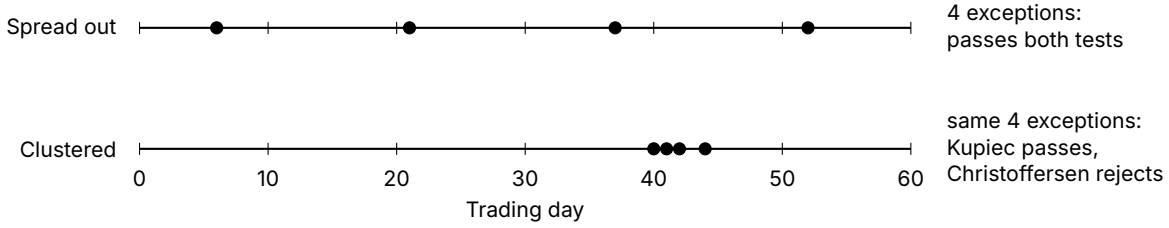


Figure 3. Two histories with the same number of exceptions (filled dots). The unconditional test cannot tell them apart; the independence test flags the clustered one.

2. Christoffersen's test

Each day set an indicator to 0 (no exception) or 1 (exception). Let T_{ij} be the number of days in state j when the previous day was in state i, and π_i the probability of an exception given state i the day before:

$$\pi_0 = \frac{T_{01}}{T_{00} + T_{01}} \qquad \pi_1 = \frac{T_{11}}{T_{10} + T_{11}} \qquad \pi = \frac{T_{01} + T_{11}}{T}$$

If exceptions are independent, $\pi_0 = \pi_1$. The independence statistic compares the likelihood with a common π to the likelihood with separate π_0, π_1:

$$LR_{ind} = -2\ln\left[(1-\pi)^{T_{00}+T_{10}}\pi^{T_{01}+T_{11}}\right] + 2\ln\left[(1-\pi_0)^{T_{00}}\pi_0^{T_{01}}(1-\pi_1)^{T_{10}}\pi_1^{T_{11}}\right]$$

$$\boxed{LR_{cc} = LR_{uc} + LR_{ind}}$$

Each component is $\chi^2(1)$; the sum is $\chi^2(2)$.

Statistic	Tests	Reject at 95% if
LR_{uc} (Kupiec)	right *number* of exceptions	> 3.841
LR_{ind} (Christoffersen)	exceptions *independent* (no bunching)	> 3.841
LR_{cc}	both together (conditional coverage)	$>$ **5.991**

JPMorgan, 1998. 20 exceptions in 252 days ($\pi = 7.9\%$): 14 followed a day with no exception, 6 followed an exception. So $\pi_0 = 14/232 = 6.0\%$ and $\pi_1 = 6/20 = 30.0\%$: an exception is five times as likely after an exception. $LR_{ind} = 9.53 > 3.841$: reject independence; exceptions cluster. With $LR_{uc} = 3.91$, $LR_{cc} = 13.44 > 5.991$.

Limits. The test is a first-order Markov chain: it looks one day back only, so breaches a week apart are invisible to it (Ljung-Box or the Christoffersen and Pelletier duration test address that, see MR-7). Standard exception tests are the most powerful in their class; more powerful tests need *more information*, such as the full distribution (Crnkovic and Drachman's Kuiper test), the time between exceptions (duration), or parametric volatility forecasts.

Exam traps

- **Kupiec versus Christoffersen (sibling swap).** Kupiec cannot see clustering if the count is right; detecting bunching is the independence test's job.
- **Two critical values.** LR_{uc} and LR_{ind} against 3.841; LR_{cc} against 5.991. Using 3.841 for LR_{cc} is the classic slip.
- **The statistics add.** LR_{cc} is the *sum*, never a difference.
- **Denominators of π_0 and π_1.** Condition on the previous day's state ($T_{00}+T_{01}$ and $T_{10}+T_{11}$), not on all T days.
- **One lag only.** Christoffersen does not detect dependence at distances greater than one day.
- **Independence is the null.** Serially independent exceptions are exactly the case where the unconditional test suffices.

Practice questions: MR-4 e

Question MR-4 e.1

Bank Q236 • Schweser Module Quiz 4.1 Q2

Unconditional testing does not reflect the:

A. size of the portfolio.

B. number of exceptions.

C. confidence level chosen.

D. timing of the exceptions.

Question MR-4 e.2

Bank Q231 • AP Mock 2 Q25 (rebuilt)

A bank backtests its 99% one-day VaR model and observes that several exceptions occurred within a short span of consecutive days, although the total number of exceptions over the sample is close to the expected number. What does this clustering of exceptions MOST likely indicate?

A. The confidence level is set too low, so the model produces too many exceptions overall.

B. The exceptions are not independent over time, so the model does not adequately capture time-varying risk such as volatility clustering.

C. The total number of exceptions is too large, so the model fails the unconditional coverage test.

D. The sample is too small, so a Type II error is certain to occur.

Question MR-4 e.3

Bank Q233 • BT L2.T5.59.3

A conditional coverage model would be appropriate for EACH of the following EXCEPT in the case of:

A. Serial independence of exceptions

B. Serial correlation of exceptions

C. Time variation of exceptions

D. Recent volatility clustering of exceptions

Question MR-4 e.4

Bank Q234 • BT P2.T5.713.3

An analyst evaluates the latest backtest results of his firm's VaR model. He concludes that the firm should shift to a VaR model that uses filtered historical simulation with GARCH volatility, and his primary reason is that his statistical test shows that a conditional coverage framework is necessary. Which of the following is the MOST likely proximate justification for a shift to a conditional coverage model?

A. Extreme losses are bunching (i.e., not independent) in the sample.

B. His test provides evidence of a long-run (unconditional) volatility in the series.

C. The historical sample is not large enough to give sufficiently low Type I and Type II error rates.

D. His Kupiec test implies a confidence region that extends to zero exceptions, so he cannot evaluate outcomes with few exceptions.

Question MR-4 e.5

Bank Q235 • BT P2.T5.22.8.3

An analyst backtests a VaR model using Christoffersen's approach, with joint statistic $LR_{cc} = LR_{uc} + LR_{ind}$. The results are: $LR_{uc} = 2.58$, below the $\chi^2(1)$ cutoff of 3.841, so the null is not rejected; and $LR_{ind} = 15.36$, above the cutoff, so the null is rejected. The analyst hopes that the model is unbiased and that exceptions do not cluster. What do the results say?

A. Good news: the VaR model is unbiased and there is no clustering.

B. Mixed news: the VaR model is unbiased, but there is clustering.

C. Mixed news: there is no clustering, but the VaR model is biased.

D. Bad news: the VaR model is biased and there is clustering.

Question MR-4 e.6

New • built from GARP MR Ch.4 (JPM 1998 example) and Vardeez

A risk manager reviews a 95% VaR model over 252 trading days with 20 exceptions. Of these, 14 occurred on a day following a day with no exception (232 such days) and 6 occurred on a day following an exception (20 such days). The unconditional coverage statistic is $LR_{uc} = 3.91$ and the independence statistic is $LR_{ind} = 9.53$. Which of the following statements is correct?

A. $\pi_1 = 30.0\%$ versus $\pi_0 = 6.0\%$ and $LR_{ind} = 9.53 > 3.841$ indicate clustering, and $LR_{cc} = 13.44 > 5.991$ rejects conditional coverage.

B. $\pi_1 = 6/252 = 2.4\%$ is below $\pi_0 = 14/252 = 5.6\%$, so exceptions are less likely after an exception day and independence is not rejected.

C. $LR_{cc} = 13.44$ is compared with the $\chi^2(1)$ cut-off 3.841; since LR_{uc} alone already exceeds 3.841, the independence test adds no information.

D. $LR_{cc} = 9.53 - 3.91 = 5.62$ is below 5.991, so the model passes conditional coverage even though it fails unconditional coverage.

Question MR-4 e.7

Workbook C2.3 • S3, S4 • Map #21

An internal auditor at a Singapore bank is reviewing the backtesting report for a desk's 99% VaR model. Over 250 days the desk recorded 3 exceptions, on three consecutive trading days of a volatile month. The auditor drafts comments for the risk committee. Which of the following comments is correct?

A. Kupiec's proportion of failures test should reject the model, because exceptions bunched within two weeks raise the failure rate measured by the test.

B. If the model is in fact correct and is rejected because of bad luck, the error committed is a Type 2 error, the risk that the Basel zones are designed to keep low.

C. Christoffersen's independence test would also reject a model whose exceptions occur exactly one week apart, because it detects dependence between exceptions at any lag.

D. The conditional coverage statistic adds Kupiec's LR_{uc} to Christoffersen's LR_{ind} and can reject this model even though its exception count is acceptable.

Solutions • MR-4 e

Solution MR-4 e.1 — Answer: D

Bank Q236 • Schweser Module Quiz 4.1 Q2

Step 1 – What the unconditional test uses

It compares the observed number of exceptions with pT, where p comes from the VaR confidence level. Portfolio size is embedded in the VaR and P&L that define an exception.

Step 2 – What it ignores

Each day is treated alike: *when* the exceptions happen is invisible to it.
$\Rightarrow$ Timing is what unconditional testing does not reflect.

Interpretation

Adding the independence test is what brings timing in, giving conditional coverage.

Why the other options are wrong

A is incorrect. Portfolio size is reflected through the VaR and the P&L.
B is incorrect. The number of exceptions is exactly what the test examines.
C is incorrect. The confidence level sets the expected exception rate under the null.

LO: MR-4 e: Explain the need to consider conditional coverage in the backtesting framework. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 Conditional Coverage Models

Solution MR-4 e.2 — Answer: B

Bank Q231 • AP Mock 2 Q25

Step 1 – Separate count from timing

The total is close to the expected number, so unconditional coverage is satisfied.

Step 2 – Read the clustering

Exceptions should be spread evenly. Bunching means they are not independent: the model is not responding to changing conditions such as volatility clustering.
$\Rightarrow$ A failure of the independence component of conditional coverage.

Interpretation

A model can have exactly the right number of exceptions and still be wrong at the moments that matter.

Why the other options are wrong

A is incorrect. A wrong confidence level changes the *number* of exceptions, which the stem says is about right.
C is incorrect. Contradicts the stem: the count is near expectation.
D is incorrect. Sample size affects power, but nothing makes an error "certain", and it does not explain the bunching.

LO: MR-4 e: Explain the need to consider conditional coverage in the backtesting framework. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 Conditional Coverage Models

Solution MR-4 e.3 — Answer: A

Bank Q233 • BT L2.T5.59.3

Step 1 – What conditional coverage adds

It extends the unconditional test to allow for time variation and dependence of exceptions.

Step 2 – Classify each case

Serial correlation, time variation and volatility clustering are all forms of dependence: conditioning is needed. Serial *independence* is the null itself: exceptions are already unrelated over time.
⇒ The unconditional framework suffices under independence.

Interpretation

Independence is the assumption that the standard binomial test already makes.

Why the other options are wrong

B is incorrect. Serial correlation is precisely what conditional coverage addresses.
C is incorrect. Time variation is the feature conditioning captures.
D is incorrect. Volatility clustering produces bunched exceptions, the classic reason to condition.

LO: MR-4 e: Explain the need to consider conditional coverage in the backtesting framework. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 Conditional Coverage Models

Solution MR-4 e.4 Answer: A

Bank Q234 • BT P2.T5.713.3

Step 1 – What the unconditional test assumes

Daily exceptions are independent Bernoulli trials.

Step 2 – What would justify conditioning

Evidence that extreme losses bunch contradicts independence and signals time-varying risk. A model that updates volatility through time (FHS with GARCH) is the remedy.
⇒ Bunching is the proximate justification.

Interpretation

The move to a conditional model is driven by the timing of exceptions, not by sample size or power.

Why the other options are wrong

B is incorrect. A long-run volatility is an unconditional, constant feature; it does not call for a conditional model.
C is incorrect. A small sample is a power problem (Type I/II), unrelated to timing.
D is incorrect. A Kupiec region reaching zero is a low-p limitation of unconditional testing, not a reason to condition.

LO: MR-4 e: Explain the need to consider conditional coverage in the backtesting framework. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 Conditional Coverage Models

Solution MR-4 e.5 Answer: B

Bank Q235 • BT P2.T5.22.8.3

Step 1 – Unconditional coverage

$LR_{uc} = 2.58 < 3.841$: do not reject correct coverage. The number of exceptions is fine (the model is unbiased).

Step 2 – Independence

$LR_{ind} = 15.36 > 3.841$: reject independence. Exceptions cluster.

Step 3 – Joint test (check)

$LR_{cc} = 2.58 + 15.36 = 17.94 > 5.991$: conditional coverage fails, driven by the independence term.
⇒ Mixed news: unbiased, but clustered.

Interpretation

This is exactly the pattern conditional coverage exists to catch: right count, wrong timing.

Why the other options are wrong

A is incorrect. Ignores the rejected independence test.
C is incorrect. Reverses the two results.
D is incorrect. Treats the non-rejected LR_{uc} as a rejection.

LO: MR-4 e: Explain the need to consider conditional coverage in the backtesting framework. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2, eqs. 4.4 and 4.5

Solution MR-4 e.6

Answer: A

New • built from GARP MR Ch.4 and Vardeez

Step 1 – Conditional probabilities

$$\pi_0 = \frac{T_{01}}{T_{00}+T_{01}} = \frac{14}{232} = 6.0\%, \qquad \pi_1 = \frac{T_{11}}{T_{10}+T_{11}} = \frac{6}{20} = 30.0\%$$

$\Rightarrow$ An exception is five times as likely after an exception day.

Step 2 – Independence

$LR_{ind} = 9.53 > 3.841$: reject independence; exceptions cluster.

Step 3 – Conditional coverage

$$LR_{cc} = LR_{uc} + LR_{ind} = 3.91 + 9.53 = 13.44 > 5.991 \ \ (\chi^2(2))$$

Interpretation

The model fails on both counts: too many exceptions ($LR_{uc} = 3.91 > 3.841$) and bunched ones. GARP's conclusion is to explore models that allow for time variation in risk.

Why the other options are wrong

B is incorrect. Divided by all 252 days instead of conditioning on the previous day's state.
C is incorrect. Used the $\chi^2(1)$ cut-off for the $\chi^2(2)$ joint statistic, and wrongly dismissed the independence test, which carries different information from the count.
D is incorrect. Subtracted instead of adding: LR_{cc} is the sum of the two statistics.

LO: MR-4 e: Explain the need to consider conditional coverage in the backtesting framework. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 Conditional Coverage Models, Table 4.5, eqs. 4.4 and 4.5

Solution MR-4 e.7

Answer: D

Workbook C2.3 • S3, S4

Step 1 – Unconditional coverage

Expected exceptions: $0.01 \times 250 = 2.5$. Three exceptions lies inside the nonrejection region ($N < 7$ at $T = 252$), so Kupiec's test does not reject: $LR_{uc} = 0.095 < 3.841$. The count is fine.

Step 2 – Independence

Exceptions "could cluster or bunch closely in time, which also should invalidate the model". Here $T_{00} = 245, T_{01} = 1, T_{10} = 1, T_{11} = 2$: $\pi_0 = 1/246 = 0.4\%$ but $\pi_1 = 2/3 = 66.7\%$, giving $LR_{ind} = 15.65 > 3.841$.

Step 3 – Conditional coverage

$$LR_{cc} = LR_{uc} + LR_{ind} = 0.095 + 15.651 = 15.75 > 5.991 = \chi^2_{0.95}(2)$$

Interpretation

D is correct. The joint test checks whether the proportion of violations following an exceedance equals the proportion following a non-exceedance, while also checking the level. It catches a model whose exceptions are right in number but wrong in timing: the signal that VaR is not responding to changing volatility.

Why the other options are wrong

A is incorrect. Sibling swap. Kupiec counts exceptions only and "ignores the time variation in the data"; three consecutive exceptions have the same failure rate as three spread-out ones.

B is incorrect. Polarity flip. Rejecting a correct model is a Type 1 error (the 10.8% probability of five or more exceptions under a correct model); Type 2 is failing to reject an incorrect model.

C is incorrect. Scope error. Christoffersen's test is a first-order Markov chain and "does not allow for the possibility that independence may be violated at a distance greater than one lag"; breaches a week apart need Ljung-Box or the duration test.

LO: MR-4 c, d, e; MR-7 a: Identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest. • **Reference:** Jorion, Value at Risk 3e, Ch.6, 4.2 (Model Verification, Conditional Coverage Models), Tables 4.1 and 4.2; Iercosan et al., Ch.4, 7.5

MR-4 • Backtesting VaR • Vardeez: IMPORTANT (refer to crash-course notes)

MR-4 f

P3 CONCEPT

Describe the Basel rules for backtesting.

Workbook: C2 Backtesting VaR: Exceedance vs PIT • Tier C • Map #21 (4/16)

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C2) hits this LO.
Vardeez: Basel penalty zones.

Reported exam shape

- **C2 S5** Basel penalty zones and the multiplier they set.

What the questions look like: zone and plus factor for a given number of exceptions (read off the table), which causes attract a penalty, which exceptions may be excluded, and the design trade-off of the zones. One reported shape feeds the multiplier into the capital charge.

1. The Basel (1996) backtest

- Record daily exceptions of the **99% one-day VaR** over the **last 250 days** (one year). Expected: $0.01 \times 250 = 2.5$.
- The rules are derived directly from the failure-rate test; first choose a low Type I error rate, because a bank that suffers bad luck should not be penalised unduly.
- Up to four exceptions is acceptable: the **green** zone. From five, the bank is in the **yellow** or **red** zone and the multiplicative factor k rises progressively from 3 to 4.

Zone	Exceptions	Increase in k (plus factor)	k
Green	0 to 4	0.00	3.00
Yellow	5	0.40	3.40
	6	0.50	3.50
	7	0.65	3.65
	8	0.75	3.75
	9	0.85	3.85
Red	10 or more	1.00 (automatic)	4.00



Figure 4. Basel traffic-light zones for a 99% VaR over 250 days, with the multiplier k under each exception count. Thick lines mark the zone boundaries at 5 and 10 exceptions. Shades: green lightest, yellow mid grey, red darkest; each zone is also labelled.

2. Inside the yellow zone: why did the exceptions happen?

The plus factors are *specified* by the table, but whether they are applied in the yellow zone is up to the supervisor, depending on the cause:

Cause	What happened	Penalty
Basic integrity of the model	Positions reported incorrectly, or an error in the program code	"should" apply
Model accuracy could be improved	The model does not measure risk precisely enough (e.g. too few maturity buckets, median duration for a bond book)	"should" apply
Intraday trading	Positions changed during the day	"should be considered"
Bad luck	Markets particularly volatile or correlations changed	no guidance; "expected to occur at least some of the time"

Exclusions. Exceptions may be excluded if they are the "result of such occurrences as sudden abnormal changes in interest rates or exchange rates, major political events, or natural disasters". Ordinary bad luck (volatile markets, changed correlations) is *not* excluded: it still counts toward the zone. Supervisors keep the flexibility to adjust the rules in turbulent times.

3. Where the multiplier goes, and the design trade-off

The multiplier enters the 1996 market risk charge (ORR-21 c), with 10-day 99% VaR:

$$\text{MR} = \max\left(\text{VaR}_{t-1},\ k \times \overline{\text{VaR}}_{60}\right)$$

- **Error rates of the zones (MR-4 d).** With a cut-off of 5, a correct model is penalised 10.8% of the time (Type I) and a model with only 97% coverage passes 12.8% of the time (Type II). The framework is not very powerful.
- **Moving the boundaries.** A smaller green zone or a lower red threshold catches more banks that understate risk (less Type II) but penalises more unlucky banks (more Type I); a larger green zone or a higher red threshold does the reverse.
- Carey's summary ("five or less exceptions enabled $k = 3$") is looser than Jorion's table: for MR-4 f questions use the table above.
- Under FRTB (MR-18 c), backtesting moves to the desk level: one-day VaR at 99% *and* 97.5% over 12 months; more than 12 exceptions at 99% or 30 at 97.5% sends the desk to the standardized approach.

Correction to the source notes. Vardeez writes "Economic capital = VaR $\times(3 + k)$". Three fixes: it is *regulatory* market risk capital, not economic capital; k is the *whole* multiplier (3 plus the plus factor), so the expression should read $k = 3 +$ plus factor; and the charge is $\max(\text{VaR}_{t-1},\ k \times \overline{\text{VaR}}_{60})$ on a 10-day 99% VaR, not a simple product. Also note that his table numbers the four Basel causes I to IV: these are *not* the statistical Type I and Type II errors of MR-4 d.

Exam traps

- **Plus-factor row off by one.** 5 is already yellow (+0.40); 6 is +0.50, 7 is +0.65, 8 is +0.75, 9 is +0.85.
- **Excluding bad luck.** Only sudden abnormal market moves, major political events and natural

disasters may be excluded; "volatile markets, changed correlations" still count.

- **Yellow is automatic.** It is not: the size is specified, the application is discretionary. Only red is automatic.
- **Multiplier below 3.** k never falls below 3, whatever the backtest.
- **Sample size as a Basel cause.** The four causes are integrity, accuracy, intraday trading and bad luck.
- **Zone-change direction.** Expanding the red zone *raises* the chance of penalising an unlucky bank.

Practice questions: MR-4 f

Question MR-4 f.1

Bank Q244 • Schweser Module Quiz 62.1 Q2

Saugatuck National Bank uses the internal model-based approach to set market risk capital as prescribed by the 1996 Amendment to the 1988 Basel Accord. The bank has backtested its 99% one-day VaRs against the actual losses over the last 250 trading days. Based on the results of the backtesting, the bank recorded 11 exceptions. Based on these results, the multiplicative factor (m) in the model should be set:

A. less than 3.

B. equal to 3.

C. between 3.1 and 3.9.

D. equal to 4.

Question MR-4 f.2

Bank Q237 • BT L2.T5.60.4 (rebuilt)

If a bank's backtest produces exceptions in the yellow zone (five to nine exceptions out of 250 days at 99% VaR), how is the VaR multiplier treated?

A. The multiplier remains at 3.0.

B. The multiplier might be increased by between +0.40 and +0.85 (to between 3.40 and 3.85), at the supervisor's discretion depending on the cause of the exceptions.

C. The multiplier must be increased to 4.0 or greater.

D. An additional stressed VaR must be added to the standard VaR.

Question MR-4 f.3

Bank Q238 • BT L2.T5.60.5

In regard to backtest outcomes in the yellow zone, which of the following is MOST accurate?

A. The increase to the multiplier is specified (e.g., +0.65 at 7 exceptions), and the supervisor must apply it in all cases.

B. The increase to the multiplier is specified, but the supervisor has discretion in whether to apply it.

C. The increase to the multiplier is subjective, but the supervisor must apply it in all cases.

D. The increase to the multiplier is subjective and the supervisor has discretion in whether to apply it.

Question MR-4 f.4

Bank Q243 • Schweser Module Quiz 4.2 Q1

The Basel Committee has established four categories of causes for exceptions. Which of the following does not apply to one of those categories?

A. The sample is small.

B. Intraday trading activity.

C. Model accuracy needs improvement.

D. The basic integrity of the model is lacking.

Question MR-4 f.5

Bank Q240 • GARP PE1 Q58 • Mock 3 Q23 (key corrected)

A regulatory analyst at an investment bank is reviewing the Basel rules for backtesting VaR models. Under the Basel framework, a penalty can be given to banks that have more than four exceptions to their 1-day 99% VaR over the last 250 trading days. Which of the following scenarios is most likely to result in a penalty?

A. A large move in interest rates occurs in conjunction with a small move in correlations.

B. The bank's model calculates interest rate risk based on the median duration of the bonds in the portfolio.

C. A sudden market crisis in an emerging market leads to losses in the bank's equity positions in that country.

D. A sudden devastating earthquake causes major losses in the bank's key area of operation.

Question MR-4 f.6

Bank Q239 • BT P2.T5.57.3 (rebuilt)

Under Basel's backtest, a 99% one-day VaR is tested over 250 days (green 0–4, yellow 5–9, red 10 or more exceptions). Regulators want to catch banks that understate risk without unduly penalizing banks with a correct model that are merely unlucky. EACH of the following is true EXCEPT:

A. Shrinking the green zone (e.g., to 0–2 exceptions) increases the probability of catching banks that understate their risk.

B. Raising the red-zone threshold (e.g., to 13 or more exceptions) makes it less likely that a bank with a correct model is penalized through bad luck.

C. Lowering the red-zone threshold (e.g., to 7 or more exceptions) makes it less likely that a bank with a correct model is penalized through bad luck.

D. Lowering the red-zone threshold (e.g., to 7 or more exceptions) increases the probability of catching banks that understate their risk.

Question MR-4 f.7 Workbook C2.1 • S1, S5 • Map #21

A regulatory analyst at a German bank is computing the market risk capital charge for a trading book under the internal models approach. Over the last 250 trading days the one-day 99% VaR was exceeded on 8 days, with the following causes identified:

- 4 days: markets were particularly volatile and correlations changed
- 2 days: the model uses too few maturity buckets for the yield curve
- 1 day: positions changed materially during the day through intraday trading
- 1 day: a major earthquake caused large losses across local markets

The supervisor will exclude any exception that the Basel rules allow to be excluded. The 10-day 99% VaR for the previous day is EUR 52 million and the average 10-day 99% VaR over the last 60 days is EUR 40 million. Which of the following is closest to the multiplier k and the market risk capital charge?

A. $k = 3.00$; EUR 120 million

B. $k = 3.50$; EUR 140 million

C. $k = 3.65$; EUR 146 million

D. $k = 3.75$; EUR 150 million

Solutions • MR-4 f

Solution MR-4 f.1 — Answer: D

Bank Q244 • Schweser Module Quiz 62.1 Q2

Step 1 – Zone

Expected exceptions $0.01 \times 250 = 2.5$. Eleven is "10 or more": the red zone.

Step 2 – Multiplier

The red zone carries an automatic plus factor of 1.00: $m = 3 + 1 = 4$.
$\Rightarrow$ The multiplier is at its maximum.

Interpretation

In the red zone there is no discretion and no cause analysis: the penalty is automatic.

Why the other options are wrong

A is incorrect. The multiplier never falls below 3.
B is incorrect. 3 applies only to the green zone (0 to 4).
C is incorrect. An approximation of the yellow zone (3.40 to 3.85); 11 exceptions is beyond it.

LO: MR-4 f: Describe the Basel rules for backtesting. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules, Table 4.3

Solution MR-4 f.2 — Answer: B

Bank Q237 • BT L2.T5.60.4

Step 1 – Yellow-zone plus factors

From +0.40 (5 exceptions) to +0.85 (9 exceptions), so k ranges from 3.40 to 3.85.

Step 2 – Application

Depends on the cause: integrity and accuracy problems "should" be penalised, intraday trading "should be considered", bad luck gets no guidance.
$\Rightarrow$ A graduated increase, applied at the supervisor's discretion.

Interpretation

Only the red zone makes the increase to 4.0 automatic.

Why the other options are wrong

A is incorrect. 3.0 is the green-zone multiplier; in the yellow zone a plus factor is presumed unless not warranted.
C is incorrect. Describes the red zone.
D is incorrect. Stressed VaR (Basel 2.5) is a separate capital add-on, not the backtesting penalty.

LO: MR-4 f: Describe the Basel rules for backtesting. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules, Table 4.3

Solution MR-4 f.3 — Answer: B

Bank Q238 • BT L2.T5.60.5

Step 1 – Is the size specified?

Yes: +0.40, +0.50, +0.65, +0.75, +0.85 for 5 to 9 exceptions (Table 4.3).

Step 2 – Is the application mandatory?

No: in the yellow zone it depends on the supervisor's judgement of the cause.
$\Rightarrow$ Specified size, discretionary application.

Interpretation

GARP calls the description of the penalty "suitably vague": the vagueness is in *whether* to apply, not *how much*.

Why the other options are wrong

A is incorrect. Right on size, wrong on automatic application (only red is automatic).
C is incorrect. Reverses both features.
D is incorrect. Right on discretion, wrong that the size is subjective.

LO: MR-4 f: Describe the Basel rules for backtesting. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules

Solution MR-4 f.4 Answer: A

Bank Q243 • Schweser Module Quiz 4.2 Q1

Step 1 – Recall the four categories

Basic integrity of the model; model accuracy could be improved; intraday trading; bad luck.

Step 2 – Find the odd one out

B, C and D are three of the four. A small sample is a statistical (power) issue, not a Basel cause of exceptions.

Interpretation

The categories describe *why* an exception happened, which decides whether the penalty applies.

Why the other options are wrong

B is incorrect. Intraday trading is a category (penalty "should be considered").
C is incorrect. Model accuracy is a category (penalty "should" apply).
D is incorrect. Basic integrity is a category (penalty "should" apply).

LO: MR-4 f: Describe the Basel rules for backtesting. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules

Solution MR-4 f.5 Answer: B

Bank Q240 • GARP PE1 Q58

Step 1 – Sort each scenario into a Basel cause

A large rate move with a small correlation move, an emerging-market crisis, and an earthquake are unforeseeable market or natural events: bad luck (and the earthquake may even be excluded).

Step 2 – Find the model flaw

Measuring interest rate risk with the *median* duration of the bonds ignores the spread of maturities: "model accuracy could be improved". The penalty "should" apply.
$\Rightarrow$ B is the scenario most likely to be penalised.

Interpretation

Penalties target avoidable model deficiencies the bank could fix, not exceptions caused by events no model would have predicted.

Why the other options are wrong

A is incorrect. Unusual market moves and correlation changes are bad luck.
C is incorrect. A sudden market crisis is bad luck.
D is incorrect. A natural disaster is bad luck, and its exceptions may be excluded.

LO: MR-4 f: Describe the Basel rules for backtesting. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules

Solution MR-4 f.6 — Answer: C

Bank Q239 • BT P2.T5.57.3

Step 1 – Error rates for a correct model ($p = 1\%$, $T = 250$)

$P(N \geq 10) \approx 0.025\%$, $P(N \geq 7) \approx 1.4\%$, $P(N \geq 13) \approx 0.0002\%$.

Step 2 – Effect of lowering the red threshold to 7

More correct models land in the red zone: the Type I rate rises from about 0.025% to 1.4%. More unlucky banks are penalised, not fewer.

$\Rightarrow$ C is false; D (it catches more understating banks) is true.

Interpretation

Every tightening of the zones buys less Type II at the cost of more Type I, and every loosening does the reverse.

Why the other options are wrong

A is incorrect. True: a smaller green zone catches more faulty models.

B is incorrect. True: a higher red threshold lowers the bad-luck penalty probability.

D is incorrect. True: a lower red threshold catches more understating banks.

LO: MR-4 f: Describe the Basel rules for backtesting. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules, Table 4.4

Solution MR-4 f.7 — Answer: C

Workbook C2.1 • S1, S5

Step 1 – Count the exceptions that remain

Exceptions "may be excluded if they are the result of such occurrences as sudden abnormal changes in interest rates or exchange rates, major political events, or natural disasters". Only the earthquake qualifies. Volatile markets and changed correlations are bad luck that should "be expected to occur at least some of the time"; model accuracy and intraday trading exceptions are penalised.

$$N = 8 - 1 = 7$$

Step 2 – Zone and plus factor

Seven exceptions fall in the yellow zone (5 to 9); Jorion's Table 4.3 gives an increase in k of 0.65:

$$k = 3 + 0.65 = 3.65$$

$\Rightarrow$ With a correct 99% model the expected count is 2.5; the probability of seven or more under a correct model is only 1.4% (Table 4.4), so a penalty is appropriate.

Step 3 – Capital charge

$$\text{MR} = \max\left(\text{VaR}_{t-1},\ k \times \overline{\text{VaR}}_{60}\right) = \max(52,\ 3.65 \times 40) = \max(52,\ 146) = 146$$

Interpretation

The bank holds EUR 146 million, EUR 26 million more than with $k = 3$. Fixing the maturity buckets ("model accuracy could be improved") is the actionable item: those exceptions carry a penalty that "should" apply.

Why the other options are wrong

A is incorrect. Excluded every bad-luck exception as well as the earthquake: $8 - 1 - 4 = 3$, green zone, $3 \times 40 = 120$.

B is incorrect. Read the plus factor one row too high (0.50 belongs to 6 exceptions): $3.50 \times 40 = 140$.

D is incorrect. Did not exclude the earthquake: 8 exceptions, $k = 3.75$, $3.75 \times 40 = 150$.

LO: MR-4 f: Describe the Basel rules for backtesting. ORR-21 c: Describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book. • **Reference:** Jorion, Value at Risk 3e, Ch.6, The Basel Rules, Tables 4.3 and 4.4; Carey, Capital Regulation Before the Global Financial Crisis, 21.1

MR-5 • VaR Mapping • Vardeez: IMPORTANT

MR-5 c

P3 CONCEPT

Differentiate among the three methods for mapping portfolios of fixed-income securities.

Workbook: C3 VaR Mapping • Tier C • Map #22

WHAT IS TESTED HERE

Why P3. Tier C workbook topic (C3, VaR Mapping): "difference in VaR mapping: principal, duration, cash flow" was reported twice in identical wording.
Vardeez: differentiate principal, duration and cash-flow mapping. Do not worry about mapping linear and non-linear derivatives (MR-5 g, out of this book).

Reported exam shapes

- **C3·S1** The difference among principal, duration and cash-flow mapping.
- **C3·S3** Ranking the three methods by how faithfully they capture risk.

What the questions look like: "which statement is correct" items built by polarity flips (principal mapping *under*states risk), sibling swaps (a feature of cash-flow mapping credited to duration mapping) and scope errors ("only cash-flow mapping preserves market value"). The numbers you need are the ranking, not the matrix algebra.

1. What mapping does

Mapping replaces the current value of every position by **exposures on a small set of risk factors**. For fixed income the risk factors are **zero-coupon bonds at standard maturities** ("vertices": 1, 2, 3, 4, 5 years and so on), whose VaRs and correlations the risk system already holds.

- Mapping **must preserve the market value** of the position. "Ideally, it also should preserve its market risk." All three methods preserve value; they differ in how well they preserve risk.
- The methods differ only in *where* on the maturity grid the value is placed, and *how many* vertices are used.

2. The three methods side by side

	Principal mapping	Duration mapping	Cash-flow mapping
Where the value goes	All of it at the **average maturity of principal** (one vertex)	All of it at the **portfolio duration** (one vertex, interpolated)	**PV of each cash flow** on the vertex of its own date (many vertices)
Uses coupons?	No: timing of redemptions only	Yes, through duration	Yes, each one separately
Correlations?	No (one factor)	No (one factor)	Yes: correlation matrix of the vertices
Accuracy	Crudest; **overstates** risk	Better	Most precise, most complex
Catch	Ignores intervening coupons	A zero with exactly that maturity rarely exists, so its risk is interpolated	Needs the vertex VaRs and the full correlation matrix

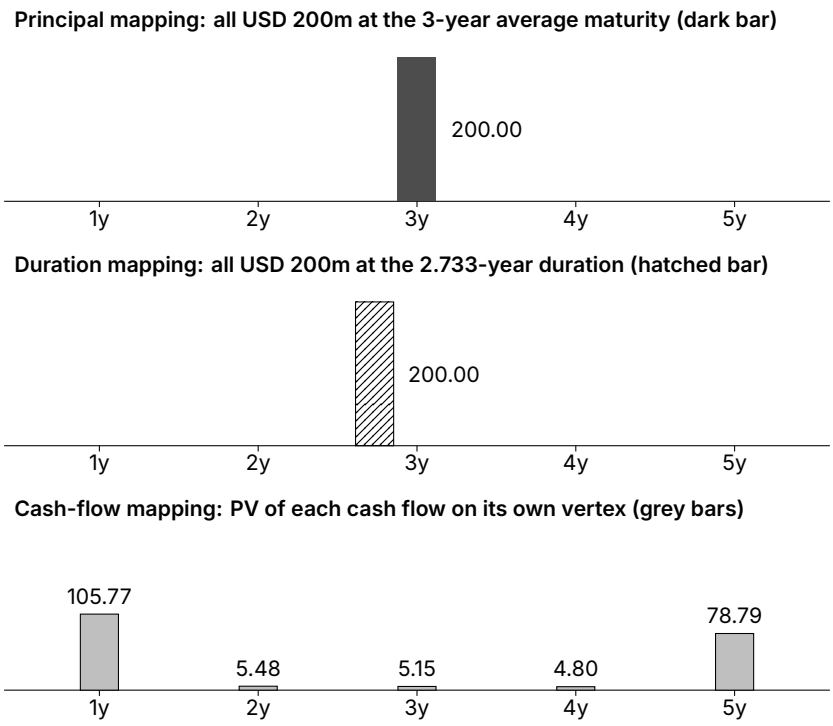


Figure 1. Jorion's two-bond portfolio (USD 100m 5-year 6% and USD 100m 1-year 4%, both at par) mapped three ways. Every panel carries the same USD 200m of value; only the placement changes. Principal mapping puts it all at the 3-year average maturity, duration mapping at the 2.733-year duration, cash-flow mapping spreads the present values over the five vertices.

3. The ranking, and why it runs that way

Jorion's two-bond example (monthly VaR at 95%):

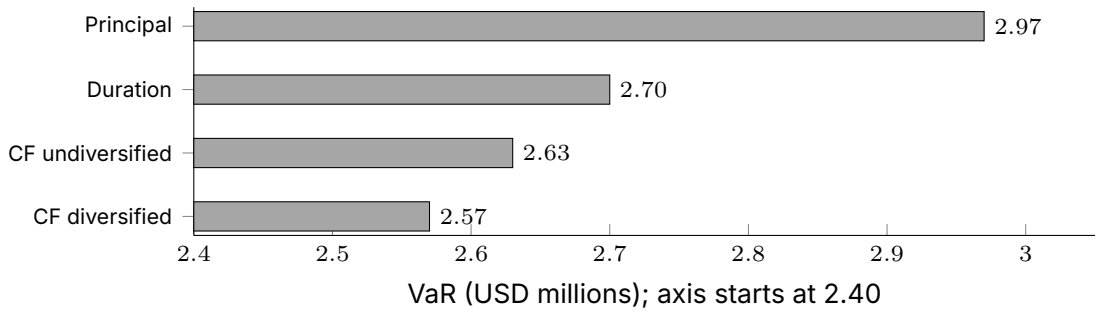


Figure 2. Principal > duration > cash flow (undiversified) > cash flow (diversified). Of the USD 0.13m between duration and diversified cash-flow VaR, USD 0.07m (2.70 to 2.63) comes from risk not being linear in maturity and USD 0.06m (2.63 to 2.57) from correlations below one.

- **Principal mapping overstates risk.** By ignoring the coupons it places the value at the average maturity of principal, which is longer than the duration, and longer maturities carry more risk.
- **Duration to cash flow, two effects.** (1) VaR is not exactly linear in maturity, so one interpolated vertex misprices the spread of cash flows. (2) Vertex correlations are below one, so spreading the cash flows diversifies. Only cash-flow mapping captures either.
- **Undiversified vs diversified.** Summing the vertex VaRs (correlation 1) gives the undiversified cash-flow VaR; using the correlation matrix gives the diversified figure, which is always lower when correlations are below one.

Two versions of the example. GARP (Jorion) uses a 5-year 6% bond and a 1-year 4% bond: 2.97, 2.70, 2.63, 2.57. Schweser and Vardeez use a 5-year 5% bond and a 1-year 3.5% bond, duration 2.768: 2.968, 2.737, 2.674 and 2.615. Both are correct for their inputs and give the same ranking. Questions may use either.

Exam traps

- **Polarity.** Principal mapping *overstates* risk (highest VaR), it does not understate it.
- **Sibling swap.** Placing each cash flow on its own vertex is *cash-flow* mapping. Duration mapping uses one vertex at the portfolio duration.
- **Scope.** All three methods preserve market value. What separates them is how well they preserve *risk*.
- **Correlations.** Principal and duration mapping use a single factor, so no correlation adjustment enters either of them.
- **"Duration mapping is exact for any shift."** False: a single factor captures parallel moves only.
- **Not a fourth method.** "Present value mapping" is a distractor; the three are principal, duration, cash flow.

Practice questions: MR-5 c

Question MR-5 c.1

Bank Q270 • AnalystPrep Q.1520

Which method of mapping fixed-income portfolios decomposes the risk of a bond into the risk of each of the bond's cash flows?

A. Principal mapping

B. Duration mapping

C. Cash-flow mapping

D. Present value mapping

Question MR-5 c.2

Bank Q279 • GARP PE2 Q79

An analyst at a fixed-income investment company is evaluating different ways the company uses to estimate the VaR of its corporate bond portfolios. The portfolios consist of a large number of bonds with a wide range of maturities. The analyst examines the possibility of using a mapping approach to simplify the estimation process. Which of the following statements would the analyst be correct to make regarding the approaches to mapping fixed-income portfolios?

A. The VaR estimated using the principal mapping approach understates the true risk of a portfolio since it ignores coupon payments and any risk associated with them.

B. The VaR estimated using the duration mapping approach replaces the portfolio with a zero-coupon bond whose maturity equals the duration of the portfolio.

C. The VaR estimated using the principal mapping approach differs from the undiversified VaR estimated using the duration mapping approach due to an adjustment made for correlations.

D. The VaR estimated using the cash-flow mapping approach is less accurate than the VaR estimated using the duration mapping approach since it does not account for the timing of cash flows.

Question MR-5 c.3

Bank Q280 • Mock 4 Q80

A risk analyst is choosing among the three approaches to mapping a portfolio of coupon-bearing bonds onto a set of zero-coupon risk factors: principal mapping, duration mapping and cash-flow mapping. Which of the following correctly describes the relationship between these approaches?

A. Cash-flow mapping produces the most accurate VaR estimate because it assigns every individual payment to the risk factor matching its own maturity.

B. Principal mapping produces the most accurate VaR estimate because it avoids the estimation error introduced by decomposing the portfolio into many cash flows.

C. Duration mapping produces a more accurate VaR estimate than cash-flow mapping because it captures the convexity of the portfolio.

D. All three approaches produce identical VaR estimates for a portfolio of option-free bonds, since the underlying cash flows are the same.

Question MR-5 c.4

Bank Q269 • AnalystPrep Q.1516

In a two-bond portfolio example, the VaR computed with duration mapping is 2.80 while the VaR computed with cash-flow mapping is 2.67. The difference between these values is due to the fact that:

A. risk measures are not perfectly linear with maturity and correlations between vertices are below unity.

B. risk measures are perfectly linear with maturity and correlations are below unity.

C. risk measures are perfectly linear with maturity and correlations are above unity.

D. risk measures are not perfectly linear with maturity and correlations are equal to unity.

Question MR-5 c.5

Bank Q272 • BT L2.T5.63.1

Assume a flat yield curve with a spot rate of 4.0% at all maturities and a normally distributed yield volatility of 1.0%. A two-bond portfolio is being mapped: both bonds have USD 100 million face value and pay an annual 4% coupon; one matures in one year, the other in five years. The returns VaR of each vertex at 95% is $1.645 \times 1.0\% \times$ modified duration:

Term (years)	1	2	3	4	5
Modified duration	0.962	1.923	2.885	3.846	4.808
Returns VaR	1.58%	3.16%	4.74%	6.33%	7.91%

Under these assumptions (flat yield curve, constant 1.0% yield volatility), why can we expect cash-flow mapping to produce a lower diversified VaR than either duration or principal mapping?

A. The risk measures are non-linear.

B. Due to imperfect correlations between pairwise risk factors.

C. Fewer total cash flows will be mapped.

D. We cannot expect a lower diversified VaR.

Question MR-5 c.6

Bank Q278 • BT P2.T5.22.10.1

Ralph manages a fixed-income portfolio of high-quality corporate bonds (mostly coupon-bearing, some zero-coupon). His colleague Mary is designing a VaR mapping process using three approaches: principal mapping, duration mapping, and cash-flow mapping. In regard to these three approaches, each of the following statements is true EXCEPT which is false?

A. In all three approaches the primitive instruments are zero-coupon bonds.

B. Principal mapping is the simplest approach, but it tends to overstate the risk of a coupon bond because it ignores the earlier coupon payments.

C. Duration mapping gives the exact change in portfolio value for any change in the term structure, including non-parallel shifts.

D. The diversified VaR from cash-flow mapping rises as the pairwise correlations among the risk factors rise, and equals the undiversified VaR when all correlations are 1.

Question MR-5 c.7 Workbook C3.2 • S1, S3 • Map #22

A junior risk analyst at a Japanese life insurer is preparing a report comparing principal, duration and cash-flow mapping for a bond portfolio, using the textbook example of a 5-year 6% bond and a 1-year 4% bond held in equal amounts at par. Which of the following statements is correct for the analyst to include in the report?

A. Principal mapping understates the true risk of the portfolio, because it maps the market value to the average maturity of principal and ignores the intervening coupon payments.

B. Duration mapping places the present value of each cash flow on the vertex nearest its payment date, which allows it to capture imperfect correlation along the curve.

C. Cash-flow mapping is preferred because it preserves the market value of the portfolio, a property that principal mapping and duration mapping do not have.

D. The diversified cash-flow VaR is below the duration VaR, because risk is not linear in maturity and because correlations between the vertices are below one.

Solutions • MR-5 c

Solution MR-5 c.1 — Answer: C

Bank Q270 • AnalystPrep Q.1520

Step 1 – What each method does with the cash flows

Principal mapping keeps only the principal repayments, placed at their average maturity. Duration mapping collapses all cash flows into one zero at the portfolio duration. Cash-flow mapping takes the present value of *each* coupon and principal payment and puts it on the vertex of its own date.

$\Rightarrow$ Only cash-flow mapping treats every payment as its own risk exposure.

Interpretation

This is the defining feature that makes cash-flow mapping the most precise of the three, and the reason it needs a correlation matrix.

Why the other options are wrong

A is incorrect. Principal mapping ignores coupons and uses only the timing of principal.

B is incorrect. Duration mapping replaces the bond by a single zero at its duration.

D is incorrect. "Present value mapping" is not one of the three standard methods.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Mapping Approaches

Solution MR-5 c.2 — Answer: B

Bank Q279 • GARP PE2 Q79

Step 1 – Test each statement against Jorion's definitions

Duration mapping: "we replace the portfolio by a zero-coupon bond with maturity equal to the duration of the portfolio". That is statement B, word for word.

$\Rightarrow$ B is correct.

Step 2 – Check the others

Principal mapping *overstates* risk (A flips the polarity). Principal and duration mapping are both single-factor, so neither involves a correlation adjustment (C). Cash-flow mapping is the *most* accurate because it keeps the timing of every cash flow (D reverses the ranking).

Interpretation

GARP built three distractors from the three classic errors: a polarity flip (A), a feature of cash-flow mapping attributed to the single-factor methods (C) and a reversed ranking (D).

Why the other options are wrong

A is incorrect. Polarity flip: ignoring coupons places the value at a longer maturity than the duration, so principal mapping overstates risk.

C is incorrect. Correlation adjustments arise only in multi-vertex cash-flow mapping (diversified vs undiversified VaR); principal and duration VaR differ because they use different mapping maturities.

D is incorrect. Cash-flow mapping maps each payment at its own maturity, so it accounts for timing best.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Mapping Approaches

Solution MR-5 c.3

Answer: A

Bank Q280 • Mock 4 Q80

Step 1 – Rank the methods by fidelity

Principal mapping keeps only the principal and its maturity (crudest). Duration mapping uses one zero at the portfolio duration (good for parallel shifts only). Cash-flow mapping maps each payment to its own vertex and uses the correlations (most accurate, most complex).
⇒ A states the ranking correctly and gives the right reason.

Interpretation

Decomposing contractual cash flows adds no estimation error: the cash flows are known. The extra precision comes from capturing the non-linearity of risk in maturity and the imperfect correlation of the vertices.

Why the other options are wrong

B is incorrect. Principal mapping is the least precise, because it ignores coupons.
C is incorrect. Duration mapping is a linear, single-factor approximation; it does not capture convexity and is less accurate than cash-flow mapping.
D is incorrect. The methods give different VaRs because they place the value on different risk factors (average maturity, duration, each vertex): 2.97, 2.70 and 2.57 in Jorion's example.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2, Tables 5.2 to 5.4

Solution MR-5 c.4

Answer: A

Bank Q269 • AnalystPrep Q.1516

Step 1 – What duration mapping misses

Duration mapping represents the portfolio by one zero at the portfolio duration, with its risk interpolated linearly between two vertices. It assumes risk is linear in maturity and implicitly treats all cash flows as perfectly correlated.

Step 2 – What cash-flow mapping adds

It uses the actual vertex risks (so non-linearity in maturity is captured) and the correlation matrix (so correlations below one diversify).
⇒ Both effects push the cash-flow VaR (2.67) below the duration VaR (2.80).

Interpretation

In Jorion's version the USD 0.13m gap splits into USD 0.07m from non-linearity and USD 0.06m from imperfect correlation. Correlations can never exceed one.

Why the other options are wrong

B is incorrect. If risk were perfectly linear in maturity, interpolation at the duration would be exact and that part of the gap would vanish.
C is incorrect. Correlations cannot exceed unity, and linearity is not perfect.
D is incorrect. Correlations equal to one would remove the diversification benefit, so cash-flow mapping would not come out lower on that account.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 (decomposition of the USD 130,000 difference)

Solution MR-5 c.5 — Answer: B

Bank Q272 • BT L2.T5.63.1

Step 1 – Is risk linear in maturity here?

With a flat curve and constant yield volatility, vertex risk $= 1.645 \times 1.0\% \times M/1.04$, which is exactly linear in maturity M. So the non-linearity effect is switched off.

Step 2 – What is left

Principal and duration mapping use one risk factor, which treats all cash flows as perfectly correlated. Cash-flow mapping combines the vertex VaRs with a correlation matrix below one.
$\Rightarrow$ The only source of a lower diversified VaR here is imperfect correlation between the vertices.

Interpretation

The question isolates one of the two effects in Jorion's decomposition. Under the usual upward-sloping, non-linear risk profile both effects operate; in this constructed case only correlation does.

Why the other options are wrong

A is incorrect. By construction risk is linear in maturity here, so non-linearity cannot explain the gap.
C is incorrect. The same USD 200m of present value is mapped; cash flows are spread over more vertices, not fewer.
D is incorrect. A lower diversified VaR is expected because correlations are below one.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2, eq. 5.7

Solution MR-5 c.6 — Answer: C

Bank Q278 • BT P2.T5.22.10.1

Step 1 – Check each statement

A: all three methods map onto zero-coupon bonds at standard maturities. True.
B: principal mapping ignores coupons and overstates risk. True.
C: duration mapping is a single-factor linear approximation; it captures parallel shifts only, approximately. **False.**
D: diversified VaR $= \sqrt{(x \times V)' R (x \times V)}$ rises with the correlations and equals the undiversified sum when all correlations are 1. True.
$\Rightarrow$ C is the false statement.

Interpretation

Non-parallel moves in the term structure are exactly what duration mapping cannot see; they are also the source of tracking error in MR-5 f.

Why the other options are wrong

A is incorrect. True: zero-coupon bonds are the primitive instruments in all three approaches.
B is incorrect. True: placing the value at the average maturity of principal overstates risk.
D is incorrect. True: higher correlations reduce diversification, and at correlation 1 diversified equals undiversified VaR.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2

Solution MR-5 c.7 **Answer: D**

Workbook C3.2 • S1, S3

Step 1 – The three numbers

Principal (3-year vertex): USD 2.97 million. Duration (2.733 years, interpolated risk 1.351%): USD 2.70 million. Cash flow: USD 2.63 million undiversified, USD 2.57 million diversified.

Step 2 – Decompose the duration to cash-flow gap

$$2.70 - 2.57 = 0.13 = \underbrace{(2.70 - 2.63)}_{0.07 \text{ non-linear risk in maturity}} + \underbrace{(2.63 - 2.57)}_{0.06 \text{ correlations below one}}$$

Interpretation

D is correct. The ranking of fidelity runs from cash-flow mapping (finest, captures both effects), through duration mapping (one vertex, ignores the curvature of risk across maturities and assumes perfect correlation), to principal mapping (crudest, ignores coupons).

Why the other options are wrong

A is incorrect. Polarity flip. Principal mapping "overstates the true risk because it ignores intervening coupon payments": it gives the highest VaR, USD 2.97 million.

B is incorrect. Sibling swap. Grouping each cash flow on term-structure vertices is cash-flow mapping; duration mapping replaces the whole portfolio by one zero-coupon bond with maturity equal to the portfolio duration.

C is incorrect. Scope/condition error. All three methods preserve market value (USD 200 million is mapped in each case); "ideally, it also should preserve its market risk", and it is risk preservation that separates them.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Mapping Approaches, Tables 5.2 to 5.4

MR-5 • VaR Mapping • Vardeez: IMPORTANT

MR-5 d

P3 BOTH

Summarize how to map a fixed-income portfolio into positions of standard instruments.

Workbook: C3 VaR Mapping • Tier C • Map #22 • C3.2 worked as MR-5 c.7

WHAT IS TESTED HERE

Why P3. Vardeez flags it inside an Important chapter: *calculation of principal and duration mapping*. The C3 workbook section tags this LO through C3.2 (worked as Question MR-5 c.7).
Vardeez: calculation of principal and duration mapping. Skip the mapping of linear and non-linear derivatives (heavy calculations are unlikely).

What the questions look like: a VaR-% table for zeros at 1 to 5 years and a two-bond portfolio; you place the value at the right maturity (average maturity of principal, or the duration with linear interpolation) and multiply by the portfolio value. Cash-flow mapping appears as mechanics (how many vertices, present values not face values, undiversified vs diversified), not as matrix algebra. Schweser's own note: the full cash-flow calculation is "highly unlikely to show up on the exam".

1. The recipe for each method

Inputs: the VaR % of a zero at each vertex $(V_1, \dots, V_5)$, the portfolio's cash flows, and the spot curve.

Principal mapping.

1. Average maturity of principal: $\bar{M} = \sum_i w_i M_i$, with w_i the market-value weights of the bonds.
2. Read the VaR % of a zero at $\bar{M}$ (interpolate if $\bar{M}$ falls between vertices).
3. VaR = total market value $\times V_{\bar{M}}$.

Duration mapping.

1. Portfolio (Macaulay) duration $D = \sum_t t\,\text{PV}(CF_t) \big/ \sum_t \text{PV}(CF_t)$.
2. Interpolate linearly between the two vertices that bracket D: $V_D = V_j + (V_{j+1} - V_j)(D - j)$.
3. VaR = total market value $\times V_D$.

Cash-flow mapping.

1. Discount each cash flow at the spot rate for its date and place the *present value* on that vertex; cash flows on the same date share one vertex.
2. Individual VaRs $x_i V_i$; undiversified VaR $= \sum_i |x_i| V_i$.
3. Diversified VaR $= \sqrt{(x \times V)'\, R\, (x \times V)}$ with R the vertex correlation matrix.

2. Worked example (Schweser and Vardeez version)

USD 100m 1-year 3.5% bond and USD 100m 5-year 5% bond, both at par; portfolio duration 2.768 years. Monthly 95% VaR % of zeros:

Maturity (years)	1	2	3	4	5
VaR (%)	0.4696	0.9868	1.4841	1.9714	2.4261

- **Principal:** $\bar{M} = 0.5(1) + 0.5(5) = 3$ years, so VaR $= 200 \times 1.4841\% =$ USD 2.968m.
- **Duration:** $V_{2.768} = 0.9868 + (1.4841 - 0.9868)(0.768) = 1.3687\%$, so VaR $= 200 \times 1.3687\% =$ USD 2.737m.
- **Cash flow:** present values 104.83, 4.63, 4.42, 4.24, 81.88 (sum 200.00). Undiversified $= \sum x_i V_i =$ USD 2.674m; with the correlation matrix, diversified $=$ USD 2.615m.

Link to stress testing. Shocking every zero by its own VaR and repricing (all correlations 1) reproduces the undiversified VaR exactly: GARP's version falls from USD 200.00m to 197.37m, a loss of USD 2.63m.

Formula card: MR-5 d

Average maturity	$\bar{M} = \sum_i w_i M_i$ (market-value weights)
Principal VaR	$W \times V_{\bar{M}}$
Interpolation	$V_D = V_j + (V_{j+1} - V_j)(D - j),\ j < D < j+1$
Duration VaR	$W \times V_D$
Returns VaR of a vertex	$z \times \sigma_{\text{yield}} \times D^{\text{mod}},\ D^{\text{mod}} = M/(1+y)$ for a zero
Undiversified CF VaR	$\sum_i \lvert x_i \rvert V_i$ (x_i = PV of the cash flow on vertex i)
Diversified CF VaR	$\sqrt{(x \times V)' R (x \times V)}$

Exam traps

- **Half the portfolio.** Applying the VaR % to one bond (USD 100m) instead of the whole USD 200m.
- **Wrong vertex for principal mapping.** Using the shorter or the longer bond's maturity instead of the *average* maturity of principal.
- **No interpolation.** Reading the nearest vertex instead of interpolating at the duration.
- **Macaulay vs modified.** The mapping point is a maturity (Macaulay duration); modified duration enters only when a returns VaR is built from yield volatility.
- **Face values instead of present values** in cash-flow mapping.
- **Counting vertices.** Cash flows on the same date share one vertex.
- **Undiversified vs diversified.** A straight sum of vertex VaRs is the undiversified (correlation 1) figure.

Practice questions: MR-5 d

Question MR-5 d.1

Bank Q268 • AP Mock 2 Q69 (rebuilt)

The table gives 95% VaR percentages for zero-coupon bonds of various maturities:

Maturity (years)	1	2	3	4	5
VaR (%)	0.4777	0.9961	1.4264	1.9618	2.4120

A portfolio consists of a USD 100 million bond maturing in one year and a USD 100 million bond maturing in three years. Using the principal mapping method, what is the VaR of this portfolio?

A. USD 1.9041 million

B. USD 1.5043 million

C. USD 1.9922 million

D. USD 0.9961 million

Question MR-5 d.2

Bank Q275 • BT P2.T5.715.1 (rebuilt)

Consider a USD 200.00 million portfolio equally allocated between two bonds: the first pays a 5.0% annual coupon and matures in five years; the second pays a 3.0% annual coupon and matures in three years. Both are priced at par (USD 100.00 million each). The returns VaR (risk) is based on a yield volatility of 100 basis points at all maturities and a 99.0% confidence level; for example, the risk at one year is $1.0\% \times 2.33 \times (1/1.0120) = 2.30\%$.

Term	CF bond 1	CF bond 2	Spot rate	Risk (returns VaR)
1	5.00	3.00	1.20%	2.30%
2	5.00	3.00	2.00%	4.57%
3	5.00	103.00	3.04%	6.78%
4	5.00	–	4.50%	8.92%
5	105.00	–	5.20%	11.07%

Under the principal mapping approach, which is nearest to the portfolio VaR?

A. USD 8.92 million

B. USD 13.57 million

C. USD 17.84 million

D. USD 22.15 million

Question MR-5 d.3

New • built from GARP MR Ch.5 (Jorion 11.5.2) and Vardeez MR-5 d

A risk analyst at a Mumbai-based insurer maps a USD 200 million bond portfolio onto zero-coupon vertices. It holds USD 100 million of a 2-year 5% annual-coupon bond and USD 100 million of a 4-year 5% annual-coupon bond, both priced at par on a flat 5% curve. The analyst has computed:

- Average maturity of principal: 3.00 years
- Portfolio Macaulay duration: 2.84 years

Monthly 95% VaR percentages for zeros:

Maturity (years)	1	2	3	4	5
VaR (%)	0.4696	0.9868	1.4841	1.9714	2.4261

Using duration mapping with linear interpolation, which of the following is closest to the portfolio's monthly 95% VaR?

A. USD 1.40 million

B. USD 1.97 million

C. USD 2.81 million

D. USD 2.97 million

Question MR-5 d.4

Bank Q273 • BT L2.T5.63.2 (rebuilt, key corrected in the bank)

Assume a flat yield curve with a spot rate of 4.0% at all maturities and a normally distributed yield volatility of 1.0%. A two-bond portfolio is being mapped: both bonds have USD 100 million face value and pay an annual 4% coupon; one matures in one year, the other in five years, so both are at par. Returns VaR at 95% for each vertex:

Term (years)	1	2	3	4	5
Macaulay duration	1.0	2.0	3.0	4.0	5.0
Modified duration	0.962	1.923	2.885	3.846	4.808
Returns VaR	1.58%	3.16%	4.74%	6.33%	7.91%

If we use duration mapping with linear interpolation, what is the portfolio's 95% VaR?

A. USD 8.56 million

B. USD 9.26 million

C. USD 8.90 million

D. USD 9.49 million

Question MR-5 d.5

Bank Q274 • BT L2.T5.63.3

Using the same two-bond portfolio as the previous question (USD 100 million 1-year 4% annual-coupon bond and USD 100 million 5-year 4% annual-coupon bond), if we use cash-flow mapping, the portfolio is mapped to how many vertices?

A. Five

B. Six

C. Ten

D. Eleven

Question MR-5 d.6

Bank Q277 • BT P2.T5.715.3

Consider a USD 200.00 million portfolio equally allocated between a USD 100.00 million 5-year 6.0% annual-coupon bond and a USD 100.00 million 1-year 4.0% annual-coupon bond, both priced at par.

Term	FV CF bond 1	FV CF bond 2	PV of CF (x)	Spot rate	Risk (returns VaR)
1	6.00	104.00	105.77	4.000%	1.550%
2	6.00	0.00	5.48	4.618%	3.110%
3	6.00	0.00	5.15	5.192%	4.660%
4	6.00	0.00	4.80	5.716%	6.210%
5	106.00	0.00	78.79	6.112%	7.760%
Total			200.00		

If we use the cash-flow mapping approach, which of the following is TRUE about the resulting portfolio VaR? Assume the correlations among the risk factors are less than 1.

A. The portfolio is mapped to a single risk factor, the 5-year vertex, so VaR is approximately $200.00 \times 7.760\% =$ USD 15.52 million.

B. The undiscounted (future value) cash flows are mapped to the five vertices, so VaR is approximately $110(1.550\%) + 6(3.110\%) + 6(4.660\%) + 6(6.210\%) + 106(7.760\%) =$ USD 10.77 million.

C. The present values are mapped to the five vertices and $105.77(1.550\%) + 5.48(3.110\%) + 5.15(4.660\%) + 4.80(6.210\%) + 78.79(7.760\%) =$ USD 8.46 million is the diversified portfolio VaR.

D. The present values are mapped to the five vertices and $105.77(1.550\%) + 5.48(3.110\%) + 5.15(4.660\%) + 4.80(6.210\%) + 78.79(7.760\%) =$ USD 8.46 million is the undiversified portfolio VaR.

Solutions • MR-5 d

Solution MR-5 d.1

Answer: C

Bank Q268 • AP Mock 2 Q69

Step 1 – Average maturity of principal

$$\bar{M} = 0.5 \times 1 + 0.5 \times 3 = 2 \text{ years}$$

$\Rightarrow$ Principal mapping places the whole USD 200m at the 2-year vertex.

Step 2 – VaR

$$\text{VaR} = 200 \times 0.9961\% = \text{USD } 1.9922 \text{ million}$$

Interpretation

Principal mapping treats the entire portfolio as one zero-coupon position at the average maturity of principal. The 1-year and 3-year VaR figures in the table are not used.

Why the other options are wrong

A is incorrect. Added the separate 1-year and 3-year VaRs, $0.4777 + 1.4264 = 1.9041$: that maps each bond on its own vertex with perfect correlation, which is not principal mapping.

B is incorrect. Combined the 1-year and 3-year VaRs with zero correlation: $\sqrt{0.4777^2 + 1.4264^2} = 1.5043$. Again not principal mapping.

D is incorrect. Applied the 2-year VaR % to USD 100m only: $100 \times 0.9961\% = 0.9961$.

LO: MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Mapping Approaches

Solution MR-5 d.2

Answer: C

Bank Q275 • BT P2.T5.715.1

Step 1 – Average maturity of principal

$$\bar{M} = 0.5 \times 5 + 0.5 \times 3 = 4.0 \text{ years}$$

Step 2 – Read the 4-year risk

$1.0\% \times 2.33 \times 4/1.045 = 8.92\%$ (given in the table).

Step 3 – VaR on the whole portfolio

$$\text{VaR} = 200 \times 8.92\% = \text{USD } 17.84 \text{ million}$$

Interpretation

The coupons, the spot curve and the per-vertex cash flows are red herrings for principal mapping: only the principal amounts and their maturities matter. They would be needed for duration or cash-flow mapping.

Why the other options are wrong

A is incorrect. Applied the 4-year risk to USD 100m only: $100 \times 8.92\% = 8.92$.

B is incorrect. Mapped to the 3-year vertex (the shorter bond's maturity): $200 \times 6.78\% = 13.57$.

D is incorrect. Mapped to the 5-year vertex (the longest maturity): $200 \times 11.07\% = 22.15$.

LO: MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Mapping Approaches

Solution MR-5 d.3 — Answer: C

New • built from GARP MR Ch.5 and Vardeez MR-5 d

Step 1 – Locate the duration on the grid

$D = 2.84$ years lies between the 2-year and 3-year vertices.

Step 2 – Interpolate the VaR %

$$V_{2.84} = 0.9868 + (1.4841 - 0.9868) \times (2.84 - 2) = 0.9868 + 0.4973 \times 0.84 = 1.4045\%$$

Step 3 – Duration-mapping VaR

$$\text{VaR} = 200 \times 1.4045\% = \text{USD } 2.81 \text{ million}$$

$\Rightarrow$ Slightly below the principal-mapping figure, as the ranking predicts.

Interpretation

The average maturity of principal (3.00 years) is the input for principal mapping, not duration mapping; it is there to lure you to option D. (For reference: the 2-year 5% par bond has Macaulay duration 1.952 years and the 4-year one 3.723 years; their equal-weighted average is 2.838.)

Why the other options are wrong

A is incorrect. Correct interpolated VaR %, but applied to one bond only: $100 \times 1.4045\% = 1.40$.
B is incorrect. Did not interpolate; read the 2-year vertex: $200 \times 0.9868\% = 1.97$.
D is incorrect. Used the average maturity of principal (3 years), which is principal mapping: $200 \times 1.4841\% = 2.97$.

LO: MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 (duration mapping by linear interpolation); SchweserNotes 2026 Book 1, LO 5.d

Solution MR-5 d.4 — Answer: C

Bank Q273 • BT L2.T5.63.2

Step 1 – Duration of each bond

The 1-year bond has Macaulay duration 1.0. The 5-year 4% bond at a 4% yield:

$$D_5 = \frac{1.04}{0.04}\left(1 - 1.04^{-5}\right) = 4.630 \text{ years}$$

Step 2 – Portfolio duration (equal market values)

$$D = \frac{4.630 + 1.0}{2} = 2.815 \text{ years}$$

Step 3 – Interpolate the returns VaR between 2 and 3 years

$$V_{2.815} = 3.163\% + 0.815 \times (4.746\% - 3.163\%) = 4.453\%$$

Step 4 – VaR

$$\text{VaR} = 200 \times 4.453\% = \text{USD } 8.90 \text{ million}$$

Interpretation

The mapping point is a maturity, so the Macaulay duration (2.815) is the right coordinate on the grid; modified duration is already built into each vertex's returns VaR.

Why the other options are wrong

A is incorrect. Interpolated at the modified duration $2.815/1.04 = 2.707$, double-counting the $1/(1+y)$ adjustment: $200 \times 4.282\% = 8.56$.
B is incorrect. Multiplied the Macaulay duration directly by 1.645% without the $1/1.04$ adjustment: $200 \times 2.815 \times 1.645\% = 9.26$.
D is incorrect. The principal-mapping answer (3-year vertex): $200 \times 4.745\% = 9.49$.

LO: MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 (duration mapping)

Solution MR-5 d.5 Answer: A

Bank Q274 • BT L2.T5.63.3

Step 1 – List the cash-flow dates

The 5-year annual-coupon bond pays at years 1, 2, 3, 4 and 5. The 1-year bond pays once, at year 1.

Step 2 – Aggregate by date

Cash flows on the same date share one vertex, so the year-1 flows of both bonds sit together.
$\Rightarrow$ Distinct dates: years 1 to 5, five vertices.

Interpretation

Cash-flow mapping maps exposures by date, not by instrument: the vector x holds one present value per vertex.

Why the other options are wrong

B is incorrect. Counted six cash flows $(5+1)$ without merging the two year-1 flows.
C is incorrect. Treated each bond as having five cash flows on separate vertices $(5+5)$.
D is incorrect. Added a time-0 vertex; vertices are only the cash-flow dates.

LO: MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2, Table 5.2

Solution MR-5 d.6 Answer: D

Bank Q277 • BT P2.T5.715.3

Step 1 – What gets mapped

Cash-flow mapping maps the *present value* of each cash flow onto the zero-coupon vertex of the same maturity: $x = (105.77,\ 5.48,\ 5.15,\ 4.80,\ 78.79)$.

Step 2 – Individual VaRs and their sum

$1.639 + 0.170 + 0.240 + 0.298 + 6.114 =$ USD 8.46 million.
$\Rightarrow$ Adding individual VaRs assumes all correlations equal 1: this is the **undiversified** VaR. The diversified VaR, $\sqrt{(x \times V)' R (x \times V)}$, is lower when correlations are below 1.

Why the other options are wrong

A is incorrect. Maps everything to one factor at the longest vertex; that is none of the three methods and overstates VaR.
B is incorrect. Applies the VaR % to undiscounted (future value) cash flows instead of present values: 10.77.
C is incorrect. Right figure, wrong label: the sum of vertex VaRs is the undiversified VaR.

LO: MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2, eq. 5.7 and Table 5.4

MR-5 • VaR Mapping • Vardeez: IMPORTANT

MR-5 f

P3 BOTH

Explain how VaR can be calculated and used relative to a performance benchmark.

Workbook: C3 VaR Mapping • Tier C • Map #22

WHAT IS TESTED HERE

Why P3. Tier C workbook topic (C3). The sheet singles out this LO: "especially interpretation of variance improvement".
Vardeez: interpretation of variance improvement (his "LO 5.f").

Reported exam shapes

- **C3·S2** VaR relative to a benchmark: tracking error VaR and the interpretation of variance improvement.

What the questions look like: pick the portfolio that tracks the index best from a table of absolute and tracking error VaRs (lowest TE-VaR, not lowest or closest absolute VaR); compute the variance improvement $1-(\text{TE-VaR}/\text{VaR}_{\text{index}})^2$; interpret TE-VaR as a VaR of relative performance.

1. Absolute risk versus relative risk

A manager measured against an index cares about **relative** risk: how far the portfolio can fall behind the benchmark. With x the vector of the portfolio's exposures on the vertices and x_0 the benchmark's,

$$\text{Absolute VaR} = \alpha\sqrt{x'\Sigma x} \qquad \boxed{\text{Tracking error VaR} = \alpha\sqrt{(x-x_0)'\,\Sigma\,(x-x_0)}}$$

TE-VaR is the VaR of the *difference in positions*. "With 95% confidence, under normal market conditions, the portfolio will not underperform the index by more than TE-VaR over the horizon." When only the active risk (tracking error volatility) is given: TE-VaR $= z \times \sigma_{\text{active}} \times$ value.

Correction to the source notes. Vardeez (and Schweser's prose) describe tracking error VaR as "the difference between your portfolio's VaR and the benchmark's VaR". It is not: it is the VaR of the position difference $x - x_0$. In Jorion's table portfolio 3 has absolute VaR 2.04 against the index's 1.99, a difference of 0.05, yet its TE-VaR is 0.16; portfolio 1's difference is 0.26 but its TE-VaR is 0.43.

2. Jorion's benchmarking exercise

Benchmark: USD 100m J.P. Morgan US bond index, duration 4.62, monthly 95% absolute VaR USD 1.99m (about the risk of a 4-year note). Five two-zero portfolios are each **duration-matched** at 4.62:

	Index	1	2	3	4	5
Vertices used	all	4Y, 5Y	3Y, 7Y	2Y, 9Y	1Y, 10Y	<1M, 30Y
Absolute VaR	1.99	2.25	2.16	2.04	1.94	1.71
TE-VaR	0.00	0.43	0.29	**0.16**	0.20	0.81

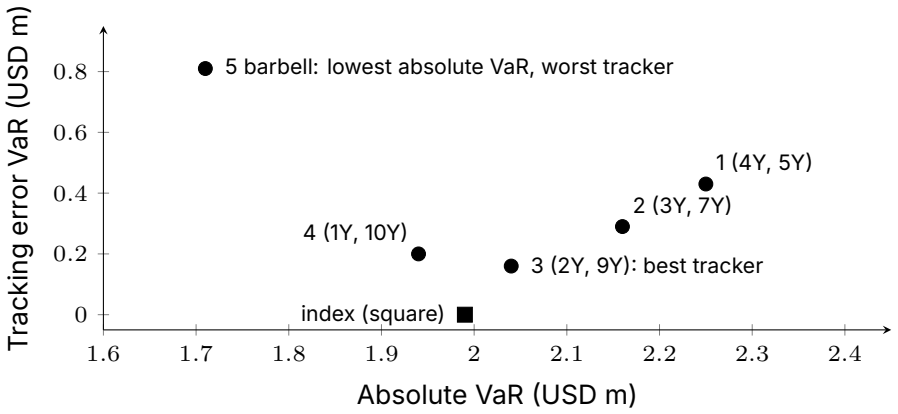


Figure 3. Same duration, very different tracking. Moving down the chart means tracking better; moving left means less absolute risk. The barbell (portfolio 5) is furthest left and highest: minimizing absolute risk is not minimizing relative risk.

Lessons

- **Duration matching is only a first approximation.** All five match the index duration, yet TE-VaR ranges from 0.16 to 0.81. What remains is due to **non-parallel moves** in the term structure.
- **Track by matching cash flows.** Portfolio 3 (2 and 9 years) wins because the index has its largest weight at the 2-year vertex: it comes closest to the index's cash-flow profile.
- **Absolute is not relative.** The barbell has the lowest absolute VaR (distant maturities are less correlated) and the worst tracking error.

3. Variance improvement

$$\text{Variance improvement} = 1 - \left(\frac{\text{TE-VaR}}{\text{VaR}_{\text{benchmark}}}\right)^2$$

- Read it like an R^2: the share of the benchmark's variance the tracking portfolio reproduces. The rest, $(\text{TE-VaR}/\text{VaR})^2$, is unexplained variance.
- The ratio of VaRs is a ratio of *standard deviations*, so it must be **squared** to become a ratio of variances.
- Higher variance improvement means lower TE-VaR: good for a tracker.
- Jorion, portfolio 1: $1 - (0.43/1.99)^2 \approx 95.3\%$ (GARP prints 95.4% from unrounded figures), "in line with the explanatory power of the first factor". Portfolio 3: $1 - (0.16/1.99)^2 = 99.4\%$.

Formula card: MR-5 f

Absolute VaR	$\alpha\sqrt{x'\Sigma x}$
Tracking error VaR	$\alpha\sqrt{(x-x_0)'\Sigma(x-x_0)}$; or $z \times \sigma_{\text{active}} \times$ value
Variance improvement	$1-(\text{TE-VaR}/\text{VaR}_{\text{benchmark}})^2$
Unexplained share	$(\text{TE-VaR}/\text{VaR}_{\text{benchmark}})^2$

Exam traps

- **Not squaring.** $1-\text{TE-VaR}/\text{VaR}$ understates the improvement.
- **Stopping at the square.** $(\text{TE-VaR}/\text{VaR})^2$ is what is left *unexplained*, not the improvement.
- **Picking the lowest absolute VaR** (the barbell) or the **absolute VaR closest to the index** for a tracking mandate. The criterion is the lowest TE-VaR.
- **Same duration, same tracking.** False: equal-duration portfolios can have very different TE-VaR.
- **TE-VaR is not a guaranteed maximum,** and it is not the difference between two absolute VaRs.
- **Using the fund or benchmark volatility** instead of the active risk when computing TE-VaR from volatilities.

Practice questions: MR-5 f

Question MR-5 f.1

Bank Q289 • AnalystPrep Q.1517 (rebuilt)

A portfolio manager benchmarks portfolio A against an index. The tracking error VaR, $\alpha\sqrt{(x-x_0)'\Sigma(x-x_0)}$, is computed at the 95% confidence level over a one-month horizon and equals USD 0.63 million. Which interpretation is most appropriate?

A. Over one month, there is a 5% chance that portfolio A will underperform the index by more than USD 0.63 million, under normal market conditions.

B. Over one month, portfolio A will never deviate from the index by more than USD 0.63 million, in either direction.

C. Portfolio A has a one-month 95% absolute VaR of USD 0.63 million.

D. Over one month, the minimum deviation between portfolio A and the index is USD 0.63 million with 95% confidence.

Question MR-5 f.2

Bank Q294 • BT L2.T5.65.1

A fund with a value of USD 10 million is benchmarked against an index. The fund's volatility is 10.0% and normally distributed. The benchmark volatility is 5.0% and normally distributed. The fund's active risk (tracking error) is 8.0% and normally distributed. What is the 95% tracking error value at risk?

A. USD 820,000

B. USD 1.32 million

C. USD 1.65 million

D. USD 1.96 million

Question MR-5 f.3

Bank Q290 • AnalystPrep Q.2826 (rebuilt)

A portfolio of zero-coupon bonds is benchmarked against a bond index. At the 95% confidence level the index has an absolute VaR of USD 2.080 million, and the tracking error VaR of the portfolio relative to the index is USD 1.09 million. What is the variance improvement of the portfolio relative to the original index?

A. 27.5%

B. 47.6%

C. 72.5%

D. 52.4%

Question MR-5 f.4

Bank Q291 • BT L2.T5.65.2

Which of the following is a true statement if we are benchmarking a fixed-income portfolio relative to a fixed-income index?

A. Minimizing absolute VaR will also tend to minimize tracking error VaR.

B. Zero tracking error implies no underperformance vis-à-vis the benchmark.

C. Duration mapping is insufficient to minimize tracking error.

D. Portfolios with the same duration should have similar or the same tracking error VaRs.

Question MR-5 f.5

Bank Q293 • Schweser Module Quiz 5.3 Q1 (rebuilt)

A USD 100 million benchmark bond portfolio has the following market value cash flows by maturity (zero-coupon equivalents):

Maturity (years)	1	2	3	4	5	7	10
Market value (USD m)	5	10	30	10	30	10	5

Each candidate portfolio consists of two zero-coupon bonds weighted so that its duration equals the benchmark's duration of 4.25 years. Which combination would most likely have the smallest tracking error VaR?

A. 1 year and 10 year

B. 2 year and 7 year

C. 3 year and 5 year

D. 4 year and 7 year

Question MR-5 f.6

Bank Q292 • BT P2.T5.716.3 (rebuilt)

The table recreates Jorion's example of benchmarking a USD 100.0 million bond portfolio against a bond index (monthly tracking error VaR at the 95% level; all five portfolios are duration-matched to the index at 4.62 years):

	Index	#1	#2	#3	#4	#5
Vertices	all	4Y, 5Y	3Y, 7Y	2Y, 9Y	1Y, 10Y	<1M, 30Y
Absolute VaR	1.99	2.25	2.16	2.04	1.94	1.71
Tracking error VaR	0.00	0.43	0.29	0.16	0.20	0.81

All of the following statements about this exercise are true EXCEPT:

A. If the goal is to minimize risk, portfolios #1 and #2 are superior to the others.

B. The barbell portfolio (#5) has the lowest absolute VaR, consistent with imperfect correlation between distant maturities.

C. The variance improvement of portfolio #1 relative to the index is about $1 - (0.43/1.99)^2 = 95.3\%$.

D. If the goal is to minimize tracking error, the best portfolio is #3.

Question MR-5 f.7 Bank Q288 • BT P2.T5.22.10.2 (rebuilt; option D made precise)

VaR mapping lends itself to stress testing and to performance benchmarking against an index. Consider both extensions:

I. Stress testing a fixed-income portfolio by shocking the zero-coupon vertices (risk factors) of the spot rate term structure by their VaRs.

II. Performance benchmarking a fixed-income portfolio against a well-known bond index.

Which of the following statements is TRUE?

A. The stress test is equivalent to the undiversified VaR at a sufficiently high confidence level, whatever the correlations between vertices.

B. Shocking each zero by its own VaR remains a valid way to obtain the portfolio VaR when correlations between maturities are less than one.

C. In performance benchmarking, minimizing the absolute VaR necessarily minimizes the relative VaR (tracking error VaR).

D. The variance improvement from benchmarking can be measured as $1 - (\text{TE-VaR}/\text{absolute VaR of the index})^2$, analogous to R^2 in a regression.

Question MR-5 f.8 Workbook C3.1 • S2 • Map #22

A portfolio manager at a German pension fund must track a government bond index with EUR 250 million. The index's monthly 95% absolute VaR is EUR 4.80 million. The risk team has built five candidate portfolios, all with the same duration as the index, and computed their monthly 95% VaRs (EUR million):

Portfolio	1	2	3	4	5
Vertices used	4Y, 5Y	3Y, 7Y	2Y, 9Y	1Y, 10Y	1M, 30Y
Absolute VaR	5.40	5.10	4.95	4.75	4.10
Tracking error VaR	1.05	0.62	0.38	0.55	1.95

The manager's mandate is to track the index as closely as possible. Which portfolio should the manager choose, and what is its variance improvement relative to the index?

A. Portfolio 3; variance improvement 99.4%

B. Portfolio 3; variance improvement 92.1%

C. Portfolio 4; variance improvement 98.7%

D. Portfolio 5; variance improvement 83.5%

Solutions • MR-5 f

Solution MR-5 f.1

Answer: A

Bank Q289 • AnalystPrep Q.1517

Step 1 – What TE-VaR measures

TE-VaR is a VaR applied to the position difference $x - x_0$: a quantile of the portfolio's *relative* P&L against the index.

Step 2 – Translate to words

With 95% confidence the shortfall versus the index over one month will not exceed USD 0.63m; equivalently, a 5% chance it will, under normal market conditions.

Interpretation

Everything you know about reading an absolute VaR carries over, with "loss" replaced by "underperformance against the benchmark".

Why the other options are wrong

B is incorrect. VaR is not a guaranteed maximum; it is exceeded with probability 5%.
C is incorrect. Confuses relative with absolute VaR: absolute VaR uses x, TE-VaR uses $x - x_0$.
D is incorrect. VaR is a worst-case quantile, not a minimum deviation; the deviation could well be zero.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Benchmarking, eq. 5.8

Solution MR-5 f.2

Answer: B

Bank Q294 • BT L2.T5.65.1

Step 1 – Which volatility

TE-VaR is the VaR of the active (fund minus benchmark) return, so it uses the active risk of 8.0%.

Step 2 – TE-VaR

$$\text{TE-VaR} = 1.645 \times 8.0\% \times \text{USD } 10\text{m} = \text{USD } 1.316\text{m} \approx \text{USD } 1.32 \text{ million}$$

Interpretation

The fund and benchmark volatilities are red herrings: they describe absolute risk. (The fund's absolute 95% VaR would be USD 1.645m.)

Why the other options are wrong

A is incorrect. Used the benchmark volatility: $1.645 \times 5\% \times 10 = 0.82$.
C is incorrect. Used the fund's own volatility, which gives its absolute VaR: $1.645 \times 10\% \times 10 = 1.645$.
D is incorrect. Used the fund volatility with the two-tailed 1.96: $1.96 \times 10\% \times 10 = 1.96$.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Benchmarking

Solution MR-5 f.3

Answer: C

Bank Q290 • AnalystPrep Q.2826

Step 1 – Ratio of VaRs

$$\frac{1.09}{2.080} = 0.5240$$

Step 2 – Square it (ratio of variances)

$$0.5240^2 = 0.2746$$

$\Rightarrow$ 27.5% of the index's variance is *not* reproduced by the portfolio.

Step 3 – Variance improvement

$$1 - 0.2746 = 0.7254 \approx 72.5\%$$

Interpretation

VaR scales with standard deviation, so the VaR ratio must be squared before it can be read as a share of variance. A 72.5% improvement is a poor tracker by Jorion's standards (95% to 99%).

Why the other options are wrong

A is incorrect. Stopped at the squared ratio, 27.5%: that is the unexplained share.
B is incorrect. Forgot to square: $1 - 0.524 = 47.6\%$.
D is incorrect. Reported the unsquared ratio itself, 52.4%.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Benchmarking

Solution MR-5 f.4 **Answer: C**

Bank Q291 • BT L2.T5.65.2

Step 1 – What drives tracking error

Tracking error comes from mismatched exposures along the term structure. Matching duration only neutralizes parallel shifts; non-parallel moves still create tracking error.
$\Rightarrow$ Duration matching alone is insufficient; the cash-flow profile must be matched.

Step 2 – Test the other statements

A fails: the barbell has the lowest absolute VaR and the highest TE-VaR. B fails: TE measures the dispersion of the difference, not its sign; a constant lag has zero tracking error. D fails: Jorion's five equal-duration portfolios have TE-VaRs from 0.16 to 0.81.

Interpretation

This is Jorion's conclusion in one line: "duration hedging is only a first approximation".

Why the other options are wrong

A is incorrect. Absolute and tracking error VaR measure different things; the barbell is the counterexample.
B is incorrect. Tracking error is about variability, not direction; a portfolio that always lags by the same amount has zero tracking error.
D is incorrect. Same duration matches parallel shifts only; equal-duration portfolios differ greatly in TE-VaR.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2, Table 5.5

Solution MR-5 f.5 — Answer: C

Bank Q293 • Schweser Module Quiz 5.3 Q1

Step 1 – Confirm the benchmark duration

$(5 \times 1 + 10 \times 2 + 30 \times 3 + 10 \times 4 + 30 \times 5 + 10 \times 7 + 5 \times 10)/100 = 4.25$ years. Every pair brackets 4.25, so duration can be matched with any of them.

Step 2 – Match the cash-flow profile

With duration matched, TE-VaR is smallest when the zeros sit where the benchmark's largest cash flows are: USD 30m at 3 years and USD 30m at 5 years (60% of the value).
$\Rightarrow$ The 3- and 5-year pair comes closest to the benchmark's profile.

Interpretation

Same logic as Jorion's portfolio 3, which won because it sat on the index's heaviest vertex.

Why the other options are wrong

A is incorrect. The 1/10-year barbell is furthest from the benchmark's concentrated 3- and 5-year cash flows, leaving the most non-parallel exposure.
B is incorrect. The 2- and 7-year buckets hold only USD 10m each and miss both large cash flows.
D is incorrect. The 4- and 7-year buckets also miss the USD 30m flows at 3 and 5 years.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Benchmarking; SchweserNotes 2026 Book 1, LO 5.f

Solution MR-5 f.6 — Answer: A

Bank Q292 • BT P2.T5.716.3

Step 1 – Check each statement

A: #1 and #2 have the *highest* absolute VaRs (2.25, 2.16) and higher TE-VaRs than #3 and #4. They are dominated, not superior. **False.**
B: #5 has the lowest absolute VaR (1.71); distant maturities are less correlated. True.
C: $0.43/1.99 = 0.2161$, squared 0.0467, so $1 - 0.0467 = 95.3\%$. True.
D: #3 has the lowest TE-VaR (0.16). True.

Interpretation

The table shows the trade-off between absolute and relative risk at a glance: the best tracker (#3) is neither the safest nor the riskiest in absolute terms.

Why the other options are wrong

B is incorrect. True: lowest absolute VaR, but also the highest TE-VaR (0.81).
C is incorrect. True: the variance improvement is about 95.3% (GARP quotes 95.4% from unrounded figures).
D is incorrect. True: 0.16 is the smallest TE-VaR.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2, Table 5.5

Solution MR-5 f.7 — Answer: D

Bank Q288 • BT P2.T5.22.10.2

Step 1 – Stress test (I)

Shocking every zero by its own VaR and repricing equals the undiversified VaR only when all correlations are 1. It "ceases to be viable if correlations are anything but perfect". So A and B are false.

Step 2 – Benchmarking (II)
A low absolute VaR portfolio can deviate strongly from the index (the barbell), so C is false. Variance improvement $= 1 - (\text{TE-VaR}/\text{VaR}_{\text{index}})^2$, the share of the index's variance reproduced, like an R^2. D is true.

Interpretation
The equivalence of the VaR-shock stress test with undiversified VaR comes from the perfect-correlation assumption, not from the confidence level.

Why the other options are wrong
A is incorrect. The equivalence depends on perfect correlation, not on raising the confidence level.
B is incorrect. With imperfect correlations the VaR-shock method no longer gives the portfolio VaR.
C is incorrect. Confuses absolute with relative risk.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Stress Test and Benchmarking

Solution MR-5 f.8 Answer: A

Workbook C3.1 • S2

Step 1 – Choose the right risk measure
The mandate is relative, so the selection criterion is the tracking error VaR, $\alpha\sqrt{(x-x_0)'\Sigma(x-x_0)}$, not the absolute VaR. Portfolio 3 has the lowest TE-VaR, EUR 0.38 million.

Step 2 – Variance improvement for each candidate

$$\text{VI} = 1 - \left(\frac{\text{TE-VaR}}{\text{VaR}_{\text{index}}}\right)^2$$

Portfolio 1: $1 - (1.05/4.80)^2 = 95.2\%$; portfolio 2: 98.3%; portfolio 3: $1 - (0.38/4.80)^2 = 1 - 0.00627 = 99.4\%$; portfolio 4: 98.7%; portfolio 5: 83.5%.
$\Rightarrow$ The ranking by variance improvement is the same as the ranking by TE-VaR, and portfolio 3 wins.

Interpretation
Like an R^2, a 99.4% variance improvement means the portfolio reproduces all but 0.6% of the variance of the index's returns; the residual comes from non-parallel moves in the term structure. Spreading the two positions to 2 and 9 years comes closest to the index's cash-flow profile, which is why it beats the narrow 4- and 5-year bracket even though all five have the same duration. The EUR 250 million size is not needed.

Why the other options are wrong
B is incorrect. Right portfolio, but used the ratio of VaRs instead of the ratio of variances: $1 - 0.38/4.80 = 92.1\%$.
C is incorrect. Picked the portfolio whose absolute VaR is closest to the index's (4.75 against 4.80) instead of the lowest tracking error; matching absolute risk is not matching the index.
D is incorrect. Picked the lowest absolute VaR (the barbell of 1-month bills and 30-year zeros). "Minimizing absolute market risk is not the same as minimizing relative market risk": its TE-VaR is the worst, $1 - (1.95/4.80)^2 = 83.5\%$.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. • **Reference:** Jorion, Value at Risk 3e, Ch.11, 5.2 Benchmarking, Table 5.5 and eq. 5.8

MR-6 • Validating Bank Holding Companies' Value-at-Risk Models for Market Risk • Vardeez: IMPORTANT (new topic, highly testable)

MR-6 a

P3 CONCEPT

Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process.

Workbook: D10 BHC Capital Planning and VaR Model Validation • Tier D • Map #53

WHAT IS TESTED HERE

Why P3. A Tier D workbook topic hits this LO: D10 (Map #53, reported on 2 of 16 sources, "BHC"). The reading is new in the syllabus, which historically means heavier sampling.
Vardeez: new topic, highly testable. No specific pointer inside the LO.

Reported exam shape

- **D10·S3** Validation standards for BHC VaR models: conceptual soundness, sensitivity analysis, confidence intervals, benchmarking, all tested in one "which statement is correct" question (worked as Question MR-6 a.7).

What the questions look like: statement questions. The key idea is *fit for purpose*: a VaR model that does not move when positions move fails conceptual soundness, even if it forecasts P&L well. Distractors swap "accurate" for "fit for purpose", or swap pseudo history for actual P&L.

1. Where conceptual soundness sits in VaR validation

US supervisors expect three kinds of validation evidence for a VaR model. The reading (Lynch, Ch.6) shows how each is done in practice:

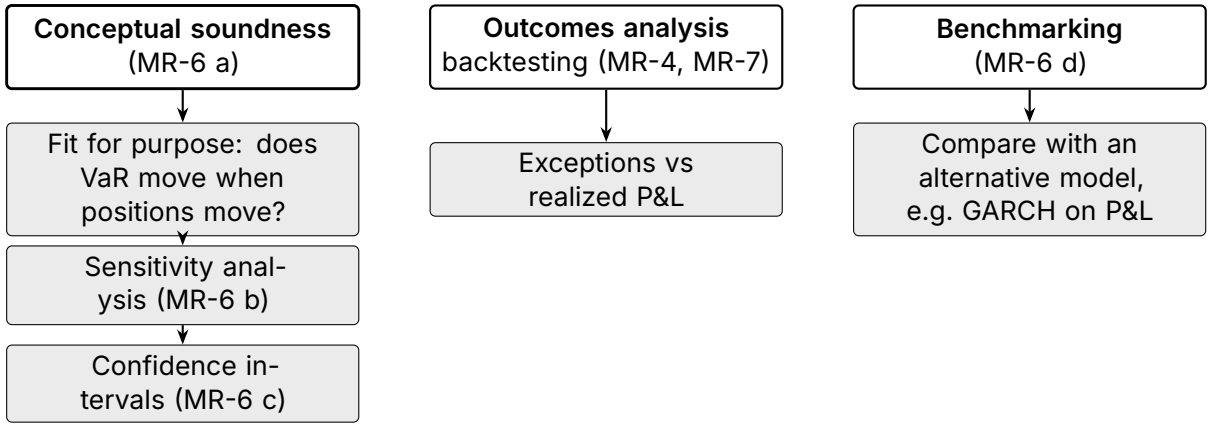


Figure 1. The three validation components US regulators expect for VaR models. Conceptual soundness (heavy-bordered box) is judged by a narrative on assumptions, data and techniques, backed by the checks stacked below it: the fit-for-purpose test and two quantitative tests, sensitivity analysis and confidence intervals.

2. What the validator checks

Conceptual soundness asks whether the model's **assumptions, techniques and data** are appropriate. It is usually delivered as a *narrative* by the validator, reviewing the many modelling choices made in building a large VaR model. Schweser and Vardeez list the checks as:

- **Methodology:** is the approach robust and appropriate?
- **Input data:** is the data reliable and complete? Large trading desks often have *missing* return data for some positions (too short a history, or no clear valuation).
- **Assumptions:** do they match the portfolio's risk characteristics (for example, normality for a book of options)?
- **Fit for purpose:** does the model serve the bank's risk management objective? This is the crucial test.

The crucial test: VaR must show how risk changes when positions change

A VaR model's basic purpose is to help the firm manage the risk of its *positions*. Capital calculation is a further, somewhat ancillary use. So the key question is whether VaR responds when the positions change.

- **Pseudo history.** Most VaR models revalue the *current* positions under past market moves, producing a "pseudo history" of P&L. If a trader halves a position, every pseudo P&L shrinks and VaR falls at once. Supervisors spend a lot of effort checking that this pseudo history is built correctly.
- **Actual P&L history.** A VaR estimated on the bank's *actual* past P&L does not change when positions are cut, because the past P&L does not change. Such a model is **not fit for purpose** and fails conceptual soundness.
- **Accurate is not the same as sound.** Berkowitz and O'Brien (2002) found that a simple GARCH model fitted to banks' actual P&L *outperformed* the banks' own VaR models as a forecast. It is tempting to prefer it, but it cannot show the effect of a position change, so it is a useful *benchmark* (MR-6 d), not a risk management model.
- **Lo versus Jorion.** Lo's hypothetical fund, "Capital Decimation Partners", sells deep out-of-the-money puts. Its P&L shows good returns at low risk. Jorion shows that a VaR recomputed on the current positions reveals the true risk and rises when the positions become riskier.
- **Hedging overlays.** If a manager adds a derivatives overlay to cut risk, VaR on historical returns does not change, but VaR on pseudo returns falls (Schweser's example).

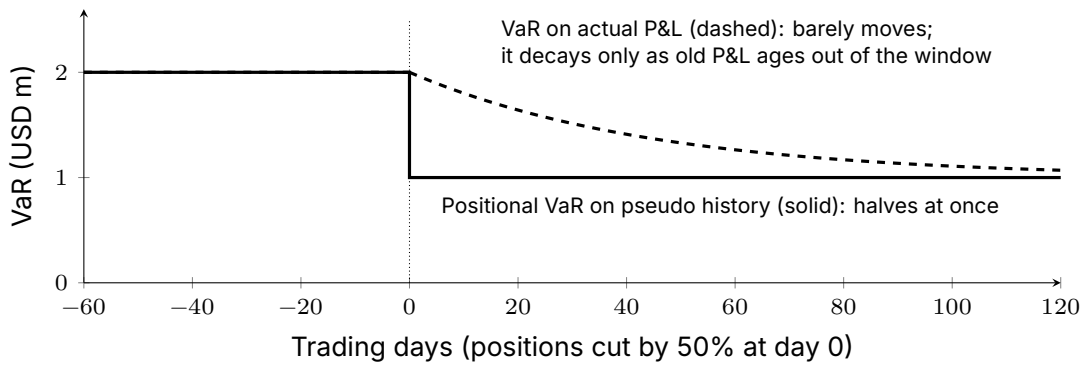


Figure 2. Illustrative only. A trader halves every position at day 0. Positional VaR, built on the pseudo history of the current positions (solid), halves immediately. A VaR estimated on actual P&L (dashed) is unaffected by the cut, because past P&L does not change. That is why it fails the fit-for-purpose test.

Other points the reading makes

- **Historical simulation** is the most widely used VaR model at commercial banks, but it is not very responsive to recent volatility (Pritsker). GARCH VaR, with RiskMetrics as a special case, and **filtered historical simulation** (GARCH volatility times the empirical quantile of standardised shocks) address that. Note that GARCH *on the pseudo history* is fine; GARCH *on actual P&L* is what fails the fit-for-purpose test.
- **Two quantitative tests** support the narrative: sensitivity analysis (MR-6 b) shows the effect of data limitations and modelling choices; confidence intervals (MR-6 c) show estimation error.
- **Backtesting alone is not enough.** It compares VaR with realized P&L, but it says nothing about whether VaR links position changes to risk.

	VaR on pseudo history (positional)	VaR on actual P&L
Input	Current positions revalued under past market moves	The bank's realized past P&L
Position cut	VaR falls at once	VaR unchanged
Fit for purpose	Yes: shows how risk changes with positions	No: fails conceptual soundness
Forecast accuracy	Often beaten (conservative)	GARCH version often more accurate
Role in validation	The risk management model	A ready benchmark (MR-6 d)

Exam traps

- **Accurate versus fit for purpose (sibling swap).** A GARCH VaR on actual P&L can be the more accurate forecast *and* still fail conceptual soundness, because it ignores position changes.
- **Pseudo history is not synthetic data.** It is real past market moves applied to today's positions.
- **GARCH is not the cure for everything.** GARCH fixes slow response to volatility; it does not fix the position-change problem unless it is run on the pseudo history.
- **Backtesting does not test soundness.** It is outcomes analysis, a separate validation component.
- **Normality is not a requirement.** Assuming normal returns is a weakness, not a criterion for soundness.
- **Cost of alternatives is not a weakness of the model.** Computing burden is an implementation issue, not a conceptual flaw.

Practice questions: MR-6 a

Question MR-6 a.1

Bank Q324 • Schweser Module Quiz 6.1 Q1

Which of the following items is crucial when evaluating the conceptual soundness of a VaR model?

A. The portfolio return distribution should be normal.

B. Inputs need to be adjusted to determine the change in VaR.

C. The model should use actual historical returns to calculate VaR.

D. The model should be designed to meet specific risk management objectives.

Question MR-6 a.2

Bank Q319 • AnalystPrep Q.6438

Zenith Securities is validating a new simulation-based VaR model for a trading portfolio of equities, fixed income, and derivatives. The validation team is assessing conceptual soundness, in particular how the model reflects changes in risk caused by trading activity. A junior analyst proposes to rely primarily on backtesting against historical profit and loss (P&L). Why is relying solely on backtesting insufficient to assess conceptual soundness?

A. Backtesting primarily evaluates the model's ability to predict future losses rather than its ability to simulate historical risks.

B. Backtesting assesses historical performance but does not evaluate how the model links position changes to risk.

C. Backtesting primarily focuses on tail risks, which may not represent the model's overall behavior.

D. Backtesting is primarily a regulatory requirement and not a useful tool for internal model validation.

Question MR-6 a.3

Bank Q321 • AnalystPrep Q.6467

A portfolio manager is told by the risk department to reduce the portfolio's VaR and responds by cutting holdings in all assets by 25%. Which approach would MOST effectively demonstrate the reduction in risk?

A. Recalculating VaR from the portfolio's historical P&L before the position reduction

B. Recalculating VaR from the portfolio's historical P&L after the position reduction

C. Recalculating VaR from a pseudo history of P&L that applies the reduced positions to all historical scenarios

D. Comparing the current VaR with the average VaR over the previous year

Question MR-6 a.4 Bank Q323 • BT P2.T5.25.5.2 (rebuilt)

A mid-sized investment bank trades structured credit and commodity derivatives. Its VaR model is parametric (variance-covariance), assumes normally distributed returns, and uses a one-year window of historical portfolio returns without adjusting for changes in portfolio composition. Backtesting shows exceedances in market shocks. Management is considering Monte Carlo or historical simulation but worries about computing cost. Which of the following is NOT a conceptual weakness of the current model?

A. Assuming normally distributed returns understates fat-tailed risk in structured credit and commodities.

B. A one-year window may omit extreme market conditions and past crises.

C. Monte Carlo and historical simulation require more computing power than a variance-covariance model.

D. Using unadjusted historical portfolio returns ignores changes in portfolio composition over the window.

Question MR-6 a.5 Bank Q318 • AnalystPrep Q.6207 (rebuilt)

ABC Financial uses historical simulation VaR with an equally weighted 10-year window of portfolio returns. In a recent period of unusually high volatility, daily losses exceeded the 99% VaR on several occasions. A validation team is assessing the conceptual soundness of the model. Which change would MOST directly address the cause of the exceedances?

A. Raising the confidence level while keeping the same equally weighted history

B. Adding a stress-testing overlay and using the stress results in place of the VaR figure

C. Replacing the equally weighted history with a volatility-updating approach (e.g., GARCH-based VaR) that reflects recent changes in volatility

D. Extending the historical window from 10 to 20 years to obtain more tail observations

Question MR-6 a.6 New • built from GARP MR Ch.6, 6.3 (Lo 2001; Jorion 2007)

A risk manager at a multi-strategy fund reviews a desk that has earned small, steady profits for three years by selling deep out-of-the-money index puts. The desk's 99% VaR is estimated by fitting a GARCH(1,1) model to the desk's daily actual P&L, and it has stayed low throughout. The validation team has flagged the model. Which of the following is the most appropriate reason?

A. A GARCH model fitted to actual P&L cannot respond to volatility clustering, so the reported VaR is too stable over time.

B. The P&L history does not reflect the risk of the positions now held, so VaR would neither reveal the tail risk in the puts nor fall when positions are cut.

C. A 99% confidence level is too low for an option-selling desk, so raising it to 99.9% on the same P&L history would make the model sound.

D. A P&L-based GARCH VaR is known to be less accurate than banks' positional VaR models, so it would fail a backtest.

Question MR-6 a.7 Workbook D10.2 • S3 • Map #53

A model validation analyst at a US bank holding company is writing the validation report for the trading desk's historical-simulation VaR model, covering conceptual soundness, sensitivity analysis, confidence intervals and benchmarking. Which of the following statements is correct for the analyst to include in the report?

A. A VaR model fitted by applying a GARCH model to the desk's actual P&L history would fail the conceptual soundness test, because its VaR would not fall when the trader cuts positions.

B. Confidence intervals for historical-simulation VaR are typically narrower than those for filtered historical simulation, and they become narrower still when estimated over stress periods.

C. Under the regulatory loss function, which penalizes only exceedances, a P&L-based GARCH VaR outperformed the banks' positional VaR models at most of the banks studied.

D. The component VaR of a position with scarce valuation data is best estimated by regressing changes in portfolio value on the full time series of that position's value changes.

Solutions • MR-6 a

Solution MR-6 a.1 — Answer: D

Bank Q324 • Schweser Module Quiz 6.1 Q1

Step 1 – List what conceptual soundness reviews

Methodology, input data and assumptions, judged against the model's purpose.
⇒ The model must be designed for the bank's specific risk management objectives (regulatory capital, internal risk assessment).

Step 2 – Test each option against that list

Normality is not required; adjusting inputs is sensitivity analysis; actual historical returns are the weakness, not the standard.

Interpretation

"Fit for purpose" is the crucial conceptual soundness test in the reading: a VaR model must serve its intended use, and its central use is managing the risk of the current positions.

Why the other options are wrong

A is incorrect. Returns are generally non-normal; assuming normality is a weakness, not a criterion.

B is incorrect. Adjusting inputs one at a time to see the change in VaR is sensitivity analysis (MR-6 b), a supporting test, not the crucial criterion.

C is incorrect. Actual historical returns ignore changes in portfolio composition; pseudo-historical returns are preferred.

LO: MR-6 a: Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process. • **Reference:** Lynch, Validating BHCs' VaR Models, Section 6.3; SchweserNotes 2026 Book 1, Reading 6, LO 6.a

Solution MR-6 a.2 — Answer: B

Bank Q319 • AnalystPrep Q.6438

Step 1 – What conceptual soundness requires

VaR must respond correctly when positions change; a model that cannot show this fails a crucial conceptual soundness test.

Step 2 – What backtesting measures

Backtesting compares ex ante VaR with realized P&L. It is *outcomes analysis*: it can be passed by chance and says nothing about the position-to-risk link.
⇒ Separate tests (pseudo-history checks, sensitivity analysis) are needed.

Interpretation

Backtesting and conceptual soundness are different validation components (Figure 1). The question tests that you do not let an outcome test stand in for a review of the model's structure.

Why the other options are wrong

A is incorrect. Mischaracterizes backtesting: it compares ex ante VaR with subsequent realized P&L. It is also not the reason it is insufficient here.

C is incorrect. Exceedance backtests do look at the tail, but that is not the gap. The gap is the missing link between positions and risk.

D is incorrect. Backtesting is a useful internal validation tool, not merely a regulatory formality.

LO: MR-6 a: Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process. • **Reference:** Lynch, Section 6.3

Solution MR-6 a.3 Answer: C

Bank Q321 • AnalystPrep Q.6467

Step 1 – Why actual P&L fails

VaR from actual historical P&L reflects the old positions. Cutting positions does not change past P&L, so VaR shows no reduction.

Step 2 – Apply the cut to the pseudo history

Revaluing the 25% smaller positions under every historical scenario scales each pseudo P&L by 0.75.

$\Rightarrow$ For a proportional cut, VaR falls by 25% immediately (Figure 2).

Interpretation

This is the reading's own illustration of "fit for purpose": a trader told to cut risk duly cuts positions, and only a VaR built on the pseudo history shows the reduction.

Why the other options are wrong

A is incorrect. Pre-reduction actual P&L embeds the old positions, so VaR does not change.

B is incorrect. Post-reduction actual P&L is only a short sample, and most of the window still reflects the old positions.

D is incorrect. Comparing with last year's average VaR mixes the position change with changes in market volatility.

LO: MR-6 a: Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process. • **Reference:** Lynch, Section 6.3

Solution MR-6 a.4 Answer: C

Bank Q323 • BT P2.T5.25.5.2

Step 1 – Sort the statements into the three review areas

A is an *assumption* weakness (normality understates fat tails). B is a *data* weakness (a short window can miss crises). D is an *input* weakness (no pseudo history for the changed positions).

Step 2 – Identify the odd one out

C is about the operating cost of the *alternative* models, not about the current model's logic.

$\Rightarrow$ C is not a conceptual weakness.

Interpretation

Conceptual soundness reviews methodology, data and assumptions. Run time is an implementation consideration. Note the "NOT" in the stem: three options are genuine weaknesses.

Why the other options are wrong

A is incorrect. A genuine assumption weakness, so it cannot be the "NOT" answer.

B is incorrect. A genuine data weakness.

D is incorrect. A genuine input weakness, and the one the reading emphasizes most.

LO: MR-6 a: Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process. • **Reference:** Lynch, Section 6.3; SchweserNotes 2026 Book 1, LO 6.a

Solution MR-6 a.5 Answer: C

Bank Q318 • AnalystPrep Q.6207

Step 1 – Diagnose the exceedances

Exceptions clustered in a high-volatility period mean VaR reacts too slowly to changing volatility.

Step 2 – Find the cause

An equally weighted 10-year history gives the recent turbulent days a tiny weight, so VaR lags the current risk level.
⇒ A volatility-updating method (GARCH VaR, or filtered historical simulation) addresses the cause.

Interpretation

The reading notes that historical simulation "is not very responsive to recent volatility", and GARCH and FHS were developed for that. Keep the distinction from MR-6 a.6: GARCH applied to the pseudo history of current positions is sound; GARCH applied to actual P&L is not fit for purpose.

Why the other options are wrong

A is incorrect. A higher confidence level scales VaR up but still lags current volatility.
B is incorrect. Stress testing complements VaR; replacing VaR with stress results does not repair the model.
D is incorrect. A longer window dilutes recent volatility even more, making the lag worse.

LO: MR-6 a: Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process. • **Reference:** Lynch, Sections 6.2 and 6.3; SchweserNotes 2026 Book 1, Module 6.1

Solution MR-6 a.6 Answer: B

New • built from GARP MR Ch.6, 6.3

Step 1 – Identify the model

A GARCH VaR on the desk's *actual* P&L history: the same type of model Berkowitz and O'Brien found to be accurate.

Step 2 – Apply the fit-for-purpose test

The P&L of a short deep out-of-the-money put book is a run of small gains until the market moves. That history does not show the tail risk of the puts now held, and it does not change when the positions change.
⇒ This is Lo's "Capital Decimation Partners" case: only a VaR on the current positions (Jorion) shows the risk and responds to position changes.

Interpretation

The model fails conceptual soundness because it is not fit for purpose, not because it is inaccurate. The confidence level and GARCH's volatility dynamics are not the issue.

Why the other options are wrong

A is incorrect. Wrong model fact: GARCH is designed to capture volatility clustering. The problem is the input (actual P&L), not the volatility dynamics.
C is incorrect. Raising the confidence level on the same P&L history leaves the position-change problem in place.
D is incorrect. Sibling swap: Berkowitz and O'Brien found the GARCH model on actual P&L *more* accurate than banks' VaR models.

LO: MR-6 a: Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process. • **Reference:** Lynch, Section 6.3 (Berkowitz and O'Brien 2002; Lo 2001; Jorion 2007)

Solution MR-6 a.7 Answer: A

Workbook D10.2 • S3

Step 1 – Purpose test (Section 6.3)

"A crucial aspect of VaR modeling; its key purpose is to show how risk changes when positions

change." A VaR built on actual P&L history is unaffected when positions are cut, because the history does not change; a VaR on the pseudo history of the current portfolio does fall.
⇒ Such a model "is not fit for purpose and thus fails a crucial conceptual soundness test", even though Berkowitz and O'Brien found the P&L GARCH model more accurate as a forecast.

Step 2 – Check the remaining validation claims

Confidence intervals (Section 6.5, Table 6.1): FHS narrowest; HS "particularly wide" in stress periods. Benchmarking (Section 6.8): positional VaR wins under the regulatory loss function at every bank; P&L VaR wins under the check loss at 16 of 19. Sensitivity (Section 6.4): with scarce data the regression approach "is difficult to implement"; Tasche and Hallerbach use the position's change on the day that determines the HS VaR.

Interpretation

A is correct. It is the reason a validator may use the P&L GARCH VaR as a benchmark but not as the risk management model. The other three options are tested again in MR-6 b, c and d.

Why the other options are wrong

B is incorrect. Polarity flip: HS intervals are wider than FHS intervals (FHS "provides more efficient estimates of VaR"), and HS intervals widen, not narrow, in stress periods.
C is incorrect. Sibling swap: the P&L VaR outperformed on accuracy under the check loss function; under the regulatory loss function the positional VaR "significantly outperforms the P&L VaR in every case", reflecting conservatism.
D is incorrect. Scope/condition error: the regression of Equation 6.5 "requires that you have a full time series of the change in value of the positions", which is what is missing; the component is instead estimated from the position's loss on the VaR-determining day, or an average of ordered observations near it.

LO: MR-6 a, b, c, d (all four LOs of the reading). • **Reference:** Lynch, Validating Bank Holding Companies' Value-at-Risk Models for Market Risk, Sections 6.3, 6.4, 6.5, 6.8

MR-6 • Validating Bank Holding Companies' Value-at-Risk Models for Market Risk • Vardeez: IMPORTANT (new topic, highly testable)

MR-6 b

P3 CONCEPT

Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis.

Workbook: D10 BHC Capital Planning and VaR Model Validation • Tier D • Map #53

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D10 (Map #53) hits this LO; Vardeez flags the reading as a new, highly testable topic.
Vardeez: no specific pointer inside the LO.

Reported exam shape

- **D10·S3** Validation standards, with sensitivity analysis as one option: "with scarce data the regression approach is difficult to implement". Worked as option D of Question MR-6 a.7.

What the questions look like: the three-step procedure (identify, adjust one input, compare), the list of benefits and challenges, and the data-scarcity problem with its Tasche and Hallerbach fix.

1. Why sensitivity analysis, and how to do it

Because a VaR model exists to show how risk changes when positions change, the validator checks how sensitive VaR is to the *positions* and to the simplifications and omissions made in building the pseudo history. Supervisors increasingly ask banks to **track data proxies and quantify the risks not in VaR**; sensitivity analysis gives a consistent framework for that, where most practice is ad hoc.

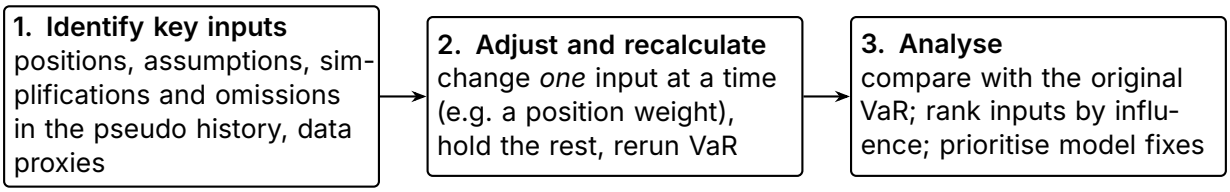


Figure 1. The three-step procedure (Schweser and Vardeez). Changing many inputs at once, or swapping the whole model, is benchmarking, not sensitivity analysis.

2. The Euler decomposition behind it

VaR is **linear homogeneous in positions** (double every position and VaR doubles), so it satisfies Euler's equation:

$$\mathrm{VaR}(V_P) = \sum_{i \in P} \frac{\partial\, \mathrm{VaR}}{\partial V_i} V_i, \qquad \text{Component VaR}_i = \mathrm{VaR}(V_P)\, w_i\, \beta_i$$

- $\partial \mathrm{VaR}/\partial V_i$ is the **marginal VaR**: how VaR changes for a small change in the size of position i. Each term in the sum is a **component VaR**, and the components add up to total VaR.
- β_i comes from regressing the change in the position's value on the change in the portfolio's value, $\Delta V_{i,t} = \alpha + \beta_i\, \Delta V_{P,t} + \varepsilon_t$.
- So the sensitivity of VaR to a position depends on two things: the position's **share** of the portfolio, w_i, and the **sensitivity of its value to the portfolio value**, β_i.

Precision note on the source notes. Vardeez (following Schweser) defines marginal VaR as "the change in portfolio *value* due to a small change in a position's weight". It is the change in *VaR*. Schweser's prose also states the regression the other way round (portfolio on position); Vardeez's equation, with the position's value change as the dependent variable, is the one consistent with Component $\text{VaR}_i = \text{VaR}\, w_i \beta_i$.

3. The data-scarcity problem

- The regression needs a **full time series** of the position's value changes. But the positions a bank most wants to test are usually the ones with **missing or scarce valuation data**.
- A **proxy** may be *less volatile* or *less correlated* with the portfolio than the real position, and so **understates** its contribution to VaR.
- **Tasche and Hallerbach** fix (easy with historical simulation): linear homogeneity implies that, on the scenario that sets the HS VaR, the portfolio loss is the sum of each position's expected change conditional on that portfolio loss. So a position's component VaR can be estimated by **its change in value on the day that determines the HS VaR**. A single day is noisy, so an **average of the ordered observations near the HS VaR** may be used instead.
- **The paradox of omitted risk.** If you had enough data to estimate the omitted risk accurately, it would probably already be in VaR. If you do not, it is hard to measure precisely how much the omission matters.
- Because portfolios change fast, many supervisors ask for risks not in VaR on **both a component and a stand-alone basis**; stand-alone VaR is independent of the portfolio's composition.
- The same approach handles other omissions, for example the next, omitted term of a Taylor-series valuation. It tests the assumption that an omitted position or risk factor is **immaterial**, and gives a rigorous way to **prioritise** adding risk factors or higher-order terms (Jarrow's view of validation as testing sensitivity to assumptions).

Benefits	Challenges
Risk assessment: finds the inputs that drive the risk profile	**Data limitations:** scarce data for new or illiquid instruments
Model validation: tests robustness to assumptions, adds credibility	**Computational burden:** large books, Monte Carlo models
Regulatory compliance: shows the model's limits are understood	**Proxy issues:** a proxy can understate volatility or correlation
Better decisions: on position changes and risk mitigation	**Changing composition:** sensitivities shift as the book changes

Not a benefit: *reducing regulatory capital*. The analysis informs; it does not change the book's risk.

Exam traps

- **One input at a time.** Swapping the model or changing many inputs together is benchmarking or re-estimation, not sensitivity analysis.
- **Backtesting is not sensitivity analysis.** Counting exceptions tests outcomes, not inputs.
- **Proxies understate.** A proxy is usually *less* volatile or *less* correlated, not more.
- **The regression needs full data.** With scarce data, use the VaR-day change (Tasche and Hallerbach), not the regression.

- **Stand-alone vs component.** Stand-alone VaR is independent of the portfolio; supervisors ask for both.
- **Marginal VaR is a change in VaR**, not in portfolio value.

Practice questions: MR-6 b

Question MR-6 b.1

Bank Q326 • AnalystPrep Q.6464

A financial analyst is performing sensitivity analysis on a VaR model. What is the primary purpose of such an exercise?

A. To determine the model's backtesting performance.

B. To evaluate the accuracy of the VaR model in predicting portfolio losses.

C. To identify the key drivers of VaR and assess model robustness.

D. To assess whether the model's VaR estimates meet regulatory requirements under varying assumptions.

Question MR-6 b.2

Bank Q328 • Schweser Module Quiz 6.1 Q2

Which of the following actions is least likely a benefit of sensitivity analysis?

A. Model validation.

B. Regulatory compliance.

C. Improved decision-making.

D. Reducing regulatory capital.

Question MR-6 b.3

Bank Q329 • AnalystPrep Q.6209 (rebuilt)

A global investment bank holds large foreign exchange and interest rate derivative positions. As part of validating the conceptual soundness of its VaR model, the risk team wants to find out how much the VaR estimate depends on the model's assumed correlation between FX and interest rate risk factors, which could shift sharply during geopolitical stress. Which of the following approaches is most consistent with conducting a sensitivity analysis of the VaR model for this purpose?

A. Replace the current model with a historical simulation that uses only data from past geopolitical crises, and compare the number of backtesting exceptions of the two models.

B. Lower the VaR confidence level from 99% to 95% so that the VaR estimate is less affected by uncertainty in the correlation assumption.

C. Identify the FX/interest rate correlation as a key input, change it by a set amount while holding all other inputs fixed, recalculate VaR, and compare the revised VaR with the original estimate.

D. Lengthen the historical window used to estimate correlations, and adopt whichever window length produces the lowest VaR.

Question MR-6 b.4 Bank Q327 • BT P2.T5.25.5.3

A global systemically important bank (G-SIB) is conducting a sensitivity analysis on its VaR model as part of its validation process. The bank uses a hybrid approach that combines parametric methods with Monte Carlo simulations. During testing, analysts observe that small variations in input volatility assumptions lead to large fluctuations in VaR estimates, particularly for certain illiquid fixed-income securities and complex derivatives. At the same time, the risk oversight committee is concerned that stress scenarios designed to evaluate the model's stability may not fully capture non-linear risks inherent in the bank's structured product holdings. Some senior risk managers argue that the model's heavy reliance on recent historical correlations may produce overly stable VaR estimates that do not reflect true tail risk exposure during market stress. Which of the following actions would best address the concerns raised while maintaining a robust and conceptually sound sensitivity analysis framework?

A. Adjust the VaR model to use a constant historical correlation matrix, ensuring that sensitivity tests show stable risk estimates over time and avoiding excessive model recalibrations.

B. Introduce factor perturbation sensitivity analysis by systematically shifting key risk factors such as implied volatility, correlations, and liquidity spreads to assess VaR stability across extreme but plausible scenarios.

C. Reduce the Monte Carlo simulation iterations to limit computational burden while maintaining historical return distributions, as excessive simulations could distort sensitivity analysis findings.

D. Rely on ex-post backtesting results to determine whether sensitivity analysis is necessary, as the primary focus should be on validating real-world P&L alignment rather than artificially constructed stress scenarios.

Question MR-6 b.5 New • built from GARP MR Ch.6, 6.4

A risk manager explains to a trading head how the VaR model's sensitivity to a single position is measured. Which of the following statements is correct?

A. Because VaR is linear homogeneous in positions, component VaRs sum to total VaR, and a position's sensitivity depends on its portfolio share and on how its value moves with the portfolio.

B. Because VaR is subadditive, component VaRs sum to more than total VaR, and the excess measures each position's contribution to diversification.

C. Marginal VaR measures the change in portfolio value caused by a small change in a position's price, so it does not depend on the position's share of the portfolio.

D. Supervisors ask for risks not in VaR on a component basis only, because stand-alone VaR changes whenever the portfolio's composition changes.

Question MR-6 b.6 New • built from GARP MR Ch.6, 6.4 (Tasche 2000; Hallerbach 2003)

A validation team wants to measure how much a thinly traded structured note, which was valued on only a few days in the past year, contributes to the desk's 99% historical-simulation VaR. Which approach is most consistent with the methods described for this situation?

A. Regress the note's value changes on the portfolio's value changes over the full year, estimate β, and set component VaR equal to VaR times the note's weight times β.

B. Replace the note with a liquid proxy and use the proxy's beta, since a proxy is at least as volatile and as correlated with the portfolio as the note.

C. Use the change in the note's value on the day that determines the historical-simulation VaR, or an average of ordered observations near that day.

D. Exclude the note, since a material omission would already have produced an unexpected cluster of exceptions in the desk's backtest.

Question MR-6 b.7 New • built from GARP MR Ch.6, 6.4

A bank's VaR pseudo history values an options book with a first-order (delta) Taylor approximation and omits basis risk on several emerging-market bonds. The head of validation asks what sensitivity analysis can achieve in this situation. Which response is most accurate?

A. It tests whether the assumption that the omitted terms and positions are immaterial holds, and gives a consistent way to prioritise which risk factors or higher-order terms to add.

B. It proves that the omissions are immaterial whenever the VaR model passes its backtest with fewer than five exceptions over the past 250 days.

C. It removes the need to track data proxies, because supervisors accept sensitivity results in place of an inventory of the risks not in VaR.

D. It estimates the omitted risks precisely, because the scarcer the data on a position, the easier its effect on VaR is to isolate.

Solutions • MR-6 b

Solution MR-6 b.1 — Answer: C

Bank Q326 • AnalystPrep Q.6464

Step 1 – What the exercise does

Change key inputs (positions, volatilities, correlations) one at a time and recalculate VaR.
⇒ The size of each change shows which inputs drive VaR and how robust the model is to its assumptions.

Interpretation

Sensitivity analysis is a test of the model's *inputs and assumptions*. Accuracy against realized losses is a backtesting question.

Why the other options are wrong

A is incorrect. Backtesting performance is judged by comparing VaR with realized P&L, not by varying inputs.
B is incorrect. Predictive accuracy is also an outcomes (backtesting) question.
D is incorrect. Regulatory compliance is a secondary benefit, not the primary purpose.

LO: MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. • **Reference:** Lynch, Section 6.4; SchweserNotes 2026 Book 1, LO 6.b

Solution MR-6 b.2 — Answer: D

Bank Q328 • Schweser Module Quiz 6.1 Q2

Step 1 – Recall the four benefits

Risk assessment, model validation, regulatory compliance and improved decision-making.

Step 2 – Test the odd one

Regulatory capital depends on the level of VaR, which is driven by the positions. Running the analysis does not change the positions.
⇒ Reducing regulatory capital is not a benefit.

Interpretation

The analysis informs management; any capital saving would come from decisions taken afterwards, not from the analysis itself.

Why the other options are wrong

A is incorrect. A genuine benefit: it tests robustness to assumptions.
B is incorrect. A genuine benefit: it shows regulators the model's limits are understood.
C is incorrect. A genuine benefit: it supports position and mitigation decisions.

LO: MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. • **Reference:** SchweserNotes 2026 Book 1, LO 6.b

Solution MR-6 b.3 — Answer: C

Bank Q329 • AnalystPrep Q.6209

Step 1 – Identify the key input

The assumed correlation ρ between the FX and rate factors.

Step 2 – Adjust it alone and recalculate

Illustration with stand-alone VaRs of 10 (FX) and 8 (rates): at $\rho = 0.2$, VaR $= \sqrt{10^2 + 8^2 + 2(0.2)(10)(8)} = 14.0$; at $\rho = 0.6$, VaR $= \sqrt{260} = 16.12$.

⇒ VaR rises by about 15.2% for this correlation shift, all else equal.

Step 3 – Analyse

Compare with the original estimate and judge whether the correlation assumption needs closer validation.

Interpretation

Only C isolates one input and follows the identify, adjust, compare sequence (Figure 1).

Why the other options are wrong

A is incorrect. Swapping in a different model and counting exceptions is benchmarking or back-testing, and it changes many inputs at once.

B is incorrect. Lowering the confidence level just shrinks the reported VaR; it says nothing about the correlation sensitivity.

D is incorrect. Changing the window changes many estimates at once, and choosing the lowest VaR is result-shopping, not validation.

LO: MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. • **Reference:** SchweserNotes 2026 Book 1, LO 6.b

Solution MR-6 b.4 — Answer: B

Bank Q327 • BT P2.T5.25.5.3

Step 1 – List the concerns

VaR is very sensitive to volatility inputs; stress scenarios may miss non-linear risk; historical correlations may make VaR look too stable.

Step 2 – Match the remedy

Sensitivity analysis shifts the key inputs (implied volatility, correlations, liquidity spreads) and recalculates VaR across extreme but plausible ranges.

⇒ This exposes which assumptions drive VaR and where non-linear risk hides.

Interpretation

B strengthens the testing; each of the other options removes or weakens it.

Why the other options are wrong

A is incorrect. Fixing the correlation matrix hides the very sensitivity being tested and makes VaR look stable artificially.

C is incorrect. Fewer simulations add sampling error, especially for non-linear products, and do not test input sensitivity.

D is incorrect. Backtesting is ex post and cannot replace sensitivity analysis, which exposes weak assumptions before losses occur.

LO: MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. • **Reference:** Lynch, Section 6.4

Solution MR-6 b.5 — Answer: A

New • built from GARP MR Ch.6, 6.4

Step 1 – Linear homogeneity and Euler

VaR is linear homogeneous in positions, so $\text{VaR}(V_P) = \sum_i (\partial \text{VaR}/\partial V_i) V_i$: the component VaRs add up to total VaR.

Step 2 – What drives a position's component

Component $\text{VaR}_i = \text{VaR}\, w_i \beta_i$: the position's share w_i and the sensitivity β_i of its value to the portfolio value.

⇒ A is the reading's statement.

Interpretation

This decomposition is what makes sensitivity analysis to positions possible. It also explains why supervisors ask for a stand-alone view too: component VaR moves with the portfolio's composition.

Why the other options are wrong

B is incorrect. Wrong property: Euler's equation makes the components sum *exactly* to total VaR; subadditivity is a different property of risk measures.
C is incorrect. Marginal VaR is the change in *VaR* for a small change in the size of the position, and it does depend on the position's weight through β_i and the portfolio.
D is incorrect. Reversed: supervisors ask for *both* component and stand-alone measures, because stand-alone VaR is *independent* of the portfolio's composition.

LO: MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. • **Reference:** Lynch, Section 6.4, Euler equation and Equation 6.5

Solution MR-6 b.6 Answer: C

New • built from GARP MR Ch.6, 6.4

Step 1 – Why the regression fails here

Equation 6.5 needs a full time series of the position's value changes. The note was valued on only a few days, so that series does not exist.

Step 2 – Apply Tasche and Hallerbach

On the scenario that sets the HS VaR, the portfolio loss equals the sum of the positions' changes. So a position's component VaR can be estimated by its change in value on that day.
⇒ One day is noisy, so an average of the ordered observations near the HS VaR may be used instead.

Interpretation

The method is "especially easy to implement when using historic simulation", which is the model in the stem.

Why the other options are wrong

A is incorrect. The regression needs exactly the full time series that is missing.
B is incorrect. Polarity flip: a proxy may be *less* volatile or *less* correlated, and so understate the contribution.
D is incorrect. A backtest does not show whether a specific position's omission is material; that is what the sensitivity analysis is for.

LO: MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. • **Reference:** Lynch, Section 6.4 (Tasche 2000; Hallerbach 2003)

Solution MR-6 b.7 Answer: A

New • built from GARP MR Ch.6, 6.4

Step 1 – What the analysis tests

It tests the assumption that the omitted positions or risk factors (here, basis risk and the next Taylor term) are immaterial.

Step 2 – What it delivers

A consistent framework for quantifying omitted risks and a more rigorous way to prioritise model changes: more risk factors or higher-order terms.

⇒ This is Jarrow's view of validation: check whether the model is sensitive to its assumptions.

Interpretation

Sensitivity analysis complements, rather than replaces, backtesting and the tracking of proxies that supervisors require.

Why the other options are wrong

B is incorrect. A backtest result does not prove that a particular omission is immaterial.
C is incorrect. Supervisors increasingly require banks to *track* their data proxies and risks not in VaR; sensitivity analysis supports that, it does not replace it.
D is incorrect. Reversed: with scarce data it is *hard* to make a precise estimate of the omission's effect.

LO: MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. • **Reference:** Lynch, Section 6.4

MR-6 • Validating Bank Holding Companies' Value-at-Risk Models for Market Risk • Vardeez: IMPORTANT (new topic, highly testable)

MR-6 c

P3 CONCEPT

Describe the challenges a financial institution could face when calculating confidence intervals for VaR.

Workbook: D10 BHC Capital Planning and VaR Model Validation • Tier D • Map #53

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D10 (Map #53) hits this LO; Vardeez flags the reading as a new, highly testable topic.
Vardeez: no specific pointer inside the LO.

Reported exam shape

- **D10·S3** Validation standards, with the confidence-interval option "FHS narrowest, HS particularly wide in stress". Worked as option B of Question MR-6 a.7.

What the questions look like: the empirical findings (asymmetry, more data tighter, GARCH and FHS tighter than HS, HS wider in stress, no winner between order statistics and bootstrap) and the method challenges (density at the VaR, non-normality, independence, computational cost).

1. The idea and the gap

A confidence interval around a VaR estimate says how precise it is: the true VaR should lie inside it a stated percentage of the time. This measures **estimation risk**. Although VaR is an inherently statistical framework, **few bank holding companies routinely calculate confidence intervals** for their VaR, even though the methods are well established.

2. Methods and where each one struggles

- **Jorion's asymptotic standard error of a quantile:**

$$SE(\text{VaR}) = \frac{1}{f(\text{VaR})}\sqrt{\frac{c(1-c)}{T}}$$

  c and T are known. The difficulty is $f(\text{VaR})$, the **density at the VaR**. Imposing a normal density "would probably be a mistake" because returns are non-normal. Jorion also suggests using more than the single quantile point, basing the interval on the variance of the distribution of pseudo returns.

- **Order statistics (Dowd):** order the pseudo history and use the binomial distribution of order statistics to pick the ranks that bracket the VaR (for a 90% interval, set the probabilities to 0.05 and 0.95). Non-parametric, like HS itself.

- **Bootstrap (Christoffersen and Goncalves):** resample the pseudo history with replacement, recompute VaR each time, and read the interval off the ordered bootstrapped VaRs.
  - For **HS**, resampling assumes the value changes are **independent over time**, so dependence is not reflected.
  - For **GARCH**, resample the IID standardised residuals, rebuild the series and **re-estimate the GARCH model on each sample** so that estimation risk enters the interval. Straightforward but **computationally expensive**.

– For **FHS**, keep the re-estimated residuals and recompute the FHS VaR for each sample.

3. What the evidence shows (Table 6.1, S&P 500, one-day 99% VaR)

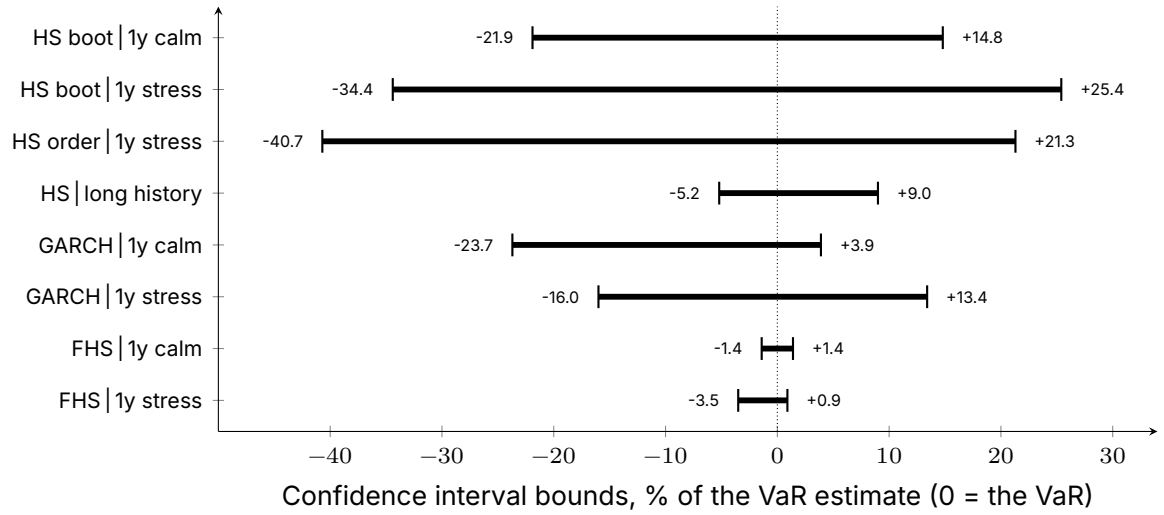


Figure 1. Selected rows of GARP Table 6.1 (lower and upper 95% bounds as a percentage of the VaR). Bars are asymmetric around the VaR; HS bars are long, and longest in the one-year stress period; the long history (9,424 days) is far tighter than one year; FHS bars are the shortest.

- Intervals are **not symmetric**.
- **More data, tighter intervals.**
- HS: **no convincing evidence** that order statistics or the bootstrap gives the tighter interval.
- HS intervals are **particularly wide in stress periods**.
- **FHS gives the narrowest intervals**: more efficient VaR estimates.
- On banks' P&L (Table 6.2), **GARCH VaR intervals are much narrower than HS VaR intervals**, which suggests FHS would improve banks' estimates.

Precision note on the source notes. Schweser writes that "in most cases, a normal distribution is assumed" for the density at the VaR. That describes common practice, not a recommendation: the GARP text says imposing normality "would probably be a mistake" because returns are non-normal.

Exam traps

- **Polarity of width.** FHS narrowest; HS widest, and wider still in stress. More data narrows, it does not widen.
- **No winner** between order statistics and bootstrap. Any option claiming one is "consistently tighter" is wrong.
- **Asymmetric**, not symmetric.
- **The hard input is** $f(\text{VaR})$, not c or T.
- **Bootstrap assumptions.** HS bootstrap assumes independence; GARCH bootstrap resamples standardised residuals and re-estimates the model; neither requires normality.
- **"Always" is wrong.** A longer lookback tightens the interval statistically but does not *always*

improve accuracy: old data can be stale.

Practice questions: MR-6 c

Question MR-6 c.1

Bank Q337 • Schweser Module Quiz 6.1 Q3

Which of the following findings is incorrect regarding the empirical analysis of VaR confidence intervals?

A. Confidence intervals are not symmetric.

B. Larger datasets lead to tighter confidence intervals.

C. Order statistics produces tighter confidence intervals for VaR compared to bootstrap techniques.

D. GARCH VaR tends to produce much tighter confidence intervals compared to historical simulation VaR.

Question MR-6 c.2

Bank Q334 • AnalystPrep Q.6468 (rebuilt)

A risk analyst compares confidence intervals for one-day 99% VaR of a bank's P&L estimated by historical simulation, GARCH, order statistics and bootstrapping. Which of the following statements about the width of the confidence intervals is correct?

A. Confidence intervals for VaR are symmetric around the point estimate.

B. GARCH VaR tends to produce tighter confidence intervals than historical simulation VaR.

C. Using more data generally leads to wider confidence intervals.

D. Order statistics consistently produces tighter confidence intervals than bootstrapping.

Question MR-6 c.3

Bank Q335 • AnalystPrep Q.6469 (rebuilt)

A risk manager uses historical simulation to estimate one-day 99% VaR for a portfolio from (i) the most recent calm one-year period, (ii) a one-year stress period and (iii) a much longer history. Which statement about confidence interval width is correct?

A. The interval is widest for the longer history, because it contains more regimes.

B. The interval is narrowest for the stress period, because large losses make the quantile easy to locate.

C. The interval is the same for all three periods, because historical simulation makes no distributional assumption.

D. The interval for the stress period is wider than for the calm one-year period, while the longer history gives a tighter interval than a one-year sample.

Question MR-6 c.4 New • built from GARP MR Ch.6, 6.5 (Jorion 1996)

A bank holding company's validation team proposes to place a 90% confidence interval around its one-day 99% historical-simulation VaR using Jorion's asymptotic standard error of a quantile, $SE = \frac{1}{f(\text{VaR})}\sqrt{c(1-c)/T}$. Which of the following is the main challenge in applying this formula?

A. The coverage level c and the sample size T are unknown and must first be estimated from the pseudo history.

B. The density at the VaR, $f(\text{VaR})$, must be specified, and assuming a normal density is probably a mistake because returns are non-normal.

C. The formula gives intervals that are too narrow whenever more data are used, so T must be kept to one year or less.

D. The formula cannot be applied to VaR at all, because VaR is not a quantile of the distribution of portfolio value changes.

Question MR-6 c.5 New • built from GARP MR Ch.6, 6.5 (Christoffersen and Goncalves)

A risk analyst bootstraps confidence intervals for two VaR models of the same trading book: a historical-simulation VaR and a GARCH(1,1) VaR. Which of the following statements is correct?

A. For the historical-simulation VaR, resampling the pseudo history with replacement assumes daily value changes are independent, so dependence over time is not reflected.

B. For the GARCH VaR, the analyst should resample the raw daily value changes directly, because the GARCH model already accounts for their dependence.

C. For the GARCH VaR, the parameters estimated on the original sample should be held fixed in every bootstrap sample, so that estimation risk is kept out.

D. Filtered historical simulation typically yields the widest intervals of the three, because it combines GARCH volatility with an empirical distribution.

Question MR-6 c.6 Bank Q331 • BT P2.T5.25.6.1 (rebuilt, key corrected)

A bank's historical-simulation 99% VaR confidence intervals were found to be too narrow in a regulatory stress test. A group in the risk team proposes simply extending the historical lookback period. Which of the following is NOT a valid reason why extending the lookback period is not a foolproof solution?

A. Longer historical periods may dilute the impact of recent extreme events, leading to an underestimation of current risk.

B. Increasing the lookback period will always improve model accuracy by incorporating a wider range of tail risk scenarios.

C. Market dynamics evolve over time, and older data may not reflect current volatility regimes, leading to misleading risk estimates.

D. Historical simulation assumes that past returns are representative of future risks, which may not hold in rapidly changing environments.

Question MR-6 c.7 Bank Q330 • AP Mock 3 Q1 (rebuilt)

A bank builds a confidence interval for its GARCH VaR by bootstrapping: it resamples the standardized residuals with replacement, rebuilds a return series, re-estimates the GARCH parameters and recomputes VaR, repeating this thousands of times. Which of the following is the main practical challenge of this approach?

A. Resampling with replacement systematically biases the VaR estimates toward zero.

B. The standardized residuals must be assumed to be normally distributed before they can be resampled.

C. Re-estimating the GARCH model by numerical likelihood maximization for every bootstrap sample is computationally intensive.

D. Resampled histories cannot contain the extreme observations that appear in the original data.

Solutions • MR-6 c

Solution MR-6 c.1 — Answer: C

Bank Q337 • Schweser Module Quiz 6.1 Q3

Step 1 – Recall the findings

Intervals are asymmetric; more data tightens them; GARCH VaR gives tighter intervals than HS VaR; neither order statistics nor the bootstrap is consistently tighter; HS is wider in stress.

Step 2 – Find the statement that contradicts them

C claims a consistent winner between the two non-parametric methods.
$\Rightarrow$ C is the incorrect finding.

Interpretation

Note the "incorrect" in the stem: A, B and D are all findings of the reading.

Why the other options are wrong

A is incorrect. A correct finding: the intervals are asymmetric (Figure 1).
B is incorrect. A correct finding: more data, more precise quantile estimates.
D is incorrect. A correct finding: GARCH reflects current volatility and gives tighter intervals than HS.

LO: MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR.
• **Reference:** Lynch, Section 6.5, Tables 6.1 and 6.2

Solution MR-6 c.2 — Answer: B

Bank Q334 • AnalystPrep Q.6468

Step 1 – Match each statement to the evidence

On banks' P&L, "much narrower confidence intervals for the GARCH VaR than for historic simulation VaR" (Table 6.2).
$\Rightarrow$ B matches.

Interpretation

GARCH reflects current volatility, so its VaR estimates are more precise than plain historical simulation.

Why the other options are wrong

A is incorrect. Intervals are not symmetric.
C is incorrect. Polarity flip: more data gives tighter intervals.
D is incorrect. Neither non-parametric method is consistently tighter.

LO: MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR.
• **Reference:** Lynch, Section 6.5, Table 6.2

Solution MR-6 c.3 — Answer: D

Bank Q335 • AnalystPrep Q.6469

Step 1 – Stress effect

HS VaR has particularly wide intervals in stress periods: in Table 6.1 the one-year stress interval runs from -34.4% to $+25.4\%$ of the VaR, against -21.9% to $+14.8\%$ in the calm year.

Step 2 – Sample-size effect

The long history (9,424 days) gives about -5.2% to $+9.0\%$: far tighter than either one-year sample.

⇒ D states both findings.

Interpretation

Two forces: stress raises estimation uncertainty, more data reduces it. Being non-parametric does not make HS immune to either.

Why the other options are wrong

A is incorrect. Reverses the sample-size effect.
B is incorrect. Reverses the stress effect.
C is incorrect. Non-parametric does not mean insensitive: sample size and the dispersion of tail observations still drive width.

LO: MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR.
• **Reference:** Lynch, Section 6.5, Table 6.1

Solution MR-6 c.4 Answer: B

New • built from GARP MR Ch.6, 6.5

Step 1 – Which inputs are known

VaR, T and c are known, so the calculation is "seemingly straightforward".

Step 2 – Which input is hard

$f(\text{VaR})$, the probability density at the VaR. Imposing a distribution, usually normal, is convenient, but financial returns are non-normal, so "imposing a normal distribution would probably be a mistake".
⇒ The challenge is specifying $f(\cdot)$; non-parametric methods (order statistics, bootstrap) avoid it.

Interpretation

Jorion also suggests basing the interval on more than the single quantile point, using the variance of the distribution of pseudo returns.

Why the other options are wrong

A is incorrect. c and T are chosen and known, not estimated.
C is incorrect. More data narrows the interval correctly; nothing in the formula makes it "too narrow".
D is incorrect. VaR *is* a quantile of the distribution of value changes; that is why a quantile standard error applies.

LO: MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR.
• **Reference:** Lynch, Section 6.5 (Jorion 1996)

Solution MR-6 c.5 Answer: A

New • built from GARP MR Ch.6, 6.5

Step 1 – HS bootstrap

Drawing observations from the pseudo history with replacement treats the changes as independent over time.
⇒ It is non-parametric, like HS itself, but it does not reflect dependence.

Step 2 – GARCH bootstrap

The standardised residuals z_t are IID, so resample *them*, rebuild the series, and *re-estimate* the GARCH equation on each sample so that estimation risk enters the interval.

Interpretation

The two bootstraps are designed differently because the bootstrap must mimic the dependence

of the data. FHS, which keeps the re-estimated residuals, gives the narrowest intervals in Table 6.1.

Why the other options are wrong

B is incorrect. Raw value changes are not IID under GARCH; it is the standardised residuals that are resampled.
C is incorrect. Reversed: the equation is re-estimated precisely to *include* estimation risk.
D is incorrect. Polarity flip: FHS gives the *narrowest* intervals.

LO: MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR.
• **Reference:** Lynch, Section 6.5 (Christoffersen and Goncalves 2005)

Solution MR-6 c.6 Answer: B

Bank Q331 • BT P2.T5.25.6.1 (key corrected)

Step 1 – Read the double negative

The stem asks for the statement that is *not* a valid reason, i.e. the false statement.

Step 2 – Test each

A (dilution of recent extremes), C (stale regimes) and D (past may not represent the future) are valid reasons. B claims a longer window will *always* improve accuracy.
$\Rightarrow$ B is false: more data tightens the interval statistically, but stale data can make the estimate worse.

Interpretation

The absolute word "always" is the GARP-style giveaway. The bank's source key had A; the corrected key is B.

Why the other options are wrong

A is incorrect. A valid reason, so it cannot be the "NOT" answer.
C is incorrect. A valid reason: regime change.
D is incorrect. A valid reason: the stationarity assumption of historical simulation.

LO: MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR.
• **Reference:** Lynch, Section 6.5

Solution MR-6 c.7 Answer: C

Bank Q330 • AP Mock 3 Q1

Step 1 – Count the work

Each bootstrap sample needs a full GARCH re-estimation by numerical maximum likelihood, then a new volatility path and VaR.
$\Rightarrow$ With thousands of resamples, run-time is the practical obstacle.

Interpretation

The reading calls these procedures "fairly straightforward to apply for univariate estimates, although potentially computationally expensive".

Why the other options are wrong

A is incorrect. The bootstrap is not systematically biased toward zero; this confuses sampling variability with bias.
B is incorrect. The bootstrap uses the empirical residual distribution; no normality is needed.
D is incorrect. Extreme observations in the data can be redrawn; only losses beyond those observed cannot be generated.

LO: MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR.

- **Reference:** Lynch, Section 6.5

MR-6 • Validating Bank Holding Companies' Value-at-Risk Models for Market Risk • Vardeez: IMPORTANT (new topic, highly testable)

MR-6 d

P3 CONCEPT

Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them.

Workbook: D10 BHC Capital Planning and VaR Model Validation • Tier D • Map #53

WHAT IS TESTED HERE

Why P3. Tier D workbook topic D10 (Map #53) hits this LO; Vardeez flags the reading as a new, highly testable topic.
Vardeez: no specific pointer inside the LO.

Reported exam shape

- **D10·S3** Validation standards, with the benchmarking option "positional VaR wins under the regulatory loss; P&L VaR wins on accuracy at 16 of 19". Worked as option C of Question MR-6 a.7.

What the questions look like: the two hindrances (non-IID errors, no second model), the GARCH-on-P&L benchmark, the two loss functions and which model each favours, and the sign test.

1. The most neglected component

Benchmarking compares a bank's VaR model with an alternative model. It is **the most neglected part** of market risk model validation. Banks may run a new model in parallel with the old one for a short time during a transition, but there is rarely a formal comparison: building even one VaR model is time-consuming and resource-intensive. The usual practice is to **plot the two VaR series**, which shows little beyond which model is *more conservative*. Formal statistical tests would be an improvement.

2. Two hindrances, two answers

Challenge	Approach proposed
Errors are not IID. Trading portfolios change frequently, so forecast errors are not independent and identically distributed; especially a problem for *regression-based* tests.	Christoffersen et al. (2001) offer methods, restricted to location-scale models. A **sign test** on loss differentials needs no distributional assumption.
No second model. Banks rarely have two VaR models to compare.	Berkowitz and O'Brien: fit a **GARCH(1,1) VaR to the bank's own trading P&L**. The data are available to every bank, and it is "a difficult benchmark to beat".

The P&L GARCH VaR is *not* suitable for risk management (it fails conceptual soundness, MR-6 a), but it is a ready benchmark.

3. Comparing the models: loss functions and the sign test

Following Diebold and Mariano (1995), compare the models' daily **losses** under a chosen loss function:

- **Regulatory loss function** (Lopez): only exceedances are penalised, because capital depends on VaR and the regulator cares about losses that exceed VaR. It **favours conservatism**.
- **Check (quantile) loss function**: the same function used to estimate a quantile regression; it penalises errors on both sides and **rewards accuracy**.
- **Sign test** (Sarma et al. 2003): with loss differential $z_t = l_t^i - l_t^j$, test whether the median differential is zero:

$$\frac{\sum_{t=1}^{N} I(z_t > 0) - 0.5N}{\sqrt{0.25N}} \sim N(0,1)$$

Rejecting the null means one model is significantly better under that loss function. It is easy to implement, so there is "little reason for banks not to make their comparisons more formal".

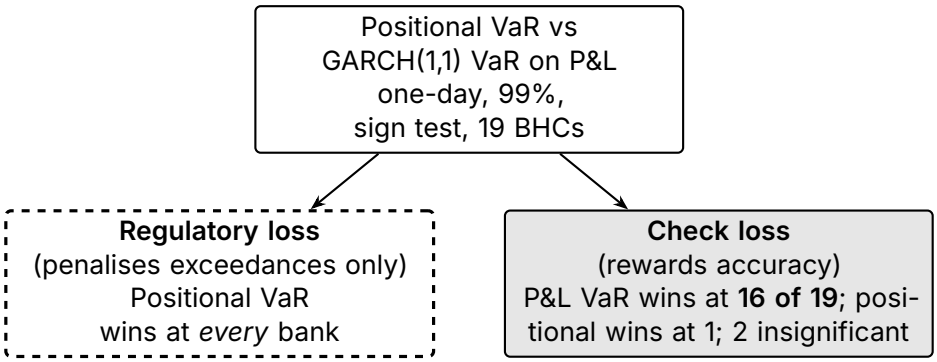


Figure 1. The two results to keep apart. Left (dashed border): conservatism wins. Right (grey fill): accuracy wins. Banks' positional VaR models are designed to be conservative, often attributed to regulatory oversight, and that conservatism stops them beating the P&L VaR on accuracy.

Exam traps

- **Which loss, which winner (sibling swap).** Regulatory loss: positional VaR wins every time (conservatism). Check loss: P&L VaR wins 16 of 19 (accuracy).
- **A plot shows conservatism, not accuracy.**
- **Errors are not IID**, because portfolios change. Options saying "independent but not identically distributed" are wrong too.
- **Banks rarely have even two models**, let alone several; benchmarking is usually a short exercise during a model transition.
- **The P&L GARCH VaR is a benchmark, not a replacement**; it fails the fit-for-purpose test.
- **The sign test uses signs only**, not the size of the differentials, and needs no normality.

Practice questions: MR-6 d

Question MR-6 d.1

Bank Q345 • Schweser Module Quiz 6.1 Q4

Which of the following statements regarding benchmarking VaR models is most accurate?

A. Benchmarking VaR models is not used for validating their performance.

B. Banks routinely benchmark their VaR model against several competing models.

C. In the statistical backtesting of VaR models, the errors are independently, but not identically, distributed.

D. Benchmarking is usually conducted for only a short time period during a bank's transition to a new model.

Question MR-6 d.2

Bank Q338 • BT P2.T5.25.6.3 (rebuilt)

A bank is moving from its current VaR model to a new one and wants to benchmark the two. The risk team's proposal is to plot the daily VaR of both models over the same period and compare the lines. Which statement best describes the information gained from this approach and what a more formal evaluation would require?

A. The plot shows which model is more accurate; no further testing is needed.

B. The plot mainly reveals which model is more conservative; a formal evaluation requires statistical tests.

C. The plot proves that the models' errors are independent and identically distributed.

D. The plot is the only valid approach because banks routinely maintain several competing models.

Question MR-6 d.3

Bank Q343 • AnalystPrep Q.6437

Which of the following is a key challenge in benchmarking VaR models?

A. The difficulty of accessing granular historical data for accurate backtesting.

B. The assumption of independent and identically distributed (i.i.d.) errors.

C. The inability of VaR models to capture tail risk accurately.

D. The consistency of model performance across various market conditions.

Question MR-6 d.4

Bank Q344 • AnalystPrep Q.6441 (rebuilt)

Banks often do not have a second VaR model to benchmark against. An institution therefore benchmarks its positional VaR model against a GARCH(1,1) VaR fitted to the bank's daily trading P&L. Which is the primary reason this benchmark is useful?

A. It uses detailed position-level risk factors and so measures positional risk more precisely than the internal model.

B. It is simple to build from readily available P&L data yet is hard for the more conservative positional VaR to beat on accuracy.

C. It requires no distributional or model assumptions at all.

D. It is more conservative than positional VaR, so it lowers regulatory capital.

Question MR-6 d.5 Bank Q340 • AnalystPrep Q.6444

A bank is comparing its positional VaR model to a P&L-based VaR model using a loss function-based approach. They are using the regulatory loss function. What is the primary focus of the regulatory loss function in this comparison?

A. Ensuring that the VaR model aligns with regulatory guidelines for acceptable confidence levels.

B. Penalizing underestimation of risk (exceptions) more heavily than overestimation.

C. Ensuring that the VaR estimates are consistent across different time horizons.

D. Balancing the trade-off between false positives (overestimations) and false negatives (underestimations) in risk predictions.

Question MR-6 d.6 Bank Q341 • AnalystPrep Q.6461 (rebuilt)

A financial institution is comparing two VaR models, A and B, using the sign test on daily loss differentials (loss of A minus loss of B). Which statement correctly describes what the sign test examines?

A. Whether each model's exception rate equals one minus its VaR confidence level.

B. Whether the two models' exceptions are clustered in time.

C. Whether the number of positive loss differentials is consistent with a 50% probability, i.e., neither model has systematically lower loss.

D. Whether the average size of the loss differential is different from zero, assuming normally distributed differentials.

Question MR-6 d.7 New • built from GARP MR Ch.6, 6.8

A validation analyst compares the bank's positional VaR with a GARCH(1,1) VaR estimated on the bank's trading P&L, both one-day 99%, using the sign test under two loss functions. Based on the evidence reported for US bank holding companies, which of the following conclusions is correct?

A. Under the regulatory loss function the P&L VaR outperformed the positional VaR at most banks, showing that positional VaR models are too aggressive.

B. Under the regulatory loss function the positional VaR outperformed in every case, while under the check loss function the P&L VaR was more accurate at 16 of 19 banks.

C. Under the check loss function the positional VaR outperformed at 16 of 19 banks, because conservatism is rewarded when accuracy is measured.

D. Under both loss functions the two models were statistically indistinguishable, so a plot of the two VaR series is sufficient benchmarking.

Solutions • MR-6 d

Solution MR-6 d.1 — Answer: D

Bank Q345 • Schweser Module Quiz 6.1 Q4

Step 1 – Recall how benchmarking is done in practice

Banks may run a new model in parallel with the old one for a short time during a transition; outside that, formal benchmarking is rare because building even one model is costly.
⇒ D matches.

Interpretation

That is why the reading calls benchmarking "perhaps the most neglected aspect of model validation in market risk".

Why the other options are wrong

A is incorrect. Benchmarking is one of the three validation components regulators expect.
B is incorrect. Banks rarely have even two models, let alone several.
C is incorrect. Because portfolios change frequently, errors are *neither* independent *nor* identically distributed.

LO: MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. • **Reference:** Lynch, Section 6.8; SchweserNotes 2026 Book 1, LO 6.d

Solution MR-6 d.2 — Answer: B

Bank Q338 • BT P2.T5.25.6.3

Step 1 – What a plot shows

"There rarely is anything useful to say about the models from these plots other than one model seems more conservative than the other."

Step 2 – What is needed

Formal tests on forecast accuracy, such as a loss-function comparison with the sign test.
⇒ B.

Interpretation

A higher line is a more conservative model, not a more accurate one.

Why the other options are wrong

A is incorrect. Accuracy cannot be read off a chart.
C is incorrect. A plot proves nothing about the error distribution; changing portfolios make errors non-IID.
D is incorrect. Banks seldom maintain even one competing model.

LO: MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. • **Reference:** Lynch, Section 6.8

Solution MR-6 d.3 — Answer: B

Bank Q343 • AnalystPrep Q.6437

Step 1 – Name the two hindrances

(1) Trading portfolios change frequently, so errors are not IID, especially a problem for regression-based tests. (2) Banks rarely have two models.
⇒ B is the first hindrance: the IID assumption of the tests is violated.

Interpretation

The other options are general VaR or validation concerns, not the specific benchmarking challenges in the reading.

Why the other options are wrong

A is incorrect. Data access is a backtesting practicality, not the benchmarking challenge cited.
C is incorrect. Tail-risk limitations are a general criticism of VaR.
D is incorrect. Robustness across conditions is a general validation concern.

LO: MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. • **Reference:** Lynch, Section 6.8

Solution MR-6 d.4 Answer: B

Bank Q344 • AnalystPrep Q.6441

Step 1 – Why this benchmark

It needs only the bank's own P&L history, which every bank has (Berkowitz and O'Brien).

Step 2 – Why it is demanding

Positional VaR is conservative because of its link to regulatory capital, so it rarely beats the P&L GARCH VaR on accuracy: "a difficult benchmark to beat".
$\Rightarrow$ B.

Interpretation

Cheap to build, hard to beat: a good benchmark, but still not a risk management model.

Why the other options are wrong

A is incorrect. It ignores positions entirely; it is not a positional model.
C is incorrect. GARCH assumes a variance process and an innovation distribution.
D is incorrect. Positional VaR is the more conservative model, and benchmarking does not set capital.

LO: MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. • **Reference:** Lynch, Section 6.8

Solution MR-6 d.5 Answer: B

Bank Q340 • AnalystPrep Q.6444

Step 1 – What the regulator cares about

Capital is based on VaR, so the regulator is more concerned with losses that exceed VaR than with VaR exceeding the loss.

Step 2 – The loss function that follows

Only exceedances are penalised.
$\Rightarrow$ Underestimation of risk is penalised more than overestimation; the function favours conservatism.

Interpretation

That is why positional VaR wins under this loss function at every bank (MR-6 d.7).

Why the other options are wrong

A is incorrect. It does not test the choice of confidence level.
C is incorrect. Horizon consistency is not what it measures.
D is incorrect. A balanced trade-off describes accuracy-type (check) loss, not the regulatory one.

LO: MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. • **Reference:** Lynch, Section 6.8

Solution MR-6 d.6 — Answer: C

Bank Q341 • AnalystPrep Q.6461

Step 1 – Form the differentials

$z_t = l_t^A - l_t^B$ each day.

Step 2 – What is tested

Under the null of equal performance, positive and negative differentials are equally likely ($p = 0.5$). The test counts positive differentials: $\left[\sum I(z_t > 0) - 0.5N\right] / \sqrt{0.25N} \sim N(0,1)$.
$\Rightarrow$ It is a test of the *median* differential, using signs only.

Interpretation

Because it uses only signs, it needs no normality assumption and is easy to run.

Why the other options are wrong

A is incorrect. That is a Kupiec-type coverage backtest of one model.
B is incorrect. Clustering of exceptions is tested by independence tests (Christoffersen).
D is incorrect. That describes a t-test on the mean differential.

LO: MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. • **Reference:** Lynch, Section 6.8 (Diebold and Mariano 1995; Sarma et al. 2003)

Solution MR-6 d.7 — Answer: B

New • built from GARP MR Ch.6, 6.8

Step 1 – Regulatory loss function

"The positional VaR model significantly outperforms the P&L VaR in every case."
$\Rightarrow$ This shows the conservative nature of banks' positional VaR models, often attributed to regulatory oversight.

Step 2 – Check loss function

"The P&L VaR outperforms the positional VaR at sixteen out of nineteen banks." One bank's positional VaR won; two were insignificant.

Interpretation

Conservatism wins under a loss that penalises only exceedances; accuracy wins under the check loss (Figure 1). The same pair of facts is option C of Question MR-6 a.7.

Why the other options are wrong

A is incorrect. Reversed: under the regulatory loss the *positional* VaR won every time, which shows conservatism, not aggressiveness.
C is incorrect. Sibling swap: the 16 of 19 result belongs to the P&L VaR under the check loss.
D is incorrect. The differences were significant in most cases; and a plot only shows relative conservatism.

LO: MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. • **Reference:** Lynch, Section 6.8

MR-7 • Beyond Exceedance-Based Backtesting of Value-at-Risk Models • Vardeez: IMPORTANT

MR-7 a

P3 CONCEPT

Identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest.

Workbook: C2 Backtesting VaR: Exceedance vs PIT • Tier C • Map #21 (4/16)

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C2, "PIT vs exceedance", Map #21, 4/16) hits this LO.
Vardeez: new topic (remember backtesting is important), highly testable.

Reported exam shape

- **C2·S2** PIT-based versus exceedance-based backtesting: which property of one approach corresponds to which property of the other.

What the questions look like: statement questions. "Which property does a PIT test add?", "What does conditional coverage mean?", "Which portfolio's exceedances are acceptable?". Distractors swap Kupiec and Christoffersen, or claim that a model passing the count test is accurate everywhere.

1. The exceedance (hit) series

Backtesting compares realised outcomes with the model's forecast. The standard practice counts the days on which the daily P&L is worse than the previous day's VaR, a VaR *exceedance* (exception, breach, hit):

$$I_{t+1} = \begin{cases} 1 & \text{if } P\&L_{t+1} < -\text{VaR}_t \\ 0 & \text{otherwise} \end{cases}$$

For a correct 99% VaR, I_{t+1} is i.i.d. Bernoulli with $p = 1\%$. Regulatory VaR is one-day 99%, so a correct model is exceeded on 1% of days. **Too few** exceedances means the model *overestimates* risk (conservative, excess capital); **too many** means it *underestimates* risk (aggressive).

2. The three properties of an accurate VaR model

Property	What it asks	Test and decision
Unconditional coverage	Is the *number* of exceedances consistent with p? Timing is ignored.	Kupiec (1995) proportion of failures; $LR_{uc} \sim \chi^2(1)$, reject if > 3.84
Independence	Do exceedances *cluster*? Is an exceedance today more likely after one yesterday?	Christoffersen (1998) Markov chain, $H_0 : \pi_{01} = \pi_{11}$; $LR_{ind} \sim \chi^2(1)$, reject if > 3.84
Conditional coverage	Both at once: right rate *and* independent.	$H_0 : \pi_{01} = \pi_{11} = p$; $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$, reject if > 5.99

π_{01} is the probability of an exceedance after a day without one; π_{11} the probability of an exceedance after a day with one. Christoffersen's two properties (unconditional coverage and independence) are the core; conditional coverage is their joint test.

Counting is binomial. For x exceedances in n days, $\Pr(x) = \binom{n}{x} p^x (1-p)^{n-x}$. With $n = 100$

and $p = 5\%$, $\Pr(3) = 13.96\%$. Kupiec: for a 95% VaR over 252 days the non-rejection region is $6 < N < 20$ (expected 12.6).

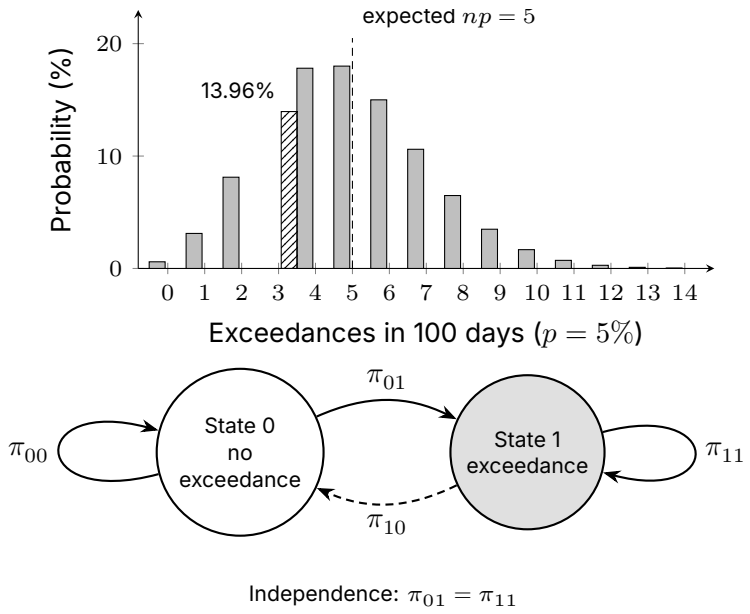


Figure 1. Left: binomial distribution of the exceedance count for a correct 95% VaR over 100 days; the hatched bar is $\Pr(3) = 13.96\%$, the dashed line the expected count. Counts of 2 to 8 are unremarkable. Right: Christoffersen's two-state Markov chain. If yesterday's state changes the chance of an exceedance today ($\pi_{11} \neq \pi_{01}$), exceedances are not independent.

3. Limits of exceedance tests

- **One percentile only.** The hit series reduces each day to 0 or 1 at a single preselected cut-off. A model can pass while the rest of its forecast distribution is wrong.
- **Low power.** At 99% and 250 days only 2.5 exceedances are expected, so a wrong model is hard to reject.
- **Kupiec ignores timing.** Three exceedances on consecutive days give the same LR_{uc} as three spread over the year.
- **Christoffersen sees one lag only.** It is a first-order Markov test: it says nothing about breaches one week or one month apart. (Workbook question C2.3, worked in the MR-4 block, tests this.)
- **Joint tests are weak when only one property fails.** The conditional coverage test is good when both properties are violated, but has limited ability to detect a violation of just one.
- **Other exceedance tests.** Ljung-Box tests m lags of autocorrelation of the hits jointly, $Q = T(T+2)\sum_{k=1}^{m} r_k^2/(T-k) \sim \chi^2(m)$; logit and linear probability regressions of hits on lagged hits, lagged VaR or lagged P&L; and the **Christoffersen–Pelletier duration test**: the time between exceedances, $\Pr(D = d) = (1-p)^{d-1}p$, is memoryless (exponential) under a correct model, tested against a Weibull alternative. Clustering produces too many short and too many long durations.

4. How the properties carry over to a PIT backtest

The PIT for day $t+1$ is the previous day's forecast CDF evaluated at the realised P&L (MR-7 b). Christoffersen's two properties map one-for-one:

	Exceedance-based backtest	PIT-based backtest
Coverage	Unconditional coverage: exceedances at the VaR level occur with probability p	**Uniformity**: PITs are $U(0,1)$, so the 1% VaR is exceeded 1% of the time, the 3% VaR 3% of the time, *at every level*
Independence	Exceedances do not cluster	PITs are **independent** over time
Joint	Conditional coverage	PITs are **i.i.d.** $U(0,1)$ (Rosenblatt, 1952)
Information used	One percentile (the VaR cut-off)	The whole forecast distribution

What the GARP study found (Iercosan et al., 597 US bank subportfolios, 2013 to 2015, one-day 99% VaR): 67% passed Kupiec, 92% passed Christoffersen's independence test and 71% passed conditional coverage, but only 21% passed Kolmogorov-Smirnov, 11% Anderson-Darling and 21% Cramér-von Mises. *Banks do better on exceedance tests than on tests of the whole distribution.*

Exam traps

- **Kupiec vs Christoffersen swap.** Kupiec counts (unconditional coverage); Christoffersen checks clustering (independence). Kupiec cannot see bunching if the count is right.
- **Direction.** Too few exceedances = conservative (overestimates risk); too many = aggressive (underestimates risk).
- **Mapping swap.** Uniformity of PITs mirrors *unconditional coverage*; independence of PITs mirrors *independence*. Not the other way round.
- **"Passes Kupiec, so accurate".** Passing a count test says nothing about the rest of the distribution or about clustering.
- **Critical values.** 3.84 for LR_{uc} and LR_{ind} (one degree of freedom); 5.99 for LR_{cc} (two).

Formulas to recognise: MR-7 a

Binomial count	$\Pr(x) = \dfrac{n!}{(n-x)!\,x!}\,p^x(1-p)^{n-x}$
Kupiec	$LR_{uc} = -2\ln[(1-p)^{n-x}p^x] + 2\ln\left[\left(1-\frac{x}{n}\right)^{n-x}\left(\frac{x}{n}\right)^{x}\right] \sim \chi^2(1)$
Christoffersen	$\pi_{01} = \dfrac{T_{01}}{T_{00}+T_{01}},\ \pi_{11} = \dfrac{T_{11}}{T_{10}+T_{11}};\ LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$
Duration	$\Pr(D=d) = (1-p)^{d-1}p$ (exponential under H_0)

Practice questions: MR-7 a

Question MR-7 a.1

Bank Q355 • Schweser Module Quiz 7.1 Q1

Which of the following characteristics must a VaR model satisfy to ensure it passes the unconditional coverage test?

A. Exceedances must occur in clusters.

B. Exceedances must be independent of prior exceedances.

C. Exceedances must follow a uniform distribution over $[0,1]$.

D. The total number of exceedances must match the expected frequency within a confidence level.

Question MR-7 a.2

Bank Q356 • Schweser Module Quiz 7.1 Q2

What does it mean when exceedance-based backtest analysis on VaR models suggests that the model has conditional coverage at a given confidence level?

A. The model performs well only on a subset of positions.

B. The model requires specific distribution assumptions to adequately deliver results.

C. The model cannot be rejected on the basis of observed exceedances, nor can it be rejected for potential lack of independence.

D. The suggested rate of exceedance incidents cannot be rejected; however, there is serial correlation among the exceedance observations.

Question MR-7 a.3

Bank Q349 • AP Mock 3 Q37

A risk management team is conducting an exceedance-based backtest on its VaR model and wants to reflect the model's key properties through a PIT-based backtest. Which attributes of the probability integral transforms are essential for validating that the exceedance-based properties are maintained?

A. Skewness and heavy tails.

B. Uniform distribution and independence.

C. Periodicity and cyclicality.

D. Concentration and clustering.

Question MR-7 a.4

Bank Q352 • AnalystPrep Q.6479 (rebuilt)

Historically, regulators have focused on backtests based on the indicator variable for a VaR exceedance. What is the main limitation of relying only on such exceedance-based tests?

A. They use only whether the loss exceeded the VaR threshold, so a model can pass while misrepresenting the rest of the forecast distribution.

B. They can only detect models that are too conservative, not models that are too aggressive.

C. They require the P&L to be normally distributed.

D. They test independence of exceedances but not their total number.

Question MR-7 a.5

Bank Q346 • AnalystPrep Q.6475 (rebuilt)

Exceedance-based backtests of a VaR model often have low statistical power when the sample is small and the VaR confidence level is high, because exceedances are rare. Which of the following is the MOST effective way to address this power problem?

A. Use overlapping windows to create more backtest observations from the same data.

B. Tighten the significance level of the test from 5% to 1%.

C. Rely on the unconditional coverage test alone, since it uses the total number of exceedances.

D. Use a PIT-based backtest, which draws on the entire forecast distribution rather than only the exceedance indicator.

Question MR-7 a.6

New • built from GARP MR Ch.7 (Iercosan et al.), 7.5

A validation team has confirmed that a desk's one-day 99% VaR model passes both Kupiec's unconditional coverage test and Christoffersen's independence test. A senior validator asks what further exceedance-based evidence could still reveal a misspecified model. Which of the following statements is correct?

A. Kupiec's test would have rejected the model if exceedances had bunched together, so clustering can be ruled out.

B. The Christoffersen–Pelletier duration test models the time between exceedances, which is memoryless under a correct model, so it can flag clustering beyond a one-day lag.

C. The Ljung-Box test examines a single lag of the hit series, so it adds nothing beyond Christoffersen's independence test.

D. The joint conditional coverage test is more powerful than either individual test when only one of the two properties is violated.

Question MR-7 a.7

Bank Q354 • BT P2.T5.25.3.1 (rebuilt)

A risk manager backtests the 99% one-day VaR of two portfolios over 750 trading days. Both portfolios have 8 exceedances (expected: 7.5). Critical values at 95% confidence are 3.84 for LR_{uc} and LR_{ind} and 5.99 for LR_{cc}.

Portfolio	Exceedances	Pattern of exceedances	Pattern of tail PITs
A	8	All isolated; never on consecutive days	Evenly scattered, no serial pattern
B	8	Four pairs of back-to-back days	Tail PITs appear in consecutive-day pairs

Day-to-day transition counts: A has $n_{00} = 733$, $n_{01} = 8$, $n_{10} = 8$, $n_{11} = 0$; B has $n_{00} = 737$, $n_{01} = 4$, $n_{10} = 4$, $n_{11} = 4$. Which portfolio's backtest indicates an accurate VaR model?

A. Both A and B, since both have the same number of exceedances.

B. Only B.

C. Only A.

D. Neither A nor B.

Question MR-7 a.8 Workbook C2.2 • S2 • Map #21

A model validation analyst at a US bank holding company is presenting to newly hired analysts on how the bank's subportfolio VaR models are validated. The bank stores, for each day, the one-day 99% VaR, the clean P&L and the corresponding probability integral transform (PIT). Which of the following statements is correct for the analyst to make?

A. A hump-shaped PIT histogram, with realised P&L clustering in the middle of the forecast distribution, signals aggressive misspecification because the model distribution is too narrow.

B. Uniformity of the PITs on the interval from 0 to 1 is the counterpart of unconditional coverage, so a PIT test uses information from every percentile of the forecast distribution.

C. The Kolmogorov-Smirnov test is the preferred uniformity test for a 99% VaR model, because it places more weight on the tails of the PIT distribution than the Anderson-Darling test.

D. Because PITs use the whole forecast distribution, a subportfolio that passes Kupiec's unconditional coverage test will also pass a Kolmogorov-Smirnov test of PIT uniformity.

Solutions • MR-7 a

Solution MR-7 a.1 — Answer: D

Bank Q355 • Schweser Module Quiz 7.1 Q1

Step 1 – What unconditional coverage tests

Kupiec's proportion of failures test looks only at the *total* number of exceedances x in n days. Under H_0, x is binomial with probability p.

$\Rightarrow$ When and in what pattern the exceedances happen is ignored.

Step 2 – The decision rule

$LR_{uc} \sim \chi^2(1)$: reject if $LR_{uc} > 3.84$. For a 95% VaR over 252 days, any count with $6 < N < 20$ passes.

$\Rightarrow$ The model passes when the count is statistically consistent with the expected frequency.

Interpretation

Passing unconditional coverage is a statement about the *rate* of exceedances only. Clustering is checked separately, by the independence test.

Why the other options are wrong

A is incorrect. Clustering is a failure of the independence property; it is never a requirement.

B is incorrect. Independence is the second, separate property (Christoffersen), needed for conditional coverage, not for unconditional coverage.

C is incorrect. Uniformity over [0,1] is a property of PITs, the PIT counterpart of unconditional coverage; exceedances are 0/1 indicators.

LO: MR-7 a: Identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest. • **Reference:** Iercosan et al., GARP MR Ch.7, 7.5; SchweserNotes 2026 Book 1, Reading 7, LO 7.a

Solution MR-7 a.2 — Answer: C

Bank Q356 • Schweser Module Quiz 7.1 Q2

Step 1 – The joint hypothesis

Conditional coverage tests $H_0 : \pi_{01} = \pi_{11} = p$: exceedances occur at the right rate *and* do not depend on yesterday.

Step 2 – The statistic

$LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$; reject if $LR_{cc} > 5.99$.

$\Rightarrow$ "Having conditional coverage" means neither the exceedance rate nor the independence of exceedances is rejected.

Interpretation

Conditional coverage is the combination of Christoffersen's two properties. A model with the right count but clustered exceedances does not have it.

Why the other options are wrong

A is incorrect. The test is run on the exceedance series of the whole portfolio; it says nothing about subsets of positions.

B is incorrect. The test uses binomial and Markov-chain likelihoods for the hit series; it needs no assumption about the P&L distribution.

D is incorrect. Serial correlation among exceedances means independence is rejected, so conditional coverage fails. Only unconditional coverage would hold.

LO: MR-7 a • **Reference:** Iercosan et al., GARP MR Ch.7, 7.5, eq. 7.11; SchweserNotes 2026 Book 1, Reading 7, LO 7.a

Solution MR-7 a.3 Answer: B

Bank Q349 • AP Mock 3 Q37

Step 1 – Map unconditional coverage

Unconditional coverage "is analogous to the uniformity distributional property of a series of PITs": the 1% VaR is exceeded 1% of the time, the 3% VaR 3% of the time, and so on, so the PITs are $U(0,1)$.

Step 2 – Map independence

Independence of exceedances maps to independence of the PITs over time.

$\Rightarrow$ A perfectly accurate model produces PITs that are i.i.d. $U(0,1)$ (Rosenblatt).

Interpretation

Uniformity plus independence is the PIT version of conditional coverage, checked at every percentile instead of one.

Why the other options are wrong

A is incorrect. Skewness and heavy tails describe returns. A heavy-tailed PIT histogram is a symptom of a wrong model.

C is incorrect. Periodicity and cyclicality are forms of serial dependence, the opposite of independence.

D is incorrect. Concentration (non-uniformity) and clustering (dependence) are the failure symptoms, not the required properties.

LO: MR-7 a • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1

Solution MR-7 a.4 Answer: A

Bank Q352 • AnalystPrep Q.6479

Step 1 – What an exceedance test uses

Each day becomes a 0/1 indicator at one preselected cut-off (the VaR level).

Step 2 – What it therefore misses

Kupiec, Christoffersen and conditional coverage all use only that indicator, so they test the model at a single percentile.

$\Rightarrow$ A model can pass every exceedance test while the rest of its forecast distribution is badly wrong. PIT tests address exactly this.

Interpretation

This is the motivation for the whole reading: PITs "provide information on the accuracy of the risk model at any percentile of the forecast distribution".

Why the other options are wrong

B is incorrect. Unconditional coverage rejects too many exceedances (aggressive models) as readily as too few.

C is incorrect. Exceedance tests rely on binomial and Markov-chain likelihoods of the hit series; no normality assumption is needed.

D is incorrect. They test both: Kupiec tests the count and Christoffersen tests clustering.

LO: MR-7 a • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1 and 7.6

Solution MR-7 a.5 — Answer: D

Bank Q346 • AnalystPrep Q.6475

Step 1 – Why power is low

With a 99% VaR over about 250 days, only 2.5 exceedances are expected. The 0/1 indicator throws away how far inside or outside the VaR each P&L fell.

Step 2 – What adds information

A PIT converts every day's P&L into its position in that day's full forecast distribution, so every observation carries information.

$\Rightarrow$ Using the whole distribution, not one percentile, raises power.

Interpretation

The remedy for low power is more information per observation, which is what a PIT backtest delivers.

Why the other options are wrong

A is incorrect. Overlapping windows create serially dependent observations and add no independent information.

B is incorrect. A stricter significance level lowers the Type I error rate but raises the Type II error rate, so power falls.

C is incorrect. Unconditional coverage uses only the count, which is the source of the low power.

LO: MR-7 a • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1 and 7.6

Solution MR-7 a.6 — Answer: B

New • built from GARP MR Ch.7, 7.5

Step 1 – What the two passed tests cannot see

Kupiec ignores timing. Christoffersen uses a first-order Markov chain, so it "does not allow for the possibility that independence may be violated at a distance greater than one lag".

Step 2 – Tests that look further

The Christoffersen–Pelletier duration test models $D_i = t_i - t_{i-1}$. Under a correct model $\Pr(D = d) = (1-p)^{d-1}p$, an exponential (memoryless) distribution; it is tested against a Weibull alternative. Clustering produces too many short and too many long durations, whatever the lag.

$\Rightarrow$ Statement B is correct.

Interpretation

GARP stresses that passing Kupiec and Christoffersen is not the end of the story: the duration test, Ljung-Box and logit regressions of hits on lagged information can all reveal a model that does not respond to market changes.

Why the other options are wrong

A is incorrect. Sibling swap: Kupiec counts exceedances only; bunched exceedances give the same LR_{uc} as spread-out ones.

C is incorrect. Ljung-Box tests m lags jointly, $Q = T(T+2)\sum_{k=1}^{m} r_k^2/(T-k) \sim \chi^2(m)$; that is precisely what it adds beyond one lag.

D is incorrect. GARP notes the opposite: joint tests "have significant limitations in their ability to detect misspecifications when only one of the two properties is violated".

LO: MR-7 a • **Reference:** Iercosan et al., GARP MR Ch.7, 7.5, eqs. 7.12 to 7.21

Solution MR-7 a.7 **Answer: C**

Bank Q354 • BT P2.T5.25.3.1

Step 1 – Coverage is the same for both

8 exceedances against 7.5 expected: $LR_{uc} = 0.033 < 3.84$ for both portfolios.
$\Rightarrow$ The count cannot separate A from B.

Step 2 – Independence separates them

A: $\pi_{01} = 8/741 = 1.08\%$, $\pi_{11} = 0/8 = 0$, $LR_{ind} = 0.17 < 3.84$.
B: $\pi_{01} = 4/741 = 0.54\%$, $\pi_{11} = 4/8 = 50\%$, $LR_{ind} = 27.70 > 3.84$.
$\Rightarrow$ In B, an exceedance after an exceedance is almost 100 times as likely as after a quiet day: clustering.

Step 3 – Conditional coverage

A: $0.033 + 0.17 = 0.21 < 5.99$ (pass). B: $0.033 + 27.70 = 27.73 > 5.99$ (reject).

Interpretation

The pattern alone gives the answer: B's exceedances come in back-to-back pairs, and its tail PITs show the same pairing, so B fails independence. Only A shows both properties. The numbers confirm what the pattern says.

Why the other options are wrong

A is incorrect. Looks only at the count (unconditional coverage) and ignores clustering.
B is incorrect. Reverses the conclusion: B is the portfolio with clustered exceedances.
D is incorrect. Coverage passes for both, and A also passes independence.

LO: MR-7 a • **Reference:** Iercosan et al., GARP MR Ch.7, 7.5; SchweserNotes 2026 Book 1, Reading 7, LO 7.a

Solution MR-7 a.8 **Answer: B**

Workbook C2.2 • S2

Step 1 – What the PIT is

$p_{t+1} = F_t(P\&L_{t+1})$: the previous day's forecast distribution evaluated at the realised P&L. A correctly specified model produces i.i.d. $U(0,1)$ PITs.

Step 2 – Map the two exceedance properties

Unconditional coverage (exceedances at 1% VaR observed 1% of the time, at 3% VaR 3% of the time, and so on) "is analogous to the uniformity distributional property of a series of PITs"; independence of exceedances maps to independence of the PITs.

Interpretation

B is correct. An exceedance count tests the model at a single percentile; PITs, when well estimated, "provide information on the accuracy of the risk model at any percentile of the forecast distribution". That is why PIT tests can reveal misspecifications that pure exceedance analysis misses.

Why the other options are wrong

A is incorrect. Polarity flip. P&L clustering in the middle is conservative misspecification: the model distribution is too wide. Aggressive misspecification (too narrow) shows realised P&L in the tails.
C is incorrect. Sibling swap. The KS test lacks sensitivity at the extremes; Anderson-Darling is the modification that puts more weight on the tails and so better detects tail misspecification of a 99% VaR model.
D is incorrect. Absolute claim. In the study 67% of subportfolios passed Kupiec but only 21% passed KS: banks perform better on exceedance tests than on tests of the distribution.

LO: MR-7 a, b, c, d • **Reference:** Iercosan, Shcherbakova, McArthur and Alper, GARP MR Ch.7, 7.1, 7.2, 7.6, 7.7

MR-7 • Beyond Exceedance-Based Backtesting of Value-at-Risk Models • Vardeez: IMPORTANT

MR-7 b

P3 CONCEPT

Explain how to derive probability integral transforms (PITs) in the context of validating a VaR model.

Workbook: No dedicated workbook shape • see C2 (Tier C, Map #21)

WHAT IS TESTED HERE

Why P3. Vardeez flags the chapter (new topic, highly testable) and the LO sits in the C2 workbook section; no shape is reported for this LO on its own.
Vardeez: the four-step derivation with the worked observation ($N(0, \$500)$, actual $-\$250$, PIT 0.308).

What the questions look like: the order of the steps, why each day uses its own forecast, what a PIT is (and is not), and at most a one-line reading of a normal CDF.

1. Why a transform is needed

The forecast distribution of P&L changes every day, because volatility moves and positions change. Raw P&L from different days therefore comes from different distributions and cannot be pooled. The PIT puts every day on the same footing: it records *where* the realised outcome fell inside that day's own forecast.

2. The four steps

1. **Forecast** the P&L distribution for period T. Example: $N(\$0,\ \$500)$.
2. **Observe** the realised P&L for period T. Example: $-\$250$.
3. **Transform**: read the observation off the forecast CDF. $-\$250$ is -0.5 standard deviations, and $N(-0.5) = 0.3085$, so the PIT is about **0.309** (Schweser and Vardeez print 0.308).
4. **Analyse** the series of PITs for uniformity, independence and conditional coverage, just as for exceedances (MR-7 a, c, d).

$$\boxed{p_{t+1} = F_t(P\&L_{t+1})}$$ the previous day's forecast CDF at the realised P&L

GARP describes the PIT as the model's probability of observing a loss below the actual P&L, and notes it is often called the *p-value*. **Low PITs are losses** deep in the left tail; high PITs are gains in the right tail. The approach places no requirement on the shape of the forecast distribution: any model that delivers a CDF will do (Schweser's CDF example: a P&L of $+0.005$ sits at the 69.1% mark).

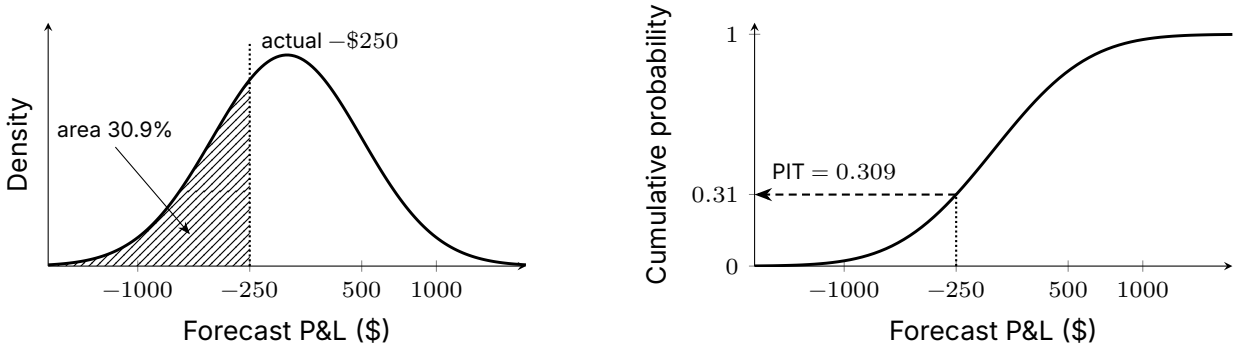


Figure 2. Deriving one PIT. Left: the day's forecast density $N(0, \$500)$; the hatched area to the left of the realised $-\$250$ (dotted line) is 30.9%. Right: the same forecast as a CDF; the PIT is the height of the CDF at the realised P&L (dotted up, dashed across), 0.309.

3. A series of PITs with changing forecasts

With zero-mean normal forecasts whose standard deviation changes daily, each PIT is $N(\text{P\&L}/\sigma_t)$:

Day	Forecast s.d.	Actual P&L	z	PIT	Standard normal equivalent
1	2.00%	4.24%	2.12	0.983	2.12
2	2.25%	0.95%	0.42	0.664	0.42
3	1.65%	−0.70%	−0.42	0.336	−0.42
4	1.00%	0.40%	0.40	0.655	0.40
5	3.25%	−5.48%	−1.69	0.046	−1.69
6	2.50%	−0.65%	−0.26	0.397	−0.26
7	1.85%	−3.03%	−1.64	0.051	−1.64

Day 5's loss of 5.48% and Day 7's loss of 3.03% land at almost the same PIT, because Day 5's forecast was much wider. The last column (Berkowitz's transform, MR-7 d) is used to test independence.

Correction to a source key. Schweser's Module Quiz 7.2 Q1 ("PIT values: ...") prints the answer as C ("analyze exceedance independence by forming a Markov chain"), but its own explanation describes D ("convert observed data points into a uniform distribution for the entire forecasted distribution"). D is correct; the question bank has already corrected the key. Worked as Question MR-7 b.6.

Exam traps

- **PIT is not "point in time".** It is the probability integral transform.
- **PIT is not the z-score.** The z of -0.5 becomes a PIT of 0.309 only after passing it through the CDF. The density value 0.352 is a third, wrong number.
- **Sign.** A loss maps to a PIT below 0.5; $N(+0.5) = 0.691$ is the reading for a gain of \$250.
- **Timing.** The PIT uses the forecast made *before* the outcome (day t's model for day $t+1$).
- **Purpose.** Not counting exceedances, not a Markov chain: it maps every outcome to $[0, 1]$ so the whole distribution can be tested.

Practice questions: MR-7 b

Question MR-7 b.1

New • built from GARP MR Ch.7, 7.1 and 7.2

A junior analyst is drafting the definition of a probability integral transform (PIT) for the bank's VaR validation manual. Which of the following definitions is correct?

A. The PIT for day $t+1$ is the realised P&L for day $t+1$ divided by the standard deviation forecast for that day.

B. The PIT for day $t+1$ is the previous day's forecast CDF evaluated at the realised P&L of day $t+1$, i.e. the model's probability of a P&L at or below the one realised.

C. The PIT is a point-in-time version of the exceedance indicator, equal to 1 when the loss exceeds VaR and 0 otherwise.

D. The PIT for day $t+1$ is the forecast CDF re-estimated after observing day $t+1$'s P&L, evaluated at the VaR threshold.

Question MR-7 b.2

Bank Q357 • AnalystPrep Q.6488 (rebuilt)

In the procedure for deriving a probability integral transform (PIT) for a single day, which step comes first?

A. Locate the realised P&L on the cumulative distribution function to obtain a value in $[0,1]$.

B. Obtain the forecast distribution of P&L for the period (e.g. normal with mean 0 and standard deviation 500).

C. Test the set of PIT values for uniformity and independence.

D. Count the days on which the realised loss exceeded the VaR.

Question MR-7 b.3

Bank Q358 • AnalystPrep Q.6489 (rebuilt)

Forecast distributions of daily P&L change from day to day because volatility and positions change. Why does the PIT method pass each day's realised P&L through that day's own forecast CDF?

A. To remove outlier days before computing the VaR.

B. To re-estimate the model parameters from historical data.

C. To count how many days exceeded the VaR threshold.

D. To put observations drawn from different forecast distributions on one common $[0,1]$ scale so they can be tested together.

Question MR-7 b.4

Bank Q359 • AnalystPrep Q.6490 (rebuilt)

A model forecasts tomorrow's P&L as normal with mean \$0 and standard deviation \$500. The realised P&L is $-\$250$. What is the PIT value for this observation?

A. 0.692

B. -0.500

C. 0.352

D. 0.309

Question MR-7 b.5 New • built from GARP MR Ch.7, 7.2 and Vardeez's PIT table

A validation analyst reviews two recent days for a desk whose VaR model forecasts normally distributed daily returns with a mean of zero:

Day	Forecast s.d.	Realised return	z	PIT
14	3.25%	-3.00%	-0.92	0.178
15	1.00%	-2.00%	-2.00	0.023

Which of the following statements is correct?

A. Day 14 is the more extreme outcome for the model, because its loss of 3.00% is the larger of the two.

B. Day 15 is the more extreme outcome relative to its own forecast; the PIT puts days with different forecast distributions on one comparable $[0,1]$ scale.

C. Neither PIT is informative, because PITs can only be compared when the forecast standard deviation is the same every day.

D. A PIT close to zero indicates a large gain relative to the forecast, so Day 15 was an unusually good day for the desk.

Question MR-7 b.6 Bank Q361 • Schweser Module Quiz 7.2 Q1 (key corrected)

Which of the following statements correctly describes the purpose of probability integral transform (PIT) values for backtesting VaR models? PIT values:

A. are used to determine the total number of exceedances for a VaR model.

B. measure the conditional coverage property of exceedance-based models.

C. analyze exceedance independence by forming a Markov chain of observations.

D. convert observed data points into a uniform distribution for the entire forecasted distribution.

Solutions • MR-7 b

Solution MR-7 b.1 — Answer: B

New • built from GARP MR Ch.7, 7.1 and 7.2

Step 1 – The definition

$p_{t+1} = F_t(P\&L_{t+1})$: the forecast distribution made on day t, evaluated at day $t+1$'s realised P&L. GARP: the model's probability of observing a loss below the actual P&L, "often called the p-value".

$\Rightarrow$ It is a cumulative probability, so it always lies in $[0,1]$.

Interpretation

Two details carry the definition: the CDF (not the density, not the z-score) and the forecast made *before* the outcome.

Why the other options are wrong

A is incorrect. That is the z-score (for a zero-mean forecast). It still has to go through the CDF to become a PIT.

C is incorrect. PIT is not "point in time", and the 0/1 indicator is the exceedance (hit) series of the exceedance-based approach.

D is incorrect. A forecast re-estimated after the outcome is no longer a forecast, and the CDF is evaluated at the realised P&L, not at the VaR threshold.

LO: MR-7 b: Explain how to derive probability integral transforms (PITs) in the context of validating a VaR model. • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1 and 7.2, eq. 7.1

Solution MR-7 b.2 — Answer: B

Bank Q357 • AnalystPrep Q.6488

Step 1 – The order

(1) obtain the forecast distribution for the period; (2) collect the realised P&L; (3) locate it on the forecast CDF to get the PIT; (4) analyse the PITs for uniformity, independence and conditional coverage.

$\Rightarrow$ The forecast must exist before an observation can be located in it.

Interpretation

A PIT is a position *within* a forecast, so the forecast always comes first.

Why the other options are wrong

A is incorrect. This is Step 3; it needs both the forecast and the realisation.

C is incorrect. This is Step 4, the analysis of the collected PITs.

D is incorrect. Counting exceedances belongs to the exceedance-based approach, not to the PIT method.

LO: MR-7 b • **Reference:** SchweserNotes 2026 Book 1, Reading 7, LO 7.b; Iercosan et al., GARP MR Ch.7, 7.2

Solution MR-7 b.3 — Answer: D

Bank Q358 • AnalystPrep Q.6489

Step 1 – The problem

Each day's P&L is drawn from a different forecast distribution, so raw P&L cannot be pooled.

Step 2 – The fix

$F_t(P\&L_{t+1})$ expresses each outcome as a cumulative probability under its own forecast. If every

forecast is right, all PITs share the same $U(0,1)$ distribution.
$\Rightarrow$ The PITs can then be tested together as one sample.

Interpretation

This is the whole point of the transform: it standardises each observation relative to its own forecast.

Why the other options are wrong

A is incorrect. The PIT removes nothing; extreme outcomes are kept because they reveal tail misforecasts.
B is incorrect. The PIT is a validation tool applied to forecasts; it does not re-estimate the model.
C is incorrect. Counting exceedances is the exceedance-based approach, which uses only one cut-off.

LO: MR-7 b • **Reference:** SchweserNotes 2026 Book 1, Reading 7, LO 7.b; Iercosan et al., GARP MR Ch.7, 7.2

Solution MR-7 b.4 — Answer: D

Bank Q359 • AnalystPrep Q.6490

Step 1 – Standardise

$$z = \frac{-250 - 0}{500} = -0.5$$

Step 2 – Read the forecast CDF

$$\text{PIT} = N(-0.5) = 0.3085 \approx 0.309$$

$\Rightarrow$ About 30.9% of the forecast distribution lies at or below the realised loss.

Interpretation

This is the Schweser and Vardeez worked observation (printed there as 0.308). The PIT is the CDF value, not the z-score and not the density.

Why the other options are wrong

A is incorrect. $N(+0.5) = 0.692$: ignored the sign of the loss and read the gain side.
B is incorrect. -0.5 is the z-score; the CDF has not been applied.
C is incorrect. 0.352 is the standard normal *density* at 0.5, confusing the PDF with the CDF.

LO: MR-7 b • **Reference:** SchweserNotes 2026 Book 1, Reading 7, LO 7.b, Step 3

Solution MR-7 b.5 — Answer: B

New • built from GARP MR Ch.7, 7.2

Step 1 – Compare each loss with its own forecast

Day 14: $-3.00/3.25 = -0.92$, PIT $= N(-0.92) = 0.178$. Day 15: $-2.00/1.00 = -2.00$, PIT $= N(-2.00) = 0.023$.
$\Rightarrow$ Day 15's smaller loss sits much deeper in the left tail of its forecast.

Step 2 – Why that is the right comparison

The forecasts differ (3.25% vs 1.00%), so raw returns are not comparable. The PIT maps each outcome to [0,1] under its own forecast.

Interpretation

A PIT of 0.023 means the model gave only a 2.3% chance to a day at least that bad; Day 14's loss was a roughly one-in-six event for its wider forecast.

Why the other options are wrong

A is incorrect. Compares raw losses across days with different forecasts, exactly what the PIT is designed to avoid.

C is incorrect. Reverses the purpose: the PIT exists *because* forecasts change day to day.

D is incorrect. Polarity flip: low PITs are losses in the left tail; a large gain would give a PIT near 1.

LO: MR-7 b • **Reference:** Iercosan et al., GARP MR Ch.7, 7.2; SchweserNotes 2026 Book 1, Reading 7, LO 7.b

Solution MR-7 b.6 — Answer: D

Bank Q361 • Schweser Module Quiz 7.2 Q1

Step 1 – What PITs do

Each realised P&L is located on its forecast CDF, giving a value in $[0,1]$. Under a correct model these values are $U(0,1)$ across the *entire* distribution, not only at the VaR tail.

$\Rightarrow$ The PIT turns every observation into a uniform draw, so the whole forecast distribution can be backtested.

Interpretation

Schweser prints C as the key, but its own explanation is D. D is correct (see the correction box).

Why the other options are wrong

A is incorrect. Counting exceedances is the exceedance-based method.

B is incorrect. Conditional coverage is a property of exceedance tests; in PIT terms it becomes i.i.d. uniformity, which PITs are *tested* for, not a measurement of it.

C is incorrect. A Markov-chain independence test is Christoffersen's exceedance-based technique.

LO: MR-7 b • **Reference:** SchweserNotes 2026 Book 1, Reading 7, Module Quiz 7.2 Q1 (key corrected)

MR-7 • Beyond Exceedance-Based Backtesting of Value-at-Risk Models • Vardeez: IMPORTANT

MR-7 c

P3 CONCEPT

Describe how the shape of the distribution of PITs can be used as an indicator of the quality of a VaR model.

Workbook: No dedicated workbook shape • see C2 trap "PIT histogram polarity"

WHAT IS TESTED HERE

Why P3. Vardeez flags the chapter (new topic, highly testable); the C2 workbook lists "PIT histogram polarity" as an exam trap.
Vardeez: perfect forecasting gives 10% in every decile; a forecast that is too narrow piles PITs up at both ends; too wide bunches them in the middle.

What the questions look like: a PIT histogram or decile counts are described; you name the shape and its diagnosis (conservative, aggressive, location bias, or fine but dependent).

1. The benchmark: flat

If the model forecasts the next period's P&L distribution correctly, a large set of PITs is **uniform on [0,1] and independent**: 10% of PITs in each decile. This generalises the exceedance count: a 95% VaR wants 5% of PITs below 0.05, but a PIT test asks the same question at *every* level. A uniform distribution has mean 0.5, standard deviation 0.289, skewness 0 and kurtosis 1.8.

2. Reading the shape

PIT histogram	What happened	Diagnosis
Flat	Outcomes land at every percentile equally often	Correct (check independence too)
Hump in the middle	Realised P&L clusters near the centre of the forecast; tails rarely hit	Forecast **too wide**: **conservative**, overstates risk
U-shape (spikes at 0 and 1)	Realised P&L lands in both tails far more often than forecast	Forecast **too narrow**: **aggressive**, understates risk
Pile-up near 0 only	Large losses happen more often than forecast	Left tail too thin: losses underestimated
Steady slope	Outcomes fall systematically on one side	**Location bias**: forecast mean too low or too high
Flat but serially correlated	Uniform overall, but high PITs follow high PITs	Fails **independence**: model too slow to adapt

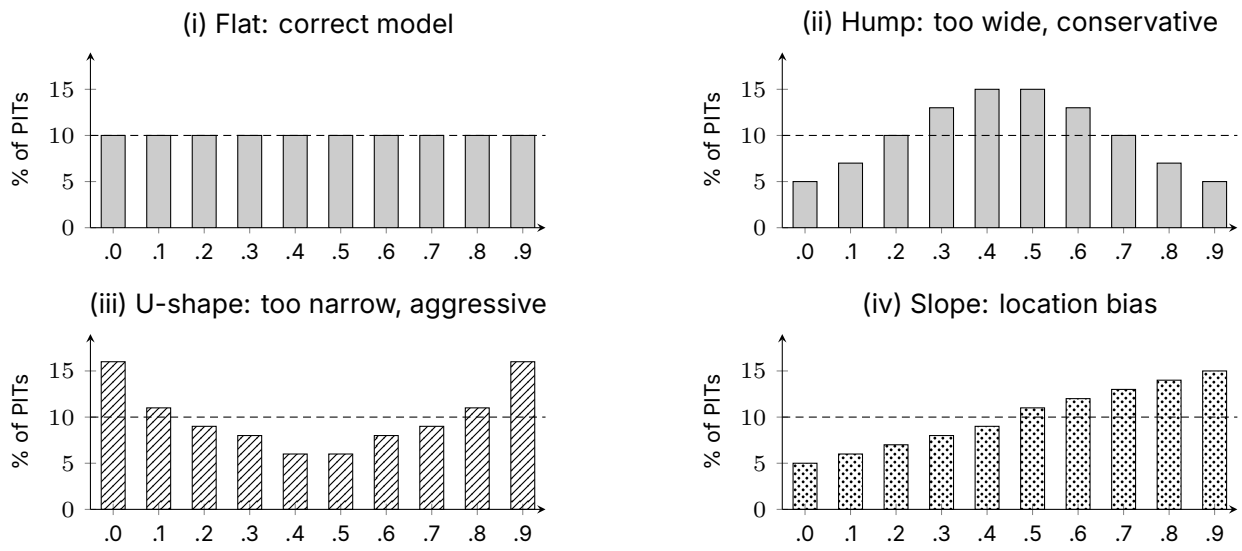


Figure 3. Four PIT histograms by decile (illustrative). The dashed line is the 10% that every decile holds under a correct model. (ii) A forecast that is too wide puts outcomes in the middle. (iii) A forecast that is too narrow sends outcomes into both tails, so PITs pile up near 0 and 1. (iv) More mass towards 1 means outcomes were systematically better than forecast: the forecast is centred too low.

3. QQ plots and what GARP's banks looked like

A QQ plot of PITs graphs the share of PITs at or below each value u against u. For uniform PITs that share equals u, so the plot is the **straight line from (0,0) to (1,1)**; departures show where the forecast is wrong. In the GARP study (Iercosan et al.):

- Pooled subportfolio PITs were **humped in the body** (models conservative in the middle) but with **visible spikes at both ends**: tails *thinner* than realised P&L required.
- In the 5% left tail the estimate at 1% was conservative, but extreme realisations were understated. Firm-wide (top of the house) models deviated less.
- In terms of moments, VaR models generally **overstate variance and understate kurtosis**.

Exam traps

- **Polarity.** Hump = too *wide* = conservative. U-shape = too *narrow* = aggressive. The most common swap on the exam.
- **Slope is not width.** A one-sided slope is a location (mean) problem, not a tails problem.
- **Flat is not enough.** Uniformity is necessary but not sufficient; serially correlated PITs fail independence.
- **Normal moments.** Skewness 0 and kurtosis 3 describe a normal, not a uniform (kurtosis 1.8).
- **QQ line.** The ideal is the 45-degree line from (0,0) to (1,1), not a horizontal line or a flat scatter.

Practice questions: MR-7 c

Question MR-7 c.1

Bank Q362 • AP Mock 1 Q6 (rebuilt)

A risk manager computes a probability integral transform (PIT) for each day's realised P&L using that day's forecast distribution. If the VaR model's forecast distributions are correct, which description of the resulting PIT values is correct?

A. They are skewed toward 0 and 1, reflecting the fat tails of P&L.

B. They cluster around 0.5, reflecting the central tendency of P&L.

C. They are uniformly distributed over $[0,1]$ and independent over time.

D. They are normally distributed with mean 0 and standard deviation 1.

Question MR-7 c.2

Bank Q369 • Schweser Module Quiz 7.2 Q2

Ideally, when backtesting VaR models, a QQ plot of probability integral transforms (PITs) should be close to a(n):

A. horizontal line.

B. inverted parabola.

C. straight line from $[0,0]$ to $[1,1]$.

D. uniformly distributed scatterplot.

Question MR-7 c.3

Bank Q366 • AnalystPrep Q.6494

In the evaluation of a VaR model, an analyst detects clustering in the PIT distribution concentrated around the middle of the $[0,1]$ interval, with few PITs near 0 or 1. What does this suggest about the model's risk assumptions?

A. The model is conservative: forecast distributions are too wide, overstating risk.

B. The model is aggressive: forecast distributions are too narrow, underestimating risk.

C. Middle clustering indicates a perfectly balanced risk model.

D. Middle clustering indicates randomness, so the model is precise in normal states.

Question MR-7 c.4

Bank Q365 • AnalystPrep Q.6492 (rebuilt)

A risk manager computes 1,000 PITs for a VaR model's daily P&L forecasts. Expected frequency is 100 per decile. The observed counts are 190 in the 0.0–0.1 decile, 180 in the 0.9–1.0 decile, and a combined 630 across the eight middle deciles (about 79 each). What does this indicate, and what is the appropriate adjustment?

A. The forecast distributions are too wide and risk is overstated; reduce the volatility estimate.

B. The forecast distributions are too narrow and risk is understated in both tails; widen the forecast distribution (e.g., higher volatility or fatter tails).

C. The forecasts are biased toward gains; shift the forecast mean P&L downward.

D. The model is well calibrated, since the middle deciles are close to the expected 100 and extra capital is not needed.

Question MR-7 c.5

Bank Q368 • BT P2.T5.25.3.3 (rebuilt)

The risk team at Horizon Capital reviews its VaR model after a period of unexpected volatility. It computes 200 PITs, where a PIT is the forecast CDF evaluated at the realised P&L (so losses map to low values). The decile counts are:

PIT bin	.0–.1	.1–.2	.2–.3	.3–.4	.4–.5	.5–.6	.6–.7	.7–.8	.8–.9	.9–1.0
Count	62	28	22	18	17	15	12	10	9	7

Which observation best indicates that the model is poorly calibrated?

A. The PIT distribution is uniform on [0,1], with no clustering in the tails or centre.

B. The PIT distribution has a concentration of values near 0 (31% of PITs in the lowest decile versus 10% expected), indicating frequent underestimation of large losses.

C. The PIT distribution shows slight deviations around the centre but aligns well with a uniform distribution in the tails.

D. The PIT distribution has skewness near 0 and kurtosis near 3, indicating balanced calibration.

Question MR-7 c.6

Bank Q367 • AnalystPrep Q.6496 (rebuilt)

An analyst computes the PITs of a VaR model's P&L forecasts over a long sample. The PIT histogram is skewed: frequencies rise steadily from the 0.0–0.1 decile (about 5% of PITs) to the 0.9–1.0 decile (about 15% of PITs). What does this indicate?

A. The forecast distributions are too narrow, so risk is understated in both tails.

B. The forecast distributions are too wide, so risk is overstated.

C. The forecast distributions are biased in location: realised P&L is systematically higher than forecast, so the forecast is too pessimistic and the location (mean) should be adjusted.

D. The PITs are serially dependent, so exceedances cluster in time.

Question MR-7 c.7

Bank Q364 • AnalystPrep Q.6487 (rebuilt)

A fund backtests its VaR model with PITs. A histogram of 1,000 PITs is essentially flat, with about 100 observations in each decile, and a Kolmogorov-Smirnov test does not reject uniformity. However, the PITs show strong positive serial correlation: runs of high PITs tend to be followed by high PITs, and runs of low PITs by low PITs. Which conclusion is most appropriate?

A. The model is well calibrated, because uniformity of the PITs is the only requirement.

B. The model fails the independence requirement; forecasts are not adapting to changing conditions such as volatility clustering, so the dynamics of the model should be revised.

C. The forecast distributions are too narrow, so the model underestimates risk in the tails.

D. The forecast distributions are biased in location, so the mean P&L forecast should be shifted.

Question MR-7 c.8 New • built from GARP MR Ch.7, 7.3 and 7.4

Iercosan et al. pooled the PITs of hundreds of US bank subportfolios, each with a one-day 99% regulatory VaR model, into a single histogram. The histogram was humped across the middle deciles but also showed visible spikes in the 0.0–0.1 and 0.9–1.0 bins. Which interpretation is most consistent with this shape?

A. The models are uniformly conservative: the forecast distributions are too wide everywhere, including the tails.

B. The models are conservative in the body of the distribution, but their tails are thinner than realised P&L requires, so extreme outcomes are understated.

C. The models are aggressive in the body but conservative in the tails, so the 99% VaR overstates extreme losses.

D. The shape reflects serial dependence of the PITs rather than any miscalibration of the forecast distribution.

Solutions • MR-7 c

Solution MR-7 c.1 — Answer: C

Bank Q362 • AP Mock 1 Q6

Step 1 – A variable through its own CDF

Each PIT is $u_t = F_t(x_t)$. If F_t is the true distribution of x_t, then u_t is $U(0,1)$: any continuous variable passed through its own CDF is uniform.

Step 2 – Over time

Correct forecasts also leave no information in yesterday's PIT about today's, so the PITs are independent.

$\Rightarrow$ A correct model gives a flat histogram (10% per decile) of independent PITs.

Interpretation

Uniform and independent is the PIT version of conditional coverage.

Why the other options are wrong

A is incorrect. Mass near 0 and 1 is the U-shape of a forecast that is too narrow, a flawed model.
B is incorrect. Mass around 0.5 is the hump of a forecast that is too wide, also a flawed model.
D is incorrect. Confuses the PIT with the z-score; standard normal values appear only after Berkowitz's transform.

LO: MR-7 c: Describe how the shape of the distribution of PITs can be used as an indicator of the quality of a VaR model. • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1; SchweserNotes 2026 Book 1, Reading 7, LO 7.c

Solution MR-7 c.2 — Answer: C

Bank Q369 • Schweser Module Quiz 7.2 Q2

Step 1 – What the QQ plot graphs

The share of PITs at or below each value $u \in [0,1]$, against u.

Step 2 – Under uniformity

For $U(0,1)$ PITs that share equals u, so the plot is the 45-degree line from (0,0) to (1,1).

$\Rightarrow$ The closer the PITs are to that line, the more confidence that the model forecasts the whole distribution well.

Interpretation

Deviations from the line show which part of the forecast distribution is mis-specified.

Why the other options are wrong

A is incorrect. A horizontal line would mean almost all PITs sit at one value, an extreme failure.
B is incorrect. A curved shape signals over- or under-populated regions, i.e. non-uniform PITs.
D is incorrect. "Uniformly distributed" describes the PIT histogram, not the shape of a QQ plot.

LO: MR-7 c • **Reference:** SchweserNotes 2026 Book 1, Reading 7, LO 7.c, Figure 7.6

Solution MR-7 c.3 — Answer: A

Bank Q366 • AnalystPrep Q.6494

Step 1 – What a PIT near 0.5 means

The realised P&L is close to the median of its forecast distribution.

Step 2 – Many PITs in the middle, few at the ends

Outcomes are less dispersed than forecast; tail outcomes happen less often than the model says.

$\Rightarrow$ The forecast distribution is too wide: risk is overstated, capital is excessive.

Interpretation

GARP calls this *conservative* misspecification: "the model distribution is too wide, and P&L observed is small, clustering in the middle".

Why the other options are wrong

B is incorrect. Polarity flip: a too-narrow (aggressive) model produces the U-shape, mass near 0 and 1.

C is incorrect. Only a flat histogram indicates good calibration.

D is incorrect. A hump is a systematic departure from uniformity, not randomness.

LO: MR-7 c • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1

Solution MR-7 c.4 Answer: B

Bank Q365 • AnalystPrep Q.6492

Step 1 – Compare with 10% per decile

Ends: 190 and 180 out of 1,000, i.e. 19% and 18% instead of 10%. Middle: $630/8 \approx 79$ per decile, below 100.

$\Rightarrow$ A U-shaped histogram.

Step 2 – Diagnose

PITs near 0 or 1 mean realised P&L landed far out in the tails more often than forecast: the forecast is too narrow.

$\Rightarrow$ Risk is understated in both tails; widen the forecast (higher volatility or fatter tails).

Interpretation

This is GARP's aggressive misspecification: "the model distribution is too narrow and we often see realized P&L in the tails".

Why the other options are wrong

A is incorrect. Reverses the diagnosis: too-wide forecasts produce a central hump with thin ends.

C is incorrect. Location bias gives a one-sided slope, not near-equal excess at both ends (190 vs 180).

D is incorrect. The middle deciles are about 79, not 100, and the ends are far above 100: clearly not uniform.

LO: MR-7 c • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1

Solution MR-7 c.5 Answer: B

Bank Q368 • BT P2.T5.25.3.3

Step 1 – Expected count

$200/10 = 20$ per decile.

Step 2 – Where the mass is

Lowest decile: $62/200 = 31\%$ against 10%, falling steadily to 7 in the top decile.

$\Rightarrow$ Low PITs are losses deep in the left tail; large losses happen about three times as often as the model expects.

Interpretation

The model underestimates large losses: its left tail is too thin. The histogram is far from uniform.

Why the other options are wrong

A is incorrect. The counts range from 62 down to 7 against 20 expected: nowhere near uniform.

C is incorrect. The main deviation is in the tail (lowest decile), not around the centre.

D is incorrect. Skewness 0 and kurtosis 3 describe a normal distribution; a uniform has kurtosis 1.8, and here the mean PIT is about 0.32, not 0.5.

LO: MR-7 c • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1 and 7.3

Solution MR-7 c.6 — Answer: C

Bank Q367 • AnalystPrep Q.6496

Step 1 – What a high PIT means

A PIT near 1 is an outcome better than most of the forecast distribution.

Step 2 – Read the shape

A steady rise from 5% to 15% is one-sided: outcomes systematically fall in the upper part of the forecast. It is neither a symmetric U nor a central hump.

$\Rightarrow$ The forecast is centred too low (too pessimistic): a location bias.

Interpretation

Width problems are symmetric; a monotone slope points to the mean (or skew) of the forecast.

Why the other options are wrong

A is incorrect. Too-narrow forecasts give excess mass at *both* 0 and 1; here the lowest decile is under-populated.

B is incorrect. Too-wide forecasts give a central hump; this histogram is monotone.

D is incorrect. Serial dependence is judged from PITs over time (e.g. PIT vs lagged PIT), not from the histogram's shape.

LO: MR-7 c • **Reference:** SchweserNotes 2026 Book 1, Reading 7, LO 7.c

Solution MR-7 c.7 — Answer: B

Bank Q364 • AnalystPrep Q.6487

Step 1 – Two requirements

A perfect forecast gives PITs that are both uniform *and* independent.

Step 2 – Which one fails

The flat histogram (and the KS result) satisfies uniformity, the counterpart of unconditional coverage. Runs of high PITs after high PITs violate independence, the counterpart of clustered exceedances.

$\Rightarrow$ The model is slow to update for changing volatility; its dynamics need revising.

Interpretation

Uniformity is necessary, not sufficient. Independence is checked separately, for example by Berkowitz's transform and regression (MR-7 d).

Why the other options are wrong

A is incorrect. Ignores independence.

C is incorrect. A too-narrow forecast would show a U-shape, but the histogram is flat.

D is incorrect. Location bias would show a slope, which is not observed.

LO: MR-7 c • **Reference:** Iercosan et al., GARP MR Ch.7, 7.1 and 7.6

Solution MR-7 c.8 Answer: B

New • built from GARP MR Ch.7, 7.3 and 7.4

Step 1 – Read the body

A hump across the middle deciles: outcomes cluster near the centre of the forecasts, so in the body the forecasts are too wide (conservative).

Step 2 – Read the ends

Spikes at both ends: extreme outcomes land beyond the forecast tails more often than predicted, so the tails are too thin.

$\Rightarrow$ Conservative in the body, but tails thinner than realised P&L requires.

Interpretation

This is exactly GARP's finding for subportfolio PITs: models are "fairly conservative in the body" while "the tails of these risk models are thinner than is required by the realizations of P&L". It also matches the moment results: models overstate variance and understate kurtosis.

Why the other options are wrong

A is incorrect. Ignores the spikes: a model too wide everywhere would have thin ends, not spikes.

C is incorrect. Reverses both halves.

D is incorrect. A histogram describes the distribution of PITs, not their order in time; dependence needs a separate test.

LO: MR-7 c • **Reference:** Iercosan et al., GARP MR Ch.7, 7.3 and 7.4

MR-7 • Beyond Exceedance-Based Backtesting of Value-at-Risk Models • Vardeez: IMPORTANT

MR-7 d

P3 CONCEPT

Describe backtesting using PITs, and compare the various goodness-of-fit tests that can be used to evaluate the distribution of PITs: the Kolmogorov-Smirnov test, the Anderson-Darling test, and the Cramér-von Mises test.

Workbook: C2 Backtesting VaR: Exceedance vs PIT • Tier C • Map #21 (4/16)

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C2, "PIT vs exceedance", Map #21) hits this LO.
Vardeez: new topic, highly testable; the three tests side by side.

Reported exam shape

- **C2·S2** PIT-based versus exceedance-based backtesting (worked as Question MR-7 a.8).

What the questions look like: "which test weights the tails?", definitions swapped between KS and CvM, reading a statistic against its critical value or p-value, and how independence of PITs is tested. The workbook trap: KS vs Anderson-Darling is a sibling swap.

1. What a goodness-of-fit test on PITs does

All three tests compare the **empirical CDF of the PITs** with the **uniform CDF** on [0,1], which is the 45-degree line. H_0: the PITs are $U(0,1)$. Each statistic measures the gap between the two curves, so:

- for a perfect forecast the statistic **converges to zero** as observations accumulate; **smaller is better**;
- reject uniformity when the statistic exceeds its critical value (equivalently, when its p-value is small);
- not rejecting does *not* prove the PITs are exactly uniform.

The three tests differ only in *how they summarise the gap*.

	Kolmogorov-Smirnov (KS)	Anderson-Darling (AD)	Cramér-von Mises (CvM)
Summary of the gap	Maximum vertical distance D	Squared gaps, **tails weighted up**	Mean squared deviation over the whole range
Strength	Simple, widely used; good at the central mass	Most power for tail misspecification (99% VaR)	Uses every observation, so a pattern of small misfits registers
Weakness	**Insensitive in the tails**: treats all discrepancies equally	More computationally intensive	Like KS, no extra weight on the tails
GARP pass rate	21%	**11%** (most powerful, fewest pass)	21%

Formulas to recognise: MR-7 d

Kolmogorov-Smirnov	$D = \max_j \lvert z_j - A_j \rvert$, z_j theoretical (uniform) CDF, A_j empirical CDF
Anderson-Darling	$A^2 = -n - \frac{1}{n}\sum_{j=1}^{n}(2j-1)\left[\ln z_j + \ln\left(1 - z_{n+1-j}\right)\right]$, z_j the jth ordered PIT
Cramér-von Mises	$W^2 = \sum_{j=1}^{n}\left[z_j - \frac{2j-1}{2n}\right]^2 + \frac{1}{12n}$

The $\ln z_j$ and $\ln(1-z)$ terms explode as PITs approach 0 or 1, which is why AD weights the tails. $(2j-1)/2n$ is where the jth PIT would sit under perfect uniformity.

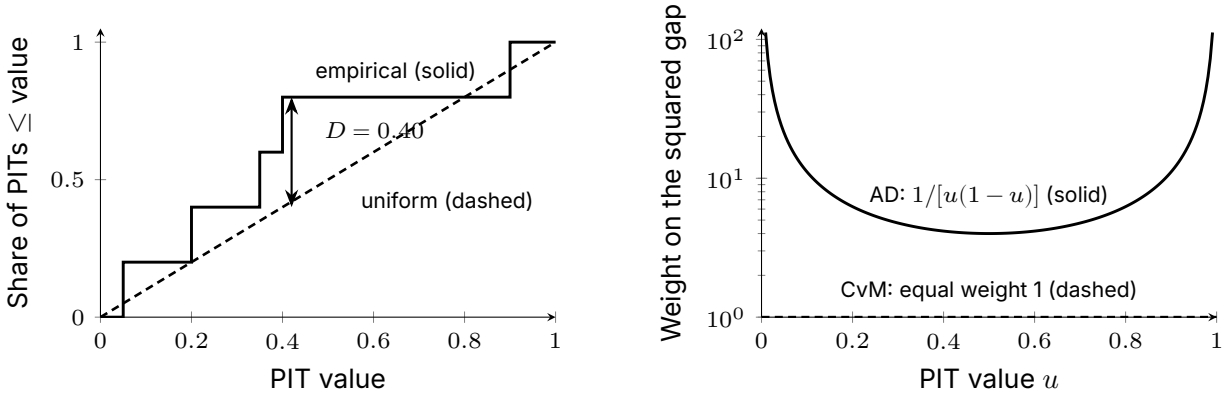


Figure 4. Left: five sorted PITs (0.05, 0.20, 0.35, 0.40, 0.90). The empirical CDF (solid steps) against the uniform CDF (dashed 45-degree line); the KS statistic is the single widest vertical gap, $D = 0.40$ at PIT 0.40. Right: in integral form CvM weights every squared gap equally, while AD divides by $u(1-u)$, which is 4 at the centre and about 100 near 0 or 1, so tail gaps dominate.

2. Testing independence of PITs

Uniformity is only half the story. Following **Berkowitz (2001)**:

1. Transform each PIT into a **standard normal** value, $\Phi^{-1}(p_t)$ (e.g. PIT 0.046 becomes -1.69). This avoids inference problems from the *bounded support* of [0,1].
2. Test the transformed series for dependence using standard log-likelihood methods. GARP regresses the transformed PIT on its own lag and on lagged P&L (significant coefficients mean the model does not react fast enough).
3. **Conditional coverage** can be checked visually: a scatterplot of PIT against lagged PIT should be an even cloud, with no diagonal streak and no dense corners.

GARP's scatter showed extra mass in the centre and at the extremes: dependence and non-uniformity.

3. The bottom line from the GARP study

- Exceedance tests: Kupiec 67%, Christoffersen independence 92%, conditional coverage 71% of subportfolios pass.
- Distribution tests: KS 21%, AD 11%, CvM 21% pass. "As the power of the test increases, we are able to find fewer subportfolios which exhibit the desired properties."

- Banks perform better on exceedance tests than on tests of the whole distribution: PIT tests uncover misspecification that counts miss.

Exam traps

- **KS vs AD swap.** Anderson-Darling, not Kolmogorov-Smirnov, weights the tails.
- **KS vs CvM swap.** KS is the *maximum* gap; CvM is the *mean squared* deviation.
- **Direction of the statistic.** Large statistic and small p-value = poor fit, uniformity rejected. A p-value of 0.0000 is the worst fit, not a perfect one.
- **Majority vote.** One rejection by the tail-sensitive AD test is enough to question a 99% VaR model, even if KS and CvM pass.
- **Independence needs its own test.** Goodness-of-fit tests check uniformity only; Berkowitz's normal transform handles dependence, and Kupiec is not a PIT test at all.

Practice questions: MR-7 d

Question MR-7 d.1

Bank Q370 • AP Mock 2 Q64 (rebuilt)

A risk manager compares goodness-of-fit statistics for a PIT-based VaR backtest. Which statement is correct?

A. The KS statistic is the mean squared deviation between the actual and theoretical distributions, while the Cramér-von Mises statistic is the maximum vertical distance.

B. For a perfect distribution forecast, each of the KS, Anderson-Darling and Cramér-von Mises statistics is expected to converge toward zero as the number of observations grows.

C. The Anderson-Darling test treats all discrepancies between the distributions equally, which is why it is preferred over KS for risk management.

D. A statistic that is larger than zero but below its critical value means the PITs are exactly uniform.

Question MR-7 d.2

Bank Q374 • AnalystPrep Q.6500 (rebuilt)

A validator contrasts the Cramér-von Mises (CvM) test with the Kolmogorov-Smirnov (KS) test when evaluating the distribution of PITs. Which statement is correct?

A. CvM uses the maximum vertical distance between distributions, like KS, but applies a tail weighting.

B. CvM is very similar to KS but, instead of the maximum difference, it considers the mean squared deviation between the distributions.

C. CvM is the test with the greatest emphasis on tail observations among the three.

D. CvM tests the independence of successive PIT values rather than their distribution.

Question MR-7 d.3

Bank Q375 • BT P2.T5.25.4.1 (rebuilt)

Adam Knight, a risk manager, backtests a 99% one-day VaR model over 250 trading days. Losses exceeded VaR 8 times versus about 2 to 3 expected, mostly in volatile periods in 2022. A histogram of the PITs shows a concentration of values near zero (well above the uniform density), indicating a systematic underestimation of extreme losses. Which goodness-of-fit test is most appropriate to identify this tail problem?

A. Kolmogorov-Smirnov, because it measures overall deviation between the empirical and theoretical CDFs and so ensures calibration across all quantiles.

B. Anderson-Darling, because it places more emphasis on the tails, making it useful for detecting underestimation of extreme losses.

C. Cramér-von Mises, because it focuses equally on the centre and the tails of the PIT distribution.

D. Kupiec test, because it evaluates the full distribution of PITs.

Question MR-7 d.4

Bank Q376 • BT P2.T5.25.4.2 (rebuilt)

A risk analyst backtests a VaR model by testing the uniformity of PITs at the 99% confidence level. A PIT histogram shows clustering near 0 and 1 relative to the uniform baseline. Results:

Test	Critical value (99%)	Observed statistic
Kolmogorov-Smirnov	0.136	0.09
Anderson-Darling	4.0	4.8
Cramér-von Mises	0.115	0.075

Which conclusion is best supported?

A. The KS test shows the model is well calibrated, so no further concern exists.

B. The Anderson-Darling test rejects uniformity because its statistic exceeds the critical value, signalling tail miscalibration that KS and CvM do not detect.

C. The Cramér-von Mises test proves the model is well calibrated in the tails.

D. All three tests indicate the model is well calibrated because two of three statistics are below critical values.

Question MR-7 d.5

Bank Q377 • BT P2.T5.25.4.3 (rebuilt)

A risk team backtests three VaR models by computing PITs and testing them against the uniform [0,1] distribution. The maximum gap between the empirical and uniform CDFs is about 0.05 for Model X, 0.14 for Model Y and 0.20 for Model Z. The test results are:

Model	KS statistic	KS p-value	CvM statistic	CvM p-value
X	0.0538	0.4478	0.0910	0.6312
Y	0.1398	0.0001	1.5855	0.0001
Z	0.1954	0.0000	4.0694	0.0000

Which of the following statements correctly interprets these results?

A. Model X has the most uniform PITs: its small statistics and high p-values mean uniformity is not rejected, whereas it is rejected for Models Y and Z.

B. Model Y passes both tests, because its large test statistics indicate a close fit to the uniform distribution.

C. Model Z is overly conservative, because its p-values of 0.0000 show its PITs are almost perfectly uniform.

D. Model X is the only model whose PITs show tail clustering, because it has the highest p-values.

Question MR-7 d.6

New • built from GARP MR Ch.7, 7.6 (Berkowitz)

A validator wants to test whether a desk's daily PITs are independent over time, following the approach described by Berkowitz (2001). Which procedure is correct?

A. Apply the Anderson-Darling test to the PITs, because it weights the tails where clustered losses show up.

B. Transform each PIT into a standard normal value with the inverse normal CDF, then test the transformed series for dependence, for example by regressing it on its own lag and on lagged P&L.

C. Count the exceedances of the 99% VaR and apply Kupiec's proportion of failures test to the count.

D. Regress the raw PITs directly on their lags, because their [0,1] support makes the normal transformation unnecessary.

Question MR-7 d.7 New • built from GARP MR Ch.7, 7.5 and 7.6

In Iercosan et al.'s study of 597 US bank subportfolios, the share whose PITs passed (did not reject) each uniformity test at the 1% level was about 21% for Kolmogorov-Smirnov, 11% for Anderson-Darling and 21% for Cramér-von Mises, against 67% passing Kupiec's unconditional coverage test. Which conclusion is best supported?

A. Because the most tail-sensitive test rejected the most subportfolios, the models' main weakness lies in the centre of the forecast distribution.

B. The models perform better on exceedance tests than on tests of the whole distribution, and Anderson-Darling, which weights the tails, has the most power to detect the misspecification.

C. The Kolmogorov-Smirnov and Cramér-von Mises tests are the more reliable, because they agree with each other and reject fewer models.

D. The PIT tests must be wrong, because a model that passes Kupiec's test has a correctly specified forecast distribution.

Solutions • MR-7 d

Solution MR-7 d.1 — Answer: B

Bank Q370 • AP Mock 2 Q64

Step 1 – What every statistic measures

The distance between the empirical PIT distribution and the uniform $[0,1]$ distribution.

Step 2 – Under a perfect forecast

The PITs are truly uniform, so the distance shrinks toward zero as observations accumulate.
$\Rightarrow$ Lower is better for all three; B is correct.

Interpretation

The tests differ in *how* they measure the distance: KS the maximum gap, CvM the mean squared deviation, AD squared gaps weighted toward the tails.

Why the other options are wrong

A is incorrect. Swaps the definitions: KS is the maximum vertical distance, CvM the mean squared deviation.
C is incorrect. Reverses the logic: KS treats all discrepancies equally; AD places more emphasis on the tails, which is why risk managers prefer it.
D is incorrect. Below the critical value only means uniformity is not rejected; it does not prove exact uniformity.

LO: MR-7 d: Describe backtesting using PITs, and compare the various goodness-of-fit tests that can be used to evaluate the distribution of PITs: the Kolmogorov-Smirnov test, the Anderson-Darling test, and the Cramér-von Mises test.
• **Reference:** SchweserNotes 2026 Book 1, Reading 7, LO 7.d; Iercosan et al., GARP MR Ch.7, 7.6

Solution MR-7 d.2 — Answer: B

Bank Q374 • AnalystPrep Q.6500

Step 1 – Common ground

Both KS and CvM compare the empirical PIT CDF with the uniform CDF.

Step 2 – The difference

KS takes the *maximum* difference; CvM uses the *mean squared* deviation across the distribution.
$\Rightarrow$ CvM is "very similar to the KS test" but averages instead of taking the worst gap.

Interpretation

Tail emphasis belongs to AD; independence is tested separately (Berkowitz).

Why the other options are wrong

A is incorrect. The maximum distance is the KS definition, and CvM has no tail weighting.
C is incorrect. Anderson-Darling, not CvM, puts the most weight on the tails.
D is incorrect. CvM is a goodness-of-fit test of the PIT distribution, not an independence test.

LO: MR-7 d • **Reference:** Iercosan et al., GARP MR Ch.7, 7.6, eq. 7.24

Solution MR-7 d.3 — Answer: B

Bank Q375 • BT P2.T5.25.4.1

Step 1 – Name the problem

8 exceedances against 2.5 expected and PITs bunched near zero: a *tail* problem (extreme losses underestimated).

Step 2 – Pick the tool

The test that weights tail discrepancies more heavily is Anderson-Darling.
⇒ B is correct.

Interpretation

GARP: "any misspecification in the tail of the distribution of the risk model will be better identified by the A-D test".

Why the other options are wrong

A is incorrect. KS treats all discrepancies equally and lacks sensitivity at the extremes.
C is incorrect. CvM averages squared deviations with equal weight; it does not single out the tails.
D is incorrect. Kupiec counts exceedances; it is not a test of the PIT distribution.

LO: MR-7 d • **Reference:** Iercosan et al., GARP MR Ch.7, 7.6

Solution MR-7 d.4 **Answer: B**

Bank Q376 • BT P2.T5.25.4.2

Step 1 – Compare each statistic with its critical value

KS: $0.09 < 0.136$, not rejected. CvM: $0.075 < 0.115$, not rejected. AD: $4.8 > 4.0$, **rejected**.

Step 2 – Why only AD rejects

The histogram shows clustering at 0 and 1, a tail problem. AD weights the tails; KS and CvM give tail gaps no extra weight, so they can pass.
⇒ Uniformity is rejected by the tail-sensitive test: tail miscalibration.

Interpretation

For a 99% VaR model, the AD result is the one that matters; it is not outvoted by the other two.

Why the other options are wrong

A is incorrect. KS passing means no large gap overall; it does not override the AD rejection.
C is incorrect. CvM passing says little about the tails, since it averages across the distribution.
D is incorrect. Majority voting is wrong: one rejection by the tail-sensitive test is enough to question the model.

LO: MR-7 d • **Reference:** Iercosan et al., GARP MR Ch.7, 7.6

Solution MR-7 d.5 **Answer: A**

Bank Q377 • BT P2.T5.25.4.3

Step 1 – Direction of the statistics

Under H_0 the PITs are uniform. Larger KS (maximum gap) and CvM (mean squared deviation) statistics mean a worse fit.

Step 2 – Read the p-values

X: p-values 0.4478 and 0.6312, far above 5%, so uniformity is not rejected. Y and Z: p-values about 0, so uniformity is rejected; Z (KS 0.1954, CvM 4.0694) is the worst.
⇒ Model X is the best calibrated.

Interpretation

Small statistics with high p-values are what a good model looks like; large statistics with tiny p-values are a rejection.

Why the other options are wrong

B is incorrect. Reverses the direction: large statistics and tiny p-values mean a large deviation and a rejection.
C is incorrect. A p-value of 0.0000 means uniformity is strongly rejected; Z is the worst fit, not

a near-perfect one.
D is incorrect. High p-values mean the PITs are consistent with uniformity; tail clustering would show up as large statistics and low p-values.

LO: MR-7 d • **Reference:** Iercosan et al., GARP MR Ch.7, 7.6

Solution MR-7 d.6 — Answer: B

New • built from GARP MR Ch.7, 7.6

Step 1 – Why transform

PITs live on the bounded interval $[0,1]$. Berkowitz notes it is easier to test independence outside that range, so each PIT is mapped to a standard normal value, $\Phi^{-1}(p_t)$ (e.g. 0.046 becomes -1.69), to "avoid problems of inference due to bounded support".

Step 2 – Then test dependence

The transformed series is tested with log-likelihood methods; GARP regresses it on its lag and on lagged P&L. Significant coefficients mean the forecasts ignore information already available.
$\Rightarrow$ B is correct.

Interpretation

Uniformity tests (KS, AD, CvM) and independence tests are separate steps; together they check i.i.d. uniformity, the PIT version of conditional coverage.

Why the other options are wrong

A is incorrect. AD is a goodness-of-fit test of the PIT distribution; it says nothing about the order of PITs in time.
C is incorrect. Kupiec tests the exceedance count (unconditional coverage), not independence of PITs.
D is incorrect. Reverses the reason: the bounded $[0,1]$ support is exactly why the normal transformation is used.

LO: MR-7 d • **Reference:** Iercosan et al., GARP MR Ch.7, 7.6, Table 7.10; SchweserNotes 2026 Book 1, Reading 7, LO 7.d

Solution MR-7 d.7 — Answer: B

New • built from GARP MR Ch.7, 7.5 and 7.6

Step 1 – Exceedance tests vs distribution tests

67% pass Kupiec, but only 11% to 21% pass the PIT uniformity tests.
$\Rightarrow$ The models do better at the single VaR percentile than across the whole distribution.

Step 2 – Rank the tests

AD rejects the most (only 11% pass). GARP: "as the power of the test increases, we are able to find fewer subportfolios which exhibit the desired properties".
$\Rightarrow$ AD, the tail-weighted test, is the most powerful here.

Interpretation

Passing an exceedance test is weak evidence that the whole forecast distribution is right; PIT tests, especially AD, reveal the tail misspecification the counts miss.

Why the other options are wrong

A is incorrect. The most tail-sensitive test rejecting the most points to the *tails*, not the centre.
C is incorrect. Agreement between two less powerful tests is not reliability; they reject less because they are less sensitive.
D is incorrect. Kupiec checks one percentile only; passing it does not validate the full distribu-

tion.

LO: MR-7 d ● **Reference:** Iercosan et al., GARP MR Ch.7, 7.5, 7.6 and 7.7

MR-9 • Empirical Properties of Correlation: How Do Correlations Behave in the Real World? • Vardeez: MODERATE

MR-9 b

P3 CALC

Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation.

Workbook: D11 Empirical Properties of Correlation • Tier D • Map #54

WHAT IS TESTED HERE

Why P3. Tier D workbook topic (D11). The sheet names "know calculation of mean reversion and autocorrelation".
Vardeez: know the calculation of mean reversion and the autocorrelation concept.

Reported exam shapes

- **D11·S2** Mean reversion rate from regressing the correlation change on the lagged level.
- **D11·S3** The corresponding autocorrelation.

What the questions look like: a regression output $Y = \alpha + \beta X$ (sometimes with an R^2 you do not need) and a current correlation; read $a = -\beta$, get the long-run mean from the intercept, forecast next month's correlation, or give the one-lag autocorrelation $1 - a$. Short one-line items are common.

Note on the LO letters. Vardeez's notes letter this reading a (economic states), b (best-fit distribution), c (mean reversion). The official 2026 LO list in the Master LO Map, Schweser and the question bank use b = mean reversion and autocorrelation, c = best-fit distribution. This book follows the official order, so his "MR-9 c" material is here under MR-9 b.

1. Mean reversion

Mean reversion is "the tendency of a variable to be pulled back to its long-term mean". Statistically: the change $S_t - S_{t-1}$ is *negatively* related to the level S_{t-1}. Interest rates, volatilities, credit spreads and (as Meissner shows) correlations all display it. The discrete Vasicek form:

$$S_t - S_{t-1} = a\,(\mu_S - S_{t-1})\,\Delta t + \sigma_S\,\varepsilon\sqrt{\Delta t}, \qquad 0 \le a \le 1$$

Take expectations and set $\Delta t = 1$: the stochastic term drops out, leaving

$$\boxed{E[S_t] - S_{t-1} = a\,(\mu_S - S_{t-1})}$$

a is the fraction of the gap to the mean that is closed each period: $a = 1$ jumps straight to the mean, $a = 0.5$ closes half the gap, $a = 0$ never moves.

2. Estimating a by regression

Expand the deterministic form and match it term by term to a regression line:

$$\underbrace{S_t - S_{t-1}}_{Y} = \underbrace{a\mu_S}_{\alpha\ \text{(intercept)}} \underbrace{-\,a}_{\beta\ \text{(slope)}} \underbrace{S_{t-1}}_{X} \qquad \Longrightarrow \qquad \boxed{a = -\beta, \qquad \mu_S = \frac{\alpha}{a}}$$

Meissner's result (monthly average correlation of the 30 Dow stocks, 1972 to 2017): $y = 0.256 - 0.7903x$, $R^2 = 0.3948$. So $a = 79.03\%$, a very strong pull, and $\mu = 0.256/0.7903 = 32.38\%$.

Forecast (Example 9.1). February 2017 correlation 26.15%, mean 32.38%:

$$E[S_{\text{Mar}}] = 0.2615 + 0.7903 \times (0.3238 - 0.2615) = 31.07\%$$

Other rearrangements: solve for the rate, $a = \dfrac{S_t - S_{t-1}}{\mu - S_{t-1}}$ (Vardeez: from 22% to 35% with mean 45%: $0.13/0.23 = 56.5\%$); or for the mean, $\mu = S_{t-1} + \dfrac{S_t - S_{t-1}}{a}$.

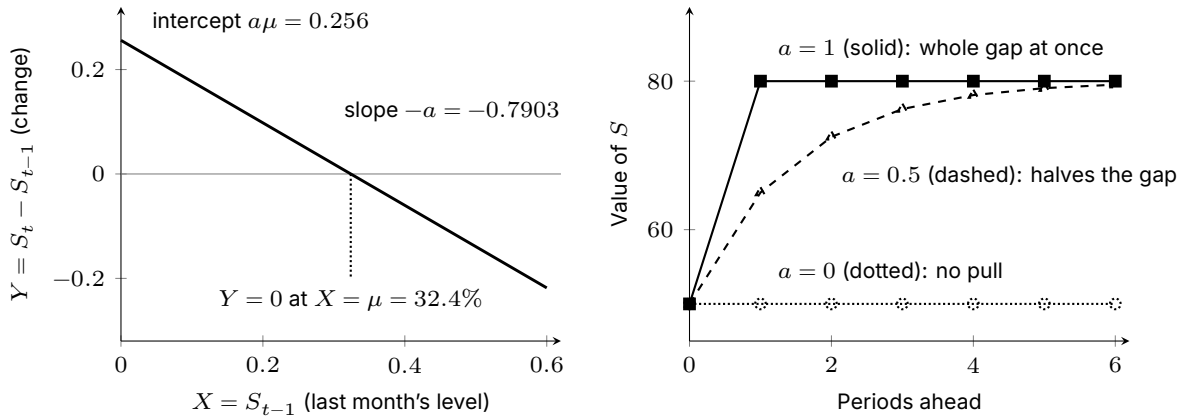


Figure 1. Left: Meissner's regression line. The change is positive below the mean and negative above it; the line crosses zero exactly at the long-run mean, α/a. Right: starting at 50 with a long-run mean of 80, the speed of the pull depends only on a.

3. Autocorrelation: the reverse property

Autocorrelation is "the degree to which a variable is correlated to its past values" (positive autocorrelation is also called persistence). For one lag it is the Pearson coefficient between the series and itself one period back:

$$AC(\rho_t, \rho_{t-1}) = \frac{\text{cov}(\rho_t, \rho_{t-1})}{\sigma(\rho_t)\,\sigma(\rho_{t-1})}$$

It can be estimated with ARCH/GARCH-type models or by regression. In Meissner's study the one-lag autocorrelation was 20.97%: **mean reversion and one-lag autocorrelation add up to 1**.

$$\boxed{AC_{1\text{ lag}} = 1 - a} \qquad 79.03\% + 20.97\% = 100\%$$

Strong mean reversion means low autocorrelation, and vice versa. Autocorrelation decays as the lag lengthens (the 2-month lag was highest, about 26%, falling to about 10% at ten months): the memory of the correlation process fades.

Correlation type	Mean reversion a	One-lag AC $= 1 - a$	Reading
Equity (Dow, monthly)	79.03%*	about 21%	strong pull, little persistence
Bond	25.79%	about 74%	weak pull, persistent
Default probability	29.97%	about 70%	weak pull, persistent

*77.51% in the sample used for the bond and default comparison.

Formula card: MR-9 b

Model	$S_t - S_{t-1} = a(\mu - S_{t-1})\Delta t + \sigma\varepsilon\sqrt{\Delta t}$
Regression	$Y = S_t - S_{t-1}$ on $X = S_{t-1}$: slope $= -a$, intercept $= a\mu$
Rate and mean	$a = -\beta, \quad \mu = \alpha/a$
Forecast	$E[S_t] = S_{t-1} + a(\mu - S_{t-1})$
Solve for a	$a = (S_t - S_{t-1})/(\mu - S_{t-1})$
Solve for μ	$\mu = S_{t-1} + (S_t - S_{t-1})/a$
Autocorrelation	$AC = \text{cov}(\rho_t, \rho_{t-1})/(\sigma_t\sigma_{t-1})$; one lag: $AC = 1 - a$

Exam traps

- **Sign of the slope.** The slope is $-a$; the mean reversion rate is its absolute value. Using β with its sign pushes the forecast away from the mean.
- **Intercept is not the mean.** The intercept is $a\mu$; divide by a.
- **Sibling swap.** Using the autocorrelation $(1 - a)$ as the speed of reversion in the forecast, or reporting a as the autocorrelation.
- **Change versus level.** $a(\mu - S_{t-1})$ is the expected *change*; add it to S_{t-1} for the level.
- **Direction.** Above the mean the expected change is negative, below it positive.
- **Gap measured from the wrong point.** The gap is $\mu - S_{t-1}$, not $\mu - S_t$.
- R^2 **and the current level** are often red herrings when only the autocorrelation is asked.

Practice questions: MR-9 b

Question MR-9 b.1

Bank Q487 • Mock 2 Q37 (rebuilt)

Suppose mean reversion exists for a variable with a value of 100 at time $t-1$. The long-run mean of the variable is 120. Ignoring the stochastic term, what is the expected change in the variable for the next period if the mean reversion rate is 0.6?

A. -12

B. 8

C. 12

D. 20

Question MR-9 b.2

Bank Q480 • AP Mock 1 Q9 (option A rebuilt)

Given the following data about a variable S:

$S(t)$	50
$S(t-1)$	30
Mean reversion rate	0.4

Calculate the long-run mean value for the variable.

A. 50

B. 75

C. 80

D. 100

Question MR-9 b.3

Bank Q486 • Mock 1 Q27 (rebuilt; intercept made consistent with the 35% mean)

In September 2021, the average monthly correlation for all the Dow stocks was 30% and the long-run mean correlation of Dow stocks was 35%. A regression of the change in correlation on the lagged level gives $Y = 0.273 - 0.78X$. What is the expected correlation for October 2021, given the mean reversion rate estimated in the regression?

A. 3.9%

B. 5.0%

C. 31.1%

D. 33.9%

Question MR-9 b.4 Bank Q490 • Schweser Module Quiz 9.1 Q3

A risk manager uses 480 months of correlation data from the Dow to estimate the long-run mean correlation and the mean reversion rate. The long-run mean correlation of Dow stocks was 32%, and the regression gives $Y = 0.24 - 0.75X$. In April 20X4 the average monthly correlation for all Dow stocks was 36%. What is the expected correlation for May 20X4 using the estimated mean reversion rate?

A. 32%

B. 33%

C. 35%

D. 37%

Question MR-9 b.5 Bank Q489 • Mock 2 Q59 (rebuilt; intercept made consistent with the 44% mean)

A risk analyst uses 600 months of correlation data from the S&P 500 to estimate the long-run mean correlation of common stocks and the mean reversion rate. The long-run mean correlation was 44%, and the regression gives $Y = 0.348 - 0.79X$. In July 2017 the average monthly correlation for all S&P stocks was 37%. What is the estimated one-period autocorrelation based on the mean reversion rate from the regression?

A. 21%

B. 37%

C. 56%

D. 65%

Question MR-9 b.6 Bank Q485 • BT P2.T5.22.12.1

Albert has retrieved the monthly average correlation for stocks in the Dow Jones Industrial Average for the last ten years: each month's $\rho(t)$ is the average of the 30×30 pairwise correlation matrix. With this time series he computes the one-period lag autocorrelation, $AC[\rho(t), \rho(t-1)]$. If Albert's one-period lag autocorrelation is low, which of the following is most likely?

A. The regression coefficient is positive.

B. The Pearson correlation coefficient is high.

C. The mean reversion rate (aka gravity) is high.

D. The volatilities of correlation(t) and correlation$(t-1)$ are low.

Question MR-9 b.7 New • built from GARP MR Ch.9 (Meissner 9.2, 9.3)

A risk analyst replicates Meissner's study of the 30 Dow stocks. Regressing the monthly change in the average correlation on the previous month's level gives $y = 0.256 - 0.7903x$, with $R^2 = 0.3948$. Which of the following statements is correct?

A. The long-run mean correlation implied by the regression is 25.6%, the value of the intercept.

B. The one-month-lag autocorrelation is about 79.0%, and the mean reversion rate is about 21.0%.

C. The negative slope of -0.7903 means correlations tend to move away from their long-run mean, so the series shows no mean reversion.

D. The mean reversion rate is 79.03%, the implied long-run mean is about 32.4%, and the one-month-lag autocorrelation is about 21.0%.

Question MR-9 b.8 Workbook D11.1 • S2, S3 • Map #54

A quantitative analyst at a Hong Kong asset manager models the monthly average pairwise correlation of a regional equity basket. Regressing the monthly change in the average correlation, $\rho_t - \rho_{t-1}$, on the previous month's level, ρ_{t-1}, gives:

- Intercept: 0.217; slope coefficient: -0.62; R^2: 0.31
- Current (last month's) average correlation: 24%

Ignoring the stochastic term, which of the following is closest to the one-lag autocorrelation of the correlation series implied by the regression and the expected average correlation for next month?

A. Autocorrelation 38%; expected correlation 30.8%

B. Autocorrelation 38%; expected correlation 28.2%

C. Autocorrelation 38%; expected correlation 22.6%

D. Autocorrelation 62%; expected correlation 30.8%

Solutions • MR-9 b

Solution MR-9 b.1

Answer: C

Bank Q487 • Mock 2 Q37

Step 1 – Expected change

$$E[S_t] - S_{t-1} = a(\mu - S_{t-1}) = 0.6 \times (120 - 100) = +12$$

$\Rightarrow$ The variable is expected to rise to 112: it closes 60% of the 20-point gap.

Interpretation

Below the mean the expected change is positive; above it, negative. The question asks for the change, not the new level.

Why the other options are wrong

A is incorrect. Sign error: the variable is below its mean, so it is pulled up, not down.
B is incorrect. Used the autocorrelation $1 - a = 0.4$ as the reversion fraction: $0.4 \times 20 = 8$.
D is incorrect. The full gap of 20 applies only if $a = 1$.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.2, eq. 9.2

Solution MR-9 b.2

Answer: C

Bank Q480 • AP Mock 1 Q9

Step 1 – Write the model

$$S_t - S_{t-1} = a(\mu - S_{t-1}) \quad \Rightarrow \quad 50 - 30 = 0.4(\mu - 30)$$

Step 2 – Solve for the mean

$$\mu - 30 = \frac{20}{0.4} = 50 \qquad \Rightarrow \qquad \mu = 80$$

$\Rightarrow$ The observed change of 20 is 40% of the gap, so the gap was 50 and the mean is 80.

Interpretation

The gap is always measured from S_{t-1}, the level the variable reverted *from*.

Why the other options are wrong

A is incorrect. Reported the gap $(20/0.4 = 50)$ instead of adding it to S_{t-1}.
B is incorrect. Divided the starting level by the rate, $30/0.4 = 75$, ignoring the observed change.
D is incorrect. Measured the gap from $S(t) = 50$ instead of $S(t-1)$: $50 + 20/0.4 = 100$.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.2

Solution MR-9 b.3

Answer: D

Bank Q486 • Mock 1 Q27

Step 1 – Mean reversion rate from the regression

$$a = -\beta = 0.78$$

(Check: intercept $= a\mu = 0.78 \times 0.35 = 0.273$, consistent with the stated 35% mean.)

Step 2 – Expected change and level

$$E[\rho_{\text{Oct}}] = 0.30 + 0.78 \times (0.35 - 0.30) = 0.30 + 0.039 = 33.9\%$$

$\Rightarrow$ The correlation is pulled from 30% towards 35%, closing 78% of the 5-point gap.

Interpretation

This is Vardeez's Example 1 with new dates. The intercept is only a check: the mean is given directly.

Why the other options are wrong

A is incorrect. Reported the expected change (3.9%) and forgot to add it to 30%.
B is incorrect. Reported the raw gap $35\% - 30\%$, which is the change only if $a = 1$.
C is incorrect. Used the autocorrelation $1 - a = 0.22$ as the speed: $30\% + 0.22 \times 5\% = 31.1\%$.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.2, Example 9.1

Solution MR-9 b.4

Answer: B

Bank Q490 • Schweser Module Quiz 9.1 Q3

Step 1 – Rate

$a = -\beta = 0.75$ (check: $0.24/0.75 = 0.32$, the stated mean).

Step 2 – Forecast

$$E[\rho_{\text{May}}] = 0.36 + 0.75 \times (0.32 - 0.36) = 0.36 - 0.03 = 33\%$$

$\Rightarrow$ Above the mean, the correlation is expected to fall.

Interpretation

The direction trap is built in: the current level (36%) is *above* the long-run mean (32%), so the pull is downward.

Why the other options are wrong

A is incorrect. Assumed full reversion ($a = 1$) to the mean.
C is incorrect. Used $1 - a = 0.25$: $36\% + 0.25 \times (-4\%) = 35\%$.
D is incorrect. Moved away from the mean: $36\% + 0.25 \times 4\% = 37\%$, wrong direction and wrong rate.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.2; SchweserNotes 2026 Book 1, LO 9.b

Solution MR-9 b.5

Answer: A

Bank Q489 • Mock 2 Q59

Step 1 – Mean reversion rate

$$a = -\beta = 0.79$$

Step 2 – One-period autocorrelation

$$AC = 1 - a = 1 - 0.79 = 21\%$$

Interpretation

The long-run mean (44%), the July level (37%) and the intercept are red herrings: the one-lag autocorrelation depends only on the slope.

Why the other options are wrong

B is incorrect. Picked the July correlation level, irrelevant to the autocorrelation.
C is incorrect. Used the long-run mean instead of the rate: $1 - 0.44 = 56\%$.
D is incorrect. Confused the intercept with the slope: $1 - 0.348 \approx 65\%$.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.3, eq. 9.6

Solution MR-9 b.6 — Answer: C

Bank Q485 • BT P2.T5.22.12.1

Step 1 – Link autocorrelation and mean reversion

For one lag, $AC = 1 - a$, so $a = 1 - AC$.
$\Rightarrow$ A low autocorrelation means a high mean reversion rate.

Interpretation

They are reverse properties: a series that is pulled hard to its mean every month cannot also remember last month's level.

Why the other options are wrong

A is incorrect. With mean reversion the regression slope is $-a$, negative.
B is incorrect. The autocorrelation *is* the Pearson coefficient between $\rho(t)$ and $\rho(t-1)$; if it is low, that coefficient is low.
D is incorrect. The volatility levels do not determine the autocorrelation; they only scale the covariance.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.3

Solution MR-9 b.7 — Answer: D

New • built from GARP MR Ch.9

Step 1 – Mean reversion rate

$$a = -\beta = 0.7903 = 79.03\%$$

Step 2 – Long-run mean

$$\mu = \frac{\alpha}{a} = \frac{0.256}{0.7903} = 32.4\%$$

Step 3 – One-lag autocorrelation

$$AC = 1 - a = 20.97\% \approx 21.0\%$$

Interpretation

These are exactly Meissner's figures for the Dow, 1972 to 2017. The R^2 describes how well the regression fits; it is not a parameter of the mean reversion model.

Why the other options are wrong

A is incorrect. The intercept is $a\mu$, not μ: dividing by a gives 32.4%.

B is incorrect. Sibling swap: 79.03% is the mean reversion rate and 20.97% the autocorrelation, not the other way round.
C is incorrect. A negative slope of the change on the level is exactly what mean reversion means: above the mean, the expected change is negative.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.2 and 9.3, eqs. 9.4 to 9.6

Solution MR-9 b.8 — Answer: A

Workbook D11.1 • S2, S3

Step 1 – Mean reversion rate

In $\rho_t - \rho_{t-1} = a\mu_\rho - a\rho_{t-1}$, the slope is $-a$:

$$a = -\beta = -(-0.62) = 0.62$$

$\Rightarrow$ Each month, 62% of the gap between the current level and the long-term mean is closed.

Step 2 – Long-term mean from the intercept

$$\mu_\rho = \frac{\text{intercept}}{a} = \frac{0.217}{0.62} = 0.35$$

Step 3 – Autocorrelation

$$AC = 1 - a = 1 - 0.62 = 0.38$$

$\Rightarrow$ Mean reversion and one-lag autocorrelation are reverse properties; in Meissner's study they sum to 1 (79.03% and 20.97%).

Step 4 – Expected correlation next month

$$E[\rho_t] = \rho_{t-1} + a(\mu_\rho - \rho_{t-1}) = 0.24 + 0.62 \times (0.35 - 0.24) = 0.24 + 0.0682 = 0.3082$$

Interpretation

The basket's correlation, currently well below its long-term mean of 35%, is expected to jump to about 30.8% next month. For a book that is short correlation (for example, sold basket options), that strong pull back towards the mean is the risk to hedge. The R^2 of 0.31 describes the regression's fit and is not an input.

Why the other options are wrong

B is incorrect. Used the autocorrelation (0.38) as the speed of mean reversion: $0.24 + 0.38 \times 0.11 = 0.2818$.
C is incorrect. Took the intercept (0.217) as the long-term mean instead of $a\mu_\rho$: $0.24 + 0.62 \times (0.217 - 0.24) = 0.2257$.
D is incorrect. Reported the absolute slope (0.62), which is the mean reversion rate, as the autocorrelation; the forecast is right but the autocorrelation is the sibling quantity.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.2 and 9.3, eqs. 9.2 to 9.6, Example 9.1

MR-9 • Empirical Properties of Correlation: How Do Correlations Behave in the Real World? • Vardeez: MODERATE

MR-9 c

P3 CONCEPT

Identify the best-fit distribution for equity, bond, and default correlations.

Workbook: D11 Empirical Properties of Correlation • Tier D • Map #54

WHAT IS TESTED HERE

Why P3. Tier D workbook topic (D11). The sheet names "identifying best-fit distribution (equity, bond and default correlation)".
Vardeez: identifying the best-fit distribution for equity, bond and default correlations (his "MR-9 b"; see the note on letters under MR-9 b).

Reported exam shapes

- **D11·S1** Best-fit distribution for equity, bond and default probability correlations: a three-way sibling swap.

What the questions look like: one recall fact tested by swapping bond and default, by absolute claims ("Johnson SB for all three") and by tempting normal, lognormal or beta answers for equity correlations.

1. The equity correlation study

- **Data.** Daily correlations between all 30 Dow stocks, 1972 to 2017: 464,580 correlation values. 77.23% of them were positive (the remaining 22.77% were zero or negative: correlations are not bounded below by zero).
- **Tests.** Sixty-one distributions were fitted and ranked with the Kolmogorov-Smirnov, Anderson-Darling and chi-squared tests.
- **Best fit: Johnson SB.** A four-parameter distribution: two shape parameters (γ, δ), one location (μ) and one scale (σ). It is bounded and flexible, which suits a variable that lives inside an interval and is skewed.
- **Poor fits:** normal, lognormal and beta.
- **Robust across states.** Fitting by state of the economy changed the result "slightly but not significantly": Johnson SB remains the best fit.

2. Bond and default probability correlations

Preliminary studies (7,645 bond correlations and 4,655 default probability correlations, "currently being verified with a larger sample") found:

- **Default probability correlations** look like equity correlations and are "replicated best by the **Johnson SB** distribution".
- **Bond correlations** show "a more normal shape" and are "best fitted with the **generalised extreme value** distribution and quite well with the normal distribution".

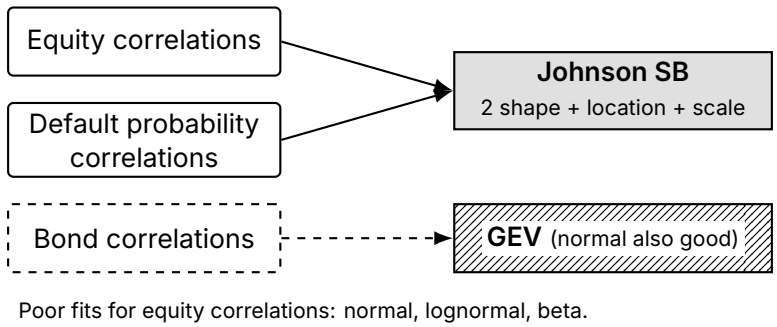


Figure 2. Two of the three types share Johnson SB (grey box); bonds are the odd one out (dashed arrow, hatched box). Memory hook: default probability correlations behave like equity correlations, bonds look more normal.

	Avg level	Volatility	Mean reversion	Best fit
Equity	34.83%	79.73%	77.51%	Johnson SB
Bond	41.67% (highest)	63.74% (lowest)	25.79% (slowest)	GEV (normal also good)
Default prob.	30.43% (lowest)	87.74% (highest)	29.97%	Johnson SB

Exam traps

- **Bond and default swapped.** It is *default probability* correlations that resemble equity (Johnson SB); bonds are GEV.
- **Absolute claim.** "Johnson SB for all three" is false.
- **Normal for equity correlations.** Normal, lognormal and beta all fit equity correlations poorly.
- **State dependence.** The economic state does *not* significantly change the equity fit.
- **GEV is not only an EVT tool here.** In MR-3 GEV models block maxima; in this reading it is simply the best-fitting distribution for bond correlations.
- **Normal for bonds is second best,** not the best fit.

Practice questions: MR-9 c

Question MR-9 c.1

Bank Q494 • AnalystPrep Q.2655

Which of the following distributions best fits the empirical distribution of equity correlations?

A. Chi-squared

B. Generalized extreme value

C. Pareto

D. Johnson SB

Question MR-9 c.2

Bank Q493 • GARP book Ch.9 Q9.10 (converted to multiple choice)

When modelling stocks, bonds, commodities, exchange rates and volatilities, risk managers typically assume a normal or lognormal distribution. A quant proposes doing the same for the monthly equity correlations that drive a basket-option book. Based on Meissner's empirical study of Dow correlations, which response is most appropriate?

A. Accept the normal: equity correlations, like equity returns, are approximately normal over monthly horizons.

B. Accept the lognormal: correlations are bounded below by zero, which the lognormal distribution respects.

C. Reject both: normal, lognormal and beta distributions fit equity correlations poorly; the Johnson SB fits best, and the fit holds across economic states.

D. Reject for recessions only: in recessions correlations are high and need a separate distribution, while the normal is adequate in expansions.

Question MR-9 c.3

New • built from GARP MR Ch.9 (Meissner 9.4)

A model validator asks what kind of distribution the Johnson SB is, since it was identified as the best fit for equity correlations in Meissner's study. Which description is correct?

A. A one-parameter distribution defined on positive values only, like the chi-squared distribution.

B. A three-parameter distribution (location, scale and shape) used to model block maxima in extreme value theory.

C. An unbounded, heavy-tailed distribution like the Pareto, chosen because correlations have fat right tails.

D. A four-parameter distribution with two shape parameters, one location and one scale parameter, flexible enough to fit the bounded, skewed shape of correlations.

Question MR-9 c.4 Bank Q497 • Schweser Module Quiz 9.1 Q5

In estimating correlation matrices, risk managers often assume an underlying distribution for the correlations. Which of the following statements most accurately describes the best-fit distributions for equity correlation distributions, bond correlation distributions, and default probability correlation distributions? The best-fit distributions for the equity, bond, and default probability correlation distributions, respectively, are:

A. lognormal, generalized extreme value, and normal.

B. Johnson SB, generalized extreme value, and Johnson SB.

C. beta, normal, and beta.

D. Johnson SB, normal, and beta.

Question MR-9 c.5 Bank Q495 • AnalystPrep Q.1566

Bonds and their default probabilities also have correlation distributions, just like equity. Which of the following best describes the default probability correlation distribution and the bond correlation distribution?

A. Both the default probability correlation distribution and the bond correlation distribution are best modeled using the Johnson SB distribution.

B. The default probability correlation distribution shows a normal shape and is best modeled using the generalized extreme value distribution, whereas the bond correlation distribution is best modeled using the Johnson SB distribution.

C. The default probability correlation distribution is best modeled using the Johnson SB distribution, whereas the bond correlation distribution is best modeled using the generalized extreme value distribution.

D. Both the default probability correlation distribution and the bond correlation distribution are best modeled using the generalized extreme value distribution.

Question MR-9 c.6 Bank Q496 • BT P2.T5.503.3

Consider the following four statements about the best distributional fit to equity prices, equity correlations, bond correlations, and default probability correlations:

I. Assuming a sharp volatility skew (i.e., continuously decreasing), equity prices are best fit by a lognormal distribution.

II. Equity correlations can be fit by a normal or lognormal distribution.

III. Bond correlations can be fit by a normal or generalized extreme value (GEV) distribution.

IV. Default probability correlations can be fit by a normal or generalized extreme value (GEV) distribution.

Which of the above is (are) TRUE?

A. None are true

B. III only

C. I and II only

D. All are true

Question MR-9 c.7 Workbook D11.2 • S1 • Map #54

A risk analyst at a UK bank is calibrating a correlation simulation engine that will draw correlation scenarios for three books: an equity basket options book, a corporate bond portfolio, and a credit book whose losses depend on default probability correlations. Following Meissner's empirical study, which of the following recommendations should the analyst make?

A. Use the Johnson SB distribution for all three correlation types, since it gave the best fit for equity, bond and default probability correlations alike.

B. Use the Johnson SB distribution for equity and default probability correlations, and the generalised extreme value distribution for bond correlations.

C. Use the Johnson SB distribution for equity and bond correlations, and the generalised extreme value distribution for default probability correlations.

D. Use the Johnson SB distribution for equity correlations in expansions, but switch to the normal distribution in recessions when correlations are high.

Solutions • MR-9 c

Solution MR-9 c.1

Answer: D

Bank Q494 • AnalystPrep Q.2655

Step 1 – Recall the study

Sixty-one distributions were ranked with the Kolmogorov-Smirnov, Anderson-Darling and chi-squared tests on 464,580 Dow correlations. The Johnson SB came out best and stayed best across economic states.
$\Rightarrow$ D.

Interpretation

Note that "chi-squared" appears in the study as a *test*, not as a fitted distribution: a planted association.

Why the other options are wrong

A is incorrect. The chi-squared distribution lives on positive values only, while correlations can be negative; the name is borrowed from the goodness-of-fit test.
B is incorrect. GEV is the best fit for bond correlations, not equity correlations.
C is incorrect. The Pareto is a one-sided, unbounded heavy-tailed distribution; correlations are bounded.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.4

Solution MR-9 c.2

Answer: C

Bank Q493 • GARP book Ch.9 Q9.10

Step 1 – What the tests showed

Normal, lognormal and beta distributions all gave a poor fit to equity correlations; the Johnson SB gave the best fit, and the result was robust to splitting the data by state of the economy.
$\Rightarrow$ The usual normal or lognormal assumption should not be carried over to equity correlations.

Interpretation

GARP's own model answer: "Equity correlations are not well described by normal or lognormal distributions". Correlations are bounded and have a distinctive skewed shape.

Why the other options are wrong

A is incorrect. Equity correlations fit the normal poorly at any horizon in the study.
B is incorrect. Correlations are not bounded below by zero: 22.77% of the Dow values were not positive, and the lognormal fit poorly.
D is incorrect. The economic state changed the fit "slightly but not significantly"; and the normal is poor in every state.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.4 and 9.5

Solution MR-9 c.3

Answer: D

New • built from GARP MR Ch.9

Step 1 – The parameters

Meissner describes the Johnson SB as a versatile distribution with four parameters: γ and δ (shape), μ (location) and σ (scale).

Step 2 – Why it suits correlations
It is bounded and can take skewed shapes, so it can match a variable that lives in an interval and leans to one side, which is what the histogram of Dow correlations looks like.

Interpretation
The distractors describe the distributions the Johnson SB is set against in this and other readings: chi-squared (a test name), GEV (the bond-correlation fit and the EVT block-maxima distribution) and Pareto (EVT tails).

Why the other options are wrong
A is incorrect. Describes a chi-squared distribution: one parameter, positive values only. Correlations can be negative.
B is incorrect. Describes the GEV distribution (the best fit for bond correlations, and the block-maxima distribution of MR-3).
C is incorrect. Describes a Pareto-type tail; correlations are bounded, not unbounded.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.4

Solution MR-9 c.4 Answer: B

Bank Q497 • Schweser Module Quiz 9.1 Q5

Step 1 – One fact per type
Equity: Johnson SB. Bond: generalised extreme value (normal also good). Default probability: Johnson SB.
⇒ Only B lists Johnson SB, GEV, Johnson SB in that order.

Interpretation
Two of three are Johnson SB; bonds are the odd one out.

Why the other options are wrong
A is incorrect. The lognormal fits equity correlations poorly, and the normal is not the best fit for default correlations.
C is incorrect. The beta fits equity correlations poorly; normal is only the second-best fit for bonds.
D is incorrect. Right for equity, but normal is second best for bonds and beta is wrong for default correlations.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.6 and Summary

Solution MR-9 c.5 Answer: C

Bank Q495 • AnalystPrep Q.1566

Step 1 – Default probability correlations
"Similar to equity correlation distribution … replicated best by the Johnson SB distribution."

Step 2 – Bond correlations
"A more normal shape … best fitted with the generalised extreme value distribution and quite well with the normal distribution."
⇒ C matches both.

Interpretation
The question tests exactly the swap the workbook flags: which of bond and default correlations behaves like equity.

Why the other options are wrong

A is incorrect. Applies Johnson SB to bonds too; bond correlations are GEV.
B is incorrect. Reverses the two assignments: it is the *bond* distribution that has the more normal shape and the GEV fit.
D is incorrect. Applies GEV to default probability correlations, which are Johnson SB like equity.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.6

Solution MR-9 c.6 Answer: B

Bank Q496 • BT P2.T5.503.3

Step 1 – Test each statement

I: a lognormal price distribution corresponds to a flat volatility smile; a sharp skew implies a fatter left tail than the lognormal. False.
II: equity correlations fit the normal and lognormal poorly; Johnson SB fits best. False.
III: bond correlations are best fit by GEV and quite well by the normal. **True.**
IV: default probability correlations are best fit by Johnson SB, like equity. False.
$\Rightarrow$ Only III is true.

Interpretation

Statement I borrows from MR-17 (the equity skew); it is there to test whether you keep the two readings apart.

Why the other options are wrong

A is incorrect. Overlooks that III is true.
C is incorrect. Treats a skewed-volatility world as lognormal and equity correlations as normal or lognormal; both are false.
D is incorrect. Applies the bond result to every statement.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.4 to 9.6

Solution MR-9 c.7 Answer: B

Workbook D11.2 • S1

Step 1 – Equity correlations

Of 61 distributions tested (Kolmogorov-Smirnov, Anderson-Darling, chi-squared), the four-parameter Johnson SB gave the best fit; normal, lognormal and beta fit poorly.

Step 2 – Default probability correlations

"The default probability correlation distribution is similar to equity correlation distribution ... and can be replicated best by the Johnson SB distribution."

Step 3 – Bond correlations

"The bond correlation distribution shows a more normal shape and can be best fitted with the generalised extreme value distribution and quite well with the normal distribution."

Interpretation

B is correct. The engine should use Johnson SB for the equity and credit books and the generalised extreme value distribution for the bond portfolio. The bond and default results are preliminary and "currently being verified with a larger sample", so the analyst should flag them for periodic review.

Why the other options are wrong

A is incorrect. Absolute claim. Johnson SB is not the best fit for every type: bond correlations have a more normal shape and are best fitted by the generalised extreme value distribution.
C is incorrect. Sibling swap. The bond and default probability fits are exchanged; it is default probability correlations that resemble equity correlations.
D is incorrect. Scope/condition error. Fitting equity correlations by state of the economy gave results that were "slightly but not significantly different", and the normal distribution fits equity correlations poorly in any state.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. MR-9 a: Describe how equity correlations and correlation volatilities behave throughout various economic states. • **Reference:** Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.4 and 9.6, Summary

MR-11 • Regression Hedging and Principal Component Analysis • Vardeez: IMPORTANT

MR-11 a

P3 CONCEPT

Explain the drawbacks to using a DV01-neutral hedge for a bond position.

Workbook: C1 Regression Hedging, DV01 and PCA • Tier C • Map #23 (5/16)

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C1, Map #23, reported 5 of 16) hits this LO. It is the qualitative counterpart of the regression hedge: the reason the regression hedge exists.
Vardeez (reading level): regression hedge calculation; the same formula gives beta and the face value. No separate pointer for this LO.

Reported exam shapes

- **C1·S3** DV01 hedging: the DV01-neutral amount appears as the planted distractor against the regression amount.
- **C1·S4** The drawbacks of a DV01-neutral hedge and why the regression hedge improves it.

What the questions look like: "which is the drawback" statement questions. Wrong options describe a regression-hedge caveat, a convexity point, or claim the DV01 hedge minimises variance.

1. The DV01-neutral hedge

A trader long a bond (the *position*, P) sells a second bond (the *hedge*, H) so that the two DV01s offset:

$$\boxed{F^H = -F^P \frac{\text{DV01}^P}{\text{DV01}^H}} \qquad \text{risk weight} = \frac{-F^H\,\text{DV01}^H}{F^P\,\text{DV01}^P} = 100\% \text{ always}$$

GARP's example (Tuckman, Ch.6). On 14 May 2021 a market maker buys \$100 million of JNJ 2.450s of 2060 (DV01 0.2124) and hedges with the Treasury 1.625s of 2050 (DV01 0.1910), since there is no 40-year Treasury:

$$F^{30} = -100 \times \frac{0.2124}{0.1910} = -\$111.2 \text{ million (sell)}$$

2. The drawbacks

- **It assumes a one-for-one (parallel) yield move.** The hedge is exact only if both yields rise or fall by the same number of basis points, i.e. a slope of 1 between the two yield changes.
- **That assumption fails in practice.** The JNJ bond trades at a *changing corporate spread* to Treasuries, and 40- and 30-year rates are *not perfectly correlated*. In Vardeez's TIPS version, nominal yields move *more* than one basis point per basis point of real yield, with dispersion around that relationship.
- **The data reject it.** Regressing daily JNJ yield changes on Treasury yield changes gives a slope of 0.842 with standard error 0.051: the 95% interval $(0.740, 0.944)$ excludes 1. On average the JNJ yield moves only 0.842 bp per 1 bp Treasury move, so the DV01 hedge *over-hedges* (sells \$111.2m when \$93.6m is enough, MR-11 d).
- **It says nothing about the risk left over.** A DV01 hedge gives no estimate of the volatility of the hedged P&L. The regression framework does (MR-11 b).

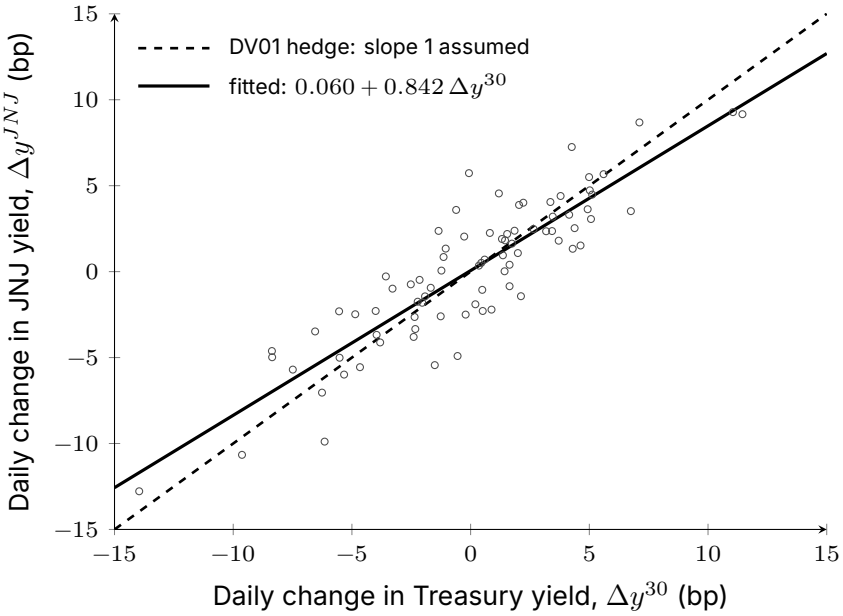


Figure 1. Daily yield changes of a 40-year corporate bond against a 30-year Treasury (82 days, simulated to reproduce GARP's fitted line: $\hat{\beta} = 0.842$, $\hat{\alpha} = 0.060$, standard error 2 bp, $R^2 \approx 78\%$). The cloud is flatter than the dashed 45-degree line the DV01 hedge implicitly assumes, and wide around the solid fitted line: both the slope and the scatter are information the DV01 hedge ignores.

	DV01-neutral hedge	Regression hedge (MR-11 b)
Yield relationship	Assumed one-for-one ($\beta = 1$)	Estimated from history ($\hat{\beta}$)
Face amount	$F^P\,\mathrm{DV01}^P/\mathrm{DV01}^H$	DV01 amount $\times\,\hat{\beta}$
Risk weight	Always 100%	$\hat{\beta}$ (84.2% in GARP's case)
Variance of hedged P&L	Not minimised unless $\beta = 1$	Minimised (in sample)
Estimate of hedged risk	None	$\lvert F^P\mathrm{DV01}^P/100\rvert\times$ regression standard error
Weakness	Parallel-shift assumption	Beta may be unstable; window choice

Exam traps

- **Sibling swap.** "Minimises the variance of the hedged P&L" is the regression hedge's property, not the DV01 hedge's.
- **Caveat swap.** "Relies on a historical regression that may be unstable" is a drawback of the *regression* hedge. The DV01 hedge uses no historical data at all.
- **Convexity distractor.** The LO's drawback is the parallel-shift assumption, which bites even on small daily moves; it is not "fails only for large moves".
- **Face-amount confusion.** A DV01 hedge equates DV01s, not face amounts.
- **Direction.** With $\hat{\beta} < 1$ the DV01 hedge over-hedges; with $\hat{\beta} > 1$ (nominal vs TIPS) it under-hedges.

Practice questions: MR-11 a

Question MR-11 a.1

Bank Q524 • AnalystPrep Q.6502

A large investment fund uses a DV01-neutral hedge to manage the interest rate exposure of its bond portfolio. Over time, market dynamics have shown this approach to be unrealistic. What is a critical drawback of relying on a DV01-neutral hedge?

A. It captures non-parallel yield curve shifts across multiple bond maturities.

B. The strategy inherently assumes that all spread-related risks are neutralized.

C. It assumes yield changes are parallel across maturities, which is rarely the case.

D. The accuracy of the hedge greatly improves with non-static historical correlations.

Question MR-11 a.2

Bank Q525 • AnalystPrep Q.6503 (rebuilt, key corrected)

A trader hedges a corporate bond position with a DV01-neutral position in another fixed-income instrument but finds that the hedge performs poorly when market conditions change. Which of the following is the fundamental limitation of the DV01-neutral approach?

A. It assumes the yields on the position and the hedging instrument change by the same number of basis points, whereas the actual changes can be proportionally larger or smaller and dispersed.

B. It requires estimating a regression of historical yield changes, which may be unstable over time.

C. It accounts for convexity and so over-hedges when yields change by large amounts.

D. It can be implemented only with complex derivative instruments, which limits its use.

Question MR-11 a.3

Bank Q527 • BT L2.T5.32.1 (rebuilt)

A trader hedges a long position in a nominal Treasury bond with TIPS, using a standard DV01-neutral hedge. Which of the following is the main drawback of this hedge?

A. It assumes the nominal yield and the real yield change by the same number of basis points, whereas empirically the nominal yield tends to move by more than one basis point per basis point of real-yield change.

B. It requires the hedger to estimate the historical volatility of yield differences, which makes it harder to compute than a regression hedge.

C. It can only be applied when the bond and the hedging instrument have identical maturities.

D. It ignores the face value of the hedging instrument, so the hedge cannot be sized.

Question MR-11 a.4

Bank Q526 • Schweser Module Quiz 11.1 Q1

If a trader is creating a fixed-income hedge, which hedging methodology would be least effective if the trader is concerned about the dispersion of the change in the nominal yield for a particular change in the real yield?

A. One-variable regression hedge.

B. DV01 hedge.

C. Two-variable regression hedge.

D. Principal components hedge.

Question MR-11 a.5

New • built from GARP Ch.11 (Tuckman), Tables 11.1 and 11.2

A market maker buys USD 100 million face of the JNJ 2.450s of 2060 (DV01 0.2124) from a client and plans to sell USD 111.2 million of the Treasury 1.625s of 2050 (DV01 0.1910), a DV01-neutral hedge. A risk analyst regresses daily changes in the JNJ yield on daily changes in the Treasury yield over the past four months:

	Value	Std. error
Constant ($\hat{\alpha}$)	0.060	0.223
Slope ($\hat{\beta}$)	0.842	0.051
R^2	77.5%	
Standard error of regression	2.00 bp	

Which of the following statements best describes the planned DV01-neutral hedge?

A. It over-hedges: the 95% interval for the slope, about 0.740 to 0.944, excludes one, so the data reject the parallel-shift assumption on which the hedge rests.

B. It under-hedges: because the R^2 is only 77.5%, the Treasury position should be scaled up by $1/0.775$ to cover the unexplained variance.

C. It is appropriate: the 95% interval for the constant includes zero, so the data are consistent with parallel shifts in the two yields.

D. It is appropriate: equal DV01s on the two legs make it the hedge that minimizes the variance of the hedged P&L.

Question MR-11 a.6 New • built from GARP Ch.11 (Tuckman), 11.1

A trader at a dealer hedges a long position in a 40-year corporate bond by selling a 30-year Treasury bond in a DV01-neutral amount. The hedge has lost money on several days when Treasury yields moved by only a few basis points. Which of the following is the most accurate explanation?

A. DV01 is a first-order measure, so the hedge fails only when yields move by large amounts; for small daily moves it is exact.

B. The hedge assumes the two yields change by the same number of basis points, but the corporate bond trades at a changing spread to Treasuries and 40- and 30-year rates are not perfectly correlated.

C. The hedge sets the face amounts of the two bonds equal, so it ignores the difference in their price sensitivities to yield changes.

D. The hedge relies on a regression estimated over a historical window, so it becomes unreliable when the relationship between the two yields changes.

Solutions • MR-11 a

Solution MR-11 a.1

Answer: C

Bank Q524 • AnalystPrep Q.6502

Step 1 – What the DV01 hedge equates

The hedge sizes the offsetting position so that its DV01 equals the DV01 of the position.
$\Rightarrow$ That neutralises risk only if both yields move by the same number of basis points.

Step 2 – Does that hold?

Yields of different maturities (or nominal and real yields) do not move one-for-one, so the hedge leaves residual risk.

Interpretation

The drawback is the built-in assumption of parallel yield changes.

Why the other options are wrong

A is incorrect. Reversed: a DV01 hedge cannot capture non-parallel shifts. That is its weakness.
B is incorrect. A DV01 hedge addresses yield (rate) risk only; nothing about it neutralises spread risk, so this confuses it with a full-risk hedge.
D is incorrect. A DV01 hedge uses no historical correlations at all; time-varying relationships are a concern for regression hedges.

LO: MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1

Solution MR-11 a.2

Answer: A

Bank Q525 • AnalystPrep Q.6503

Step 1 – The DV01 hedge's implicit slope

Hedge DV01 equals position DV01. This is exact only if $\Delta y^P = \Delta y^H$, a slope of one.

Step 2 – Reality

Actual yield changes can be proportionally larger or smaller (slope $\neq 1$) and scattered around that slope, so the position ends up under- or over-hedged.

Interpretation

A states both halves of the problem: the size of the move differs (beta) and the relationship is noisy (dispersion).

Why the other options are wrong

B is incorrect. Estimating a regression on historical data is the *improvement* (regression hedge); instability of beta is that method's caveat.
C is incorrect. DV01 is a first-order measure and contains no convexity adjustment.
D is incorrect. A DV01 hedge works with ordinary bonds or swaps; no derivatives are needed.

LO: MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1

Solution MR-11 a.3

Answer: A

Bank Q527 • BT L2.T5.32.1

Step 1 – Implicit assumption

DV01-neutral means $\Delta y^{\text{nominal}} = \Delta y^{\text{real}}$, i.e. $\beta = 1$.

Step 2 – Evidence

Nominal yields tend to move by more than one basis point per basis point change in real yields, and the relationship is noisy.
$\Rightarrow$ The DV01 hedge buys too few TIPS; a regression hedge scales it up by $\hat{\beta} > 1$.

Interpretation

This is Vardeez's version of the same drawback: nominal yield changes are larger and more dispersed than real yield changes.

Why the other options are wrong

B is incorrect. Reversed: the DV01 hedge needs no volatility estimate; the regression hedge is the one that uses historical data.
C is incorrect. DV01 hedges are used across different maturities and instruments all the time (the TIPS hedge itself is an example).
D is incorrect. The face amount of the hedge is exactly what the DV01 calculation solves for.

LO: MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. • **Reference:** Tuckman and Serrat 4e, Ch.6; SchweserNotes 2026 Book 1, LO 11.a

Solution MR-11 a.4 — Answer: B

Bank Q526 • Schweser Module Quiz 11.1 Q1

Step 1 – The concern

Nominal yield changes are not one-for-one with real yield changes and are dispersed around the relationship.

Step 2 – Which method ignores it

Only the DV01 hedge hard-wires a one-for-one move. One- and two-variable regression hedges estimate the relationship from data, and a PCA hedge uses the empirical factor structure of yield changes.

Interpretation

The least effective method is the one that assumes the relationship away.

Why the other options are wrong

A is incorrect. A one-variable regression hedge adjusts the DV01 hedge by the estimated beta, addressing the nominal/real relationship directly.
C is incorrect. A two-variable regression hedge also uses estimated sensitivities.
D is incorrect. A PCA hedge is built from the empirical covariance structure of rates.

LO: MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. • **Reference:** SchweserNotes 2026 Book 1, Reading 11, LO 11.a

Solution MR-11 a.5 — Answer: A

New • GARP Ch.11, Tables 11.1 and 11.2

Step 1 – Test the slope against one

95% interval: $0.842 \pm 2 \times 0.051 = (0.740,\ 0.944)$. One lies outside it.
$\Rightarrow$ The hypothesis of parallel shifts ($\beta = 1$) is rejected: the JNJ yield moves only about 0.842 bp per Treasury basis point.

Step 2 – Direction of the error

The regression hedge sells $111.2 \times 0.842 = \$93.6$ million. The DV01 hedge sells \$111.2 million, about \$17.6 million too much.
$\Rightarrow$ With $\hat{\beta} < 1$ the DV01-neutral hedge over-hedges.

Interpretation

The constant's interval, 0.060 ± 0.446, does include zero, but the constant has nothing to do with parallel shifts and does not enter the hedge. The R^2 and standard error describe how much risk is left, not the hedge size.

Why the other options are wrong

B is incorrect. R^2 measures the share of variance explained; it never scales the hedge. Dividing by 0.775 would only increase the over-hedge.

C is incorrect. Tests the wrong coefficient: parallel shifts concern the slope, not the constant.

D is incorrect. Sibling swap: variance minimisation is the property of the regression hedge. Equal DV01s minimise variance only if $\beta = 1$, which the data reject.

LO: MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, Tables 11.1 and 11.2, eq. 11.9

Solution MR-11 a.6 Answer: B

New • GARP Ch.11, 11.1

Step 1 – The assumption

A DV01-neutral hedge is exact only if both yields move by the same number of basis points.

Step 2 – Why it fails here

GARP gives two reasons for this exact pair: the corporate bond trades at a changing spread to Treasuries, and 40-year and 30-year rates are not perfectly correlated.

$\Rightarrow$ Losses on small moves are the signature of a wrong relationship between the yields, not of a second-order (convexity) effect.

Interpretation

The losses on days of small Treasury moves point to basis risk between the two yields, which a DV01 hedge cannot address.

Why the other options are wrong

A is incorrect. The parallel-shift assumption fails on small moves too; convexity is not this LO's drawback.

C is incorrect. A DV01 hedge equates DV01s, not face amounts; here it sells \$111.2m against \$100m.

D is incorrect. Caveat swap: dependence on a historical regression is a weakness of the regression hedge, not of the DV01 hedge.

LO: MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1

MR-11 • Regression Hedging and Principal Component Analysis • Vardeez: IMPORTANT

MR-11 b

P3 CONCEPT

Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge.

Workbook: C1 Regression Hedging, DV01 and PCA • Tier C • Map #23 (5/16)

WHAT IS TESTED HERE

Why P3. Tier C workbook topic (C1, Map #23).
Vardeez (reading level): regression hedge calculation. For this LO the point is *why* the regression hedge is better.

Reported exam shape

- **C1·S4** Drawbacks of a DV01-neutral hedge and why the regression hedge improves it.

What the questions look like: "best argument for the regression hedge", "what is the risk weight", "what does the hedge assume about beta". The two calculations (beta and the face amount) are tested under MR-11 c and d.

1. The regression model

Regress daily changes in the yield of the bond being hedged on daily changes in the yield of the hedging bond:

$$\Delta y_t^P = \alpha + \beta\, \Delta y_t^H + \varepsilon_t$$

- **Least squares** picks $\hat{\alpha}, \hat{\beta}$ to minimise $\sum_t \hat{\varepsilon}_t^2$. Squaring stops positive and negative errors offsetting and penalises large errors heavily.
- **Assumptions:** the model is a true description; errors are identically distributed, independent of each other and uncorrelated with the independent variable. Then the estimates are linear, unbiased, consistent and efficient.
- **The dependent variable is the yield of the bond being hedged.** Reversing the regression gives a different hedge (MR-11 g, P4).

2. Reading the output (GARP Table 11.2, JNJ on 30-year Treasury)

Statistic	Value	What it tells you
$\hat{\beta}$ (std error)	0.842 (0.051)	JNJ yield moves 0.842 bp per Treasury bp; 95% CI (0.740, 0.944) excludes 1
$\hat{\alpha}$ (std error)	0.060 (0.223)	Not significantly different from 0; plays no part in the hedge
R^2	77.5%	Share of JNJ yield-change variance explained; $\sqrt{R^2} = 88.0\%$ correlation
Standard error	2.00 bp	Volatility of the residual; drives hedged P&L volatility

3. How the regression hedge improves the DV01 hedge

1. **It uses the estimated relationship instead of assuming one-for-one.** The hedge is the DV01 hedge scaled by $\hat{\beta}$, and the **risk weight** (hedge DV01 as a share of position DV01) equals $\hat{\beta}$, not 100%.

$$F^H = -F^P \frac{\text{DV01}^P}{\text{DV01}^H}\hat{\beta} \quad \Rightarrow \quad \text{JNJ: } -100 \times \frac{0.2124}{0.1910} \times 0.842 = -\$93.6\text{m (vs } \$111.2\text{m)}$$

2. **It minimises the variance of the hedged P&L.** This is GARP's "best argument": the hedge is optimal in a variance sense whether or not yields move exactly along the regression line.
3. **It gives an estimate of the hedged position's risk:** $\sigma_{P\&L} = \left|F^P \,\mathrm{DV01}^P/100\right| \times \sigma_\varepsilon = \$212{,}400 \times 2 = \$424{,}800$ per day. The trader can compare this with the expected gain: five basis points over a day may justify 2 bp/day of risk; 1.5 bp over a week probably does not.

Vardeez's TIPS version. Selling \$100m T-bonds (DV01 0.068) against TIPS (DV01 0.084): DV01 hedge $100 \times 0.068/0.084 = \$80.95$m TIPS; with $\hat{\beta} = 1.0198$ the regression hedge buys \$82.55m. Beta above 1 means *more* of the hedging bond.

4. What the regression hedge still assumes

- **A constant beta** over the hedging horizon. In practice beta varies, so estimate it over several periods and compare.
- **The estimation window is a judgement.** Too short a window may not give statistically reliable estimates; too long a window may include less relevant data; sometimes an earlier episode (a past Fed tightening) is more relevant than the recent past.
- **In-sample results.** Check out of sample. GARP's JNJ hedge held up: out-of-sample residual standard error 1.5 bp versus 2.0 in sample. Poor out-of-sample performance questions the stability of the coefficients.
- **Hedging remains imperfect.** An R^2 of 77.5% leaves much of the variance unhedged.

Exam traps

- **Risk weight.** DV01 hedge: always 100%. Regression hedge: $\hat{\beta}$. Not R^2, not the standard error.
- **Constant beta is the assumption,** not $\beta = 1$ (that is the DV01 hedge) and not a daily-changing beta.
- **Residual risk remains.** The regression does not force the residual to zero; it minimises its variance.
- **Historical, not forecast.** The hedge uses past yield changes; it cannot be "always exactly right".
- **Absolutes about windows** ("longer is always better", "the latest data are always most relevant") are wrong.

Practice questions: MR-11 b

Question MR-11 b.1

Bank Q528 • AP Mock 2 Q15 (rebuilt)

A fixed-income analyst currently hedges a long bond position with a DV01-neutral hedge, which implicitly assumes that the hedging instrument's yield and the bond's yield change by the same number of basis points. How does a regression hedge improve on this approach?

A. It sets the hedge amount using the historical relationship between the two yield changes (the regression slope), rather than assuming a one-for-one move.

B. It replaces the historical data with a forecast of tomorrow's yield changes, so the hedge is always exactly right.

C. It eliminates basis risk completely, because the regression residual is forced to zero.

D. It sizes the hedge from the average historical price of the two securities rather than from yield changes.

Question MR-11 b.2

Bank Q529 • AnalystPrep Q.6505 (rebuilt)

Compared with a DV01-neutral hedge, which of the following best describes the "risk weight" applied to the hedging instrument in a single-variable regression hedge?

A. It is fixed at 1.0 because the hedge is built to be DV01-neutral.

B. It equals the regression slope coefficient β from historical yield changes, which can differ from 1.0.

C. It equals the regression R-squared, the proportion of variance explained.

D. It equals the regression standard error, expressed in basis points.

Question MR-11 b.3

Bank Q530 • BT Mock Q31

In comparison to a yield-based DV01 hedge, which of the following is the best argument in favor of a regression hedge?

A. The regression hedge minimizes the profit and loss (P&L) variance of the hedged position.

B. The regression hedge is based on economic theory and future developments rather than empirical analysis.

C. The regression hedge does not require any out-of-sample tests, which are often unreliable if the sample is small.

D. The regression hedge solves for a risk-weight which qualifies as a valid risk-weighted asset under Basel regulations.

Question MR-11 b.4

Bank Q532 • Schweser Module Quiz 11.1 Q3

What is a key advantage of using a regression hedge to fine-tune a DV01 hedge?

A. It assumes that term structure changes are driven by one factor.

B. The proper hedge amount may be computed for any assumed change in the term structure.

C. Bond price changes and returns can be estimated with proper measures of price sensitivity.

D. It gives an estimate of the hedged portfolio's volatility over time.

Question MR-11 b.5 Bank Q533 • Schweser Module Quiz 11.1 Q4 (rebuilt)

A trader uses a regression of changes in nominal yields on changes in real yields to set a single-variable regression hedge. Which assumption about the slope coefficient (beta) underlies the hedge?

A. Beta is constant over time, so the historical estimate is assumed to hold over the hedging horizon.

B. Beta equals 1.0, so the hedge is the same as the DV01-neutral hedge.

C. Beta is zero, so real yield changes do not affect nominal yields.

D. Beta changes each day to match the latest yield move.

Question MR-11 b.6 New • built from GARP Ch.11 (Tuckman), 11.1

The head of a rates desk is reviewing an analyst's regression hedge, which was estimated from four months of daily yield changes. Which of the following statements should the head of the desk accept?

A. A longer estimation window is always preferable, because more observations reduce the standard errors of the regression coefficients.

B. Because least-squares estimates are unbiased, the in-sample standard error of the regression is a reliable guide to future hedged P&L volatility, so out-of-sample tests add little.

C. Too short a window may not give statistically reliable estimates and too long a window may include less relevant data, and an earlier episode can be more relevant than the recent past.

D. The most recent period is the relevant one by definition, because a regression hedge assumes that beta changes each day with the latest data.

Solutions • MR-11 b

Solution MR-11 b.1 — Answer: A

Bank Q528 • AP Mock 2 Q15

Step 1 – DV01 hedge

$F^H = F^P \times \text{DV01}^P/\text{DV01}^H$, which assumes $\Delta y^P = \Delta y^H$.

Step 2 – Regression hedge

Estimate $\Delta y^P = \alpha + \beta \Delta y^H + \varepsilon$ by least squares and scale the DV01 hedge by $\hat{\beta}$.

$\Rightarrow$ The hedge reflects how the two yields have actually moved together.

Interpretation

The R^2 and standard error then say how much risk is removed and how much residual (basis) risk remains.

Why the other options are wrong

B is incorrect. The method uses history, not forecasts, and cannot guarantee an exact hedge.

C is incorrect. The residual remains; the standard error of the regression measures it. Least squares minimises its variance, it does not force it to zero.

D is incorrect. The regression is on yield changes, not on average prices.

LO: MR-11 b: Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eqs. 11.1 and 11.8

Solution MR-11 b.2 — Answer: B

Bank Q529 • AnalystPrep Q.6505

Step 1 – Define the risk weight

Risk weight $= \dfrac{-F^H \,\text{DV01}^H}{F^P \,\text{DV01}^P}$: the DV01 of the hedge as a share of the DV01 of the position.

Step 2 – Substitute the regression hedge

With $F^H = -F^P\,(\text{DV01}^P/\text{DV01}^H)\,\hat{\beta}$ the ratio collapses to $\hat{\beta}$.

$\Rightarrow$ GARP's JNJ hedge has a risk weight of 84.2%; a DV01 hedge always has 100%.

Interpretation

The DV01-neutral hedge is the special case $\hat{\beta} = 1$.

Why the other options are wrong

A is incorrect. A weight fixed at 1.0 describes the DV01-neutral hedge.

C is incorrect. R^2 measures fit, not hedge size.

D is incorrect. The standard error measures residual dispersion and drives hedged P&L volatility, not the hedge size.

LO: MR-11 b: Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eq. 11.10

Solution MR-11 b.3 — Answer: A

Bank Q530 • BT Mock Q31

Step 1 – What least squares minimises

The sum of squared residuals.

Step 2 – What the residual is

With the position's yield regressed on the hedge's yield, the residual times the position DV01 is

the hedged P&L.
$\Rightarrow$ Minimising squared residuals minimises the variance of hedged P&L (GARP Appendix A6.1).

Interpretation

GARP calls this the best argument for the regression hedge: it holds whether or not yields move exactly by the regression model.

Why the other options are wrong

B is incorrect. The method is empirical, built on historical data.
C is incorrect. Out-of-sample tests matter *more*, because beta may be unstable.
D is incorrect. The risk weight is the regression beta; it has nothing to do with Basel risk-weighted assets.

LO: MR-11 b: Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eq. 11.11 and Appendix A6.1

Solution MR-11 b.4 — Answer: D

Bank Q532 • Schweser Module Quiz 11.1 Q3

Step 1 – What the regression produces

A residual standard error, in basis points.

Step 2 – Turn it into money

$\sigma_{P\&L} = \left|F^P \, \text{DV01}^P / 100\right| \times \sigma_\varepsilon$.
$\Rightarrow$ The trader can compare the expected gain with this volatility before doing the trade.

Interpretation

Vardeez: "The advantage of a regression framework is that it provides an estimate of a hedged portfolio's volatility."

Why the other options are wrong

A is incorrect. A one-factor assumption is a limitation shared with DV01 hedging, not an advantage.
B is incorrect. The regression is estimated from history; it cannot give the proper hedge for any assumed term structure change.
C is incorrect. This describes duration or DV01 price sensitivity in general, not a benefit specific to regression hedging.

LO: MR-11 b: Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge. • **Reference:** SchweserNotes 2026 Book 1, Reading 11, LO 11.b; Tuckman and Serrat 4e, Ch.6, eq. 11.12

Solution MR-11 b.5 — Answer: A

Bank Q533 • Schweser Module Quiz 11.1 Q4

Step 1 – How the hedge is set

The face amount is scaled by $\hat{\beta}$ estimated from past yield changes.

Step 2 – What using it going forward implies

Applying a historical estimate to the future presumes the relationship is stable: beta is constant.

Interpretation

Because beta can vary, practitioners estimate it over several periods and test it out of sample.

Why the other options are wrong

B is incorrect. Beta equal to 1 is the DV01-neutral special case.
C is incorrect. A beta of zero would mean no relationship and no point hedging with TIPS at all.

D is incorrect. A daily-changing beta would make historical estimation pointless; stability is the assumption.

LO: MR-11 b: Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge. • **Reference:** SchweserNotes 2026 Book 1, Reading 11, LO 11.b

Solution MR-11 b.6 Answer: C

New • GARP Ch.11, 11.1

Step 1 – GARP on the estimation window

"Choosing the length of the observation or estimation window is also part of the art of regression hedging. Too short a window might fail to furnish statistically reliable estimates, but too long a window might include less relevant historical data."

Step 2 – GARP on which period is relevant

If the Fed is expected to raise rates, an earlier tightening episode may be more relevant than a recent period of unchanged rates.

Interpretation

C states the trade-off and the possibility that an earlier period is more relevant. The other three statements each turn a judgement into an absolute.

Why the other options are wrong

A is incorrect. Absolute claim: more data reduce standard errors but may bring in irrelevant regimes.

B is incorrect. Unbiasedness is an in-sample property under the model's assumptions; poor out-of-sample performance is exactly how unstable coefficients show up, so out-of-sample tests matter.

D is incorrect. The regression hedge assumes a *constant* beta, and the recent past is not relevant "by definition".

LO: MR-11 b: Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1 (estimation window, out-of-sample analysis)

MR-11 • Regression Hedging and Principal Component Analysis • Vardeez: IMPORTANT

MR-11 c

P3 CALC

Calculate the regression hedge adjustment factor, beta.

Workbook: C1 Regression Hedging, DV01 and PCA • Tier C • Map #23 (5/16)

WHAT IS TESTED HERE

Why P3. Tier C workbook topic (C1, Map #23, reported: "hedge ratio given beta, alpha, DV01"). **Vardeez:** regression hedge: beta (the same formula gives beta and the face value).

Reported exam shape

- **C1·S1** Hedge ratio from a regression output that lists alpha, beta and both DV01s, with alpha as the tempting extra input.

What the questions look like: beta read from a regression table, or computed from data (covariance over variance, or correlation times a volatility ratio), and tests of whether beta differs from one.

1. Three ways to get beta

1. **Read it off the regression output.** $\hat{\beta}$ is the slope coefficient on the hedge yield change. Ignore the constant: it does not enter the hedge (C1·S1).
2. **From data:** the least-squares slope of Δy^P on Δy^H,

$$\hat{\beta} = \frac{\text{Cov}(\Delta y^P, \Delta y^H)}{\text{Var}(\Delta y^H)} = \rho \frac{\sigma_P}{\sigma_H}$$

   Use the same convention (n or $n-1$) for both covariance and variance. The denominator is the variance of the *hedge* yield change.
3. **From R^2 and volatilities:** in a one-variable regression $R^2 = \rho^2$, so $\rho = \sqrt{R^2}$ (sign from the relationship) and $\hat{\beta} = \sqrt{R^2}\,\sigma_P/\sigma_H$.

2. Is beta different from one?

A DV01 hedge is the special case $\beta = 1$. Test it with the coefficient's standard error:

$$t = \frac{\hat{\beta} - 1}{\text{se}(\hat{\beta})}, \qquad 95\%\text{ interval} \approx \hat{\beta} \pm 2\,\text{se}(\hat{\beta})$$

GARP: $0.842 \pm 2(0.051) = (0.740,\ 0.944)$ excludes 1, so parallel shifts are rejected. If the standard error were 0.193 instead, the interval $(0.46,\ 1.23)$ would include 1 and the DV01 hedge could not be rejected.

Correction to the source text. GARP's chapter gives the constant's 95% interval as "0.060 plus or minus 2 times 0.223, or $(-0.386,\ 0.446)$". The upper bound is $0.060 + 0.446 = 0.506$, so the interval is $(-0.386,\ 0.506)$. The conclusion (it includes zero) is unchanged.

Formula card: MR-11 c

Regression	$\Delta y_t^P = \alpha + \beta\,\Delta y_t^H + \varepsilon_t$ (P = bond hedged, H = hedge)
Beta from data	$\hat{\beta} = \mathrm{Cov}(\Delta y^P, \Delta y^H)/\mathrm{Var}(\Delta y^H) = \rho\,\sigma_P/\sigma_H$
Beta from R^2	$\rho = \sqrt{R^2}$, $\hat{\beta} = \sqrt{R^2}\,\sigma_P/\sigma_H$
Test of $\beta = 1$	$t = (\hat{\beta} - 1)/\mathrm{se}(\hat{\beta})$; reject if $\lvert t\rvert > 2$ (about 1.96 to 1.99)
Risk weight	$= \hat{\beta}$ (DV01 hedge: 100%)
Hedge (MR-11 d)	$F^H = -F^P\,(\mathrm{DV01}^P/\mathrm{DV01}^H)\,\hat{\beta}$

Exam traps

- **Alpha as red herring.** Adding $\hat{\alpha}$ to $\hat{\beta}$ (or using it at all) is a named mistake.
- **Wrong denominator.** Dividing the covariance by the variance of the bond being hedged gives the reverse-regression slope.
- **Mixed conventions.** Population covariance with sample variance (or the reverse).
- R^2 **is not** ρ. Take the square root before multiplying by the volatility ratio.
- **Inverted volatility ratio.** $\rho\,\sigma_H/\sigma_P$ instead of $\rho\,\sigma_P/\sigma_H$.
- **Testing the wrong coefficient.** Parallel shifts are about $\beta = 1$, not $\alpha = 0$.

Practice questions: MR-11 c

Question MR-11 c.1

Bank Q537 • AP Mock 3 Q56 (rebuilt)

An analyst is calculating the hedge ratio for a bond portfolio using bond futures:

Correlation between portfolio and futures yield changes, ρ	0.60
Standard deviation of bond portfolio yield changes, σ_p	0.08
Standard deviation of bond futures yield changes, σ_h	0.05

What is the minimum-variance (regression) hedge ratio, β?

A. 0.38

B. 0.96

C. 1.60

D. 2.67

Question MR-11 c.2 Bank Q534 • AnalystPrep Q.6508 (rebuilt, key corrected)

A portfolio manager hedges a corporate bond portfolio with a hedging instrument using a regression hedge. Daily changes in yield (in bps) over four days are:

Day	Y_p (corporate bond portfolio)	Y_h (hedging instrument)
1	1.2	2.0
2	1.8	2.5
3	0.8	1.5
4	2.4	3.0

Using sample $(n-1)$ statistics, what is the regression hedge adjustment factor, β, of the portfolio yield change on the hedge yield change?

A. 0.81

B. 0.92

C. 1.08

D. 1.44

Question MR-11 c.3 New • built from GARP Ch.11 (Tuckman), 11.1

A risk analyst has four months of daily yield changes for a 40-year corporate bond, which is to be hedged, and for a 30-year Treasury bond, the hedging instrument. The summary statistics are:

Standard deviation of corporate yield changes	4.20 bp
Standard deviation of Treasury yield changes	4.40 bp
Mean daily change, corporate / Treasury	0.10 bp / 0.05 bp
R^2 of the regression of corporate on Treasury yield changes (positive slope)	77.5%

Which of the following is closest to the regression hedge adjustment factor, β?

A. 0.740

B. 0.840

C. 0.880

D. 0.922

Question MR-11 c.4

Workbook C1.1 • S1, S2, S3 • Map #23

A relative value trader at a Canadian dealer buys CAD 50 million face amount of a 30-year provincial bond for a client and hedges the interest rate risk by selling the Government of Canada 30-year benchmark bond. The desk's risk analyst supplies the following:

- DV01 per 100 face: provincial bond 0.1835; Government of Canada bond 0.2010
- Regression of daily changes in the provincial yield on daily changes in the Government of Canada yield (bp), 90 observations: constant 0.08 (std. error 0.21); slope 0.91 (std. error 0.04); R^2 81%; standard error of the regression 1.6 bp

Which of the following is closest to the face amount of the Government of Canada bond the trader should sell under a regression hedge, and to the standard deviation of the daily P&L of the hedged position?

A. CAD 41.54 million; CAD 133,588

B. CAD 41.54 million; CAD 146,800

C. CAD 45.19 million; CAD 146,800

D. CAD 45.65 million; CAD 146,800

Question MR-11 c.5

Bank Q535 • BT P2.T5.22.14.3

Erica works for a market maker who is going to hedge the purchase of Johnson & Johnson (JNJ) bonds by selling Treasury bonds. To compare a yield-based DV01 hedge to a regression hedge, Erica conducts a single-variable regression; the summary results are:

No. of observations	82	
R-squared	77.5%	
Standard error	2.10	
Regression coefficients	**Value**	**Std. error**
Constant (α)	0.060	0.223
Change in 30y Treasury yield (β)	0.842	0.193

Erica is interested in the advisability of a yield-based DV01 hedge, which assumes parallel shifts in the yields of the two bonds. Her hypothesis test assumes a two-sided 95.0% confidence level. Can she accept the null hypothesis of parallel shifts in yields of the two bonds?

A. She rejects the hypothesis of parallel shifts in the yields of the two bonds because the true beta is not one.

B. She accepts (i.e., cannot reject) the hypothesis of parallel shifts in the yields because the true beta might be one.

C. She rejects the hypothesis of parallel shifts in the yields of the two bonds because the alpha is not zero.

D. From a risk-return perspective, the trade is not advised because the standard error of the regression exceeds two.

Question MR-11 c.6 Bank Q536 • BT P2.T5.321.3

A trader shorts $100.0 million of nominal US Treasury 3 5/8s bonds and hedges with a purchase of a face amount of 1 7/8s US TIPS based on the regression results of a single-variable hedge, shown here:

Bond	Yield	DV01
Nominal 3 5/8s	3.275%	0.0400
TIPS 1 7/8s	1.237%	0.0700
Regression of Δ nominal yield (bp) on Δ real yield (bp), 229 observations		
R-squared	68.0%	
Standard error (bp)	2.200	
Constant (α)	0.0200	(std. error 0.2529)
Change in real yield (β)	1.3000	(std. error 0.0510)

Each of the following is true about this regression hedge EXCEPT which is false?

A. The regression model estimates a correlation between changes in the nominal and real yields of about 82.5%.

B. We can be 95% confident that the true value of beta is unity (1.0).

C. The daily volatility of the P&L of the hedged portfolio is about $88,000.

D. The difference between a DV01 hedge and a regression hedge is about $17.14 million face amount of the TIPS.

Solutions • MR-11 c

Solution MR-11 c.1

Answer: B

Bank Q537 • AP Mock 3 Q56

Step 1 – Beta as correlation times volatility ratio

$$\hat{\beta} = \frac{\text{Cov}}{\sigma_h^2} = \frac{\rho\,\sigma_p\,\sigma_h}{\sigma_h^2} = \rho\,\frac{\sigma_p}{\sigma_h}$$

Step 2 – Plug in

$$\hat{\beta} = 0.60 \times \frac{0.08}{0.05} = 0.60 \times 1.60 = 0.96$$

$\Rightarrow$ The portfolio yield moves 0.96 bp per basis point move in the futures yield.

Interpretation

The volatility of the position being hedged goes on top; the volatility of the hedge goes underneath.

Why the other options are wrong

A is incorrect. Inverted the volatility ratio: $0.60 \times 0.05/0.08 = 0.375$.
C is incorrect. Ignored the correlation: $0.08/0.05 = 1.60$.
D is incorrect. Divided the volatility ratio by the correlation: $1.60/0.60 = 2.67$.

LO: MR-11 c: Calculate the regression hedge adjustment factor, beta. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1

Solution MR-11 c.2

Answer: C

Bank Q534 • AnalystPrep Q.6508

Step 1 – Means

$\bar{Y}_h = 2.25$, $\bar{Y}_p = 1.55$.

Step 2 – Deviations and products

Y_h: $-0.25,\ 0.25,\ -0.75,\ 0.75$; Y_p: $-0.35,\ 0.25,\ -0.75,\ 0.85$.

$$\text{Cov} = \frac{0.0875 + 0.0625 + 0.5625 + 0.6375}{3} = \frac{1.35}{3} = 0.45, \qquad \text{Var}(Y_h) = \frac{1.25}{3} = 0.4167$$

Step 3 – Slope

$$\hat{\beta} = \frac{0.45}{0.4167} = 1.08$$

$\Rightarrow$ The portfolio yield moves 1.08 bp per bp of the hedge yield.

Interpretation

The $n-1$ convention cancels when it is used for both moments; only mixing conventions changes the answer.

Why the other options are wrong

A is incorrect. Population covariance ($1.35/4 = 0.3375$) over the sample variance: 0.81.
B is incorrect. Reverse regression: covariance over $\text{Var}(Y_p) = 0.49$: 0.92.
D is incorrect. Sample covariance over the population variance ($1.25/4 = 0.3125$): 1.44.

LO: MR-11 c: Calculate the regression hedge adjustment factor, beta. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eq. 11.5

Solution MR-11 c.3

Answer: B

New • GARP Ch.11, 11.1

Step 1 – Correlation from R^2

In a one-variable regression $R^2 = \rho^2$: $\rho = \sqrt{0.775} = 0.8803$ (positive slope).

Step 2 – Beta

$$\hat{\beta} = \rho \frac{\sigma_{\text{corp}}}{\sigma_{\text{Tsy}}} = 0.8803 \times \frac{4.20}{4.40} = 0.840$$

$\Rightarrow$ The corporate yield moves about 0.84 bp per Treasury basis point, close to GARP's 0.842 for JNJ.

Interpretation

The mean daily changes are a red herring: they determine the intercept, not the slope.

Why the other options are wrong

A is incorrect. Used R^2 instead of ρ: $0.775 \times 4.20/4.40 = 0.740$.
C is incorrect. Took the correlation as the hedge ratio: $\sqrt{0.775} = 0.880$.
D is incorrect. Inverted the volatility ratio: $0.8803 \times 4.40/4.20 = 0.922$, which is what a reverse regression of the Treasury on the corporate would give.

LO: MR-11 c: Calculate the regression hedge adjustment factor, beta. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1 (in a one-variable regression R^2 is the squared correlation)

Solution MR-11 c.4

Answer: B

Workbook C1.1 • S1, S2, S3

Step 1 – DV01 ratio

$$\frac{\text{DV01}^P}{\text{DV01}^H} = \frac{0.1835}{0.2010} = 0.91294$$

$\Rightarrow$ Per unit of face, the provincial bond moves 0.913 times as much as the benchmark for the same yield change.

Step 2 – Regression hedge face amount

$$F^H = -F^P \frac{\text{DV01}^P}{\text{DV01}^H} \hat{\beta} = -50 \times 0.91294 \times 0.91 = -41.54$$

$\Rightarrow$ Sell CAD 41.54 million. The hedge ratio per unit of face is $0.91294 \times 0.91 = 0.831$, and the risk weight (hedge DV01 over position DV01) equals $\hat{\beta} = 91\%$.

Step 3 – DV01 of the hedged position

$$\frac{F^P \, \text{DV01}^P}{100} = \frac{50{,}000{,}000 \times 0.1835}{100} = 91{,}750$$

Step 4 – Hedged P&L volatility

$$\sigma_{P\&L} = 91{,}750 \times 1.6 = 146{,}800$$

$\Rightarrow$ Each 1 bp residual (the provincial yield moving 1 bp more than 0.91 times the benchmark change) costs CAD 91,750, so a 1.6 bp residual volatility gives about CAD 146,800 per day.

Interpretation

The slope's 95% interval, $0.91 \pm 2(0.04) = (0.83,\ 0.99)$, excludes one, so the data reject parallel shifts and the DV01-neutral hedge over-hedges. The constant's interval, 0.08 ± 0.42, includes zero and the constant plays no part in the hedge. Whether CAD 146,800 per day is acceptable depends on the spread the trader earns and the holding period.

Why the other options are wrong

A is incorrect. Correct face amount, but computed the P&L volatility with the DV01 of the hedge leg: $41.54\text{m} \times 0.2010/100 \times 1.6 = 133{,}588$, which is $\hat{\beta}$ times the right answer.

C is incorrect. Added the intercept to the slope: $50 \times 0.91294 \times (0.08 + 0.91) = 45.19$. The constant is a red herring for the hedge ratio.

D is incorrect. Dropped $\hat{\beta}$: $50 \times 0.91294 = 45.65$ is the DV01-neutral hedge, which assumes parallel yield shifts.

LO: MR-11 c: Calculate the regression hedge adjustment factor, beta. MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. MR-11 a. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eqs. 11.8 to 11.12, Tables 11.1 and 11.2

Solution MR-11 c.5

Answer: B

Bank Q535 • BT P2.T5.22.14.3

Step 1 – State the hypothesis

A DV01 hedge implicitly assumes $\beta = 1$: $H_0 : \beta = 1$.

Step 2 – t-statistic

$$t = \frac{0.842 - 1}{0.193} = -0.82$$

Step 3 – Compare with the critical value

Two-sided 95%, about 80 degrees of freedom: about 1.99. $|-0.82| < 1.99$. Equivalently, $0.842 \pm 1.99 \times 0.193 = (0.458,\ 1.226)$ contains 1.

$\Rightarrow$ She cannot reject parallel shifts.

Interpretation

Same point estimate as GARP's (0.842), but with a standard error of 0.193 instead of 0.051 the data no longer reject $\beta = 1$. The conclusion depends on the standard error of the slope, not on the point estimate alone.

Why the other options are wrong

A is incorrect. Concludes beta is not one from the point estimate, ignoring sampling error.

C is incorrect. Tests the wrong coefficient; the constant is insignificant anyway $(0.060/0.223 = 0.27)$ and parallel shifts concern beta.

D is incorrect. The regression standard error measures fit in basis points; it is not a test of $\beta = 1$.

LO: MR-11 c: Calculate the regression hedge adjustment factor, beta. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, Table 11.2

Solution MR-11 c.6

Answer: B

Bank Q536 • BT P2.T5.321.3

Step 1 – Check A

$\rho = \sqrt{0.680} = 0.825$. True.

Step 2 – Check B

$t = (1.30-1)/0.051 = 5.88$; 95% interval $\approx 1.30 \pm 1.97 \times 0.051 = (1.20,\ 1.40)$, which excludes 1. False.

Step 3 – Check C

Position DV01 $= 100\text{m} \times 0.0400/100 = \$40{,}000$ per bp; $\times\ 2.2$ bp $= \$88{,}000$. True.

Step 4 – Check D

DV01 hedge $= 100 \times 0.04/0.07 = \$57.14\text{m}$; regression hedge $= 57.14 \times 1.3 = \$74.29\text{m}$; difference $\$17.14\text{m}$. True.

Interpretation

Beta is almost six standard errors above one, so the data strongly reject the DV01 hedge's assumption, and the regression hedge buys about 30% more TIPS.

Why the other options are wrong

A is incorrect. True statement, so not the exception.
C is incorrect. True statement, so not the exception.
D is incorrect. True statement, so not the exception.

LO: MR-11 c: Calculate the regression hedge adjustment factor, beta. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eqs. 11.8 and 11.12

MR-11 • Regression Hedging and Principal Component Analysis • Vardeez: IMPORTANT

MR-11 d

P3 CALC

Calculate the face value of an offsetting position needed to carry out a regression hedge.

Workbook: C1 Regression Hedging, DV01 and PCA • Tier C • Map #23 (5/16)

WHAT IS TESTED HERE

Why P3. Tier C workbook topic (C1, Map #23, reported: "regression hedge: face amount if offsetting").
Vardeez: regression hedge face value (the same formula as for beta).

Reported exam shapes

- **C1·S2** Face amount of the offsetting position in a regression hedge, paired with the hedged P&L volatility (worked as Question MR-11 c.4).
- **C1·S3** The DV01-neutral amount as the planted distractor against the regression amount.

What the questions look like: one formula, four distractors: DV01 hedge (beta dropped), inverted DV01 ratio, dividing by beta, and the wrong direction (buy versus sell).

1. One formula, two corrections

$$F^H = -F^P \times \frac{\text{DV01}^P}{\text{DV01}^H} \times \hat{\beta}$$

The DV01 ratio corrects for *size* (price sensitivity per unit of face); beta corrects for the fact that the two yields do *not* move one-for-one. The minus sign says the hedge is the opposite side: long the bond means sell the hedge; short the bond means buy the hedge.

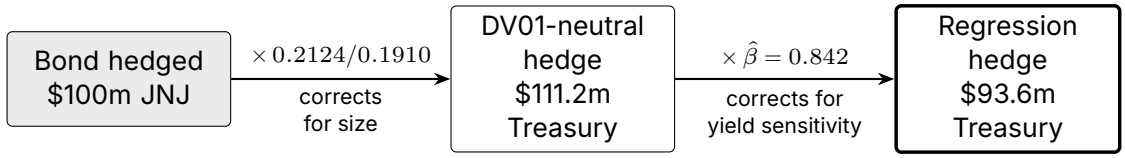


Figure 2. Two corrections applied in sequence (GARP's JNJ example). The DV01 ratio scales for price sensitivity; beta then scales for the less-than-one-for-one yield relationship. With $\hat{\beta} > 1$ (Vardeez's TIPS case, 1.0198) the second step increases the amount instead: \$80.95m to \$82.55m.

2. Worked examples

GARP (JNJ against the 30-year Treasury). $F^{30} = -100 \times (0.2124/0.1910) \times 0.842 = -\$93.6$ million, against \$111.2 million for the DV01 hedge. Hedged P&L volatility: DV01 of the JNJ position $= \$100\text{m} \times 0.2124/100 = \$212{,}400$; $\times$ 2.00 bp standard error $= \$424{,}800$ per day.

Vardeez (short T-bond, long TIPS). $F^R = 100 \times (0.068/0.084) \times 1.0198 = \$82.55$ million of TIPS bought (DV01 hedge \$80.95 million).

Hedged P&L volatility. Always the DV01 of the *position being hedged* times the regression standard error (in bp):

$$\sigma_{P\&L} = \left| \frac{F^P \, \text{DV01}^P}{100} \right| \sigma_\varepsilon$$

Using the hedge leg's DV01 instead gives $\hat{\beta}$ times the right answer.

Formula card: MR-11 d

DV01-neutral hedge	$F^H = -F^P \,\mathrm{DV01}^P/\mathrm{DV01}^H$ (risk weight 100%)
Regression hedge	$F^H = -F^P \,(\mathrm{DV01}^P/\mathrm{DV01}^H)\,\hat{\beta}$ (risk weight $\hat{\beta}$)
Adjustment	Regression hedge − DV01 hedge = DV01 hedge $\times(\hat{\beta}-1)$
Hedged P&L volatility	$\lvert F^P \,\mathrm{DV01}^P/100\rvert \times \sigma_\varepsilon$ (position's DV01, regression standard error)
Not used	$\hat{\alpha}$, R^2, the standard error of $\hat{\beta}$, yields

Exam traps

- **Beta dropped:** gives the DV01 hedge, the planted distractor (C1·S3).
- **DV01 ratio inverted:** the DV01 of the bond being hedged goes on top.
- **Divided by beta** instead of multiplying.
- **Direction:** long position, sell the hedge; short position, buy it. With $\hat{\beta} > 1$ you need *more* of the hedge, with $\hat{\beta} < 1$ less.
- **Wrong base for an adjustment:** compare the regression hedge with the existing hedge, not with the face of the bond.
- **P&L volatility:** uses the regression's standard error (bp), not the standard error of beta and not beta.

Practice questions: MR-11 d

Question MR-11 d.1

Bank Q539 • AnalystPrep Q.6509 (rebuilt)

An asset management firm is establishing a regression hedge for its bond portfolio. The face value of the bond to be hedged is \$100 million with a DV01 of 0.300 (per \$100 face). The hedging instrument has a DV01 of 0.250 (per \$100 face), and the regression hedge adjustment factor, $\hat{\beta}$, is 0.800. What is the face value of the offsetting position required for the regression hedge?

A. −\$66.7 million

B. −\$96.0 million

C. −\$120.0 million

D. −\$150.0 million

Question MR-11 d.2

Bank Q543 • Mock 1 Q70 (rebuilt)

A hedge fund sells USD 150 million face of U.S. Treasury bonds and buys TIPS as a relative value trade. Yields and DV01s (per 100 face):

Bond	Yield	DV01
TIPS	1.475%	0.092
T-bond	3.325%	0.063

Given a yield beta of 1.012 (nominal yield change on real yield change), what is the face value of TIPS in the regression hedge?

A. USD 102.72 million

B. USD 103.95 million

C. USD 219.05 million

D. USD 221.67 million

Question MR-11 d.3

Bank Q541 • BT P2.T5.22.14.2 (rebuilt)

A market maker purchases \$100.0 million face of a 30-year JNJ bond and hedges the interest rate risk by selling 30-year Treasury bonds. Yields and DV01s (per \$100 face):

Bond	Yield	DV01
JNJ bond	3.952%	0.3124
Treasury bond	3.764%	0.2910

The market maker regresses the JNJ yield change on the Treasury yield change and obtains intercept $\alpha = 0.0240$, slope $\beta = 0.7150$, $R^2 = 73.4\%$. Using a single-variable regression hedge, which is nearest to the face amount of Treasury bonds sold?

A. \$66.60 million

B. \$76.76 million

C. \$107.35 million

D. \$150.15 million

Question MR-11 d.4

New • built from GARP Ch.11 (Tuckman), eq. 11.8

A portfolio manager is short USD 80 million face of a 10-year agency bond (DV01 0.0870 per 100 face) and hedges with the 10-year Treasury note (DV01 0.0905 per 100 face). A regression of daily changes in the agency yield on daily changes in the Treasury yield gives a constant of -0.03, a slope of 1.15, an R^2 of 88% and a standard error of 1.4 bp. Which trade implements the regression hedge?

A. Buy USD 66.88 million of the Treasury note.

B. Buy USD 76.91 million of the Treasury note.

C. Buy USD 88.44 million of the Treasury note.

D. Sell USD 88.44 million of the Treasury note.

Question MR-11 d.5

Bank Q545 • Schweser Module Quiz 11.1 Q2 (rebuilt)

A trader is shorting $100 million of a nominal Treasury and buying $89.8 million of TIPS, a position sized on DV01s (no beta adjustment). A regression shows the nominal yield changes by 1.0274 bp per 1 bp change in the real yield. Should the trader adjust the hedge, and by how much?

A. No adjustment is needed.

B. Yes, buy an additional $2.46 million of TIPS.

C. Yes, sell $2.39 million of TIPS.

D. Yes, sell $7.74 million of TIPS.

Question MR-11 d.6

Bank Q531 • BT P2.T5.321.2 (moved from LO 11.b)

A trader shorts $100.0 million of nominal US Treasury 3 5/8s bonds and, employing a single-variable regression hedge, buys a face amount of 1 7/8s US TIPS. The bonds and the results of a regression of the nominal against the real yield are:

Bond	Yield	DV01
Nominal 3 5/8s	3.275%	0.0400
TIPS 1 7/8s	1.237%	0.0650
Regression of Δ nominal yield (bp) on Δ real yield (bp), 229 observations		
R-squared	68.0%	
Standard error (bps)	5.300	
Constant (α)	0.0200	(std. error 0.2529)
Change in real yield (β)	1.0940	(std. error 0.0595)

Which is nearest to an estimate of the daily volatility of the P&L of the hedged portfolio?

A. $2,380

B. $43,760

C. $67,323

D. $212,000

Solutions • MR-11 d

Solution MR-11 d.1

Answer: B

Bank Q539 • AnalystPrep Q.6509

Step 1 – DV01 ratio

$0.300/0.250 = 1.2$.

Step 2 – Scale by beta

$$F^H = -100 \times 1.2 \times 0.800 = -\$96.0 \text{ million}$$

$\Rightarrow$ Short \$96.0 million of the hedging instrument against the long \$100 million.

Interpretation

The DV01 hedge would be \$120 million; beta below one cuts it by 20%.

Why the other options are wrong

A is incorrect. Inverted the DV01 ratio: $100 \times (0.250/0.300) \times 0.8 = 66.7$.
C is incorrect. Dropped beta (the DV01 hedge): $100 \times 1.2 = 120$.
D is incorrect. Divided by beta: $120/0.8 = 150$.

LO: MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eq. 11.8

Solution MR-11 d.2

Answer: B

Bank Q543 • Mock 1 Q70

Step 1 – Identify the position and the hedge

The position is the T-bond sold (DV01 0.063); the hedge is the TIPS bought (DV01 0.092).

Step 2 – DV01 hedge

$150 \times 0.063/0.092 = 102.72$.

Step 3 – Regression hedge

$102.72 \times 1.012 =$ USD 103.95 million of TIPS.
$\Rightarrow$ The TIPS have the larger DV01, so less than USD 150 million face is needed; beta slightly above one scales it up.

Interpretation

The yields are red herrings.

Why the other options are wrong

A is incorrect. Dropped beta: 102.72.
C is incorrect. Inverted the DV01 ratio and dropped beta: $150 \times 0.092/0.063 = 219.05$.
D is incorrect. Inverted the DV01 ratio: $219.05 \times 1.012 = 221.67$.

LO: MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1; Vardeez MR-11 d (TIPS example)

Solution MR-11 d.3

Answer: B

Bank Q541 • BT P2.T5.22.14.2

Step 1 – DV01 ratio

$0.3124/0.2910 = 1.07354$; DV01 hedge = \$107.35 million.

Step 2 – Regression hedge

$107.35 \times 0.7150 = \$76.76$ million sold.
$\Rightarrow$ The JNJ yield moves only 0.715 bp per Treasury basis point, so far fewer Treasuries are needed.

Interpretation
The intercept, R^2 and yields are not used in the face amount.

Why the other options are wrong
A is incorrect. Inverted the DV01 ratio: $100 \times 0.9315 \times 0.715 = 66.60$.
C is incorrect. Dropped beta: 107.35, the DV01 hedge.
D is incorrect. Divided by beta: $107.35/0.715 = 150.15$.

LO: MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eq. 11.8

Solution MR-11 d.4 Answer: C

New • GARP Ch.11, eq. 11.8

Step 1 – Sign of the position
$F^P = -80$ (short), so $F^H = -F^P \times \cdots$ is positive: **buy** the Treasury.

Step 2 – DV01 hedge
$80 \times 0.0870/0.0905 = 76.91$.

Step 3 – Regression hedge
$76.91 \times 1.15 =$ USD 88.44 million.
$\Rightarrow$ The agency yield moves 1.15 bp per Treasury basis point, so the hedge needs 15% more Treasury DV01 than the position's DV01.

Interpretation
The constant, R^2 and standard error do not enter the face amount (the standard error gives the hedged P&L volatility: $80\text{m} \times 0.0870/100 \times 1.4 = \$97{,}440$ per day).

Why the other options are wrong
A is incorrect. Divided by beta: $76.91/1.15 = 66.88$.
B is incorrect. Dropped beta: 76.91, the DV01 hedge.
D is incorrect. Right size, wrong direction: a short position is hedged by buying.

LO: MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eqs. 11.7 and 11.8

Solution MR-11 d.5 Answer: B

Bank Q545 • Schweser Module Quiz 11.1 Q2

Step 1 – Regression hedge
$89.8 \times 1.0274 = 92.26$ million of TIPS.

Step 2 – Adjustment
$92.26 - 89.8 = +2.46$: buy \$2.46 million more.
$\Rightarrow$ Beta above one means nominal yields move more than real yields, so more TIPS are needed.

Interpretation
The adjustment is the existing DV01 hedge times $(\hat{\beta} - 1)$: $89.8 \times 0.0274 = 2.46$.

Why the other options are wrong
A is incorrect. Ignored a beta that differs from one.
C is incorrect. Divided by beta: $89.8/1.0274 = 87.41$, a reduction of 2.39 (wrong direction).
D is incorrect. Compared the regression hedge with the \$100m nominal face instead of the

existing \$89.8m TIPS position: $100 - 92.26 = 7.74$.

LO: MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. • **Reference:** SchweserNotes 2026 Book 1, Reading 11, LO 11.d

Solution MR-11 d.6

Answer: D

Bank Q531 • BT P2.T5.321.2

Step 1 – DV01 of the position being hedged

$100{,}000{,}000 \times 0.0400/100 = \$40{,}000$ per bp.

Step 2 – Hedged P&L volatility

$\$40{,}000 \times 5.300$ bp $= \$212{,}000$ per day.

$\Rightarrow$ Each basis point of residual (nominal yield moving 1 bp more than $1.094\times$ the real-yield change) costs \$40,000.

Interpretation

Beta sizes the hedge (\$67.32m TIPS); the standard error of the regression measures what is left over.

Why the other options are wrong

A is incorrect. Used the standard error of beta: $40{,}000 \times 0.0595 = 2{,}380$.

B is incorrect. Multiplied by beta: $40{,}000 \times 1.094 = 43{,}760$.

C is incorrect. This is the TIPS face amount in thousands, $100 \times (0.04/0.065) \times 1.094 = 67.32$m: a hedge size, not a volatility.

LO: MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1, eq. 11.12

MR-11 • Regression Hedging and Principal Component Analysis • Vardeez: IMPORTANT

MR-11 h

P3 CONCEPT

Describe principal component analysis and explain how it is applied to constructing a hedging portfolio.

Workbook: C1 Regression Hedging, DV01 and PCA • Tier C • Map #23 (5/16)

WHAT IS TESTED HERE

Why P3. Tier C workbook topic (C1, Map #23): PCA as a hedging-portfolio construction is the item next to the reported regression-hedge shapes.

Reported exam shape

- **C1·S5** PCA as the construction of a hedging portfolio: how many PCs two hedging securities can neutralize, and the risk-weight sum.

What the questions look like: statement questions on the three properties of PCs, level/slope/short rate, and the hedging logic. GARP's own practice exam has set one PCA hedge calculation on this LO (Question MR-11 h.6); it is the only numerical question here.

1. Why PCA

Regression hedging is ad hoc: each trader picks bonds, windows and the number of variables, and each regression is effectively a different model of the term structure. PCA gives **one empirical description of how the whole curve moves**, usable consistently across a portfolio.

PCA turns n correlated rate changes (GARP: 13 USD swap rates, 1 to 30 years) into n **principal components**, each a fixed pattern of changes across all rates, with three properties:

1. The sum of the PC variances equals the sum of the rate variances: the PCs capture the volatility of the set of rates.
2. The PCs are **uncorrelated** with each other.
3. Each PC has the **maximum possible variance** given the earlier PCs, so they are ranked by variance explained.

Empirically, the first three PCs explain an overwhelming share: 99.84% across the 13 USD rates (level over 90%, slope about 8%, short rate under 1%). So three factors, not a 13-by-13 covariance matrix, describe the curve. The PCs are statistical constructs; the names are given afterwards from their shapes.

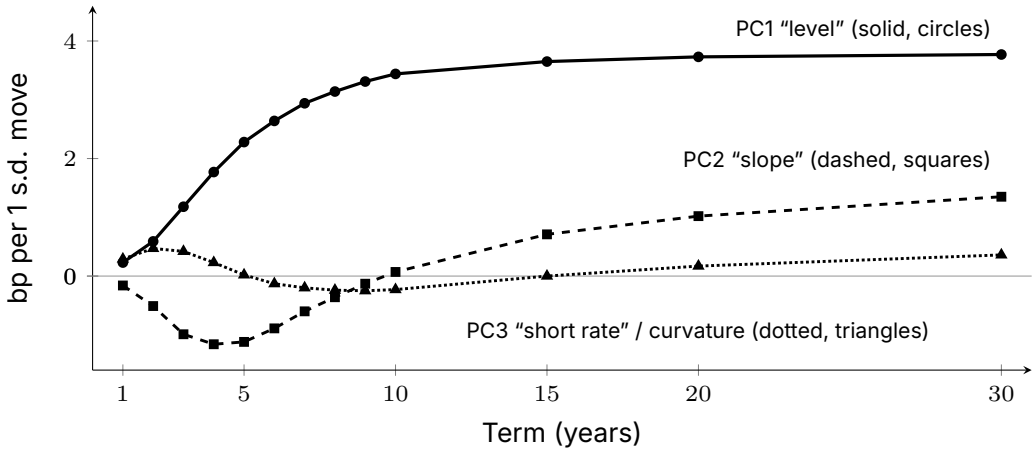


Figure 3. The first three principal components of USD LIBOR swap rates, June 2020 to July 2021 (GARP Table 11.5). Level: every rate up, rising steeply to about seven years and flat beyond. Slope: short rates down, long rates up. Short rate: small, adds explanatory power at the very short end.

2. Reading the PCs (GARP Table 11.5)

- A one-standard-deviation move in the level PC raises the 1-year rate 0.23 bp, the 10-year 3.44 bp and the 30-year 3.77 bp. It is not particularly “level” from 1 to 7 years in this sample (the Fed pinned short rates).
- **PC volatility of a rate** (PCs uncorrelated, so variances add): 5-year $\sqrt{2.28^2 + (-1.12)^2 + 0.02^2} = 2.54$ bp, essentially its total volatility.
- Three PCs explain 74.54% of the 1-year rate’s variation, 97.47% of the 2-year, and over 99% beyond three years.
- **Implication for hedging:** 8- to 10-year (even 7- to 15-year) traders can defend a one-factor (level) approach; the 3- to 6-year sector and the long end may need two factors; the very short end may need three.

3. Building a PCA hedge

1. Price the portfolio on today’s curve.
2. Shift the curve by each PC in turn, reprice, and compute a “’01” of the portfolio for each PC.
3. Choose hedging positions that set the net ’01 to each chosen PC to zero: **one equation per PC**, so **one hedging instrument per PC hedged**.

GARP’s butterfly. Receive fixed on 100 of 20-year swaps; pay fixed in 10s and 30s, neutral to level and slope:

$$\begin{aligned} \text{Level:} \quad & -F^{10}\,\tfrac{0.09347}{100}(3.44) - F^{30}\,\tfrac{0.23600}{100}(3.77) - 100\,\tfrac{0.17064}{100}(3.73) = 0 \\ \text{Slope:} \quad & -F^{10}\,\tfrac{0.09347}{100}(0.07) - F^{30}\,\tfrac{0.23600}{100}(1.35) - 100\,\tfrac{0.17064}{100}(1.02) = 0 \end{aligned}$$

giving $F^{10} = -49.56$, $F^{30} = -53.60$; risk weights 27.1% and 74.1%, summing to **101.2%**. Only parallel-shift hedges always sum to 100%. With two hedging swaps only two PCs are hedged; the residual short-rate exposure is 0.007 per 100 (under one cent), so the trader may skip a third (15-year) swap given transaction costs and the relative value view.

	Regression hedge	PCA hedge
Description	Pairwise, ad hoc: chosen bonds, window, variables	One empirical description of the whole curve
Factors	The yields of the chosen hedging bonds	Uncorrelated PCs ranked by variance
Hedge condition	Minimise variance given the position	Zero exposure to each chosen PC
Instruments needed	One per explanatory yield	One per PC neutralised
Risk weights	$\hat{\beta}$ (sum of $\hat{\beta}$'s in two-variable case)	Need not sum to 100% (101.2% in GARP)

Exam traps

- **Polarity:** PCs are *uncorrelated*, not highly correlated; that is why each needs its own equation.
- **Counting rule:** two hedging securities neutralise two PCs only; measure the residual third-PC exposure.
- **Absolute:** PCA risk weights need not sum to 100%.
- **Equal variance:** PCs are ranked by variance; the small ones can be ignored.
- **Pre-specified factors:** PCs come out of the data; "level, slope, short rate" are labels attached afterwards.
- **Scaling:** the number of PCs needed stays small regardless of portfolio size; that is PCA's advantage.
- **PC volatility:** square, add, take the root (the PCs are uncorrelated); do not add loadings directly.

Practice questions: MR-11 h

Question MR-11 h.1

Bank Q559 • AP Mock 2 Q68 (rebuilt)

An analyst is exploring principal component analysis (PCA) for constructing a hedging portfolio. What primary advantage does PCA offer when describing the risk of a portfolio of fixed-income instruments?

A. It assumes that all rates are perfectly correlated, so a single factor describes every rate change.

B. It decomposes yield curve changes into uncorrelated components, ordered by the variance they explain, so a few components capture most of the term structure's volatility.

C. It guarantees that the slope of the yield curve stays constant, so hedge ratios never need updating.

D. It requires one component per rate, so each rate is hedged separately and no simplification is needed.

Question MR-11 h.2 Bank Q560 • AnalystPrep Q.6516 (rebuilt)

An asset manager wants to hedge a bond portfolio using principal component analysis (PCA). Which approach correctly applies PCA to construct the hedge?

A. Estimate the portfolio's sensitivity to every principal component, including the smallest ones, because each component carries the same amount of variance.

B. Compute only the portfolio DV01 and offset it with a single instrument, assuming all rates move in parallel.

C. Compute the portfolio's exposure to the first few (dominant) principal components, such as level and slope, and choose hedge positions that make the net exposure to each of them zero.

D. Regress the portfolio's yield changes on the hedging instrument's yield changes and set the hedge ratio equal to the regression beta.

Question MR-11 h.3 Bank Q561 • BT P2.T5.22.15.3 (rebuilt)

A PCA of 13 USD swap rate factors (Tuckman) shows that, for the 1-year rate, the loadings in basis points are 0.23 on level, -0.16 on slope and 0.29 on short rate, and that the first three PCs explain almost all of the total variance. Which of the following statements about this PCA is FALSE?

A. The rate covariance matrix has 78 distinct covariances (plus 13 variances), whereas the PC covariance matrix is diagonal.

B. The PCs are fundamental economic factors that the analyst specifies in advance, which is why the names level, slope and short rate are assigned before the analysis is run.

C. PCA produces 13 principal components, but only a few (here three) are needed to characterize term structure fluctuations.

D. The PC volatility of the 1-year rate is about 0.40 bp, equal to $\sqrt{0.23^2 + (-0.16)^2 + 0.29^2}$, because the PCs are uncorrelated.

Question MR-11 h.4 New • built from GARP Ch.11 (Tuckman), Table 11.5

A risk manager reviews the PCA of USD swap rates below (share of each rate's variance, in percent):

Term	Level	Slope	Short rate	Three PCs / total
1-year	32.7	15.0	52.3	74.54
4-year	69.1	29.7	1.2	99.72
9-year	99.3	0.2	0.6	99.92
30-year	87.9	11.3	0.8	99.93
All 13 rates	91.3	7.8	0.8	99.84

Which of the following conclusions is most consistent with these results?

A. A trader of 8- to 10-year swaps can reasonably defend a one-factor (level) hedge, while traders in the 3- to 6-year sector or at the long end may need two factors and the very short end may need three.

B. Because the level PC explains over 90% of the total variance, a single-factor hedge is adequate at every point of the curve, including the one-year sector.

C. The short-rate PC is most important at the long end of the curve, where it explains the largest share of each rate's variance.

D. Because three PCs explain 99.84% of the total variance, hedging a position with any three swaps removes all but 0.16% of its risk.

Question MR-11 h.5 Workbook C1.2 • S3, S4, S5 • Map #23

A risk consultant is reviewing the hedging methodology of a bank's swap desk. Single-bond positions are currently hedged DV01-neutral, and the desk proposes to hedge its 10s/20s/30s butterflies using principal component analysis (PCA) of swap rates, with the 10-year and 30-year swaps as the two hedging instruments. Which of the following statements should the consultant include in the report?

A. A DV01-neutral hedge minimizes the variance of the hedged P&L, because by construction the DV01 of the hedge equals the DV01 of the position being hedged.

B. A PCA hedge is efficient because the principal components are constructed to be highly correlated, so neutralizing the level PC also neutralizes the others.

C. A PCA hedge using two hedging swaps can neutralize exposure to only two PCs, so the residual exposure to the third PC must be measured and either accepted or hedged.

D. A PCA hedge of the butterfly sets the risk weights of the two hedging swaps so that they always sum to 100% of the DV01 of the position being hedged.

Question MR-11 h.6 Bank Q563 • GARP PE1 Q12 • Mock 3 Q68 (rebuilt, key corrected)

A trader enters a butterfly in which the bank pays fixed on EUR 100 million notional of a 5-year swap and receives fixed on 2-year and 10-year swaps. The 2-year and 10-year notionals are chosen so the position is neutral to both the level and slope principal components. Data (change in rate in bp per 1 standard deviation move in the PC):

Swap tenor	DV01 (per EUR 100 notional)	Level PC	Slope PC
2-year	0.0285	5.06	−2.93
5-year	0.0496	5.97	−1.28
10-year	0.0731	5.43	0.02

What are the required notionals, and what are the risk weights of the 2-year and 10-year legs, defined as each leg's DV01 relative to the DV01 of the 5-year swap?

A. 2-year EUR 76.85m; 10-year EUR 46.68m; risk weights 39.1% and 60.9%.

B. 2-year EUR 46.68m; 10-year EUR 76.85m; risk weights 44.2% and 68.8%.

C. 2-year EUR 76.85m; 10-year EUR 46.68m; risk weights 44.2% and 68.8%.

D. 2-year EUR 87.02m; 10-year EUR 33.93m; risk weights 50.0% and 50.0%.

Solutions • MR-11 h

Solution MR-11 h.1 — Answer: B

Bank Q559 • AP Mock 2 Q68

Step 1 – What PCA produces

As many uncorrelated PCs as rates, each chosen to have the maximum variance given the earlier ones; their variances sum to the rates' total variance.

Step 2 – Why that helps

The first three PCs (level, slope, short rate) capture almost all the volatility, so many rates are described by three factors.

Interpretation

PCA gives a single, compact, empirical description of how the curve moves.

Why the other options are wrong

A is incorrect. PCA estimates correlations from data; if all rates were perfectly correlated there would be only one PC.
C is incorrect. The slope is itself a PC that moves; PCA guarantees nothing about it.
D is incorrect. PCA has as many PCs as rates, but only the first few are needed: that is the simplification.

LO: MR-11 h: Describe principal component analysis and explain how it is applied to constructing a hedging portfolio.
• **Reference:** Tuckman and Serrat 4e, Ch.6, 11.5

Solution MR-11 h.2 — Answer: C

Bank Q560 • AnalystPrep Q.6516

Step 1 – Exposures

Compute how much the portfolio and each candidate hedge change for a one-standard-deviation move in each dominant PC.

Step 2 – Neutralise

Solve for hedge notionals so that the net exposure to each dominant PC is zero: one equation, and one hedging instrument, per PC.

Interpretation

This is GARP's recipe: shift by each PC, reprice, compute a '01 per PC, and neutralise those '01s.

Why the other options are wrong

A is incorrect. PCs are ranked by variance; the small ones can be ignored.
B is incorrect. This is a DV01 hedge, which assumes a parallel shift.
D is incorrect. This is a single-variable regression hedge.

LO: MR-11 h: Describe principal component analysis and explain how it is applied to constructing a hedging portfolio.
• **Reference:** Tuckman and Serrat 4e, Ch.6, 11.5 (hedging with PCA)

Solution MR-11 h.3 — Answer: B

Bank Q561 • BT P2.T5.22.15.3

Step 1 – Check A

$13 \times 12/2 = 78$ covariances; the PCs are uncorrelated, so their covariance matrix is diagonal. True.

Step 2 – Check C

PCA yields 13 PCs, but three explain nearly all the variance. True.

Step 3 – Check D

$0.23^2 + 0.16^2 + 0.29^2 = 0.1626$, $\sqrt{0.1626} = 0.40$ bp. Adding variances is valid because the PCs are uncorrelated. True.

Step 4 – Check B

PCs come out of the covariance structure of the data; the labels are attached afterwards from their shapes. False.

Interpretation

PCA is purely statistical: it finds the patterns; the economics is interpretation.

Why the other options are wrong

A is incorrect. True statement.
C is incorrect. True statement.
D is incorrect. True statement (GARP's table shows 0.41 for the 1-year PC volatility using more decimals).

LO: MR-11 h: Describe principal component analysis and explain how it is applied to constructing a hedging portfolio.
• **Reference:** Tuckman and Serrat 4e, Ch.6, 11.5, Table 11.5, eq. 11.26

Solution MR-11 h.4 Answer: A

New • GARP Ch.11, Table 11.5

Step 1 – Where one factor is enough

At 9 years the level PC explains 99.3% of the variance (GARP: 98.2 to 99.5% from 8 to 10 years), so a one-factor hedge is defensible there.

Step 2 – Where it is not

At 4 years the slope PC explains 29.7%; at 30 years 11.3%: two factors may be needed. At 1 year the short-rate PC explains 52.3% and three PCs together only 74.54%: the very short end may need three.

Interpretation

A is GARP's own conclusion from Table 11.5.

Why the other options are wrong

B is incorrect. The 91.3% is a total across all rates; at the one-year point the level PC explains only 32.7%.
C is incorrect. The short-rate PC matters at the *short* end (52.3% at 1 year, 0.8% at 30 years).
D is incorrect. The 99.84% describes the rates' variance, not the risk left after an arbitrary three-swap hedge; a hedge must be solved to neutralise the PC exposures.

LO: MR-11 h: Describe principal component analysis and explain how it is applied to constructing a hedging portfolio.
• **Reference:** Tuckman and Serrat 4e, Ch.6, 11.5, Table 11.5

Solution MR-11 h.5 Answer: C

Workbook C1.2 • S3, S4, S5

Step 1 – What a DV01-neutral hedge assumes

Equal DV01s on both sides (risk weight 100%) is the right hedge only if the two yields move in parallel. The book rejects $\beta = 1$ for JNJ versus the 30-year Treasury (interval 0.740 to 0.944).

Step 2 – Which hedge minimises variance

The regression hedge, with risk weight $\hat{\beta}$, minimises the variance of the hedged P&L (Appendix

A6.1). The DV01 hedge does so only in the special case $\hat{\beta} = 1$.

Step 3 – How many PCs two securities can hedge

Each hedged PC adds one equation (11.30, 11.31); two hedging swaps solve two equations. The book measures the residual short-rate exposure at 0.007 per 100 and notes the trader may accept it or add a 15-year swap and a third equation.

Interpretation

C is correct: it states the counting rule and the follow-up decision that the book models.

Why the other options are wrong

A is incorrect. Sibling swap. Variance minimisation is the property of the regression hedge; equal DV01s minimise variance only if yields shift in parallel.

B is incorrect. Polarity flip. PCs are constructed to be uncorrelated; that is why each needs its own neutralising equation.

D is incorrect. Absolute claim. Only under parallel shifts do risk weights always sum to 100%; the book's PCA butterfly needs $27.1\% + 74.1\% = 101.2\%$.

LO: MR-11 a, MR-11 b; MR-11 h: Describe principal component analysis and explain how it is applied to constructing a hedging portfolio. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.1 and 11.5, eqs. 11.30 to 11.34

Solution MR-11 h.6 — Answer: C

Bank Q563 • GARP PE1 Q12

Step 1 – Exposure of the 5-year leg (EUR 100m)

Per one-standard-deviation move: level $100 \times \frac{0.0496}{100} \times 5.97 = 0.29611$; slope $100 \times \frac{0.0496}{100} \times (-1.28) = -0.063488$ (EUR millions per 100).

Step 2 – Two equations, two unknowns (notionals in EUR m)

$$\begin{aligned} \text{Level:} \quad & 0.0014421\, F_2 + 0.0039693\, F_{10} = 0.29611 \\ \text{Slope:} \quad & -0.00083505\, F_2 + 0.0000146\, F_{10} = -0.063488 \end{aligned}$$

$$\Rightarrow \quad F_2 = 76.85, \qquad F_{10} = 46.68$$

$\Rightarrow$ Both wings are received fixed, opposite to the 5-year paid leg.

Step 3 – Risk weights against the 5-year DV01

5-year DV01 $= 100 \times 0.0496/100 = 0.0496$. 2-year: $76.85 \times 0.0285/100/0.0496 = 44.2\%$; 10-year: $46.68 \times 0.0731/100/0.0496 = 68.8\%$.

Interpretation

The weights sum to 113.0%: as in GARP's butterfly (101.2%), a PCA hedge's risk weights need not sum to 100%. The 2-year leg carries most of the slope hedge because its slope loading (-2.93) is the only large one.

Why the other options are wrong

A is incorrect. Right notionals, but weights normalised to the two-leg total (39.1% and 60.9%) instead of to the 5-year DV01.

B is incorrect. Right weights, notionals assigned to the wrong tenors.

D is incorrect. Split the 5-year DV01 equally between the wings, ignoring the PC loadings: 87.02m and 33.93m.

LO: MR-11 h: Describe principal component analysis and explain how it is applied to constructing a hedging portfolio. • **Reference:** Tuckman and Serrat 4e, Ch.6, 11.5, eqs. 11.30 to 11.33

MR-12 • Arbitrage Pricing with Term Structure Models • Vardeez: MODERATE

MR-12 d

P3 BOTH

Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift.

Workbook: C7 CMT Swap Valuation and OAS on a Rate Tree • Tier C • Map #33

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C7) hits this LO: the reported "decision tree question with OAS of CMT swap" carries the true-versus-risk-neutral distinction as its standing trap.
Vardeez: the chapter pointer is the CMT swap; this LO is its conceptual partner.

Reported exam shape

- **C7·S3** True versus risk-neutral probabilities and the drift they imply, with the real-world probabilities supplied as a temptation.

What the questions look like: a tree with both sets of probabilities in the stem (use the risk-neutral ones), a "solve for p from a bond price" step, and a drift in basis points. Statement questions test why risk-neutral probabilities are not forecasts.

1. The set-up: a binomial tree for the six-month rate

GARP's tree (Tuckman): the six-month rate is 2.00% today (date 0) and, six months later, 2.50% (up) or 1.50% (down). The six-month and one-year spot rates are 2.00% and 2.15%, so the one-year zero trades at

$$P_{1y} = \frac{1{,}000}{(1 + 0.0215/2)^2} = \$978.842.$$

On date 1 that bond is a six-month zero worth $1{,}000/1.0125 = \$987.654$ (up) or $1{,}000/1.0075 = \$992.556$ (down). Every node value is the probability-weighted average of the next two values, discounted one period at that node's own rate (backward induction).

2. True versus risk-neutral probabilities

- **True (real-world) probabilities** describe how rates actually move. In GARP's example they are 0.5 and 0.5. The expected discounted value under them is

$$\frac{0.5(987.654) + 0.5(992.556)}{1.01} = \$980.302 \;>\; \$978.842 \text{ (market)}.$$

  Investors dislike the price risk of the one-year zero, so they pay *less* than its expected discounted value.

- **Risk-neutral probabilities** are the probabilities that make expected discounted value *equal* the market price:

$$\frac{p\,(987.654) + (1-p)\,(992.556)}{1.01} = 978.842 \quad \Rightarrow \quad p = 0.8009.$$

  They are a pricing device, not a forecast. Once found from the underlying bonds, every contingent claim on the same rate process is priced by expected discounted value under them.

- **Why the option price ignores the true probabilities.** The replicating portfolio (long \$521.4375 face of one-year zeros, short \$515.00 of six-month zeros) is the same whatever the true probabilities

are. They enter only *indirectly*, through the market price of the one-year zero. The 990 call on the six-month zero is worth \$0.504 by arbitrage and by risk-neutral pricing, against \$1.2653 if you (wrongly) use the true 0.5.

3. Applying the difference to drift

Drift is the **expected change in the short rate** over a step. With symmetric moves of $\pm\Delta$:

$$\boxed{\text{drift} = p\,\Delta - (1-p)\,\Delta = (2p-1)\,\Delta}$$

- Under the true 0.5: drift $= 0$. Under $p = 0.8009$ with $\Delta = 50$ bp: drift $= 0.6018 \times 50 = +30.09$ bp.
- Date 1 to date 2 needs its *own* probability, solved from the 1.5-year zero (spot 2.25%, price \$966.995): $q = 0.6520$, so drift $= (2 \times 0.6520 - 1) \times 50 = +15.20$ bp. Reusing p at date 1 is a named mistake.
- Risk-neutral probabilities put more weight on the **high-rate, low-price** states. The excess of risk-neutral over true drift is how a tree carries the risk premium (MR-13: a drift of x bp acts like a risk premium of x bp per year of duration).
- Two equivalent ways to build a risk-neutral tree: adjust the *rates* on the tree to match on-the-run prices (keeping the given probabilities), or keep the rates and adjust the *probabilities*. Either way the derivative value is the same.

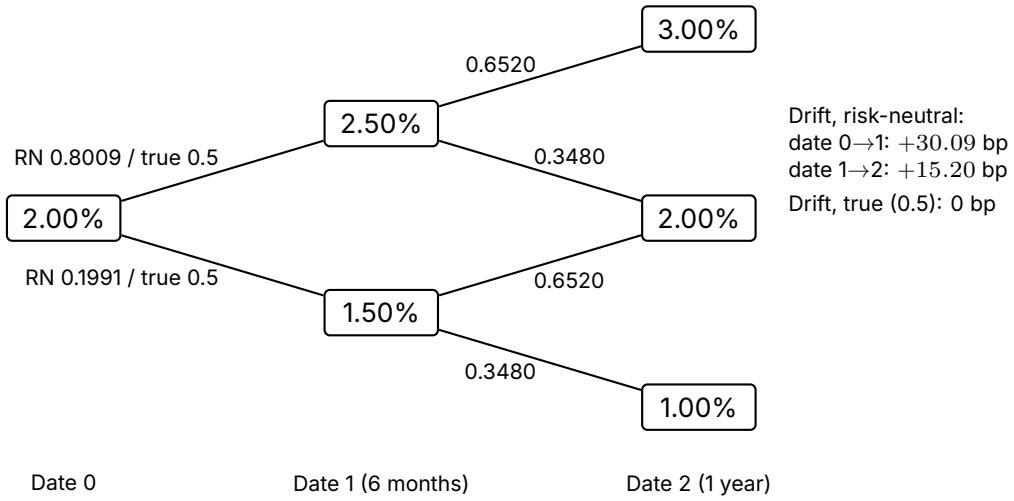


Figure 1. GARP's recombining tree for the six-month rate. Branch labels give the risk-neutral probability (and the true 0.5 on the first step; the true probability is 0.5 on every branch). Up-down and down-up both land on 2.00%.

Correction to the source notes.

- Vardeez writes: "If $q = 0.5$ it is a real-world probability. If $q \neq 0.5$ it is risk-neutral." That is true of the textbook example only, not a rule. A probability is risk-neutral because it makes expected discounted values recover market prices, whatever its value; true probabilities need not be 0.5. Question MR-12 d.2 tests this.
- Schweser says "any difference between the risk-neutral and true probabilities is referred to as the interest rate drift." Drift is the expected change in the rate, $(2p-1)\Delta$. The *difference* between risk-neutral and true drift is what the probability gap produces.

Formula card: MR-12 d

Risk-neutral p from a zero	$\dfrac{p\,P_{\text{up}} + (1-p)\,P_{\text{down}}}{1 + r_0/2} = P_{\text{market}} \;\Rightarrow\; p = \dfrac{P_{\text{market}}(1 + r_0/2) - P_{\text{down}}}{P_{\text{up}} - P_{\text{down}}}$
Drift over a $\pm\Delta$ step	$(2p-1)\,\Delta$; zero under true 0.5
Expected rate after two steps	$r_0 + (2p-1)\Delta + (2q-1)\Delta$ (different p and q)
GARP numbers	$p = 0.8009$, $q = 0.6520$; drifts 30.09 bp and 15.20 bp

Exam traps

- **Wrong-input twin.** The stem gives the true probabilities as well. They do not recover market prices and give the wrong price and a zero drift.
- **Half the drift.** $(p - 0.5)\Delta$ counts only the extra weight on the up move and ignores the matching loss on the down move. The drift is $(2p-1)\Delta$.
- **Probability read as drift.** A 70% up probability is not a 70 bp drift.
- **Same probability at every date.** q at date 1 is solved separately from a longer bond (0.6520 versus 0.8009).
- **Forecast fallacy.** Risk-neutral probabilities are not the market's forecast; they embed the risk premium.

Practice questions: MR-12 d

Question MR-12 d.1

New • built from GARP MR Ch.12, 12.2 to 12.3

A fixed-income analyst values a call option that expires in six months and gives the right to buy USD 1,000 face value of a then six-month zero-coupon bond at USD 990. The six-month rate is 2.00% today and will be 2.50% or 1.50% in six months, each with a true probability of 0.50. The one-year zero trades at USD 978.842, and on date 1 a six-month zero is worth USD 987.654 in the up state and USD 992.556 in the down state. Which of the following is closest to the option's price, with the correct reason?

A. USD 1.27, because expected discounted value under the true probabilities is the arbitrage price.

B. USD 0.50, because the risk-neutral up probability of 0.8009 leaves a weight of 0.1991 on the only state where the option pays.

C. USD 0.50, because investors' risk aversion lowers the true probability of the down state below 0.50.

D. USD 2.53, because the option pays USD 2.556 in the down state, discounted at the six-month rate.

Question MR-12 d.2

New • built from GARP MR Ch.12, 12.2 to 12.4

A junior analyst is reviewing a binomial tree for the six-month rate that a desk uses to price interest rate derivatives. Which of the following statements about true and risk-neutral probabilities is correct?

A. Risk-neutral probabilities are the market's best forecasts of future rate moves, which is why they recover market prices.

B. Any tree whose up probability differs from 0.50 is a risk-neutral tree, since true probabilities are always 0.50.

C. Risk-neutral probabilities put extra weight on high-rate, low-price states, so the risk-neutral drift exceeds the true drift.

D. The arbitrage price of an option on a zero depends directly on the true probabilities of the up and down moves.

Question MR-12 d.3

Bank Q589 • BT L2.T5.42.1

Assume a binomial interest rate tree where the six-month rate either jumps up 50 basis points or jumps down 50 basis points. The real-world (true) probabilities are 50% and 50% for the up- and down-states. If the interest rate drift is +20 basis points under risk-neutral probabilities, what is the risk-neutral probability of an up-state (p)?

A. 50%

B. 60%

C. 70%

D. 80%

Question MR-12 d.4

Bank Q591 • BT L2.T5.41.7 (rebuilt)

A two-step, one-year binomial tree (six-month steps) prices a one-year zero-coupon bond with face value USD 1,000. The current six-month rate is 4.0% (semiannual compounding); in six months the rate is 4.5% (bond worth USD 978.00) in the up-state or 3.5% (bond worth USD 982.80) in the down-state. The bond's current market price is USD 960.23 and the true probabilities are 50% up and 50% down. Using risk-neutral probabilities calibrated to the market price, what is the implied interest rate drift (rate moves are $\pm$50 bp)?

A. 0 bp

B. +10 bp

C. +20 bp

D. +70 bp

Question MR-12 d.5

New • built from GARP MR Ch.12, 12.4 (Figure 12.6)

A risk manager uses a recombining tree in which the six-month rate is 2.00% today and moves up or down by 50 bp every six months. The true probability of an up move is 0.50 at every node. The risk-neutral probability of an up move, solved to recover the one-year and 1.5-year zero prices, is 0.8009 from date 0 and 0.6520 from either date-1 node. Under the risk-neutral process, which of the following is closest to the expected six-month rate one year from now (date 2) and its cumulative drift relative to the true process?

A. 2.00%; zero drift, because the up and down moves are symmetric

B. 2.30%; drift of 30 bp

C. 2.45%; drift of 45 bp

D. 2.60%; drift of 60 bp

Question MR-12 d.6

Bank Q592 • BT P2.T5.23.2.1

Below is a multi-period price tree for a zero-coupon bond with 1.5 years to maturity. It is similar to Tuckman's; the only difference is that the spot rate term structure has been shifted up in parallel by 100 basis points.

Spot rates	$s(0.5) = 3.00\%$, $s(1.0) = 3.15\%$, $s(1.5) = 3.25\%$; PV of \$1,000: \$985.22, \$969.23, \$952.79
Six-month rate tree	date 0: 3.00%; date 1: 3.50% (up), 2.50% (down); date 2: 4.00%, 3.00%, 2.00%
Risk-neutral probabilities	date 0 to 1: 80.09% up, 19.91% down; date 1 to 2: 65.19% up, 34.81% down
Price tree	date 0: \$952.79; date 1: \$965.18 (up), \$974.73 (down); date 2: \$980.39, \$985.22, \$990.10; date 3: \$1,000

In regard to this multi-period tree, each of the following statements is true EXCEPT which is false?

A. The bond's current market price of \$952.79 assumes real-world probabilities applied to its expected discounted value (EDV).

B. If the 1.5-year spot rate, $s(1.5)$, rises from 3.25% to 3.50%, this will change at least some of the risk-neutral probabilities.

C. The drifts (expected changes) in the six-month rate are +30.1 basis points (date 0 to 1) and +15.2 basis points (date 1 to 2).

D. Extending the tree to further dates (e.g., 2 years, 2.5 years) requires assumptions about future short-rate values and risk-neutral probabilities that recover a given set of bond prices.

Solutions • MR-12 d

Solution MR-12 d.1 — Answer: B

New • built from GARP MR Ch.12, 12.2 to 12.3

Step 1 – Option payoffs on date 1

Up: $\max(987.654 - 990,\ 0) = 0$. Down: $992.556 - 990 = 2.556$.

Step 2 – Risk-neutral probability from the one-year zero

$$\frac{p(987.654) + (1-p)(992.556)}{1.01} = 978.842 \ \Rightarrow\ p = 0.8009, \quad 1 - p = 0.1991$$

$\Rightarrow$ The market prices the one-year zero below its true expected discounted value (980.302), and p captures that penalty.

Step 3 – Price by expected discounted value under p

$$\frac{0.8009 \times 0 + 0.1991 \times 2.556}{1.01} = \$0.504$$

Interpretation

The same \$0.504 comes from the replicating portfolio (long 521.4375 face of one-year zeros, short 515.00 of six-month zeros). The true probabilities of 0.50 never enter the arbitrage price; they matter only through the one-year zero's market price.

Why the other options are wrong

A is incorrect. Used the true probabilities: $0.5 \times 2.556/1.01 = 1.27$. That is expected discounted value, not the arbitrage price.

C is incorrect. Right number, wrong reason: risk aversion does not change the true probabilities. It is the risk-neutral probabilities that shift weight to the up (high-rate) state.

D is incorrect. Discounted the down-state payoff as if it were certain: $2.556/1.01 = 2.53$.

LO: MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. • **Reference:** Tuckman and Serrat, Fixed Income Securities 4e (GARP MR Ch.12), 12.2 and 12.3, eqs. 12.5 to 12.8

Solution MR-12 d.2 — Answer: C

New • built from GARP MR Ch.12, 12.2 to 12.4

Step 1 – What risk-neutral probabilities are

They are the probabilities that make expected discounted values equal market prices. Because investors pay less than expected discounted value for risky bonds, these probabilities must put more weight on states where bond prices are low, that is, where rates are high.

$\Rightarrow$ Higher weight on the up-rate move means a positive risk-neutral drift: GARP's 30.09 bp against 0 under the true 0.5.

Step 2 – Test each statement

A calls them forecasts; B turns the textbook's 0.5 into a rule; D says option prices depend directly on true probabilities. Only C survives.

Interpretation

The gap between risk-neutral and true drift is the tree's way of carrying the risk premium for bearing rate risk.

Why the other options are wrong

A is incorrect. Risk-neutral probabilities are not forecasts. They embed investors' aversion to price risk, which is exactly why they differ from true probabilities.
B is incorrect. Absolute claim. True probabilities need not be 0.5; a probability is risk-neutral because it recovers market prices, not because of its value.
D is incorrect. The replicating portfolio, and so the arbitrage price, does not depend directly on true probabilities; they enter only through current bond prices.

LO: MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.2 to 12.4

Solution MR-12 d.3 — Answer: C

Bank Q589 • BT L2.T5.42.1

Step 1 – True drift
$0.5(+50) + 0.5(-50) = 0$ bp.

Step 2 – Risk-neutral drift as a function of p

$$p(50) + (1-p)(-50) = (2p-1) \times 50 = 100p - 50$$

Step 3 – Solve

$$100p - 50 = 20 \;\Rightarrow\; p = 0.70$$

$\Rightarrow$ A 70% up probability produces a +20 bp expected change in the six-month rate.

Interpretation
Because the true drift is zero, the whole +20 bp is the risk-neutral adjustment.

Why the other options are wrong
A is incorrect. 50% is the true probability; it gives zero drift.
B is incorrect. 60% gives $(2 \times 0.6 - 1) \times 50 = +10$ bp.
D is incorrect. 80% gives $(2 \times 0.8 - 1) \times 50 = +30$ bp.

LO: MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.3

Solution MR-12 d.4 — Answer: C

Bank Q591 • BT L2.T5.41.7

Step 1 – Calibrate p to the market price

$$960.23 = \frac{p(978.00) + (1-p)(982.80)}{1.02} \;\Rightarrow\; 979.43 = 982.80 - 4.80p \;\Rightarrow\; p = 0.7011$$

Step 2 – Risk-neutral drift

$$(2 \times 0.7011 - 1) \times 50 = +20.1 \text{ bp}$$

$\Rightarrow$ Against a true drift of $0.5(50) + 0.5(-50) = 0$, the drift implied by the market price is about +20 bp.

Interpretation
The 50/50 expected discounted value is 961.18, above the market's 960.23; the gap is what forces the extra weight on the up-rate state.

Why the other options are wrong

A is incorrect. 0 bp is the drift under the true 50/50 probabilities.
B is incorrect. Used $(p - 0.5) \times 50 = 10.1$: counts only the higher weight on the up move, not the matching lower weight on the down move.
D is incorrect. Read the 70% probability as 70 bp.

LO: MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.3, eq. 12.7

Solution MR-12 d.5 Answer: C

New • built from GARP MR Ch.12, 12.4

Step 1 – Drift from date 0 to date 1

$$(2 \times 0.8009 - 1) \times 50 = +30.09 \text{ bp} \Rightarrow E[r_1] = 2.3009\%$$

Step 2 – Drift from date 1 to date 2 (its own probability)

$$(2 \times 0.6520 - 1) \times 50 = +15.20 \text{ bp from either date-1 node}$$

Step 3 – Expected rate on date 2

$$E[r_2] = 2.00\% + 0.3009\% + 0.1520\% = 2.4529\% \approx 2.45\%$$

$\Rightarrow$ Cumulative risk-neutral drift of about 45 bp, against zero under the true probabilities.

Interpretation

The true probabilities (0.50) are in the stem only as a temptation. The second-step probability is solved from the 1.5-year zero and is lower than the first.

Why the other options are wrong

A is incorrect. Used the true probabilities: symmetric moves, zero drift.
B is incorrect. Stopped after the first step: $2.00\% + 0.30\%$.
D is incorrect. Reused 0.8009 at date 1: $2.00\% + 2 \times 0.3009\% = 2.60\%$.

LO: MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.4, Figure 12.6

Solution MR-12 d.6 Answer: A

Bank Q592 • BT P2.T5.23.2.1

Step 1 – Which probabilities price the bond

$$\frac{0.8009 \times 965.18 + 0.1991 \times 974.73}{1.015} = \$952.79$$

$\Rightarrow$ The market price is recovered with the *risk-neutral* probabilities, so A is false.

Step 2 – Check the other statements

B: probabilities are solved to fit bond prices, so a new $s(1.5)$ changes them. C: $(2 \times 0.8009 - 1) \times 50 = 30.1$ bp and $(2 \times 0.6519 - 1) \times 50 = 15.2$ bp. D: each new date needs rate assumptions and probabilities calibrated to more bond prices.

Interpretation

This is GARP's own tree shifted up 100 bp; the probabilities and drifts carry over.

Why the other options are wrong

B is incorrect. True: the risk-neutral probabilities are calibrated to the term structure.
C is incorrect. True: those are the risk-neutral drifts.
D is incorrect. True: this is how the tree is extended.

LO: MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.4, Figures 12.6 and 12.7

MR-12 • Arbitrage Pricing with Term Structure Models • Vardeez: MODERATE

MR-12 f

P3 BOTH

Define option-adjusted spread (OAS) and apply it to security pricing.

Workbook: C7 CMT Swap Valuation and OAS on a Rate Tree • Tier C • Map #33

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C7) hits this LO: the reported question valued a CMT swap "with OAS".
Vardeez: CMT swap (the OAS is applied to it).

Reported exam shapes

- **C7·S1** Value a CMT swap backward through a two-period tree with risk-neutral probabilities, inside an OAS wrapper (worked under MR-12 h).
- **C7·S2** OAS: shift the discount rates only, then read cheap versus rich from the sign.

What the questions look like: the sign and meaning of OAS, which rates it shifts, a value at a given OAS, and the expected return of a security that trades at an OAS.

1. Definition

The **option-adjusted spread** is the spread that, *added to the discount rates in the model*, makes the model value equal the market price. It measures a security's mispricing relative to the model, and the extra return a hedged position earns if the model is right.

- **Shift discounting only.** Rates are shifted only for discounting. The rates that *set the cash flows* (for a CMT swap, the yields in the payoff) are not touched.
- **Sign.** Market price below model value needs heavier discounting: **OAS > 0, the security trades cheap**. Market price above model value: **OAS < 0, rich**.
- **GARP's example.** The CMT swap of MR-12 h has a model value of \$3,702.11. Adding 10 bp to every discount rate (2.10% at date 0, 2.60% and 1.60% at date 1) gives \$3,699.18: a market price \$2.92 below the model value means an OAS of +10 bp, cheap.

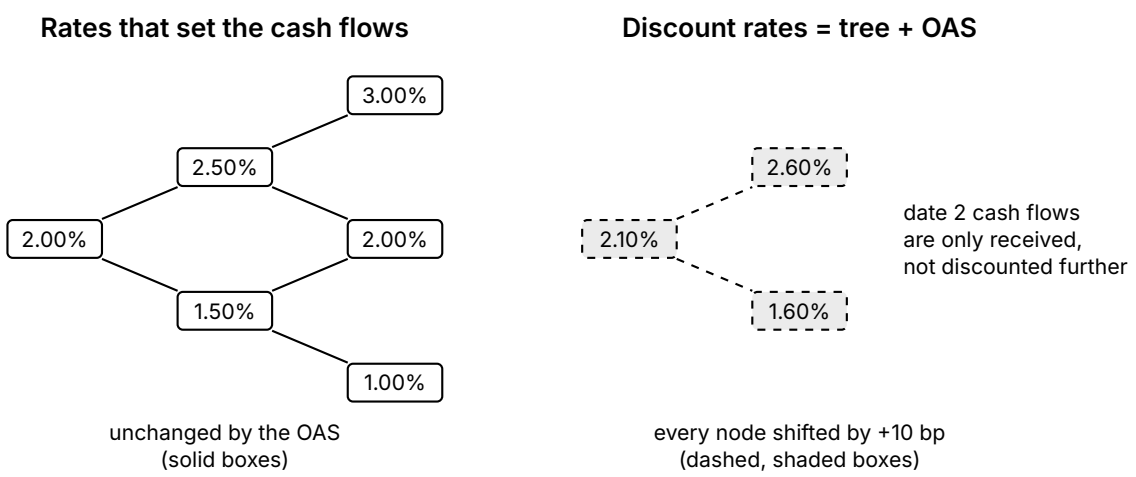


Figure 2. OAS as a parallel shift of the discount tree only. The left tree (solid boxes) still sets the CMT payments; the right tree (dashed, shaded) is used to discount. GARP's CMT swap: OAS +10 bp takes the value from \$3,702.11 to \$3,699.18.

2. OAS, expected return and P&L

- **Expected return.** Under the risk-neutral process a fairly priced security earns the short rate; one with an OAS earns the short rate *plus* the OAS:

$$E\left[\frac{dP}{P}\right] = (r_0 + \text{OAS})\,dt.$$

  GARP: the fairly priced swap earns 1.00% over six months (half of 2%); at an OAS of 10 bp it earns 1.05% (half of 2.10%).

- **P&L attribution.**

$$dP = (r_0 + \text{OAS})P\,dt + \frac{\partial P}{\partial r}(dr - E[dr]) + \frac{\partial P}{\partial\,\text{OAS}}\,d\text{OAS}$$

  In order: carry-roll-down, gains or losses from rate changes, gains or losses from spread changes.

- **Relative value trade.** Buy the cheap security, hedge the factor risk ($\partial P/\partial r = 0$) and finance at the short rate: the expected return is **the OAS plus any convergence of the OAS toward zero**. Risk premium terms cancel because factor risk is hedged.

- **Two worked versions.** The 2026 GARP text uses the 2%-strike swap above. Schweser and Vardeez use an older version: model \$1,466.63, market \$1,464.40, OAS +20 bp (cheap). Same mechanics.

Formula card: MR-12 f

Value at an OAS	discount at $r_{\text{node}} + \text{OAS}$; cash flows still from r_{node}
Sign	$\text{OAS} > 0$: market $<$ model, cheap. $\text{OAS} < 0$: market $>$ model, rich
Expected return	$E[dP/P] = (r_0 + \text{OAS})\,dt$; per six months $= (r_0 + \text{OAS})/2$
Hedged, financed trade	earns OAS + convergence

Exam traps

- **Shifting the cash flows.** Adding the OAS to the rates that set the payments changes the cash flows and moves the value far more than a spread adjustment should.
- **Cheap versus rich.** Positive OAS means cheap; GARP puts "positive, rich" next to the right answer.
- **Changing probabilities.** The OAS leaves the risk-neutral probabilities alone.
- **Intermediate result.** The zero-OAS model value is a step, not the market price when an OAS is quoted.
- **Annual versus per period.** The expected return over a six-month step is half of $r_0 + \text{OAS}$.

Practice questions: MR-12 f

Question MR-12 f.1

Bank Q602 • AnalystPrep Q.1633

The P&L of a security can be decomposed into a component due to the passage of time, a component due to changes in the factor (short rate), and a component due to changes in the option-adjusted spread (OAS). If a security is not priced in accordance with the model (its OAS is not zero), its cash flows in the tree are discounted at:

A. the short-term rate.

B. the long-term rate.

C. the short-term rate plus the OAS.

D. the short-term rate minus the OAS.

Question MR-12 f.2

Bank Q601 • AnalystPrep Q.1632 (rebuilt)

A binomial interest rate tree gives a constant-maturity Treasury swap a model value of USD 1,466.63, while its observed market price is USD 1,464.40. The option-adjusted spread (OAS) is the constant spread that, when added to the short rates used for discounting in the tree, equates the model value to the market price. Which statement is correct?

A. The OAS is positive; the security is trading cheap relative to the model.

B. The OAS is negative; the security is trading cheap relative to the model.

C. The OAS is positive; the security is trading rich relative to the model.

D. The OAS is zero, because the model already prices the security.

Question MR-12 f.3

Bank Q604 • BT Mock Q26

A stylized constant-maturity Treasury (CMT) swap has notional USD 1,000,000 and strike 2.00%, with a semiannual rate tree: date 0 = 2.00%; date 1 = 2.50% (up) or 1.50% (down); date 2 = 3.00%, 2.00%, 1.00%. Risk-neutral up-probabilities are 80.09% (date 0 to 1) and 65.20% (date 1 to 2). The expected discounted value (model value) is about USD 3,702. The observed market price is USD 3,694.81, which is USD 7.30 below the model value. This implies an OAS of about 25 basis points. Which is the best interpretation?

A. The model returns the market price if the risk-neutral probabilities are changed to 55.09% and 44.91%.

B. The model returns the market price if the cash flow at the top date-2 node is cut to USD 3,750 from USD 5,000.

C. The model returns the market price if the discount rates are 2.25% (date 0), 2.75% (date 1, up node) and 1.75% (date 1, down node), i.e., each 25 bp higher.

D. The OAS cannot be interpreted because this is a CMT swap, and an OAS requires an embedded option.

Question MR-12 f.4 New • built from GARP MR Ch.12, 12.5 to 12.6

A trader is marking the USD 1,000,000 stylized CMT swap struck at 2.00% on GARP's tree: six-month rate 2.00% today, 2.50% or 1.50% on date 1, and 3.00%, 2.00% or 1.00% on date 2, with risk-neutral up probabilities of 0.8009 (date 0) and 0.6520 (date 1) and a model value of USD 3,702.11. The swap is quoted in the market at an OAS of -10 bp. Which of the following is closest to its market price, and how does it trade?

A. USD 2,719.83; the swap trades rich

B. USD 3,699.18; the swap trades rich

C. USD 3,705.03; the swap trades cheap

D. USD 3,705.03; the swap trades rich

Question MR-12 f.5 New • built from GARP MR Ch.12, 12.6 to 12.7

A relative value analyst models a mortgage-related security on a semiannual binomial tree. The six-month rate today is 4.00%, and the security trades at an option-adjusted spread of -20 bp. Assuming the OAS is expected to stay constant, which of the following is closest to the security's expected return over the next six months under the risk-neutral process?

A. 1.90%

B. 2.00%

C. 2.10%

D. 3.80%

Question MR-12 f.6 New • built from GARP MR Ch.12, 12.7

A relative value trader identifies a bond that trades at an OAS of +15 bp under the desk's term structure model. She buys it, hedges all of its factor (interest rate) risk so that $\partial P/\partial r = 0$, and finances the purchase by borrowing at the short-term rate. If the model is correct, what does the position's expected return consist of?

A. The OAS plus any gain from the OAS converging toward zero.

B. The short-term rate plus the OAS, whatever the financing cost.

C. The risk premium on interest rate risk plus the OAS.

D. The short-term rate only, because the factor risk is hedged.

Solutions • MR-12 f

Solution MR-12 f.1 — Answer: C

Bank Q602 • AnalystPrep Q.1633

Step 1 – Definition

The OAS is the spread added to the model's discount rates so that model value equals market price.

$\Rightarrow$ A security with a non-zero OAS is discounted node by node at the short rate plus the OAS; its cash flows are unchanged.

Interpretation

A security priced exactly by the model (OAS $= 0$) is discounted at the short rate.

Why the other options are wrong

A is incorrect. The short rate alone applies only when OAS $= 0$.

B is incorrect. A tree discounts one period at a time at the node's short rate, not at a long-term rate.

D is incorrect. The OAS is added. If the market price is above the model value the OAS itself is negative, but it is still added.

LO: MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.6 and 12.7, eq. 12.35

Solution MR-12 f.2 — Answer: A

Bank Q601 • AnalystPrep Q.1632

Step 1 – Compare market and model

Market USD 1,464.40 is USD 2.23 below the model's USD 1,466.63.

Step 2 – Sign of the spread

To bring the model value down, discount at higher rates: the spread is positive (20 bp in the Schweser example).

$\Rightarrow$ Positive OAS, market below model: cheap.

Interpretation

The security offers more than the model says it should, so a buyer expects to earn the short rate plus the OAS.

Why the other options are wrong

B is incorrect. A negative spread lowers the discount rates and raises the model value further above the market price.

C is incorrect. Right sign, wrong label: market below model is cheap, not rich.

D is incorrect. OAS is zero only when model and market agree.

LO: MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. • **Reference:** SchweserNotes 2026 Book 1, Reading 12, LO 12.f; Tuckman and Serrat 4e, 12.6

Solution MR-12 f.3 — Answer: C

Bank Q604 • BT Mock Q26

Step 1 – What an OAS changes

Only the discount rates: each node rate plus the spread. Cash flows and risk-neutral probabilities are unchanged.

Step 2 – Check the size

Discounting at 2.25%, 2.75% and 1.75% gives date-1 values 5,715.55 and $-4{,}225.15$ and a date-0 value of 3,694.81, the market price.

$\Rightarrow$ A uniform 25 bp shift in the discount rates is exactly what the OAS means.

Interpretation

Positive OAS, market below model: the swap trades cheap.

Why the other options are wrong

A is incorrect. OAS does not change the risk-neutral probabilities; moving them to 55.09% would shift the value by hundreds of dollars.

B is incorrect. OAS does not change cash flows; cutting a payment by USD 1,250 would also move the value by hundreds of dollars.

D is incorrect. An OAS can be computed for any security valued on a tree, with or without an embedded option.

LO: MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.6, eqs. 12.27 to 12.29

Solution MR-12 f.4

Answer: D

New • built from GARP MR Ch.12, 12.5 to 12.6

Step 1 – Payments (unshifted rates)

Date 1: $+2{,}500$ (2.50%) and $-2{,}500$ (1.50%). Date 2: $+5{,}000$, 0, $-5{,}000$.

Step 2 – Date-1 values, discounting at the node rate minus 10 bp

$$V_{1,u} = \frac{0.6520(5{,}000) + 0.3480(0)}{1 + 0.0240/2} + 2{,}500 = 5{,}721.11$$

$$V_{1,d} = \frac{0.6520(0) + 0.3480(-5{,}000)}{1 + 0.0140/2} - 2{,}500 = -4{,}228.14$$

Step 3 – Date-0 value at 1.90%

$$V_0 = \frac{0.8009(5{,}721.11) + 0.1991(-4{,}228.14)}{1 + 0.0190/2} = 3{,}705.03$$

$\Rightarrow$ The market price is USD 2.92 above the model value: negative OAS, rich.

Interpretation

GARP's model and Figure 12.8 use the unrounded $q = 0.65195$; with $q = 0.6520$ each value is about USD 0.2 higher. Either way the answer is closest to D.

Why the other options are wrong

A is incorrect. Shifted the payment rates as well as the discount rates: 2,719.83.

B is incorrect. Added +10 bp instead of -10 bp: 3,699.18 is GARP's value at a positive OAS.

C is incorrect. Right price, wrong label: a negative OAS means the market price is above the model value, so rich.

LO: MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.5 and 12.6, eqs. 12.24 to 12.29

Solution MR-12 f.5 **Answer: A**

New • built from GARP MR Ch.12, 12.6 to 12.7

Step 1 – Expected return with an OAS

$$E\left[\frac{dP}{P}\right] = (r_0 + \text{OAS})\,dt = (4.00\% - 0.20\%) \times \tfrac{1}{2} = 1.90\%$$

$\Rightarrow$ A rich security (negative OAS) is expected to earn less than the short rate.

Interpretation

GARP's version: 2% short rate and +10 bp OAS give 1.05% over six months. "Mortgage-related" is background only.

Why the other options are wrong

B is incorrect. Ignored the OAS: $4.00\%/2$.
C is incorrect. Added the spread with the wrong sign: $(4.00\% + 0.20\%)/2$.
D is incorrect. Gave the annual rate $r_0 + \text{OAS} = 3.80\%$, not the six-month return.

LO: MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.6 and 12.7, eqs. 12.30 to 12.35

Solution MR-12 f.6 **Answer: A**

New • built from GARP MR Ch.12, 12.7

Step 1 – Attribution

$dP = (r_0 + \text{OAS})P\,dt + \frac{\partial P}{\partial r}(dr - E[dr]) + \frac{\partial P}{\partial \text{OAS}}d\text{OAS}$.

Step 2 – Apply the trade

Hedged: $\partial P/\partial r = 0$, so the rate term drops. Financed at r_0: the short-rate part of the carry is paid away.
$\Rightarrow$ What is left is the OAS, plus the gain $\frac{\partial P}{\partial \text{OAS}}\,d\text{OAS}$ if the OAS converges toward zero.

Interpretation

This is why the OAS is used as a relative value measure: a model with predictive power implies cheap securities converge to fair value.

Why the other options are wrong

B is incorrect. Financing at the short rate cancels the short-rate component.
C is incorrect. Risk premium terms cancel when factor risk is hedged away.
D is incorrect. Ignores the OAS that the cheap security earns.

LO: MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.7, eqs. 12.36 and 12.37

MR-12 • Arbitrage Pricing with Term Structure Models • Vardeez: MODERATE

MR-12 h

P3 CALC

Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities.

Workbook: C7 CMT Swap Valuation and OAS on a Rate Tree • Tier C • Map #33

WHAT IS TESTED HERE

Why P3. A Tier C workbook topic (C7): "decision tree question with OAS of CMT swap" was reported, and this is the exact LO.
Vardeez: CMT swap.

Reported exam shapes

- **C7·S1** Value a CMT swap backward through a two-period rate tree with risk-neutral probabilities, inside an OAS wrapper.
- **C7·S2** Shift the discount rates only for the OAS, then read cheap or rich (see MR-12 f).
- **C7·S3** True probabilities supplied as a temptation (see MR-12 d).

What the questions look like: a two-date tree, payoffs at each node, backward induction with node-specific discount rates, and distractors built from dropped date-1 payments, true probabilities or reused probabilities.

1. The CMT swap

A constant-maturity Treasury swap exchanges a fixed rate for a Treasury yield of a fixed maturity. Every six months it pays

$$\boxed{\text{Payment} = N \times \frac{y_{\text{CMT}} - K}{2}}$$

where y_{CMT} is the semiannually compounded yield of the predetermined maturity *on the payment date* and K the strike. In practice CMT swaps trade on 2-, 5- and 10-year yields; the textbook uses the six-month yield, which equals the six-month rate on the tree. A negative amount is paid.

2. Recipe: backward induction

1. **Payments at every node** from that node's rate (date 1 and date 2).
2. **Date-1 values** = discounted risk-neutral expectation of the date-2 payments (at the date-1 node's own rate) **plus that node's own date-1 payment**.
3. **Date-0 value** = discounted risk-neutral expectation of the date-1 values at the date-0 rate.

$$V_{\text{node}} = \frac{p\,V_{\text{up}} + (1-p)\,V_{\text{down}}}{1 + r_{\text{node}}/2} + \text{Payment}_{\text{node}}$$

GARP's example. $N = \$1{,}000{,}000$, $K = 2\%$, tree of Figure 1 ($p = 0.8009$, $q = 0.6520$):

- Date 1 payments: $1{,}000{,}000 \times (2.50\% - 2\%)/2 = +\$2{,}500$; $1{,}000{,}000 \times (1.50\% - 2\%)/2 = -\$2{,}500$.
- Date 2 payments: $+\$5{,}000$ (3%), \$0 (2%), $-\$5{,}000$ (1%).

$$V_{1,u} = \frac{0.6520(5{,}000) + 0.3480(0)}{1.0125} + 2{,}500 = 5{,}719.52$$
$$V_{1,d} = \frac{0.6520(0) + 0.3480(-5{,}000)}{1.0075} - 2{,}500 = -4{,}227.29$$
$$V_0 = \frac{0.8009(5{,}719.52) + 0.1991(-4{,}227.29)}{1.01} = \$3{,}702.11$$

- **Why so large when 2% is the average rate?** Under the *true* 0.5 probabilities the value is close to zero ($-\$6.07$). The risk-neutral probabilities, with their upward drift, set the arbitrage price.
- **Rounding.** GARP's figures use the unrounded $q = 0.65195$; with $q = 0.6520$ you get 5,719.75, $-4{,}227.05$ and 3,702.32. Answers are "closest to", so either works.

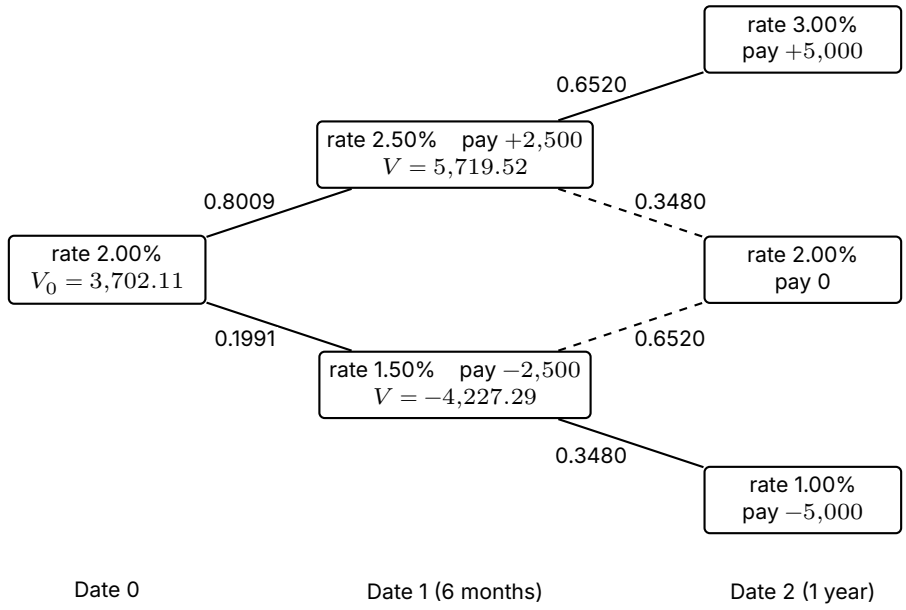


Figure 3. GARP's CMT swap price tree (USD 1,000,000, strike 2%). Branch labels are risk-neutral probabilities. The dashed branches lead to the middle node, whose $0 payment still enters both date-1 values. Each date-1 value V includes that node's own payment.

Formula card: MR-12 h

CMT payment	$N\,(y_{\text{CMT}} - K)/2$ every six months, from the node's own rate
Node value	$\dfrac{p\,V_{\text{up}} + (1-p)\,V_{\text{down}}}{1 + r_{\text{node}}/2} + \text{Payment}_{\text{node}}$
With an OAS	discount at r_{node} + OAS; payments unchanged
GARP check	5,719.52; $-4{,}227.29$; 3,702.11; 3,699.18 at OAS 10 bp; -6.07 with true 0.5
Schweser/Vardeez check	$K = 7\%$, steps of 25 bp, $p = 0.76$, $q = 0.60$: 2,697.53; $-2{,}217.35$; 1,466.63 (1,464.40 at OAS 20 bp)

Exam traps

- **Dropped term.** Each date-1 value includes that node's own payment as well as the discounted date-2 expectation.

- **The middle node.** Its \$0 payment still carries a probability into both date-1 values; it is not the same as leaving the branch out.
- **Payment size.** Divide by 2 (semiannual), once. Date-2 rates are two steps from today.
- **One discount only.** Date-2 payments are discounted twice: to date 1 at the node rate, then to date 0.
- **True or reused probabilities.** Use the risk-neutral probabilities, and the date-1 probability where it differs from date 0.
- **OAS on the payments.** The OAS shifts discount rates only.

Practice questions: MR-12 h

Question MR-12 h.1

Bank Q612 • AnalystPrep Q.2851 (rebuilt)

A \$7 million face value stylized constant-maturity Treasury (CMT) swap is struck at 7%. It pays $7{,}000{,}000 \times (y_{\text{CMT}} - 7\%)/2$ every six months for one year, where y_{CMT} is the six-month yield. The current six-month yield is 7% and moves up or down 0.5% each six months on a recombining tree. What are the possible payoffs of the swap at 6 months and at one year?

A. 6 months: +\$17,500 and −\$17,500; one year: +\$35,000, \$0 and −\$35,000

B. 6 months: +\$35,000 and −\$35,000; one year: +\$70,000, \$0 and −\$70,000

C. 6 months: +\$8,750 and −\$8,750; one year: +\$17,500, \$0 and −\$17,500

D. 6 months: +\$17,500 and −\$17,500; one year: +\$17,500, \$0 and −\$17,500

Question MR-12 h.2

Bank Q613 • BT L2.T5.44.1 (rebuilt)

A CMT swap has notional \$1 million, pays every six months notional/2 × (yield − 3%) on the six-month yield, and matures in one year. The current six-month yield is 3%; it moves +20 bp or −20 bp each six months on a recombining binomial tree. Real-world probabilities are 50%/50%; risk-neutral probabilities are 70% up / 30% down at every node. What is the price of the swap under risk-neutral pricing?

A. \$0

B. \$775

C. \$1,170

D. \$1,182

Question MR-12 h.3 Bank Q611 • AnalystPrep Q.2660 (rebuilt)

A constant-maturity Treasury (CMT) swap with face value \$1 million is struck at 6%. Every six months it pays $1{,}000{,}000 \times (y_{\text{CMT}} - 6\%)/2$, where y_{CMT} is the semiannually compounded six-month yield on the payment date. The six-month rate today is 6%. The risk-neutral probability of an up move is 45% on date 0 (to 6.5% or 5.5% at 6 months) and 70% at each 6-month node (to 7%, 6% or 5% at 1 year, recombining). The swap matures at 1 year. What is the value of the swap?

A. \$500.00

B. \$458.74

C. \$701.46

D. \$472.50

Question MR-12 h.4 New • built from GARP MR Ch.12, 12.5

A treasury analyst at a European bank values a one-year stylized CMT swap on the six-month yield with notional USD 2,000,000 and strike 4.00%. The bank receives USD $2{,}000{,}000 \times (y_{\text{CMT}} - 4.00\%)/2$ every six months (a negative amount is paid). The model's recombining tree has the six-month rate at 4.00% today, moving up or down by 40 bp each six months. The risk-neutral probability of an up move is 0.65 from date 0 and 0.58 from either date-1 node; the desk's estimate of the true probability is 0.50 at every node. Which of the following is closest to the value of the swap?

A. USD -7.54

B. USD 1,760.66

C. USD 2,937.13

D. USD 3,475.07

Question MR-12 h.5 Bank Q614 • BT L2.T5.44.2 (rebuilt)

A CMT swap (notional \$1 million, strike 3%, pays notional/2 $\times$ (six-month yield $-$ 3%) every six months for one year) is valued on a recombining tree: six-month yield today 3%, moving $\pm$20 bp each six months. Valued with the real-world 50/50 probabilities and node-by-node discounting, the swap's value is about $-$\$0.96 rather than exactly zero, even though the up and down moves are symmetric. What best explains the non-zero value?

A. Convexity: discount factors are convex in the rate, so the symmetric payments are discounted at different node rates and do not offset (Jensen's inequality).

B. Mean reversion in the interest-rate process.

C. The risk-neutral probabilities (70/30) differ from the 50/50 probabilities used.

D. Nothing; the discounted expected value with 50/50 probabilities is exactly zero.

Question MR-12 h.6 Workbook C7.1 • S1, S2, S3 • Map #33

A risk analyst at a US insurer is checking the desk's mark on a one-year stylized constant-maturity Treasury (CMT) swap on the six-month yield. The insurer receives, every six months for one year, USD $5{,}000{,}000 \times (y_{\text{CMT}} - 2.80\%)/2$, where y_{CMT} is the six-month rate on the payment date (a negative amount is paid). The desk's recombining tree of six-month rates (semiannually compounded) and its probabilities are:

- Date 0: 3.00%. Date 1: 3.60% (up) or 2.40% (down). Date 2: 4.20%, 3.00% or 1.80%
- Risk-neutral probability of an up move: 0.70 from date 0; 0.60 from either date-1 node
- Estimated true probability of an up move at every node: 0.50

The pricing system reports that the swap trades in the market at an option-adjusted spread (OAS) of +25 bp. Which of the following is closest to the swap's market price, and how does it trade relative to the model?

A. USD 24,328; the swap trades cheap

B. USD 24,328; the swap trades rich

C. USD 9,719; the swap trades cheap

D. USD 36,523; the swap trades cheap

Solutions • MR-12 h

Solution MR-12 h.1 — Answer: A

Bank Q612 • AnalystPrep Q.2851

Step 1 – Six-month payments

Yields 7.5% and 6.5%: $7{,}000{,}000 \times (\pm 0.5\%)/2 = \pm\$17{,}500$.

Step 2 – One-year payments

Yields 8%, 7%, 6% (two steps): $7{,}000{,}000 \times (+1\%,\ 0,\ -1\%)/2 = +\$35{,}000,\ \$0,\ -\$35{,}000$.

$\Rightarrow$ Notional times the yield gap over the strike, halved for the semiannual accrual.

Interpretation

Getting the payments right at every node is the first step of every CMT valuation.

Why the other options are wrong

B is incorrect. Forgot to divide by 2, doubling every payment.

C is incorrect. Divided by 2 twice, halving every payment.

D is incorrect. Used a single 0.5% step at one year instead of two cumulative steps.

LO: MR-12 h: Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.5, eqs. 12.18 to 12.23

Solution MR-12 h.2 — Answer: C

Bank Q613 • BT L2.T5.44.1

Step 1 – Payments

Date 1: $\pm\$1{,}000$ (3.2%, 2.8%). Date 2: $+\$2{,}000$ (3.4%), \$0 (3.0%), $-\$2{,}000$ (2.6%).

Step 2 – Date-1 values

$$V_{1,u} = \frac{0.7(2{,}000) + 0.3(0)}{1.016} + 1{,}000 = 2{,}377.95$$

$$V_{1,d} = \frac{0.7(0) + 0.3(-2{,}000)}{1.014} - 1{,}000 = -1{,}591.72$$

Step 3 – Date-0 value

$$V_0 = \frac{0.7(2{,}377.95) + 0.3(-1{,}591.72)}{1.015} = \frac{1{,}187.05}{1.015} = \$1{,}169.51 \approx \$1{,}170$$

Interpretation

With 70/30 risk-neutral probabilities the swap is worth about \$1,170 although the true moves are symmetric. The real-world probabilities are a red herring.

Why the other options are wrong

A is incorrect. Assumed the symmetric moves cancel, as they would (almost) under the true 50/50.

B is incorrect. Left out the date-1 payments of $\pm\$1{,}000$: \$775.

D is incorrect. Discounted the date-2 payments only once, at 1.5%, skipping the date-1 discount: \$1,182.

LO: MR-12 h: Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.5

Solution MR-12 h.3

Answer: B

Bank Q611 • AnalystPrep Q.2660

Step 1 – Payments

Date 1: $\pm\$2{,}500$. Date 2: $+\$5{,}000$ (7%), \$0 (6%), $-\$5{,}000$ (5%).

Step 2 – Date-1 values (probability 0.70)

$$V_{1,u} = \frac{0.7(5{,}000) + 0.3(0)}{1.0325} + 2{,}500 = 5{,}889.83$$

$$V_{1,d} = \frac{0.7(0) + 0.3(-5{,}000)}{1.0275} - 2{,}500 = -3{,}959.85$$

Step 3 – Date-0 value (probability 0.45)

$$V_0 = \frac{0.45(5{,}889.83) + 0.55(-3{,}959.85)}{1.03} = \frac{472.50}{1.03} = \$458.74$$

Interpretation

Here the date-0 probability is below 0.5 and differs from the date-1 probability; each step uses its own.

Why the other options are wrong

A is incorrect. The undiscounted risk-neutral expectation of all payments: \$500.00.
C is incorrect. Left out the date-1 payments: \$701.46.
D is incorrect. Stopped at the date-1 expectation, \$472.50, without the final discount by 1.03.

LO: MR-12 h: Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.5

Solution MR-12 h.4

Answer: C

New • built from GARP MR Ch.12, 12.5

Step 1 – Payments

Date 1: $2{,}000{,}000 \times (\pm 0.40\%)/2 = \pm\$4{,}000$. Date 2: $+\$8{,}000$ (4.80%), \$0 (4.00%), $-\$8{,}000$ (3.20%).

Step 2 – Date-1 values (probability 0.58)

$$V_{1,u} = \frac{0.58(8{,}000) + 0.42(0)}{1.022} + 4{,}000 = 8{,}540.12$$

$$V_{1,d} = \frac{0.58(0) + 0.42(-8{,}000)}{1.018} - 4{,}000 = -7{,}300.59$$

Step 3 – Date-0 value (probability 0.65)

$$V_0 = \frac{0.65(8{,}540.12) + 0.35(-7{,}300.59)}{1.02} = \$2{,}937.13$$

Interpretation

The swap is struck at today's rate, yet worth about USD 2,900 to the receiver of the floating yield, because the risk-neutral process drifts up (+12 bp, then +6.4 bp). The true probabilities are a red herring.

Why the other options are wrong

A is incorrect. Used the true 0.50 at every node: $-\$7.54$, close to zero.

B is incorrect. Left out the date-1 payments of $\pm\$4{,}000$: \$1,760.66.
D is incorrect. Reused 0.65 at date 1 instead of 0.58: \$3,475.07.

LO: MR-12 h: Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.5, eqs. 12.18 to 12.26

Solution MR-12 h.5 — Answer: A

Bank Q614 • BT L2.T5.44.2

Step 1 – Value with 50/50

$$V_{1,u} = \frac{0.5(2{,}000)}{1.016} + 1{,}000 = 1{,}984.25$$

$$V_{1,d} = \frac{0.5(-2{,}000)}{1.014} - 1{,}000 = -1{,}986.19$$

$$V_0 = \frac{0.5(1{,}984.25) + 0.5(-1{,}986.19)}{1.015} = -\$0.96$$

Step 2 – Why not zero

Positive payments occur on up paths, where rates and so discounting are higher; negative payments occur on down paths, where discounting is lighter. The negative payments get slightly more weight.
$\Rightarrow$ A convexity (Jensen's inequality) effect of discounting node by node.

Interpretation

GARP's own example shows the same effect: $-\$6.07$ under the true probabilities. The large positive values come from the risk-neutral drift, not from this effect.

Why the other options are wrong

B is incorrect. The tree has symmetric steps and no mean reversion.
C is incorrect. The 70/30 probabilities are not used here; they would give about \$1,170 (Question MR-12 h.2).
D is incorrect. The calculation gives $-\$0.96$, not zero.

LO: MR-12 h: Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. • **Reference:** Tuckman and Serrat 4e (GARP MR Ch.12), 12.5

Solution MR-12 h.6 — Answer: A

Workbook C7.1 • S1, S2, S3

Step 1 – CMT payments (from the unshifted tree rates)

Date 1: $5{,}000{,}000 \times \frac{3.60\%-2.80\%}{2} = 20{,}000$; $5{,}000{,}000 \times \frac{2.40\%-2.80\%}{2} = -10{,}000$.
Date 2: 35,000 (4.20%); 5,000 (3.00%); $-25{,}000$ (1.80%).
$\Rightarrow$ The OAS does not touch these: rates are shifted only for discounting.

Step 2 – Date-1 values, discounting at the tree rate plus 25 bp

$$V_{1,u} = \frac{0.60(35{,}000) + 0.40(5{,}000)}{1 + 0.0385/2} + 20{,}000 = \frac{23{,}000}{1.01925} + 20{,}000 = 42{,}565.61$$

$$V_{1,d} = \frac{0.60(5{,}000) + 0.40(-25{,}000)}{1 + 0.0265/2} - 10{,}000 = \frac{-7{,}000}{1.01325} - 10{,}000 = -16{,}908.46$$

$\Rightarrow$ Each node value is the discounted risk-neutral expectation of the next payment plus the payment made at that node.

Step 3 – Date-0 value at the OAS

$$V_0 = \frac{0.70(42{,}565.61) + 0.30(-16{,}908.46)}{1 + 0.0325/2} = \frac{24{,}723.39}{1.01625} = 24{,}328.06$$

Step 4 – Compare with the model value (zero OAS)

Repeating Steps 2 and 3 at the unshifted rates gives date-1 values 42,593.32 and $-16{,}916.99$ and a model value of USD 24,374.61. The market price, 24,328, is USD 46.55 below the model value, consistent with a positive OAS: the swap trades cheap.

Interpretation

The swap is worth far more than zero although its strike, 2.80%, sits below a 3.00% starting rate and the true probabilities are symmetric. The reason is the risk-neutral drift: $(2\times0.70-1)\times60 = 24$ bp from date 0 and $(2 \times 0.60 - 1) \times 60 = 12$ bp from date 1, against zero under the true probabilities. If the model is right, a hedged, financed long position earns the 25 bp OAS plus any convergence of the OAS to zero; under the risk-neutral process the expected six-month return is $(3.00\% + 0.25\%)/2 = 1.625\%$.

Why the other options are wrong

B is incorrect. Polarity flip on the OAS sign. A positive OAS means the market price is below the model price, so the swap is cheap, not rich.

C is incorrect. Used the true probabilities of 0.50 at every node (with the OAS): date-1 values 39,622.27 and $-19{,}869.23$, date-0 value 9,718.59. Real-world probabilities do not recover market prices.

D is incorrect. Added the OAS to the CMT yields that set the payments as well as to the discount rates: payments become 26,250, $-3{,}750$, 41,250, 11,250 and $-18{,}750$, giving 36,522.75.

LO: MR-12 h: Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. • **Reference:** Tuckman and Serrat 4e, Ch.7 (GARP MR Ch.12), 12.3, 12.5 and 12.6, eqs. 12.18 to 12.31, Figure 12.8

MR-13 • Expectations, Risk Premium, Convexity, and the Shape of the Term Structure • Vardeez: MODERATE

MR-13 a

P3 CONCEPT

Explain the role of interest rate expectations in determining the shape of the term structure.

Workbook: D18 Inflation and Market Stress Across Bond Maturities • Tier D

WHAT IS TESTED HERE

Why P3. A Tier D workbook topic hits this LO: D18, inflation and market stress across bond maturities (Map #72, 2/16).
Vardeez (whole reading): Jensen's inequality, and "the value of convexity increases with bond volatility and time to maturity". Both are worked under MR-13 b and c.

Reported exam shapes

- **D18 S1** Higher expected inflation as higher expected short rates: the effect on forward rates and bond values grows with duration.
- **D18 S3** A climate-related shock framed as an inflation and stress scenario for short versus long bonds.

What the questions look like: "which statement is correct" questions. What shape a given expected path produces when rates are certain; what expectations can and cannot say about the long end; and a scenario (inflation, stress) read through the decomposition of forward rates.

1. Expectations alone determine the curve

Take annual periods, a one-year rate of 8%, and investors who know *with certainty* that the one-year rate will be 7% in one year and 6% in two years. Zero prices are then

$$P(1) = \frac{1}{1.08}, \qquad P(2) = \frac{1}{1.08 \times 1.07}, \qquad P(3) = \frac{1}{1.08 \times 1.07 \times 1.06}.$$

By the definition of forward rates, the first three forward rates are 8%, 7% and 6%: **with no volatility around expectations, forward rates equal expected future rates and the term structure is completely determined by expectations.** It can take any shape: flat, upward-sloping, downward-sloping, or a mix.

Spot rates are then geometric averages of the expected one-year rates: $(1 + \hat{r}(n))^n = (1 + r_1)(1 + r_2) \cdots (1 + r_n)$.

Expected one-year rates (now, in 1 yr, in 2 yrs)	1-yr spot	2-yr spot	3-yr spot
8%, 8%, 8% (flat)	8.000%	8.000%	8.000%
8%, 10%, 12% (rising)	8.000%	8.995%	9.988%
8%, 6%, 4% (falling)	8.000%	6.995%	5.987%

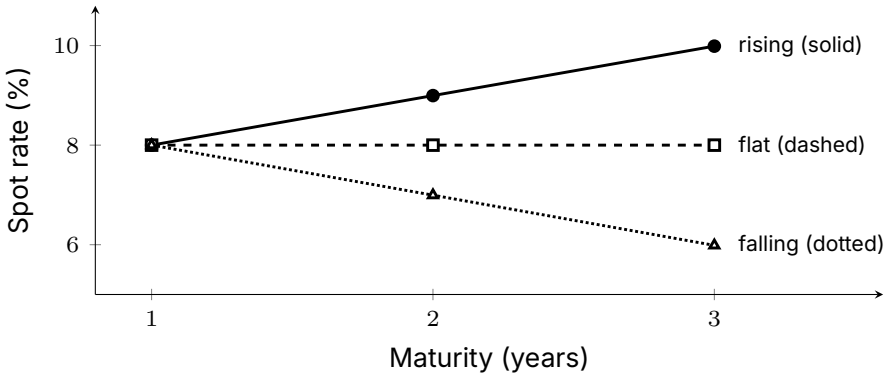


Figure 1. All three curves start at the same 8% one-year rate. Under pure expectations with no volatility, everything after that is driven by what the market expects future one-year rates to be: nothing else is in the model.

2. What expectations can and cannot do

- **Short horizons: specific views, so they shape the front end.** Views come from central bank meeting dates and from the supply and demand for funds (tax payment dates, the bond issuance calendar, quarterly balance sheet management).
- **Long horizons: not granular.** Money market analysis cannot say that the expected one-year rate in 29 years differs from that in 30 years.
- **Long horizons: macroeconomics anchors the level.** GARP's example: a long-run expected short rate of 4%, being 1% long-run real rate plus 3% long-run inflation.
- **So:** expectations describe the shape and level of the curve over short horizons, but mainly the *level* over long horizons.
- **In practice** expectations cannot take any arbitrary pattern, and pure expectations cannot fully explain the real curve: volatility (convexity) and the risk premium also act on it (MR-13 b, c, d).

3. Expectations inside the full decomposition

GARP's general result (derived under MR-13 d) splits every forward rate into three pieces:

$$f(T) = \underbrace{r_0 + D\frac{E[dr]}{dt}}_{\text{expectations}} + \underbrace{\lambda D}_{\text{risk premium}} - \underbrace{\tfrac{1}{2}C\sigma^2}_{\text{convexity}}$$

The expectations term says: the higher the short rate, the higher the forward rate; the more rates are expected to rise, the higher the forward rate; and **the greater the duration, the greater the effect** of expected rate changes. So a rise in expected inflation (higher expected short rates) lifts forwards most at the long end, and long bonds lose most value (shape D18 S1).

	Pure expectations hypothesis	Pure risk premium hypothesis
Assumption	Risk premium $\lambda = 0$	No expected rate change, $E[dr/dt] = 0$
Forward rates set by	Expected rate changes	The risk premium
Natural "no-change" scenario	Short rates evolve as expected, so **forward rates are realized**	Short rates stay where they are, so the **term structure is unchanged**
A downward-sloping curve means	Investors expect short rates to fall	(Cannot come from expectations)

Reality lies between the two. And the split cannot be observed: **a drift of x basis points has the same pricing effect as a risk premium of x basis points per year of duration**, so forward rates alone cannot separate expected rate changes from risk premium. Only the risk-neutral process matters for pricing.

Exam traps

- **Horizon swap.** Expectations shape the curve at *short* horizons and set only the *level* at long horizons. A statement that they dominate the long-end shape reverses this.
- **Convexity where there is none.** With certainty there is no volatility, so forwards equal expected rates exactly: no convexity effect, no premium.
- **Parallel-shift assumption.** The expectations term and the risk premium both scale with duration; neither moves all forwards by the same amount.
- **Hypothesis swap.** Pure expectations: $\lambda = 0$, forwards realized. Pure risk premium: $E[dr/dt] = 0$, curve unchanged.
- **Absolute claim.** Forward rates cannot, on their own, split a move into expectations and risk premium.

Practice questions: MR-13 a

Question MR-13 a.1

New • built from GARP MR Ch.13, 13.1

A rates strategist assumes that the current one-year rate is 8%, that investors know with certainty that the one-year rate will be 7% in one year and 6% in two years, and that investors are risk neutral. Which of the following statements about the term structure implied by these assumptions is correct?

A. The one-year forward rates are 8%, 7% and 6%, so the forward curve slopes downward and is completely determined by expectations.

B. The spot curve is flat, because each spot rate is an average of the same three expected one-year rates.

C. Forward rates lie below the expected one-year rates, because bond prices are convex in interest rates.

D. Forward rates lie above the expected one-year rates, because investors demand compensation for holding longer bonds.

Question MR-13 a.2

Bank Q623 • BT P2.T5.300.2

Each of the following is TRUE about the shape of the term structure of spot interest rates EXCEPT for:

A. If all investors agree with certainty that the current one-year spot rate of 1.0% will increase to 2.0% next year and again to 3.0% in two years, the three-year spot rate should be almost 2.0% (with annual compounding).

B. It is unreasonable and impractical to expect pure (and strict or narrow) expectations theory to fully characterize the term structure in practice.

C. Although investors' expectations about short-term rates have little or no impact on the SHAPE of the term structure, they tend to dominate its SHAPE over longer time horizons (e.g., 20 or 30 years).

D. Under an assumption of pure expectations, it is valid to hypothesize (infer) from a downward-sloping term structure that investors are forecasting a drop in future short-term (e.g., one-year) interest rates.

Question MR-13 a.3 New • built from GARP MR Ch.13, 13.1

A strategist at a pension fund is building a 30-year forward curve. Her team has detailed views on the next four central bank meetings, on tax payment dates and on quarter-end funding pressure. Her economist estimates a long-run real rate of 1% and long-run inflation of 3%. Which of the following is the most appropriate use of these views?

A. Use the short-horizon views to shape the front of the curve, and the 4% macroeconomic estimate to anchor the level of long-horizon expected rates.

B. Use the short-horizon views to project how the expected one-year rate in 29 years differs from the expected one-year rate in 30 years.

C. Use the macroeconomic estimate to set both the level and the detailed shape of the long end, because inflation forecasts are as granular as meeting-date views.

D. Ignore expectations beyond one year, because the long end of the curve is determined by convexity alone.

Question MR-13 a.4 New • built from GARP MR Ch.13, 13.3

An analyst explains two theories of the term structure using the decomposition $f(T) = r_0 + D\,E[dr]/dt + \lambda D - \frac{1}{2}C\sigma^2$, setting the convexity term aside. Which of the following statements is correct?

A. Under the pure expectations hypothesis, $\lambda = 0$, and the natural no-change scenario is that short-term rates evolve as expected so that forward rates are realized.

B. Under the pure expectations hypothesis, $E[dr/dt] = 0$, and the natural no-change scenario is that the term structure stays unchanged.

C. Under the pure risk premium hypothesis, forward rates are set by expected rate changes, so an upward-sloping curve signals that rates are expected to rise.

D. Under the pure risk premium hypothesis, the natural no-change scenario is that short-term rates evolve so that forward rates are realized.

Question MR-13 a.5 New • built from GARP MR Ch.13, 13.3

After a central bank announcement, a portfolio manager observes that forward rates have risen, by more at longer terms. She wants to attribute the move between a higher expected path of short-term rates and a higher risk premium. Which of the following statements is correct?

A. Forward rates alone cannot split the move, because a drift of a given number of basis points has the same pricing effect as a risk premium of that number per year of duration.

B. The move must be a higher risk premium, because a higher expected path of short-term rates would raise all forward rates by the same amount.

C. The move must reflect higher expected rates, because a risk premium depends on the characteristics of individual bonds rather than on duration.

D. The two can be separated by comparing forward rates with the risk-neutral probabilities used in a pricing tree.

Question MR-13 a.6 Workbook D18.1 • S1, S2, S3 • Map #72

A treasury analyst at a UK bank is preparing a report for the asset-liability committee on how a stress scenario would affect the bank's short-dated and long-dated gilt holdings. In the scenario, a climate-related supply shock raises expected future short-term rates through higher expected inflation, while market stress raises interest rate volatility and the risk premium investors demand per unit of duration. Using the decomposition of forward rates into expectations, risk premium and convexity, which of the following statements should the analyst include in the report?

A. The rise in rate volatility raises long-term forward rates through the convexity term, so long-dated gilts cheapen relative to short-dated gilts for that reason.

B. The higher risk premium raises all forward rates by the same number of basis points, because the premium per unit of duration does not depend on any individual bond.

C. Observed forward rates allow the analyst to fully separate the part of the move due to higher expected inflation from the part due to a higher risk premium.

D. Higher expected short-term rates and a higher risk premium each raise forward rates in proportion to duration, so long-dated gilts lose more value than short-dated gilts.

Solutions • MR-13 a

Solution MR-13 a.1

Answer: A

New • GARP MR Ch.13, 13.1

Step 1 – Price the zeros on the certain path

$P(1) = 1/1.08$, $P(2) = 1/(1.08 \times 1.07)$, $P(3) = 1/(1.08 \times 1.07 \times 1.06)$.

Step 2 – Read off the forward rates

$1 + f_t = P(t-1)/P(t)$ gives forwards of 8%, 7% and 6%: exactly the expected one-year rates.
$\Rightarrow$ With certainty, the term structure is completely determined by expectations; here it slopes downward.

Interpretation

The spot rates are 8%, $\sqrt{1.08 \times 1.07} - 1 = 7.50\%$ and $(1.08 \times 1.07 \times 1.06)^{1/3} - 1 = 7.00\%$: also falling. The assumptions remove both other forces on the curve: certainty removes volatility (so no convexity effect), and risk neutrality removes the premium.

Why the other options are wrong

B is incorrect. Each spot rate averages a *different* set of expected rates (one, two or three of them), so the spot curve falls: 8.00%, 7.50%, 7.00%.
C is incorrect. The convexity effect needs volatility; with certain rates there is none, so forwards equal expected rates exactly.
D is incorrect. A risk premium is excluded by the risk-neutrality assumption, and in any case it is not what makes this curve slope.

LO: MR-13 a: Explain the role of interest rate expectations in determining the shape of the term structure. • **Reference:** Tuckman and Serrat, Fixed Income Securities 4e, Ch.8, 13.1, eq. 13.1

Solution MR-13 a.2

Answer: C

Bank Q623 • BT P2.T5.300.2

Step 1 – Check A

$(1.01 \times 1.02 \times 1.03)^{1/3} - 1 = 1.9967\%$, "almost 2.0%". True.

Step 2 – Check B and D

Pure expectations ignores the risk premium and convexity, so it cannot fully explain the real curve (B true). Under pure expectations forwards equal expected rates, so a falling curve implies expected falling short rates (D true).

Step 3 – Check C

Expectations are specific, and shape-determining, over short horizons. Over very long horizons they are not granular and mainly set the level.
$\Rightarrow$ C reverses the relationship, so it is the false statement.

Interpretation

This is the granularity point in Section 2: views on meeting dates and funding calendars shape the front end; the expected one-year rate in 29 years is unlikely to differ from that in 30 years.

Why the other options are wrong

A is incorrect. True: the geometric average of 1%, 2% and 3% is 1.9967%.
B is incorrect. True: the real curve also reflects risk premium and convexity.
D is incorrect. True: under pure expectations a declining curve signals expected lower short rates.

LO: MR-13 a: Explain the role of interest rate expectations in determining the shape of the term structure. • **Reference:**

Tuckman and Serrat 4e, Ch.8, 13.1

Solution MR-13 a.3 Answer: A

New • GARP MR Ch.13, 13.1

Step 1 – What short-horizon views can do

Views tied to central bank meeting dates and the supply and demand for funds (tax dates, issuance calendar, quarter-end balance sheets) are specific, so they can shape the front of the curve.

Step 2 – What long-horizon views can do

Expectations far out are not granular, but macroeconomics can anchor the long-run level: $1\% + 3\% = 4\%$ for the expected short rate.

$\Rightarrow$ Short-horizon views for shape at the front, the macro anchor for the level at the back.

Interpretation

This is GARP's own example: the long-run expected short rate of 4% is "1% due to the long-run real rate of interest and 3% due to long-run inflation".

Why the other options are wrong

B is incorrect. Money market analysis is "unlikely to reveal" a difference between the expected one-year rate in 29 and in 30 years.

C is incorrect. Macroeconomic estimates anchor the level; they are not granular enough to give the long end a detailed shape.

D is incorrect. Absolute claim: expectations still set the level of the long end, and the risk premium also acts there.

LO: MR-13 a: Explain the role of interest rate expectations in determining the shape of the term structure. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.1

Solution MR-13 a.4 Answer: A

New • GARP MR Ch.13, 13.3

Step 1 – Pure expectations

$\lambda = 0$, so forwards are set by expectations, $E[dr/dt]$. The natural no-change scenario: short rates evolve as expected and **forward rates are realized**.

Step 2 – Pure risk premium

$E[dr/dt] = 0$, so forwards are set by the risk premium. The natural no-change scenario: short rates stay the same and the **term structure is unchanged**.

$\Rightarrow$ A matches Step 1 exactly.

Interpretation

GARP notes that reality lies between the two extremes: the curve is set by a mix of expectations and risk premium (plus convexity).

Why the other options are wrong

B is incorrect. Sibling swap: $E[dr/dt] = 0$ and an unchanged curve belong to the pure risk premium hypothesis.

C is incorrect. Under pure risk premium there are no expected rate changes, so an upward slope reflects the premium, not expected rises.

D is incorrect. Sibling swap: "forward rates are realized" is the pure expectations no-change scenario.

LO: MR-13 a: Explain the role of interest rate expectations in determining the shape of the term structure. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eq. 13.20

Solution MR-13 a.5 — Answer: A

New • GARP MR Ch.13, 13.3

Step 1 – Both terms scale with duration

Expectations enter $f(T)$ as $D\,E[dr]/dt$ and the risk premium as λD. Both raise forwards more at longer terms.

Step 2 – They are indistinguishable in prices

"Increasing the risk premium or increasing the expected short-term rate by the same amount are indistinguishable from the observation of forward rates."

$\Rightarrow$ The split is not observable from a cross-section of bond prices alone; only the risk-neutral drift is.

Interpretation

The split may be useful for economic analysis or macro trading, but it cannot be read from the curve. From a modelling perspective, only the risk-neutral process matters for pricing.

Why the other options are wrong

B is incorrect. The expectations term is scaled by duration too, so expected rate rises do not move all forwards equally.

C is incorrect. The premium depends on time and the level of rates, "but not on the characteristics of any individual bond"; it enters as λD.

D is incorrect. Risk-neutral probabilities embed the combined drift; they do not reveal how much of it is expectation and how much premium.

LO: MR-13 a: Explain the role of interest rate expectations in determining the shape of the term structure. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eqs. 13.18 and 13.20

Solution MR-13 a.6 — Answer: D

Workbook D18.1 • S1, S2, S3

Step 1 – Expectations term

Expectations part of $f(T)$: $r_0 + D\,E[dr]/dt$.

$\Rightarrow$ Higher expected inflation raises expected short rates; the effect on the forward rate is scaled by the duration of the zero of that term, so it is largest at the long end.

Step 2 – Risk premium term

Risk premium part of $f(T)$: λD.

$\Rightarrow$ A higher λ raises every forward rate by λ times its duration: again largest at the long end.

Step 3 – Convexity term

Convexity part of $f(T)$: $-\frac{1}{2}C\sigma^2$.

$\Rightarrow$ Higher volatility pushes long forwards down, partly offsetting Steps 1 and 2 for long bonds; it does not add to their losses.

Step 4 – Effect on bond values

From $dP/P = f(T)\,dt - D\,dr + \frac{1}{2}C\sigma^2\,dt$, a rise in rates reduces value in proportion to duration, so long-dated gilts lose more than short-dated gilts. IM-2 b reaches the same direction from the asset-pricing side: high inflation is bad for bonds because their payments are fixed.

Interpretation

A climate shock has no formula of its own in the 2026 readings; it is read through the same three terms: expected inflation (expectations), uncertainty (volatility, convexity) and required compensation (risk premium).

Why the other options are wrong

A is incorrect. Polarity flip. The convexity term $-\frac{1}{2}C\sigma^2$ lowers long forward rates when volatility rises; volatility supports long bond prices rather than cheapening them.
B is incorrect. Scope/condition error. λ is common to all bonds, but the premium enters as λD; dropping the duration condition makes the effect look parallel.
C is incorrect. Absolute claim. The term structure cannot, on its own, separate expectations of rate changes from risk premium; only their combined risk-neutral drift is observable.

LO: MR-13 a: Explain the role of interest rate expectations in determining the shape of the term structure. MR-13 b, c, d. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.1 to 13.3, eqs. 13.15 to 13.20; Ang, Ch.7 (IM-2 b), Table 2.1

MR-13 • Expectations, Risk Premium, Convexity, and the Shape of the Term Structure • Vardeez: MODERATE

MR-13 b

P3 BOTH

Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

Workbook: D12 Convexity and Jensen's Inequality D18 Inflation and Market Stress • Tier D

WHAT IS TESTED HERE

Why P3. Tier D workbook topics hit this LO: D12, convexity and Jensen's inequality (Map #55), and D18, inflation and market stress (Map #72).
Vardeez (whole reading): Jensen's inequality; the value of convexity increases with bond volatility and time to maturity.

Reported exam shapes

- **D12 S2** The value of convexity increases with volatility.
- **D18 S2** Market stress as a higher risk premium and higher rate volatility: the premium term grows with duration, the convexity term pulls the other way.

What the questions look like: a two- or three-period tree priced by expected discounted value, then the implied spot rate compared with the expected rate; or a statement question on the direction and size of the volatility effect.

1. Pricing a zero on a risk-neutral tree

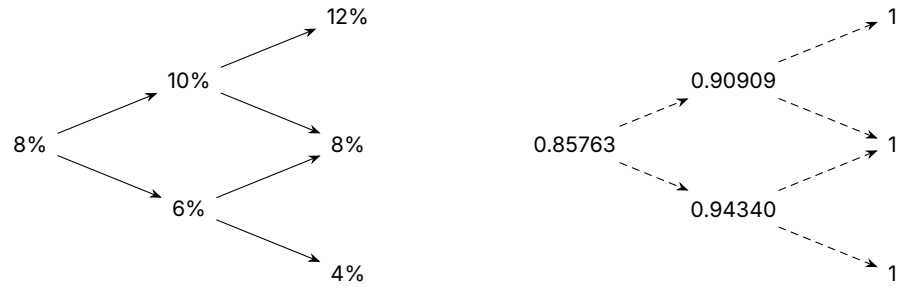
Risk neutral means each bond is priced at its **expected discounted value**: work backwards node by node, averaging the *prices* (not the rates) and discounting at the rate of the node you are at.

Example. One-year rate 8% today; in one year 10% or 6% (50% each); in two years 12%, 8% or 4%. The expected one-year rate is 8% at every date.

$$P_u = \frac{1}{1.10} = 0.90909, \qquad P_d = \frac{1}{1.06} = 0.94340$$

$$P(2) = \frac{0.5 \times 0.90909 + 0.5 \times 0.94340}{1.08} = 0.85763$$

Implied two-year spot rate: $(1/0.85763)^{1/2} - 1 = 7.9815\%$, below the expected 8%. The difference, about **1.85 bp**, is the **value of convexity** (Vardeez and Schweser show 1.84 bp, from the rounded price). The year-2 rates do not enter the two-year price at all: they only matter for the three-year zero.



Rate tree (50% each branch) Price tree of the two-year zero

Figure 2. Left: the rate tree (solid arrows). Right: the two-year zero priced backwards (dashed arrows): $1/1.10$ and $1/1.06$ at date 1, then the average discounted at 8%. The result, 0.85763, exceeds $1/1.08^2 = 0.85734$, so the implied yield is below 8%.

2. Volatility bends the curve down

With flat expectations and no risk premium, the only force left is volatility interacting with convexity, and it **lowers** yields.

GARP's tree (exaggerated on purpose): 9% today; 13% or 5%; 17%, 9% or 1%; volatility of every transition 400 bp; expected rate 9% at every date.

Term	Zero price	Forward rate
1	0.917431	9.0000%
2	0.842815	8.8532%
3	0.776366	8.5590%

The forward curve **slopes downward although expectations are flat at 9%**, and it falls further at longer terms.

More volatility, bigger effect. Widen the 8% tree's year-1 nodes from 10%/6% to 11%/5% (same 8% expectation): the two-year zero rises to 0.858000 and its implied spot falls to 7.958%, a value of convexity of about 4.2 bp instead of 1.85 bp.
Longer maturity, bigger effect. In the 8% tree, the three-year zero's value of convexity is about 4.9 bp against 1.85 bp for the two-year (see MR-13 c).

3. Adding a risk premium: the hump

Under stress, investors demand a higher premium and rates are more volatile (shape D18 S2). The two pull in opposite directions:

- The **risk premium** raises forwards by λD: roughly linear in maturity, so it dominates the *short* end.
- The **convexity effect** lowers forwards by $\frac{1}{2}C\sigma^2$; since convexity grows roughly with the square of maturity, it dominates the *long* end.

With flat expectations, significant volatility and a premium, the curve therefore rises at first and then falls: a **humped** shape.

Formula card: MR-13 b

Node price (risk neutral)	$P_{\text{node}} = \dfrac{0.5\,P_{\text{up}} + 0.5\,P_{\text{down}}}{1 + r_{\text{node}}}$
Implied spot rate	$\hat{r}(n) = (1/P(n))^{1/n} - 1$
Value of convexity	expected rate − implied spot rate (flat expectations, no premium)
Forward from zeros	$1 + f(n) = P(n-1)/P(n)$
Direction	more volatility ⇒ higher prices ⇒ lower yields; effect grows with maturity

Exam traps

- **Discounting at the expected rate.** $1/1.08^2$ ignores Jensen's inequality and gives the no-volatility price, not the tree price.
- **Using the wrong nodes.** A two-year zero uses the date-0 and date-1 rates only; the date-2 rates are irrelevant to it.
- **Stopping a period early.** For a three-year zero, roll back all three periods; stopping at two gives the two-year rate.
- **Sign of the effect.** In a risk-neutral tree volatility can only lower yields. "Volatility raises required returns" imports a risk premium that the tree does not have.
- **One-sided path.** Discounting along only the up path or only the down path is not an expectation.

Practice questions: MR-13 b

Question MR-13 b.1

Bank Q625 • BT P2.T5.300.1 (rebuilt)

The true (real-world) process for the one-year interest rate is a recombining binomial tree with equal 50% probabilities of an up or down move:

Date	Node values
Year 0	8.0%
Year 1	11.0% (up) or 5.0% (down)
Year 2	14.0%, 8.0% or 2.0%

Assume investors are risk-neutral, so a bond's price equals its expected discounted value under these probabilities with no risk premium. Which is nearest to the price of a two-year zero-coupon bond with a $1,000 par value?

A. $834.17

B. $857.34

C. $858.00

D. $881.83

Question MR-13 b.2

Bank Q630 • Mock 1 Q23 (rebuilt)

The current one-year spot rate is 10.00%. One year from now the one-year rate will be either 18.00% or 2.00%, each with probability 0.5, so the expected one-year rate is also 10.00%. There is no risk premium. What is the implied two-year spot rate?

A. 9.42%

B. 9.71%

C. 10.00%

D. 10.29%

Question MR-13 b.3

Bank Q628 • BT P2.T5.23.2.3 (rebuilt)

The one-year rate is 6.00% today, and the expected one-year rate is 6.00% in each of the next two years. The risk-neutral process is a recombining tree with equal 50% probabilities and no risk premium:

Date	One-year rate nodes
Year 0	6.00%
Year 1	8.00% (up) or 4.00% (down)
Year 2	10.00%, 6.00% or 2.00%

Which is nearest to the three-year spot rate implied by this tree?

A. 6.00%

B. 5.98%

C. 5.95%

D. 6.05%

Question MR-13 b.4

New • built from GARP MR Ch.13, 13.2; Schweser LO 13.c

A risk analyst prices a two-year zero-coupon bond on a risk-neutral tree with no risk premium. The one-year rate is 8% today and will be 10% or 6% in one year with equal probability. To stress volatility she widens the year-1 nodes to 11% or 5%, keeping the expected one-year rate at 8%. Which of the following is closest to the change in the implied two-year spot rate caused by the stress?

A. -4.2 bp

B. -2.3 bp

C. -1.9 bp

D. $+2.3$ bp

Question MR-13 b.5

Bank Q629 • GARP PE2 Q74 • Mock 4 Q75 (rebuilt)

An analyst on an emerging-markets fixed-income desk builds a term structure model. The 1-year spot rate is expected to be 17.00% in each of the next 3 years, and the interest rate process is a recombining risk-neutral tree with moves of plus or minus 2% each year and equal 50% probabilities:

Date	One-year rate nodes
Year 0	17.00%
Year 1	19.00% or 15.00%
Year 2	21.00%, 17.00% or 13.00%

Which correctly describes the shape of the resulting term structure?

A. Slightly downward sloping, since discounting the terminal cash flow over longer time periods when interest rates exhibit volatility results in slightly lower spot rates.

B. Flat, since the interest rate tree is based on the assumption that all expected 1-year spot rates are equal.

C. Slightly upward sloping, since there is an opportunity cost associated with long-term interest rates.

D. Steeply upward sloping, since investors will demand higher long-term interest rates as compensation for interest rate volatility.

Question MR-13 b.6

Bank Q631 • Mock 1 Q24

An analyst is comparing two government bond markets. Both have a flat expected path of short-term interest rates and neither prices a risk premium. Market A exhibits substantially higher interest rate volatility than Market B. Which correctly describes the difference between the two markets' term structures?

A. Market A's long-dated spot rates will sit further above the expected short rate than Market B's, because volatility raises required returns.

B. The two term structures will be identical, since expectations and risk premiums are the same in both markets.

C. Market A's term structure will slope upward and Market B's will be flat, because volatility steepens the curve.

D. Market A's long-dated spot rates will sit further below the expected short rate than Market B's, and the gap will widen with maturity in both markets.

Question MR-13 b.7 Bank Q627 • BT P2.T5.301.3

Robert is asked to graph a five-year term structure of interest rates from a binomial interest rate tree. The tree recombines, with up- and down-jumps equal in size (plus or minus X basis points) and likelihood (50%), so that the expected one-year rate in each of years two through five equals today's one-year spot rate. The tree reflects significant rate volatility, and the model also includes a risk premium reflecting risk-averse investors. Which is the most plausible prediction about the term structure he produces?

A. The term structure is upward-sloping throughout.

B. In the shorter end, the yield increases, but in the longer end, the yield decreases.

C. In the shorter end, the yield decreases, but in the longer end, the yield increases.

D. The term structure is downward-sloping throughout.

Solutions • MR-13 b

Solution MR-13 b.1

Answer: C

Bank Q625 • BT P2.T5.300.1

Step 1 – Date-1 values of the $1,000 payoff

Up node: $1000/1.11 = 900.90$; down node: $1000/1.05 = 952.38$. The year-2 rates are irrelevant: the bond matures at the end of year 2.

Step 2 – Expected value, discounted at today's rate

$$P = \frac{0.5 \times 900.90 + 0.5 \times 952.38}{1.08} = \frac{926.64}{1.08} = \$858.00$$

$\Rightarrow$ This is above $1000/1.08^2 = \$857.34$, the price at the expected rate: convexity at work.

Interpretation

Averaging discount factors (not rates) across the states, then discounting at 8%, is what "expected discounted value" means. The real-world label on the tree does not matter here because the question tells you to treat investors as risk neutral.

Why the other options are wrong

A is incorrect. Discounted only along the up path: $1000/(1.08 \times 1.11) = 834.17$.

B is incorrect. Discounted at the expected 8% for both years, $1000/1.08^2 = 857.34$, ignoring the convexity effect.

D is incorrect. Discounted only along the down path: $1000/(1.08 \times 1.05) = 881.83$.

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

• **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2; SchweserNotes 2026 Book 1, LO 13.c (Figures 13.5 and 13.6)

Solution MR-13 b.2

Answer: B

Bank Q630 • Mock 1 Q23

Step 1 – First year is certain

Year-1 discount factor: $1/1.10 = 0.909091$.

Step 2 – Expected year-2 discount factor

$0.5(1/1.18) + 0.5(1/1.02) = 0.5(0.847458 + 0.980392) = 0.913925$.

Step 3 – Two-year price and spot

$P(2) = 0.909091 \times 0.913925 = 0.830841$; $\quad (1/0.830841)^{1/2} - 1 = 9.71\%$.

$\Rightarrow$ About 29 bp below the 10% expected rate: a large effect because the volatility is extreme.

Interpretation

Jensen's inequality acts only over the uncertain second year; the first year contributes a certain 10%, which is why the two-year effect is about half the one-year effect.

Why the other options are wrong

A is incorrect. $1/0.913925 - 1 = 9.42\%$ is the full one-period effect; it forgets that the first year is certain.

C is incorrect. Assumes the spot rate equals the expected rate, ignoring convexity.

D is incorrect. Wrong sign: $10\% + 29$ bp. Volatility lowers yields in a risk-neutral tree.

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

• **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2, eqs. 13.5 to 13.8

Solution MR-13 b.3 — Answer: C

Bank Q628 • BT P2.T5.23.2.3

Step 1 – Date-2 values of $1

$1/1.10 = 0.909091,\ 1/1.06 = 0.943396,\ 1/1.02 = 0.980392.$

Step 2 – Date-1 values

Up (8%): $0.5(0.909091 + 0.943396)/1.08 = 0.857633$; down (4%): $0.5(0.943396 + 0.980392)/1.04 = 0.924898.$

Step 3 – Date 0 and the spot rate

$P(3) = 0.5(0.857633 + 0.924898)/1.06 = 0.840817; \quad (1/0.840817)^{1/3} - 1 = 5.95\%.$

$\Rightarrow$ Below the flat 6.00% expectation, and further below than the two-year rate (5.98%).

Interpretation

With expectations flat and no premium, convexity is the only force on the curve, and it grows with maturity: 6.00%, 5.98%, 5.95% for one, two and three years.

Why the other options are wrong

A is incorrect. Assumes a flat curve because all expected rates are 6%; it ignores Jensen's inequality.

B is incorrect. 5.98% is the *two-year* spot rate; the roll-back stopped one period early.

D is incorrect. Wrong sign: volatility lowers yields in a risk-neutral tree.

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

• **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2, Figure 13.1 and Table 13.1

Solution MR-13 b.4 — Answer: B

New • GARP MR Ch.13, 13.2; Schweser LO 13.c

Step 1 – Base tree (10% or 6%)

$P(2) = 0.5(1/1.10 + 1/1.06)/1.08 = 0.857633;$ spot $= (1/0.857633)^{1/2} - 1 = 7.9815\%.$

Step 2 – Stressed tree (11% or 5%)

$P(2) = 0.5(1/1.11 + 1/1.05)/1.08 = 0.858000;$ spot $= (1/0.858000)^{1/2} - 1 = 7.9583\%.$

Step 3 – Change

$7.9583\% - 7.9815\% = -0.023\% = -2.3$ bp.

$\Rightarrow$ More volatility, same expectations: higher price, lower yield.

Interpretation

The value of convexity rises from about 1.85 bp to about 4.2 bp. The question asks for the *change* in the spot rate, which is the difference between the two.

Why the other options are wrong

A is incorrect. Reported the stressed value of convexity ($8\% - 7.958\% = 4.2$ bp) instead of the change from the base case.

C is incorrect. Reported the base-case value of convexity (about 1.85 bp) as the change.

D is incorrect. Wrong sign: in a risk-neutral tree, more volatility lowers yields.

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

• **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2; SchweserNotes 2026 Book 1, LO 13.b and 13.c

Solution MR-13 b.5 — Answer: A

Bank Q629 • GARP PE2 Q74

Step 1 – One- and two-year zeros

$P(1) = 1/1.17 = 0.854701$ (17.00%). $P(2) = 0.854701 \times 0.5(1/1.19 + 1/1.15) = 0.730727$, spot $= 16.98\%$.

Step 2 – Three-year zero

Rolling back with year-2 rates of 21%, 17% and 13%: $P(3) = 0.625101$, spot $= 16.95\%$.
$\Rightarrow$ Spot rates of 17.00%, 16.98%, 16.95%: slightly downward sloping.

Interpretation

Expectations are flat and the tree is risk neutral, so the only influence is convexity: averaging discount factors gives higher prices (lower yields) than discounting at the mean rate, and the effect grows with maturity. With $\pm 2\%$ moves it is small.

Why the other options are wrong

B is incorrect. Equal expected rates give a flat curve only without volatility.
C is incorrect. An "opportunity cost" or premium story has no place in a risk-neutral tree, where convexity can only lower yields.
D is incorrect. Imports a risk premium for volatility; the tree is risk neutral, and the effect is not steep.

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.
• **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2, Table 13.1

Solution MR-13 b.6 — Answer: D

Bank Q631 • Mock 1 Q24

Step 1 – What differs between the markets

Spot rates reflect expectations, risk premium and convexity. The first two are identical here, so only convexity differs.

Step 2 – Direction and size

The convexity effect always lowers yields, more so with higher volatility and with longer maturity.
$\Rightarrow$ Market A's long spot rates sit further below the expected short rate, and in both markets the gap widens with maturity.

Interpretation

This is shape D12 S2 in market language: the value of convexity increases with volatility (Market A versus B) and with maturity (along each curve).

Why the other options are wrong

A is incorrect. Wrong sign: treats volatility as a source of a positive premium. Without a premium, volatility only lowers yields.
B is incorrect. Ignores convexity, which depends on volatility and is part of neither expectations nor the premium.
C is incorrect. Convexity makes the curve slope down, not up, and Market B also has a (smaller) convexity effect.

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.
• **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2 and 13.3

Solution MR-13 b.7 Answer: B

Bank Q627 • BT P2.T5.301.3

Step 1 – Expectations

The expected one-year rate equals today's rate in every year: on their own, expectations give a flat curve.

Step 2 – Risk premium

The premium raises yields roughly in proportion to duration (maturity): it dominates the short end and creates an initial upward slope.

Step 3 – Convexity

The convexity effect lowers yields and grows roughly with the square of maturity; with significant volatility it eventually overwhelms the premium at the long end.

$\Rightarrow$ Up at first, then down: a humped curve.

Interpretation

This is shape D18 S2: the premium term grows with duration, the convexity term pulls the other way and wins at long maturities.

Why the other options are wrong

A is incorrect. Ignores convexity, which with significant volatility outweighs the premium at long maturities.

C is incorrect. Reverses the order: convexity is negligible at short maturities, so yields cannot fall first.

D is incorrect. Ignores the risk premium, which dominates the short end because convexity is tiny there.

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

• **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eq. 13.20

MR-13 • Expectations, Risk Premium, Convexity, and the Shape of the Term Structure • Vardeez: MODERATE

MR-13 c

P3 CALC

Estimate the convexity effect using Jensen's inequality.

Workbook: D12 Convexity and Jensen's Inequality • Tier D

WHAT IS TESTED HERE

Why P3. A Tier D workbook topic hits this LO (D12, Map #55, 2/16), and Vardeez flags it.
Vardeez: Jensen's inequality. Remember: the value of convexity increases with bond volatility and time to maturity.

Reported exam shapes

- **D12 S1** Jensen's inequality applied to bond prices: the expected price exceeds the price at the expected rate.
- **D12 S2** The value of convexity increases with volatility.
- **D12 S3** The value of convexity increases with time to maturity, so forward rates bend down more at longer terms.

What the questions look like: one map sheet says calculations here are unlikely and calls it a pure polarity item: get the direction right and the question is free. Practise the direction first, then the small two-period calculation.

1. Jensen's inequality for bond prices

For a random one-year rate r,

$$\boxed{E\left[\frac{1}{1+r}\right] > \frac{1}{1+E[r]}}$$

In words: **the expected price of a bond is greater than the price of the bond at the expected rate.** The price-rate curve $1/(1+r)$ is convex, so the chord joining the prices at r_d and r_u lies above the curve at $E[r]$ (Figure 3).

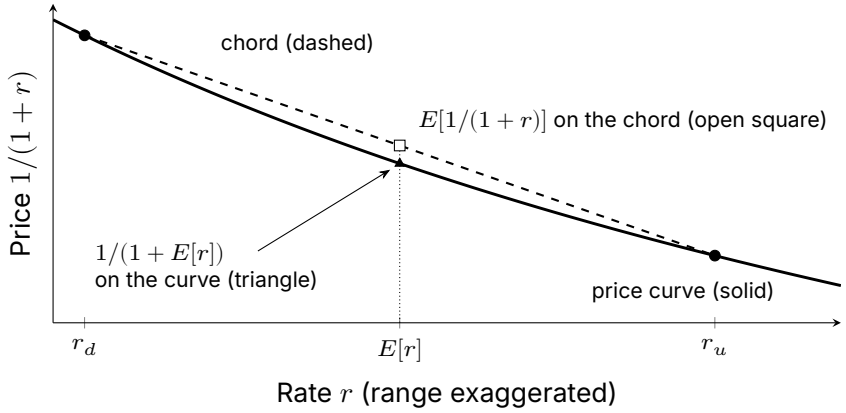


Figure 3. Jensen's inequality, with the rate range exaggerated so the shape is visible. The midpoint of the chord (the expected price) lies above the curve at the expected rate. That vertical gap is the convexity effect in price terms.

Vardeez's example. Next year's one-year rate is 10% or 6%, 50% each; today's rate is 8%.

- One year ahead: $E[1/(1+r)] = 0.5/1.10 + 0.5/1.06 = 0.92624 \;>\; 1/1.08 = 0.92593$.

- Discount one more year at 8%: $0.92624/1.08 = 0.85763 > 1/1.08^2 = 0.85734$.

The two-year price exceeds the price from straight discounting, so its implied rate is below 8%.

GARP's version, through the forward rate. With 9% today and 13% or 5% next year: $0.5/1.13 + 0.5/1.05 = 0.918668 > 1/1.09 = 0.917431$. Since the two-year zero equals $\dfrac{1}{1.09(1+f)} = \dfrac{1}{1.09}\left[\dfrac{0.5}{1.13} + \dfrac{0.5}{1.05}\right]$, Jensen gives $\dfrac{1}{1+f} > \dfrac{1}{1.09}$, so $\boxed{f < 9\%}$: **the one-year rate one year forward is less than the expected one-year rate in one year.**

2. Three ways to express the convexity effect

	Definition	Example (8%; 10% or 6%)
In price	$E[P] - P(E[r])$	$0.85763 - 0.85734 = 0.00029$
In yield (value of convexity)	expected rate − implied spot rate	$8\% - 7.9815\% \approx 1.85$ bp
In the forward rate	expected rate − forward rate	GARP: $9\% - 8.8532\% = 14.7$ bp

Read the question for which one it wants, and for the sign convention: GARP and Schweser quote the value of convexity as a positive number (the size of the yield reduction).

3. What makes it bigger: volatility and maturity

Case (expected rate 8%)	Implied spot	Value of convexity
Two-year zero, nodes 10% / 6%	7.9815%	1.85 bp
Two-year zero, nodes 11% / 5% (more volatility)	7.9583%	4.2 bp
Three-year zero, nodes 10/6, then 12/8/4 (longer maturity)	7.9506%	4.9 bp

In continuous time the forward rate is reduced by $\frac{1}{2}C\sigma^2$: larger with rate volatility σ, and larger with the convexity C of the zero of that term, which grows with maturity (for a zero, $C \approx T^2$). Hence **as maturity increases, the price-yield curve becomes more convex**, and forward rates bend down more at long terms (Figure 4).

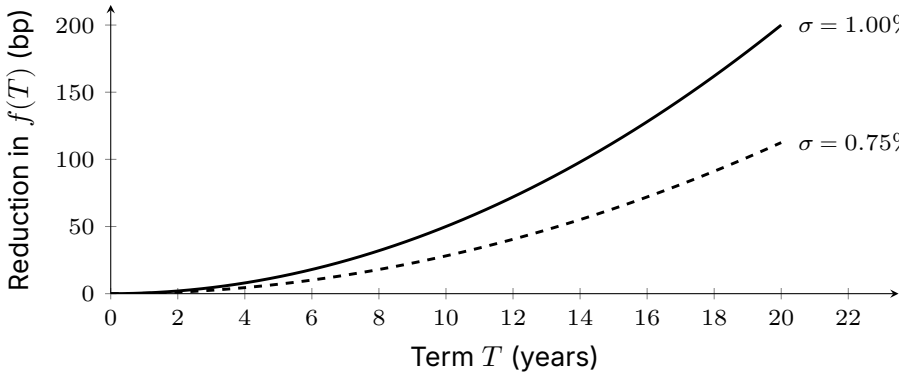


Figure 4. Illustration of the convexity term $\frac{1}{2}C\sigma^2$ for a model with normally distributed rates, no drift and no mean reversion ($f(T) = r_0 - \frac{1}{2}\sigma^2 T^2$). The reduction grows with maturity and with volatility. Mean reversion damps it at the long end: GARP's tables give -135.3 bp (no mean reversion) against -74.7 bp (Vasicek) for the 30-year par rate.

Remember the polarity. Value of convexity ↑ when volatility ↑ and when maturity ↑. It *lowers* yields and forward rates, and *raises* bond prices.

Formula card: MR-13 c

Jensen's inequality	$E\left[1/(1+r)\right] > 1/(1+E[r])$
Two-year zero	$P(2) = \dfrac{1}{1+r_0}\, E\left[\dfrac{1}{1+r_1}\right]$, compare with $\dfrac{1}{(1+r_0)(1+E[r_1])}$
Implied spot rate	$(1/P(n))^{1/n} - 1$
Value of convexity	expected rate − implied spot rate (a positive number)
Forward rate	$1 + f(n) = P(n-1)/P(n)$; with volatility $f < E[r]$
Continuous version	forward rate reduced by $\frac{1}{2}C\sigma^2$

Exam traps

- **Polarity of Jensen.** The expected price exceeds the price at the expected rate, so forwards and spot rates lie *below* the expected rate once volatility is added.
- **Polarity of the value of convexity.** It increases with volatility and with maturity. "Falls with maturity" or "independent of volatility" is a flip.
- **Not a parallel shift.** Equal per-step volatility still gives a bigger effect at longer terms, because C grows with term.
- **Mixed maturities.** Compare a two-year price with a two-year price: discount both sides one more year.
- **Units.** A price difference (0.00029) is not a yield difference (1.85 bp); a discount-factor gap is not a yield gap.
- **No free lunch.** Being long convexity does not earn a riskless excess return: the lower forward rate is the price paid for it (MR-13 d).

Practice questions: MR-13 c

Question MR-13 c.1

Bank Q638 • Schweser Module Quiz 13.2 Q1

An investor expects the one-year rate to be 6% for the first year of a two-year zero-coupon bond, and projects a 50% probability that the one-year rate in one year is 8% and a 50% probability that it is 4%. Which inequality most accurately reflects the convexity effect for this two-year bond using Jensen's inequality? (Prices are per 1 unit of face value.)

A. $0.89031 > 0.89000$.

B. $0.89000 > 0.80000$.

C. $0.94340 > 0.89031$.

D. $0.94373 > 0.94340$.

Question MR-13 c.2

Bank Q634 • BT P2.T5.25.2.3 (rebuilt)

A two-year zero-coupon bond has a face value of \$100. The one-year spot rate is 5%. The one-year forward rate one year from now is 9% in expectation, but will actually be either 7% or 11% with equal probability. Assuming risk neutrality, the convexity effect (Jensen's inequality) is defined as the expected price of the bond minus the price computed using the expected 9% rate. Which of the following is closest to this convexity effect?

A. -0.029

B. 0.029

C. 1.604

D. 1.634

Question MR-13 c.3

Bank Q637 • Mock 1 Q22

A fixed-income analyst is illustrating the effect of Jensen's inequality on the pricing of a zero-coupon bond. The analyst assumes that the one-year interest rate one year from now will be either 18% or 2%, each with probability 0.5, so that the expected one-year rate is 10%. The analyst compares the expected value of the one-year discount factor prevailing in one year with the discount factor evaluated at the expected 10% rate, and expresses the difference in yield terms (the expected 10% rate less the one-year rate implied by the expected discount factor). What is the implied convexity effect demonstrated by Jensen's inequality?

A. 0.48%.

B. 0.58%.

C. 1.16%.

D. 8.00%.

Question MR-13 c.4 Bank Q635 • BT P2.T5.301.2

The current one-year rate is 10.0% and a binomial tree is known to investors. Next year the one-year rate will be either 14.0% or 6.0%, each with 50% probability, so the expected one-year rate in one year is 10.0%. Investors are risk-neutral and rates compound annually. If 10.0% is the two-year yield in the absence of uncertainty, what is the yield change implied by Jensen's inequality due to volatility?

A. An increase of about 25 basis points to about 10.25%

B. None

C. A decrease of about 7 basis points to 9.93%

D. A decrease of about 25 basis points to 9.75%

Question MR-13 c.5 New • built from GARP MR Ch.13, Figure 13.1 and Table 13.1

A portfolio manager uses a risk-neutral binomial tree with annual steps and 50% transition probabilities. The one-year rate is 9% today; in one year it is 13% or 5%; in two years it is 17%, 9% or 1%. The expected one-year rate is therefore 9% at every date. The tree prices the two-year zero at 0.842815 and the three-year zero at 0.776366. Which of the following is closest to the one-year rate one year forward implied by the tree?

A. 8.56%

B. 8.85%

C. 8.93%

D. 9.00%

Question MR-13 c.6 New • built from Schweser LO 13.c, Figures 13.3 and 13.4

A risk-neutral tree has a one-year rate of 8% today; 10% or 6% in one year; and 12%, 8% or 4% in two years, with 50% probabilities on every branch. On this tree the value of convexity for the two-year zero-coupon bond is about 1.85 basis points. Which of the following is closest to the value of convexity for the three-year zero-coupon bond?

A. 1.9 bp

B. 2.8 bp

C. 4.9 bp

D. 17.3 bp

Question MR-13 c.7 Bank Q639 • Schweser Module Quiz 13.2 Q2

What is the impact on the bond price-yield curve if, all other factors held constant, the maturity of a zero-coupon bond increases? The pricing curve becomes:

A. less concave.

B. more concave.

C. less convex.

D. more convex.

Question MR-13 c.8 Workbook D12.1 • S1, S2, S3 • Map #55

A portfolio manager at a German insurer is presenting to a group of newly hired analysts. She uses a risk-neutral binomial tree with annual steps in which the expected one-year rate is the same at every future date, the one-year rate is volatile, and investors require no risk premium. She then compares the forward rates implied by the tree's zero-coupon bond prices with the flat expected rate. Which of the following statements is correct for her to make?

A. Because bond prices are convex in rates, the price of a zero at the expected rate exceeds its expected price, so forward rates lie above the expected short rate.

B. Convexity lowers every forward rate by the same number of basis points, because the volatility of each one-year rate transition in the tree is the same at every node.

C. The value of convexity increases with rate volatility and with maturity, so forward rates lie below the expected short rate and fall further at longer terms.

D. A position that is long convexity earns an expected return above the short rate in every scenario, because convexity adds to return whether rates rise or fall.

Solutions • MR-13 c

Solution MR-13 c.1

Answer: A

Bank Q638 • Schweser Module Quiz 13.2 Q1

Step 1 – Left-hand side: expected price

One year ahead: $0.5(1/1.08 + 1/1.04) = 0.94373$; one more year at 6%: $0.94373/1.06 = 0.89031$.

Step 2 – Right-hand side: price at the expected rate

$1/1.06^2 = 0.89000$.

$\Rightarrow 0.89031 > 0.89000$: the expected price exceeds the price at the expected rate.

Interpretation

For a two-year bond, both sides must be two-year prices. Option D is the same inequality one year ahead, before the final discounting.

Why the other options are wrong

B is incorrect. 0.80000 is not a price from this problem.

C is incorrect. Compares a one-year price (0.94340) with a two-year price; different maturities say nothing about convexity.

D is incorrect. The one-year-ahead inequality $(0.94373 > 0.94340)$, not discounted one more year to two-year prices.

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. • **Reference:** SchweserNotes 2026 Book 1, LO 13.c; Tuckman and Serrat 4e, Ch.8, eq. 13.5

Solution MR-13 c.2

Answer: B

Bank Q634 • BT P2.T5.25.2.3

Step 1 – Scenario prices

At 11%: $100/(1.05 \times 1.11) = 85.8001$. At 7%: $100/(1.05 \times 1.07) = 89.0076$.

Step 2 – Expected price

$0.5(85.8001 + 89.0076) = 87.4038$.

Step 3 – Price at the expected rate, and the difference

$100/(1.05 \times 1.09) = 87.3744$; $\quad 87.4038 - 87.3744 = 0.0294 \approx 0.029$.

$\Rightarrow$ Small and positive: expected price above the price at the expected rate.

Interpretation

The effect is tiny in price terms (about 3 cents per \$100) at these volatilities; it is the direction that GARP tests.

Why the other options are wrong

A is incorrect. Sign error: for a convex price function the expected price exceeds the price at the expected rate.

C is incorrect. Compared the expected price with the 11% scenario price: $87.4038 - 85.8001 = 1.604$.

D is incorrect. Compared the 7% scenario price with the price at the expected rate: $89.0076 - 87.3744 = 1.633$.

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2, eq. 13.5

Solution MR-13 c.3

Answer: B

Bank Q637 • Mock 1 Q22

Step 1 – Expected discount factor

$0.5(1/1.18 + 1/1.02) = 0.5(0.847458 + 0.980392) = 0.913925.$

Step 2 – Discount factor at the expected rate

$1/1.10 = 0.909091$, lower, as Jensen says.

Step 3 – Convert to yield

Implied one-year rate: $1/0.913925 - 1 = 9.418\%$; effect $= 10.00\% - 9.418\% = 0.58\%$.
$\Rightarrow$ The extreme 18%/2% spread makes the effect large: 58 bp over one period.

Interpretation

This is the one-period effect. Over a two-year bond with a certain first year it is roughly halved (Question MR-13 b.2 gives 29 bp).

Why the other options are wrong

A is incorrect. Reported the discount-factor difference $(0.913925 - 0.909091 = 0.0048)$ as a percentage instead of converting it to a yield.
C is incorrect. Doubled the effect.
D is incorrect. Used the rate shock $(18\% - 10\%)$ instead of the yield difference.

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2, eq. 13.5

Solution MR-13 c.4

Answer: C

Bank Q635 • BT P2.T5.301.2

Step 1 – Year-1 prices and their average

$1/1.14 = 0.877193$, $1/1.06 = 0.943396$; average $= 0.910294$.

Step 2 – Two-year price

$0.910294/1.10 = 0.827541$, against the no-volatility price $1/1.10^2 = 0.826446$.

Step 3 – Implied yield and change

$(1/0.827541)^{1/2} - 1 = 9.927\%$; change $= 9.927\% - 10.000\% \approx -7$ bp.
$\Rightarrow$ Volatility lowers the two-year yield by about 7 bp.

Interpretation

The price difference is only about 0.0011, which is why the yield effect is single-digit basis points.

Why the other options are wrong

A is incorrect. Wrong direction: convexity raises prices and lowers yields.
B is incorrect. Ignores Jensen's inequality: $0.827541 > 0.826446$, so the yield changes.
D is incorrect. Right direction, magnitude far too large.

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2

Solution MR-13 c.5

Answer: B

New • GARP MR Ch.13, Table 13.1

Step 1 – One-year zero

$P(1) = 1/1.09 = 0.917431.$

Step 2 – Forward from zero prices

$$1 + f(2) = \frac{P(1)}{P(2)} = \frac{0.917431}{0.842815} = 1.088532 \quad \Rightarrow \quad f(2) = 8.85\%$$

$\Rightarrow$ Below the 9% expected rate, exactly as Jensen's inequality requires ($f < 9\%$).

Interpretation

The forward curve is 9.00%, 8.85%, 8.56%: it slopes down although expectations are flat, purely because volatility interacts with the convexity of bond prices. The 400 bp per-step volatility is exaggerated on purpose to make this visible.

Why the other options are wrong

A is incorrect. 8.56% is the next forward, $0.842815/0.776366 - 1$: the rate two years forward, not one.

C is incorrect. 8.93% is the two-year spot rate, $(1/0.842815)^{1/2} - 1$, an average of 9.00% and 8.85%, not the forward.

D is incorrect. The expected rate: it ignores Jensen's inequality, which puts the forward below the expectation.

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2, Figure 13.1, Table 13.1, eqs. 13.6 to 13.8

Solution MR-13 c.6

Answer: C

New • Schweser LO 13.c, Figures 13.3 and 13.4

Step 1 – Date-2 prices of a one-year zero

$1/1.12 = 0.89286,\ 1/1.08 = 0.92593,\ 1/1.04 = 0.96154.$

Step 2 – Date-1 prices of the then two-year zero

Up (10%): $0.5(0.89286 + 0.92593)/1.10 = 0.82672$; down (6%): $0.5(0.92593 + 0.96154)/1.06 = 0.89031$.

Step 3 – Today's price and spot rate

$P(3) = 0.5(0.82672 + 0.89031)/1.08 = 0.79492;\ (1/0.79492)^{1/3} - 1 = 7.9506\%.$

Step 4 – Value of convexity

$8\% - 7.9506\% = 0.0494\% \approx 4.9$ bp.

$\Rightarrow$ Against 1.85 bp for the two-year zero: the value of convexity increases with maturity (shape D12 S3).

Interpretation

Schweser rounds this to "5 basis points". Going from two to three years more than doubles the effect (1.85 bp to 4.9 bp): convexity grows faster than maturity.

Why the other options are wrong

A is incorrect. Reported the two-year value of convexity.

B is incorrect. Scaled the two-year value linearly by maturity: $1.85 \times 3/2 = 2.8$ bp. Convexity grows faster than that.

D is incorrect. Forgot the cube root and compared cumulative growth factors: $1.08^3 - 1/0.79492 = 0.173\%$, or 17.3 bp, a three-year total, not an annual rate.

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. • **Reference:** SchweserNotes 2026 Book 1, LO 13.c (Figures 13.3 and 13.4); Tuckman and Serrat 4e, Ch.8, 13.2

Solution MR-13 c.7 — Answer: D

Bank Q639 • Schweser Module Quiz 13.2 Q2

Step 1 – Size of the effect by maturity

Same tree: the value of convexity is about 1.85 bp for the two-year zero and about 4.9 bp for the three-year zero (Question MR-13 c.6).

Step 2 – Read it as curvature

A larger convexity effect at longer maturity means the price-yield relationship is more convex.
$\Rightarrow$ The pricing curve becomes more convex.

Interpretation

Zero-coupon convexity grows roughly with the square of maturity, which is why the effect dominates the long end.

Why the other options are wrong

A is incorrect. Bond price-yield curves are convex, not concave.
B is incorrect. Same curvature error.
C is incorrect. Direction reversed: convexity increases with maturity.

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. • **Reference:** SchweserNotes 2026 Book 1, LO 13.c

Solution MR-13 c.8 — Answer: C

Workbook D12.1 • S1, S2, S3

Step 1 – Jensen's inequality

$$E\left[\frac{1}{1+r}\right] > \frac{1}{1+E[r]}: \qquad 0.5 \times \frac{1}{1.13} + 0.5 \times \frac{1}{1.05} = 0.918668 > \frac{1}{1.09} = 0.917431$$

$\Rightarrow$ Averaging over volatile rates raises bond prices above the price at the average rate, which lowers the yields and forward rates they imply.

Step 2 – Effect grows with maturity

In the book's tree (expected rate 9% at every horizon, 400 bp per transition) the forward rates are 9.0000%, 8.8532% and 8.5590%: about 15 bp below 9% for year two and 44 bp for year three. In the decomposition the reduction is $\frac{1}{2}C\sigma^2$, and C rises with term.

Step 3 – Effect grows with volatility

The same $\frac{1}{2}C\sigma^2$ term rises with σ. The MR-14 tables show it: lowering long-rate volatility through mean reversion cuts the 30-year par-rate convexity effect from -135.3 bp (Model 1) to -74.7 bp (Vasicek).

Interpretation

With flat expectations and no risk premium, the forward curve slopes down purely because of convexity, more at longer terms and more in more volatile markets.

Why the other options are wrong

A is incorrect. Polarity flip of Jensen's inequality: the expected price exceeds the price at the expected rate, so forwards lie below the expected rate.
B is incorrect. Scope/condition error: equal per-transition volatility does not make the effect equal across terms; $\frac{1}{2}C\sigma^2$ grows with the convexity of longer zeros (15 bp versus 44 bp in the book).
D is incorrect. Absolute claim: the forward rate is reduced by exactly the convexity gain, so the expected return condition leaves no net advantage of convexity; nothing is earned "in every scenario".

LO: MR-13 c: Estimate the convexity effect using Jensen's inequality. MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.2 and 13.3, eqs. 13.5 to 13.8 and 13.20, Table 13.1; Tuckman and Serrat 3e, Ch.9, Tables 14.1 and 14.2

MR-13 • Expectations, Risk Premium, Convexity, and the Shape of the Term Structure • Vardeez: MODERATE

MR-13 d

P3 BOTH

Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor.

Workbook: D18 Inflation and Market Stress Across Bond Maturities • Tier D

WHAT IS TESTED HERE

Why P3. A Tier D workbook topic hits this LO: D18, inflation and market stress (Map #72, 2/16). **Vardeez** (whole reading): Jensen's inequality; the value of convexity increases with volatility and maturity. His notes cover the risk-premium calculation but not the three return components; those are written below from the GARP text.

Reported exam shapes

- **D18 S2** Market stress as a higher risk premium and higher rate volatility: the premium term grows with duration, the convexity term pulls the other way.
- **D18 S3** A climate-related shock framed as an inflation and stress scenario for short versus long bonds (worked as Question MR-13 a.6).

What the questions look like: name the components; compute $r_0 + \lambda D$ (often with a convexity input as a red herring); price a zero on a tree with a risk premium; or a "NOT correct" statement on the forward-rate decomposition.

1. The three components of bond return

By Ito's lemma, the instantaneous return on a zero-coupon bond of term T decomposes into exactly three pieces:

$$\frac{dP}{P} = \underbrace{f(T)\,dt}_{\text{passage of time}} - \underbrace{D\,dr}_{\text{rate change}} + \underbrace{\tfrac{1}{2}C\sigma^2\,dt}_{\text{volatility (convexity)}}$$

- **Passage of time.** With the term structure unchanged, the bond earns the forward rate $f(T)$ of its term.
- **Rate change.** A rise in rates reduces return in proportion to duration D.
- **Volatility.** Movement of rates, up *or* down, adds return in proportion to convexity C: across portfolios with the same duration, the more convex one gains more whichever way rates move.

Taking expectations replaces dr by $E[dr]$: $E[dP/P] = f(T)\,dt - D\,E[dr] + \frac{1}{2}C\sigma^2\,dt$.

2. Expected return: risk neutral versus risk averse

- **Risk neutral:** every bond must offer the short rate, $E[dP/P] = r_0\,dt$, whatever its maturity.
- **Risk averse:** investors demand more for more interest rate risk, measured by duration:

$$\boxed{E\left[\frac{dP}{P}\right] = r_0\,dt + \lambda D\,dt}$$

λ is the premium per year of duration risk. It may depend on time and on the level of rates, but **not on the characteristics of any individual bond**.

Examples. GARP: $r_0 = 1\%$, $D = 5$, $\lambda = 10$ bp $\Rightarrow 1\% + 5 \times 0.10\% = 1.5\%$. Schweser: $r_0 = 2.5\%$, $D = 7$, $\lambda = 15$ bp $\Rightarrow 3.55\%$ (risk neutral: 2.5%).
Sharpe ratio view: $SR = \lambda/\sigma$; 10 bp per year of duration over 100 bp rate volatility is a Sharpe ratio of 10%.

On a one-period tree. The premium is earned for the rate risk carried over the period, which is set by what the bond becomes at the end of it: a two-year zero becomes a one-year zero, so it carries one year of duration risk.

- Vardeez's example: 8% today, 10% or 6% next year, premium 20 bp.

$$E[P_1] = 0.5(0.90909 + 0.94340) = 0.92624, \qquad P_0 = \frac{0.92624}{1.082} = 0.85605$$

$$\text{expected return} = \frac{0.92624}{0.85605} - 1 = 8.2\%$$

  The expected return is $8\% + 0.20\%$. (Vardeez reaches the same 0.85605 by adding 20 bp to each date-1 rate.)
- A three-year zero becomes a two-year zero, so it carries two years of duration risk over the first year (Question MR-13 d.5).

3. Forward rates: expectations, risk premium, convexity

Equating the expected-return decomposition with the risk-averse requirement gives GARP's eq. 13.20:

$$f(T) = r_0 + D\,\frac{E[dr]}{dt} + \lambda D - \tfrac{1}{2}C\sigma^2$$

Example (Schweser). $r_0 = 2.5\%$, $E[dr]/dt = 0.3\%$, $\lambda = 0.15\%$, $D = 7$, convexity effect 0.65%: $f = 2.5\% + 2.1\% + 1.05\% - 0.65\% = 5.00\%$ (Figure 5).

No net advantage of convexity. Convexity raises return directly by $\frac{1}{2}C\sigma^2$ but lowers the forward rate (the passage-of-time return) by exactly the same amount. Net: $E[dP/P] = r_0 + \lambda D$, with no convexity term. That is why a convexity input in an expected-return question is a red herring.

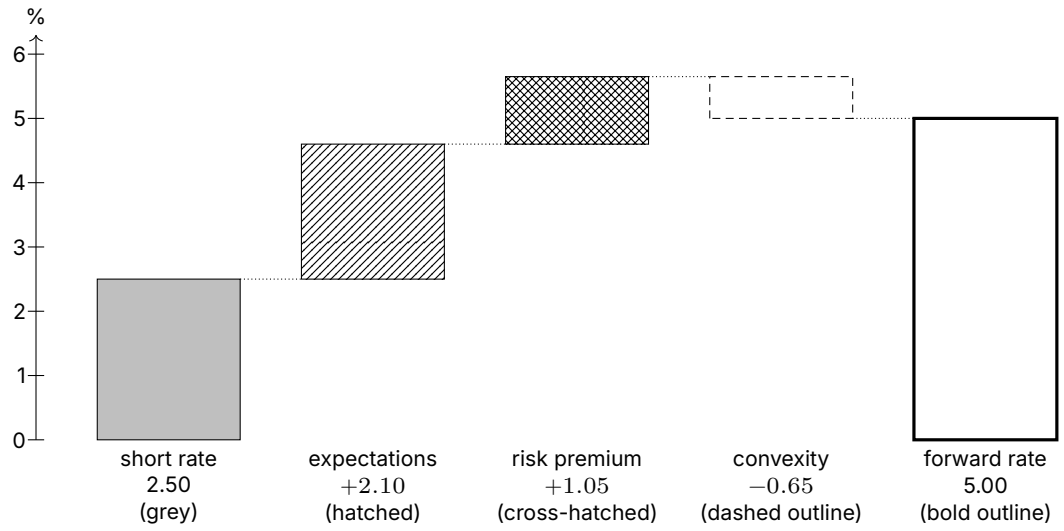


Figure 5. The forward rate as a waterfall, in %: short rate 2.50, expectations $7 \times 0.30 = 2.10$, risk premium $7 \times 0.15 = 1.05$, convexity -0.65, forward rate 5.00. Expectations and the risk premium add in proportion to duration; convexity subtracts. Under stress the premium bar grows and the convexity bar deepens, pulling in opposite directions (shape D18 S2).

Formula card: MR-13 d

Return decomposition	$dP/P = f(T)\,dt - D\,dr + \frac{1}{2}C\sigma^2\,dt$
Risk-neutral expected return	$E[dP/P] = r_0\,dt$
Risk-averse expected return	$E[dP/P] = r_0\,dt + \lambda D\,dt$ (no convexity term)
Sharpe ratio of a bond	$SR = \lambda/\sigma$
Forward rate	$f(T) = r_0 + D\,E[dr]/dt + \lambda D - \frac{1}{2}C\sigma^2$
Tree with a premium	price $= E[\text{date-1 price}]/(1 + r_0 + \lambda \times \text{duration carried})$

Exam traps

- **Adding convexity to expected return.** The convexity gain is offset by the lower forward rate: $E[dP/P] = r_0 + \lambda D$. Adding or subtracting $\frac{1}{2}C\sigma^2$ is the planted error.
- **Premium once, not times duration.** The premium is λD, not λ.
- **Sign in the forward-rate decomposition.** Expectations and premium enter with $+$, convexity with $-$. Convexity raises *returns* but lowers *forward rates*.
- **Component swap.** The forward rate is the passage-of-time return; volatility drives the convexity term; the risk premium is compensation for bearing interest rate risk.
- **Risk neutral means the same expected return for every bond**, long or short: the short rate.
- **Duration on a tree.** Over one period a bond carries the rate risk of what it becomes (a three-year zero carries two years).

Practice questions: MR-13 d

Question MR-13 d.1 Bank Q646 • Schweser Module Quiz 13.2 Q3

Which of the following statements represents the risk premium component in the decomposition of forward rates?

A. The return due to the passage of time.

B. The expected volatility in interest rates.

C. The compensation investors demand for convexity effects.

D. The compensation investors demand for bearing interest rate risk.

Question MR-13 d.2 Bank Q644 • BT notes P2.T5.23.3.3 (rebuilt)

The forward rate for term T can be approximated as $f(T) \approx r_0 + E[\Delta r] \times D + \lambda \times D -$ (convexity effect), where D is duration and the convexity effect increases with volatility. Which of the following statements about this decomposition is NOT correct?

A. The forward rate increases with the expected change in the short-term rate, scaled by the bond's duration.

B. The forward rate increases with the risk premium in proportion to duration.

C. Greater interest rate volatility lowers the forward rate through the convexity effect.

D. Because convexity raises a bond's return, it also raises the forward rate, so greater volatility steepens the forward curve.

Question MR-13 d.3 New • built from GARP MR Ch.13, eq. 13.20; Schweser LO 13.d example

A fixed-income strategist decomposes the forward rate for a term whose zero-coupon bond has a duration of 7. The short-term rate is 2.5%, the expected change in the short-term rate is 0.3% per year, the risk premium is 15 basis points per year of duration risk, and the convexity effect is 0.65%. Which of the following is closest to the forward rate?

A. 3.20%

B. 4.10%

C. 5.00%

D. 6.30%

Question MR-13 d.4 Bank Q641 • BT P2.T5.25.2.1 (rebuilt)

The short-term interest rate is 3.0%, the risk premium is 15 basis points per unit of duration, and a bond has a duration of 6 and a convexity of 50. The volatility of interest rate changes is 1.5% per year. What is the expected return on the bond for a risk-averse investor over the next year?

A. 3.15%

B. 3.34%

C. 3.90%

D. 4.46%

Question MR-13 d.5 Bank Q640 • BT Mock Q21 (rebuilt)

The one-year interest rate is 8.0%. Risk-averse investors require compensation of 50 basis points per year of duration risk, where the duration risk borne over a one-year holding period is that of the bond as it will exist at the end of the period (for example, a three-year zero-coupon bond becomes a two-year zero-coupon bond). What is the expected return over the next year on a three-year zero-coupon bond?

A. 8.00%

B. 8.50%

C. 9.00%

D. 9.50%

Question MR-13 d.6

Bank Q642 • BT P2.T5.25.2.2 (rebuilt, key corrected)

The expected one-year return on a bond can be decomposed into the forward rate, the effect of expected rate changes (proportional to duration), and a convexity effect (proportional to volatility). Suppose investors are risk neutral (risk premium $\lambda = 0$, as under the pure expectations hypothesis). How does the expected one-year return on a 10-year zero-coupon bond compare with that on a 1-year zero-coupon bond?

A. It is higher by $\lambda\times$ duration, because the 10-year bond has more interest rate risk.

B. It is higher, because the 10-year bond has greater convexity that earns extra return from rate volatility.

C. It is the same: both have an expected return equal to the short-term rate r_0.

D. It is lower, because the 10-year bond's forward rate is reduced by the convexity effect.

Question MR-13 d.7

Bank Q645 • Mock 2 Q64

A risk analyst uses a decision tree framework to estimate the price of zero-coupon bonds. The analyst assumes that investors are risk averse and require a risk premium of 50 basis points for each year of interest rate risk. What is the price that he should calculate for a 2-year zero-coupon bond with a face value of USD 100, given expected 1-year rates in one year of 10% in the up node and 6% in the down node (equal probabilities)? Assume the 1-year spot rate is 8%.

A. USD 85.37

B. USD 92.59

C. USD 92.17

D. USD 85.76

Solutions • MR-13 d

Solution MR-13 d.1

Answer: D

Bank Q646 • Schweser Module Quiz 13.2 Q3

Step 1 – Name the pieces

Return = forward rate (passage of time) − duration × rate change + convexity × volatility. Forward rates = expectations + risk premium − convexity.

Step 2 – Identify the premium

Risk-averse investors require $r_0 + \lambda D$: λ is the compensation per unit of duration for bearing interest rate risk.

$\Rightarrow$ D.

Interpretation

Each distractor names a real component, just not this one: a sibling-swap question.

Why the other options are wrong

A is incorrect. The passage-of-time return is the forward rate itself.

B is incorrect. Rate volatility drives the convexity component.

C is incorrect. Convexity is a separate component, a Jensen's inequality effect; there is no premium "for convexity".

LO: MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eqs. 13.15, 13.18 and 13.20

Solution MR-13 d.2

Answer: D

Bank Q644 • BT notes P2.T5.23.3.3

Step 1 – Read each term's sign

Forward rate = short rate + expectations term + risk premium term − convexity term.

Step 2 – Test the statements

A: expectations term $E[\Delta r] \times D$, true. B: risk premium λD, true. C: convexity is subtracted and grows with volatility, true. D: convexity raises *returns* but enters the forward rate with a minus sign.

$\Rightarrow$ D is the false statement.

Interpretation

More volatility makes the forward curve bend *down*, not steepen. This is the D18 S2 tension: the premium pushes long forwards up, convexity pulls them down.

Why the other options are wrong

A is incorrect. True statement (expectations term scaled by duration).

B is incorrect. True statement (risk premium in proportion to duration).

C is incorrect. True statement (convexity lowers forwards, more with volatility).

LO: MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eq. 13.20

Solution MR-13 d.3

Answer: C

New • GARP eq. 13.20; Schweser LO 13.d

Step 1 – Expectations term

$D \times E[dr]/dt = 7 \times 0.3\% = 2.10\%$.

Step 2 – Risk premium term
$\lambda D = 7 \times 0.15\% = 1.05\%$.

Step 3 – Assemble

$$f = 2.50\% + 2.10\% + 1.05\% - 0.65\% = 5.00\%$$

$\Rightarrow$ Both expectations and premium scale with duration; convexity comes off.

Interpretation
Figure 5 shows the same numbers as a waterfall. Convexity enters the forward rate with a minus sign even though it adds to bond returns.

Why the other options are wrong
A is incorrect. Did not scale the expected rate change by duration: $2.5 + 0.3 + 1.05 - 0.65 = 3.20\%$.
B is incorrect. Did not scale the risk premium by duration: $2.5 + 2.1 + 0.15 - 0.65 = 4.10\%$.
D is incorrect. Added the convexity effect instead of subtracting it: $2.5 + 2.1 + 1.05 + 0.65 = 6.30\%$.

LO: MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eq. 13.20; SchweserNotes 2026 Book 1, LO 13.d

Solution MR-13 d.4 — Answer: C

Bank Q641 • BT P2.T5.25.2.1

Step 1 – Risk-averse expected return
$E[\text{return}] = r_0 + \lambda D = 3.00\% + 0.15\% \times 6 = 3.00\% + 0.90\% = 3.90\%$.

Step 2 – What about convexity?
$\frac{1}{2}C\sigma^2 = 0.5 \times 50 \times 0.015^2 = 0.5625\%$ is *not* added or subtracted: the convexity gain in return is offset by the lower forward rate it implies.
$\Rightarrow$ 3.90%. Convexity and volatility are red herrings.

Interpretation
"No net advantage of convexity": the expected return condition has only the short rate and the duration premium.

Why the other options are wrong
A is incorrect. Added the premium once, ignoring duration: $3.00\% + 0.15\% = 3.15\%$.
B is incorrect. Subtracted the convexity term: $3.90\% - 0.56\% = 3.34\%$.
D is incorrect. Added the convexity term: $3.90\% + 0.56\% = 4.46\%$, double counting.

LO: MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eqs. 13.18 and 13.20

Solution MR-13 d.5 — Answer: C

Bank Q640 • BT Mock Q21

Step 1 – Duration risk carried
After one year the three-year zero is a two-year zero, so the rate risk carried over the year is $D = 2$.

Step 2 – Expected return
$8.00\% + 2 \times 0.50\% = 9.00\%$.

⇒ Two units of duration risk, priced at 50 bp each.

Interpretation

The stem defines the convention explicitly; it matches the tree logic in Section 2 (a two-year zero carries one year of risk and earns $r_0 + \lambda$).

Why the other options are wrong

A is incorrect. The risk-neutral return r_0, with no premium.
B is incorrect. One year of duration risk, as if the bond became a one-year zero.
D is incorrect. Used the current three-year maturity instead of the two years remaining at the end of the period.

LO: MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eq. 13.18

Solution MR-13 d.6 — Answer: C

Bank Q642 • BT P2.T5.25.2.2

Step 1 – Expected return for any investor

$E[\text{return}] = r_0 + \lambda D$.

Step 2 – Risk neutral

$\lambda = 0$, so $E[\text{return}] = r_0$ regardless of duration or maturity.

Step 3 – Convexity nets out

The 10-year zero's convexity gain is exactly offset by the lower forward rate convexity implies.
⇒ Both bonds have expected return r_0.

Interpretation

Risk neutrality means every bond offers the short rate. Convexity changes how the return is split between components, not its expected total.

Why the other options are wrong

A is incorrect. With $\lambda = 0$, λD is zero.
B is incorrect. Convexity raises the return component but lowers the forward rate by the same amount.
D is incorrect. Convexity lowers the forward rate but raises the return component; the expected return is unchanged.

LO: MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eqs. 13.17 and 13.20

Solution MR-13 d.7 — Answer: A

Bank Q645 • Mock 2 Q64

Step 1 – Date-1 prices of the then one-year zero

Up: $100/1.10 = 90.909$; down: $100/1.06 = 94.340$; expected $= 92.625$.

Step 2 – Required return over the first year

The bond carries one year of duration risk over the year: $8\% + 0.50\% = 8.50\%$.

Step 3 – Price

$92.625/1.085 = \text{USD } 85.37$.
⇒ Below the risk-neutral price of 85.76: a premium lowers prices.

Interpretation

Risk-averse pricing discounts the expected future price at the short rate plus the premium for the

duration risk carried.

Why the other options are wrong

B is incorrect. $100/1.08$: ignores the tree and discounts only one period.
C is incorrect. $100/1.085$: one period at the risk-adjusted rate, omitting the second period.
D is incorrect. $92.625/1.08 = 85.76$: the risk-neutral price, ignoring the premium.

LO: MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. • **Reference:** Tuckman and Serrat 4e, Ch.8, 13.3, eq. 13.18

MR-14 • The Art of Term Structure Models: Drift • Vardeez: VERY IMPORTANT (calculations super important; this LO is concept only)

MR-14 c

P3 CONCEPT

Describe methods for addressing the possibility of negative short-term rates in term structure models.

Workbook: D1 CIR and Lognormal Short-Rate Models • Tier D • Map #39

WHAT IS TESTED HERE

Why P3. A Tier D workbook topic hits it: D1 (Map #39, 4 of 16 sources), shape S3 on non-negativity. Vardeez's pointers for MR-14 sit on the Ho-Lee and Vasicek calculations, not on this LO.

Reported exam shape

- **D1·S3** Non-negativity: CIR and lognormal versus normal models. Tested inside the sibling-set question worked as Question MR-14 f.7 in this book (workbook D1.2).

What the questions look like: why negative rates are a problem, when the problem matters, and the two remedies with their costs: change the distribution (lognormal, CIR) or floor the tree at zero (shadow rates).

1. Why negative rates are a problem for normal models

In normal (Gaussian) models such as Model 1, $dr = \sigma\, dw$, the terminal distribution of the short rate is normal, so there is always some probability of a negative rate.

- **Economics.** People would not lend at a materially negative rate when they can hold cash and earn zero. (A slightly negative rate is possible if a deposit or security is safer or more convenient than holding cash.)
- **Horizon.** The probability grows with the horizon, because the terminal standard deviation is $\sigma\sqrt{T}$. With $r_0 = 6.18\%$ and $\sigma = 1.13\%$: after one year the distribution, three standard deviations either side, stays well above zero. After 10 years the standard deviation is $1.13\% \times \sqrt{10} = 3.573\%$, so a negative shock of only $6.18/3.573 = 1.73$ standard deviations takes the rate below zero (about a 4.2% chance under the normal).
- **Level of rates.** The problem is much more acute when rates are low.
- **The application decides.** For securities that depend mostly on the *average* rate path, such as coupon bonds, a small probability of negative rates usually does not rule a model out. For securities *asymmetrically* sensitive to low rates it is dangerous: a 10-year call to buy a long bond at a yield of 0% would be valued far too high, because the model puts too much probability on negative rates.

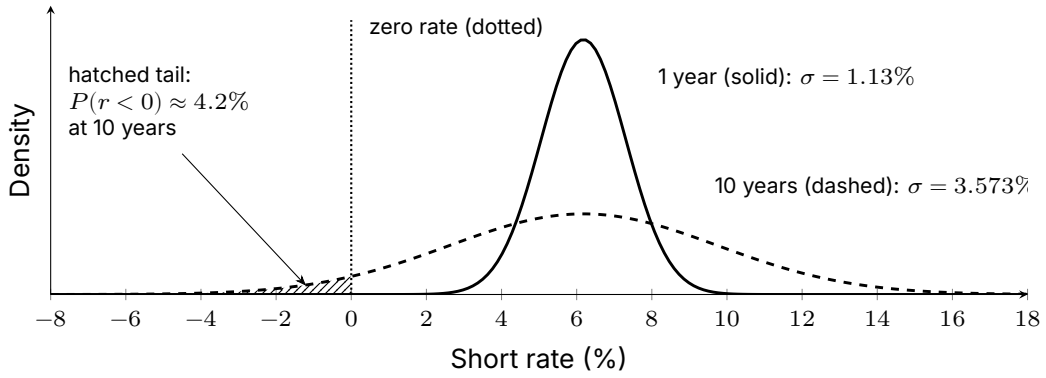


Figure 1. Model 1 terminal distributions from $r_0 = 6.18\%$, $\sigma = 1.13\%$ per year (GARP's example). At one year (solid) essentially no mass lies below zero; at ten years (dashed) the hatched tail below zero is about 4.2%.

2. The two remedies

	1. Change the distribution	2. Floor the tree at zero (shadow rates)
How	Use a distribution that is non-negative: lognormal rates, or CIR (chi-squared terminal distribution)	Build the tree with any desired distribution, then set every negative rate to zero (Black 1995)
Effect	Changes the dynamics across the *entire* range of rates	Change is *localised* around zero and negative rates
Cost	May give volatilities (and skew) that are unacceptable for the purpose	The observed rate's distribution is no longer the postulated one near zero
Best when	The new distribution suits the whole range of strikes and rates	Rates are very low (Vardeez: preferred there)

Shadow rates

In the floored tree the original rates are **shadow rates**; the floored rates are **observed rates**. When the observed rate hits zero it can **stay there for a while** until the shadow rate crosses back above zero. The justification: the *observed* rate is constrained to be positive only because investors can hold cash; the *shadow* rate, the result of aggregate saving, investment and consumption decisions, may well be negative.

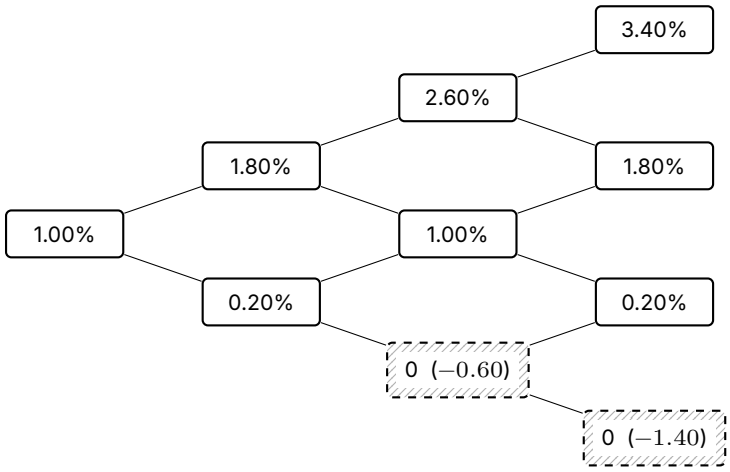


Figure 2. Illustrative normal tree with a low starting rate. Two nodes would be negative; flooring sets the observed rate to zero there while the shadow rate stays negative. Only the nodes near and below zero change.

How the distribution remedy works (link to MR-15)

- **CIR**, $dr = k(\theta - r)dt + \sigma\sqrt{r}\,dw$: basis-point volatility $\sigma\sqrt{r}$ is zero when the rate is zero, and the drift $k\theta$ is positive there, so the rate cannot go negative.
- **Lognormal** (Model 4, Salomon Brothers, Black-Karasinski): the rate moves multiplicatively, $r\,e^x > 0$, so every node stays positive.
- The price: basis-point volatility now depends on the level of rates, and the shape of the terminal distribution changes everywhere, which matters for out-of-the-money options.
- A trader may still *accept* a normal model if constant volatility suits the market, negative rates barely affect the securities, and computational simplicity is valued.

Exam traps

- **Local versus global (sibling swap).** Flooring changes the distribution only near zero; changing the distribution changes dynamics, including volatilities, everywhere.
- **Horizon polarity.** Negative rates become *more* likely as the horizon lengthens and as the starting rate falls.
- **Shadow versus observed (name swap).** The original tree holds the shadow rates; the floored tree holds the observed rates. The shadow rate can be negative.
- **More time steps do not help.** A finer normal tree converges to a normal distribution, which still has mass below zero.
- **Which securities suffer.** Options sensitive to very low rates are mispriced most; coupon bonds, which depend on the average path, much less.
- **Recombining is not mean reversion.** Model 1's recombining tree does not pull rates back toward r_0.

Practice questions: MR-14 c

Question MR-14 c.1

Bank Q669 • AnalystPrep Q.1652 (rebuilt)

In term structure models with normally distributed rates, the short-term rate can become negative. Which of the following statements best explains why negative rates are a drawback of such models?

A. Lenders would accept negative rates because lending at a negative rate removes all credit risk.

B. Investors can instead hold cash and earn a zero rate, so lending at a materially negative rate makes little economic sense.

C. Negative rates are harmless because they always revert to positive levels within one period.

D. Negative rates occur only for short horizons, so the problem is smaller as the investment horizon lengthens.

Question MR-14 c.2

New • built from GARP MR Ch.14, 14.2

A fixed-income analyst calibrates Tuckman's Model 1 (normal rates, no drift) with $r_0 = 6.18\%$ and $\sigma = 1.13\%$ per year, so the terminal standard deviation of the short rate is 1.13% after one year and about 3.57% after ten years. She will use the model to value (i) a 10-year coupon bond and (ii) a 10-year option to buy a long-term bond at a yield of 0%. Which of the following statements is correct?

A. Negative rates can be ignored for both securities, because the one-year terminal distribution, three standard deviations either side of 6.18%, does not reach zero.

B. At ten years a negative shock of only about 1.73 standard deviations takes the rate below zero; this matters most for the option, which the model overvalues, and much less for the bond.

C. At ten years a negative shock of only about 1.73 standard deviations takes the rate below zero; this matters most for the bond, whose value depends on the average rate path, and little for the option.

D. The probability of negative rates falls as the horizon lengthens, because the recombining tree pulls the short rate back toward its starting level of 6.18%.

Question MR-14 c.3

Bank Q668 • BT P2.T5.303.3 (rebuilt)

Analyst Susan is evaluating a normally distributed short-rate model and wants to avoid negative short-term rates in the model. Each of the following approaches can help her EXCEPT:

A. Using a lognormal distribution for the short rate.

B. Using a chi-squared distribution for the short rate.

C. Adjusting the interest rate tree so that any negative rate is set to zero.

D. Increasing the number of time steps so the terminal distribution approaches a continuous normal distribution.

Question MR-14 c.4 Bank Q717 • AnalystPrep Q.1676 (rebuilt; re-filed from LO 15.d)

In the CIR model, basis-point volatility equals $\sigma\sqrt{r}$, which is zero when the short rate is zero; together with a positive drift at $r = 0$, this prevents negative rates. What is the main weakness of models with constant basis-point volatility (such as Vasicek or Ho-Lee) that this CIR feature addresses?

A. Shocks have the same size near zero as at high rate levels, so the short rate can become negative.

B. Constant basis-point volatility makes the short rate deterministic, so no interest rate risk is modelled.

C. Constant basis-point volatility prevents the model from including mean reversion toward a long-run rate.

D. Basis-point volatility rises roughly in proportion to the rate level, overstating risk when rates are high.

Question MR-14 c.5 Bank Q667 • AnalystPrep Q.1653 (rebuilt)

A normally distributed short-rate tree allows negative rates. A common fix is to construct the tree with the desired distribution and then force every negative rate to zero. Which of the following statements about this method is correct?

A. It replaces the normal distribution with a lognormal distribution at every node, so all rates in the tree change.

B. It changes the original distribution only where rates are very low, leaving rates away from zero unchanged.

C. It introduces skewness and inappropriate volatilities across the entire distribution of rates.

D. It removes the negative rate problem only when the investment horizon is long and the current rate is high.

Question MR-14 c.6 New • built from GARP MR Ch.14, 14.2 (Black 1995)

A rates strategist builds a normal short-rate tree and then sets every negative node to zero, following Black's shadow-rate approach. Which of the following statements about this method is correct?

A. The rates in the original tree are the observed rates and the floored rates are the shadow rates, so the shadow rate can never fall below zero.

B. When the observed rate reaches zero it can stay there for several periods, until the shadow rate moves back above zero.

C. The justification is that aggregate saving and investment decisions can never produce a negative equilibrium rate, so the floor recovers the true rate.

D. Flooring changes the dynamics of the rate across the entire range of rates, which is why it is less attractive than switching to a lognormal model.

Question MR-14 c.7 New • built from GARP MR Ch.14, 14.2 and Vardeez

Rates are close to zero and a desk needs a short-rate model to price interest rate options across a wide range of strikes. One modeller proposes switching from a normal model to a lognormal model to remove negative rates; a colleague proposes keeping the normal model and flooring negative rates at zero. Which assessment is most appropriate?

A. The lognormal switch is preferable, because it rules out negative rates without changing the model's volatility at any level of rates.

B. Flooring is preferable here, because its effect is localised around zero, whereas changing the distribution alters rate dynamics, including volatilities, across all rates.

C. Neither is needed, because adding time steps to the normal tree makes the terminal distribution continuous and so removes the negative rates.

D. The two are equivalent, because any distribution that rules out negative rates has the same volatility structure as a floored normal model.

Solutions • MR-14 c

Solution MR-14 c.1 — Answer: B

Bank Q669 • AnalystPrep Q.1652

Step 1 – Compare lending with holding cash

Lending 100 and receiving less than 100 back is dominated by holding cash at a 0% return.
$\Rightarrow$ A materially negative rate makes little economic sense; a model that generates one is unrealistic.

Step 2 – Allow for the footnote

A slightly negative rate can be rationalised if a deposit or security is safer or more convenient than cash.

Interpretation

The problem grows with the horizon and with lower starting rates, which rules out D.

Why the other options are wrong

A is incorrect. A negative rate does not remove credit risk; the lender simply loses principal for certain.
C is incorrect. Nothing in a normal model forces a return to positive rates within one period.
D is incorrect. Reversed: the terminal standard deviation grows with $\sqrt{T}$, so negative rates become *more* likely at long horizons.

LO: MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models.
• **Reference:** Tuckman and Serrat, Fixed Income Securities 3e, Ch.9 (GARP MR Ch.14), Model 1 and footnote 2

Solution MR-14 c.2 — Answer: B

New • built from GARP MR Ch.14

Step 1 – Horizon

At one year, 6.18% is $6.18/1.13 = 5.5$ standard deviations above zero: negligible. At ten years, $6.18/3.573 = 1.73$ standard deviations.
$\Rightarrow$ A modest negative shock now takes the rate below zero (Figure 1).

Step 2 – Which security suffers

The coupon bond depends mostly on the average rate path, so a small probability of negative rates does not rule the model out. The option to buy at a 0% yield is asymmetrically sensitive to very low rates, so the model assigns it much too high a value.

Interpretation

Whether negative rates make a model unusable "depends on the application". This is GARP's own pair of examples.

Why the other options are wrong

A is incorrect. Uses only the one-year horizon; both securities run for ten years, where the probability is material.
C is incorrect. Sibling swap: it is the option, not the coupon bond, that is asymmetrically sensitive to low rates.
D is incorrect. Reversed, and recombining is not mean reversion: Model 1 has no force pulling rates back to r_0.

LO: MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models.
• **Reference:** Tuckman and Serrat, Ch.9 (GARP MR Ch.14), Model 1 and Figure 14.1

Solution MR-14 c.3 — Answer: D

Bank Q668 • BT P2.T5.303.3

Step 1 – List the two remedies

(1) A non-negative distribution: lognormal, or chi-squared (CIR). (2) Force negative tree rates to zero.

$\Rightarrow$ A and B are remedy (1); C is remedy (2).

Step 2 – Test D

A finer tree converges to a continuous *normal* distribution, which still has mass below zero.

Interpretation

Note the "EXCEPT": three options are valid remedies; more time steps only improve the approximation to the same normal distribution.

Why the other options are wrong

A is incorrect. A valid remedy: lognormal rates cannot be negative (at the cost of different volatility behaviour).

B is incorrect. A valid remedy: the CIR terminal distribution (non-central chi-squared) is non-negative.

C is incorrect. A valid remedy: the shadow-rate floor.

LO: MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models.

• **Reference:** Tuckman and Serrat, Ch.9; SchweserNotes 2026 Book 1, Reading 14

Solution MR-14 c.4 — Answer: A

Bank Q717 • AnalystPrep Q.1676

Step 1 – Constant basis-point volatility

In Vasicek or Ho-Lee the standard deviation of dr is σ whatever the level of r. Near zero, a normal shock of size $\sigma\sqrt{dt}$ can still push the rate below zero.

Step 2 – What CIR changes

Volatility $\sigma\sqrt{r}$ vanishes at $r = 0$ while the drift $k\theta$ is positive, so the rate cannot turn negative.

$\Rightarrow$ CIR is an example of remedy (1): change the distribution.

Interpretation

This is the link between MR-14 c and MR-15 d: CIR and lognormal models solve negative rates by making volatility depend on the rate level.

Why the other options are wrong

B is incorrect. Constant does not mean zero; these models are stochastic.

C is incorrect. Vasicek has constant volatility *and* mean reversion, so the two are compatible.

D is incorrect. That describes the lognormal model (volatility σr), not constant-volatility models.

LO: MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models.

• **Reference:** Tuckman and Serrat, Ch.10 (GARP MR Ch.15), CIR model

Solution MR-14 c.5 — Answer: B

Bank Q667 • AnalystPrep Q.1653

Step 1 – What the floor does

Every node with a positive rate keeps its original value; only nodes that would be below zero are reset to 0 (Figure 2).

⇒ The distribution changes only in a very low-rate region.

Step 2 – Contrast with remedy (1)

Changing the distribution (lognormal, CIR) reshapes the whole distribution, which is the cost GARP attaches to it.

Interpretation

That localisation is why the floor is attractive when rates are very low.

Why the other options are wrong

A is incorrect. Describes remedy (1), not the zero floor.

C is incorrect. Skewness and inappropriate volatilities are the drawbacks of non-negative distributions, not of the floor.

D is incorrect. Reversed: the problem is worst for long horizons and *low* current rates, which is when the floor matters most.

LO: MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models.

• **Reference:** Tuckman and Serrat, Ch.9 (GARP MR Ch.14), Section on negative rates

Solution MR-14 c.6 **Answer: B**

New • built from GARP MR Ch.14 (Black 1995)

Step 1 – Names

The original tree holds the *shadow* rates; the adjusted tree holds the *observed* rates.

Step 2 – Dynamics at zero

When the observed rate hits zero it can remain there until the shadow rate crosses back to a positive value.

⇒ B is GARP's description.

Step 3 – Justification

The observed rate is constrained to be positive only because investors can hold cash; the shadow rate, the result of aggregate saving, investment and consumption, may well be negative.

Interpretation

The change from flooring is localised around zero and negative rates; changing the distribution changes dynamics across the entire range.

Why the other options are wrong

A is incorrect. Name swap: the floored rates are the observed rates; the shadow rate *can* be negative.

C is incorrect. Reversed justification: the shadow (equilibrium) rate may well be negative; the floor reflects the cash alternative.

D is incorrect. Sibling swap: global change in dynamics is the cost of changing the distribution, not of flooring.

LO: MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models.

• **Reference:** Tuckman and Serrat, Ch.9 (GARP MR Ch.14); Black, "Interest Rates as Options", 1995

Solution MR-14 c.7 **Answer: B**

New • built from GARP MR Ch.14 and Vardeez

Step 1 – Cost of the lognormal switch

Building the model around a distribution that rules out negative rates "may result in volatilities that are unacceptable for the purpose at hand", and it changes dynamics across the entire range

of rates, which matters when pricing options at many strikes.

Step 2 – Cost of the floor

Only rates near and below zero change.

$\Rightarrow$ With rates close to zero, the floor is the less disruptive fix; Vardeez notes it is preferred in very low-rate environments.

Interpretation

The choice depends on purpose. Here the desk prices options across strikes, so it should avoid a fix that distorts volatility at every rate level.

Why the other options are wrong

A is incorrect. Absolute claim: a lognormal model does change volatility behaviour (basis-point volatility proportional to r).

C is incorrect. A finer normal tree still has mass below zero.

D is incorrect. The two methods have different volatility structures; that is the whole trade-off.

LO: MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models.

• **Reference:** Tuckman and Serrat, Ch.9 (GARP MR Ch.14); Vardeez notes MR-14 c